A Dichotomy for Complex Boolean Holant with Binary Disequality

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Abstract

We classify Boolean Holant problems with a supplied binary disequality signature and an arbitrary fixed finite set of algebraic complex-valued signatures. Each problem is polynomial-time computable or #P-hard, and the tractability criterion is decidable from exact algebraic descriptions. After a fixed change of coordinates, the criterion has four alternatives: tensor products of signatures of arity at most two, or one common affine, product, or local-affine presentation. The proof follows the smallest nonbinary tensor-prime signature together with the finite projective group of realizable binary transfers. Pair contractions connect this local information to a factorization, a smaller prime, or a common tractable presentation. The resulting cyclic, dihedral, Klein-four, and Platonic cases combine structural arguments with specified exact finite classifications. We also prove a signature-set dichotomy when an endpoint-nondegenerate, tensor-prime eight-vertex signature is available; this result supplies the four-port boundary of the main proof.

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Code Availability

The accompanying Lean formalization contains the public mathematical specifications in `Holant/Statements.lean`, their proofs in `Holant/Proofs.lean`, and exact-input recognition results in `Holant/Decidability.lean`. Supplementary lemmas and exact finite certificates are included through `Holant/Library.lean`. The statement-only reference `Verification/Challenge.lean` and `Verification/comparator.json` support independent comparison using Lean's Comparator. Build and verification instructions are given in the repository README.

The original Python companion repository is *Holant-X-Dichotomy*. References below to `certificates/` and its Python scripts refer to that companion repository. The Lean release supplies its own exact certificates under `Holant/Certificates/`; the Python companion is not required to build or check the Lean proofs.

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1 Introduction

A Boolean signature of arity n is a function $f : \{0,1\}^n \to \mathbb{C}$. In a Holant instance, a graph vertex of degree n carries such a function, with its incident edges ordered. For a fixed finite set $\mathcal{F}$ of signatures, the problem Holant($\mathcal{F}$) evaluates

$$\sum_{\sigma : E \to \{0,1\}} \prod_{v \in V} f_v(\sigma|_{E(v)}), \qquad f_v \in \mathcal{F}.$$

Here $E(v)$ is the ordered list of edge incidences at v; Section 2.3 gives the conventions for loops and multiple edges. The graph and its vertex labels form the input; the set $\mathcal{F}$ is fixed. This framework originated in holographic algorithms [Val08, CLX09]. A complexity dichotomy specifies which fixed signature sets admit polynomial-time evaluation and proves #P-hardness for all remaining sets.

We study arbitrary algebraic complex-valued signatures with the binary disequality relation supplied:

$$X = \mathrm{NEQ}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad I = \mathrm{EQ}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Supplying X means that any number of degree-two vertices may carry this signature. It enforces opposite values on its two incident edges. Our main result is a decidable dichotomy for Holant($\mathcal{F}, X$), where Holant($\mathcal{F}, X$) abbreviates Holant($\mathcal{F} \cup \{X\}$).

Background and scope. Several earlier classifications prescribe auxiliary signatures. The model Holant* supplies every unary signature [CLX11], whereas Holant^c supplies the two pinnings $\Delta_0 = [1,0]^{\mathsf{T}}$ and $\Delta_1 = [0,1]^{\mathsf{T}}$. The latter model was classified for symmetric complex signatures by Cai, Huang, and Lu [CHL12], for arbitrary real signatures by Cai, Lu, and Xia [CLX18], and for algebraic complex signatures by Backens [Bac21, Bac25]. The real-valued classification introduced the local-affine tractable family $\mathscr{L}$, which also occurs here. Binary disequality supplies a relation between two bits, while a pinning fixes one bit. Accordingly, our reductions account explicitly for any unary information extracted from the supplied binary relation.

Without a prescribed auxiliary signature, dichotomies are known for complex symmetric signatures [CGW16], nonnegative signatures [LW18], and arbitrary real Boolean signatures [SC20]. Meng, Wang, Xia, and Zheng classified the complex case when a nonzero odd-arity signature is present and established a tensor-factor extraction theorem [MWXZ25]. Their two support-defined tractable alternatives have polynomial-time algorithms by Guan, Shao, and Shi [GSS26]. The main remaining work in our proof concerns sets whose nonbinary tensor-prime signatures all have even arity.

There is also a precedent for prescribing disequality in counting CSP. In #CSP$_d$, each variable occurs a number of times divisible by d. Cai, Fu, and Shao classified #CSP$_d$(NEQ$_2$, $\mathcal{F}$) for complex signatures [CFS20b, full version, Section 5.1 and Theorem 5.3]; Shao and Cai used its hardness conclusion in their real Holant dichotomy [SC20, full version, Theorem 2.24]. Here the instance is an arbitrary Holant graph: the supplied resource is NEQ$_2$, with no degree-divisible family of equality constraints.

Two presentations and the main theorems. It is useful to place X on every edge and give each incident vertex its own bit. We denote this presentation by K-Holant. Joining two bits a, b then contributes X_{ab}; in ordinary Holant the corresponding edge contributes I_{ab}. Let $\equiv_{\mathrm{T}}$ denote

polynomial-time Turing equivalence and fix

$$K = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \qquad K^{\mathsf{T}} K = X, \qquad K(\mathcal{S}) = \{K^{\otimes \operatorname{arity}(f)} f : f \in \mathcal{S}\}.$$

The basis change acts on every vertex and on the edge pairing, giving K-Holant($\mathcal{S}$) $\equiv_{\mathrm{T}}$ Holant($K(\mathcal{S})$). A separate wire substitution gives

$$\text{Holant}(\mathcal{F}, X) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}, I). \tag{1}$$

In this second equivalence the signatures in $\mathcal{F}$ retain their entries: paths of degree-two X-vertices are replaced according to their parity, and closed components contribute known scalars. Lemma 5.1 proves both substitutions.

The predicate Tract in Section 2.10.3 records the seven tractable alternatives in the odd-arity classification, with the polynomial-time improvements just cited. On a set containing $Z = K^{\otimes 2} I = \operatorname{diag}(1, -1)$, it reduces to the four-alternative predicate Tract_X by Lemma 2.12. These alternatives are tensor products of signatures of arity at most two, or transformability into the affine, product, or local-affine class. For a transformable set, one basis must work for the edge and every signature; the precise definitions and decision procedure are in Sections 2.10.1 and 2.10.6.

Theorem 1.1 (Binary-disequality Holant dichotomy)**.** *For every finite set of algebraic complex-valued Boolean signatures $\mathcal{F}$, the problem* Holant($\mathcal{F}, X$) *is polynomial-time computable when* $\mathsf{Tract}_X(K(\mathcal{F} \cup \{I\}))$ *holds and is* #P*-hard otherwise. The tractability criterion is decidable from exact algebraic input.*

By Equation (1), the ordinary-Holant statement is equivalent to the following formulation with native disequality edges.

Theorem 1.2 (Literal-equality K-Holant dichotomy)**.** *For every finite set $\mathcal{F}$ of algebraic complex-valued Boolean signatures with $I = \mathrm{EQ}_2 \in \mathcal{F}$, the problem* K-Holant($\mathcal{F}$) *is polynomial-time computable when* $\mathsf{Tract}_X(K(\mathcal{F}))$ *holds and is* #P*-hard otherwise. The tractability criterion is decidable from exact algebraic input.*

The proof also needs a four-port classification without a supplied I-vertex. A quaternary signature q is *endpoint-nondegenerate* when

$$q(x) = 0 \quad \text{for odd } \operatorname{wt}(x), \qquad q(0000)q(1111) \neq 0.$$

Here $\operatorname{wt}(x)$ is the number of ones in the Boolean word x. A nonzero signature is *tensor-prime* if it has no factorization $a \otimes b$ on two nonempty blocks of ports, allowing a port permutation and a nonzero scalar.

Theorem 1.3 (Quaternary K-Holant dichotomy)**.** *Let $\mathcal{F}$ be a finite set of algebraic complex-valued Boolean signatures. Suppose that $\mathcal{F}$ contains an endpoint-nondegenerate tensor-prime quaternary signature. Then* K-Holant($\mathcal{F}$) *is polynomial-time computable if* $\mathsf{Tract}(K(\mathcal{F}))$ *holds and is* #P*-hard otherwise. The tractability criterion is decidable from exact algebraic input.*

Section 4 proves this signature-set theorem using the single-signature eight-vertex boundary of Cai and Fu [CF23, Theorem 3.1]. Tensor primality removes its factorization outcome. The full predicate Tract is needed here because the signature set need not contain the binary anchor I. For the other two theorems, Section 10.1 proves the structural classification used in the hardness argument, and Section 10 then gives a separate proof of each stated dichotomy. Its literal-equality proof uses the given set $\mathcal{F}$; its binary-disequality proof applies that result to $\mathcal{F} \cup \{I\}$ through Equation (1). The decision assertion in all three statements is supplied by Theorem 2.19.

The proof's main obstacle. A contraction may reveal a useful small tensor while giving little information about the other signatures. A tractable conclusion, however, requires one presentation of the entire signature set. We therefore retain the signatures already available whenever a gadget or a tensor factor is added. A realized gadget gives a direct reduction; the tensor-factor theorem gives a Turing reduction to the retained problem. Section 2.15 records these two operations and the common tractability witness they must preserve.

The first stage handles nonzero odd signatures, singular binaries, infinite-order binary transfers, and suitable equalities using Theorems 2.15 and 2.16 and Lemmas 3.1, 3.8 and 3.9. The weighted Eulerian-orientation and eight-vertex classifications provide further reductions [CFX18, CF23, CFS20a, MWX25, GSS26]. Weighted Eulerian orientations use X-edges and signatures supported on their central Hamming layer, so their statements can be applied in the same coordinates. At the four-port boundary the resulting signature is checked within the whole retained set. A common local-affine presentation is kept as such unless the hypotheses of Lemma 4.4 give an affine one.

For the remaining cases, let Λ be the retained signature set and let $f \in \Lambda$ have the smallest arity among its nonbinary tensor-prime signatures. An ordered binary with matrix B has transfer matrix $T_B = BX$, so chaining binaries multiplies their transfers. After the preceding reductions, the directly realizable projective transfer classes form a finite group

$$G \le \mathrm{PGL}_2(\overline{\mathbb{Q}}), \qquad x = [X] \in G, \qquad j = [Z] \in N_{\mathrm{PGL}_2(\overline{\mathbb{Q}})}(G).$$

Square brackets identify matrices differing by a nonzero scalar; $N(G)$ consists of projective matrices whose conjugation preserves G. The first mark comes from the supplied I-vertex, and the second describes reversal of the two ordered ports. This uses Xia's finite-group approach [Xia26, Section 4 and Table 2], together with the arity descent used in real Holant [SC20]. The relative position of x, j, as well as the abstract group, determines which matrices the subsequent constructions may use.

We examine the *cards* of f: tensors obtained by contracting two selected ports against available binary kernels, with allowed binary transformations on the remaining ports. Closing all such contractions and their factor decompositions either resolves the problem, lowers the chosen prime's arity, enlarges the complete group G, or leaves every card zero or a tensor product of group binaries on a matching of its ports. In the last case the pair (f, G) is stable in the precise sense of Section 5. The coefficient-field bound on finite groups and the finite factor worklist prove termination; Lemma 10.1 accounts for the additional constructions used in the individual cases.

Recovering a tensor from its cards. At a stable pair, every proper two-port contraction has a binary matching decomposition, although f itself is tensor-prime. This discrepancy supplies the rigidity argument. If the matchings agree, their common factors constrain the original tensor; if they disagree, a small alternating cycle localizes the discrepancy to four or six ports. The four marked group families use this information in different ways.

For dihedral groups, Section 6 combines the geometry of root-of-unity pencils with Boolean support constraints. A pencil is a family $u + tv$ generated by two fixed tensors. The permitted values of t constrain both its support and its coefficient ratios. A six-port classification isolates the possible small configurations; alternating-cycle localization and Laurent interpolation then recover binary factors at larger arities.

For proper cyclic groups, Section 7 first constructs a four-port output or a binary reflection outside the current group. At higher arity, the contraction supports reconstruct the first-order Reed–Muller code $RM(1, 3)$, whose words are evaluations of affine Boolean functions on $\mathbb{F}_2^3$. The subsequent network construction realizes sums over specified affine fibres. These sums test whether every retained signature has the same tractable presentation; Lemma 7.15 supplies their actual Holant realization.

For the two Klein-four forms, Section 8 uses different contraction bases. Standard Pauli contractions determine a tensor line above arity four, allowing a non-real tensor to be reduced to a non-real four-port tensor. If all retained signatures are real up to individual scalars, the real Holant dichotomy applies to the whole set. In the other form, the six-port classification and explicit crossed contractions exclude a stable nonbinary prime. The four-copy construction from H_6 to RM_8 connects this analysis to the tetrahedral case.

For Platonic groups, Section 9 represents binaries by four quaternion coordinates. Incidences of the resulting finite root configurations control the linear maps that send a kernel to its contracted card. A common matching gives a tensor factor; three different matchings on a projective line identify a four-port block and reduce the parent to six ports. Octahedral and icosahedral outputs then reach a four-port boundary or produce a new binary. For the tetrahedral group, one permutation class reaches that boundary; the other uses stabilizer codes to shorten the tensor while preserving failure of affine membership. An eight-port classification completes the induction to a common affine presentation of Λ.

The low-arity classifications that still require exact finite enumeration are stated in Appendix A, with their mathematical inputs, outputs, and completeness arguments. The general arity arguments and the reductions linking them are proved in the text.

Organization. Section 2 defines the signatures, coordinate changes, tractable classes, and mathematical tools used throughout. Sections 3 and 4 establish the unary, equality, and four-port reductions. Section 5 sets up the finite-group and contraction analysis; Sections 6 to 9 prove its four branches. Section 10 completes the main proofs.

2 Preliminaries

We fix the tensor and reduction conventions first, then state the tractable classes and imported dichotomies. The later subsections introduce the specialized geometric and stabilizer tools at the level used in the proofs.

2.1 Foundational algebraic conventions

2.1.1 Sets, maps, and exact fields

We write $[n] = \{1, \dots, n\}$, $[0] = \varnothing$, and $\mathbf{1}[\mathsf{P}] \in \{0, 1\}$ for the indicator of a proposition. For a finite set S, $|S|$ is its cardinality and S^c its complement in the stated ambient set. The symbols $\dot\cup$ and $\bigsqcup$ denote disjoint unions. A restriction is written $f|_S$, and $\pi^{-1}(y) = \{x : \pi(x) = y\}$ is the fibre of a map π. Tuples and Boolean words use lexicographic order.

For a field L, write

$$L^\times := L \setminus \{0\}, \qquad \mathrm{Mat}_{m\times n}(L) := \{m\text{-by-}n \text{ matrices over } L\}, \qquad \mathrm{Mat}_n(L) := \mathrm{Mat}_{n\times n}(L).$$

We use $\mathbb{Z}_m = \mathbb{Z}/m\mathbb{Z}$ and the binary field $\mathbb{F}_2 = \mathbb{Z}_2$. The function $\varphi(m)$ counts the residue classes modulo m coprime to m.

Fix the field $\overline{\mathbb{Q}} = \overline{\mathbb{Q}} \subseteq \mathbb{C}$ of algebraic numbers. An algebraic closure $\overline{L}$ of a field L contains only elements algebraic over L, and every nonconstant polynomial over $\overline{L}$ has a root there. We write $[L' : L] = \dim_L L'$ for a finite field-extension degree and $L(\alpha)$ for the field generated by L and α. A number field is a finite extension of $\mathbb{Q}$; the compositum of finitely many subfields of $\mathbb{C}$ is the field they generate. An L-conjugate of α is its image under an embedding $L(\alpha) \hookrightarrow \overline{L}$ fixing L, whereas $\bar z$ always denotes complex conjugation. An effective number field supports exact arithmetic and

equality tests. One representation is $\mathbb{Q}(\theta)$, with the monic minimal polynomial of θ and a rational isolating region selecting its complex root [Coh93].

2.1.2 Linear, polynomial, and tensor conventions

All vector spaces are finite-dimensional unless stated otherwise. The notation $\operatorname{span}_L S$ specifies the field of linear combinations; $W^\vee = \operatorname{Hom}_L(W, L)$ is the dual space. For a linear map $T : U \to V$, we write

$$\ker T := \{u : T(u) = 0\}, \qquad \operatorname{im} T := \{T(u) : u \in U\},$$
$$\operatorname{rank} T := \dim \operatorname{im} T, \qquad \operatorname{Hom}_L(U, V) := \{T : U \to V : T \text{ is } L\text{-linear}\}.$$

For subspaces U, V of W, the notation $W = U \oplus V$ means that each $w \in W$ has a unique expression $u + v$ with $u \in U$ and $v \in V$. For $w \in W$, put $w + U := \{w + u : u \in U\}$. The quotient is $W/U := \{w + U : w \in W\}$, and, when W is finite-dimensional, $\operatorname{codim}_W U := \dim W - \dim U$. An *affine subspace* is a coset $a + U$, its translation space is U, and an affine hyperplane has codimension one.

For a square matrix A, nonsingular means $\det A \neq 0$. Define $\mathrm{GL}_n(L) := \{A \in \mathrm{Mat}_n(L) : \det A \neq 0\}$ and, for an L-space V, $\mathrm{GL}(V)$ as the group of invertible L-linear maps $V \to V$. Matrix rank is the dimension of the column space. Also A^{T} is the transpose, $\operatorname{tr} A := \sum_i a_{ii}$, $I_n = (\mathbf{1}[i = j])_{i,j=1}^n$, and a *minor* is the determinant of a square submatrix. A *pivot* is a selected nonzero entry used in elimination or normalization. An eigenpair satisfies $Av = \lambda v$ with $v \neq 0$; an eigenbasis is a basis of eigenvectors, and a matrix is diagonalizable when it has one. A matrix N is nilpotent when $N^k = 0$ for some $k \geq 1$, and $J_k(\lambda) = (\lambda \mathbf{1}[i = j] + \mathbf{1}[j = i + 1])_{i,j=1}^k$ is a Jordan block. An L-matrix algebra is an L-subspace of $\mathrm{Mat}_n(L)$ closed under multiplication; it is unital when it contains I_n, and transpose-stable when it is closed under transpose. For a matrix whose rows and columns have an explicitly ordered index set J, the matrix unit E_{ij}, $i, j \in J$, has its sole 1 in position (i, j). Thus Boolean-indexed matrices use $E_{00}, E_{01}, E_{10}, E_{11}$, while ordinary one-based indexing uses $E_{11}, E_{12}, \ldots$; the stated index set distinguishes the convention.

A monomial is $x^u = \prod_j x_j^{u_j}$ for $u \in \mathbb{Z}_{\geq 0}^n$, with total degree $\sum_j u_j$. The polynomial ring $L[x_1, \ldots, x_n]$ consists of the finite sums $\sum_u c_u x^u$ with $c_u \in L$. A polynomial is *homogeneous* when all its nonzero monomials have one total degree. For $p(t) = \sum_j a_j t^j$, its formal derivative is $p'(t) = \sum_{j \geq 1} j a_j t^{j-1}$; a repeated root is a common root of p and p', and p is square-free when it has none.

The word *multilinear* means linear in each argument with the others fixed. The tensor product $V_1 \otimes \cdots \otimes V_r$ comes with a multilinear map $\tau(v_1, \ldots, v_r) = v_1 \otimes \cdots \otimes v_r$ such that every multilinear map $m : V_1 \times \cdots \times V_r \to W$ factors uniquely as $m = \widetilde{m} \circ \tau$ through a linear map $\widetilde{m}$. The displayed tensor $v_1 \otimes \cdots \otimes v_r$ is a *simple tensor*. The tensor power $A^{\otimes r}$ applies A to every factor. A bipartite *flattening* groups selected tensor coordinates into a row index and the complementary coordinates into a column index; its flattening rank is the resulting matrix rank.

2.1.3 Groups and projective spaces

A homomorphism preserves the displayed group multiplication, and an isomorphism is a bijective homomorphism. A subgroup is written $H \leq G$, and $H \lhd G$ means $gHg^{-1} = H$ for all $g \in G$. A left coset is gH, its index $[G : H]$ is the number of left cosets, and, for $N \lhd G$, the quotient group $G/N = \{gN : g \in G\}$ has multiplication $(gN)(hN) = ghN$. The subgroup generated by S is $\langle S \rangle$; $\langle S : \mathcal{R} \rangle$ denotes a presentation by generators and relations.

The order of $g \in G$ is the least positive m with $g^m = 1$, or is infinite; an involution has order two. The order of a finite group is its cardinality, and its exponent is the least common multiple

of its element orders. The center is $Z(G) := \{z : zg = gz \text{ for all } g \in G\}$. A conjugacy class is $\{gxg^{-1} : g \in G\}$, and a nontrivial group is simple when its only normal subgroups are 1 and G. An automorphism is an isomorphism $G \to G$; the automorphisms form $\mathrm{Aut}(G)$. An automorphism is inner when it has the form $g \mapsto hgh^{-1}$; the inner automorphisms form the normal subgroup $\mathrm{Inn}(G)$ of $\mathrm{Aut}(G)$, and $\mathrm{Out}(G) := \mathrm{Aut}(G)/\mathrm{Inn}(G)$.

A group action of G on X is a map $(g, x) \mapsto gx$ satisfying $1x = x$ and $(gh)x = g(hx)$. Its orbit and stabilizer are $Gx := \{gx : g \in G\}$ and $G_x := \{g : gx = x\}$. The action is transitive when it has one orbit and two-transitive when it is transitive on ordered pairs of distinct points. A representation is a group action by linear maps, equivalently a homomorphism $G \to \mathrm{GL}(V)$, and is faithful when that homomorphism is injective. Given an action of H on N for which every map $n \mapsto h \cdot n$ is an automorphism, the semidirect product $N \rtimes H$ has multiplication $(n, h)(n', h') = (n(h \cdot n'), hh')$. An element is torsion when it has finite order; a group has bounded torsion when there is an $m \geq 1$ such that $g^m = 1$ for every element g. For $H \leq G$, its normalizer and centralizer are

$$N_G(H) := \{g : gHg^{-1} = H\}, \qquad C_G(H) := \{g : gh = hg \text{ for every } h \in H\}.$$

For groups G and H, the direct product $G \times H$ has componentwise multiplication $(g, h)(g', h') = (gg', hh')$. The symbols S_m, A_m, C_m, D_{2m}, and V_4 denote the symmetric group, its subgroup of permutations with an even number of inversions, the cyclic group $\langle r : r^m = 1\rangle$, the dihedral group

$$D_{2m} = \langle r, s : r^m = s^2 = 1,\ srs = r^{-1}\rangle,$$

and $C_2 \times C_2$, respectively. In D_{2m}, elements of $\langle r\rangle$ are rotations and those in its other coset are reflections. For any finite set V, $\mathrm{Sym}(V)$ is the group of all permutations of V under composition. A *transposition* swaps two points and fixes the others; a *double transposition* is the product of two transpositions with disjoint moved points.

For a finite-dimensional L-space V, its projective space is $\mathbb{P}_L(V) := (V \setminus \{0\})/L^\times$; when $L = \mathbb{C}$ or is clear we write $\mathbb{P}(V)$. The class of $v \neq 0$ is $[v]$, and the line through distinct $[u], [v]$ is $\mathbb{P}(\mathrm{span}_L\{u, v\})$. A projective subspace $\mathbb{P}(U)$ has dimension $\dim U - 1$. An invertible linear map induces a projective automorphism, and scalar multiples induce the same one. We put

$$\mathrm{PGL}(V) := \mathrm{GL}(V)/(L^\times I_V), \qquad \mathrm{PGL}_n(L) := \mathrm{GL}_n(L)/(L^\times I_n).$$

Here I_V is the identity linear map on V. The diagonal torus in $\mathrm{PGL}_n(L)$ is the subgroup of projective classes of nonsingular diagonal matrices. Thus brackets denote a projective class when their argument is a nonzero vector or matrix; group-index brackets have two group arguments.

2.2 Complexity conventions

We use polynomial-time Turing reductions $\leq_{\mathrm{T}}$, allowing adaptive exact oracle queries; $\equiv_{\mathrm{T}}$ means reduction in both directions. The classes FP, $\#\mathsf{P}$, and $\mathsf{FP}^{\mathsf{NP}}$ have their usual meanings. For algebraic-valued problems, $\#\mathsf{P}$-hardness means that every integer-valued $\#\mathsf{P}$ function reduces to the exact algebraic oracle, including polynomial-time postprocessing of its answers.

Signature sets, their coefficient fields, and fixed gadgets are fixed independently of the input network. Thus polynomial time measures the network's encoding length; exact answers use the chosen effective number field. Recognition of a tractable signature set is a separate task whose input consists of finite exact signature tables. We prove decidability for that task without an encoding-size complexity bound.

2.3 Signatures, tensors, and Holant instances

We use the fields $\overline{\mathbb{Q}}$ and $\mathbb{F}_2$ fixed above. For $r \geq 0$, an arity-r Boolean signature is a function $f : \{0,1\}^r \to \overline{\mathbb{Q}}$, identified with its lexicographically ordered column tensor; its r argument positions are its *ports*, and their order is part of the data. Arity zero, one, two, three, and four are called nullary, unary, binary, ternary, and quaternary. We write $\mathrm{arity}(f) = r$ and $\mathrm{supp}(f) = \{x : f(x) \neq 0\}$. For a Boolean word $x \in \{0,1\}^r$, we also write $\mathrm{supp}(x) := \{j \in [r] : x_j = 1\}$; the argument type distinguishes word support from signature support. All binary-vector arithmetic is over $\mathbb{F}_2$, with $\oplus$ used when bitwise addition should be explicit. The symbols $\mathbf{0}^r$ and $\mathbf{1}^r$ denote the all-zero and all-one words, respectively; at $r = 0$, both denote the unique empty word. For scalars a, b, define the diagonal matrix

$$\mathrm{diag}(a,b) = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}.$$

For $\tau \in S_r$, define the port reordering $(\tau f)(x) = f(x_{\tau^{-1}(1)}, \ldots, x_{\tau^{-1}(r)})$. With this convention, $\tau(\sigma f) = (\sigma \circ \tau) f$. On disjoint ordered ports, $(f \otimes g)(x,y) = f(x)g(y)$. For $T \subseteq [r]$, $\mathbf{1}_T \in \{0,1\}^r$ denotes the incidence vector of T; in particular $\mathbf{1}_\varnothing = \mathbf{0}^r$. We write $e_j = \mathbf{1}_{\{j\}}$ for the j-th standard basis vector whenever the ambient arity is clear. A nonzero positive-arity signature is *tensor-prime* if, up to port permutation and nonzero scalar, it cannot be written as $a \otimes b$ with both a and b of positive arity. Prime factorization is unique up to these operations and factor order [CFS20a, full version, Lemma 2.13]. For any class $\mathscr{C}$ of signatures, its tensor closure $\langle\mathscr{C}\rangle$ consists of tensor products of members of $\mathscr{C}$, up to port permutation and nonzero scalar. By convention, it also contains zero and the arity-zero empty product. Thus $\langle\mathscr{C}\rangle$ denotes tensor closure, whereas $\langle S\rangle$ denotes the subgroup generated by S. A *tensor factorization* partitions the ordered ports into blocks and writes the tensor, up to a recorded nonzero scalar and port permutation, as a tensor product on those blocks. For example, a $2+2$ factorization of a quaternary has two binary factors.

For $a \in \{0,1\}^r$, define $\delta_a : \{0,1\}^r \to \overline{\mathbb{Q}}$ by

$$\delta_a(x) = \mathbf{1}[x = a].$$

For $r \geq 1$, a *weighted generalized equality* is a signature

$$c_0\delta_a + c_1\delta_{\bar{a}}, \qquad c_0, c_1 \in \overline{\mathbb{Q}}, \quad c_0 c_1 \neq 0,$$

where $\bar{a}$ is the bitwise complement. It is *pure* when $a = \mathbf{0}^r$. Replacing each argument bit x_j by $x_j \oplus a_j$ makes it pure, but these portwise bit flips are used only when actually available or inside a stated global transform.

The binary equality and disequality tensors are

$$I = \mathrm{EQ}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad X = \mathrm{NEQ}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Graphs are finite multigraphs with ordered incidences at each vertex. Loops and parallel edges are allowed; a loop contributes two incidences to the degree. Cycles include loops and two-cycles formed by parallel edges, and are identified under reversal and cyclic changes of starting point. A unicyclic component has exactly one cycle. Subdivision inserts degree-two vertices on an edge; suppression reverses this operation. Splicing joins two exposed incidences and removes their boundary designations. A factor graph records signature vertices and their incidences with edge kernels. A matching is a family of disjoint two-element subsets, and is perfect when it covers its ground set.

We use a half-edge semantics throughout. A (possibly open) signature network Ω consists of a finite set $H(\Omega)$ of port half-edges, a finite set of ordinary tensor vertices, and, for each such

vertex v, an assigned signature f_v and a subset $H_v \subseteq H(\Omega)$ equipped with a linear order. These subsets are pairwise disjoint, satisfy $|H_v| = \mathrm{arity}(f_v)$, and obey $H(\Omega) = \big(\bigsqcup_v H_v\big) \sqcup H_{\mathrm{free}}(\Omega)$, where the free stubs $H_{\mathrm{free}}(\Omega)$ are half-edges not attached to a tensor vertex. Every free stub is a paired boundary endpoint; no isolated unpaired free stub is allowed. A symbol E_W is a zero-cost structural edge box for the model's default edge kernel W ($W = X$ in the native model and $W = I$ in the ordinary model): it replaces one wire pair of half-edges (whose endpoints may be ordinary or free) and contributes the single factor $W(a, b)$ on their endpoint bits a, b. Thus E_W is not an additional tensor vertex, has no separate H_v, and is never multiplied again as a vertex factor. Any displayed I or X supplied in the signature set (including the tensor equal to the default kernel) is instead an ordinary tensor vertex and is transformed and counted like the other input signatures. An explicitly ordered tuple $\partial\Omega$ of distinct exposed ports is drawn from $H(\Omega)$. The set $E_{\mathrm{int}}(\Omega)$ is a matching of unordered pairs of half-edges. An ordinary vertex port is unpaired exactly when it is in the boundary; every other ordinary port belongs to exactly one pair. Every free stub belongs to the boundary and to exactly one pair. Thus a boundary port is either an unpaired ordinary port or a paired free endpoint of an edge box, and no half-edge belongs to two pairs. We call these wire pairs $E_{\mathrm{int}}(\Omega)$ (for a closed network they are exactly the usual internal edges). A pair may join two ports of the same vertex (a loop), and parallel pairs are allowed. For each wire pair $e = \{h, h'\}$ choose a kernel $W_e \in \{I, X\}$, with I and X as fixed above, and put $\mathbf{W} = (W_e)_{e \in E_{\mathrm{int}}(\Omega)}$. For a boundary word $z_{\partial\Omega} \in \{0,1\}^{|\partial\Omega|}$, interpreted in the fixed order of $\partial\Omega$, first form the raw table

$$R_\Omega(z_{\partial\Omega}) = \sum_{\substack{\tau : H(\Omega) \to \{0,1\} \\ \tau|_{\partial\Omega} = z_{\partial\Omega}}} \prod_v f_v(\tau|_{H_v}) \prod_{\{h,h'\} \in E_{\mathrm{int}}(\Omega)} W_{\{h,h'\}}(\tau_h, \tau_{h'}).$$

For the j-th boundary port, put $A_j = I$ if it is an ordinary port, and $A_j = W_e$ if it is a free endpoint of the wire pair e. The boundary tensor, in the coordinates used to replace a tensor vertex, and the closed partition function are

$$T_\Omega = \left(\bigotimes_{j=1}^{|\partial\Omega|} A_j \right) R_\Omega, \tag{2}$$

$$Z_{\mathbf{W}}(\Omega) := T_\Omega(\varnothing) \qquad (\partial\Omega = \varnothing). \tag{3}$$

An open signature network, equipped with this boundary order and viewed through its boundary tensor T_Ω, is an *open gadget*. If every boundary port is ordinary, then $T_\Omega = R_\Omega$. For a wire with one free endpoint, the boundary action cancels that wire because $W_e^2 = I$: deleting the free endpoint and its wire and exposing its ordinary mate gives the same T_Ω, in the inherited boundary order. A wire with two free endpoints is an isolated E_W, whose boundary tensor is $WWW = W$. These operations reduce every open network to ordinary exposed ports and isolated edge boxes without adding resources. When every wire kernel is the same matrix W, we abbreviate $Z_{\mathbf{W}}$ by Z_W. In the product, f_v always means the assigned signature at an ordinary vertex; global signature transformations act only on ordinary signature vertices. The edge-box notation is shorthand for one occurrence of the default edge kernel: when an E_W is spliced into a network, its two ports are connected by the displayed W exactly once (equivalently, one wire pair's kernel is replaced by W); the same W is not multiplied again. The edge boxes just described are the formal realization of the native two-port primitive (and of the ordinary equality primitive); the empty network is the nullary unit 1, not a nontrivial two-port tensor. The order of every H_v is part of the input, and the order of the boundary ports is part of an open gadget. In particular, a native X-loop has two independent endpoint bits; it is not a single shared edge variable. For example, an E_X with two

exposed endpoints h, h' has boundary tensor $T_{\mathsf{E}_X}(a,b) = X(a,b)$. Closing its two endpoints by the native contraction gives the known scalar $\sum_{a,b} X(a,b)^2 = 2$.

The notation $Z_{\mathbf{W}}$ is a bookkeeping device for the specified edge labelling; an instance is allowed to use only the homogeneous ordinary choice Z_I or the homogeneous native choice Z_X, unless a displayed I or X is supplied as an explicit degree-two signature/gadget. The ordinary problem Holant($\mathcal{F}$) is Z_I on closed networks over $\mathcal{F}$, and the native-edge problem is K-Holant($\mathcal{F}$) := Z_X; problem instances have no free stubs (free stubs occur only in open gadgets). When all wire kernels are I, summing the half-edge variables forces the two endpoints of each edge to agree and recovers the usual one-bit-per-edge multigraph definition (including loops). For signature sets $\mathscr{E}$ and $\mathcal{F}$, the bipartite problem Holant($\mathscr{E} \mid \mathcal{F}$) is the ordinary ($I$-edge) half-edge problem on a bipartite network: vertices in the left part are assigned signatures from $\mathscr{E}$, and vertices in the right part are assigned signatures from $\mathcal{F}$. For X-edges no one-bit compression is made. Subdividing every native X-edge by a degree-two X-vertex, with ordinary equality wires on its two sides, gives the equivalent bipartite presentation

$$\text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \text{Holant}(X \mid \mathcal{F}).$$

The empty network is the nullary unit. Throughout, comma-separated signature or signature-set arguments mean adjoining their union; for example, Holant($\mathcal{F}, a, b$) abbreviates Holant($\mathcal{F} \cup \{a, b\}$).

We use the unary pinnings

$$\Delta_0 = [1,0]^{\mathsf{T}}, \qquad \Delta_1 = [0,1]^{\mathsf{T}},$$

which are point masses forcing the incident bit to 0 and 1, respectively. We also fix the sign and skew matrices

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Y = ZX = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

A *weighted equality* is a binary table $\mathrm{diag}(p, q)$, with $p, q \in \overline{\mathbb{Q}}$; it is nonsingular when $pq \neq 0$, and literal equality is $I = \mathrm{diag}(1, 1)$. A weighted equality is never a free auxiliary signature in this paper. In Section 4 it may instead be produced by an actual gadget and retained with its reduction provenance; after that interface returns, the equality-accessible analysis uses the supplied literal I in the original coordinates. Define the nullary equality/unit by $\mathrm{EQ}_0(\varnothing) := 1$. More generally, for $r \geq 1$, define the arity-r equality signature by

$$\mathrm{EQ}_r(x_1, \ldots, x_r) = \mathbf{1}[x_1 = \cdots = x_r].$$

We write

$$\text{Holant}(\mathcal{F}, X) = \text{Holant}(\mathcal{F} \cup \{X\})$$

for the same ordinary equality-wire problem with the additional binary signature X available as a vertex. Subdividing every ordinary edge by an EQ_2-vertex gives

$$\text{Holant}(\mathcal{F}) = \text{Holant}(\mathrm{EQ}_2 \mid \mathcal{F}).$$

2.4 Holographic transformations and the K-basis

Let $i \in \mathbb{C}$ satisfy $i^2 = -1$, and fix the K-coordinate matrix

$$K = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \qquad K^{-1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}. \tag{4}$$

The transformations below rewrite every tensor and its incident edge consistently, giving exact identities between partition functions. Their fixed algebraic coefficients lie in a finite extension of the input coefficient field. Known nonzero rescalings are restored by exact postprocessing, as in Section 2.2.

For tensors or matrices A and B, write

$$A \doteq B$$

when

$$A = \lambda B \qquad \text{for some } \lambda \in \overline{\mathbb{Q}}^{\times}.$$

In every gadget or reduction identity, this scalar is known, tracked, and restored by exact algebraic postprocessing.

For $M \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ and an arity-r right-side signature f, write

$$Mf = M^{\otimes r} f, \qquad M\mathcal{F} = \{Mf : f \in \mathcal{F}\}.$$

For any signature set $\mathcal{S}$, $M(\mathcal{S})$ means the same tensorwise image $M\mathcal{S}$. In particular, the expressions $K(\mathcal{F})$ and $K(\mathcal{F} \cup \{I\})$ in the two main-theorem predicates are not matrix products; they are these set images. In the native theorem the hypothesis $I \in \mathcal{F}$ supplies the equality signature, whereas the ordinary-Holant formulation adjoins I through the equality-wire equivalence. For a signature in ordinary coordinates, a hat denotes $\widehat{f} = K^{-1} f$, and $\widehat{\mathcal{S}} = K^{-1}\mathcal{S}$. For an arity-$r$ left-side signature g, juxtaposition denotes the dual action $gM = (M^{\mathsf{T}})^{\otimes r} g$. In particular, $(\mathrm{EQ}_2)M$ has matrix $M^{\mathsf{T}}M$, and the holographic identity is

$$\mathrm{Holant}(\mathrm{EQ}_2 \mid \mathcal{F}) = \mathrm{Holant}((\mathrm{EQ}_2)M \mid M^{-1}\mathcal{F}), \qquad (\mathrm{EQ}_2)M = M^{\mathsf{T}}M.$$

This exact identity makes the transformed edge a genuine weighted binary, not a componentwise scalar.

We use $Z_X(\Omega)$ for the homogeneous native-X partition function defined in Equation (3). We also write

$$GO(X) = \{D \in \mathrm{GL}_2(\overline{\mathbb{Q}}) : D^{\mathsf{T}} X D = \lambda X \text{ for some } \lambda \in \overline{\mathbb{Q}}^{\times}\}$$

for legal global transformations preserving the native disequality edge X projectively. For a closed network, if $D\Omega$ applies D at every ordinary vertex and $D^{\mathsf{T}} X D = \lambda X$, then

$$Z_X(D\Omega) = \lambda^{|E_{\mathrm{int}}(\Omega)|} Z_X(\Omega). \tag{5}$$

Thus the two problems differ only by the known edge-count scalar. The matrix acts globally and is not thereby a local gadget. For $m \geq 1$, define the group of m-th roots of unity by

$$\mu_m = \{z \in \overline{\mathbb{Q}} : z^m = 1\}.$$

An element $\omega \in \mu_m$ is a *primitive* m-th root of unity when m is the least positive integer satisfying $\omega^m = 1$. For a nonsingular 2-by-2 matrix T, its class $[T] \in \mathrm{PGL}_2(\overline{\mathbb{Q}})$ uses the projective convention of Section 2.1.3. The *projective order* of $[T]$ is the least $m \geq 1$ for which $T^m = \lambda I$ for some $\lambda \in \overline{\mathbb{Q}}^{\times}$, and is infinite if no such m exists. We use the quaternion matrix basis

$$Q_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad Q_1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad Q_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Q_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}. \tag{6}$$

It is a basis of $\mathrm{Mat}_2(\mathbb{C})$; moreover, $Q_0 = I$, $Q_j^2 = -I$ for $1 \leq j \leq 3$, and $Q_j Q_k = -Q_k Q_j$ for distinct $j, k \in \{1, 2, 3\}$. These relations are what the adjective *quaternion* records.

Proposition 2.1 (Finite subgroups of $\mathrm{PGL}_2(\mathbb{C})$)**.** *Every finite subgroup of* $\mathrm{PGL}_2(\mathbb{C})$ *is conjugate to exactly one of the following types:*

$$C_m, \qquad D_{2m}, \qquad A_4, \qquad S_4, \qquad A_5,$$

where $m \geq 1$ *for* C_m*,* $m \geq 2$ *for* D_{2m}*,* $|C_m| = m$*, and* $|D_{2m}| = 2m$*. If* ω *is a primitive* m*-th root of unity, explicit cyclic and dihedral models are*

$$C_m^{\mathrm{std}} = \left\{ \left[\begin{pmatrix} 1 & 0 \\ 0 & \omega^k \end{pmatrix} \right] : 0 \leq k < m \right\},$$

and

$$D_{2m}^{\mathrm{std}} = C_m^{\mathrm{std}} \cup \left\{ \left[X \begin{pmatrix} 1 & 0 \\ 0 & \omega^k \end{pmatrix} \right] : 0 \leq k < m \right\}.$$

The diagonal classes are the rotations, and the anti-diagonal coset consists of the reflections.

In the basis Equation (6)*, take the tetrahedral model*

$$\begin{aligned} \mathcal{T} = \{&[Q_0], [Q_1], [Q_2], [Q_3]\} \\ &\cup \{[Q_0 + \epsilon_1 Q_1 + \epsilon_2 Q_2 + \epsilon_3 Q_3] : \epsilon_j \in \{1, -1\}\}, \end{aligned} \tag{7}$$

and the octahedral model

$$\mathcal{O} = \mathcal{T} \cup \{[Q_\mu + \epsilon Q_\nu] : 0 \leq \mu < \nu \leq 3,\ \epsilon \in \{1, -1\}\}. \tag{8}$$

Finally, define the golden-ratio scalar and its reciprocal by

$$\phi := \frac{1 + \sqrt{5}}{2}, \qquad \tau := \phi^{-1},$$

and define the icosahedral model by

$$\begin{aligned} \mathcal{I} = \mathcal{T} \ \cup \ \{[v_0 Q_0 + v_1 Q_1 + v_2 Q_2 + v_3 Q_3] : (v_0, v_1, v_2, v_3) \\ = \pi(0, \epsilon_1, \epsilon_2 \phi, \epsilon_3 \tau), \quad \pi \in A_4,\ \epsilon_j \in \{1, -1\}\}, \end{aligned} \tag{9}$$

where A_4 *acts by even coordinate permutations and projective duplicates are removed. The additional part of* $\mathcal{I}$ *has* 48 *projective classes. Thus*

$$\mathcal{T} \cong A_4, \qquad \mathcal{O} \cong S_4, \qquad \mathcal{I} \cong A_5,$$

of orders $12, 24, 60$*, respectively.*

Proof. The conjugacy classification is classical [Bea83, Section 5.1]. The first two displays are the standard rotation and reflection models. Direct matrix multiplication verifies closure, cardinalities, and element-order distributions of the three remaining displayed sets, identifying them respectively with A_4, S_4, A_5. □

Lemma 2.2 (Finite projective-group facts)**.** *The following properties will be used below.*

1. *A cyclic group has at most one involution, and it has one exactly when its order is even.*

2. *For*
$$D_{2m} = \langle r, s : r^m = s^2 = 1,\ srs = r^{-1} \rangle,$$
the rotations $\langle r \rangle \cong C_m$ *form a normal index-two subgroup, and every element outside it is a reflection. If* m *is odd, the rotation group is the unique index-two subgroup and the center is trivial. If* m *is even and* $m \geq 4$*, the center is* $\{1, r^{m/2}\}$*; for* $m = 2$*,* $D_4 \cong V_4$ *and all three nonidentity elements are central. Every subgroup of a cyclic or dihedral group is cyclic or dihedral.*

3. *The three involutions of* A_4*, together with the identity, form its unique normal* V_4*. The group has no index-two subgroup, and the centralizer of an involution is* V_4*.*

4. *The unique index-two subgroup of* S_4 *is* A_4*. Its involutions are six transpositions and three double transpositions. The centralizer of a transposition has order* 4*, whereas the centralizer of a double transposition is* D_8*. The three double transpositions are the nonidentity elements of the unique normal* V_4*. Conjugation is transitive both on ordered pairs of distinct double transpositions and on ordered pairs of disjoint transpositions.*

5. *The group* A_5 *is simple and has no index-two subgroup. Its fifteen involutions form one conjugacy class, and the centralizer of each is* V_4*.*

6. *The element-order distributions are*

	1	2	3	*order* 4 *or* 5
A_4	1	3	8	0
S_4	1	9	8	6 *of order* 4
A_5	1	15	20	24 *of order* 5.

7. *For the displayed models,*
$$N_{\mathrm{PGL}_2(\mathbb{C})}(\mathcal{T}) = \mathcal{O}, \qquad N_{\mathrm{PGL}_2(\mathbb{C})}(\mathcal{O}) = \mathcal{O}, \qquad N_{\mathrm{PGL}_2(\mathbb{C})}(\mathcal{I}) = \mathcal{I}.$$

8. *The projective centralizers of the distinguished involutions are*
$$C_{\mathrm{PGL}_2(\mathbb{C})}([Z]) = \left\{ \left[\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \right] : ab \neq 0 \right\} \sqcup \left\{ \left[\begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix} \right] : ab \neq 0 \right\}.$$
Thus $[Z]$ *is the only nonidentity diagonal involution, while every nonsingular anti-diagonal class is an involution. Moreover,*
$$C_{\mathrm{PGL}_2(\mathbb{C})}(\langle [X], [Z] \rangle) = \langle [X], [Z] \rangle \cong V_4.$$
Every finite subgroup of the displayed diagonal torus is cyclic.

Proof. Items 1–6 follow directly from the cyclic and dihedral presentations and the usual permutation models of A_4, S_4, A_5 [DM96]. For item 7, conjugation embeds each normalizer modulo its projective centralizer into the corresponding automorphism group. The projective centralizers are trivial; $\mathrm{Aut}(A_4) \cong S_4$, the displayed $\mathcal{O}$ normalizes $\mathcal{T}$, and every automorphism of S_4 is inner. Since $\mathrm{Aut}(A_5) \cong S_5$, an additional outer normalizer of $\mathcal{I}$ would give a finite projective group H of order 120 containing A_5. The classification forces $H \cong C_{120}$ or D_{120}; every subgroup of a cyclic or dihedral group is cyclic or dihedral, so H cannot contain A_5. Item 8 follows by solving $UZ \doteq ZU$, and then imposing the same relation with X. The ratio map $[\mathrm{diag}(a, b)] \mapsto a/b$ embeds the diagonal torus into $\mathbb{C}^\times$; every finite subgroup of $\mathbb{C}^\times$ is a cyclic group of roots of unity. □

The conjugacy classification above is group-theoretic only. Its conjugating matrix need not lie in $GO(X)$, so it does not by itself define a legal global K-Holant normalization or provide an actual binary gadget. Later applications retain the distinguished classes $[X]$ and $[Z]$.

Since $K^{\mathsf{T}}K = X$, putting $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$ gives

$$\text{Holant}(\mathcal{F}) \equiv_{\text{T}} \text{Holant}(X \mid \widehat{\mathcal{F}}) \equiv_{\text{T}} \text{K-Holant}(\widehat{\mathcal{F}}). \tag{10}$$

For original-coordinate signatures and signature sets, the hats in Equation (10) record passage to K-coordinates. Once we work with a signature set already in K-coordinates, we suppress these hats and denote its member signatures simply by f. Unless stated otherwise, signatures and signature sets appearing in a K-Holant problem are understood to be in K-coordinates, with native edge X. The Walsh coordinates used below are analytic bookkeeping, not another legal global transformation. Define the Walsh matrix P and, for a signature f already in K-coordinates, its Walsh-coordinate tensor $\widetilde{f}$ by

$$P := \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \widetilde{f} := (P^{-1})^{\otimes \text{arity}(f)} f. \tag{11}$$

Thus $\widetilde{f}$ denotes the analytic Walsh representation of a signature f. For $s \in \overline{\mathbb{Q}}$, the identities

$$P^{-1}\left[\frac{1}{2}\begin{pmatrix} 1+s & 1-s \\ 1-s & 1+s \end{pmatrix}\right] P = \text{diag}(1, s), \qquad P^{\mathsf{T}} X P = 2Z$$

explain its use in the odd cyclic and dihedral calculations.

2.5 Gadgets, interpolation, and factor reductions

For a signature f, a *card* means an output $\partial_p^C f$ of one of the two-port contractions defined below.

A *direct gadget* over $\mathcal{G}$ is a finite native-X network of $\mathcal{G}$-vertices, possibly using the primitive E_X, with an ordered exposed boundary. If its boundary tensor in the sense of Equation (2) is $T_\Omega = \rho h$, where $\rho \in \overline{\mathbb{Q}}^{\times}$ is a known nonzero scalar, then

$$\text{K-Holant}(\mathcal{G}, h) \leq_{\text{T}} \text{K-Holant}(\mathcal{G}). \tag{12}$$

A fixed gadget has constant size. A gadget realizing ρh, $\rho \neq 0$, implements the same Turing reduction after an instance with N occurrences is rescaled by ρ^{-N}. We call such an h an *actual signature* over $\mathcal{G}$. More generally, a finite set S of signatures is *Turing-exposed* over $\mathcal{G}$ if

$$\text{K-Holant}(\mathcal{G}, S) \leq_{\text{T}} \text{K-Holant}(\mathcal{G}).$$

In either homogeneous model, gluing two boundary ports contracts their boundary-tensor coordinates against the default kernel W; the remaining boundary order is inherited. Two unpaired ordinary ports receive a new W-wire. For one paired free endpoint and one unpaired ordinary port, splice the latter into the former's incidence, retaining its existing wire. For two free endpoints on distinct wires, delete those endpoints and their two wires and join their mates by one W-wire. If both endpoints belong to the same isolated E_W, remove that component and record the scalar $\sum_{a,b} W(a,b)^2 = 2$. These rules follow from $W^2 = I$ and $W^3 = W$. In particular, vertex replacement first cancels the one-free-end wires as above and then inserts the resulting ordinary-boundary gadget; each remaining isolated E_W splices into a single wire. Closed components containing only such edge boxes contribute known factors of 2. This proves Equation (12), including loops, without leaving free stubs in the oracle instance.

When transporting an open native gadget by $D \in GO(X)$, first cancel its one-free-end wires in this way. Every remaining internal wire between ordinary ports contributes the known scalar from $D^{\mathsf{T}}XD = \lambda X$. An isolated primitive remains available because $DXD^{\mathsf{T}} = \lambda X$. Thus its boundary tensor is transported by D up to a recorded nonzero scalar; the closed-network edge count in Equation (5) is used only for closed instances. Thus every actual signature is Turing-exposed, but a factor obtained from the tensor-factor theorem or a signature obtained by interpolation need not be an actual gadget. This provenance is persistent: a gadget built over an augmentation containing a merely Turing-exposed resource is, relative to the predecessor, only Turing-exposed. Slices, Walsh expansions, linear inverses, similarities, projective normalizations, tomography, and decoder coordinates are analytic arguments; they may prove a tensor identity or select an actual contraction, but they do not themselves add a signature resource. A *slice* is a partial evaluation after selected tensor coordinates have been fixed; a matrix similarity has the form $B = SAS^{-1}$ with S invertible; and *tomography* is reconstruction from a separating family of linear measurements, meaning that their joint value map is injective on the tensors under consideration. An interpolation recovers finitely many unknown coefficients from oracle evaluations of a parameterized gadget family by solving an invertible linear system. Its output is Turing-exposed unless a separate direct gadget realizes it.

Let f have arity r. For an ordered pair of distinct ports $p = (i, j)$, a binary kernel $C : \mathbb{F}_2^2 \to \overline{\mathbb{Q}}$, and a residual word z indexed by $[r] \setminus \{i, j\}$ in increasing port order, let $z^{i\leftarrow a, j\leftarrow b}$ be the arity-r word obtained by inserting a, b at ports i, j. Define the two-port contraction by

$$(\partial_p^C f)(z) = \sum_{a,b\in\mathbb{F}_2} C(a, b) f\big(z^{i\leftarrow a, j\leftarrow b}\big).$$

In the notation of Shao–Cai [SC20, full version, Section 2.4], their $\widehat{\partial}_{ij}\widehat{f}$ and $\partial_{ij}f$ are, respectively, our $\partial^X_{(i,j)}\widehat{f}$ and $\partial^I_{(i,j)}f$. Port order fixes the ordered pair (i, j). For binary matrices U, V, the Frobenius pairing is bilinear:

$$\langle U, V\rangle_F = \operatorname{tr}(U^{\mathsf{T}}V) = \sum_{a,b\in\mathbb{F}_2} U(a, b)V(a, b),$$

without conjugation; tr is the matrix trace fixed in Section 2.1.2. A *binary chain* is a serial composition of ordered binary gadgets joined by native X-edges. A *dressed card* $\partial_p^C f$ uses a kernel realized by actual binary chains over the current retained set and one native edge. Suppose the closing subnetwork has ordered boundary tensor B. Its two attachment X-edges complement both boundary bits, so

$$C(a, b) = \sum_{u,v\in\mathbb{F}_2} X(a, u)B(u, v)X(v, b) = (XBX)(a, b) = B(1 - a, 1 - b).$$

Exchanging the two ordered boundary inputs replaces B by B^{T}, and the same attachment therefore gives $XB^{\mathsf{T}}X$. Likewise, two inward-ordered binary chains L and R joined by one native X-edge have external tensor LXR^{T}. These formulas explain where transposes arise in physical gadget constructions and fix the ordered-port convention used below. Every dressed card so constructed is an actual gadget; linear combinations of cards are only analytic comparisons.

2.5.1 Two-port slices and contractions

For an arity-r tensor f, ordered ports $p = (i, j)$, and a residual word z, the *two-port slice* is the binary matrix

$$F^f_{p,z}(a, b) := f\big(z^{i\leftarrow a, j\leftarrow b}\big), \qquad a, b \in \mathbb{F}_2.$$

If $\mathscr{K} \leq \mathrm{Mat}_2(\mathbb{C})$ is a space of kernel directions, its Frobenius-orthogonal space is

$$\mathscr{K}^{\perp_F} := \{F \in \mathrm{Mat}_2(\mathbb{C}) : \langle K, F\rangle_F = 0 \text{ for every } K \in \mathscr{K}\}.$$

Thus contractions against $\mathscr{K}$ determine the slice modulo $\mathscr{K}^{\perp_F}$.

Fix the preprocessed signature set $\Gamma_\star$ at the beginning of a structural branch. A *retained signature set* is a finite set reached from $\Gamma_\star$ by recorded direct gadget augmentation, retained factor exposure, or, inside the eight-vertex interface, a legal global $GO(X)$ change of coordinates. Every such step carries its reduction direction and provenance. Generic interpolation is terminal only, and an analytic tensor identity does not by itself adjoin a retained signature.

We record the current coordinates and signature set as (A, Λ), with

$$A \in GO(X), \qquad A\Gamma_\star \subseteq \Lambda, \qquad \text{K-Holant}(\Lambda) \equiv_{\mathrm{T}} \text{K-Holant}(\Gamma_\star).$$

A literal retained augmentation is the special case $A = I$. After a global normalization the current signature set need not literally contain $\Gamma_\star$; the invariant is the transported inclusion $A\Gamma_\star \subseteq \Lambda$. The normalization data in Lemma 2.23 records the additional coefficient-field and group data. This transported form is used for the self-contained eight-vertex interface in Section 4. Before that interface hands a continuing branch to the matching-deck analysis, it pulls the state back to the caller's coordinates. From Section 5 onward, every continuing retained set contains the distinguished literal I, and no realized weighted equality replaces it as the anchor. Any continuing coordinate change must preserve both native X and literal I projectively; otherwise it is used only inside a terminal reduction and is not a state update.

A card is *proper* if its arity is strictly smaller than its parent's; *actual* always means directly realized, not analytically combined.

The zero-cost native edge box E_X is a legal two-port gadget with tensor X; the empty network is the nullary unit. Thus, when a transfer group is introduced below, we take the identity-class binary gadget $B_1 := \mathsf{E}_X = X$ and define its transfer matrix $R_1 := B_1 X = I$. This is an actual gadget identity rather than an implicit free signature.

For every ordered binary signature B, define its transfer matrix and transfer class by

$$T_B := BX, \qquad [T_B] = [BX].$$

For a finite realized projective group $G \leq \mathrm{PGL}_2(\overline{\mathbb{Q}})$, choose one nonsingular ordered gadget B_g for each class g. Define its transfer matrix $R_g := T_{B_g} = B_g X$, require $[R_g] = g$, and normalize $R_1 = I$, where $1 = [I]$. The matrix of every actual ordered binary B having transfer class g is $B = \rho B_g$ for a unique $\rho \in \overline{\mathbb{Q}}^{\times}$; we likewise record the multiplication and reversal scalars. Consequently all displayed projective identities below stand for actual gadget identities with their known scalar restored.

Whenever the proof says that an external port is X-translated, it means attaching the retained two-port chain whose transfer class is $[X]$ and recording its nonzero scalar in the port data. This is a direct binary chain, not a free unary bit flip.

For each signature, record its realizing gadget or reduction, ordered ports, discarded nonzero scalars, and coordinate changes. To express it in earlier coordinates, apply the inverses of the recorded changes.

Interpolation records parameter count, gadget size, and oracle calls. Linear combinations are not gadgets. The tensor-factor extraction result used later is stated as Theorem 2.16; its common-one-sided exception is always kept explicit.

2.6 Zero and nullary preprocessing

For a closed network Ω, define its model-dependent partition-function value by $Z(\Omega) := Z_I(\Omega)$ in the ordinary model and $Z(\Omega) := Z_X(\Omega)$ in the native model under discussion. Let Ω° be the network obtained by deleting all nullary vertices (and their scalar contributions); a zero-signature vertex instead makes the value zero. Let the reduced signature set $\mathcal{F}^\circ$ be obtained from $\mathcal{F}$ by deleting all zero signatures and all arity-zero signatures. If an instance Ω contains a zero-signature vertex, then $Z(\Omega) = 0$. Otherwise, writing $m_c(\Omega)$ for the number of occurrences of a nullary c,

$$Z(\Omega) = \left(\prod_{\substack{c \in \mathcal{F} \\ \text{arity}(c)=0}} c^{m_c(\Omega)} \right) Z(\Omega^\circ),$$

where the partition function of the empty residual instance is the empty product 1. Thus structural classification applies only to the nonzero positive-arity signature set $\mathcal{F}^\circ$, while the root evaluator restores all original nullary scalars exactly. The tractable predicates ignore the deleted signatures. We call this deletion-and-restoration step *preprocessing*; a *preprocessed signature set* contains only nonzero positive-arity signatures, with all deleted nullaries handled by the root evaluator.

Algebraic numbers are represented exactly, for example by a minimal polynomial together with an isolating rectangle selecting the intended complex root. All zero tests, complex conjugations $z \mapsto \bar{z}$, field operations, and oracle answers are performed in a finite effective compositum and returned in the chosen source embedding. For a fixed finite signature set, the number and size of oracle instances and the exact field arithmetic are polynomial in the Holant instance size. Effective recognition is asserted to be decidable for finite exact algebraic input; no polynomial bound in the encoding length of the fixed signature set is claimed.

2.7 Charge and Hamming layers

All charge terminology in this subsection is in K-coordinates. For $x \in \{0,1\}^r$, define its Hamming weight by $\text{wt}(x) := |\{j : x_j = 1\}|$ and its doubled charge by

$$q_r(x) = 2\,\text{wt}(x) - r.$$

A *Hamming layer* is the set of words of one fixed Hamming weight. A word has even or odd parity according as its weight is even or odd, and a signature has *even support* when it vanishes on every odd-parity word. We distinguish $\operatorname{supp} f \subseteq M$ from $\operatorname{supp} f = M$; coefficients in a prescribed support set may vanish unless equality is required. For even r, we also use the integer half-charge

$$\chi_r(x) = \frac{q_r(x)}{2} = \text{wt}(x) - \frac{r}{2}.$$

In the all-even analysis, gcd, divisibility, height, and interval statements use χ_r; *doubled charge* explicitly means q_r. Define the Hamming half-spaces

$$HW_r^{\geq} = \{x : q_r(x) \geq 0\}, \quad HW_r^{\leq} = \{x : q_r(x) \leq 0\}, \quad HW_r^{=} = \{x : q_r(x) = 0\},$$

and define $HW_r^{>}, HW_r^{<}$ analogously. A signature is $HW^{\geq}$ (respectively, $HW^{\leq}$) when its support lies in $HW_r^{\geq}$ (respectively, $HW_r^{\leq}$). A signature set is *common one-sided* when all members are $HW^{\geq}$ or all are $HW^{\leq}$; otherwise it is *two-sided*, equivalently both charge signs occur among its support words. An arity-$2d$ *EO signature* is supported in $HW_{2d}^{=}$.

For a homogeneous native-X network (all structural wire pairs have kernel X; any explicitly supplied I or X is an ordinary tensor vertex), let $U(\Omega) = H(\Omega) \setminus \bigcup_{e \in E_{\mathrm{int}}(\Omega)} e$ be the unpaired boundary half-edges (by definition, $U(\Omega) \subseteq \partial\Omega$). Call an assignment *edge-supported* when every structural edge-kernel factor is nonzero on the two bits assigned to its endpoints. Every edge-supported assignment has one 0-bit and one 1-bit on each X-pair; assignments violating an X-edge have zero weight and are not contributing assignments. Hence, pointwise for every edge-supported assignment τ,

$$\sum_v q_{\mathrm{arity}(f_v)}(\tau|_{H_v}) + \sum_{h \in H_{\mathrm{free}}(\Omega)} (2\tau_h - 1) = \sum_{h \in U(\Omega)} (2\tau_h - 1).$$

For a contribution to the boundary tensor $T_\Omega(z)$, an ordinary boundary bit equals τ_h, while a free boundary bit equals $1 - \tau_h$ by Equation (2). Moving the free-stub sum to the right therefore gives $q_{|\partial\Omega|}(z) = \sum_v q_{\mathrm{arity}(f_v)}(\tau|_{H_v})$. In the standard closed instances below there are no free stubs and $U(\Omega) = \varnothing$, so $\sum_v q_{\mathrm{arity}(f_v)}(\tau|_{H_v}) = 0$. Thus native X-contraction preserves the total charge in every closed contributing assignment.

Remark 2.3 (Notation relative to prior work). We follow [MWXZ25, Sections 2.1–2.3] in writing $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$, $HW^{\geq}$, $HW^{\leq}$, $HW^{=}$, and *EO signature*. Their $\neq_2$ and $=_r$ are our X and EQ_r, respectively; subscripts in $HW_r^{\bullet}$ record arity. Thus $q_r(x) = |\{j : x_j = 1\}| - |\{j : x_j = 0\}|$, while the height arguments use its integer half $\chi_r = q_r/2$.

Our span is linear span; the affine span denoted $\mathrm{Span}(f)$ in [MWXZ25] is written $\mathrm{aff}(\mathrm{supp}(f))$.

2.8 Binary codes and affine geometry

In Section 8, signature supports become binary affine spaces: contraction imposes affine equations on the support, while coefficient ratios record the remaining phase. The code and bilinear-form conventions below describe these two parts separately.

2.8.1 Boolean and affine tools

For a finite coordinate set S, put $\mathbb{F}_2^S := \{x : S \to \mathbb{F}_2\}$. A *binary code* of length n is a subset of $\mathbb{F}_2^n$; its elements are *codewords*. It is linear when it is a subspace and affine when it is a coset of a linear code. The truth table of a Boolean function lists its values in the fixed lexicographic input order. For a commutative ring R, a Boolean polynomial over R is an element of the quotient ring of $R[x_1, \ldots, x_n]$ by the ideal generated by the relations $x_j^2 - x_j$, $j \in [n]$, and hence has a unique multilinear representative. Its degree is the largest number of distinct variables in a nonzero monomial, and a mixed term contains at least two distinct variables. An affine-linear form is a constant plus a linear form. A *quadratic phase* is $i^{Q(x)}$, where $Q : \mathbb{F}_2^n \to \mathbb{Z}_4$ has multilinear degree at most two and even coefficients on mixed terms. A character of an additive group is a group homomorphism into the multiplicative group of the coefficient field. These Boolean normal forms are instances of Boolean-lattice Möbius inversion [Car21, Chapters 1–2].

For an affine subspace $a + U$, U is its *translation space*; the affine hull $\mathrm{aff}(D)$ is the smallest affine subspace containing D. An affine map $a + U \to b + V$ has the form $a + u \mapsto b + T(u)$ for a linear map $T : U \to V$. If an affine surjection $\pi : a + U \twoheadrightarrow D$ onto an affine space D is fixed, an *affine section* is an affine map $\sigma : D \to a + U$ with $\pi \circ \sigma = \mathrm{id}_D$. For $S \subseteq [n]$, in increasing coordinate order, define coordinate projection and zero extension by

$$\pi_S : \mathbb{F}_2^n \to \mathbb{F}_2^S, \quad \pi_S(x) = (x_i)_{i \in S}, \qquad \iota_S : \mathbb{F}_2^S \to \mathbb{F}_2^n, \quad (\iota_S y)_i = \begin{cases} y_i, & i \in S, \\ 0, & i \notin S. \end{cases}$$

For $d \geq 1$,

$$\mathrm{AGL}(d,2) := \mathbb{F}_2^d \rtimes \mathrm{GL}_d(\mathbb{F}_2)$$

acts by $x \mapsto Ax + b$. An *affine relabelling* of ports indexed by $\mathbb{F}_2^d$ is the corresponding permutation of those ports. Indexing the eight coordinates by $\mathbb{F}_2^3$ in lexicographic order, define the first-order Reed–Muller code by

$$RM(1,3) := \{(\ell(t))_{t\in\mathbb{F}_2^3} : \ell(t) = a + u_1t_1 + u_2t_2 + u_3t_3,\ a, u_1, u_2, u_3 \in \mathbb{F}_2\}.$$

Thus $RM(1,3)$ consists of the sixteen truth tables of affine-linear functions on $\mathbb{F}_2^3$. The natural action satisfies $\mathrm{AGL}(2,2) \cong S_4$; $\mathrm{AGL}(3,2)$ is two-transitive and preserves $RM(1,3)$, because affine precomposition preserves affine truth tables [Car21, Section 4.1].

A *spread* in a binary vector space is a family of nonzero subspaces whose nonzero vectors partition the nonzero vectors of the ambient space; a spread coset is a coset of one of its member subspaces. For a function $F : \mathbb{F}_2^d \to R$ into an abelian group, its additive derivative is

$$(\Delta_a F)(x) := F(x + a) - F(x).$$

The identities $\Delta_a\Delta_b F = 0$ for all a, b hold exactly when $F(x) = F(0) + \ell(x)$ for a group homomorphism $\ell : (\mathbb{F}_2^d, +) \to (R, +)$: the identities make every $\Delta_a F$ constant, and the resulting $\ell(x) := F(x) - F(0)$ is additive.

For $Q : \mathbb{F}_2^d \to R$, with R a commutative ring, its *Boolean Möbius expansion* is the unique multilinear expression

$$Q(x) = \sum_{S\subseteq[d]} \widehat{Q}_S \prod_{j\in S} x_j, \qquad \widehat{Q}_S = \sum_{T\subseteq S} (-1)^{|S|-|T|} Q(\mathbf{1}_T).$$

The coefficient table $(\widehat{Q}_S)_S$ is the Boolean Möbius transform; the formula is interpreted in the stated coefficient ring, including $\mathbb{F}_2$ and $\mathbb{Z}_4$. Direct substitution is the Möbius-inversion proof of uniqueness. This additive transform is distinct from the multiplicative Möbius inversion used later.

2.8.2 Symmetric binary spaces

A bilinear form $b : W \times W \to \mathbb{F}_2$ is linear in each argument and is symmetric when $b(x, y) = b(y, x)$. A *symmetric binary space* is a finite-dimensional $\mathbb{F}_2$-space with a specified symmetric form, written $x \cdot y$. On $\mathbb{F}_2^m$, the standard dot product is $x \cdot y := \sum_j x_j y_j$. For $U \leq W$, define

$$U^{\perp W} := \{y \in W : x \cdot y = 0 \text{ for every } x \in U\}, \qquad \mathrm{rad}_W(U) := U \cap U^{\perp W}.$$

We omit W when it is clear. The restricted form on U is *nondegenerate* when $\mathrm{rad}_W(U) = \{0\}$. The notation $W = U \perp P$ means $W = U \oplus P$ and $U \cdot P = 0$; then $\pi_U(u + p) = u$ and $\pi_P(u + p) = p$ are the orthogonal projections.

If $R \leq \mathrm{rad}_W(U)$, the quotient U/R has the induced form $(x + R) \cdot (y + R) := x \cdot y$. When R acts by translations on an affine space with translation space U, its orbit space has translation space U/R. A linear map $\phi : U \to U'$ is an *isometry* when it is bijective and preserves the form; the corresponding affine isometry is $a + u \mapsto a' + \phi(u)$.

A vector x is *isotropic* when $x \cdot x = 0$ and anisotropic otherwise. A subspace is *totally isotropic* when $U \subseteq U^{\perp W}$, and is *self-dual in* W when $U = U^{\perp W}$. In a nondegenerate symmetric binary space, the map $x \mapsto x \cdot x$ is linear, so nondegeneracy identifies it with a unique vector c satisfying $x \cdot x = c \cdot x$; this is the *characteristic vector*. The space is alternating when $c = 0$, and nonalternating otherwise.

For linearly independent vectors $r_1, \dots, r_s$, a *dual family* is a family $r_1^*, \dots, r_s^*$ with $r_i \cdot r_j^* = \mathbf{1}[i = j]$. Pairing the radical means adjoining such a family, chosen orthogonal to a previously fixed nondegenerate complement. Indeed, when the ambient W is nondegenerate and $R = \mathrm{span}_{\mathbb{F}_2}\{r_1, \dots, r_s\}$, the pairing map

$$W \longrightarrow R^\vee, \qquad w \longmapsto (r \mapsto r \cdot w),$$

is surjective: otherwise a nonzero element of R would pair to zero with all of W. Choosing preimages gives a dual family. If $U = P \perp R$ with P nondegenerate and $R = \mathrm{rad}_W(U)$, the same argument inside $P^{\perp_W}$ makes the family orthogonal to P. The new vectors are independent modulo U, as pairing a linear combination with each r_i shows. On R and its dual family the Gram matrix is $\left(\begin{smallmatrix} 0 & I \\ I & A \end{smallmatrix}\right)$, which is nonsingular. Adjoining the family therefore gives a nondegenerate extension of dimension $\dim U + \dim R$. We use the following facts about symmetric binary spaces: nondegenerate subspaces split off orthogonally; every even-dimensional nondegenerate symmetric binary space has a self-dual subspace; every nondegenerate nonalternating m-space is isometric to the standard dot-product space $\mathbb{F}_2^m$; and every odd-dimensional nondegenerate symmetric binary space is nonalternating and has an anisotropic vector. Here are short proofs. For a nondegenerate subspace P, its pairing map identifies P with $P^\vee$, so each ambient vector differs from a unique vector of P by an element of $P^\perp$. This gives the orthogonal splitting. A nonzero nondegenerate alternating space contains vectors u, v with $u \cdot v = 1$; their Gram matrix is $H = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Splitting off these planes shows that an alternating space has even dimension, and one vector from each plane spans a self-dual subspace. In a nonalternating space, split off norm-one vectors until the remainder is alternating. A norm-one vector e orthogonal to such a plane can absorb it: $e + u, e + v, e + u + v$ are an orthogonal norm-one basis of their three-dimensional span. Repeating this gives the standard dot-product basis. In even dimension $2k$, the vectors $e_1 + e_2, \dots, e_{2k-1} + e_{2k}$ span a self-dual subspace of that space (the empty span applies in dimension zero). Finally, odd dimension excludes the alternating case and hence supplies a norm-one vector.

2.9 Bell signatures and the two affine cores

The four Bell tensors form a basis for binary contractions. Their action on the two signatures below supplies the local data reconstructed in Section 8; the dictionary also fixes the signs inherited from Shao–Cai's conventions.

For $d, \ell, u, v \in \mathbb{F}_2$, define the ordered Bell signature

$$B_{d,\ell}(u, v) = \mathbf{1}[u \oplus v = d](-1)^{\ell u}.$$

Write $B_\gamma = B_{d,\ell}$ for $\gamma = (d, \ell)$. Port order matters for $B_{1,1}$, which is negated under reversal.

With truth-table order $00, 01, 10, 11$, these four signatures are

$$\begin{aligned} B_{0,0} &= (1, 0, 0, 1) = I, \quad B_{0,1} = (1, 0, 0, -1) = Z, \\ B_{1,0} &= (0, 1, 1, 0) = X, \quad B_{1,1} = (0, 1, -1, 0) = Y. \end{aligned} \tag{13}$$

Matrix and marked-pair dictionary. Combining Equations (6) and (13) gives the actual identities

$$I = Q_0 = B_{0,0}, \qquad iX = Q_1 = iB_{1,0}, \qquad Y = Q_2 = B_{1,1}, \qquad iZ = Q_3 = iB_{0,1}. \tag{14}$$

Consequently the marked projective notation used throughout is

$$\begin{aligned} 1 = [I] = [Q_0] = [B_{0,0}], &\quad x = [X] = [Q_1] = [B_{1,0}], \\ j = [Z] = [Q_3] = [B_{0,1}], &\quad xj = [Q_2] = [B_{1,1}]. \end{aligned} \tag{15}$$

Here the last identity is projective: $XZ = -Y$, whereas $ZX = Y$. For the transfer convention $T_B = BX$, the four Bell binaries give

B	Bell name	$[B]$	$[T_B] = [BX]$
I	$B_{0,0}$	1	x
X	$B_{1,0}$	x	1
Z	$B_{0,1}$	j	xj
Y	$B_{1,1}$	xj	j

The subscripted $B_{d,\ell}$ is a fixed ordered Bell signature, whereas B_g later denotes a chosen actual binary representative of a general transfer class g. Reversal fixes I, X, Z and sends $Y = B_{1,1}$ to $-Y$, without changing its projective class. Throughout, a *Pauli binary* means a nonzero scalar multiple of one of these four ordered Bell tensors B_γ, with its displayed input order retained; the adjective *actual* singles out a chosen representative rather than only its projective class.

Remark 2.4 (The Bell-signature dictionary of Shao–Cai). The tuple in Equation (13) is Shao–Cai's $(=_2^+, =_2^-, \neq_2^+, \neq_2^-)$ [SC20, Section 2]; $Y^{\mathsf{T}} = -Y$. Their holographic matrix named Z is our K, whereas our Z is only the diagonal sign matrix.

Use the lexicographic indexing of the eight ports by $\mathbb{F}_2^3$ fixed in Section 2.8.1. Define the eight-ary Reed–Muller indicator signature $\mathrm{RM}_8 : \mathbb{F}_2^8 \to \{0,1\}$ by

$$\mathrm{RM}_8(x) := \mathbf{1}[x \in RM(1,3)]. \tag{16}$$

Define the matrix

$$\Theta = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{F}_2), \qquad \Theta^3 = I.$$

In a tensor $A \otimes B \otimes C$ of three ordered binary signatures, A, B, and C act on port pairs $(1,2)$, $(3,4)$, and $(5,6)$, respectively. With this convention, the six-port Pauli core is

$$\begin{aligned} \mathrm{H}_6 &= \sum_{\gamma \in \mathbb{F}_2^2} B_\gamma \otimes B_{\Theta\gamma} \otimes B_{\Theta^2\gamma} \\ &= I \otimes I \otimes I + Z \otimes Y \otimes X + X \otimes Z \otimes Y + Y \otimes X \otimes Z. \end{aligned} \tag{17}$$

For an ordered binary tensor A and an ordered pair of ports (p,q), the notation $A^{(p,q)}$ places the first and second inputs of A on ports p and q, respectively. For an even finite port set R, a *Bell product* on R is a nonzero scalar multiple of

$$\bigotimes_{e=(p_e,q_e)\in M} B_{\gamma_e}^{(p_e,q_e)}, \qquad \gamma_e \in \mathbb{F}_2^2,$$

where M is a perfect matching of R and each pair of M has been given the displayed orientation. Reversing a pair carrying $B_{1,1}$ only changes the overall scalar sign, so this class is independent of the chosen orientations. For $R = \varnothing$, a Bell product is a nonzero nullary signature. A Bell product is a *same-γ Bell product* when every factor label γ_e equals one fixed γ.

An even-arity tensor-prime signature f has the *Bell property* when, for every ordered pair of distinct ports (i,j) and every $\gamma \in \mathbb{F}_2^2$, the contraction $\partial_{(i,j)}^{B_\gamma} f$ is a Bell product on the remaining ports. It has the *strong Bell property* when each such contraction is, more specifically, a same-γ Bell product. These are algebraic contraction properties and do not by themselves assert actual gadget provenance. This is the ordered-port, projective form of the standard definitions [SC20, full version, Definition 6.12] and [CFS20b, full version, Definition 1.10; conference version, Definition 12].

Remark 2.5 (The Shao–Cai f_8 and $\widehat{f_6}$ cores). With the preceding port order, RM_8 is literally the signature $f_8 = \widehat{f_8}$ of [SC20, full version, Section 8]; it has the strong Bell property: contracting any ordered port pair by B_γ gives, up to a nonzero scalar, a product of three copies of B_γ on the remaining ports [CFS20b, Theorems 13–14]. Let $\widehat{f_6}$ denote the six-ary signature of Shao–Cai given in [SC20, full version, Eq. (6.1)]. Direct comparison with Equation (17) gives the literal port-permutation identity

$$\mathrm{H}_6(x_1, \ldots, x_6) = \widehat{f_6}(x_1, x_3, x_5, x_4, x_6, x_2).$$

The same results show that H_6 has the Bell property: every ordered Bell contraction is, up to a nonzero scalar, a product of two ordered Bell binaries [CFS20b, Theorems 13–14]; see also [SC20, full version, Definition 6.12 and Lemma 6.13]. We retain $\mathrm{RM}_8, \mathrm{H}_6$ as names because their code and Pauli geometry drive the relative argument. The inverse deck classifications, the actual H_6-to-RM_8 circuit, and the signature-set-wide lifts are not supplied by these forward Bell-property results and are proved here.

The Bell tables satisfy the exact bilinear orthogonality relation

$$\langle B_\gamma, B_{\gamma'} \rangle_F = 2\,\mathbf{1}[\gamma = \gamma'].$$

2.10 Tractable signature sets and effective recognition

2.10.1 The standard signature classes

We include the zero signature in every class below. The affine class $\mathscr{A}$ consists of signatures

$$f(x) = \lambda\,\mathbf{1}[Ax = b]\, i^{Q(x)}, \tag{18}$$

where $\lambda \in \overline{\mathbb{Q}} \setminus \{0\}$ and, for $r = \mathrm{arity}(f)$, there is an $m \geq 0$ with $A \in \mathbb{F}_2^{m \times r}$ and $b \in \mathbb{F}_2^m$. The function $Q : \{0,1\}^r \to \mathbb{Z}_4$, where $\mathbb{Z}_4 = \mathbb{Z}/4\mathbb{Z}$, has a multilinear representative

$$Q(x) = q_0 + \sum_{j=1}^{r} q_j x_j + \sum_{1 \leq j < \ell \leq r} q_{j\ell} x_j x_\ell \pmod 4, \qquad q_{j\ell} \in \{0, 2\},$$

so it has degree at most two and even mixed coefficients. Equivalently, the phase is a product of fourth-root phases of affine linear forms interpreted as $\{0,1\}$-valued integers. Membership in $\mathscr{A}$ always refers to the displayed computational coordinates.

For a nowhere-zero function $f : \mathbb{F}_2^k \to \overline{\mathbb{Q}} \setminus \{0\}$, define its multiplicative third-difference ratio $\mathcal{T}_f : (\mathbb{F}_2^k)^4 \to \overline{\mathbb{Q}}^\times$ by

$$\mathcal{T}_f(z, u, v, w) = \frac{f(z+u+v+w)f(z+u)f(z+v)f(z+w)}{f(z+u+v)f(z+u+w)f(z+v+w)f(z)}.$$

Here $z, u, v, w \in \mathbb{F}_2^k$, and repeated directions are permitted. For fixed u, v, let $D_{u,v}$ be the multiplicative second-difference operator acting on the entire function f:

$$(D_{u,v} f)(z) := \frac{f(z+u+v)f(z)}{f(z+u)f(z+v)}.$$

Lemma 2.6 (Affine phase criterion)**.** *Let $f : \mathbb{F}_2^k \to \overline{\mathbb{Q}} \setminus \{0\}$. Then f belongs to $\mathscr{A}$ if and only if $\mathcal{T}_f(z, u, v, w) = 1$ for all $z, u, v, w \in \mathbb{F}_2^k$.*

Proof. Define the scalar $\lambda = f(\mathbf{0}^k)$, the scalars $A_j = f(e_j)/\lambda$ for $j \in [k]$, and, for $1 \le j < \ell \le k$, the scalars

$$B_{j\ell} = \frac{\lambda f(e_j + e_\ell)}{f(e_j) f(e_\ell)}.$$

Substitution of $z = \mathbf{0}^k$ and $u = v = w = e_j$ gives $A_j^4 = 1$; taking $u = v = e_j$ and $w = e_\ell$ gives $B_{j\ell}^2 = 1$. Moreover, the identity $\mathcal{T}_f(z, u, v, w) = 1$ says exactly that

$$(D_{u,v} f)(z + w) = (D_{u,v} f)(z).$$

Because w is arbitrary, $D_{u,v} f$ is constant on $\mathbb{F}_2^k$. In particular,

$$(D_{e_j,e_\ell} f)(z) = (D_{e_j,e_\ell} f)(\mathbf{0}^k) = B_{j\ell} \qquad (1 \le j < \ell \le k).$$

We now give the Hamming-weight induction explicitly. Extend the notation symmetrically by $B_{\ell j} = B_{j\ell}$. If $z_j = 0$, start with $f(e_j)/f(\mathbf{0}^k) = A_j$ and add the basis vectors in $\mathrm{supp}(z)$ one at a time. The identity $D_{e_j,e_\ell} f = B_{j\ell}$, rearranged after the previously added vector y, is

$$\frac{f(y + e_\ell + e_j)}{f(y + e_\ell)} = B_{j\ell} \frac{f(y + e_j)}{f(y)}.$$

It follows by induction on $\mathrm{wt}(z)$ that

$$\frac{f(z + e_j)}{f(z)} = A_j \prod_{\ell \ne j} B_{j\ell}^{z_\ell} \qquad (z_j = 0). \tag{19}$$

We next prove the desired formula for $f(z)$, again by induction on $\mathrm{wt}(z)$. It is immediate for $z = \mathbf{0}^k$. For $z \ne \mathbf{0}^k$, choose $j \in \mathrm{supp}(z)$ and define the predecessor word $y := z + e_j$, so $y_j = 0$ and $\mathrm{wt}(y) = \mathrm{wt}(z) - 1$. Applying the induction hypothesis to $f(y)$ and Equation (19) to $f(y + e_j)/f(y)$ yields

$$f(z) = \lambda \prod_j A_j^{z_j} \prod_{j<\ell} B_{j\ell}^{z_j z_\ell}, \qquad A_j \in \mu_4, \quad B_{j\ell} \in \{1, -1\},$$

because the new factors are precisely A_j and $B_{j\ell}$ for $\ell \in \mathrm{supp}(y)$. This is the full-support normal form in Equation (18). Conversely, each linear factor and each even-quadratic factor cancels in the multiplicative third difference, so $\mathcal{T}_f = 1$. □

Lemma 2.7 (Two-translate closure of the affine class)**.** *If $h \in \mathscr{A}$ and w is a Boolean translation of its input coordinates, then $h + T_w h \in \mathscr{A}$ (with the zero signature allowed), where the translation operator T_w is defined by $(T_w h)(x) := h(x + w)$.*

Proof. The zero case is immediate. Let n denote the arity of h, so $n := \mathrm{arity}(h)$. For $h \ne 0$, write

$$h(x) = c\, \mathbf{1}[x \in a + L] i^{Q(x)}, \qquad c \in \overline{\mathbb{Q}}^\times, \quad a \in \mathbb{F}_2^n, \quad L \le \mathbb{F}_2^n,$$

where $Q : \mathbb{F}_2^n \to \mathbb{Z}_4$ is a quadratic phase with even mixed coefficients. If $w \notin L$, every word in the union of the two supports has a unique form $a + y + tw$, with $y \in L$ and $t \in \mathbb{F}_2$. On this affine space,

$$(h + T_w h)(a + y + tw) = c\, i^{Q(a+y)}.$$

The inverse coordinates (y, t) are affine, so the displayed exponent is a quadratic phase with even mixed coefficients. Thus the sum is affine. For $w \in L$, expand $Q(x + w) - Q(x) = c_w + 2\ell_w(x) \pmod 4$, where $c_w \in \mathbb{Z}_4$ is constant and $\ell_w : \mathbb{F}_2^n \to \mathbb{F}_2$ is linear. This gives

$$h(x) + h(x + w) = c\, i^{Q(x)} \big(1 + i^{c_w} (-1)^{\ell_w(x)}\big) \quad (x \in a + L).$$

For even c_w the bracket is a scalar times an affine hyperplane indicator; for odd c_w it is a nonzero scalar times a fourth-root linear phase. Restriction of Q and adjoining a linear phase preserve $\mathscr{A}$, proving the claim. The displayed difference formula follows by expanding the multilinear representative of Q; all nonconstant coefficients of that difference are even. □

Let L be a linear subspace of $\mathbb{F}_2^{2k}$. Its orthogonal complement under the standard dot product $x \cdot y := \sum_{j=1}^{2k} x_j y_j \in \mathbb{F}_2$ is

$$L^{\perp} = \{y \in \mathbb{F}_2^{2k} : x \cdot y = 0 \text{ for every } x \in L\}.$$

Call L *Lagrangian* when $L = L^{\perp}$, and an arity-$2k$ nonzero signature *flat-Lagrangian* when it has the form

$$f(x) = c\,\mathbf{1}[x \in a + L], \qquad 0 \neq c \in \overline{\mathbb{Q}}, \quad a \in \mathbb{F}_2^{2k}, \quad L = L^{\perp}.$$

Let FLag denote the zero signature together with all flat-Lagrangian signatures of every even arity. We call an arity-$2k$ signature *character-flat* when a linear sign character is also allowed:

$$\mathsf{CFlat}_{2k} = \left\{c(-1)^{\ell(x)}\mathbf{1}[x \in a + L] : c \in \overline{\mathbb{Q}}^{\times},\ a \in \mathbb{F}_2^{2k},\ \ell \in (\mathbb{F}_2^{2k})^{\vee},\ L = L^{\perp}\right\} \cup \{0\}. \tag{20}$$

Define the signature class $\mathsf{CFlat} = \bigcup_{k \geq 0} \mathsf{CFlat}_{2k}$. This is a subclass of $\mathscr{A}$; it is the closure needed when a Reed–Muller core is used together with all four Pauli signs rather than only I/X.

The product class $\mathscr{P}$ consists of zero and all signatures of the form $\lambda \prod_j h_j$, where $0 \neq \lambda \in \overline{\mathbb{Q}}$ and every factor is either a nonzero unary signature $u(x_i)$, a binary equality $I(x_i, x_j)$, or a binary disequality $X(x_i, x_j)$ on designated signature variables. Empty products and repeated variables are allowed; an inconsistent product denotes the zero signature.

Define the primitive eighth root and the phase matrix by

$$\zeta_8 := e^{\pi i/4} = \frac{1+i}{\sqrt{2}}, \qquad T_2 := \operatorname{diag}(1, \zeta_8).$$

The local-affine class $\mathscr{L}$ consists of the signatures f such that

$$\left(\bigotimes_{j=1}^{\operatorname{arity}(f)} T_2^{\alpha_j}\right) f \in \mathscr{A} \qquad \text{for every } \alpha \in \operatorname{supp}(f).$$

Let $\mathscr{T}$ be the set of all unary and binary signatures, and define the signature class $\mathscr{M}$ by

$$\mathscr{M} = \{f : f(x) = 0 \text{ whenever } \operatorname{wt}(x) > 1\}.$$

Definition 2.8 (Common transformability and K-presentation)**.** Let $\mathscr{C}$ be a signature class closed under nonzero scalars and port permutations. A finite signature set $\mathcal{F}$ is $\mathscr{C}$-transformable if one $M \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ satisfies

$$M^{\mathsf{T}} M \in \mathscr{C}, \qquad M^{-1}\mathcal{F} \subseteq \mathscr{C}.$$

Equivalently, for $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$, one matrix $U = K^{-1}M$ gives the common K-presentation

$$U^{\mathsf{T}} X U \in \mathscr{C}, \qquad U^{-1}\widehat{\mathcal{F}} \subseteq \mathscr{C}.$$

We abbreviate this signature-set-wide condition by

$$\mathsf{Pres}_K(\mathcal{F}; \mathscr{C}) \iff \exists U \in \mathrm{GL}_2(\overline{\mathbb{Q}}) : \quad U^{\mathsf{T}} X U \in \mathscr{C}, \qquad U^{-1}(K^{-1}\mathcal{F}) \subseteq \mathscr{C}. \tag{21}$$

If Γ is already in K-coordinates, its common $\mathscr{C}$ K-presentation is precisely

$$\begin{aligned}\mathsf{Pres}_K(K\Gamma;\mathscr{C}) \quad &\Longleftrightarrow \exists U \in \mathrm{GL}_2(\overline{\mathbb{Q}}) : \\ &\qquad U^{\mathsf{T}} X U \in \mathscr{C}, \qquad U^{-1}\Gamma \subseteq \mathscr{C}.\end{aligned}$$

These conditions include the edge and require one basis for the entire signature set, not one basis per signature. We use this definition for $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$, and for the flat-Lagrangian and CFlat subclasses.

A common flat-Lagrangian K-presentation requires the transformed edge and every nonzero transformed signature to be flat-Lagrangian; it strengthens, rather than enlarges, the affine family. A common character-flat presentation is defined analogously and likewise implies one common affine presentation.

2.10.2 The Eulerian-orientation predicate

For an arity-r signature g, define its central restriction

$$g|_{\mathrm{EO}}(x) = g(x)\mathbf{1}[q_r(x) = 0].$$

It is zero for odd r, and zero restrictions are deleted. Define the resulting central-restriction signature set by

$$\mathcal{G}|_{\mathrm{EO}} = \{g|_{\mathrm{EO}} : g \in \mathcal{G},\ g|_{\mathrm{EO}} \not\equiv 0\}.$$

Let $g \neq 0$ be a signature of arity k supported on the single Hamming layer of weight d. Its *EO completion* is the ordered tensor

$$g_{\to\mathrm{EO}} = \begin{cases} g \otimes \Delta_0^{\otimes(2d-k)}, & 2d \geq k, \\ g \otimes \Delta_1^{\otimes(k-2d)}, & 2d < k. \end{cases} \tag{22}$$

The original ports precede the added unaries; a zeroth power is the nullary unit. The completed signature has even arity and central support, hence is EO. For a signature set of nonzero single-layer signatures, define its EO-completion signature set by

$$\mathcal{G}_{\to\mathrm{EO}} = \{g_{\to\mathrm{EO}} : g \in \mathcal{G}\}.$$

For an arity-$2d$ EO signature g, an oriented perfect pairing $\pi = ((a_1, b_1), \ldots, (a_d, b_d))$ has unordered pairs partitioning $[2d]$. For $z \in \mathbb{F}_2^d$, let $y \in \mathbb{F}_2^{2d}$ be the unique word with $y_{a_j} = z_j$ and $y_{b_j} = 1 - z_j$; the pairing induces

$$g^{\pi}(z_1, \ldots, z_d) = g(y).$$

Definition 2.9 (The predicate $\mathsf{EOTract}$)**.** For a finite EO signature set $\mathcal{G}$, $\mathsf{EOTract}(\mathcal{G})$ holds if there exist a single sign $\varepsilon \in \{+1, -1\}$ and a single class $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}\}$, both shared by the entire signature set, such that:

1. for every $g \in \mathcal{G}$ and all $\alpha, \beta, \gamma \in \mathrm{supp}(g)$, if $\delta = \alpha \oplus \beta \oplus \gamma$, then $\delta \in \mathrm{supp}(g)$ or $\varepsilon q_{\mathrm{arity}(g)}(\delta) > 0$;
2. for every $g \in \mathcal{G}$ and every oriented perfect pairing π of its ports, $g^{\pi} \in \mathscr{C}$.

We adopt the vacuous convention $\mathsf{EOTract}(\varnothing) = \mathrm{true}$.

This is the compressed form of the opposite-pair restriction used in [MWXZ25, GSS26]: substitute $y_{b_j} = 1 - y_{a_j}$. To make the interface literal, define the full zero-extended restriction by

$$R_\pi g(x) = g(x) \prod_{j=1}^{d} X(x_{a_j}, x_{b_j}).$$

After permuting ports and writing the two incidences of each pair as (z_j, w_j), this becomes

$$R_\pi g(z, w) = g^\pi(z) \prod_{j=1}^{d} X(z_j, w_j).$$

Thus $g^\pi \in \mathscr{A}$ (respectively $\mathscr{P}$) implies $R_\pi g \in \mathscr{A}$ (respectively $\mathscr{P}$) by adjoining binary affine (respectively product) factors, while contracting each w_j with the unary vector $[1, 1]$ recovers g^π. This is a closure statement about the normal forms, not an assertion that $[1, 1]$ is a free input gadget. In the native half-edge model the factors in $R_\pi g$ are realized by the structural X-edges, so the converse lift has exactly the source's value-preserving meaning. Conversely, $\mathscr{A}$ is closed under this affine-variable summation (Gaussian elimination). The product class has the analogous elementary star elimination: after multiplying all unary factors at an eliminated variable, choose one incident I/X edge as a representative, replace every other incident edge by the corresponding I/X edge between its neighbor and the representative, and absorb the remaining unary weight into that representative (an inconsistent component gives zero; an isolated variable gives a scalar). In the displayed bridge each w_j is a leaf, and explicitly

$$\sum_{w_j \in \{0,1\}} X(z_j, w_j)[1, 1](w_j) = 1.$$

Thus the contraction simply deletes that factor and preserves the product form (up to the displayed scalar). Hence $R_\pi g \in \mathscr{C}$ if and only if $g^\pi \in \mathscr{C}$, for $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}\}$.

The source papers use unoriented pairs. Reversing one pair applies the unary coordinate permutation $z_j \mapsto 1 - z_j$, and reordering pairs is a port permutation; both preserve $\mathscr{A}$ and $\mathscr{P}$. Since the signature set is fixed, enumerating orientations and pairings is constant-size. Consequently the oriented predicate above and the source predicate are equivalent, with the same exact polynomial-time tests.

2.10.3 The tractability predicates

The tractability predicate used here combines the seven alternatives of Meng, Wang, Xia, and Zheng with the polynomial-time algorithms of Guan, Shao, and Shi for the two support-defined alternatives [MWXZ25, GSS26]. We package this boundary as Tract. It is the general tractability criterion underlying Theorems 1.1 and 1.2: after the stated preprocessing, its truth specifies the polynomial-time side of the main dichotomies, and its failure specifies the #P-hard side. We retain the full seven-clause predicate because it is also applied to intermediate signature sets without a distinguished binary signature.

When either predicate below is applied to an arbitrary finite input set, its argument is first reduced to the nonzero positive-arity part according to Section 2.6. The definitions may therefore be stated for preprocessed signature sets without carrying a separate superscript in the public criteria.

For a preprocessed signature set $\mathcal{F}$, define its K-coordinate image $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$. Define $\mathsf{Tract}(\mathcal{F})$ to be the disjunction of the following seven alternatives.

(1) The signature set $\widehat{\mathcal{F}}$ is common one-sided, in the explicit sense that

$$[\forall \widehat{f} \in \widehat{\mathcal{F}},\ \mathrm{supp}(\widehat{f}) \subseteq HW^{\geq}_{\mathrm{arity}(\widehat{f})}] \quad \text{or} \quad [\forall \widehat{f} \in \widehat{\mathcal{F}},\ \mathrm{supp}(\widehat{f}) \subseteq HW^{\leq}_{\mathrm{arity}(\widehat{f})}],$$

and $\mathsf{EOTract}(\widehat{\mathcal{F}}|_{\mathrm{EO}})$ holds.

(2) Every $\widehat{f} \in \widehat{\mathcal{F}}$ is supported on one Hamming weight d_f. Across the signature set, some support word has negative charge and some (possibly other) support word has positive charge, and $\mathsf{EOTract}(\widehat{\mathcal{F}}_{\to \mathrm{EO}})$ holds, with the completion operation defined in Equation (22).

(3) $\mathcal{F} \subseteq \langle \mathscr{T} \rangle$.

(4) $\mathcal{F} \subseteq \langle K\mathscr{M} \rangle$, or $\mathcal{F} \subseteq \langle KX\mathscr{M} \rangle$.

(5) $\mathcal{F}$ is $\mathscr{A}$-transformable.

(6) $\mathcal{F}$ is $\mathscr{P}$-transformable.

(7) $\mathcal{F}$ is $\mathscr{L}$-transformable.

Remark 2.10. The matrix symbols in Item (4) act tensorwise at the arity of each signature. In particular,

$$f \in \langle K\mathscr{M} \rangle \iff \text{every tensor-prime factor of } K^{-1}f \text{ lies in } \mathscr{M},$$
$$f \in \langle KX\mathscr{M} \rangle \iff \text{every tensor-prime factor of } XK^{-1}f \text{ lies in } \mathscr{M}.$$

For the binary-disequality problems studied here, however, the tractable boundary collapses to four alternatives. A literal X in the signature set—or, in the coordinate image tested by the main theorems, the equivalent anchor $Z = K^{\otimes 2}I$—rules out the one-sided, single-weight, and matching alternatives. This motivates the following reduced predicate.

Definition 2.11 (The reduced predicate Tract_X)**.** For a preprocessed signature set $\mathcal{F}$, define

$$\begin{aligned} \mathsf{Tract}_X(\mathcal{F}) \iff\ & \mathcal{F} \subseteq \langle \mathscr{T} \rangle, \\ & \text{or } \mathcal{F} \text{ is } \mathscr{A}\text{-transformable}, \\ & \text{or } \mathcal{F} \text{ is } \mathscr{P}\text{-transformable}, \\ & \text{or } \mathcal{F} \text{ is } \mathscr{L}\text{-transformable}. \end{aligned} \tag{23}$$

Thus Tract_X is exactly the disjunction of Items (3) and (5) to (7).

Lemma 2.12 (The binary anchor removes the other three alternatives)**.** *Let $\mathcal{F}$ be a preprocessed signature set. If $X \in \mathcal{F}$, then*

$$\mathsf{Tract}(\mathcal{F}) \iff \mathsf{Tract}_X(\mathcal{F}).$$

The same conclusion holds if $Z = \mathrm{diag}(1, -1) \in \mathcal{F}$.

Proof. If $X \in \mathcal{F}$, then

$$-iZ = K^{-1}X \in \widehat{\mathcal{F}};$$

if $Z \in \mathcal{F}$, then instead $I = K^{-1}Z \in \widehat{\mathcal{F}}$. In either case $\widehat{\mathcal{F}}$ contains a nonsingular binary signature supported precisely on $\{00, 11\}$, with both endpoint values nonzero. The words 00 and 11 have opposite charges, so Item (1) is impossible, and they have different Hamming weights, so Item (2) is impossible.

The retained binary has matrix rank two and is therefore tensor-prime. It does not belong to $\mathscr{M}$, since it is nonzero at 11; applying $X^{\otimes 2}$ only exchanges its two endpoint values, so the resulting

tensor is again tensor-prime and not in $\mathscr{M}$. The two characterizations in the preceding remark therefore exclude both orientations in Item (4). Hence only Items (3) and (5) to (7) can hold, which proves the forward implication. The reverse implication follows immediately because these four alternatives are clauses of Tract. □

Both coordinate images tested in the main theorems contain $Z = K^{\otimes 2}I$. Thus the second part of Lemma 2.12 identifies their reduced Tract_X tests with the full Tract criterion.

2.10.4 Effective algebraic geometry

The transformability test in Lemma 2.13 asks for one matrix satisfying polynomial conditions for the edge and every signature. The tools below turn that existence question into ideal membership and recover an exact algebraic matrix when one exists.

Let $R = L[x_1, \dots, x_n]$ be a polynomial ring over the stated exact field. An ideal $I \lhd R$ is an additive subgroup with $rf \in I$ for all $r \in R, f \in I$. It is proper when $I \neq R$, prime when it is proper and $fg \in I$ implies $f \in I$ or $g \in I$. For $f_1, \dots, f_s \in R$, their generated ideal is

$$(f_1, \dots, f_s) := \left\{ \sum_{j=1}^{s} r_j f_j : r_1, \dots, r_s \in R \right\};$$

an ideal is finitely generated when it equals one of this form. We use *algebraically closed* in the sense of Section 2.1.1. For an algebraically closed $\Omega \supseteq L$, the affine zero locus is

$$V_\Omega(I) := \{a \in \Omega^n : f(a) = 0 \text{ for every } f \in I\}.$$

A projective zero locus is defined by homogeneous equations in the specified projective space. A Zariski-closed set is a zero locus, a Zariski-open set is its complement, a locally closed set is an intersection of one of each, and a constructible set is a finite union of locally closed sets.

A monomial order is a total order $\prec$ in which every nonempty monomial set has a least member, $1 \preceq m$ for every monomial m, and $m \prec n$ implies $mp \prec np$ for every monomial p. For $f \neq 0$, $\mathrm{LM}(f)$ is its largest monomial, and $\mathrm{LM}(I)$ is the ideal generated by all $\mathrm{LM}(f)$ with $0 \neq f \in I$. A finite $\mathcal{G} \subseteq I$ is a Gröbner basis when its leading monomials generate $\mathrm{LM}(I)$. It is reduced when every element is monic—its leading coefficient is 1—and no monomial of one element is divisible by the leading monomial of another. A *monomial ideal* is an ideal generated by monomials. For a fixed order, the reduced basis is unique; division by a Gröbner basis decides ideal membership, and Buchberger's algorithm constructs one. It terminates by the ascending-chain condition: there is no infinite strictly increasing chain of monomial ideals [CLO15, Chapters 2–3].

For $\Delta \in R$, abbreviate $V = V_{\overline{L}}$ and $V(\Delta) := V((\Delta))$. The weak Nullstellensatz and the geometric effect of saturation are

$$V_{\overline{L}}(I) = \varnothing \Longleftrightarrow 1 \in I, \qquad I : \Delta^\infty := \{f : \Delta^m f \in I \text{ for some } m \geq 0\},$$

$$V(I : \Delta^\infty) = \overline{V(I) \setminus V(\Delta)}^{\,\mathrm{Zar}},$$

where Zariski closure means the smallest Zariski-closed set containing the displayed set, so saturation retains precisely the irreducible components not contained in $V(\Delta)$ when I is radical [CLO15, Chapters 4 and 8]. Here an ideal is *radical* when $f^m \in I$ implies $f \in I$, and a prime decomposition expresses a radical ideal as a finite intersection of prime ideals. This ideal-theoretic radical is unrelated to the radical of a bilinear form.

A nonempty closed set is irreducible when it is not the union of two proper closed subsets, and an irreducible component is a maximal irreducible closed subset. For a nonempty algebraic set X, its dimension $\dim X$ is the largest d for which there is a chain $Z_0 \subsetneq \cdots \subsetneq Z_d \subseteq X$ of nonempty irreducible closed subsets. For an ordered variable split, the elimination ideal is $I \cap L[x_{k+1}, \ldots, x_n]$. In a triangular set, each successive equation has a new largest variable, called its main variable, and its initial is its leading coefficient in that variable. An element a is a non-zero-divisor modulo J when $ab \in J \Rightarrow b \in J$. A regular chain is a triangular set in which every successive initial is a non-zero-divisor modulo the ideal obtained by saturating the preceding equations by their initials. Its constructible branch requires all initials to be nonzero. A triangular or regular-chain decomposition splits a solution set into finitely many such constructible branches. A *generic algebraic specialization* assigns algebraic values to the remaining free coordinates outside the finite union of exceptional zero loci on which a required initial or denominator vanishes. Regular-chain decomposition, generic specialization, and minimal polynomials with isolating data produce exact algebraic sample points [ALMM99, Coh93].

For a field extension L'/L, base change sends I to $I_{L'} := I\,L'[x_1, \ldots, x_n]$ and extends the associated solution spaces. When $L \subseteq \mathbb{C}$, *complexification* means this base change with $L' = \mathbb{C}$. In finite certificates, RREF means reduced row-echelon form: every nonzero row has a leading 1, pivot columns have no other nonzero entry, pivots move strictly rightward down the rows, and zero rows are last.

Lemma 2.13 (Effective algebraic transformability witnesses)**.** *Let $E \in \mathrm{Mat}_2(\overline{\mathbb{Q}})$, let $\mathcal{F}$ be a finite algebraic signature set, and let $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$. Suppose there is a matrix $U \in \mathrm{GL}_2(\mathbb{C})$ such that*

$$U^{\mathsf{T}} E U \in \mathscr{C}, \qquad U^{-1}\mathcal{F} \subseteq \mathscr{C},$$

where membership in $\mathscr{C}$ allows complex values for the free scalar and unary parameters in its normal forms. Then such a witness exists in $\mathrm{GL}_2(\overline{\mathbb{Q}})$. Moreover, for exact algebraic input, the existence of a witness is decidable and an algebraic witness can be produced.

Proof. Let L be a number field containing the entries of E and $\mathcal{F}$, as well as i and $\zeta_8 = (1+i)/\sqrt{2}$ (equivalently i and $\sqrt{2}$). There are only finitely many combinatorial normal-form types compatible with the arities: affine supports and phase coefficients, product partitions with equality or disequality signs and unary weights, and, for $\mathscr{L}$, the finitely many prescribed T_2-twisted affine conditions. Enumerate one shared type choice for the transformed edge and all transformed signatures.

For $\mathscr{P}$, repeated factors cause no infinitude: at fixed arity we canonically enumerate the support set, tensor partition, factor multiplicities, equality/disequality signs, unary phase type, and port permutation. On a fixed pair of variables, repeated copies of the same I- or X-factor are idempotent and are retained as a single factor; the simultaneous presence of an I- and an X-factor makes the whole product zero. A loop $I(x, x)$ is the scalar one and a loop $X(x, x)$ is zero. Unary factors on one variable are multiplied and combined into one unary (or zero). This gives a finite canonical enumeration without using a mod–two rule for product factors. For $\mathscr{L}$, enumerate the prospective transformed support set and affine phase data. Vanishing outside that mask, nonvanishing on it, and the corresponding twist conditions then become finitely many polynomial equalities and inequations. Thus every candidate, including one whose transformed support is not known in advance, is one of finitely many constructible systems.

Fix one such branch, and let $\mathbf{x}$ be the tuple of all its scalar variables: the four entries of U, the scalar and unary parameters in the chosen normal forms, and the auxiliary variables introduced below. Introduce an auxiliary scalar variable z and impose

$$z \det U = 1.$$

Here $\mathrm{adj}(U)$ denotes the classical adjugate, characterized by $U\,\mathrm{adj}(U) = (\det U)I$; hence on this solution set $U^{-1} = z\,\mathrm{adj}(U)$. After this substitution, equality of every transformed tensor entry with its proposed normal form is a polynomial equation over L. If the branch also has the nonvanishing polynomial constraints $p_1, \ldots, p_s \neq 0$, introduce one more auxiliary scalar variable t and replace all these inequations by the single equation

$$t \prod_{\ell=1}^{s} p_\ell = 1.$$

Thus the branch is exactly the zero set of a finitely generated ideal $I \subseteq L[\mathbf{x}]$.

Compute a Gröbner basis of I over the exact number field L, using the terminating ideal-membership algorithm described above. Let $\overline{L} \subseteq \mathbb{C}$ be the algebraic closure of the embedded number field L inside $\mathbb{C}$. Writing $V_{\overline{L}}(I)$ for the affine zero locus of I over $\overline{L}$, the standard ideal-membership and weak-Nullstellensatz theorems give

$$V_{\overline{L}}(I) = \varnothing \quad \Longleftrightarrow \quad 1 \in I,$$

and the latter is equivalent to the reduced Gröbner basis being $\{1\}$. Hence emptiness of each branch is decidable. Moreover, a complex solution makes I proper, so the weak Nullstellensatz supplies a point over $\overline{L} = \overline{\mathbb{Q}}$; conversely every such point is a complex solution. In a zero-dimensional branch, standard elimination followed by triangular decomposition produces exact minimal polynomials and isolating data for one point. A nonempty positive-dimensional branch is also effective: enumerate algebraic tuples fairly, first by total exact encoding height and then lexicographically, and test every displayed equation exactly. The weak Nullstellensatz guarantees that this search eventually reaches a point of the branch. Equivalently one may use a regular-chain decomposition with a generic algebraic specialization of its free coordinates. Reading the U-coordinates gives the required algebraic witness. These are standard effective consequences of Gröbner-basis theory [Coh93, CLO15]; no encoding-size or running-time bound is asserted here. □

2.10.5 Imported dichotomies and decomposition

Unless an ordinary Holant symbol or an explicit source presentation $\mathrm{Holant}(X \mid \cdot)$ is displayed, the imported K-Holant statements below use native K-coordinates (so the edge tensor is X); applying the fixed K-change gives the corresponding ordinary statement. All signature sets in these statements are fixed and preprocessed. In particular, every later use of Theorem 2.15 on a native set Γ means the typed bridge

$$\text{K-Holant}(\Gamma) \equiv_{\mathrm{T}} \mathrm{Holant}(K\Gamma), \qquad K\Gamma = \{K^{\otimes \mathrm{arity}(g)} g : g \in \Gamma\},$$

followed by the ordinary theorem on $K\Gamma$ and the inverse bridge. The known nonzero edge and gadget scalars are restored by the record.

Theorem 2.14 (Weighted Eulerian orientations [MWX25]; [GSS26, Theorems 1.1 and 2.17])**.** *For each fixed finite preprocessed algebraic EO signature set $\mathcal{G}$ (zero signatures and nullary scalar factors are deleted and their known scalar contributions are restored separately), if* $\mathsf{EOTract}(\mathcal{G})$ *holds, then* K-Holant$(\mathcal{G}) \in \mathsf{FP}$*; otherwise* K-Holant$(\mathcal{G})$ *is* #P*-hard.*

Theorem 2.15 (Odd-arity Holant [MWXZ25, full version, Theorem 33; conference version, Theorem 30])**.** *Let $\mathcal{F}$ be a finite preprocessed algebraic signature set containing a nonzero odd-arity signature. Then* $\mathrm{Holant}(\mathcal{F}) \in \mathsf{FP}$ *if* $\mathsf{Tract}(\mathcal{F})$ *holds, and is* #P*-hard otherwise.*

Its first two alternatives were originally in $\mathsf{FP}^{\mathsf{NP}}$; the stated FP conclusion follows from [GSS26, Theorem 1.1 and Corollary 1.2].

Theorem 2.16 (Tensor-factor extraction [MWXZ25, full version, Theorem 32 and Remark 39; conference version, Theorem 29])**.** *Let Λ be a finite preprocessed algebraic signature set in native K-coordinates, and let $h = a \otimes b$ be a nonzero tensor product with algebraic factors a, b of positive arity (equivalently, apply the source theorem to the ordinary set $K(\Lambda \cup \{h\})$). Any nullary scalar factor is absorbed during preprocessing. Unless $\Lambda \cup \{h\}$ is common one-sided,*

$$\text{K-Holant}(\Lambda, a, b) \equiv_{\mathrm{T}} \text{K-Holant}(\Lambda, h).$$

In the common-one-sided exception, K-Holant(Λ, h) *is in* FP *when* $\mathsf{EOTract}((\Lambda \cup \{h\})|_{\mathrm{EO}})$ *holds and is* $\#\mathsf{P}$*-hard otherwise, by the one-sided clause of the cited theorem and Theorem 2.14.*

Theorem 2.17 (Quaternary eight-vertex boundary [CF23, Theorem 3.1])**.** *Let M be an algebraic quaternary signature satisfying*

$$\mathrm{supp}(M) \subseteq \{x \in \{0,1\}^4 : \mathrm{wt}(x) = 2\} \cup \{0000, 1111\}, \qquad M(0000)M(1111) \neq 0.$$

Then Holant$(X \mid M)$ *is* $\#\mathsf{P}$*-hard unless there exist $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$ and $U \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ such that*

$$U^{\mathsf{T}} X U \in \mathscr{C}, \qquad U^{-1} M \in \mathscr{C}.$$

Every exception is in FP*. The source permits U over $\mathbb{C}$; Lemma 2.13 makes it algebraic.*

Lemma 2.18 (One-hot matching evaluation)**.** *Fix a finite algebraic signature set, and let Γ be a closed factor-expanded finite native-X network over it (each occurrence has first been replaced by its fixed tensor-prime factors, with the induced port wiring and recorded scalar). If a vertex signature is zero, return zero; otherwise, for every vertex w, let h_w denote its nonzero signature, define its arity by $d_w := \mathrm{arity}(h_w)$, and suppose $\mathrm{supp}(h_w) \subseteq \{\mathbf{0}^{d_w}, e_1, \ldots, e_{d_w}\}$. Define its zero-word weight $a_w := h_w(\mathbf{0}^{d_w})$ and weight-one port coefficients $b_{w,p} := h_w(e_p)$. Then $Z_X(\Gamma)$ is computable using $O(|V(\Gamma)| + |E_{\mathrm{int}}(\Gamma)|)$ arithmetic operations in the fixed coefficient field, and hence in polynomial bit time; $V(\Gamma)$ is the vertex set.*

Proof. For an X-edge exactly one of its two half-edges receives bit 1. Consequently assignments are in bijection with choices of one endpoint of each edge, subject to at most one chosen endpoint at each vertex. For such a feasible endpoint-choice assignment ω, write $\mathrm{occ}_\omega(w) \in \{0,1\}$ for the number of selected endpoints at w, and let $s_w(\omega)$ be the selected port when it exists. The weight of ω is

$$\prod_{\mathrm{occ}_\omega(w)=0} a_w \prod_{\mathrm{occ}_\omega(w)=1} b_{w,s_w(\omega)}.$$

The product of the fixed nonzero scalar factors recorded during the factor expansion is multiplied into the final component product (and a zero expansion scalar was already handled by returning zero), so it does not change the recurrence below. For a connected component, let e and v denote its edge and vertex counts. A component with $e > v$ has value zero. Every surviving component has $e = v - 1$ (a tree) or $e = v$ (unicyclic, with loops and parallel edges included). On a rooted tree, let p_w be the parent-edge port at w, and let p_{wc} be the port at w of the child edge wc. If $F_w(1)$ (resp. $F_w(0)$) denotes the subtree sum when the parent edge selects w (resp. its parent), then

$$F_w(1) = b_{w,p_w} \prod_c F_c(1),$$

$$F_w(0) = a_w \prod_c F_c(1) + \sum_c b_{w,p_{wc}} F_c(0) \prod_{d \neq c} F_d(1).$$

At the root use the second line without a parent term. Order the children as $c_1, \dots, c_m$. Products and sums are evaluated without division by the prefix products $P_j := \prod_{\ell<j} F_{c_\ell}(1)$ and suffix products $S_j := \prod_{\ell>j} F_{c_\ell}(1)$, for which $P_j S_j = \prod_{\ell\neq j} F_{c_\ell}(1)$ even when some factors vanish. For a unicyclic component delete one cycle edge e, with endpoint ports p_u at u and p_v at v. Enumerate the two legal choices $p \in \{p_u, p_v\}$, and let $v(p)$ be the vertex incident to the chosen port. The ports are distinct even when e is a loop. For each choice, root the resulting tree at $v(p)$ and impose that p is already occupied through e: at that root use

$$b_{v(p),p} \prod_c F_c(1).$$

This forces every remaining incident edge at that root to select its child. Use the ordinary tree recurrence on all proper subtrees and perform this same forced-root calculation separately for each p; sum the two values. The same boundary-condition description handles a loop (the two ports are distinct incidences at one vertex) and parallel cycle edges. The prefix/suffix evaluations use linearly many field operations. Every intermediate entry is a sum of at most exponentially many products of $O(|V| + |E_{\text{int}}|)$ fixed coefficients. In a fixed rational basis of their number field, a common denominator and the numerator sizes therefore have $O(|V| + |E_{\text{int}}|)$ bits, giving polynomial bit time. □

2.10.6 Decidability and polynomial-time evaluation

Theorem 2.19 (Recognition and evaluation)**.** *For every finite algebraic signature set $\mathcal{F}$, the predicate* $\mathsf{Tract}(\mathcal{F})$*, with the preprocessing convention of Section 2.6, is decidable. If it holds, then* $\text{Holant}(\mathcal{F}) \in \mathsf{FP}$.

Proof. The two EO alternatives are decided by finitely many exact tests of support, layers, affine subspaces, and phases; enumerate all oriented pairings and the four shared $(\varepsilon, \mathscr{C})$-choices. A signature's tensor decomposition can be found by enumerating all nontrivial bipartitions $S \mid S^c$, where $\varnothing \neq S \subsetneq [\text{arity}(f)]$ and $S^c = [\text{arity}(f)] \setminus S$, and testing whether the corresponding matrix flattening has rank one; recursively applying this test to the factors decides Items (3) and (4). Meanwhile, Lemma 2.13 with $E = I$ decides Items (5) to (7) and returns the common witness.

For evaluation, charge conservation reduces Item (1) to Theorem 2.14: in a closed contributing assignment all local charges have the common sign and sum to zero, so every local charge is zero and only the central restrictions remain. Thus the off-central entries may be deleted in this evaluator (this is an identity on closed instances, not a claim that the restrictions are free gadgets). In Item (2), an occurrence of arity k_f and weight d_f has charge $c_f = 2d_f - k_f$; each connected component C must satisfy

$$\sum_{v \in V(C)} c_{f_v} = 0.$$

Failure gives zero. Otherwise construct an auxiliary EO instance solely for the evaluator: replace each occurrence f_v by its fixed completion $(f_v)_{\to\text{EO}}$, and in each component pair the newly appended Δ_0-ports (from $c_{f_v} > 0$) with the appended Δ_1-ports (from $c_{f_v} < 0$) by native X-edges. The charge balance makes the two numbers equal, and every original assignment has a unique extension to this auxiliary instance (and conversely), so its partition function is unchanged. These pins are explicit entries of the derived tensors in an algorithmic reduction; no pinning signature is assumed to be a free gadget of the input problem. The finite completed signature set is EO, so Theorem 2.14 supplies the polynomial-time evaluator. After tensor-prime expansion, the arity-two branch is a graph of maximum degree two and is evaluated by transfer matrices on paths and cycles. For the matching

alternatives, first apply the common K-coordinate equivalence, the possible global X-toggle, and tensor-prime expansion; the toggle is a legal native change because $X^{\mathsf{T}}XX = X$. After this change, every counted prime factor h satisfies $h(x) = 0$ for $\mathrm{wt}(x) > 1$. Only structural equality kernels, or degree-two equality representations introduced solely to display the source interface, are suppressed before the factor graph is built; an explicit arity-two signature occurrence remains a factor vertex unless it has first been eliminated by the path/cycle preprocessing of Lemma 5.1, with its known scalar recorded. Thus Lemma 2.18 gives a polynomial-time evaluator using linearly many field operations; the expansion has constant size per occurrence because the signature set is fixed. Finally, a common witness M gives

$$\mathrm{Holant}((\mathrm{EQ}_2)M \mid M^{-1}\mathcal{F}), \qquad (\mathrm{EQ}_2)M = M^{\mathsf{T}}M,$$

with both sides in the same class $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$. Forgetting the bipartition yields an ordinary $\mathrm{Holant}(\mathscr{C})$ instance, computable in FP: affine instances are quadratic Gauss sums, product instances split into unary/equality/disequality components, and local-affine instances are a restriction of the corrected complex-valued Holant^c evaluator of [Bac21, Bac25] (the real-valued presentation [CLX18, Corollary 3.1] gives the same local routine). □

2.11 Projective tensor and finite-incidence geometry

In Sections 6 and 9, a matching product is a simple tensor after its ports are grouped into pairs. Segre geometry therefore turns tensor factorization into rank equations. Incidences among the available binary kernels then constrain which pairings those equations can describe.

2.11.1 Projective tensor geometry

We use the projective-space conventions of Section 2.1.3. Whenever a projective object occurs in a literal tensor equation, a displayed or previously fixed nonzero representative is understood; changing it changes only the separately recorded scalar. We set $\mathbb{P}(0) := \varnothing$.

A nonzero tensor $v_1 \otimes \cdots \otimes v_s$ is *simple* or *decomposable*. For finite-dimensional spaces $V_1, \ldots, V_s$, the Segre map is

$$\mathbb{P}(V_1) \times \cdots \times \mathbb{P}(V_s) \longrightarrow \mathbb{P}(V_1 \otimes \cdots \otimes V_s), \qquad ([v_1], \ldots, [v_s]) \longmapsto [v_1 \otimes \cdots \otimes v_s].$$

Its image is the *Segre variety*; its affine cone consists of zero and the simple tensors. The Segre ideal is generated by the 2-by-2 minors of the bipartite flattenings. A *one-factor Segre ruling* is a projective linear family in which all factors but one are fixed, and every projective line contained in a Segre variety belongs to such a ruling [Lan12, Chapter 2].

2.11.2 Möbius geometry

The Riemann sphere is $\widehat{\mathbb{C}} := \mathbb{C} \cup \{\infty\} \cong \mathbb{P}(\mathbb{C}^2)$, with affine chart $z = [z : 1]$ and $\infty = [1 : 0]$. A Möbius transformation is a projective automorphism of this sphere; in the affine chart it is

$$z \longmapsto \frac{az + b}{cz + d}, \qquad ad - bc \neq 0,$$

where $cz + d = 0$ is sent to ∞ and ∞ is interpreted projectively. A *generalized circle* is a Euclidean circle $\{z : |z - z_0| = r\}$, $r > 0$, or a real affine line together with ∞; such a line has the form $\{z : \mathrm{Re}(\overline{\alpha}z) = \beta\}$ for $\alpha \in \mathbb{C}^\times$ and $\beta \in \mathbb{R}$. Möbius transformations carry generalized circles to generalized circles [Bea83, Chapter 3]. The unit circle is $S^1 := \{z \in \mathbb{C} : |z| = 1\}$; its counterclockwise cyclic order is the order of arguments modulo 2π.

2.11.3 Root configurations and incidence

Let E be a real vector space with a positive-definite symmetric bilinear form $\langle\ ,\ \rangle$; this is a real inner-product space. A *reduced finite root system* in E is a finite spanning set Φ of nonzero vectors such that each reflection

$$v \longmapsto v - 2\frac{\langle v,\alpha\rangle}{\langle \alpha,\alpha\rangle}\alpha \qquad (\alpha \in \Phi)$$

permutes Φ, and $\Phi \cap \mathbb{R}\alpha = \{\alpha, -\alpha\}$. Its root-line configuration is

$$\mathcal{R}(\Phi) := \{[\alpha] : \alpha \in \Phi\} \subseteq \mathbb{P}(E \otimes_{\mathbb{R}} \mathbb{C}).$$

In root-system notation, $\Phi(D_4), \Phi(F_4), \Phi(H_4)$ denote the standard systems of those types, and

$$\mathcal{R}(D_4) := \mathcal{R}(\Phi(D_4)), \quad \mathcal{R}(F_4) := \mathcal{R}(\Phi(F_4)), \quad \mathcal{R}(H_4) := \mathcal{R}(\Phi(H_4)).$$

These are root-system types [Hum90]; in particular, the type D_4 differs from the order-four dihedral group $D_4 \cong V_4$.

A finite projective point configuration is a finite subset X of a projective space. Points are collinear when they lie on one projective line. A secant line contains at least two points of X, a k-rich line contains at least k, and a plane section is $X \cap \Pi$ for a projective plane Π. A point–line incidence structure is a pair $(P, \mathcal{L})$, where $\mathcal{L}$ is a family of subsets of P; an automorphism is a permutation of P preserving $\mathcal{L}$. More generally, a finite incidence hypergraph has the named points as vertices and the named incident subsets as hyperedges, and its automorphisms preserve every distinguished hyperedge family.

For four distinct points $p_i = [v_i]$ on a projective line $\ell = \mathbb{P}(U)$, choose an ordered basis of the two-dimensional vector space U. Let $\det_U(v, w)$ be the 2-by-2 determinant of the coordinate columns of v, w in that basis, and define

$$[p_1, p_2; p_3, p_4] := \frac{\det_U(v_1, v_3)\det_U(v_2, v_4)}{\det_U(v_1, v_4)\det_U(v_2, v_3)}.$$

Changing a representative scales one numerator and one denominator factor, and changing the ordered basis scales every determinant by the same nonzero factor. Hence the scalar is well defined. The ordered quadruple is *harmonic* when this value is -1. A harmonic projectivity between two four-point lines is a projective isomorphism carrying one set to the other and preserving exactly the ordered quadruples of cross-ratio -1. This projective cross-ratio is distinct from any locally defined table cross-ratio.

2.12 Laurent gauges and product weights

The diagonal parameters in Section 6 attach independent weights to ports. Laurent monomials distinguish their contributions, so coefficient identities can recover restrictions on the original tensor without choosing a new basis for the Holant instance.

2.12.1 Laurent polynomials

For a commutative ring R, a Laurent polynomial over R is a finite sum

$$P(\lambda) = \sum_{u \in \mathbb{Z}^r} c_u \lambda^u, \qquad \lambda^u := \prod_{j=1}^{r} \lambda_j^{u_j},$$

with $c_u \in R$ and all but finitely many c_u zero; equivalently, $P \in R[\lambda_1^{\pm 1}, \dots, \lambda_r^{\pm 1}]$. Coefficient extraction is $[\lambda^u]P := c_u$. When a Boolean vector is used as an exponent, its entries are lifted from $\{0,1\}$ to $\mathbb{Z}$, so exponent arithmetic is never modulo two. For nonzero P, put

$$m_j := \max\bigl(0, -\min\{u_j : c_u \neq 0\}\bigr), \qquad m = (m_1, \dots, m_r).$$

Then $m \in \mathbb{Z}_{\geq 0}^r$ is coordinatewise minimal with $\lambda^m P \in R[\lambda_1, \dots, \lambda_r]$; thus multiplication by λ^m clears all negative exponents. The coordinate degree in λ_j of a nonzero P is

$$\deg_{\lambda_j} P := \max\{u_j : c_u \neq 0\} - \min\{u_j : c_u \neq 0\}.$$

By convention, $\deg_{\lambda_j} 0 := -\infty$; consequently, every finite upper degree bound used below includes the zero Laurent polynomial.

2.12.2 Gauges and product weights

A *portwise gauge* is an invertible diagonal rescaling on specified tensor ports; on a binary flattening it gives invertible diagonal row or column scaling. It is an analytic coordinate operation, not an available gadget unless separately realized. A genuine character of $(\mathbb{F}_2^r, +)$ has the form $z \mapsto \prod_j \epsilon_j^{z_j}$ with $\epsilon_j \in \{1, -1\}$. More generally,

$$z \longmapsto \prod_j \gamma_j^{z_j}, \qquad \gamma_j \in \mathbb{C}^\times,$$

is a *product weight*; it is a character exactly when every $\gamma_j^2 = 1$.

2.13 Hilbert-space, Pauli, and stabilizer conventions

The stabilizer arguments in Sections 8 and 9 view signature coefficients as amplitudes and Pauli contractions as linear measurements. Commuting Pauli labels describe affine support and phase constraints; an encoder normal form recovers tensor factors from those constraints. These are analytic descriptions of signatures. Each gadget realization is established separately with the native bilinear contraction.

2.13.1 Hilbert-space conventions

A finite-dimensional complex Hilbert space is a complex vector space with a Hermitian inner product, conjugate-linear in its first argument, linear in its second, satisfying $\langle v, u\rangle = \overline{\langle u, v\rangle}$, and positive definite in the sense that $\langle u, u\rangle > 0$ for $u \neq 0$. For a matrix A, $A^\dagger$ is its conjugate transpose and $\langle u, v\rangle := u^\dagger v$. Put $\mathcal{H}_n := (\mathbb{C}^2)^{\otimes n}$, and write

$$|x\rangle := |x_1\rangle \otimes \cdots \otimes |x_n\rangle, \qquad \langle x| := |x\rangle^\dagger \quad (x \in \mathbb{F}_2^n).$$

A unitary on $\mathcal{H}_n$ satisfies $U^\dagger U = UU^\dagger = I_{\mathcal{H}_n}$, while an isometry $V : \mathcal{H}_k \to \mathcal{H}_n$ satisfies $V^\dagger V = I_{\mathcal{H}_k}$. These Hermitian conventions are distinct from the bilinear Frobenius pairing used for Holant contractions and from isometries of symmetric binary forms.

A density operator is a positive-semidefinite operator of trace one, where positive semidefinite means $u^\dagger \rho u \geq 0$ for all u. A nonzero vector $|\psi\rangle$ has normalized density operator

$$\rho_\psi = \frac{|\psi\rangle\langle\psi|}{\langle\psi, \psi\rangle}.$$

Thus a nonzero ket, modulo multiplication by a nonzero scalar, determines a *pure state* through ρ_ψ. For a density operator ρ on $\mathcal{H}_A \otimes \mathcal{H}_B$, fix an orthonormal basis $\{|b_j\rangle\}_j$ of $\mathcal{H}_B$. The partial trace is specified, for every $|a\rangle, |a'\rangle \in \mathcal{H}_A$, by

$$\langle a| \operatorname{Tr}_B(\rho)|a'\rangle = \sum_j \langle a, b_j|\rho|a', b_j\rangle.$$

The result is independent of the chosen orthonormal basis of $\mathcal{H}_B$. The operator $\operatorname{Tr}_B(\rho)$ is the *marginal* on $\mathcal{H}_A$; orthonormal means $\langle u_i, u_j\rangle = \mathbf{1}[i = j]$. A one-qubit marginal is maximally mixed when it is $I_{\mathbb{C}^2}/2$. Every bipartite pure state has a Schmidt decomposition

$$|\psi\rangle = \sum_{j=1}^{s} \sigma_j |u_j\rangle \otimes |v_j\rangle, \qquad \sigma_j > 0,$$

with both displayed families orthonormal; s is the Schmidt rank [NC10, Section 2.5]. For $n \geq 1$, $r \in [n]$, and a one-qubit linear operator $R : \mathbb{C}^2 \to \mathbb{C}^2$, define

$$R_r := I_{\mathbb{C}^2}^{\otimes(r-1)} \otimes R \otimes I_{\mathbb{C}^2}^{\otimes(n-r)} : \mathcal{H}_n \to \mathcal{H}_n.$$

2.13.2 Pauli and stabilizer conventions

The actual n-qubit Pauli group is

$$\mathcal{P}_n^{(4)} := \{i^c \textstyle\bigotimes_{j=1}^n X^{a_j} Z^{b_j} : c \in \mathbb{Z}_4,\ a, b \in \mathbb{F}_2^n\}.$$

Its projectivization by the scalar subgroup $\{\pm I_{\mathcal{H}_n}, \pm i I_{\mathcal{H}_n}\}$ is identified with $\mathsf{P}([n]) = \mathbb{F}_2^n \oplus \mathbb{F}_2^n$; the label $a \mid b$ is the concatenation of the two binary words. This label space has the symplectic form

$$\omega((a \mid b), (a' \mid b')) = a \cdot b' + b \cdot a'.$$

Here symplectic means alternating $\omega(v, v) = 0$ and nondegenerate $\{v : \omega(v, w) = 0 \text{ for all } w\} = \{0\}$. For a subspace S, put $S^{\perp\omega} := \{v : \omega(v, s) = 0 \text{ for every } s \in S\}$; S is *Lagrangian* when $S = S^{\perp\omega}$. If S is Lagrangian, a *Lagrangian complement* of S is a Lagrangian subspace T satisfying $\mathsf{P}([n]) = S \oplus T$.

A *projective Clifford* is the phase-class, modulo multiplication by a scalar in S^1, of a unitary U satisfying $U\mathcal{P}_n^{(4)}U^\dagger = \mathcal{P}_n^{(4)}$. Clifford conjugation induces a symplectic linear map on Pauli labels [NC10, Chapter 10]. A stabilizer code is the common eigenspace

$$\{v : Pv = \chi(P)v \text{ for every } P \in \widetilde{S}\}$$

of a commuting Pauli subgroup $\widetilde{S}$, where $\chi : \widetilde{S} \to \mu_4$ is an eigenvalue character agreeing with the scalar by which every scalar Pauli acts. A one-dimensional stabilizer code is a stabilizer state, and a nonzero tensor spanning it is a stabilizer tensor.

A stabilizer isometry is an isometry whose image is a stabilizer code and whose logical Paulis have physical Pauli representatives; equivalently it is a Clifford encoder with fixed stabilizer-state ancillas. Here an *encoder* is the code isometry, a *physical operator* acts on its ambient output space, the corresponding *logical operator* is the induced operator on the encoded input space, and an *ancilla* is a fixed initialized tensor factor. If S is the projective stabilizer label space, the logical Pauli label space is $S^{\perp\omega}/S$; for a 2^k-dimensional code it has binary dimension $2k$. A Clifford decoder is a Clifford unitary D satisfying

$$DV|\psi\rangle = |\psi\rangle \otimes |\eta\rangle \qquad (|\psi\rangle \in \mathcal{H}_k)$$

for a fixed stabilizer ancilla $|\eta\rangle$, namely an additional tensor factor initialized in that state. These standard encoder–decoder and logical-Pauli facts follow from the stabilizer normal form [Got97, Chapters 3–4].

A stabilizer tableau is a binary symplectic generator matrix whose rows label Pauli generators, together with their phases; multiplying generators is an elementary row operation, while a Clifford coordinate change acts by a symplectic change of columns [Got97, Chapter 4]. A hyperbolic pair is a pair p, q with $\omega(p, q) = 1$; symplectic Gaussian elimination is row reduction by changes of symplectic basis. A nowhere-zero stabilizer quadratic phase is

$$\zeta(z) = \zeta_0 i^{Q(z)}, \qquad \zeta_0 \in \mathbb{C}^\times,$$

where $Q : \mathbb{F}_2^k \to \mathbb{Z}_4$ is multilinear of degree at most two with even mixed coefficients. After fixed physical and logical Clifford coordinate changes, a stabilizer isometry has amplitude normal form

$$V|z\rangle = \zeta(z)|c + M^{\mathsf{T}} z\rangle, \qquad c \in \mathbb{F}_2^n, \quad M \in \mathrm{Mat}_{k\times n}(\mathbb{F}_2), \quad \operatorname{rank} M = k,$$

a standard consequence of stabilizer-tableau reduction [Got97, Chapter 4].

2.13.3 Stabilizer error correction

Erasure of physical output r of an isometry V is correctable exactly when the single-site Knill–Laflamme condition holds:

$$V^\dagger R_r V \in \mathbb{C} I_{\mathcal{H}_k} \qquad \text{for every one-qubit operator } R.$$

This is the single-subsystem specialization of the Knill–Laflamme criterion [KL97]. In the stabilizer setting, an encoder tableau is the tableau of a Clifford encoder; its symplectic row reduction records physical representatives of logical Paulis and the residual stabilizer generators.

2.14 Finite induction and termination

The reductions in Section 5.2 retain tensor factors and sometimes enlarge the finite group of binary transfers. The orders below distinguish finite processing of a fixed family from a restart with a smaller arity or larger group.

2.14.1 Orders and termination

A strict order is irreflexive and transitive. It is *well founded* when it has no infinite descending chain. A finite multiset over a set E records elements of E with multiplicity; in this paragraph $\dot\cup$ adds multiplicities, rather than denoting a disjoint union of sets. Given a strict order $<$ on E and finite multisets A, B over E, its strict multiset extension is defined by $A <_{\mathrm{mul}} B$ when

$$B = Z \mathbin{\dot\cup} X, \qquad A = Z \mathbin{\dot\cup} Y, \qquad X \neq \varnothing,$$

for finite multisets X, Y, Z over E, and every $y \in Y$ satisfies $y < x$ for some $x \in X$. The multiset extension of a well-founded strict order is well founded [DM79]. Lexicographic order compares the first coordinate at which two tuples differ; finite lexicographic products of well-founded orders are well founded by induction on the number of coordinates. Well-founded induction proves a property at x assuming it at every strict predecessor of x.

We index unfinished signature occurrences by a finite multiset. Each occurrence retains its realizing gadget or reduction and its ordered ports. To identify duplicate signatures, normalize by the first nonzero coefficient in the fixed port order, recording the removed scalar separately. Repeated occurrences are still counted with multiplicity in the induction measures. The reductions test their listed cases in order; the first applicable case determines the next step.

2.15 Retained reductions and normalization

2.15.1 Retained factor saturation

Record each recursive tensor factorization as a tree, retaining the ordered port blocks, port permutation, and nonzero scalar at every split. The leaves are the tensor-prime factors.

For a preprocessed finite signature set Λ, choose an exact tensor-prime factorization of every member and let $\Pi(\Lambda)$ be the joint signature set of all prime factors, with ordered port blocks retained. We call Λ *factor-saturated* when $\Pi(\Lambda) \subseteq \Lambda$.

Lemma 2.20 (Retained factor saturation)**.** *If Λ is not common one-sided, then*

$$\text{K-Holant}(\Lambda) \equiv_{\mathrm{T}} \text{K-Holant}(\Lambda, \Pi(\Lambda)). \tag{24}$$

The recursive state is the literal augmentation $\Lambda \cup \Pi(\Lambda)$, retaining in particular the original two-sided charge witnesses. Moreover, every invocation of factor saturation is a finite worklist computation. Start with one root occurrence for each member of the current finite retained set and expose a factor only through the decomposition theorem. Let h_u denote the signature carried by a nonzero node u. Such a node, carrying its ordered port block, is either certified tensor-prime, or is replaced by one split

$$h_u \doteq a_u \otimes b_u, \qquad 1 \le \operatorname{arity}(a_u), \operatorname{arity}(b_u) < \operatorname{arity}(h_u),$$

with the nonzero scalar and the port permutation recorded in the provenance. Nullary scalars are recorded and discarded, and unary nodes are certified immediately and never enter the unresolved queue. Fix the representative of a node by dividing by its first nonzero entry and retaining its ordered port block; identical canonical keys may share one retained signature, while their occurrences may still be kept in the queue. Let $\mathcal{W}$ be the multiset of unresolved open occurrences, all of arity at least two. The potential

$$\Phi(\mathcal{W}) = \sum_{u \in \mathcal{W}} (\operatorname{arity}(h_u) - 1) \tag{25}$$

is a nonnegative integer and strictly decreases at every worklist step. Thus $\Pi(\Lambda)$ is the finite set of canonical prime leaves. Its independence of processing order follows from the tensor-prime factorization uniqueness stated in Section 2.3.

Proof. Apply the decomposition theorem at each queued split $h_u = a_u \otimes b_u$ to the current retained signature set [MWXZ25, full version, Theorem 32 and Remark 39]. The original non-one-sided signature set remains literally present, so the exceptional case never reappears. Iteration exposes every factor, while literal inclusion gives the reverse reduction; these two directions give Equation (24). Replacing a nonunary work item of arity $r+s$ removes $r+s-1$ and can add at most $(r-1)+(s-1) = r+s-2$; already queued or certified canonical children only lower this amount further. Certifying a prime removes its summand. Hence the factorization tree and the retained augmentation are finite, and all reduction scalars and ordered blocks are available for later gadgets. □

For a factor-saturated signature set Λ containing a tensor-prime signature of arity at least three, define its minimum nonbinary prime arity by

$$\nu(\Lambda) = \min\{\operatorname{arity}(h) : h \in \Lambda \text{ is tensor-prime and } \operatorname{arity}(h) \ge 3\}.$$

This parameter is used only after the branch in which every prime factor has arity at most two has been removed.

2.15.2 Downward inheritance and global normalization

Lemma 2.21 (Downward inheritance)**.** *If $\mathcal{G} \subseteq \mathcal{G}'$ are preprocessed signature sets and* $\mathsf{Tract}(K\mathcal{G}')$ *holds, then* $\mathsf{Tract}(K\mathcal{G})$ *holds.*

Proof. All alternatives except the single-weight one are hereditary by definition. In the latter case a signature subset either retains both charge signs and the same EO-completion witness, or becomes one-sided. In the latter event, its central members are unchanged by both restriction and EO completion, whereas all off-central members restrict to zero. Hence its nonzero central restrictions form a subset of the original completion signature set, and the same shared EO witness applies. □

Lemma 2.22 ($GO(X)$-invariance of transported tractability)**.** *Let $\mathcal{G}$ be a finite preprocessed signature set and let $D \in GO(X)$. Then*

$$\mathsf{Tract}(K\mathcal{G}) \quad \Longleftrightarrow \quad \mathsf{Tract}\big(K(D\mathcal{G})\big).$$

Proof. Write the matrix $D = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, and let $\lambda \in \overline{\mathbb{Q}}^{\times}$ be its conformal multiplier, defined by $D^{\mathsf{T}} X D = \lambda X$. Comparing entries gives $ac = bd = 0$ and $ad + bc = \lambda \neq 0$. Thus D is diagonal or anti-diagonal. The former preserves every support word; the latter complements it; on a fixed layer its multiplier is constant. Hence the two EO alternatives are preserved (with the support side and shared sign possibly exchanged), tensor-prime factor arities are unchanged, and $\langle \mathscr{M} \rangle$ and $\langle X\mathscr{M} \rangle$ are preserved or exchanged.

Define the conjugate matrix $A = KDK^{-1}$, so $A^{\mathsf{T}}A = \lambda I$. If the matrix W is a common $\mathscr{C}$-transformability witness, define the transported witness matrix $W' := AW$. It satisfies

$$(W')^{-1} A(K\mathcal{G}) = W^{-1}(K\mathcal{G}) \subseteq \mathscr{C}, \qquad (W')^{\mathsf{T}} W' = \lambda W^{\mathsf{T}} W \in \mathscr{C}$$

for $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$. Applying the same argument to D^{-1} proves the converse. □

Every reduction, basis change, group enlargement, hardness conclusion, and tractable presentation obtained inside a coordinate-transport branch can be transported correctly back to the original signature set. The following lemma records the data needed for that transport. Its continuing-state use is confined to the self-contained eight-vertex interface; later sections may invoke it to interpret a call to that interface, but do not use a realized weighted equality to replace the supplied literal I.

Lemma 2.23 (Normalization data)**.** *Let $\mathcal{G}_0$ be the signature set at the beginning of a structural branch in which coordinate transport is permitted. Let L_0 be a fixed number field containing all entries of $\mathcal{G}_0$, their complex conjugates, and all algebraic constants already fixed by the coordinate conventions. Thus L_0 is the frozen initial coefficient field, not a field enlarged after each normalization. Every continuing normalization inside that branch is an explicitly recorded $GO(X)$ change of coordinates, and tensor splits are performed after pulling back by the current A, using a nonzero pivot and row–column ratios. Thus the pulled-back data (rather than the displayed current coordinates) stay over L_0, and the pulled-back group remains inside* $\mathrm{PGL}_2(L_0)$*. The structural reductions maintain a current retained signature set Λ, a matrix $A \in GO(X)$, and its nonzero conformal factor $\mu_A \in \overline{\mathbb{Q}}^{\times}$, satisfying*

$$A\mathcal{G}_0 \subseteq \Lambda, \qquad A^{\mathsf{T}} X A = \mu_A X, \qquad \text{K-Holant}(\Lambda) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{G}_0). \tag{26}$$

Every positive-arity $h \in \Lambda$, up to a nonzero scalar, also has the field-compatible form

$$h = A^{\otimes \mathrm{arity}(h)} h_0 \quad \textit{for a tensor } h_0 \textit{ over } L_0.$$

At a residual node where every directly realizable nonsingular transfer has finite projective order, these transfers form a group; denote by $G(\Lambda)$ the group of their classes $[BX]$, where B ranges over directly realizable nonsingular ordered binaries over Λ. Then the pulled-back group

$$\overline{G}(A,\Lambda) := [A]^{-1}G(\Lambda)[A] \le \mathrm{PGL}_2(L_0)$$

is unchanged by a global normalization and is monotone under retained augmentation. Direct gadget augmentation and retained factor saturation preserve A. If $D \in GO(X)$ is applied globally to the current signature set, the record updates by

$$(A,\Lambda) \longmapsto (DA, D\Lambda). \tag{27}$$

The new state again satisfies Equation (26)*, up to the known edge scalar from Equation* (5)*. Before a continuing branch leaves Section 4, apply the inverse of the accumulated matrix A. The returned state therefore has $A = I$ in the caller's fixed coordinates and retains any supplied literal equality literally. Generic interpolation is used only as a terminal transition and is not a record-preserving update. If an interpolation output were ever allowed to re-enter a continuing finite-group state, the effective coefficient field and the pulled-back group would first have to be enlarged explicitly.*

If $\mathsf{Tract}(K\Lambda)$ *holds, then* $\mathsf{Tract}(K\mathcal{G}_0)$ *holds. Separately, if V is a common $\mathscr{C}$ K-presentation of Λ, where $\mathscr{C}$ is closed under nonzero scalar multiplication, then the pullback matrix*

$$W := A^{-1}V$$

is a common $\mathscr{C}$ K-presentation of $\mathcal{G}_0$. In particular, this applies to $\mathscr{A}, \mathscr{P}, \mathscr{L}$ and to the flat-Lagrangian subclass.

Proof. The assertions hold initially for $(I, \mathcal{G}_0)$. Direct augmentation and factor saturation preserve them by Equations (12) and (24). A global $D \in GO(X)$, with nonzero conformal factor $\mu_D \in \overline{\mathbb{Q}}^{\times}$ defined by $D^{\mathsf{T}}XD = \mu_D X$, replaces A by DA; then $(DA)^{\mathsf{T}}X(DA) = \mu_D\mu_A X$, and Equation (5) supplies the stated equivalence. Conversely, the transported inclusion $A\mathcal{G}_0 \subseteq \Lambda$, followed by the inverse global transformation, reduces the initial problem to the current one; this is why the record records equivalence rather than only one direction.

Inductively pull every construction through A. Gadget contraction stays over L_0; at a tensor split, normalizing one nonzero entry and taking row–column ratios chooses the factors over L_0. Moreover $A^{\mathsf{T}}X \doteq XA^{-1}$. Thus, if a current binary B has the record form $B = A^{\otimes 2}B_0$ for a binary tensor B_0 over L_0, its projective transfer pulls back by conjugation:

$$[BX] = [A(B_0X)A^{-1}].$$

The identity transfer is realized by the native edge, and chaining realizes products. At such a finite-projective-order residual node, the inverse of each transfer is a positive power of that transfer, so the realizable classes form a group. The displayed conjugation shows that global normalization leaves $\overline{G}(A,\Lambda)$ unchanged, while retained augmentation can only add realizable transfers. This proves the field and group assertions. Finally, if $\mathsf{Tract}(K\Lambda)$ holds, downward inheritance first gives $\mathsf{Tract}(K(A\mathcal{G}_0))$, and Lemma 2.22 then gives $\mathsf{Tract}(K\mathcal{G}_0)$. If V is an explicit common presentation, then $W = A^{-1}V$ satisfies

$$W^{-1}\mathcal{G}_0 \subseteq V^{-1}\Lambda \subseteq \mathscr{C}, \qquad W^{\mathsf{T}}XW = \mu_A^{-1}V^{\mathsf{T}}XV \in \mathscr{C},$$

which proves the last assertion. □

3 Reductions Using Unaries and Equalities

The imported odd-arity dichotomy and factor theorem (Section 2.10.5) resolve a branch once a nonzero unary is accessible. This section supplies the reductions that obtain such a unary from a singular or infinite-order binary, a single-occurrence carrier, or a higher equality. We use the distinction between direct realization and Turing access fixed in Section 2.5.

3.1 Binary Interpolation

Lemma 3.1 (Binary interpolation)**.** *Let Γ be a finite algebraic, retained, two-sided signature set, let $B \neq 0$ be an ordered binary directly realizable over Γ, and define its transfer matrix $A = BX$.*

1. *If B has the rank-one factorization $B = u \otimes v$, where u and v are nonzero algebraic unary signatures, then*
$$\text{K-Holant}(\Gamma, u, v) \leq_{\text{T}} \text{K-Holant}(\Gamma).$$
2. *If A is nonsingular and has infinite projective order, there is a nonzero rank-one binary H such that*
$$\text{K-Holant}(\Gamma, H) \leq_{\text{T}} \text{K-Holant}(\Gamma).$$
For $m \geq 1$ designated occurrences of H, the interpolation uses $m+1$ oracle queries, $O(m^2)$ added vertices per query, and hence $O(m^3)$ added vertices over all queries.

Proof. For the singular case, choose a nonzero algebraic pivot of the rank-one table and take the corresponding algebraic row and column factors (absorbing the pivot into one factor); then $B = u \otimes v$ with u, v algebraic, and Theorem 2.16 applies. Two-sidedness excludes its one-sided exception. For the other case, a length-t chain realizes the ordered binary signature $B_t := A^t X$. If A is diagonalizable, choose an eigenbasis matrix $V \in \text{GL}_2(\overline{\mathbb{Q}})$ and nonzero eigenvalues $\lambda_0, \lambda_1 \in \overline{\mathbb{Q}}^{\times}$ such that $A = V \,\text{diag}(\lambda_0, \lambda_1) V^{-1}$. For $j \in \{0, 1\}$, let E_{jj} be the 2×2 matrix with (a, b)-entry $\mathbf{1}[a = b = j]$, for $a, b \in \{0, 1\}$, and define the rank-one binary matrix $H_j = V E_{jj} V^{-1} X$. Then

$$B_t = \lambda_0^t H_0 + \lambda_1^t H_1,$$

with H_0, H_1 rank one. Replacing the m designated H_0-vertices by B_t and dividing by λ_1^{tm} yields evaluations of a polynomial $P(z)$ of degree at most m at the distinct points $z = (\lambda_0/\lambda_1)^t$, $1 \leq t \leq m+1$; the coefficient of z^m is the all-H_0 value. These points are distinct because $[A]$ has infinite projective order, so λ_0/λ_1 is not a root of unity.

In the remaining infinite-order Jordan case, write $A = \lambda I + N$, where $\lambda \in \overline{\mathbb{Q}}^{\times}$ is the eigenvalue and N is a nonzero nilpotent 2×2 matrix satisfying $N^2 = 0$. Then $B_t = \lambda^t (X + t\lambda^{-1} N X)$. After normalization the oracle value is a polynomial in t of degree at most m, whose coefficient of t^m is the value with the rank-one binary $\lambda^{-1} N X$ at every designated vertex. Indeed, $N \neq 0$ and $N^2 = 0$ in dimension two imply $\text{rank}(N) = 1$, and right multiplication by X preserves rank. Both cases use $m+1$ queries; $t \leq m+1$ gives $O(m^2)$ replacement size for each query and $O(m^3)$ in total. The eigenvalues and basis matrices are fixed algebraic numbers. Their powers and the interpolation systems have polynomial encoding length, so exact interpolation is polynomial-time postprocessing. □

3.2 Single-occurrence carriers and unary terminals

The next reductions initially permit only one occurrence of an auxiliary signature. A context that returns this signature together with a new tensor can be nested to make arbitrarily many

copies of the latter. The reserved carrier ports keep their order under nesting, with no extra port identifications. Define the ordered carrier signature

$$\Delta = \Delta_0 \otimes \Delta_1,$$

with the displayed order of its two ports. For a signature g, the superscripts $g^{=1}$ and $g^{\leq 1}$ mean that an oracle instance contains, respectively, exactly one or at most one g-vertex. They do not denote tensor powers.

Lemma 3.2 (Universal one-pair reduction)**.** *For every fixed finite algebraic signature set* Γ*,*

$$\text{K-Holant}(\Gamma, \Delta^{\leq 1}) \leq_{\text{T}} \text{K-Holant}(\Gamma).$$

No EO, parity, charge, or tractability hypothesis is required.

Proof. The zero-occurrence case is already an instance over Γ. Consider an instance with one Δ-vertex. Delete that vertex, retaining its two incident native X-attachments as wires to ordered free boundary ports. Write the raw table, before the boundary-coordinate correction in Equation (2), as

$$H = \begin{pmatrix} h_{00} & h_{01} \\ h_{10} & h_{11} \end{pmatrix}, \qquad \text{the desired one-}\Delta \text{ value is } h_{01}.$$

Closing its two exposed ports by a native X-edge gives an oracle value

$$S = h_{01} + h_{10}.$$

Let $\mathcal{B}_\Gamma$ be the set of ordered binary tensors realized by finite Γ-networks. If every $B \in \mathcal{B}_\Gamma$ satisfies $B_{01} = B_{10}$, then $XHX \in \mathcal{B}_\Gamma$ is symmetric, so H is symmetric and the desired value is $S/2$.

Otherwise fix once and for all a realized ordered binary

$$B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad b \neq c.$$

Insert its fixed network at the deleted location in the two boundary orientations. The two oracle values are

$$T_+ = ah_{00} + bh_{01} + ch_{10} + dh_{11}, \qquad T_- = ah_{00} + ch_{01} + bh_{10} + dh_{11}.$$

Thus

$$h_{01} = \frac{1}{2}\left(S + \frac{T_+ - T_-}{b - c}\right).$$

The ordered ports are essential in the subtraction. The witness network for B is fixed with Γ, hence has constant size, and the reduction uses at most three oracle queries. □

The next elementary lemma isolates the only additional ingredient needed to turn one-occurrence access into unrestricted access to a new signature. It applies verbatim to ordinary Holant and to K-Holant; write H for either setting, with its corresponding edge semantics, and write Z_{H} for the corresponding closed-instance value.

Lemma 3.3 (Single-occurrence replication)**.** *Let* Γ *be a fixed signature set and let* g, h *be nonzero signatures. Suppose that*

1. $\mathsf{H}(\Gamma, g^{=1}) \leq_{\text{T}} \mathsf{H}(\Gamma)$*;*

2. *a fixed carrier-preserving context C over Γ has a distinguished arity-*$\mathrm{arity}(g)$ *unassigned vertex and an ordered output boundary consisting of two disjoint ordered blocks P followed by H, where $|P| = \mathrm{arity}(g)$ and $|H| = \mathrm{arity}(h)$. The block P is the g-carrier, and filling the hole by g realizes $\rho\, g(P) \otimes h(H)$ for a known nonzero scalar $\rho \in \overline{\mathbb{Q}}^{\times}$. Substituting a copy of C into the same distinguished hole (plugging its carrier into the next unassigned vertex, with no other port identifications) is a legal composition; hence, for every $k \geq 1$, the k-fold nested context has exactly one g-occurrence and realizes $\rho^k g(P) \otimes \bigotimes_{\ell=1}^{k} h(H_\ell)$, with disjoint output blocks $H_1, \ldots, H_k$; and*

3. *some fixed $\mathsf{H}(\Gamma, g^{=1})$-instance J has $Z_{\mathsf{H}}(J) \neq 0$.*

Then

$$\mathsf{H}(\Gamma, h) \leq_{\mathrm{T}} \mathsf{H}(\Gamma).$$

Proof. By the designated-carrier composition in item 2, iterating the gadget produces $g \otimes h^{\otimes k}$ with one g-vertex, size $O(k)$, and a known nonzero scalar. Hence an instance containing one g and k designated h-vertices reduces to an instance containing exactly one g; divide out the recorded scalar. For an instance $\mathcal{I}$ over $\Gamma \cup \{h\}$, take the disjoint union $\mathcal{I} \sqcup J$. Its value is $Z_{\mathsf{H}}(\mathcal{I}) Z_{\mathsf{H}}(J)$, so division by the fixed nonzero number $Z_{\mathsf{H}}(J)$ inserts the one carrier required by the first reduction. The context and J are fixed, so the construction has size linear in the number of h-occurrences. Composing with item 1 proves the claim. □

Lemma 3.4 (Common unary exit)**.** *Let $\mathcal{F}$ be an ordinary-coordinate signature set, let Γ be a native K-coordinate signature set, and let $u \neq 0$ be a unary signature. If* $\mathrm{Holant}(\mathcal{F}, u) \leq_{\mathrm{T}} \mathrm{Holant}(\mathcal{F})$*, then* $\mathrm{Holant}(\mathcal{F})$ *is* #P*-hard unless* $\mathsf{Tract}(\mathcal{F})$ *holds. Likewise, if* $\text{K-Holant}(\Gamma, u) \leq_{\mathrm{T}} \text{K-Holant}(\Gamma)$*, then* $\text{K-Holant}(\Gamma)$ *is* #P*-hard unless* $\mathsf{Tract}(K\Gamma)$ *holds.*

Proof. The first assertion is Theorem 2.15 applied to the augmented signature set. A tractable presentation of that set restricts to one for $\mathcal{F}$. For the second, transform to ordinary coordinates:

$$\text{K-Holant}(\Gamma, u) \equiv_{\mathrm{T}} \mathrm{Holant}(K\Gamma, Ku).$$

The signature Ku is a nonzero unary, so the same argument applies. □

For unary columns g, u, their contraction is the bilinear scalar

$$g^{\mathsf{T}} u := \sum_{x \in \{0,1\}} g(x) u(x),$$

with no complex conjugation. For equality wires we need a carrier whose two unary factors have nonzero contraction. The following proof is included because the carrier need not itself be realizable by a gadget.

Lemma 3.5 (Unary companion)**.** *Let $\mathcal{F}$ be a fixed finite algebraic signature set and let $g \neq 0$ be unary. There is an algebraic unary u such that*

$$g^{\mathsf{T}} u \neq 0, \qquad \mathrm{Holant}(\mathcal{F}, (g \otimes u)^{=1}) \leq_{\mathrm{T}} \mathrm{Holant}(\mathcal{F}).$$

Proof. We first prove the claim for $g = \Delta_0$. Let $\mathcal{B}_{\mathcal{F}}$ be the set of ordered binary tensors realized by finite $\mathcal{F}$-networks with ordinary equality wires. Deleting one occurrence of $\Delta_0 \otimes u$, where $u = [s, t]^{\mathsf{T}}$, leaves a binary context $H = (h_{ij})$, and the desired value is $sh_{00} + th_{01}$. Equality-closing the exposed ports gives the oracle value

$$S := h_{00} + h_{11}.$$

We choose u by the following exhaustive alternatives.

If every $H \in \mathcal{B}_{\mathcal{F}}$ satisfies $h_{00} = h_{11}$, take $u = [2, 0]^{\mathsf{T}}$; then the desired value is S. Otherwise fix

$$G_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \in \mathcal{B}_{\mathcal{F}}, \qquad a_1 \neq d_1.$$

If every realized binary has equal off-diagonal entries, then $b_1 = c_1 = b$. For $b = 0$, take $u = [1, 0]^{\mathsf{T}}$. Inserting G_1 gives the oracle value $T := a_1 h_{00} + d_1 h_{11}$, whence

$$h_{00} = \frac{T - d_1 S}{a_1 - d_1}.$$

For $b \neq 0$, take $u = [a_1 - d_1, 2b]^{\mathsf{T}}$. The same insertion, now using $h_{01} = h_{10}$, gives

$$T - d_1 S = (a_1 - d_1) h_{00} + 2b h_{01}.$$

It remains that some

$$G_2 = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \in \mathcal{B}_{\mathcal{F}} \quad \text{satisfies} \quad b_2 \neq c_2.$$

Take $u = [a_1 - d_1, b_1 + c_1]^{\mathsf{T}}$. Let T_1 be the value obtained by inserting G_1, and let T_2, T_3 use the two orientations of G_2. Then

$$\frac{T_2 - T_3}{b_2 - c_2} = h_{01} - h_{10},$$

and therefore

$$T_1 - d_1 S + \frac{c_1}{b_2 - c_2}(T_2 - T_3) = (a_1 - d_1) h_{00} + (b_1 + c_1) h_{01}.$$

In every case the first entry of u is nonzero, so $\Delta_0^{\mathsf{T}} u \neq 0$.

Now write the unary column vector $g = [a, b]^{\mathsf{T}}$. If $a^2 + b^2 = 0$, then projectively $g \doteq [1, i]^{\mathsf{T}}$ or $g \doteq [1, -i]^{\mathsf{T}}$. In the first case take $u = [1, -i]^{\mathsf{T}}$, and in the second take $u = [1, i]^{\mathsf{T}}$. Under the ordinary-to-native K^{-1}-transformation the pair is, up to a known nonzero scalar and port order, $\Delta_0 \otimes \Delta_1$. The reduction follows from Lemma 3.2, and $g^{\mathsf{T}} u \neq 0$.

If $a^2 + b^2 \neq 0$, choose the scalar $r \in \overline{\mathbb{Q}}$ with $r^2 = a^2 + b^2$ and define the matrix

$$Q = \frac{1}{r} \begin{pmatrix} a & b \\ b & -a \end{pmatrix}.$$

Then $Q^{\mathsf{T}} Q = I$ and $Qg = r\Delta_0$. Apply the point-mass case to $Q\mathcal{F}$, obtaining $u' = [s, t]^{\mathsf{T}}$ with $s \neq 0$, and define the unary vector $u = Q^{-1} u'$. Because $Q^{\mathsf{T}} Q = I$, the holographic identity leaves every ordinary equality edge unchanged and pulls the reduction back to $\mathcal{F}$, while

$$g^{\mathsf{T}} u = (Qg)^{\mathsf{T}} u' = rs \neq 0.$$

□

Corollary 3.6 (Higher equality exposes unaries)**.** *Let $d \geq 4$. If EQ_d is accessible over a finite algebraic signature set in either ordinary Holant or* K-Holant*, then both unary point masses are accessible in that setting. In particular, access to one nonzero unary is enough to invoke Lemma 3.4.*

Proof. First work with ordinary equality wires and adjoin the accessible EQ_d. For $j \in \{0,1\}$, apply Lemma 3.5 to $e_j = \delta_j$, obtaining u_j with $u_j(j) \neq 0$ and the one-occurrence carrier $v_j = e_j \otimes u_j$. Attach the e_j-port of that carrier to one port of EQ_d. Up to port order the output is

$$v_j \otimes e_j^{\otimes(d-2)}.$$

Closing the carrier through equality has value $u_j(j) \neq 0$, so Lemma 3.3 gives access to $e_j^{\otimes(d-2)}$. Passing temporarily to K-coordinates, the unary $K^{-1}e_j$ has two nonzero entries; hence this tensor power is not in the common-one-sided exception of Theorem 2.16. Factor extraction therefore exposes e_j after transforming back.

Now work in K-Holant and use the carrier Δ. Attach its two factors, through native X-edges and in the two orientations, to two copies of EQ_d. The output is

$$\Delta^{\otimes(d-1)} = \Delta \otimes \Delta^{\otimes(d-2)}$$

up to port order. The native X-closure of Δ has value one, so Lemmas 3.2 and 3.3 give access to $\Delta^{\otimes(d-2)}$. The available EQ_d contains both endpoint words $\mathbf{0}^d, \mathbf{1}^d$, of doubled charges $-d, +d$, and hence excludes the common-one-sided exception; applying Theorem 2.16 exposes Δ_0 and Δ_1. Finally compose with the assumed reduction that removes EQ_d. □

3.3 Equality and generalized-equality anchors

An equality anchor is a quaternary gadget that becomes EQ_4 under the same basis change that makes the native edge an equality. This gives a whole-problem reduction to the unary result above.

Definition 3.7 (Equality anchor)**.** A signature set $\mathcal{G}$ has an *equality anchor* if it directly realizes an ordered quaternary signature g for which there exist a matrix $U \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ and nonzero algebraic scalars $\kappa, \lambda \in \overline{\mathbb{Q}}^{\times}$ satisfying

$$(U^{-1})^{\mathsf{T}} X U^{-1} = \kappa\, \mathrm{EQ}_2, \qquad U^{\otimes 4} g = \lambda\, \mathrm{EQ}_4.$$

Lemma 3.8 (Equality-anchor lifting)**.** *Let $\mathcal{G}$ be a finite algebraic signature set with an equality anchor g, and fix witnesses U, κ, λ as in Definition 3.7. Then either* K-Holant$(\mathcal{G})$ *is* #P*-hard or* $\mathsf{Tract}(K\mathcal{G})$ *holds.*

Proof. Apply the native-edge holographic identity with $M = U^{-1}$ to the whole problem. It sends each right signature f to Uf and the native edge to $(U^{-1})^{\mathsf{T}} X U^{-1} = \kappa \mathrm{EQ}_2$; for a queried instance Ω, the edge factor $\kappa^{|E_{\mathrm{int}}(\Omega)|}$ is known. Thus the direct anchor becomes $Ug = \lambda \mathrm{EQ}_4$, so EQ_4 is accessible in the ordinary-Holant problem over $U\mathcal{G}$. By Corollary 3.6, a nonzero unary u is accessible there. Pulling the reduction back exposes the nonzero unary $U^{-1}u$ over $\mathcal{G}$. The conclusion is now Lemma 3.4. □

Lemma 3.9 (Pure generalized-equality lifting)**.** *Suppose a retained finite algebraic signature set $\mathcal{G}$ directly realizes the arity-q pure generalized-equality signature*

$$G_q := c_0 \delta_{\mathbf{0}^q} + c_1 \delta_{\mathbf{1}^q}, \qquad c_0, c_1 \in \overline{\mathbb{Q}}^{\times}, \qquad q \geq 4.$$

Then either K-Holant$(\mathcal{G})$ *is* #P*-hard or* $\mathsf{Tract}(K\mathcal{G})$ *holds.*

Proof. Define the retained signature set $\Lambda = \mathcal{G} \cup \{G_q\}$. Direct realizability gives K-Holant(Λ) $\leq_{\mathrm{T}}$ K-Holant($\mathcal{G}$); work over Λ. Use the ordered carrier Δ from Section 3.2. Attach its two factors through native X-edges, in opposite orientations, to one port of each of two copies of G_q. The two copies are forced onto opposite endpoint words, and the external tensor is

$$c_0 c_1 \, \Delta^{\otimes(q-1)} = c_0 c_1 \, \Delta \otimes \Delta^{\otimes(q-2)}$$

up to port order. The native X-closure of the carrier has value one. Thus Lemmas 3.2 and 3.3 give access to $\Delta^{\otimes(q-2)}$. Since $G_q \in \Lambda$ contains both endpoint words, of doubled charges $-q, +q$, Theorem 2.16 has no common-one-sided exception and exposes Δ_0 and Δ_1. Apply Lemma 3.4 over Λ. A hard conclusion composes back to $\mathcal{G}$, while $\mathsf{Tract}(K\Lambda)$ implies $\mathsf{Tract}(K\mathcal{G})$ by Lemma 2.21. □

4 Eight-Vertex Signatures with Nonzero Endpoints

The quaternary dichotomy in Theorem 2.17 leaves product, affine, and local-affine exceptions. We analyze these exceptions over the entire retained signature set, using the transformability convention of Definition 2.8 and the unary/equality reductions of Section 3. The continuing outcome will be a weighted-equality factor of the original quaternary.

Definition 4.1 (Endpoint-nondegenerate eight-vertex signature)**.** An *endpoint-nondegenerate eight-vertex signature* is a quaternary signature q satisfying

$$q(x) = 0 \quad \text{when } \mathrm{wt}(x) \text{ is odd}, \qquad q(0000)q(1111) \neq 0.$$

Thus its support is contained in the eight even-parity words, while both endpoint values are nonzero.

We use the two entries $q(0000), q(1111)$, the six weight-two entries, and the three flattenings with row–column splits $12 \mid 34$, $13 \mid 24$, and $14 \mid 23$, respectively; for example, $12 \mid 34$ uses (x_1, x_2) as row index and (x_3, x_4) as column index. We list their ranks in this order. A complementary pair of two-port blocks is $J, J^c \subseteq [4]$ with $|J| = 2$.

We classify endpoint-nondegenerate eight-vertex signatures first in the product and local-affine cases and then in the sole finite affine base case. The resulting quaternary interface is used by the stable equality-accessible matching-deck proof and yields the independent quaternary dichotomy in Theorem 1.3. The two exact finite statements used in this section are given in Appendix A.1.

Throughout this section, weighted equality and nonsingularity have the meaning fixed in Section 2.3. For a signature class $\mathscr{C}$, the statement that one signature M is $\mathscr{C}$-transformable uses the singleton convention: the set $\{M\}$ and the transformed native edge have one common transformability witness, as in Definition 2.8.

4.1 Product and local-affine exceptions

Lemma 4.2 (Product-transformable eight-vertex signatures)**.** *Let M be an endpoint-nondegenerate eight-vertex signature. If M is $\mathscr{P}$-transformable in the sense of Theorem 2.17, then exactly one of the following structural outcomes applies (up to a port permutation):*

1. *M is a pure generalized equality;*
2. *M is a tensor product of two nonsingular weighted equalities;*
3. *M has full even support, has the dense form Equation* (28)*, and every retained signature set directly realizing M also directly realizes an equality anchor.*

In particular, every $\mathscr{P}$-transformable endpoint-nondegenerate eight-vertex signature has support size 2*,* 4*, or* 8*.*

Proof. Choose an algebraic witness matrix $T = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ by Lemma 2.13, and define the transformed signature $H = (T^{-1})^{\otimes 4} M \in \mathscr{P}$. The transformed edge is

$$T^{\mathsf{T}} X T = \begin{pmatrix} 2ac & ad+bc \\ ad+bc & 2bd \end{pmatrix}.$$

Because this transformed edge is nonsingular and belongs to $\mathscr{P}$, its support is either equality or disequality. Disequality support makes T monomial. Equality support gives $ad + bc = 0$; define the nonzero scalar ratio $\lambda := b/a$. Then

$$T = \operatorname{diag}(a, c) \begin{pmatrix} 1 & \lambda \\ 1 & -\lambda \end{pmatrix}, \qquad ac\lambda \neq 0.$$

If H is tensor-prime, product type makes it a generalized equality for some $s \in \mathbb{F}_2^4$, with nonzero coefficients $A, B \in \overline{\mathbb{Q}}^{\times}$:

$$H = A\delta_s + B\delta_{\bar{s}}, \qquad AB \neq 0.$$

In the monomial case the two nonzero endpoint values force $\{s, \bar{s}\} = \{\mathbf{0}^4, \mathbf{1}^4\}$, giving outcome 1. Otherwise, for $k = \operatorname{wt}(s)$ and $w = \operatorname{wt}(x)$,

$$M(x) = a^{4-w} c^w (-1)^{s \cdot x} (A\lambda^k + B\lambda^{4-k}(-1)^w).$$

The odd-weight zeros are equivalent to $B/A = \lambda^{2k-4}$. Hence all even entries are nonzero. Define the nonzero overall scale $\rho := 2A\lambda^k a^4$ and weight-ratio scalar $r := c/a$; the entries then have the form

$$M(x) = \rho r^{\operatorname{wt}(x)} (-1)^{s \cdot x} \mathbf{1}[\operatorname{wt}(x) \text{ is even}], \qquad \rho r \neq 0. \tag{28}$$

For this dense case choose nonzero algebraic scalars $q_0, q_1 \in \overline{\mathbb{Q}}^{\times}$ satisfying the first two relations below, define the diagonal matrix $Q := \operatorname{diag}(q_0, q_1)$, and define the normalization matrix $R := QT^{-1}$:

$$q_0^2 = 2ac, \qquad q_1/q_0 = i\lambda, \qquad R = QT^{-1}.$$

Then, since $\lambda^{2k-4}(i\lambda)^{4-2k} = (-1)^k$, there is a nonzero scalar $\eta \in \overline{\mathbb{Q}}^{\times}$ such that

$$(R^{-1})^{\mathsf{T}} X R^{-1} = I, \qquad R^{\otimes 4} M = \eta(\delta_s + (-1)^k \delta_{\bar{s}}). \tag{29}$$

Define the signature $F_s = \delta_s + (-1)^k \delta_{\bar{s}}$. In these coordinates the native edge is equality by Equation (29). A nonzero multiple of EQ_4 uses at most two copies of F_s: one for $k = 0, 4$; for $k = 2$, expose the two $s = 0$ ports of the first and the two $s = 1$ ports of the second, pairing the remaining ports; for $k = 1$, expose the three majority-zero ports of the first and the minority-one port of the second, join the first's minority port to one majority port of the second, and equality-close the second's last two majority ports. The $k = 3$ wiring is complementary. For $k = 0, 4$, the single copy is EQ_4. In each two-copy wiring, an all-equal external word has exactly one compatible internal assignment, while every other external word has zero contribution. The resulting signature is EQ_4 for $k = 2$ and $-\mathrm{EQ}_4$ for $k = 1, 3$. Pulling it back through R gives a direct gadget g over M with $R^{\otimes 4} g = \lambda' \mathrm{EQ}_4$, $\lambda' \neq 0$; together with Equation (29) this is outcome 3.

If H is not tensor-prime, neither is M. Its nonzero endpoints make every factor nonzero at both endpoints. Even parity excludes an odd-sized block, so the only proper tensor-factor partition is

$[4] = S \sqcup S^c$ with $|S| = |S^c| = 2$, where $S^c = [4] \setminus S$; the odd-weight zeros then force both binary factors to be diagonal. Thus they are nonsingular weighted equalities and

$$M(\mathbf{0}^4)M(\mathbf{1}^4) = M(\mathbf{1}_S)M(\mathbf{1}_{S^c}).$$

This is outcome 2 and completes the classification. □

Corollary 4.3 (Reductions from product-transformable eight-vertex signatures)**.** *Let Γ be a retained, factor-saturated finite algebraic signature set that directly realizes a product-transformable endpoint-nondegenerate eight-vertex signature M. Then at least one of the following holds:*

(i) K-Holant(Γ) *is #*P*-hard;*

(ii) $\mathsf{Tract}(K\Gamma)$ *holds;*

(iii) *there is a finite retained, factor-saturated augmentation $\Lambda \supseteq \Gamma$ such that*

$$\text{K-Holant}(\Lambda) \leq_{\text{T}} \text{K-Holant}(\Gamma),$$

and Λ retains both factors in the two-binary factorization of M, including a nonsingular weighted equality exposed as a proper factor. The factors are Turing-exposed by retained factor saturation, not asserted to be directly realizable over Γ.

Proof. Apply Lemma 4.2. Its pure-generalized-equality form is closed by Lemma 3.9, and its dense form is closed by Lemma 3.8. These give alternatives (i)–(ii), with hardness composing to Γ and tractability descending to Γ by Lemma 2.21.

In the remaining form, adjoin the directly realized M and retain the complete factor batch from its decomposition across two two-port blocks. The endpoint-nondegenerate eight-vertex signature is nonzero on both endpoint words $\mathbf{0}^4$ and $\mathbf{1}^4$, whose doubled charges are -4 and $+4$, respectively, so the common-one-sided exception in Lemma 2.20 is impossible. The resulting finite factor-saturated augmentation Λ reduces to Γ and contains both nonsingular weighted-equality factors. This is alternative (iii). □

Lemma 4.4 (Local-affine reduction to affine)**.** *Let M be a nonzero quaternary even-parity signature. If one matrix $T \in \mathrm{GL}_2(\mathbb{C})$ satisfies*

$$T^{\mathsf{T}} X T \in \mathscr{L}, \qquad (T^{-1})^{\otimes 4} M \in \mathscr{L},$$

then there is a matrix $T_0 \in \mathrm{GL}_2(\mathbb{C})$ with

$$T_0^{\mathsf{T}} X T_0 \in \mathscr{A}, \qquad (T_0^{-1})^{\otimes 4} M \in \mathscr{A}.$$

If $T \in \mathrm{GL}_2(\overline{\mathbb{Q}})$, then T_0 is algebraic.

Proof. For any nonempty affine set $S \subseteq \mathbb{F}_2^n$, choose a base point $s \in S$ and write $S = s + C$, where $C \leq \mathbb{F}_2^n$ is a linear subspace. The subspace

$$C = \{x \oplus y : x, y \in S\},$$

is independent of s. It is self-orthogonal if $u \cdot v = 0$ over $\mathbb{F}_2$ for all $u, v \in C$.

Define the phase scalar $\alpha = e^{\pi i/4}$. For an arity-n signature F and $z, y \in \mathbb{F}_2^n$, define the local twist selected by z as

$$\Theta_z := \bigotimes_{j=1}^{n} \mathrm{diag}(1, \alpha)^{z_j}, \qquad (\Theta_z F)(y) = \alpha^{\langle z, y \rangle_{\mathbb{Z}}} F(y), \qquad \langle z, y \rangle_{\mathbb{Z}} := \sum_{j=1}^{n} z_j y_j.$$

By the local-affine criterion [CLX18, Definition 3.1 and Theorem 3.2], the translation space of every support is self-orthogonal, and $\Theta_z F \in \mathscr{A}$ for every $z \in \operatorname{supp}(F)$ when $F \in \mathscr{L}$. Apply this first to the symmetric nonsingular binary $E = T^{\mathsf{T}} X T$. Self-orthogonality rules out full support, and nonsingularity rules out singleton support; hence $\operatorname{supp}(E)$ is equality or disequality. Disequality support would give

$$E = pX \notin \mathscr{L},$$

since $01 \in \operatorname{supp}(E)$ and the twist required by this support word has relative phase $\alpha \notin \mu_4$, so it is not affine. Thus $E = \operatorname{diag}(p, q)$, where $p, q \in \overline{\mathbb{Q}}^{\times}$ are its nonzero diagonal entries (their algebraicity follows from $E \in \mathscr{L}$); the 00-twist is the identity, so $q/p \in \mu_4$. Writing $T = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the off-diagonal equation yields $q/p = -(b/a)^2$. Hence, with the scalar parameters $\rho := b/a \in \mu_8$ and $t := c/a \in \mathbb{C}^{\times}$, and, for $z \in \mathbb{C}^{\times}$, the diagonal scaling matrix $D_z := \operatorname{diag}(1, z)$,

$$T = aD_t P D_\rho, \quad at \neq 0, \quad \rho^8 = 1.$$

Define the matrix $T_0 := aD_t P$, removing the right factor D_ρ, and define the function

$$W_t(y) = \sum_{\substack{x \in \mathbb{F}_2^4 \\ \operatorname{wt}(x) \equiv 0 \pmod 2}} t^{-\operatorname{wt}(x)} M(x) (-1)^{x \cdot y}.$$

Let $N := (T^{-1})^{\otimes 4} M$ denote the transformed local-affine signature. Since $P^{-1} = P/2$, define

$$\kappa_0 := (2a)^{-4} \neq 0, \qquad \widehat{W}_t := \kappa_0 W_t.$$

$$(T^{-1})^{\otimes 4} M(y) = \rho^{-\operatorname{wt}(y)} \widehat{W}_t(y), \qquad (T_0^{-1})^{\otimes 4} M(y) = \widehat{W}_t(y), \tag{30}$$

Since $N \in \mathscr{L}$ and $\rho \in \mu_8$, $\widehat{W}_t(y) = \rho^{\operatorname{wt}(y)} N(y)$ is algebraic for every y. Moreover,

$$T_0^{\mathsf{T}} X T_0 = 2a^2 t \operatorname{diag}(1, -1) = p \operatorname{diag}(1, -1) \in \mathscr{A},$$

and parity gives $\widehat{W}_t(\bar{y}) = \widehat{W}_t(y)$.

The support of N, which is also the support of $\widehat{W}_t$, is an affine coset $s + C$. Self-orthogonality and the identity $\widehat{W}_t(\bar{y}) = \widehat{W}_t(y)$, which makes this support closed under bitwise complement, give $\mathbf{1}^4 \in C$ and $1 \leq \dim C \leq 2$. For dimension one, the support is a complementary pair on which $\widehat{W}_t$ has equal values, so $\widehat{W}_t \in \mathscr{A}$. For dimension two, after permuting ports,

$$C = \operatorname{span}_{\mathbb{F}_2}\{\mathbf{1}^4, w\}, \qquad \operatorname{wt}(w) = 2.$$

For $u, v \in \mathbb{F}_2$, index the four support words by $x_{uv} := s \oplus u\mathbf{1}^4 \oplus vw$. For a table F that is nonzero on these four words, define its cross-ratio functional by $\operatorname{cr}(F) := F(x_{00})F(x_{11})/(F(x_{10})F(x_{01}))$. Complement symmetry gives $\operatorname{cr}(\widehat{W}_t) = 1$, while the nowhere-zero restriction of any member of $\mathscr{A}$ with this support has cross ratio ± 1. The quotient $\mathbb{F}_2^4/C$ has exactly three nonzero classes. Up to a port permutation and $w \leftrightarrow \bar{w}$, each of the two odd cosets has representative $(s, w) = (0001, 0011)$, and the unique even coset has representative $(s, w) = (0101, 0011)$. Thus the displayed cases exhaust all three nonzero quotient classes. In both representatives the $\rho^{-\operatorname{wt}(y)}$ factor has cross ratio one and the direct four-point check gives

$$\langle s, x_{00} \rangle_{\mathbb{Z}} + \langle s, x_{11} \rangle_{\mathbb{Z}} - \langle s, x_{10} \rangle_{\mathbb{Z}} - \langle s, x_{01} \rangle_{\mathbb{Z}} = 2.$$

The local twist selected by s is $N_s := \Theta_s N$, so $N_s(y) = \alpha^{\langle s, y \rangle_{\mathbb{Z}}} N(y)$. It would therefore have cross ratio $\alpha^2 = i$, contradicting $N_s \in \mathscr{A}$. Hence $s + C = C$.

If the complement-symmetric values of $\widehat{W}_t$ are A on the endpoints and B on the central pair, then N has values $(A, \rho^{-4}A, \rho^{-2}B, \rho^{-2}B)$. Since $0 \in \operatorname{supp}(N)$, its selected local twist is the identity, so $N \in \mathscr{A}$ and $\rho^{-2}B/A \in \mu_4$. As $\rho^2 \in \mu_4$, also $B/A \in \mu_4$, whence $\widehat{W}_t \in \mathscr{A}$. Now Equation (30) proves the claim. If the original witness is algebraic, so are t and T_0. □

4.2 The finite affine quaternary certificate

The affine branch uses the finite classification in Appendix A.1, checked by `certificates/verify_affine_q4.py`. The next proposition derives the complete list of possible basis matrices algebraically and invokes that classification for the resulting quaternary tables.

Proposition 4.5 (Affine quaternary certificate)**.** *Let M be an algebraic affine-transformable endpoint-nondegenerate eight-vertex signature. Then M admits an algebraic invertible diagonal matrix $D_{\mathrm{aff}} = \mathrm{diag}(d_0, d_1)$, with $d_0, d_1 \in \overline{\mathbb{Q}}^{\times}$, such that*

$$(D_{\mathrm{aff}}^{-1})^{\otimes 4} M \in \mathscr{A}, \qquad D_{\mathrm{aff}}^{\mathsf{T}} X D_{\mathrm{aff}} \doteq X.$$

For some $J \subseteq [4]$ with $|J| = 2$, write $J^c := [4] \setminus J$. In these coordinates its support is exactly one of

$$\{\mathbf{0}^4, \mathbf{1}^4\}, \qquad \{\mathbf{0}^4, \mathbf{1}^4, \mathbf{1}_J, \mathbf{1}_{J^c}\}, \qquad \{x : \mathrm{wt}(x) \text{ is even}\}.$$

The first support is a pure generalized equality.

Proof. By Lemma 2.13, choose an algebraic affine witness matrix $U = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, and define the transformed binary edge $E := U^{\mathsf{T}} X U$. A nonsingular symmetric affine binary has disequality support, equality support, or full support. In the first case,

$$U = D_0 K_0, \qquad K_0 \in \{I, X\},$$

for an invertible diagonal D_0. In the second, $ad + bc = 0$, so

$$U = D_0 \begin{pmatrix} 1 & r \\ 1 & -r \end{pmatrix}, \qquad r \in \mu_8,$$

for an invertible diagonal D_0. In the full-support case, after normalizing its 00-entry, E has the affine form

$$\begin{pmatrix} 1 & u \\ u & -u^2 \end{pmatrix}, \qquad u \in \mu_4.$$

Writing $U = D_0 \left(\begin{smallmatrix} 1 & r \\ 1 & s \end{smallmatrix}\right)$ gives $r + s = 2u$ and $rs = -u^2$; hence $\{r, s\} = \{u(1+\sqrt{2}), u(1-\sqrt{2})\}$. Denote the resulting eighteen-element set of cores by $\mathscr{K}_{\mathrm{aff}}$; its explicit list appears in Theorem A.4. These three calculations give precisely

$$U = D_0 K_0, \qquad K_0 \in \mathscr{K}_{\mathrm{aff}},$$

so completeness of the list is analytic rather than computer-assumed.

Now define the affine signature $f = (U^{-1})^{\otimes 4} M \in \mathscr{A}$. Since M is an endpoint-nondegenerate eight-vertex signature and D_0 is diagonal, $K_0^{\otimes 4} f$ satisfies the hypotheses of Theorem A.4. The certificate is projective, so the normalization factor for f can be absorbed into a global scalar. The same interface theorem therefore yields

$$\gamma, t \in \overline{\mathbb{Q}}^{\times}$$

and a normalized affine signature $h \in \mathscr{A}$, with $h(\mathbf{0}^4) = 1$, such that

$$K_0^{\otimes 4} f = \gamma \, \mathrm{diag}(1, t)^{\otimes 4} h.$$

Moreover, $\mathrm{supp}(h)$ is one of the three supports displayed above. Consequently the single algebraic diagonal matrix $D_\star = D_0 \, \mathrm{diag}(1, t)$ satisfies $(D_\star^{-1})^{\otimes 4} M \doteq h$, and $D_\star^{\mathsf{T}} X D_\star \doteq X$. Taking $D_{\mathrm{aff}} = D_\star$ proves the asserted diagonal normalization.

□

Corollary 4.6 (Reductions from affine quaternaries)**.** *Let Γ be a retained, factor-saturated finite algebraic signature set that directly realizes an affine-transformable endpoint-nondegenerate eight-vertex signature M. Then at least one of the following holds:*

(i) K-Holant(Γ) *is #*P*-hard;*

(ii) $\mathsf{Tract}(K\Gamma)$ *holds;*

(iii) *there is a finite retained, factor-saturated augmentation $\Lambda \supseteq \Gamma$ such that*

$$\text{K-Holant}(\Lambda) \leq_{\mathrm{T}} \text{K-Holant}(\Gamma),$$

and Λ retains the complete factor batch from the physical construction. The original endpoint-nondegenerate eight-vertex signature M itself has, up to a port permutation and a nonzero scalar, a genuine factorization across two two-port blocks, and Λ contains a nonsingular weighted-equality matrix $E := \mathrm{diag}(p, q)$, where $p, q \in \overline{\mathbb{Q}}^{\times}$, exposed as one of its proper binary factors. This factor is Turing-exposed by retained factor saturation, not asserted to be directly realizable over Γ.

Proof. Choose $D_\star = D_{\mathrm{aff}}$ from Proposition 4.5. Apply the global change $D_\star^{-1}$, and normalize the resulting quaternary to an affine signature h with $h(\mathbf{0}^4) = 1$. Thus

$$M \doteq D_\star^{\otimes 4} h.$$

Keep every direct output over the current retained set, and adjoin proper tensor factors only through retained factor saturation. Record these constructions and the chosen normal form in the normalization data of Lemma 2.23. The transported copy of M remains nonzero at both endpoint words, so every augmentation is two-sided and Lemma 2.20 applies.

The normal form has three possible supports. Clause (ii) of Theorem A.5 gives the support-four construction. The support-two form is a pure generalized equality. On support four, let $J \sqcup J^c = [4]$ be the two two-port blocks. For $u, v \in \mathbb{F}_2$, put $x_{uv} := u\mathbf{1}_J \oplus v\mathbf{1}_{J^c}$, and define the logical table by

$$F := \left(h(x_{uv})\right)_{u,v\in\mathbb{F}_2} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad ABCD \neq 0.$$

Here u is the row bit and v is the column bit. For an ordered binary tensor E, write $E^{(J)}$ for its placement on the two ports of J in their inherited order, and define $E^{(J^c)}$ analogously. If $AD = -BC$, the two-copy X-edge gadget in Theorem A.5(ii) directly realizes a nonzero pure generalized equality. If $AD = BC$, then, on these two blocks,

$$h = E_1^{(J)} \otimes E_2^{(J^c)}$$

for nonsingular weighted equalities E_1, E_2. Since $D_\star$ is diagonal,

$$M \doteq (D_\star E_1 D_\star^{\mathsf{T}})^{(J)} \otimes (D_\star E_2 D_\star^{\mathsf{T}})^{(J^c)}.$$

Thus the original endpoint-nondegenerate eight-vertex signature itself has a genuine factorization across two two-port blocks into nonsingular weighted equalities. Adjoin M and apply retained factor saturation to Turing-expose both factors.

On full even support, Theorem A.5(iii), together with Lemma 4.2, gives, over the current retained set, either a directly realized equality anchor, a directly realized nonzero rank-one binary, or a directly realized product

$$cE_r \otimes E_r, \qquad c \in \overline{\mathbb{Q}}^{\times}, \qquad E_r := \mathrm{diag}(1, r), \quad r \in \mu_4,$$

where E_r is a nonsingular weighted-equality matrix. In the last case, Lemma 2.20 Turing-exposes E_r. Choose the scalar $s \in \mu_8$ with $s^2 = r^{-1}$, and define the diagonal equality-normalization matrix $D_r^{\mathrm{eq}} := \mathrm{diag}(1, s)$. Then

$$(D_r^{\mathrm{eq}})^{\mathsf{T}} X D_r^{\mathrm{eq}} = sX, \qquad D_r^{\mathrm{eq}} E_r (D_r^{\mathrm{eq}})^{\mathsf{T}} = I.$$

Apply this global normalization and record its matrix and scalar. Define the transformed signature $h_r = (D_r^{\mathrm{eq}})^{\otimes 4} h$. Attach one port of h_r by a native X-edge to one port of the retained equality I, leaving the other equality port external. This is an actual gadget over the current retained augmentation, and it complements that input of h_r, producing, up to a known nonzero scalar, an odd-parity quaternary p that is nonzero on all eight odd-weight words and is used in Theorem A.6. Because $s \in \mu_8$, its normalized nonzero entries are fourth roots of unity.

The twelve native and equality-mediated binary gadgets of Theorem A.6 are therefore actual at this node. They expose either a nonzero rank-one binary, a nonsingular binary of infinite projective order, or a table for which neither of these two outcomes occurs in the twelve-loop test of Theorem A.6. This includes the simultaneous-zero case of the theorem's loop tests. The exceptional remaining family directly realizes the certified endpoint-nondegenerate eight-vertex signature with support of cardinality six. For every other remaining table, choose a certified $\delta \in \mathbb{Z}_4$, choose $\tau_\delta \in \mu_8$ with $\tau_\delta^2 = i^\delta$, and apply the further legal global-normalization matrix $T_\delta := \mathrm{diag}(1, \tau_\delta)$. Let p_δ denote the certified phase-shifted odd table from Theorem A.6. Since p is supported only at Hamming weights one and three,

$$T_\delta^{\otimes 4} p \doteq p_\delta.$$

Record this global change in the normalization data. Theorem A.6 now gives, through its literal native-loop and two-copy constructions, either a nonzero rank-one binary or a product-transformable endpoint-nondegenerate eight-vertex quaternary in the dense form Equation (28); the latter directly yields an equality anchor by Lemma 4.2.

The infinite-order case is handled by Lemma 3.1. The eight-vertex signature with support of cardinality six is outside the $\mathscr{A}$-, $\mathscr{P}$-, and $\mathscr{L}$-transformable classes by Theorem A.6 and Lemmas 4.2 and 4.4, and is hard by Theorem 2.17. Thus every direct gadget is formed over the current retained set, every proper factor is only Turing-exposed, and all global diagonal changes belong to one normalization-record history.

The support-two case and the support-four case $AD = -BC$ are resolved by Lemma 3.9; equality anchors are resolved by Lemma 3.8. A nonzero rank-one binary exposes a unary by Lemma 3.1, and an infinite-order binary first interpolates such a binary. In both cases Lemma 3.4 applies. An exposed odd-arity signature is resolved by Theorem 2.15 after the K-coordinate bridge. Every full-support case above therefore gives hardness or tractability.

The sole continuing case is the support-four form $AD = BC$. Its displayed factorization gives two nonsingular weighted-equality factors of the original M, after undoing the diagonal normalization. Retain both factors and the entire finite factor-saturated augmentation as Λ. By Lemma 2.23, all hardness reductions compose back to Γ, all tractable conclusions descend to $K\Gamma$, and the continuing state returns to the caller's coordinates. This proves the three alternatives. □

4.3 Consequences for the Full Signature Set

The preceding reductions give the following interface for the entire signature set. The factor outcome is retained for the later equality-accessible analysis.

Corollary 4.7 (Reductions from eight-vertex signatures)**.** *Let $\mathcal{G}$ be a retained, factor-saturated finite algebraic signature set containing a directly realizable endpoint-nondegenerate eight-vertex signature M. Then at least one of the following holds:*

(i) K-Holant($\mathcal{G}$) *is* #P*-hard;*

(ii) $\mathsf{Tract}(K\mathcal{G})$ *holds;*

(iii) *for some finite retained, factor-saturated augmentation* $\Lambda \supseteq \mathcal{G}$ *satisfying* K-Holant(Λ) $\leq_{\mathrm{T}}$ K-Holant($\mathcal{G}$)*, the signature set* Λ *retains the complete factor batch from a genuine factorization of the original endpoint-nondegenerate eight-vertex signature* M *across two two-port blocks and contains the nonsingular weighted-equality matrix*

$$E := \mathrm{diag}(p, q), \qquad p, q \in \overline{\mathbb{Q}}^{\times},$$

Turing-exposed as a proper tensor factor.

In item (iii), the factor E *is Turing-exposed from the original* M*; secondary two-copy equality outputs are resolved within the preceding reductions. The construction uses only this quaternary analysis.*

Proof. If all six weight-two entries vanish, the signature is a pure generalized equality and is closed by Lemma 3.9. Otherwise apply the Cai–Fu quaternary boundary Theorem 2.17. Its hard branch is item (i). Its product-transformable branch is routed through Corollary 4.3. A local-affine branch first collapses to affine by Lemma 4.4 and then uses Corollary 4.6; an affine branch uses that operational corollary directly.

The two operational wrappers internally close every pure generalized equality, equality anchor, odd-arity signature, and singular binary factor. In the last case, retained factor saturation exposes a unary by Lemma 2.20; the corresponding transported diagonal dressing of the original endpoint-nondegenerate eight-vertex signature remains present, excludes the common-one-sided exception, and the common unary exit Lemma 3.4 applies. Consequently all routes merge into exactly the three displayed outcomes, and every hardness reduction composes toward $\mathcal{G}$, while every tractable presentation descends to $K\mathcal{G}$. Apply Lemma 2.23 once to pull back the legal normalizations. The sole continuing output is the retained nonsingular weighted-equality factor in item (iii). □

Proof of Theorem 1.3. Choose an endpoint-nondegenerate tensor-prime quaternary $q \in \mathcal{F}$. Preprocess as in Section 2.6 and write $\mathcal{F}^{\circ}$ for the resulting signature set. The signature q survives, and the preprocessing convention identifies the displayed criterion with $\mathsf{Tract}(K(\mathcal{F}^{\circ}))$.

If $\mathsf{Tract}(K(\mathcal{F}^{\circ}))$ holds, then Theorem 2.19 and Equation (10) gives K-Holant($\mathcal{F}^{\circ}$) $\in \mathsf{FP}$, and the root evaluator restores the deleted nullary scalars. Now suppose that $\mathsf{Tract}(K(\mathcal{F}^{\circ}))$ fails. The two nonzero endpoint values of q occur at doubled charges -4 and $+4$, so $\mathcal{F}^{\circ}$ is not common one-sided. Apply Lemma 2.20 and retain

$$\Gamma := \mathcal{F}^{\circ} \cup \Pi(\mathcal{F}^{\circ}).$$

Then Γ is finite and factor-saturated, K-Holant(Γ) $\equiv_{\mathrm{T}}$ K-Holant($\mathcal{F}^{\circ}$), and it still contains the original signature q. By the contrapositive of Lemma 2.21, $\mathsf{Tract}(K(\Gamma))$ also fails.

Apply Corollary 4.7 to q over Γ. Its tractable outcome is excluded by the preceding paragraph. Its continuing outcome gives a genuine $2 + 2$ factorization of the original q, up to the recorded invertible diagonal dressing, port permutation, and nonzero scalar. An invertible portwise dressing preserves tensor-primality: undoing it would give a positive-arity tensor factorization of q. Hence the continuing outcome contradicts the hypothesis, and the only remaining outcome is that K-Holant(Γ) is #P-hard. The retained equivalence and preprocessing transfer this hardness to K-Holant($\mathcal{F}$). Finally, decidability follows from Theorem 2.19. □

5 Equality-Accessible Matching Decks

Here equality access means that the literal binary $I = \mathrm{EQ}_2$ is available. We first control all realizable binary transfers, then contract a minimum-arity nonbinary prime and retain the factors of its outputs. Stabilization makes these outputs products on perfect matchings. A finite group classification then routes the remaining tensor to one of Sections 6 to 9.

The edge and transfer conventions are those of Sections 2.4 and 2.5; the termination argument uses the retained-factor potential from Lemma 2.20. The shared finite quaternary statement is in Appendix A.2. Throughout, the supplied I remains in the caller's fixed coordinates. A call to Section 4 returns in those coordinates; any additional weighted equality is processed as a binary signature.

A *resolved leaf* is hardness, Tract for the entire retained set, or a common tractable K-presentation in the sense of Definition 2.8.

Lemma 5.1 (Equality-wire equivalence)**.** *For every signature set $\mathcal{G}$,*

$$\text{K-Holant}(\mathcal{G}, I) \equiv_{\mathrm{T}} \text{Holant}(\mathcal{G}, X),$$

where the right side uses ordinary equality wires and has X as an available binary signature.

Proof. Forward, subdivide each native X-wire by an X-vertex and suppress degree-two I-vertices as equality wires. Conversely, splice the two incident ports of the ordinary X-vertices after first preprocessing the subgraph induced by those vertices. Every such component is a path or a cycle, because X has arity two. A path with two incidences at non-X vertices collapses to I or X according to the parity of the number of X-vertices; a cycle with no external incidence contributes zero for odd length and the scalar 2 for even length. (A self-loop is the length-one case.) These are exact identities obtained by summing the internal equality-wire bits, and they also cover adjacent X-vertices and parallel edges. For each resulting X kernel use one structural E_X edge (no additional tensor vertex); for each resulting I kernel, and for each ordinary equality wire whose endpoints are not in an X-component, use

$$\text{port} - X - I - X - \text{port}.$$

Indeed, $\sum_{u,v} X(a,u)I(u,v)X(v,b) = I(a,b)$. Multiply the final answer by the known product of the even-cycle scalars (or return zero for an odd cycle). The preprocessing and substitutions are linear-size and preserve ordered incidences. □

Consequently, it suffices to work with a retained factor-saturated signature set containing the distinguished literal equality I. We first establish the binary-transfer and finite-group control required by the matching-deck construction.

5.1 Binary transfers, complete groups, and local quaternary dispatch

Binary chains turn composition into matrix multiplication. Bounding the group they generate gives a finite set of contraction kernels for the matching deck.

For an ordered binary signature B, recall its transfer

$$T_B = BX.$$

Joining B to another ordered binary signature C in series through one X-edge realizes BXC, and therefore

$$T_{BXC} = T_B T_C.$$

For a transfer matrix T, define its reversal by the anti-involution

$$T^{\#} := XT^{\mathsf{T}}X; \qquad T^{\#} = (\det T)ZT^{-1}Z \quad \text{if } T \text{ is nonsingular.} \tag{31}$$

Here "anti-involution" means explicitly that

$$(ST)^{\#} = T^{\#}S^{\#}, \qquad (T^{\#})^{\#} = T$$

for all transfer matrices S, T.

Lemma 5.2 (Bounded torsion)**.** *Fix a number field L, and let G be the set of projective transfers of all directly realizable nonzero nonsingular ordered binary gadgets over a fixed signature set with entries in L. If $G \neq \varnothing$ and every element of G has finite projective order, then G is a finite subgroup of* $\mathrm{PGL}_2(L)$.

Proof. Define the field degree $D = [L : \mathbb{Q}]$. Let $[T] \in G$ have projective order r. By definition,

$$T^r = \eta I$$

for some $\eta \in L^{\times}$, and r is the least positive integer with this property.

The minimal polynomial m_T divides $x^r - \eta$, which is square-free because $\gcd(x^r - \eta, rx^{r-1}) = 1$ in characteristic zero. Thus T is diagonalizable over the algebraic closure. Choose an eigenbasis matrix $V \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ and eigenvalues $\lambda_1, \lambda_2 \in \overline{\mathbb{Q}}^{\times}$ such that

$$V^{-1}TV = \mathrm{diag}(\lambda_1, \lambda_2).$$

Define the eigenvalue ratio

$$\zeta := \frac{\lambda_1}{\lambda_2}.$$

The equality $T^r = \eta I$ gives $\zeta^r = 1$, and the minimality of r shows that ζ is a primitive r-th root of unity. Moreover,

$$\zeta + \zeta^{-1} = \frac{(\operatorname{tr} T)^2}{\det T} - 2 \in L.$$

For $r > 2$,

$$[\mathbb{Q}(\zeta + \zeta^{-1}) : \mathbb{Q}] = \frac{\varphi(r)}{2},$$

so $\varphi(r) \leq 2D$, also valid for $r = 1, 2$. The integers with this bound form an effectively finite set: each prime power $p^a \mid r$ satisfies $p^{a-1}(p-1) \leq 2D$, bounding both p and a. Consequently all projective orders divide the computable exponent

$$e_D = \mathrm{lcm}\{r : \varphi(r) \leq 2D\}.$$

Series composition gives product closure. A positive power of every element is the identity, so the identity and inverses also belong to G. Hence G is a subgroup. Put

$$\mathfrak{sl}_2(L) := \{A \in \mathrm{Mat}_2(L) : \operatorname{tr} A = 0\}.$$

The adjoint representation is

$$\mathrm{Ad} : \mathrm{PGL}_2(L) \hookrightarrow \mathrm{GL}(\mathfrak{sl}_2(L)) \cong \mathrm{GL}_3(L), \qquad \mathrm{Ad}([T])(A) = TAT^{-1}.$$

Its kernel is trivial: a matrix commuting with all trace-zero matrices is scalar. Burnside's finite-exponent theorem states that a subgroup $H \leq \mathrm{GL}_n(\mathbb{C})$ satisfying $h^e = I$ for one fixed e and every $h \in H$ is finite [Bur05]. Apply it to $\mathrm{Ad}(G) \leq \mathrm{GL}_3(\mathbb{C})$ with $e = e_D$. Thus all binary gadgets, despite their unbounded sizes, have only finitely many projective transfer classes. □

For a finite algebraic retained, factor-saturated, two-sided signature set $\mathcal{G}$, call a nonzero directly realizable ordered binary *safe* if it is nonsingular and its transfer has finite projective order. Call it *unsafe* if it is singular or its transfer has infinite projective order. Call the set *binary-residual* when every nonzero actual ordered binary is safe. We work in this case; otherwise Lemmas 3.1 and 3.4 resolves the branch. Series composition and Lemma 5.2 then give a finite group: take L to be the number field generated by the entries of $\mathcal{G}$, which also contains the entries of every direct gadget transfer. The group is

$$G(\mathcal{G}) = \{[BX] : B \text{ is a safe ordered binary directly realizable over } \mathcal{G}\}. \tag{32}$$

The definition uses all directly realizable binaries. Taking or recomputing this complete group below is a structural case distinction for a fixed signature set. The decision algorithm for the resulting tractability alternatives is given by Theorem 2.19; it does not enumerate gadgets of unbounded size. Since the group is finite, fix an oriented actual representative B_g satisfying $[B_g X] = g$ for every $g \in G(\mathcal{G})$.

Lemma 5.3 (Relative completeness test for actual binaries)**.** *Let $\Gamma \subseteq \Lambda$ be finite algebraic retained, factor-saturated signature sets with $I \in \Gamma$, and suppose that every nonzero binary directly realizable over Γ is safe. Put $H = G(\Gamma)$. The following exhaustive priority dispatch applies:*

1. *if Λ directly realizes a nonzero singular ordered binary, take the factor/unary exit;*
2. *otherwise, if Λ directly realizes a nonsingular ordered binary of infinite projective transfer order, take the interpolation exit;*
3. *otherwise every nonzero binary directly realizable over Λ is safe, $H \le G(\Lambda)$, and every nonsingular actual binary B over Λ with $[BX] \notin H$ witnesses the strict inclusion $H < G(\Lambda)$.*

In the third case, if $G(\Lambda) = H$, every nonzero actual binary over Λ is projectively one of the fixed representatives B_h, $h \in H$.

Proof. Retained I has both charge signs, so both signature sets are two-sided. A singular binary is handled by Lemmas 3.1 and 3.4, and an infinite-order binary by Lemma 3.1. If neither exists, Λ is binary-residual, so Lemma 5.2 defines $G(\Lambda)$. Every gadget over Γ remains a gadget over Λ, hence $H \le G(\Lambda)$. The remaining assertions follow directly from the definitions and the fixed representative choice. □

Remark 5.4 (Relation to binary-signature classifications). The singular, nonsingular infinite-projective-order, and nonsingular finite-projective-order split is the K-Holant form of the binary split used by Cai–Lu–Xia [CLX09, CLX18]. The four ordered Bell signatures $\mathcal{B} = \{B_{0,0}, B_{0,1}, B_{1,0}, B_{1,1}\}$ represent the projective Bell classes of Shao–Cai [SC20]. Their holographic matrix, denoted Z in that paper, is our K; throughout this paper, $Z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, as in Equation (31). Beyond the binary classification, we need the complete transfer group and uniform control of its higher-arity cards.

For an even finite port set R, a *perfect matching* is a partition of R into two-element subsets. An *oriented perfect matching* also chooses an order within each pair, and a matching product places one ordered binary factor on every oriented pair.

For a quaternary tensor q with ordered ports $1, 2, 3, 4$, let $F_{ij|k\ell}(q)$ denote the 4×4 flattening whose row index is (x_i, x_j) and whose column index is (x_k, x_ℓ). For a finite complete transfer group G and its fixed actual representatives B_g, $g \in G$, define

$$\mathcal{M}_4(G) = \left\{ c\, B_g(x_i, x_j) B_h(x_k, x_\ell) : \begin{array}{l} c \in \overline{\mathbb{Q}}^\times,\ ((i,j),(k,\ell)) \text{ is an oriented} \\ \text{perfect matching of } [4], \quad g, h \in G \end{array} \right\}.$$

Thus $\mathcal{M}_4(G)$ is the arity-four G-matching notion used below. Define the complete oriented one-copy binary batch as follows. For an ordered binary card C, set $C^{\mathrm{id}} = C$ and $C^{\mathrm{rev}} = C^{\mathsf{T}}$, and put

$$\mathscr{D}_G^{(1)}(q) = \{0\} \cup \left\{ \left(\partial_{(i,j)}^{XB_gX} q\right)^{\epsilon} : i,j \in [4],\ i \neq j,\ g \in G,\ \epsilon \in \{\mathrm{id}, \mathrm{rev}\} \right\}.$$

The order (i,j) records the orientation of the deleted pair, and the choice of ϵ records either orientation of the residual pair. Every displayed member is a direct gadget: XB_gX is the physical kernel obtained by attaching B_g through its two native X-wires, and residual reversal is a port permutation. In this local case analysis, a *nonbinary prime* means a tensor-prime signature of arity at least three.

Lemma 5.5 (Binary contractions of a quaternary signature)**.** *Let Λ be a finite algebraic retained, factor-saturated, binary-residual signature set containing the distinguished literal equality I, suppose every retained nonbinary prime has even arity, and put $G = G(\Lambda)$. Let q be an actual signature over Λ, with its ordered ports, realization, and nonzero scalar recorded. Assume that q is endpoint-nondegenerate, tensor-prime, and*

$$q \notin \mathcal{M}_4(G).$$

The three physical flattenings satisfy

$$\operatorname{rank} F_{12|34}(q) \geq 2, \qquad \operatorname{rank} F_{13|24}(q) \geq 2, \qquad \operatorname{rank} F_{14|23}(q) \geq 2.$$

These inequalities follow from tensor primality and specify the rank checks used by later callers.

Let Λ_q be the preprocessed literal augmentation obtained by adjoining q and all nonzero tensor cards in $\mathscr{D}_G^{(1)}(q)$, followed by retained factor saturation of this finite batch. Then

$$\text{K-Holant}(\Lambda_q) \equiv_{\mathrm{T}} \text{K-Holant}(\Lambda),$$

and one of the following holds:

(i) *an established resolved leaf or terminal applies, including an odd-arity/unary exit, a nonzero singular-binary exit, or an actual infinite-order-binary interpolation exit; alternatively, a retained lower-arity nonbinary prime gives the corresponding outer deck successor;*

(ii) *Λ_q is binary-residual and*

$$G < G_q := G(\Lambda_q),$$

with the strict inclusion witnessed by an ordered binary B actual over Λ_q such that $[BX] \in G_q \backslash G$;

(iii) *Λ_q is binary-residual, $G_q = G$, q is a minimum-arity nonbinary tensor-prime in Λ_q, and every member of $\mathscr{D}_G^{(1)}(q)$ is zero or projectively equal to one of the fixed B_g, $g \in G$. All nonzero card factors are retained; on this continuing branch every nonzero card has a trivial factorization tree.*

Only item (iii) is passed to the equality-accessible quaternary boundary.

Moreover, on branch (iii), let r be an auxiliary actual quaternary signature over Λ_q, assume $r \notin \mathcal{M}_4(G)$, and suppose that r has a genuine $2+2$ factorization. Let $\Lambda_{q,r}$ be the preprocessed retained, factor-saturated augmentation obtained by adjoining r and its nonzero factor batch. Then either an established terminal applies or

$$G < G(\Lambda_{q,r}).$$

Thus there is no same-group continuing branch. This finite local case analysis does not invoke the global matching-synthesis measure.

Proof. The batch $\mathscr{D}_G^{(1)}(q)$ is finite: there are six unordered deleted pairs, two orders for the closing kernel, two residual orders, and finitely many closing-representative choices B_g. Its tensor cards are actual over Λ, since q and every B_g have recorded direct realizations. Adjoining them therefore gives one Turing reduction by Equation (12), while inclusion gives the reverse one. Because literal I makes every retained augmentation two-sided, Lemma 2.20 may retain the finite factorization tree without changing the problem up to Turing equivalence.

Dispatch every odd-arity or unary factor and every singular binary factor. If Λ_q directly realizes any nonsingular binary of infinite projective transfer order, apply Lemma 3.1; this is item (i). Otherwise Λ_q is binary-residual, so Lemma 5.2 defines the finite group $G_q = G(\Lambda_q)$. Retained inclusion gives $G \le G_q$. If the inclusion is strict, the definition of G_q supplies an actual binary representative of a class in $G_q \setminus G$, proving item (ii). The witness is an arbitrary actual gadget over Λ_q, possibly using several copies of q.

It remains that $G_q = G$. Every nonzero card in $\mathscr{D}_G^{(1)}(q)$ is then nonsingular and has transfer class in $G_q = G$, hence is projectively one of the fixed B_g. Such a binary has no proper positive-arity tensor factorization. The tensor q remains prime intrinsically. Any nonbinary prime of arity below four would have arity three and would already have triggered the odd-arity terminal, so q has minimum nonbinary-prime arity. This proves item (iii).

Finally, write $r \doteq C_1 \otimes C_2$ for the auxiliary factorization. Factor saturation makes C_1, C_2 literal resources of $\Lambda_{q,r}$; they are not asserted to have been actual over Λ_q. If either is singular, or if the augmentation realizes an infinite-order binary, an established terminal applies. Otherwise $\Lambda_{q,r}$ is binary-residual and $G \le G(\Lambda_{q,r})$. If equality held, both factors would be nonsingular and would represent classes in G, whence $r \in \mathcal{M}_4(G)$, contradicting the stated hypothesis. The inclusion is therefore strict. □

We now choose a minimum-arity nonbinary prime and retain the factors of all its dressed contractions.

5.2 Stable matching decks and restart control

Fix a retained, factor-saturated signature set $\mathcal{G} \ni I$ and its complete transfer group $G = G(\mathcal{G})$. Deck operations use the actual binaries B_g, transfers R_g, and multiplication/reversal scalars fixed in Section 2.5 before passing to projective classes.

Definition 5.6 (Matching deck)**.** Let $\mathcal{G} \ni I$ be factor-saturated and let $G = G(\mathcal{G})$. On a fixed ordered port set, a *G-matching signature* of arity $2m$ is

$$c \bigotimes_{e \in M} B_{g_e}^{(e)}, \qquad c \neq 0, \quad g_e \in G,$$

where M is an oriented perfect matching in the preceding sense and $B_{g_e}^{(e)}$ places the chosen ordered representative B_{g_e} on the oriented edge e. We allow a simultaneous port permutation; the scalar c absorbs all representative scalars, so the notion depends only on G.

For $\omega \in \{\mathrm{id}, \mathrm{rev}\}$, put $B_g^{\mathrm{id}} = B_g$ and $B_g^{\mathrm{rev}} = B_g^{\mathsf{T}}$. Let δ specify a subset of the ports and, at each selected port, a representative B_g and one of these two orientations. Attach its recorded binary through one native X-wire and retain the other binary port in the same position; write f^δ for the resulting actual dressed signature.

For each δ, choose distinct ordered ports (i, j), a closing representative B_h with orientation ω, and a residual port order ρ, leaving at least two ports. The resulting card is

$$\rho\Big(\partial_{(i,j)}^{X B_h^\omega X} f^\delta\Big).$$

The *matching deck* $\mathfrak{D}_G(f)$ contains zero and all such one-copy proper cards. Thus one copy of f is used and exactly one ordered port pair is closed; the binary dressings and closing chain use the fixed actual representatives. Every nonzero card is a direct gadget output, never an analytic linear combination.

Adjoining a Turing-exposed family preserves the problem up to Turing equivalence: the exposure reduction removes the new signatures, and inclusion gives the reverse reduction. Direct outputs use Equation (12); their factors use Lemma 2.20.

For termination, fix (A, Λ, f) and $H := \overline{G}(A, \Lambda)$. While H and $\nu(\Lambda)$ remain unchanged, keep one fixed actual representative of each group class. Let $\mathscr{U}(A, f, H)$ be the set of canonical keys, normalized after pulling a card back by A, of zero and of every one-copy proper card. Concretely, its finite datum records a choice of binary dressings using the frozen representatives indexed by H, the ordered deleted pair, the frozen closing representative B_h, the resulting kernel $XB_h^{\omega}X$ and its orientation ω, and the residual port order. The key retains the ordered external port list and the first-nonzero-entry normalization; the discarded nonzero scalar and the gadget provenance are stored separately. Attach to each nonzero key its finite factorization tree from the worklist in Lemma 2.20, and write $\mathscr{U}^{\mathrm{fac}}(A, f, H)$ for the resulting finite set of card and factor tokens. A fixed-(H, ν) queue contains each token at most once. Write

$$Q_{\mathrm{done}} \subseteq \mathscr{U}^{\mathrm{fac}}(A, f, H), \qquad Q = \mathscr{U}^{\mathrm{fac}}(A, f, H) \setminus Q_{\mathrm{done}}$$

for the completed and unfinished indices, respectively; $|Q|$ is finite. During the local deck-stabilization construction, a card token contributes one of the already enumerated members of $\mathscr{U}(A, f, H)$, and a factor token is processed entirely inside its attached finite factorization tree from Lemma 2.20. Thus this construction cannot create a fresh fixed-(H, ν) token outside the fixed finite set. A lower ν, a strict enlargement of the pulled-back group, or a terminal dispatch stops the local construction. The literal equality I and the caller's coordinates remain fixed throughout this protocol.

Definition 5.7 (Deck-stable state)**.** A retained state (Λ, G, f) is *deck-stable* if Λ is finite, factor-saturated, and contains I; $f \in \Lambda$ is a minimum-arity tensor-prime nonbinary signature; $G = G(\Lambda)$ is the recomputed complete finite safe group, with fixed actual representatives; and the regenerated deck, with all card factors retained, consists of zero and G-matching signatures. Actuality is always relative to Λ; retained factors are only Turing-exposed over their predecessor.

The next lemma either produces a deck-stable state, resolves the complexity of the current set, or exposes a smaller nonbinary prime.

Lemma 5.8 (Deck stabilization under factor access)**.** *Let Λ_0 be a retained, factor-saturated finite algebraic signature set containing the distinguished literal equality I, and let $f \in \Lambda_0$ be an even-arity tensor-prime nonbinary signature of arity $n = \nu(\Lambda_0)$.*

Then either an established terminal or the signature-set-wide conclusion $\mathsf{Tract}(K\Lambda_0)$ *applies, or the construction returns one of the following retained successors:*

1. *a finite retained factor-saturated augmentation $\Lambda_< \supseteq \Lambda_0$ such that*

$$\text{K-Holant}(\Lambda_<) \equiv_{\mathrm{T}} \text{K-Holant}(\Lambda_0), \qquad \nu(\Lambda_<) < n;$$

   every nonbinary prime retained in this successor has even arity;

2. *a finite retained factor-saturated augmentation $\Lambda_\star \supseteq \Lambda_0$ such that*

$$\text{K-Holant}(\Lambda_\star) \equiv_{\mathrm{T}} \text{K-Holant}(\Lambda_0),$$

   and, for $G_\star = G(\Lambda_\star)$, every member of $\mathfrak{D}_{G_\star}(f)$ is zero or a $G_\star$-matching signature.

The stable successor retains the actual representatives, the regenerated complete deck, and every card factor. Such factors are Turing-exposed over, but need not be directly realizable from, their predecessor.

Proof. Construct states $(A_i, \Lambda_i, H_i, Q_i)$, where $H_i = \overline{G}(A_i, \Lambda_i) \le \mathrm{PGL}_2(L_0)$ is in the fixed pulled-back coordinates and Q_i is the finite canonical card/factor queue from the preceding convention. At stage i, dispatch unsafe binaries, recompute the complete group $G_i = G(\Lambda_i)$, retain one actual representative of every class, and generate each key in $\mathscr{U}(A_i, f, H_i)$ at most once. Preprocess and jointly factor-saturate the resulting finite batch to obtain Λ_{i+1}. Retained I excludes the one-sided exception. Inclusion supplies the reverse reduction at every exposure step, so every Λ_i is Turing-equivalent to Λ_0.

Every nonbinary card factor has arity at most $n-2$. Odd factors are terminal by Theorem 2.15; an even one gives successor 1. Otherwise only binaries remain. No global normalization is made in this fixed-I protocol. Completeness therefore gives $H_i \le H_{i+1}$; only a strict inclusion $H_i \subsetneq H_{i+1}$ is an outer group successor, and it alone triggers regeneration of the complete deck.

Use the frozen pulled-back field L_0 from Lemma 2.23. Gadget evaluation and pivot-normalized rank-one splitting are performed in pulled-back coordinates, so $H_i \le \mathrm{PGL}_2(L_0)$. Let $d_{L_0} := [L_0 : \mathbb{Q}]$ be the field degree, and define the torsion exponent and group-order bound by

$$e_{L_0} = \mathrm{lcm}\{r \ge 1 : \varphi(r) \le 2d_{L_0}\}, \qquad M_{L_0} = \max\{2e_{L_0}, 60\}, \tag{33}$$

where φ denotes Euler's totient function. Proposition 2.1 and Lemma 5.2 give $|H_i| \le M_{L_0}$. Hence the strict subgroup chain has at most $M_{L_0} - 1$ changes. Between two such changes the finite queue is exhausted: each card/factor token is removed once, and the inner factor potential Φ of Equation (25) decreases while its factorization tree is processed. This stabilization procedure invokes only the one-copy proper cards in $\mathscr{U}(A_i, f, H_i)$ and the factorization trees attached to those cards; it does not yet invoke any branch-specific multi-copy reduction. Therefore every task that can remain in the same pair (H_i, n) is a token of this locally frozen queue by construction. A lower-arity factor, a new group class, or a terminal stops the current stage instead. Consequently there is a first stage at which both $H_{i+1} = H_i$ and the queue is empty; call this the stabilized stage. At that stage define the stable retained signature set and its complete group by

$$\Lambda_\star = \Lambda_{i+1}, \qquad G_\star = G_{i+1}.$$

Every $0 \ne h \in \mathfrak{D}_{G_\star}(f)$ then has retained binary prime factors $B_1, \dots, B_m$ satisfying

$$[B_j X] \in G(\Lambda_\star) = G_\star.$$

Writing $B_j = \beta_j B_{g_j}$, their blocks form a perfect matching, so

$$h = c \bigotimes_{j=1}^{m} B_{g_j}$$

for $c \ne 0$, as required. □

Later constructions can lower ν or enlarge the complete group. We call either change an *outer deck successor*; processing a fixed family while both remain unchanged is a *within-node closure*. The following conditions specify when such processing terminates.

Definition 5.9 (Processing a finite family of signatures)**.** For deck-stable (Λ, G, f), take a finite family of signatures retained in or directly realized over Λ, with their port orders and realization scalars. Adjoin them, preprocess, jointly factor-saturate, apply the reductions for odd and unsafe factors, and recompute the group. After a strict enlargement, regenerate the complete deck. We refer to these steps as processing the family. Analytic combinations and projective tables alone do not supply signatures to this family. The closure carries the partition $\mathscr{U}^{\mathrm{fac}} = Q_{\mathrm{done}} \sqcup Q$ from the finite-set construction, where Q is the unfinished queue. A token moves from Q to Q_{done} only after its direct card and its entire factorization tree have been processed; a canonical key is never processed twice. Whether a later branch-specific batch may return to the same case is governed by the conditions in Definition 5.10 below rather than imposed by this protocol definition.

A call to the stable equality-accessible quaternary boundary stated below is legal only after the current physical quaternary card and its complete oriented one-copy deck have been regenerated, factor-saturated, and dispatched as in Lemma 5.5. Auxiliary cards used inside the finite certificate are local to that proof and are not appended to Q. Only a retained factor batch returned by the reduced interface can be passed to an outer closure.

The current case is complete once finitely many such families resolve every branch and $Q = \varnothing$. Under the conditions of Definition 5.10, a transition leaving the pulled-back group and ν unchanged completes a queue token, decreases its stored factor potential, or certifies that none remains, and hence is a within-node closure; otherwise, ν decreases or the pulled-back group enlarges.

Standing predicates for relative cases. To keep the group-specific statements short, write

$$\mathsf{Stable}(\Lambda, G, f) \iff (\Lambda, G, f) \text{ is deck-stable in the sense of Definition 5.7,} \tag{34}$$

$$\mathsf{Even}(\Lambda) \iff \forall g \in \Pi(\Lambda), \quad \mathrm{arity}(g) \geq 3 \Longrightarrow \mathrm{arity}(g) \in 2\mathbb{Z}. \tag{35}$$

For a family $\mathfrak{C}$ of tractable signature classes,

$$\mathsf{Close}_{\mathfrak{C}}(\Lambda, G, f) \tag{36}$$

means that finitely many applications of Definition 5.9 resolve all branches for (Λ, G, f), with $Q = \varnothing$, and every surviving common-presentation rigidity leaf satisfies $\mathsf{Pres}_K(K\Lambda; \mathscr{C})$ for some $\mathscr{C} \in \mathfrak{C}$. Established terminals, strict outer successors, and already resolved Tract-leaves are part of closure but are not rigidity leaves; $\mathfrak{C} = \varnothing$ means that no common-presentation rigidity leaf remains. These symbols abbreviate conclusions only and do not replace any protocol hypothesis or transition.

When H and ν do not change, a continuing step must process an index already in Q and decrease $|Q|$ or its stored factor potential. The precise conditions are as follows.

Definition 5.10 (Conditions for descent)**.** Fix a deck-stable tuple (Λ, G, f), a coordinate matrix A, the pulled-back group $H = \overline{G}(A, \Lambda)$, and a frozen partition

$$\mathscr{U}^{\mathrm{fac}}(A, f, H) = Q_{\mathrm{done}} \sqcup Q,$$

where Q is the unfinished queue. Let $\mathfrak{B}$ be a collection of admissible families. Processing any member must give, on each branch, one of:

1. a resolved terminal or Tract-leaf;
2. a successor with smaller ν;
3. a strict enlargement of the pulled-back complete group;

4. unchanged (H, ν), in which case every retained continuing resource is indexed by a token of Q, and its processing is confined to that token's stored card, provenance, and attached factorization tree. The closure either moves at least one such token to Q_{done}, or, while processing one fixed unfinished token, strictly decreases the factor potential of its stored worklist.

A token is marked done only after its full factorization tree is exhausted.

The verification is necessarily branch-by-branch and uses the reductions proved in Sections 6 to 9. It is therefore deferred to Section 10, after those four arguments are available.

The following lexicographic measure strictly decreases on every continuing step satisfying these conditions.

Lemma 5.11 (Deck-transition dichotomy and termination)**.** *Fix the initial number field L_0. At a continuing all-even, binary-residual deck-stable node, carry the state (A, Λ, f), a finite token queue Q, and the distinguished literal equality I. Let $\mathfrak{B}$ satisfy Definition 5.10, let Λ^+ be the closure of one batch in $\mathfrak{B}$ over Λ, and define the pulled-back group by*

$$\overline{G}(A, \Lambda) = [A]^{-1} G(\Lambda)[A].$$

Then

$$\text{K-Holant}(\Lambda^+) \equiv_{\mathrm{T}} \text{K-Holant}(\Lambda). \tag{37}$$

Batch outputs are direct over Λ, while their factors need only be Turing-exposed there.

After signature-set-wide odd/unsafe and unary/binary dispatch, the only continuing transitions are:

1. *$\overline{G}(A, \Lambda^+) = \overline{G}(A, \Lambda)$ and $\nu(\Lambda^+) = \nu(\Lambda)$: process a previously unfinished token of Q, complete it or strictly decrease its factor potential, or certify that Q is empty.*
2. *$\overline{G}(A, \Lambda^+) = \overline{G}(A, \Lambda)$ and $\nu(\Lambda^+) < \nu(\Lambda)$: choose a new minimum core and reapply Lemma 5.8.*
3. *$\overline{G}(A, \Lambda) \subsetneq \overline{G}(A, \Lambda^+)$: choose a minimum augmented core, regenerate its deck, and restart marked routing.*

At every continuing node define the pair by

$$\Xi(A, \Lambda) = \big(M_{L_0} - |\overline{G}(A, \Lambda)|, \nu(\Lambda)\big).$$

Refine it to the tuple

$$\Omega(A, \Lambda, f, Q) = \big(M_{L_0} - |\overline{G}(A, \Lambda)|, \nu(\Lambda), |Q|, \Phi_Q\big). \tag{38}$$

For each $u \in Q$, let $\mathcal{W}_u$ be the multiset of still-open occurrences in its attached factor worklist, and define

$$\Phi_Q := \sum_{u \in Q} \Phi(\mathcal{W}_u),$$

using the potential in Equation (25)*. The displayed tuple is ordered lexicographically. Items 2–3 strictly decrease its first two coordinates. In item 1, completing a token strictly decreases the third coordinate until the queue is empty. If a token remains unfinished during a genuine factor split, the fourth coordinate Φ_Q strictly decreases. If an exposed factor changes the pulled-back group, the operation is instead item 3 and is charged to the strict subgroup chain.*

Proof. Substitution and factor saturation prove Equation (37); I excludes the one-sided exception, and inclusion reverses the reduction. Retained augmentation cannot increase ν. The continuing state remains in fixed-I coordinates. New classes can arise only from an exposed factor; such a class is precisely a strict successor in the pulled-back group. By Lemma 2.23 and Equation (33), every continuing group lies in $\mathrm{PGL}_2(L_0)$ and has order at most M_{L_0}. Therefore a lower ν decreases the second coordinate of Ξ, strict subgroup inclusion decreases the first. A fixed-(H,ν) step either decreases the finite queue counter in Ω or, within its fixed token, decreases Φ, so no factor loop is possible. By Definitions 5.9 and 5.10, item 1 never opens a fresh fixed-(H,ν) branch: it can only process one previously frozen token of Q. □

Definition 5.12 (The conclusion required of each construction)**.** Write

$$\Sigma = (A, \Lambda, H, f, Q), \qquad H = \overline{G}(A, \Lambda),$$

with the record, deck-stability, frozen queue, and fixed-I data of Definitions 5.9 and 5.10. Write

$$\Omega(\Sigma) := \Omega(A, \Lambda, f, Q) = (M_{L_0} - |H|, \nu(\Lambda), |Q|, \Phi_Q)$$

for the lexicographic measure in Equation (38).

For each concrete construction at Σ, we prove that processing its outputs by Definition 5.9 gives either

1. a resolved complexity leaf: hardness, a Tract-leaf, or an accepted common tractable presentation; or
2. a deck-stable tuple Σ', with the same required realization and induction data, satisfying

$$\Omega(\Sigma') <_{\mathrm{lex}} \Omega(\Sigma).$$

The second outcome includes a decrease of ν, a strict enlargement of the pulled-back complete group, completion of a frozen token, or a strict decrease of that token's factor potential. Every same-(H,ν) continuation must be indexed by the already frozen queue and may create no fresh unresolved fixed-(H,ν) resource. An arbitrary changed signature set is therefore not a successor in this definition.

The first outcome resolves the complexity question; the second continues the induction at Σ'. Below, saying that a construction is resolved allows either outcome. This conclusion requires a proof of realizability, factor access, and descent for that construction. The assertion $\mathsf{Close}_{\mathfrak{C}}(\Lambda, G, f)$ additionally covers every case for the current (H,ν) and restricts the surviving common presentations to $\mathfrak{C}$, as in Equation (36).

We have therefore reached a deck-stable tuple. For the active stable tuple Σ, every continuing branch of a resolved batch has strictly smaller Ω. In a signature-set-wide completion, however, Ω controls only that active tuple. A separate outer worklist induction over unresolved tensor-prime occurrences completes the current prime and advances to the next; replacing f by another prime of the same arity is charged to that outer induction and is not asserted to decrease Ω.

5.3 The main stable-state theorem and four-way routing

The preceding reductions leave a deck-stable tuple (Λ, G, f). The next theorem is proved in Section 10 from the four branches routed below. Together with stabilization, it gives the equality-accessible main theorem Theorem 1.2.

Theorem 5.13 (Relative matching-deck classification)**.** *Assume*

$$\mathsf{Stable}(\mathcal{G}, G, f), \qquad \mathrm{arity}(f) = \nu(\mathcal{G}).$$

In particular, the deck-stable state contains the distinguished literal equality I. If $\mathsf{Tract}(K(\mathcal{G}))$ *holds, then* K-Holant$(\mathcal{G}) \in \mathsf{FP}$*; otherwise* K-Holant$(\mathcal{G})$ *is* #P*-hard. Equivalently, by Lemma 2.12, the tractable condition is* $\mathsf{Tract}_X(K(\mathcal{G}))$*.*

The proof retains the stronger structural conclusion that every nonhard rigidity survivor has one common product, affine, or flat-Lagrangian K-presentation; by Equation (21), the same basis also places the transformed edge in that class.

To route the stable tuple, we now record the marked finite-group notation in the fixed coordinates containing the distinguished literal equality.

Define the diagonal matrix family

$$R(t) = \mathrm{diag}(1, t).$$

For $m \geq 2$, define the matrix family $C(t)$ and the first two marked dihedral families

$$\begin{aligned} \mathcal{M}_m &= \{[R(t)], [XR(t)] : t \in \mu_m\}, \\ C(t) &= (1+t)I + i(1-t)Y, \\ \mathcal{B}_m &= \{[C(t)], [XC(t)] : t \in \mu_m\}. \end{aligned}$$

For even m, fix $\eta \in \mu_{2m} \setminus \mu_m$, define the matrix family

$$A(t) = (1+t)I + (1-t)X,$$

and define

$$\mathcal{A}_m = \{[A(t)] : t \in \mu_m\} \cup \{[ZA(\eta t)] : t \in \mu_m\}.$$

In the same even parameter range, define the unshifted marked form

$$\mathcal{A}_m^0 = \{[A(t)], [ZA(t)] : t \in \mu_m\}.$$

These are projective displays. Any suppressed factor of two belongs to the coordinate model and does not normalize an actual representative.

Throughout this routing discussion, retain the globally fixed marks $(x, j) = ([X], [Z])$ and put $H := \langle G, j\rangle$. For routing, the Platonic symbols record the following abstract marked types. In $\mathcal{T}$, $G = H \cong A_4$ and x, j, xj are its three double transpositions. In $\mathcal{T}_{\mathrm{ext}}$, $G \cong A_4$, $G \triangleleft H$, and $H \cong S_4$; the mark $x \in G$ is a double transposition, and $j, xj \in H \setminus G$ are disjoint transpositions. In $\mathcal{O}$, $G = H \cong S_4$ and all three marks are double transpositions. In $\mathcal{O}'$, the marks x, j are disjoint transpositions and xj is a double transposition. In $\mathcal{O}_{dt}$, x is a double transposition while j, xj are transpositions; in $\mathcal{O}_{td}$, j is a double transposition while x, xj are transpositions. Finally, $\mathcal{I}$ and $\mathcal{I}_-$ denote the two fixed-I coordinate displays of the ordered marked $V_4 = \langle x, j\rangle$ in $G = H \cong A_5$; all three nonidentity marks lie in the unique involution class. Section 9 supplies explicit fixed-equality representatives and the direct coefficient dictionaries used for the alternative displays. The router below uses only this membership and conjugacy-class data; the notation never authorizes an implicit conjugacy normalization.

The literal I supplies $x = [X]$, while binary reversal singles out $j = [Z]$. Keeping their ordered positions refines the classification of Proposition 2.1 into the following four branches.

Theorem 5.14 (Marked finite-group routing)**.** *Let Λ be a retained factor-saturated signature set containing the distinguished literal equality I, and let $G = G(\Lambda)$ be its complete finite safe transfer group. Recall the ordered physical marked pair and its generated group:*

$$(x, j) := ([X], [Z]), \qquad x \in G, \qquad H := \langle G, j \rangle.$$

In the fixed coordinates containing the distinguished literal I, exactly one of the following alternatives holds:

1. *$G \cong D_{2m}$, $m \geq 3$, and the marked configuration $(G; x, j)$ has one of the forms denoted by $\mathcal{M}_m, \mathcal{B}_m, \mathcal{A}_m, \mathcal{A}_m^0$. The last two families occur only for even m.*
2. *$G = C_N$ is cyclic, where N is even and $j \notin G$. This is the cyclic branch.*
3. *$G \cong V_4$, and G is either the standard Pauli group $\mathcal{M}_2 = \mathcal{B}_2$ or the exotic group $\mathcal{A}_2 = \mathcal{V}_{\mathrm{ex}}$.*
4. *G is one of the two marked tetrahedral forms $\mathcal{T}, \mathcal{T}_{\mathrm{ext}}$, the four fixed-equality marked octahedral forms $\mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}$, or one of the two icosahedral displays $\mathcal{I}, \mathcal{I}_-$.*

Positions of the marks. *All conjugacy statements in this paragraph are taken in H. In every form, x and j are distinct commuting involutions and $\langle x, j \rangle = \{1, x, j, xj\} \cong V_4$. In the four dihedral forms the marked positions are*

$$\mathcal{M}_m: \quad x = [XR(1)], \quad j = [R(-1)], \quad xj = [XR(-1)], \qquad H = \begin{cases} \mathcal{M}_m, & 2 \mid m, \\ \mathcal{M}_{2m}, & 2 \nmid m; \end{cases}$$

$$\mathcal{B}_m: \quad x = [XC(1)], \quad j = [XC(-1)], \quad xj = [C(-1)], \qquad H = \begin{cases} \mathcal{B}_m, & 2 \mid m, \\ \mathcal{B}_{2m}, & 2 \nmid m; \end{cases}$$

$$\mathcal{A}_m: \quad x = [A(-1)] \in G, \quad j = [ZA(1)] \in H \setminus G, \quad xj = [ZA(-1)] \in H \setminus G;$$
$$H = \{[A(t)] : t \in \mu_{2m}\} \cup \{[ZA(t)] : t \in \mu_{2m}\} \cong D_{4m};$$

$$\mathcal{A}_m^0: \quad x = [A(-1)], \quad j = [ZA(1)], \quad xj = [ZA(-1)], \qquad H = G = \mathcal{A}_m^0.$$

In $\mathcal{M}_m$, j is the central half-turn and x, xj are reflections; in $\mathcal{B}_m$, x, j are reflections and xj is the central half-turn. In both cases $j \in G$ exactly when m is even, and x, xj in $\mathcal{M}_m$, respectively x, j in $\mathcal{B}_m$, are conjugate exactly when $4 \mid m$. In $\mathcal{A}_m$, where m is even, x is the central half-turn and j, xj are conjugate reflections in $H \setminus G$. In $\mathcal{A}_m^0$, x is again the central half-turn, while $j, xj \in G$ are reflections; they are conjugate in G exactly when $4 \mid m$. When m is odd in either $\mathcal{M}_m$ or $\mathcal{B}_m$, one has $j, xj \in H \setminus G$; when m is even, all three marks lie in G.

In the cyclic branch $G = C_N$, the order N is even, $j, xj \in H \setminus G$, $H \cong D_{2N}$, x is the central half-turn, and j, xj are reflections, conjugate exactly when $4 \mid N$.

At $m = 2$, the standard Pauli form has $H = G$, with x, j, xj the three nonidentity elements (three singleton conjugacy classes) of V_4; the exotic form has $j, xj \in H \setminus G$, $H \cong D_8$, central half-turn x, and conjugate outside reflections j, xj. In $\mathcal{T}$, $H = G \cong A_4$ and $x = [Q_1], j = [Q_3], xj = [Q_2]$ are the three double transpositions, hence one conjugacy class. In $\mathcal{T}_{\mathrm{ext}}$, $G \cong A_4$, $G \triangleleft H$, and $H \cong S_4$; $x = [Q_1] \in G$ is a double transposition, and $j = [Q_3], xj = [Q_2] \in H \setminus G$ are disjoint transpositions in one class. In $\mathcal{O}$, $H = G \cong S_4$ and x, j, xj are the three double transpositions. In $\mathcal{O}'$, x, j are disjoint transpositions and xj is a double transposition. In $\mathcal{O}_{dt}$, x is a double transposition and j, xj are transpositions; in $\mathcal{O}_{td}$, j is a double transposition and x, xj are transpositions. In all four cases $H = G \cong S_4$. Finally, in each icosahedral coordinate display $H = G \cong A_5$, and $x = [Q_1], j = [Q_3], xj = [Q_2]$ lie in its unique involution class.

Each of the four group-specific sections, Sections 6 to 9, proves a relative theorem for its corresponding stable tuple. It either determines the complexity at that tuple, or produces a physical transition whose closure yields a strictly smaller active tuple under Ω. In the second case, the deck is stabilized again and the new tuple is routed anew. Movement between distinct same-arity primes in a signature-set-wide proof remains governed by the separate outer worklist induction of Definition 5.12.

The group-specific proofs may use explicitly stated coefficient or open-network dictionaries, but these do not mutate the retained fixed-I state. If retaining an exposed factor enlarges G, the current route is invalidated; one must first reapply Lemma 5.8 and then reroute before invoking a group-specific case theorem.

Proof. The reversal identity $[T] \mapsto j[T]^{-1}j = [ZT^{-1}Z]$ (with $j = [Z]$) and completeness give

$$jGj = G, \qquad G \triangleleft H, \qquad [H : G] \leq 2.$$

The two distinct projective involutions x, j commute, so $\langle x, j\rangle \cong V_4 \leq H$. By Proposition 2.1, this leaves an even dihedral group or one of A_4, S_4, A_5.

Suppose first that $G \cong D_{2m}$ with $m \geq 3$. Then H is also dihedral. Indeed, if H were Platonic and $[H : G] = 2$, then G would be an index-two normal subgroup. By Lemma 2.2, the only such subgroup among the Platonic groups is $A_4 \triangleleft S_4$, which is not dihedral. The case $H = G$ is immediate. Choose a rotation generator r and a reflection generator s so that $H = \langle r, s : r^n = s^2 = 1,\ srs = r^{-1}\rangle$. Since H contains $\langle x, j\rangle \cong V_4$, n is even. Exactly one of x, j, xj is the central half-turn $r^{n/2}$, and the other two are reflections. If the central mark is j or xj, we obtain $\mathcal{M}_m$ or $\mathcal{B}_m$, respectively. If it is x, membership $j \in G$ gives $\mathcal{A}^0_m$, while $j \notin G$ gives $\mathcal{A}_m$. For $j \in G$, one has $H = G$, so $n = m$ is even. For $j \notin G$, one has $n = 2m$ and $G = \langle r^2, r^k s\rangle$ for some $k \in \{0, 1\}$. The central half-turn r^m of H belongs to G exactly when m is even. Thus m is odd in the first two outside-mark cases and even in $\mathcal{A}_m$, where $x \in G$. This exhausts the four forms and their parameter ranges.

In $\mathcal{A}^0_m$, the mark $j = [Z]$ belongs to the complete group. Its fixed actual representative B_j satisfies $[B_j X] = j$, hence $B_j \doteq Y$. This actual binary and the already available literal I give the signature-preserving edge switch used in Theorem 6.30; the distinguished equality is never replaced or re-anchored.

The marked eigenline equations give the displayed matrices, with

$$ZC(t)Z = tC(t^{-1}), \qquad x = [XC(1)],$$

while $x = [A(-1)]$ is central in both $\mathcal{A}_m$ and $\mathcal{A}^0_m$; hence the four marked configurations are legally distinct.

Next suppose that G is cyclic. By Lemma 2.2, its unique involution is x. A class commuting with $[X]$ lies in $\operatorname{span}\{I, X\}$ or $\operatorname{span}\{Z, Y\}$, and every nonsingular member of the latter is another involution. Thus all transfer classes have representatives in $\operatorname{span}\{I, X\}$. Writing $G = C_N$, it follows that N is even and $j \notin G$. Conjugation by j inverts the eigenvalue ratio of every class in this algebra, so

$$H = G \rtimes \langle j\rangle \cong D_{2N}.$$

Consequently, x is the central half-turn and j, xj are the two marked reflections.

It remains to consider $G \cong V_4$, corresponding to $m = 2$ in the dihedral notation. Here

$$\mathcal{M}_2 = \mathcal{B}_2 = \{[I], [X], [Z], [Y]\}.$$

For $\eta = i$,

$$\frac{ZA(i)}{1 - i} = Y + iZ, \qquad \frac{ZA(-i)}{1 + i} = Y - iZ,$$

which is $\mathcal{V}_{\mathrm{ex}}$. Conversely, every $V_4 \ni x$ has the form

$$\{[I], x, [aZ + bY], [bZ + aY]\}, \qquad a^2 \neq b^2,$$

and $jGj = G$ gives either $ab = 0$ (the Pauli form) or $a^2 = -b^2$ (the exotic form $\mathcal{V}_{\mathrm{ex}}$).

Thus the Pauli form has $H = G$, with x, j, xj its three nonidentity elements. In the exotic form, $j, xj \notin G$ and $H = \langle G, j\rangle \cong D_8$; relative to this H, x is the central half-turn and j, xj are conjugate reflections in the outside coset.

It remains to route the Platonic groups while retaining the ordered marked pair. Let $G \cong A_4$. If $j \in G$, then $H = G$, and the three nonidentity elements x, j, xj of their marked V_4 are precisely the double transpositions. This is $\mathcal{T}$. If $j \notin G$, then $|H| = 24$. By Proposition 2.1 and Lemma 2.2, the finite-PGL_2 classification and the subgroup structure of cyclic and dihedral groups force $H \cong S_4$ with $G = A_4 \triangleleft H$. Thus $x \in G$ is a double transposition, whereas the odd involutions $j, xj \in H \setminus G$ are transpositions. Since they are distinct and commute, they are disjoint. This is $\mathcal{T}_{\mathrm{ext}}$.

Now let $G \cong S_4$. First note that $j \in G$: otherwise $H = \langle G, j\rangle$ would be a finite group of order 48. By Proposition 2.1 and Lemma 2.2, a finite projective group of order 48 is cyclic or dihedral (here D_{48} denotes the order-48 dihedral group). Thus $H \cong C_{48}$ or D_{48}; every subgroup of a cyclic or dihedral group is cyclic or dihedral, whereas S_4 is neither. Hence $j \in G$.

The two commuting involutions may independently lie in the transposition or double-transposition class, subject to the class of their product. The four resulting ordered class patterns are

$$(dd; d), \qquad (tt; d), \qquad (dt; t), \qquad (td; t),$$

where the entries record the classes of $(x, j; xj)$. Conjugation in S_4 is transitive on ordered pairs in each pattern by Lemma 2.2. They are therefore exactly the four fixed-equality marked types

$$\mathcal{O}, \qquad \mathcal{O}', \qquad \mathcal{O}_{dt}, \qquad \mathcal{O}_{td}.$$

Their ordered marks distinguish the four types in the fixed coordinates. Concrete representatives and direct finite exits are recorded in Section 9 and Appendix A.5.

For $G \cong A_5$, necessarily $j \in G$. Otherwise $G \triangleleft H$ and $[H : G] = 2$ would give a finite subgroup $H \leq \mathrm{PGL}_2(\mathbb{C})$ of order 120 containing A_5. By Proposition 2.1 and Lemma 2.2, such an H would be cyclic or dihedral; every subgroup of a cyclic or dihedral group is cyclic or dihedral, a contradiction. Hence $H = G$. Finally, Lemma 2.2 shows that A_5 has one involution class, with x, j, xj in that class. Once an ordered pair (x, j) is fixed, two conjugators differ by

$$C_{\mathrm{PGL}_2(\mathbb{C})}(\langle x, j\rangle) = \langle x, j\rangle.$$

Each of these four projective classes has a representative in $GO(X)$. Hence the abstract marked type is unique. Fixed-equality coordinates give the two displays $\mathcal{I}, \mathcal{I}_-$; Section 9 verifies their explicit representatives and a direct coefficient-and-card dictionary that leaves the supplied literal I untouched. This deferred coordinate calculation is not used to classify the abstract group, so the four-way list is exhaustive. □

Remark 5.15 (Relation with Xia's binary-group taxonomy). Version 1 of Xia's framework [Xia26, Sections 1 and 4] organizes the finite projective binary residue into nine *unmarked* rotation-group types. Here the group consists of transfer classes $[T_B] = [BX]$, rather than projective classes $[B]$ of the underlying binary matrices, and the supplied literal equality fixes the physical marks

$$x = [X] \in G, \qquad j = [Z], \qquad jGj = G.$$

Consequently, our routing refines Xia's taxonomy by the ordered marked pair (x, j). Forgetting these marks gives the following dictionary.

Xia's unmarked type	Corresponding case in this proof
C_1	absent at a continuing node, where a nonidentity transfer is already retained
C_2	the proper-cyclic C_2 quaternary boundary
C_n, $n \geq 3$	the odd no-equality cyclic residue, or fixed-I proper cyclic $C_N \ni [X]$
D_{2m}, m odd	the odd-dihedral residue, or $\mathcal{M}_m, \mathcal{B}_m$
$V_4 = D_4$	the standard Pauli and exotic Klein forms
D_{2m}, even $m \geq 4$	$\mathcal{M}_m, \mathcal{B}_m, \mathcal{A}_m^0, \mathcal{A}_m$
A_4	$\mathcal{T}, \mathcal{T}_{\text{ext}}$
S_4	$\mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}$
A_5	the two fixed-equality coordinate displays

Thus the dihedral, Klein, and Platonic displays above are physical marked refinements, not additional abstract finite-group types. The table is only a notation dictionary: no completeness or hardness result from the cited framework is used. This paper independently proves every marked case, actual-representative statement, and signature-set-wide lift.

Work in the fixed coordinates containing the distinguished literal I. Let $\Lambda \subseteq \Lambda^+$ be a retained factor-saturated augmentation, put

$$G = G(\Lambda), \qquad G^+ = G(\Lambda^+),$$

and suppose that the unsafe-binary dispatch has not already applied. If a newly exposed ordered binary B is safe and $[BX] \notin G$, then Lemma 5.3 gives the strict inclusion $G < G^+$. If $[BX] \in G$, the binary itself is old, but the complete group is still recomputed: the augmented set may realize additional multi-copy binaries not enumerated by the batch that exposed B. Accordingly, G^+ is the complete semantic closure, and need not equal the subgroup generated by G and the first new class. Whenever $G < G^+$, regenerate the deck and reroute before invoking a group-specific theorem.

5.4 Shared methods for the four branch proofs

We use the two-port slices and bilinear Frobenius pairing of Section 2.5.1 to determine what the available contractions reveal about a tensor. The quaternary result at the end of this subsection supplies a common base case for the later branches.

Definition 5.16 (The span $\mathscr{K}_G$)**.** For a retained normalized set with complete group G, let $\mathscr{K}_G$ be the complex span of the transfer matrices $T_B = BX$ of all finite ordered two-port chains B made from the fixed actual representatives B_g, $g \in G$, the explicit I- or native-X links, and their port reversals. This is equivalently the span of independently dressed I/X kernels after the fixed right multiplication by X, which is invertible and therefore only relabels each observable fibre.

The information determined by binary contractions depends on $\mathscr{K}_G$. The following theorem gives its two possible values.

Theorem 5.17 (The span of realizable binary kernels)**.** *Let $\mathcal{G}$ be a retained signature set with normalized I, X, and let $G = G(\mathcal{G})$. With $\mathscr{K}_G$ as in Definition 5.16,*

$$\mathscr{K}_G \in \{\operatorname{span}\{I, X\}, \operatorname{Mat}_2(\mathbb{C})\}.$$

Moreover,

$$\operatorname{span}\{I, X\}^{\perp_F} = \left\{ \begin{pmatrix} u & v \\ -v & -u \end{pmatrix} : u, v \in \mathbb{C} \right\}.$$

If $\mathscr{K}_G = \operatorname{Mat}_2(\mathbb{C})$, four actual kernel evaluations determine each two-port slice. If $\mathscr{K}_G = \operatorname{span}\{I, X\}$, they determine it modulo the displayed orthogonal complement. This reconstruction is an algebraic identity and does not itself realize a gadget.

Proof. Because $B_1 = X$, the span contains I, and the explicit I-link gives X. Series composition, with the second chain taken in its inward (reversed) port order, has physical tensor BXC and satisfies $T_{BXC} = T_B T_C$, so the span is an algebra. Reversal gives $T^{\#} = XT^{\mathsf{T}}X$; since X is in the span, $T^{\mathsf{T}} = XT^{\#}X$ is also in it. Thus $\mathscr{K}_G$ is unital and transpose-stable.

Conjugate by the Walsh matrix $S := 2^{-1/2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, so $SXS = Z$. For A in the conjugated algebra, define its diagonal and off-diagonal parts by $D := (A + ZAZ)/2$ and $O := (A - ZAZ)/2$. If some $O \neq 0$, write $O = \left(\begin{smallmatrix} 0 & b \\ c & 0 \end{smallmatrix}\right)$. Since Z and O lie in the algebra, so does $ZO = \left(\begin{smallmatrix} 0 & b \\ -c & 0 \end{smallmatrix}\right)$. If $bc \neq 0$, $O + ZO = 2bE_{12}$ and $O - ZO = 2cE_{21}$, where E_{12} and E_{21} are the standard 2×2 matrix units; if one entry is zero, transpose-stability supplies the missing unit. Products supply the diagonal units, hence the algebra is $\mathrm{Mat}_2(\mathbb{C})$. If every $O = 0$, it is diagonal and contains I, Z, so it is $\mathrm{span}\{I, Z\}$, which conjugates back to $\mathrm{span}\{I, X\}$.

In the full case, four actual chain matrices can be greedily chosen as a basis; their Frobenius evaluations reconstruct every two-port slice algebraically, not as a gadget. In the proper case, orthogonality to I, X is $a + d = b + c = 0$, giving exactly $\left(\begin{smallmatrix} u & v \\ -v & -u \end{smallmatrix}\right)$. □

For the finite groups routed above, observability identifies the routing split intrinsically:

$$G \text{ is cyclic} \quad \Longleftrightarrow \quad \mathscr{K}_G = \mathrm{span}\{I, X\},$$

$$G \text{ is dihedral, Klein four, or Platonic} \quad \Longleftrightarrow \quad \mathscr{K}_G = \mathrm{Mat}_2(\mathbb{C}).$$

Indeed, the cyclic transfer classes lie in $\mathrm{span}\{I, X\}$, whereas a finite projective subgroup of the units of this proper algebra is cyclic. Thus Section 7 treats $\mathscr{K}_G = \mathrm{span}\{I, X\}$; the other three branches have $\mathscr{K}_G = \mathrm{Mat}_2(\mathbb{C})$.

At arity four, the complete stabilized deck permits the following gadget construction. Its output is an actual nonmatching eight-vertex signature to which Corollary 4.7 applies.

Theorem 5.18 (Quaternary signatures with stable matching decks)**.** *Assume*

$$\mathsf{Stable}(\Lambda, G, q), \qquad \mathrm{arity}(q) = \nu(\Lambda) = 4,$$

and that G is one of

1. $G = \{1, [X]\} \cong C_2$*;*
2. $G = \{[\mathrm{diag}(1, t)], [X\,\mathrm{diag}(1, t)] : t \in \mu_m\}$*, $m \geq 3$;*
3. *the standard Pauli V_4;*
4. *the exotic Klein group*

$$\mathcal{V}_{\mathrm{ex}} = \left\{ [I], [X], \left[\begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}\right], \left[\begin{pmatrix} -i & 1 \\ -1 & i \end{pmatrix}\right] \right\}.$$

Then an actual gadget using at most two copies of q produces an endpoint-nondegenerate eight-vertex signature that is not G-matching.

The quantifiers in this statement include zero cards, both port orientations for all six contracted pairs, every current-group representative assignment on the remaining matched edges, and the fixed representative scalars. By Corollary 4.7, the output gives a resolved hard/tractable leaf or a retained factor batch. The latter is dispatched by Lemma 5.3; only a safe factor outside G passes to the global measure as a strict outer-group successor.

Proof. By Theorem A.8, the asserted directly realized signature r satisfies

$$\text{K-Holant}(\Lambda, r) \leq_{\mathrm{T}} \text{K-Holant}(\Lambda).$$

Apply Corollary 4.7. In items 2–4 and the one-copy case of item 1, invertible dressings preserve tensor primality, so no factor outcome occurs.

In the exceptional C_2 subcase, write $a_S = \widetilde{q}(\mathbf{1}_S)$. Call it *mixed-sign* when $a_\varnothing = a_{[4]} = 0$ and two nonzero complementary weight-two blocks $\{S, \bar{S}\}$ have different ratios $a_{\bar{S}}/a_S \in \{1, -1\}$, as in Theorem A.8(i). Only this mixed-sign C_2 subcase can reach the continuing factor outcome after the two-copy construction. Keep the complete retained factor batch supplied by the reduced interface and dispatch each binary prime separately using Lemma 5.3. A singular or infinite-order prime is a terminal; a safe prime outside the current G gives a strict complete-group successor. If all safe primes were in G, the genuine $2 + 2$ product would be G-matching, contrary to the physical nonmatching certificate. Thus no same-(G, ν) successor is created; every retained factor remains Turing-exposed over the predecessor, and only the resulting outer successor is passed back to stabilization. □

Every bounded enumeration appears as an exact proposition in Appendix A; the main text supplies its reduction direction and signature-set-wide lift. The coverage map `certificates/theorem-to-script-map.md` records the manuscript anchors and finite claims checked by each entry point, and `certificates/README.md` gives the replay instructions and the package-wide computational scope. The proper- and full-kernel branches meet only at the final algebraic split.

With the stable-state program, four-way routing, and common methods in place, we begin with the marked dihedral branch.

6 Marked Dihedral Matching Decks

We prove rigidity for the four marked dihedral groups $\mathcal{M}_m, \mathcal{B}_m, \mathcal{A}_m^0, \mathcal{A}_m$ from Theorem 5.14. The input is a stable state: its proper contractions are already zero or products of available binaries. The task is to show that a tensor-prime core of arity at least six cannot have this property. At arity four, the sector-specific constructions below produce an actual eight-vertex signature and invoke its dichotomy.

For $\mathcal{M}_m$, monomial binaries turn contractions into weighted parity constraints. Comparing two such contractions reduces a disagreement to six ports; their common factors can then be recovered by linear inversion. We use the projective tensor geometry of Section 2.11.1 to identify these common factors. The four-port support classification in Theorem A.8 uses the exact certificate described there; the six-port classification in Proposition A.9 has a complete 48-case table and a normalization proof. Whole-network equivalences handle $\mathcal{B}_m$ and $\mathcal{A}_m^0$, while explicit weighted kernels handle $\mathcal{A}_m$.

For $m \geq 3$, define the diagonal and anti-diagonal matrix families

$$\mathsf{R}(t) = \mathrm{diag}(1, t), \qquad \mathsf{S}(t) = X\mathsf{R}(t), \qquad t \in \mu_m.$$

The odd-arity case is dispatched by Theorem 2.15. We also establish support and coefficient reconstruction for later occurrences: four slices determine the support, a minor proportional to $1 - t^2$ excludes the Reed–Muller alternative, and two-copy contractions test whether the remaining coefficients factor.

Theorem 6.1 (Marked-dihedral deck rigidity)**.** *Assume*

$$\mathsf{Stable}(\Lambda, G, f), \qquad \mathsf{Even}(\Lambda),$$

and suppose that the complete marked group is one of

$$G \in \{\mathcal{M}_m : m \geq 3\} \cup \{\mathcal{B}_m : m \geq 3\} \cup \{\mathcal{A}_m^0 : m \geq 4,\ 2 \mid m\} \cup \{\mathcal{A}_m : m \geq 4,\ 2 \mid m\}.$$

Then $\mathsf{Close}_{\{\mathscr{P}\}}(\Lambda, G, f)$.

The four cases are proved in Theorems 6.25, 6.29, 6.30 and 6.34; the final proof combines them with exactly the parameter ranges displayed above.

Actual-card and sector-pencil setup. Let the core arity be $n := \mathrm{arity}(f)$. We use the conventions of Section 2.5 and Definitions 5.9 and 5.12 throughout.

If $n = 4$, first regenerate and factor-saturate the complete physical deck of f, apply Theorem 5.18, and route its actual endpoint-nondegenerate output through Corollary 4.7. The retained factor batch is closed by Lemmas 2.20 and 5.3; it gives a resolved leaf or a strict stable successor. Thus the arity-four base is resolved, and throughout the remaining standard-sector proof we may assume

$$n = 2k + 2 \geq 6.$$

For every $t \in \mu_m$, the representative convention of Section 2.5 fixes actual ordered binary outputs

$$B_t^{\mathsf{R}} = \sigma_t^{\mathsf{R}} \mathsf{R}(t) X, \qquad B_t^{\mathsf{S}} = \sigma_t^{\mathsf{S}} \mathsf{S}(t) X, \qquad \sigma_t^{\mathsf{R}}, \sigma_t^{\mathsf{S}} \neq 0.$$

For a deleted set $p = \{i, j\}$ equipped with the port order (i, j), define the four analytic slices

$$A := f(00, \cdot), \quad B := f(01, \cdot), \quad C := f(10, \cdot), \quad D := f(11, \cdot),$$

and define the associated $(n-2)$-ary *sector pencils* by

$$E(t) := A + tD, \qquad O(t) := C + tB. \tag{39}$$

Here $E(t)$ is the equal/even sector associated with the deleted-pair values $00, 11$, while $O(t)$ is the unequal/odd sector associated with $10, 01$. A pencil means a one-parameter linear family $U + tV$. The corresponding kernel spaces

$$\mathrm{span}\{E_{00}, E_{11}\} \quad \text{and} \quad \mathrm{span}\{E_{01}, E_{10}\}$$

are called the even and odd *sector planes*; their displayed matrix-unit axes are the *monomial parity axes*. For $t \in \mu_m$, the native X-wires give the exact actual outputs $\sigma_t^{\mathsf{R}} O(t)$ and $\sigma_t^{\mathsf{S}} E(t)$, since

$$X B_t^{\mathsf{R}} X = \sigma_t^{\mathsf{R}} X \mathsf{R}(t), \qquad X B_t^{\mathsf{S}} X = \sigma_t^{\mathsf{S}} \mathsf{R}(t).$$

Thus $t \in \mu_m$ gives an actual card with a known nonzero scalar. We use A, B, C, D and evaluations outside μ_m only to reconstruct tensor identities. Later $K_E(t), K_O(t) \in \mathrm{Mat}_2(\mathbb{C})$ denote the single varying binary factor of $E(t), O(t)$, respectively.

Write a nonsingular monomial binary, up to scalar, as

$$M_\alpha(a)(x, y) = \mathbf{1}[x \oplus y = \alpha] a^x, \qquad \alpha \in \mathbb{F}_2, \quad a \neq 0.$$

Here α is its support parity; reversal changes only the weight and a nonzero scalar.

6.1 Trace covers and alternating-cycle localization

Consider two nonzero sampled cards of the same core. Their matchings and binary factor lines can disagree. We will contract common parts while preserving a disagreement, leaving only four or six ports to analyze.

For two perfect matchings M_0, M_1, an *alternating M_0/M_1-cycle* is an even cycle in $M_0 \triangle M_1$ whose edges alternate between the two matchings. We call a sequence of the contractions below a *trace cover*; it closes common edges and shortens alternating cycles while retaining a chosen discrepancy. Its *skeleton* records the contraction multigraph, edge orientations, and external-port order.

Definition 6.2 (Incompatible matching products)**.** Normalize the actual sampled outputs by their recorded nonzero scalars. Two nonzero cards *disagree* if their residual matchings differ, or if their ordered projective binary factor lines differ on at least two edges of the same matching. A parity change counts as a factor-line change. Variation on one designated edge, including its parity, is allowed; a single changed factor is not by itself an obstruction for the ancestor.

Here a *one-factor Segre ruling* is the projective family

$$[B_1 \otimes \cdots \otimes B_{r-1} \otimes K \otimes B_{r+1} \otimes \cdots \otimes B_s]$$

in which the matching and all factors except the displayed binary factor K are fixed. Thus two simple matching tensors lie on this ruling exactly when they differ projectively in at most that one factor. We use the simple/decomposable-tensor and Segre-line conventions recorded in Section 2.11.1.

Consider two actual nonzero sector cards

$$P_0 = \bigotimes_{e \in M_0} B_{0,e}, \qquad P_1 = \bigotimes_{e \in M_1} B_{1,e}, \tag{40}$$

where every $B_{\epsilon,e}$ is a nonsingular monomial binary from the normalized group.

An aligned common edge is *untagged* when its two ordered factors agree projectively. To *cross-close* two residual edges means to connect one endpoint of the first to one endpoint of the second by an actual ordered group binary, so that the two edges fuse in each product. A determinant or parity discrepancy is *protected* when it is designated to survive these closures; the proof below records that it is multiplied only by nonzero trace, orientation, and fusion scalars.

Definition 6.3 (Parity and coefficient discrepancies)**.** After aligning the common edges of two sector products, a residual matched factor is a *parity discrepancy* if its support parity changes between the two products. It is a *coefficient discrepancy* if the parities agree but the two ordered nonzero-entry vectors are not projectively proportional; in that case the associated nonzero 2×2 determinant is part of the tag.

Lemma 6.4 (Compatible closure and cycle shortening)**.** *Common untagged factors in Equation* (40) *can be closed by actual group binaries while keeping both products nonzero. An alternating M_0/M_1-cycle can be shortened by two ports without cancellation, and a protected coefficient determinant is only multiplied by a nonzero scalar. Two discrepancies can be cross-closed by an actual connection while preserving any distinguished remaining discrepancy.*

Proof. For ordered monomial factors one has, up to the fixed orientation scalar,

$$\sum_{x,y} M_\alpha(a)(x,y) M_\alpha(t)(x,y) = 1 + at,$$

$$\sum_{z} M_\alpha(a)(u,z) M_\beta(b)(z,v) = b^\alpha M_{\alpha+\beta}\big(ab^{1-2\alpha}\big)(u,v).$$

Thus on a common edge the two product traces are nonzero affine functions of $t \in \mu_m$; together they exclude at most two phases. Since $m \geq 3$, one actual phase keeps both nonzero. Closing an edge then splices the two incident edges in the other matching. The displayed fusion has a unique internal lift, so there is no cancellation, and every protected minor is multiplied by nonzero trace and fusion factors.

For a tagged ordered edge $q = (u_q, v_q)$, let $\alpha_{\xi,q} \in \mathbb{F}_2$ denote its support parity in product $\xi \in \mathbb{F}_2$. Put

$$\delta_q := \alpha_{0,q} + \alpha_{1,q} \in \mathbb{F}_2 \qquad (q \in \{r, s\}), \qquad \alpha_{\xi,q} = \alpha_{0,q} + \xi\delta_q.$$

One ordered connection $M_\gamma(t)$ fuses the two factors in each product. Applying the displayed fusion identity twice gives

$$M_\alpha(a)M_\gamma(t)M_\beta(b) = t^\alpha b^{\alpha\oplus\gamma} M_{\alpha\oplus\gamma\oplus\beta}\Big(a\, t^{1-2\alpha} b^{1-2(\alpha\oplus\gamma)}\Big).$$

For each supported exposed word the internal lift is unique, and the fusion scalar is nonzero. The fused parity difference is $\delta_r + \delta_s$. A mixed pair therefore leaves a parity discrepancy; equal types leave at most one coefficient discrepancy, which may also cancel. A distinguished discrepancy outside the two spliced edges is preserved: its determinant is multiplied only by the nonzero orientation and fusion scalars of the two products. Each connection removes two ports, whether or not the two spliced discrepancies cancel. All operations use only the native X-wire, the explicit normalized I-link, and fixed actual group representatives. □

Lemma 6.5 (Coefficient-line quaternary or genuine factor)**.** *For distinct $t_0, t_1 \in \mu_m$, suppose the four nonzero actual cards represented by $E(t_0), E(t_1), O(t_0), O(t_1)$ at a deleted pair p have a common residual matching edge e, with the same support parity. If all four ordered factors on e lie on one projective line, the ancestor has the exact tensor factorization*

$$f = B_e \otimes h$$

with a nonsingular monomial binary B_e. Otherwise choose two cards whose ordered factors on e are nonproportional. If their other matching edges and support parities agree, an actual closure retaining $p \cup e$ has a nonzero 2×2 flattening minor. Without this parity compatibility no simultaneous trace is asserted. The factor identity is analytic; its factors are exposed only by factor saturation.

Proof. For the nonproportional pair with the stated compatibility, close every other common matching edge successively. At each step the two traces exclude at most two roots, so Lemma 6.4 keeps both products nonzero. The two surviving e-vectors remain nonproportional. They are contractions of the resulting actual quaternary by two kernels on p; hence that quaternary has a nonzero flattening minor. The two kernels are linearly independent: two distinct phases are independent inside one sector plane, while an even and an odd kernel have different support parities. Thus nonproportional retained e-vectors really give flattening rank at least two.

For the factor alternative all four ordered e-factors lie on one line. Choose its representative

$$B_e(x_u, x_v) = \mathbf{1}[x_u \oplus x_v = \alpha] b_{x_u}, \qquad b_0 b_1 \neq 0,$$

and absorb each card's nonzero scalar into its complementary tensor. The two-card inversions

$$A = \frac{t_1 E(t_0) - t_0 E(t_1)}{t_1 - t_0}, \quad D = \frac{E(t_1) - E(t_0)}{t_1 - t_0}, \qquad C = \frac{t_1 O(t_0) - t_0 O(t_1)}{t_1 - t_0}, \quad B = \frac{O(t_1) - O(t_0)}{t_1 - t_0}$$

put every slice in $\mathbb{C}B_e \otimes V$, where V is the tensor space on the residual ports outside e. Reassembling the four slices gives the displayed tensor identity. □

Lemma 6.6 (Two-card localization to six ports)**.** *Fix the deleted pair p, and let two nonzero actual sampled p-cards of the same core disagree. There is an actual one-copy contraction h, retaining p and four further ports, whose two inherited p-cards remain nonzero and either have different residual matchings or differ on both edges of their common residual matching. The contraction closes only ports outside p and uses $O(\mathrm{arity}(f))$ fixed actual group representatives. In particular, h is not a three-factor current-group matching product.*

Proof. Only the selected two cards are preserved in this construction. The contraction may depend on that pair of cards; no common phase choice for four or more cards is required.

First suppose the two residual matchings differ. Choose one alternating cycle as distinguished. On every other alternating cycle, close an edge of the first matching by a kernel of the same support parity. The trace in the first product excludes at most one phase, whereas the other product has a unique monomial lift. Thus both products remain nonzero. Repeat until that nondistinguished cycle becomes one common edge.

Every common edge outside the distinguished cycle can now be absorbed into the cycle without any parity compatibility assumption. Write that edge as $\{u, w\}$, select a cycle vertex v, and cross-close u, v by any actual monomial group kernel. If the two cycle neighbours of v are v_0, v_1, the new matching edges are respectively $\{w, v_0\}$ and $\{w, v_1\}$. Thus the alternating cycle is unchanged except that w replaces v. Both internal lifts are unique and have nonzero weights, including when the old common edge had different support parities in the two products. Absorb every remaining common edge in this way. Finally shorten the distinguished cycle, using the same trace-and-fusion operation, until it is a four-cycle. Stop before closing an edge of that four-cycle. The two residual four-port matchings are therefore still different.

Now suppose the residual matching agrees and at least two factor lines differ. Protect two such edges. Every unprotected edge whose two support parities agree can be traced directly: the two traces exclude at most two phases, so $m \geq 3$ leaves an actual phase keeping both products nonzero. The two protected edges are untouched by these traces.

An unprotected edge with different support parities is instead cross-closed with one protected edge. The other protected edge remains untouched. If the chosen protected edge has equal parities but different coefficient lines, the fused edge has different support parities and remains a tag. If both fused edges have different support parities, the new support parities agree, but the ratio of their normalized fused weights is

$$\kappa t^{\pm 2}, \qquad \kappa \neq 0.$$

This follows directly from the three-factor fusion identity in Lemma 6.4. At most two phases make this ratio equal to one. Choose another actual phase; the fused edge is then a coefficient tag. In both cases the operation reduces the number of edges by one and leaves two protected, distinct factor lines. Continue until exactly these two edges remain.

Perform the selected contractions on the original core, leaving the ports of p open. Tensor contraction commutes with either of the two original p-kernels, so its inherited cards are precisely the two nonzero products just constructed, with their recorded nonzero scalars. Each operation removes two ports, giving the stated size bound.

If h were a product on three matching edges, deleting p would either trace out one of its matching edges, leaving the other two factors fixed, or fuse its two incident matching edges, leaving the third factor fixed. Whenever the two resulting cards are nonzero, their residual matching is therefore the same and at most the fused factor can vary. This contradicts either of the two constructed configurations. □

Lemma 6.7 (Reduction to four ports or a tensor factor)**.** *Every six-port residue in Lemma 6.6 has a complete reduction through its proper factor and physical deck batches. It cannot close as a current-group matching product. A quaternary child is processed by its own complete factor and deck batch; a safe crossed factorization completes that child only.*

Proof. Let h be the actual six-port output, with its inherited nonzero cards, port order, realization scalar, and originating frozen token retained. By Lemma 6.6, h is not a three-factor matching product. Adjoin h, jointly factor-saturate, and close every proper factor and every proper oriented deck card. Odd or unary factors and singular, unsafe, infinite-order, or outside-group binaries follow their established exits. Any proper nonbinary prime has arity at most four and is processed by the lower-arity induction, including the arity-four physical-deck boundary stated in the setup. Thus this processing does not use the six-port case being proved.

If no resolved leaf or strict successor occurs, all surviving proper factors are current-group matching products. A proper factorization of h would therefore make h itself a matching product, a contradiction. Otherwise h is tensor-prime and every nonzero proper pair card is a current-group matching product. The independent six-port argument in Lemma 6.11 excludes precisely this case. That argument uses only the independent pencil lemmas and the Boolean support atlas, not this consumer or a six-port recursive invocation.

For an actual quaternary child, first close its complete factor batch. If it factors into safe current-group binaries, this completes the child key. Otherwise use the established binary exits or retain its prime quaternary occurrence, regenerate the complete oriented deck, and apply Theorem 5.18 and Corollary 4.7. Every returned factor batch is closed by Lemmas 2.20 and 5.3. An exact factorization of this child is never used to infer a factorization of its ancestor.

In particular, for an active stable core of arity at least eight there is an immediate contradiction: the first proper pair closure outside p is zero or a matching product, and further monomial contractions preserve that property. Yet localization keeps the two disagreeing inherited cards nonzero. The same argument applies after the proper lower-arity batches of a later occurrence have been closed. □

Apply this argument separately to each pair of nonzero samples. The closures need preserve only that pair, and may differ between comparisons. On the continuing branch every two samples at a fixed deleted pair have one residual matching and differ on at most one factor. This pairwise conclusion gives a common varying edge for the entire sample family: if S, T differ on edge e, a third sample differing from S on another edge would differ from T on two edges. If every sample is proportional, designate any edge. Hence all other factor lines agree across both sectors. Two nonzero phase samples in each nonzero sector (which exist because $m \geq 3$) then reconstruct its endpoint slices with these common factors. An identically zero sector has zero slices. Thus any ancestor factor used below follows from an explicit four-slice identity, independently of a safe factorization of a single child.

Lemma 6.8 (Sector-ruling rigidity)**.** *Let $U + tV$, $t \in \mu_m$, $m \geq 3$, be a tensor pencil on an even number of ports whose samples are zero or current $\mathcal{M}_m$-matching products. Its nonzero samples have one fixed matching and all but at most one binary factor line fixed projectively. In particular this holds for each of $E(t)$ and $O(t)$.*

Proof. We use only the sampled matching hypothesis. A proportional pencil has one projective value. Otherwise its m samples are nonzero and projectively distinct. Write $N = 2^s$ for the support size of an s-edge matching product. Two different matching supports S_0, S_1 are affine cosets of the same dimension, so $|S_0 \cap S_1| \leq N/2$. Every nontrivial linear combination of the corresponding

tensors has at least $2N - 2|S_0 \cap S_1| \geq N$ supported words outside the intersection. For a third sample to have exactly N words requires equality and cancellation on the whole intersection. A single common word fixes that linear combination projectively. There is therefore at most one such third matching point, contradicting $m \geq 4$.

For $m = 3$, every ratio of two nonzero coefficients within a matching sample lies in μ_3. Use a word of $S_0 \cap S_1$ to express the required cancellation, and choose one word in each of $S_0 \setminus S_1$ and $S_1 \setminus S_0$. Their coefficient ratio in the third sample lies in $-\mu_3$, disjoint from μ_3. This is again impossible. Thus all sample supports coincide.

The weight-two words of the common support direction determine its matching uniquely, including the parity of each support coset. In that fixed tensor-product presentation every 2×2 Segre flattening minor is a polynomial in t of degree at most two. It vanishes at all $m \geq 3$ sampled phases and hence identically. The whole projective line consists of simple tensors. The Segre-line criterion in Section 2.11.1 therefore permits variation in only one factor, proving the assertion. □

6.2 Endpoint directions and the parallelogram test

By the independent Lemma 6.8, each sector has one fixed residual matching and at most one varying factor line. This subsection classifies those binary factor pencils, discharges every zero or proportional endpoint case, and isolates the complementary-axis parallelogram survivor.

The Möbius geometry of Section 2.11.2 controls the ratio of the two nonzero entries in a binary factor. That ratio is a fractional-linear function of the contraction phase. Requiring it to remain in the same finite phase group forces the two endpoint matrices onto complementary coordinate axes.

Lemma 6.9 (Möbius stabilizer of a root polygon)**.** *Let $m \geq 3$, and let the Möbius map*

$$\phi : \widehat{\mathbb{C}} \longrightarrow \widehat{\mathbb{C}}, \qquad \phi : z \longmapsto \frac{c + dz}{a + bz}, \qquad ad - bc \neq 0,$$

have no pole on μ_m and permute μ_m. Then

$$\phi(z) = \alpha z \qquad or \qquad \phi(z) = \alpha / z, \qquad \alpha \in \mu_m.$$

Proof. A Möbius map sends generalized circles to generalized circles. Three distinct roots and their images force ϕ to preserve the unit circle. Its action on the regular m-gon preserves or reverses cyclic order, so it is a rotation or a rotation followed by reflection. The corresponding Möbius maps agree with ϕ on at least three points and hence agree identically. □

The even and odd sector planes introduced above are precisely the two monomial parity planes in $\mathrm{Mat}_2(\mathbb{C})$:

$$\mathrm{span}\{E_{00}, E_{11}\} \quad \text{and} \quad \mathrm{span}\{E_{01}, E_{10}\}.$$

Their matrix-unit axes fix the following convention. For a nonsingular monomial matrix, its *ordered nonzero-entry ratio* is the coefficient of the second displayed axis divided by that of the first, in the fixed port orientation.

Lemma 6.10 (Classification of binary pencils)**.** *Let $A, B \in \mathrm{Mat}_2(\mathbb{C})$ be binary signatures, and define the binary matrix pencil*

$$K(t) := A + tB, \qquad t \in \mu_m, \qquad m \geq 3.$$

Suppose every sampled binary signature $K(t)$ is zero or a nonsingular monomial binary whose ordered nonzero-entry ratio lies in μ_m. Then either A, B are proportional, or there are a fixed parity plane with coordinate axes E_0, E_1, $\lambda \neq 0$, and $\alpha \in \mu_m$ such that

$$K(t) = \lambda(E_0 + \alpha t E_1) \qquad or \qquad K(t) = \lambda(tE_0 + \alpha E_1).$$

Here $K(1) = A + B$ is a sampled value, whereas $A = K(0)$ is an analytic endpoint because $0 \notin \mu_m$. In the application below, $K(t)$ is the one varying binary factor of a higher-arity actual matching card; it is not thereby an independently realized binary gadget.

Proof. In the nonproportional case no sample is zero and the projective samples are distinct. Two of the at least three samples lie in one of the two monomial parity planes; they span the pencil, so all samples lie in that plane. In these axes, write the matrix pencil with coefficient scalars a, b, c, d and projective coordinate ratio ϕ as

$$K(t) = (a + tb)E_0 + (c + td)E_1, \qquad \phi(t) = \frac{c + td}{a + tb}.$$

Nonproportionality gives $ad - bc \neq 0$, and nonsingularity excludes zeros and poles on μ_m. Thus ϕ permutes μ_m. By Lemma 6.9, it equals αt or α/t. Cross-multiplication gives a degree-two identity valid on at least three points. Its coefficients yield respectively $b = c = 0, d = \alpha a$, or $a = d = 0, c = \alpha b$, which are the two displayed forms. A zero sample already makes the endpoints proportional. □

Use the following names for the three binary-pencil outcomes of Lemma 6.10:

Z	:	identically zero,
P	:	nonzero proportional,
C	:	nonproportional complementary-axis.

For a deleted pair p, factor the even-sector card into the common matching factor P_E and endpoint matrix pencil $K_E(t)$, after ordering the $2k = \mathrm{arity}(f) - 2$ residual ports of a nonconstant even ruling:

$$E(t) = P_E \otimes K_E(t), \qquad K_E(t) := K_{E,0} + tK_{E,1}, \qquad K_E(t), K_{E,0}, K_{E,1} \in \mathrm{Mat}_2(\mathbb{C}),$$

with $k - 1$ fixed factors; define $O(t)$ and its binary factor pencil $K_O(t)$ analogously. To obtain one tensor identity with a fixed scalar, take two distinct nonzero samples, use their linear inversion to reconstruct the two analytic endpoint tensors, normalize the common factor P_E by a fixed linear functional $\eta_E(P_E) = 1$, and define

$$K_E(t) = (\eta_E \otimes \mathrm{id})(E(t)).$$

The normalization fixes the common factor once for the whole pencil. The whole sector inherits the type of this exact residual binary pencil; an identically zero whole pencil is type Z, and a nonzero proportional whole pencil is type P before any varying factor is selected. In a C/C branch the variable endpoints are complementary rank-one matrices.

Lemma 6.11 (Direct six-port base case)**.** *Let $h \neq 0$ be a retained six-port signature in the standard $\mathcal{M}_m$ sector, where $m \geq 3$. Suppose every actual oriented pair card of h, in both kernel parities and at every phase in μ_m, is zero or a product of two nonsingular current-group monomial binaries. Then, after a port permutation, h has an exact nontrivial factorization of type $1 + 5$ or $2 + 4$. In particular, h is not tensor-prime.*

Proof. Apply Lemmas 6.8 and 6.10 at each deleted pair p. A type-P sector is a scalar pencil times one fixed nonsingular matching product on the four residual ports. A type-C sector is one fixed nonsingular binary factor tensored with a complementary-axis pencil; its two endpoint supports are complementary singletons on the varying edge. A type-Z sector vanishes identically.

If a pair has type Z/P, then $h = U_p \otimes P$ for a nonzero binary U_p and a nonzero residual matching product P. If it has type Z/C, the fixed binary factor of the C ruling factors from h. The reversed cases are identical, and Z/Z would give $h = 0$. Each remaining case containing Z therefore proves the assertion.

Suppose a pair has type P/P. Write

$$h = U_p \otimes P_0 + V_p \otimes P_1,$$

where $U_p, V_p \neq 0$ are supported on the two opposite parities of p, and P_0, P_1 are nonsingular matching products on the four residual ports. Choose a residual pair e belonging to neither matching. Such a pair exists because the two matchings contain at most four of the six residual pairs. Contract e by either actual parity kernel at phase 1. This crosses the two factors of each P_j. For every supported p-word, the output has precisely two complementary residual words, each with one internal lift and a nonzero coefficient. Different p-words cannot cancel. The resulting actual quaternary therefore has

$$2(|\operatorname{supp} U_p| + |\operatorname{supp} V_p|)$$

nonzero entries. Six or eight entries contradict the card hypothesis. Otherwise both supports are singletons. Their opposite-parity p-words differ in exactly one bit, so the other physical port of p is fixed. Its point mass is a unary tensor factor of h.

For type P/C, choose a residual pair e belonging to neither the P matching nor the fixed/varying matching of the C ruling. Apply the same actual parity contraction. Each supported P-sector p-word leaves two residual words. Each of the two C-sector p-words leaves exactly one: the singleton varying endpoint and the kernel parity force the bit on the fixed factor, and then force both remaining residual bits. These lifts are unique and all coefficients are nonzero. If the P sector has two p-words, the card has six nonzero entries. If it has one, the card has four nonzero entries but its projection onto p has three words. A product of two nonsingular monomial binaries projects onto any two ports in exactly two words when those ports are matched, and in four words otherwise. Both alternatives contradict the card hypothesis. Type C/P is identical.

Unless one of the stated factorizations has already occurred, every deleted pair must consequently have type C/C. Fix one such pair. Its fixed binary factors and complementary-axis endpoints give exactly the eight-word four-flat support of Equation (162). The independent support atlas Proposition A.9 either gives an injective I/X-contraction with six or eight projected words, or identifies a three-edge matching coset. In the first case, injectivity gives one nonzero summand per output word, so this actual quaternary contradicts the card hypothesis. In the second case, one parity sector at any edge of that matching is identically zero, contradicting the conclusion that every deleted pair is C/C. This completes the proof.

All contractions in this proof are applied directly to h. The factors are exact tensor factors, and are exposed only through Theorem 2.16 and Lemma 2.20 when the reduction protocol uses them. □

Together with two-card localization, Lemma 6.11 proves the higher-arity part of Theorem 6.25, as shown in its proof. The reconstruction results below describe the support and coefficient constraints separately.

For a fixed deleted pair $p = \{i, j\}$, a *residual matching* of a nonzero card

$$P = \bigotimes_{\{u,v\} \in \mathcal{M}_p} B_{uv}$$

is the perfect matching $\mathcal{M}_p$ of the residual ports $[n] \setminus p$. Its matching-indicator direction is

$$M(\mathcal{M}_p) = \operatorname{span}\{e_u + e_v : \{u, v\} \in \mathcal{M}_p\}.$$

The even and odd families are *compatible at* p when their residual matchings agree as unordered edge sets, equivalently when their matching-indicator directions agree. This concerns port positions only; support-parity and ordered coefficient-line disagreements remain separate discrepancies. For example, on residual ports $3, 4, 5, 6$, the pairings $\{\{3, 4\}, \{5, 6\}\}$ in both sectors are compatible, whereas that pairing and $\{\{3, 5\}, \{4, 6\}\}$ form a matching disagreement whose union contains an alternating four-cycle. Matchings belonging to different deleted pairs use different coordinate sets and are not asserted to be literally equal; their global consistency is proved only in the support integration below.

For the following support comparison suppose that both sector pencils have type C; all ordered pairs containing Z or P are handled separately in Lemma 6.15 below.

Compare one actual nonzero even card with one actual nonzero odd card at the fixed deleted pair p. If their residual matchings differ, or their ordered factor lines differ on at least two edges (including parity changes), this is a cross-sector disagreement and is consumed by Lemmas 6.6 and 6.7. Hence the continuing branch has one common residual matching with all nondesignated factor lines aligned.

If the two sector rulings have the same residual matching but designate different varying edges, close only their common fixed factors. Up to affine translations, the remaining support, in the distinguished bits x, y and the two varying-edge bits u, v, is

$$\begin{aligned} x = y &\implies u = x + \alpha, \quad v \text{ free}, \\ x \neq y &\implies v = x + \beta, \quad u \text{ free}. \end{aligned}$$

An actual parity kernel imposing $u = x + \alpha$ has a unique lift and leaves a six-word physical quaternary, hence not a two-monomial product. Its support contains a complementary pair; actual logical X-dressings may move that pair to the two endpoints. Adjoin and factor-saturate this physical quaternary. A retained nonbinary factor has arity at most four: for the active minimum core it is a strict lower-ν successor, and for a later occurrence it is assigned to the lower-arity occurrence induction. Stabilize that quaternary token, regenerate its complete oriented deck, and only then invoke the equality-accessible quaternary boundary and reduced interface. Thus the different-edge branch is closed. We may therefore assume below that both sectors use one common varying edge.

After aligning their other common factors, the four analytic endpoint supports have the form

$$\operatorname{supp} h_\xi = a_\xi + L, \qquad h_{00} = A,\ h_{01} = B,\ h_{10} = C,\ h_{11} = D,$$

where

$$L = \operatorname{span}\{e_{u_r} + e_{v_r} : 1 \leq r < k\}, \qquad \dim L = k - 1.$$

Define the endpoint displacement vectors

$$d_E = a_{00} + a_{11}, \qquad d_O = a_{01} + a_{10}.$$

Then the actual even/odd card directions are

$$L_E = L + \operatorname{span}_{\mathbb{F}_2}\{d_E\}, \qquad L_O = L + \operatorname{span}_{\mathbb{F}_2}\{d_O\}, \qquad d_E, d_O \notin L.$$

The condition we need is

$$a_{00} + a_{01} + a_{10} + a_{11} \in L \tag{41}$$

Equivalently,

$$d_E + d_O \in L, \qquad [a_{00}] + [a_{11}] = [a_{01}] + [a_{10}] \quad \text{in } \mathbb{F}_2^{n-2}/L, \qquad L + \operatorname{span}_{\mathbb{F}_2}\{d_E\} = L + \operatorname{span}_{\mathbb{F}_2}\{d_O\}.$$

This condition is independent of the representatives a_ξ, since changing one representative changes the four-term sum by an element of L.

Lemma 6.12 (Actual-card endpoint test)**.** *In the complementary-axis C/C branch with one common varying edge, the residual matchings of the actual nonzero cards represented by $E(t)$ and $O(s)$ agree if and only if Equation* (41) *holds. Failure is localized using the actual cards, not the endpoint slices.*

Proof. A monomial matching is determined by its weight-two words, so the two matchings agree exactly when the directions agree. Since $d_E, d_O \notin L$,

$$L_E = L_O \quad \Longleftrightarrow \quad d_E + d_O \in L \quad \Longleftrightarrow \quad a_{00} + a_{01} + a_{10} + a_{11} \in L.$$

Failure therefore invokes Lemmas 6.6 and 6.7 on the actual cards and is resolved. □

Lemma 6.13 (Zero and proportional endpoint pencils)**.** *Every endpoint pair in which at least one sector has type Z or P yields at least one of the following concrete outcomes:*

1. *the branch is impossible because it would give $f = 0$;*
2. *all four slices have one common residual tensor, giving an exact binary–residual factorization of f;*
3. *all but one residual matching factor are common, giving an exact factorization $f = P \otimes q$, where q is a nonzero quaternary factor on the deleted pair and the exceptional residual edge;*
4. *actual nonzero cards have a typed disagreement, which is localized and consumed by Lemmas 6.6 and 6.7.*

In item 3 the quaternary has matching-flattening rank two whenever one of the two pencils has type C. The factor q is Turing-exposed through the factor theorem and is a strict lower-ν successor; it is not silently treated as a directly realized gadget.

Proof. For a type-Z even pencil, two-card inversion gives $A = D = 0$, hence the support relation $x_i \oplus x_j = 1$. If the odd pencil also has type Z, then $f = 0$, contrary to the active nonzero prime. If it has type P, the same inversion gives

$$f = G_p \otimes H_O, \qquad G_p = \begin{pmatrix} 0 & b \\ c & 0 \end{pmatrix},$$

where G_p is the displayed binary signature on p and H_O is the complementary $(n-2)$-ary signature. This is an exact ancestor identity, and its factors are exposed only by Theorem 2.16. If the odd pencil has type C, write its exact ruling as $O(t) = P_O \otimes K_O(t)$, where P_O is the product of the $k-1$ fixed residual factors and $K_O(t)$ is the nonproportional binary pencil on its varying edge e. Two-card inversion puts both B and C in $P_O \otimes \mathrm{Mat}_2(\mathbb{C})$; the zero slices A, D lie there as well. Hence

$$f = P_O \otimes q_{p,e}.$$

The complementary-axis endpoints of K_O are nonproportional, so the $p \mid e$ logical flattening of $q_{p,e}$ has rank two. Apply Theorem 2.16 and Lemma 2.20; the Turing-exposed quaternary is a deck-stable lower-ν successor, while every binary factor is sent through Lemma 5.3. Exchanging the sectors covers every type-Z case. A single zero sample in a type-P pencil is included in the exact proportional inversion and is never used by itself to infer a factor.

For a proportional even pencil, inversion gives

$$A = aH_E, \qquad D = dH_E.$$

Choose a nonzero actual even card as the representative of the line $\mathbb{C}H_E$, and compare it with two nonzero actual odd cards. Any matching change or variation on at least two residual factor lines, including parity changes, is a disagreement and is consumed by Lemmas 6.6 and 6.7. If no bounded output occurs, the two sectors have one common residual matching and differ on at most one designated edge e. Let P be the product of all fixed factors outside e. Four-card inversion then puts $A, B, C, D \in P \otimes \mathrm{Mat}_2(\mathbb{C})$, and therefore proves the exact identity

$$f = P \otimes q_{p,e}.$$

If no edge differs, absorb the last common factor into P; this is the stronger identity $f = G_p \otimes H$. If one edge differs, the remaining quaternary is nonzero; in a P/C or C/P case its $p \mid e$ flattening has rank two because the C-endpoints are nonproportional. In the P/P case unequal projective edge lines give the same rank-two conclusion, while equal lines give the stronger binary–residual factorization. Only after this exact identity has been proved are P and $q_{p,e}$ exposed by Theorem 2.16 and Lemma 2.20. This covers P/P, P/C, C/P, and their possible zero samples. □

By Lemma 6.13, it remains to analyze two nonproportional sector pencils.

Put the distinguished pair at $p = \{1, 2\}$ and the residual ports at $R = \{3, 4, 5, 6\}$. For each sector $\epsilon = x_1 \oplus x_2$, let $M_\epsilon = \{e_\epsilon, e'_\epsilon\}$ be its residual matching, where $e_\epsilon = \{u_\epsilon, v_\epsilon\}$ is the fixed edge and $e'_\epsilon = (r_\epsilon, s_\epsilon)$ is the ordered varying edge. There are parameters $a_\epsilon, b_\epsilon, o_\epsilon \in \mathbb{F}_2$ such that its six-port support is parametrized by

$$\begin{aligned} (x_1, x_2) &= (z, z + \epsilon), \\ x_{u_\epsilon} \oplus x_{v_\epsilon} &= a_\epsilon, \qquad\qquad \epsilon, z \in \mathbb{F}_2. \\ (x_{r_\epsilon}, x_{s_\epsilon}) &= (o_\epsilon + z, o_\epsilon + z + b_\epsilon), \end{aligned} \tag{42}$$

For fixed (ϵ, z), the parity equation on the fixed edge has two solutions and the varying edge is determined. Hence each of the four endpoint slices indexed by (ϵ, z) is a two-word affine flat, each sector has four words, and the two sectors have the eight-word support used in Equation (42). The varying-edge assignments for $z = 0, 1$ are bitwise complements. The coefficients on these slices need not be equal.

For a perfect matching M on $2r$ ports and parities $(c_e)_{e \in M}$, the associated *matching coset* is

$$\{x \in \mathbb{F}_2^{2r} : x_u + x_v = c_{\{u,v\}} \text{ for every } \{u, v\} \in M\}.$$

Lemma 6.14 (A six-to-four contraction)**.** *Suppose a normalized-dihedral localization leaves exactly the distinguished pair p and two residual monomial pairs, with support as in Equation* (42)*. Then either the six-port support is a single three-edge matching coset, or an actual I/X-contraction realizes a nonzero quaternary of support size six or eight. In the latter case, the output is not a product of current-group monomial binaries and the protocol does not re-enter a six-port state.*

Proof. The explicit support in Equation (42) is the Boolean family exhaustively covered by Theorem A.11. That interface either identifies the global matching coset or supplies a pair and parity γ with injective projection. Contracting it by the actual parity-γ kernel—I for $\gamma = 0$ and X for $\gamma = 1$—gives one nonzero internal extension per word, so there is no cancellation and the physical output has the certified six- or eight-word support. Adjoin this direct output and close its complete factor batch. A singular/unsafe factor, unary or odd factor, or a smaller prime is dispatched by the stabilized protocol; a safe binary outside the current group is a strict complete-group successor. If the output itself is tensor-prime, retain it as a lower-arity quaternary token, stabilize and regenerate its complete oriented deck, and only then invoke the equality-accessible quaternary boundary and reduced interface. Since a product of two nonsingular current-group monomial binaries has support size four, the support-six/eight output cannot close as an unrecorded matching product. This completes the reduction of the six-port tensor through its four-port output and factor batch. □

Lemma 6.15 (Reduction of the endpoint cases)**.** *For every deleted pair p, all endpoint combinations containing type Z or P are resolved. On the remaining branch, the endpoint pair is C/C, Equation* (41) *holds, and there is one compatible residual matching.*

Proof. The Z/Z entry gives $f = 0$. Every Z/P, Z/C, P/P, P/C, and transposed entry is covered by Lemma 6.13; the proof compares actual sample cards and uses the same four-card inversion as Lemma 6.5, so projective coincidence is promoted to a factor only through an exact tensor identity.

It remains to take both types equal to C. A matching change or two different ordered factor lines invokes the typed trace cover Lemmas 6.6 and 6.7. After closing common untagged factors, two different variable edges give the physical six-word quaternary constructed above; apply the lower-arity stabilized-quaternary route given there. For one common variable edge, failure of Equation (41) is again an actual-card disagreement. If the parallelogram holds, then, because $n \geq 6$, retain the varying edge and one fixed residual edge together with p. This is exactly the six-port tensor of Equation (42). Apply Lemma 6.14. Its nonmatching alternative is a physical quaternary and follows the factor-saturate, stabilize, deck-regenerate, and quaternary-boundary route in the proof of Lemma 6.14; its sole support-level survivor is a three-edge matching coset. These alternatives exhaust C/C. □

6.3 Global support integration and the Reed–Muller alternative

The $Z/P/C$ endpoint classification has resolved every case containing Z or P and every incompatible C/C card family. The remaining survivor has Equation (41) and compatible residual matchings for every deleted pair. This subsection integrates those local supports into a global code, identifies a global matching support, and excludes the sole Reed–Muller alternative by the actual $1 - t^2$ obstruction.

Lemma 6.16 (Endpoint-support saturation)**.** *Assume the normalized-dihedral analysis is on the remaining endpoint- type branch. Then, for each deleted pair $p = \{i, j\}$, there is a fixed-edge matching direction L_p on the residual coordinates and endpoint differences $d_{E,p}, d_{O,p} \notin L_p$ such that the four endpoint slices have full support on affine cosets $a_{ab} + L_p$, with every coefficient on each such coset nonzero. Moreover*

$$M_p := L_p + \operatorname{span}_{\mathbb{F}_2}\{d_{E,p}\} = L_p + \operatorname{span}_{\mathbb{F}_2}\{d_{O,p}\}$$

is the direction of the equal- and unequal-p actual cards; it is the indicator code of a perfect matching on the residual coordinates and hence $M_p = M_p^{\perp}$. This assertion is within the fixed deleted pair p; no literal equality between matchings on two different residual coordinate sets is asserted.

Proof. On the remaining C/C branch, Lemma 6.8 gives one fixed nonzero monomial factor on every residual matching edge except the designated variable edge. Two samples with nonzero output and Equation (41) reconstruct each slice by the four-card inversion used in Lemma 6.5; hence every word in the resulting affine matching coset has a nonzero coefficient. More explicitly, the two complementary-axis endpoint matrices have disjoint monomial supports, so each residual word is supplied by exactly one endpoint; its coefficient is a product of nonzero fixed monomial entries and a nonzero endpoint entry, with no cancellation between endpoints. If a word were absent, the corresponding complementary-axis pencil would have a zero sample or exactly one vanishing endpoint coordinate, producing a singular binary; this case is resolved by Lemmas 6.6, 6.7 and 6.13. A change of matching or of a fixed factor likewise gives the typed trace-cover output. Thus the only unconsumed possibility is the asserted common matching direction. For $d_{E,p} = a_{00} + a_{11}$ and $d_{O,p} = a_{01} + a_{10}$, the parallelogram equation says $d_{E,p} + d_{O,p} \in L_p$, which identifies the two displayed descriptions of M_p and the four offsets. Matchings belonging to different deleted pairs are integrated into one common code by Lemma 6.18, rather than compared directly here. □

We now regard the support as a set of binary words, using the projections and affine operations of Section 2.8.1. The local matching equations will determine a self-dual binary code. A weight-two codeword identifies a global matched pair; if no such word exists, the weight-four words identify the eight-coordinate Reed–Muller code. This reconstructs support only; coefficients are treated afterward.

Lemma 6.17 (Self-dual Steiner reconstruction)**.** *Let $D \le \mathbb{F}_2^n$ be self-dual, where $n > 4$, and suppose that D has no word of weight two. If every three distinct coordinates lie in the support of a unique weight-four word of D, then $n = 8$; after identifying the coordinates with $\mathbb{F}_2^3$, one has $D = RM(1, 3)$.*

Proof. The weight-four supports form a *Steiner quadruple system*: a family of four-element blocks in which every three coordinates belong to exactly one block. Self-orthogonality of D makes two blocks meet evenly; uniqueness through triples then makes the intersection of two distinct blocks either empty or a pair.

Fix a coordinate 0. Define a binary operation $+$ on the coordinates by $x + x = 0$, $x + 0 = x$, and, for distinct nonzero x, y, let $x + y$ be the fourth point in the block through $0, x, y$. This operation is commutative and every element has order two. For the associativity check, define the auxiliary points $u := x + y$, $v := y + z$, and $w := u + z$. The symmetric difference of the blocks $\{0, x, y, u\}$ and $\{0, y, z, v\}$ is the weight-four codeword $\{x, u, z, v\}$. Symmetric difference with $\{0, u, z, w\}$ gives the block $\{0, x, v, w\}$. Uniqueness of the block through $0, x, v$ gives

$$(x + y) + z = w = x + v = x + (y + z).$$

If an auxiliary point coincides—for example $z = x + y$—the same identity reduces immediately to $x + x = 0$; thus the displayed distinct-block calculation and these collision cases cover all inputs. The coordinates form an elementary abelian group $V \cong \mathbb{F}_2^r$, and $n = 2^r$. If $\{a, b, c, d\}$ is any block, symmetric difference with the block through $0, a, b$ shows $d = a + b + c$; hence the blocks are exactly the affine planes of V.

Let B be the binary span of the block indicator vectors. An affine plane is a translate of a two-dimensional subspace of V. Orthogonality to every such plane is equivalent to vanishing of every second additive derivative $\Delta_a \Delta_b F(x)$; hence $B^\perp$ is exactly the space of truth tables of affine-linear functions $F : V \to \mathbb{F}_2$. Thus $\dim B = 2^r - r - 1$. Since $B \subseteq D$ and $\dim D = 2^{r-1}$, the inequality $2^r - r - 1 \le 2^{r-1}$, with $r > 2$, forces $r = 3$; equality gives $D = B = RM(1, 3)$. □

Lemma 6.18 (Support integration)**.** *Assume the remaining endpoint-type survivor, so that Equation* (41) *holds for every deleted pair. Then* $\mathrm{supp}(f) = a + D$ *for a self-dual binary linear code* $D \le \mathbb{F}_2^n$*. Exactly one of the following support alternatives remains:*

1. *D is the matching-indicator direction of one perfect matching on all n ports, so $a + D$ is one* global matching coset*;*
2. *$n = 8$ and, after relabelling the ports, $D = RM(1, 3)$.*

The second alternative is incompatible with the normalized μ_m-dihedral deck for $m \ge 3$. Consequently the sole remaining support is one global matching coset.

Proof. Apply Lemma 6.16. Translation by the 00-slice gives, for any fixed deleted pair p,

$$a + D, \qquad D = (\{00\} \times L_p) + \mathrm{span}\{(01, a_{00} + a_{01}), (10, a_{00} + a_{10})\}.$$

The two displayed generators are independent modulo $\{00\} \times L_p$; hence $\dim D = n/2$. For an arbitrary deleted pair $p = \{i, j\}$, define the equal-coordinate slice $H_p := \{x \in D : x_i = x_j\}$. The endpoint saturation and the parallelogram give

$$\pi_{\{i,j\}} : D \longrightarrow \mathbb{F}_2^2 \text{ surjective}, \qquad \pi_{[n]\setminus\{i,j\}}(H_{\{i,j\}}) = M_{\{i,j\}},$$

where $M_p = M_p^\perp$ is the residual perfect-matching direction from Lemma 6.16. To see self-orthogonality, take arbitrary $x, y \in D$. Among the $n > 4$ coordinate columns $(x_\ell, y_\ell) \in \mathbb{F}_2^2$, two are equal; relabel them i, j. Then $x, y \in H_p$, so their residual projections lie in M_p and have zero mutual dot product. Their contribution on the deleted pair is $x_i y_i + x_j y_j = 0$. Hence $x \cdot y = 0$, proving $D \subseteq D^\perp$. Since $\dim D = n/2$, equality follows: $D = D^\perp$.

Suppose first that $e_u + e_v \in D$. Self-duality implies $x_u = x_v$ for every $x \in D$, so $H_{\{u,v\}} = D$. Put $E = [n] \setminus \{u, v\}$. Endpoint saturation gives

$$\pi_E(D) = M_{\{u,v\}},$$

the indicator direction of the residual perfect matching. For $y \in M_{\{u,v\}}$, choose a lift $x \in D$. Subtracting $x_u(e_u + e_v)$ gives the zero extension $\iota_E(y) \in D$. Hence

$$D = \mathrm{span}_{\mathbb{F}_2}\{e_u + e_v\} \oplus \iota_E(M_{\{u,v\}}),$$

which is exactly the indicator direction of the global perfect matching $\{\{u, v\}\} \cup \mathcal{M}_{\{u,v\}}$. This is a support identity only: it does not yet assert a tensor factorization of the coefficients.

It remains that D has no weight-two word. Fix $i \ne j$ and put $E = [n] \setminus \{i, j\}$. The projection $H_{\{i,j\}} \to \mathbb{F}_2^E$ is injective, because its only possible nonzero kernel is $e_i + e_j$. The functional $x \mapsto x_i + x_j$ is nonzero on D, since otherwise $e_i + e_j \in D^\perp = D$. Thus $\dim H_{\{i,j\}} = n/2 - 1$, and endpoint saturation identifies its image with $M_{\{i,j\}}$. For a third coordinate ℓ, the residual matching pairs ℓ with a unique q. The unique lift of $e_\ell + e_q \in M_{\{i,j\}}$ cannot have 00 on i, j, since that would be a weight-two word. It is therefore $e_i + e_j + e_\ell + e_q \in D$. Hence every three coordinates lie in a unique weight-four support, and Lemma 6.17 gives $n = 8$ and $D = RM(1, 3)$.

To exclude this residue, identify the eight coordinate positions with $\mathbb{F}_2^3$ and, after an affine relabelling, take the deleted pair to be $\{000, 001\}$. On its equal-coordinate slice the residual matching is

$$\{010, 011\}, \qquad \{100, 101\}, \qquad \{110, 111\}.$$

If y_1, y_2, y_3 are the three logical matching bits, the common bit on the deleted pair is

$$b = y_1 + y_2 + y_3.$$

Contract the deleted pair by the actual sector phase t. The projection of the equal slice is injective, so every residual support word has one internal lift and its coefficient is multiplied by t^b. Fixing y_3 and inspecting the 2×2 face indexed by y_1, y_2, the phase exponents are $(0, 1, 1, 0)$, or their complement. Thus the face has one of the raw forms

$$\begin{pmatrix} A & tB \\ tC & D \end{pmatrix}, \qquad \begin{pmatrix} tA & B \\ C & tD \end{pmatrix}, \qquad A, B, C, D \neq 0.$$

The actual $t = 1$ card lies in the fixed Segre ruling, so $AD = BC$. Nonzero row and column scalings therefore give, for some $\varepsilon \in \mathbb{F}_2$, the normalized physical block

$$\begin{aligned} H_t &= \kappa \operatorname{diag}(r_0, r_1) C_\varepsilon(t) \operatorname{diag}(s_0, s_1), \\ C_0(t) &= \begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}, \qquad C_1(t) = \begin{pmatrix} t & 1 \\ 1 & t \end{pmatrix}, \\ \det H_t &= (-1)^\varepsilon \kappa^2 r_0 r_1 s_0 s_1 (1 - t^2). \end{aligned} \tag{43}$$

with all scalar factors nonzero. If an orientation produces t^{-1}, clear its nonzero monomial factor. Since $m \geq 3$, some actual $t \in \mu_m \setminus \{\pm 1\}$ exists, and the displayed nonzero minor contradicts the fixed Segre ruling. □

Hence the support survivor is one global matching coset. A weight-two word above records a support pair only; no tensor factor is inferred until the coefficients have also separated. The following two stages first close every degenerate six-port coefficient table and then prove that the coefficients on the global matching are multiplicative.

6.4 The six-port coefficient calculation

We now return to the matching-supported six-port coefficient cases retained by the support analysis. Trace-cover six-port disagreements have already been reduced to quaternary and resolved; the only survivor is the fixed three-edge-matching coefficient pattern. This subsection first dispatches all degenerate coefficient cases and then analyzes the full-support table.

For a quaternary contraction whose support is contained in two fixed residual matching edges, write its logical coefficient table as

$$Q = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

We include the cases $0 < |\operatorname{supp} Q| < 4$. A *support-matching disagreement* means that its physical support uses a residual pairing different from the fixed pairing; this is the combinatorial disagreement of the trace-cover reduction. By contrast, *not G-matching* means that the signature has no tensor presentation by current-group binary factors. These two notions are not interchangeable.

For each directly realized quaternary, first adjoin it with its port order, scalar, and provenance; jointly factor-saturate it; test all three $2 + 2$ flattenings and dispatch every exposed factor; make an actual endpoint dressing; and invoke Lemma 5.5 only after tensor primality, endpoint nondegeneracy, non-G-matching status, and both crossed rank hypotheses have been verified.

Lemma 6.19 (Six-port coefficient dispatch)**.** *Let h be a direct six-port card whose support is exactly the eight-word coset of a fixed three-edge matching. Before a quotient of its logical coefficients is formed, close every already constructed four- or six-port batch as follows:*

1. *a zero quaternary is converted into an exact factorization of its parent six-port tensor by the proportional-slice identity below;*

2. *a support-matching disagreement is sent to Lemmas 6.6 and 6.7;*

3. *for a quaternary on the two retained matching edges, test Q: if $\det Q = 0$, factor it exactly; if $\det Q \neq 0$, use the quaternary reduction just described;*

4. *jointly factor-saturate every output and dispatch every unary, singular, unsafe, infinite-order, or new safe binary factor;*

5. *if no preceding concrete output applies, retain h only when all eight logical coefficients are nonzero.*

A quaternary with $0 < |\operatorname{supp} Q| < 4$ enters item 3. Every noncurrent factor or nonmatching quaternary outcome gives a resolved leaf or strict successor. If a nonzero child quaternary factors into two current-group binaries, close that child, record its rank-one equation, and continue the finite analysis of the same originating six-port tensor.

Proof. Let u_0, u_1 be the two standard-basis binary signatures supported on the two allowed states of a selected physical matching edge. Write

$$h = u_0 \otimes H_0 + u_1 \otimes H_1.$$

An actual phase kernel has two nonzero coefficients α_t, β_t, and its quaternary contraction is

$$q_t = \alpha_t H_0 + \beta_t H_1, \qquad \alpha_t \beta_t \neq 0.$$

If $q_t = 0$, then

$$\alpha_t H_0 + \beta_t H_1 = 0.$$

If $H_0 = 0$, then $H_1 = 0$, so $h = 0$ and its zero-parent token is completed directly. Otherwise $H_0 \neq 0$ and

$$H_1 = -\frac{\alpha_t}{\beta_t} H_0, \qquad h = \left(u_0 - \frac{\alpha_t}{\beta_t} u_1\right) \otimes H_0.$$

Thus the zero child is not itself declared a terminal: it certifies an exact binary–quaternary factorization of its parent. Close the complete factor batch by Theorem 2.16 and Lemma 2.20. A nonbinary factor gives a quaternary or lower-ν successor; unary, singular, unsafe, or infinite-order factors enter their established reductions; and a new safe binary strictly enlarges the complete group. If all prime factors are already safe current-group binaries, the parent is a current-group matching product and its frozen token is completed. Completing a zero child alone never completes an unresolved multi-copy parent without this factor-batch argument.

Now let $q_t \neq 0$ have logical table Q on the two retained matching edges. If $\det Q = 0$, then Q has rank one and the physical quaternary has an exact $2 + 2$ factorization. When $0 < |\operatorname{supp} Q| < 4$, at least one logical factor has a zero coordinate, so one exposed binary is singular. If both exposed factors are safe current-group binaries, close this child quaternary, retain the equation $\det Q = 0$, and resume the coefficient analysis of the same parent six-port tensor; the child factorization alone does not complete the parent token. If $\det Q \neq 0$, its fixed-matching flattening has rank two. Adjoin the physical card with its port order, scalar, and provenance. If Q is supported on exactly two entries, nonzero determinant forces them to be opposite logical entries. Actual logical X-translations then make the quaternary a nonzero pure generalized equality, and Lemma 3.9 applies. If exactly three entries are nonzero, actual logical flips put a nonzero opposite pair at the two endpoints, so the card is endpoint-nondegenerate. After factor saturation and removal of every crossed rank-one presentation, it is tensor-prime, non-G-matching, and has both crossed ranks at least two. The full-support case is endpoint-nondegenerate after the same endpoint choice. Thus the quaternary

reduction just described is licensed; invoke Lemma 5.5, Theorem 5.18, and Corollary 4.7 only after their hypotheses have been verified. A support-matching disagreement instead enters the typed trace-cover reduction. These routes also process every exposed factor by Lemma 5.3.

For the matching-supported six-port tensor retained from Lemma 6.18, the support is already the complete eight-word global-matching coset, so its eight logical coefficients are nonzero. The case analysis does not pre-enumerate every possible later phase contraction of this full-support table; it supplies the concrete reduction whenever the calculation below, or a later mixed two-copy batch, produces one. We may therefore write its coefficients as c_{abc}, $a, b, c \in \mathbb{F}_2$, and normalize $c_{000} = 1$. □

Lemma 6.20 (One-coordinate six-port minor localization)**.** *Let $q = (q_{ab\xi})_{a,b,\xi\in\mathbb{F}_2}$ be the logical coefficient table of a physical six-port tensor whose support is contained in a fixed three-edge matching. Close every quaternary and factor batch produced below by the quaternary reduction just described and the binary case analysis. On any branch with no resolved leaf or strict stable successor, fully close every factor batch. Each resulting physical six-port tensor is then either zero, a current-group matching product, or has rank at most one in each of*

$$a \mid (b, \xi), \qquad b \mid (a, \xi), \qquad \xi \mid (a, b)$$

The first two alternatives also have all three ranks at most one. Their rank equations are recorded before their child tokens are completed, so the conclusion holds at every physical phase assignment in any later two-copy batch to which this lemma is applied.

Proof. Consider $a \mid (b, \xi)$. Close the ξ-edge by every actual phase, obtaining the logical matrix pencil

$$A + tB$$

in the a, b coordinates. If $\det(A + tB) \neq 0$ at an actual phase, the resulting physical quaternary enters the preceding nonzero-minor reduction. If the determinant vanishes for every $t \in \mu_m$, it is a degree-two polynomial with at least three roots and is therefore identically zero.

The two-dimensional rank-one-line classification is elementary. Write nonzero rank-one endpoints as $A = uv^{\mathsf{T}}$ and $B = xy^{\mathsf{T}}$. The mixed coefficient of $\det(A + tB)$ is, up to sign, $\det[u, x] \det[v, y]$. Hence either u, x are proportional or v, y are proportional; zero or proportional endpoints satisfy the same conclusion directly. In the first case A, B have one common a-factor, so the $a \mid (b, \xi)$ flattening of q has rank at most one. In the second case they have one common b-factor, giving an exact binary factor of q; factor-saturate and dispatch that factorization. If closing this factorization gives a resolved leaf or strict stable successor, the parent branch ends. Otherwise all exposed factors are current-group matching factors, so q itself is a current-group matching product and all three logical flattenings have rank at most one; record these equations before completing its child token. Thus every branch continuing the parent analysis has the stated a-rank equation. Repeating the same argument after permuting a, b, ξ proves the other two rank bounds. □

After Lemma 6.19, all eight logical coefficients on the continuing branch are nonzero. Retain their notation and normalize the remaining global-matching six-port support to $X^{(1,2)} \otimes X^{(3,4)} \otimes X^{(5,6)}$, where the superscript records the ordered pair of ports on which the binary disequality tensor X acts. For each selected pair of matching edges, use both actual parity-zero and parity-one kernels. The parity-zero contraction on the first two edges gives

$$c_{000}c_{111} = c_{001}c_{110}.$$

The parity-one contractions on the three choices of edge pair give the remaining equalities among

$$c_{001}c_{110}, \qquad c_{010}c_{101}, \qquad c_{011}c_{100}.$$

Every nonzero determinant is routed through the quaternary reduction just described. Hence, on the continuing branch, all four products are equal:

$$c_{000}c_{111} = c_{001}c_{110} = c_{010}c_{101} = c_{011}c_{100}. \tag{44}$$

Define

$$\alpha = \frac{c_{100}}{c_{000}}, \qquad \beta = \frac{c_{010}}{c_{000}}, \qquad \gamma = \frac{c_{001}}{c_{000}}.$$

These three quantities are exposed by explicit physical factor cards. For α, join ports 3 and 5, the first ports of the second and third matching edges, by the actual parity-zero kernel $M_0(1) = I$. The remaining logical table, with the first matching edge as its row edge, is

$$\begin{pmatrix} c_{000} & c_{011} \\ c_{100} & c_{111} \end{pmatrix}.$$

Its determinant vanishes by Equation (44), so exact factor extraction exposes on the first matching edge a binary with ordered ratio $\alpha^{\pm 1}$. Cycling the three matching edges exposes $\beta^{\pm 1}$ and $\gamma^{\pm 1}$ in the same way. Every noncurrent or unsafe factor is a binary exit; hence on the continuing branch

$$\alpha, \beta, \gamma \in \mu_m.$$

Fuse the fixed actual representatives with transfers $\mathsf{R}(\alpha^{-1})$, $\mathsf{R}(\beta^{-1})$, and $\mathsf{R}(\gamma^{-1})$ onto the three oriented matching edges, and divide by the recorded common scalar c_{000}. Equation Equation (44) then gives

$$N_r(z) = \begin{cases} 1, & \mathrm{wt}(z) \le 1, \\ r, & \mathrm{wt}(z) \ge 2, \end{cases} \qquad r := \frac{c_{000}c_{110}}{c_{100}c_{010}}.$$

It remains to prove that r is also a group phase. On this physically normalized six-port card, contract ports 1 and 3, the first ports of the first and second matching edges, by the actual parity-one card $M_1(u) = M_{\alpha=1}(a = u)$, $u \in \mu_m$; in the displayed port orientation this is

$$M_1(u) = \begin{pmatrix} 0 & 1 \\ u & 0 \end{pmatrix}.$$

Write the physical bits on the six ports as $x_1, \ldots, x_6$; the support matching means $x_2 = 1 - x_1$, $x_4 = 1 - x_3$, and $x_6 = 1 - x_5$. The card joining ports 1 and 3 therefore imposes $x_1 \oplus x_3 = 1$ and contributes the factor u^{x_1}. After this contraction, use x_2 (with $x_4 = 1 - x_2$) as the row bit and x_5 (with $x_6 = 1 - x_5$) as the column bit. Ordering the rows and columns changes only the recorded nonzero representative scalar or reverses a port. The resulting quaternary table on ports $2, 4, 5, 6$, up to reversal and a nonzero representative scalar, is

$$\begin{pmatrix} 1 & r \\ u & ur \end{pmatrix} = \begin{pmatrix} 1 \\ u \end{pmatrix} \begin{pmatrix} 1 & r \end{pmatrix}.$$

This is an exact tensor factorization, and its two binary factors have ordered ratios $u^{\pm 1}$ and $r^{\pm 1}$. Expose them through Theorem 2.16 and Lemma 2.20 before testing the complete binary group. A noncurrent factor closes through the active protocol or the selected occurrence's frozen local

queue. Thus the coefficient calculation continues only when both factors are safe in the current μ_m-dihedral group. On that branch, $r^{\pm 1} \in \mu_m$, and hence $r \in \mu_m$. Thus the normalized logical coefficient function above is

$$N_r(z) = \begin{cases} 1, & \mathrm{wt}(z) \le 1, \\ r, & \mathrm{wt}(z) \ge 2, \end{cases} \qquad z \in \mathbb{F}_2^3, \quad r \in \mu_m. \tag{45}$$

Let Q_r denote the 2×2 matrix of the quaternary obtained by closing the third pair, displayed below. For $r \neq -1$, its full-support cross-ratio is

$$\mathrm{cr}(Q_r) = \frac{(Q_r)_{00}(Q_r)_{11}}{(Q_r)_{01}(Q_r)_{10}} = \frac{4r}{(1+r)^2}.$$

If this full-support quaternary enters an affine or local-affine tractable branch, Lemma 4.4 reduces the latter to the affine four-point plane, namely a coset of a two-dimensional subspace of $\mathbb{F}_2^4$. Clause (ii) of Theorem A.5 gives cross-ratio ± 1 on this plane. The value $+1$ is equivalent here to $\det Q_r = 0$, hence to $r = 1$, which is the product case removed below. Therefore a nonproduct affine residue has $\mathrm{cr}(Q_r) = -1$, giving $r + r^{-1} = -6$. Every other outcome is closed by the physical quaternary reduction. Closing the third pair gives

$$Q_r = \begin{pmatrix} 2 & 1+r \\ 1+r & 2r \end{pmatrix}, \qquad \det Q_r = -(r-1)^2. \tag{46}$$

This is a direct physical quaternary with its inherited port order and nonzero representative scalar. Adjoin it and factor-saturate its complete batch before applying the quaternary reduction. If $r = 1$, then $N_1 \equiv 1$, so the parent six-port tensor itself is the product of its three matching binaries; complete the originating frozen token after the binary dispatch. If $r \neq 1$, the displayed fixed-matching minor is nonzero, and the quaternary reduction of Lemmas 6.7 and 6.19 applies. Two actual I-dressings change its paired-disequality support to $\{0000, 0011, 1100, 1111\}$, so every nonproduct retained quaternary is endpoint-nondegenerate. For the exceptional value $r = -1$, the equality exit is the following explicit two-copy wiring. Define the six-port signature

$$F_r(z_1, \ldots, z_6) = \prod_{j=1}^{3} \mathbf{1}[z_{2j-1} \oplus z_{2j} = 1]\, N_r(z_1, z_3, z_5),$$

where the three displayed logical bits are the first ports of the three paired-disequality edges. Take two copies, leave ports $1, 2$ of each copy external, and join corresponding ports $3, 4, 5, 6$ by four actual equality links. The resulting quaternary tensor is

$$W_r(a, b, c, d) = \sum_{u,v,w,z \in \mathbb{F}_2} F_r(a, b, u, v, w, z) F_r(c, d, u, v, w, z). \tag{47}$$

Direct substitution into N_{-1} gives

$$W_{-1}(a, b, c, d) = 4\, \mathbf{1}[(a, b, c, d) \in \{0101, 1010\}].$$

Applying an actual X-translation on the second external port of each copy sends these two words to $0000, 1111$, so the output is $4\mathrm{EQ}_4$ in the displayed normalization (any previously recorded gadget scalar merely changes this to a known nonzero multiple). This terminates by Corollary 3.6 and Lemma 3.4. Otherwise the affine cross-ratio test would force $r + r^{-1} = -6$, impossible for a root of unity. Hence every nonproduct case not already terminated enters Corollary 4.7.

6.5 Polarized coefficient integration

The remaining tensor has a fixed matching support and nonzero logical coefficients. We test their factorization by the following two-copy construction, called *polarization*. Each phase assignment gives an actual six-port tensor. Its vanishing minors are polynomials in those phases; interpolation turns the finite family of physical tests into identities among the original coefficients.

Let $w : \mathbb{F}_2^s \to \mathbb{C}^\times$, $s \geq 3$, be the logical coefficient table on the global matching, with every matching edge in its fixed orientation. Fix distinct logical coordinates i, k and take two equally oriented copies. Leave the entire i-edge of each copy external. On the two k-edges, join one pair of corresponding physical ports by an actual equality link and leave the other two ports external; after one recorded X-dressing they form the third external matching edge and carry one common logical bit ξ. For every $\ell \notin \{i, k\}$, close the two ℓ-edges by one equality link and one phase-dressed equality link. Unique monomial lifts give, after division by the recorded nonzero representative scalar, the direct six-port coefficient

$$q^{i,k}_{ab\xi}(\lambda) = \sum_z \lambda^z w_i(a, \xi, z) w_i(b, \xi, z), \qquad \lambda^z = \prod_{\ell \notin \{i,k\}} \lambda_\ell^{z_\ell}. \tag{48}$$

Here $w_i(a, \xi, z)$ means that coordinates i, k of w are a, ξ, respectively, and the other coordinates are ordered as z. The fixed realization scalars may replace μ_m in each variable by a nonzero coset $c_\ell \mu_m$; the interpolation lemma below is stated in that form. Their Cartesian product is the *polarization grid*. All Laurent monomials and exponent arithmetic use the conventions of Section 2.12.1.

Lemma 6.21 (Grid interpolation for bounded Laurent degree)**.** *Let $m \geq 3$, let $c_1, \ldots, c_r \in \mathbb{C}^\times$, and let $P(\lambda_1, \ldots, \lambda_r)$ be a Laurent polynomial such that, after multiplying by a monomial, its degree in each variable is at most two. If P vanishes on the product of cosets $\prod_{j=1}^r c_j \mu_m$, then P is identically zero.*

Proof. After the monomial shift, substitute $\lambda_j = c_j t_j$. Fix all variables except the last one. The result is a univariate polynomial of degree at most two with at least three distinct roots $t_j \in \mu_m$, hence is zero. Induction on the number of variables proves that every coefficient polynomial vanishes. The nonzero monomial used to clear the Laurent exponents never affects a value on the stated cosets. □

Lemma 6.22 (Polarized determinant)**.** *Assume the global matching support has been fixed by Lemma 6.18, and use Lemma 6.20 to retain every one-coordinate rank equation across completed current-group factor batches. Form every physical card $q^{i,k}(\lambda)$ in Equation* (48)*. Suppose that, after the concrete six-port case analysis has removed every violating batch, each retained card has rank at most one in each of its three one-logical-coordinate flattenings*

$$a \mid (b, \xi), \qquad b \mid (a, \xi), \qquad \xi \mid (a, b).$$

Any concrete card violating this hypothesis is resolved before this lemma is invoked by Lemmas 5.3, 6.7 and 6.19. Then

$$w(z) = w(0) \prod_{i=1}^s \rho_i^{z_i} \tag{49}$$

for nonzero scalars ρ_i. Consequently the core is a tensor product of binary signatures on the global perfect matching. This is an exact tensor identity; the individual factors are made available only through Theorem 2.16 and Lemma 2.20 and are then sent through the binary case analysis.

Proof. Fix $i \neq k$ and $\xi \in \mathbb{F}_2$, and abbreviate

$$u_z := w_i(0, \xi, z), \qquad v_z := w_i(1, \xi, z).$$

Rank at most one of the $a \mid b$ face, including a zero card, gives the determinant polynomial

$$\Delta_\xi(\lambda) := \left(\sum_z \lambda^z u_z^2\right)\left(\sum_z \lambda^z v_z^2\right) - \left(\sum_z \lambda^z u_z v_z\right)^2.$$

This polynomial vanishes on the full product of phase cosets. Its coordinate degree is at most two, so Lemma 6.21 makes it the zero polynomial. If z, z' are adjacent in the remaining $(s-2)$-cube, the coefficient of $\lambda^{z+z'}$ is

$$(u_z v_{z'} - u_{z'} v_z)^2.$$

All values of w are nonzero; hence $w_i(1, \xi, z)/w_i(0, \xi, z)$ is independent of z for fixed ξ.

The missing cross-slice relation comes from a different $a \mid \xi$ face of the same physical six-port tensor, not from Δ_ξ. Rank at most one of $a \mid (b, \xi)$ gives the genuine face minor

$$\Gamma(\lambda) = q^{i,k}_{000}(\lambda) q^{i,k}_{101}(\lambda) - q^{i,k}_{001}(\lambda) q^{i,k}_{100}(\lambda) = 0.$$

The polynomial Γ vanishes on the full product of phase cosets and has coordinate degree at most two. Hence Lemma 6.21 makes it identically zero. At λ^{2z}, only the ordered pair (z, z) contributes, and its coefficient is

$$w_i(0,0,z) w_i(0,1,z)\big(w_i(0,0,z) w_i(1,1,z) - w_i(1,0,z) w_i(0,1,z)\big).$$

It vanishes, so the edge ratio in coordinate i is the same for $\xi = 0$ and $\xi = 1$. Combining this equality with the preceding constancy over the remaining $(s-2)$-coordinate cube shows that

$$\rho_i := \frac{w(1, x_{-i})}{w(0, x_{-i})}$$

is independent of every coordinate of x_{-i}, where x_{-i} is the ordered tuple of all coordinates except i, and $w(b, x_{-i})$ inserts b in coordinate i. Repeat for each i. Starting at 0^s and flipping the nonzero coordinates of z in any order then multiplies by the corresponding constant edge ratios, giving Equation (49). The vanishing two-face minors make the result independent of the order of the flips. □

The cross-slice minor Γ compares the ratios on the two ξ-layers, which Δ_ξ alone cannot do. The phase grid uses every element of μ_m, including all four phases when $m = 4$.

6.6 Reduction semantics and completion

Every local disagreement and exceptional endpoint batch has now been resolved. For the active prime, the remaining survivor has a global matching support and the polarized coefficient product. This final subsection checks reduction direction and provenance, then records the relative-deck hypotheses used by local constructions. The final active-core closure below resolves the entire signature set directly.

Every contraction above uses fixed actual representatives; the only analytic objects are slices, pencil endpoints, inverse evaluation formulas, and coefficient comparisons. The two-root choices are legal for $m \geq 3$. Unique-lift monomial splices are cancellation-free; phase traces and two-copy sums

may cancel and are handled by their explicit zero dispatch or by grid interpolation. Actual four- or six-port outputs therefore have the direct-gadget reduction direction. In contrast, Lemmas 6.5, 6.18 and 6.22 use analytic reconstruction only to prove tensor identities; their proper factors are adjoined through Theorem 2.16 and Lemma 2.20 with the Turing reduction supplied by the factor theorem recorded. All bounded outputs have size $O(\mathrm{arity}(f))$. Their complete concrete batches are resolved by Lemmas 5.11, 6.7 and 6.19 and Definition 5.9; in particular, no nonmatching six-port tensor is recursively re-entered.

For relative worklist invocations, let $\mathcal{W}$ be the finite multiset of unresolved tensor-prime occurrences in the retained factorization trees; occurrences, not only canonical signature keys, are counted. For each unresolved occurrence u, with underlying tensor-prime signature h_u, let A_{led} denote the current coordinate matrix and put $H_{\mathrm{pb}} := \overline{G}(A_{\mathrm{led}}, \Lambda)$. Freeze the occurrence's own finite card/factor universe

$$\mathscr{U}^{\mathrm{fac}}(A_{\mathrm{led}}, h_u, H_{\mathrm{pb}}) = Q_{u,\mathrm{done}} \sqcup Q_u$$

when that occurrence first enters the worklist. Let Φ_u be its stored factor potential. Define

$$\Theta_{\mathrm{MD}}(\mathcal{W}) = \{\{(\mathrm{arity}(h_u), |Q_u|, \Phi_u) : u \in \mathcal{W}\}\},$$

using strict lexicographic order on each triple and its well-founded multiset extension, with the conventions of Section 2.14.1.

Certifying an occurrence removes its triple. If the selected ancestor itself factors, its triple is replaced by triples having strictly smaller first coordinate. If processing one child-card key exposes lower-arity occurrences while the parent remains unresolved, that key is first consumed or its factor potential strictly decreases; hence the old parent triple is replaced by a smaller parent triple together with child triples of smaller arity. This also decreases the multiset measure. The bounded four- or six-port reductions create no fresh unresolved same-arity occurrence: their complete factorization trees are processed inside the originating Q_u.

Definition 6.23 (Occurrence-level relative deck stability)**.** For a retained occurrence h at a current state (Λ, G), write

$$\mathrm{RelStable}(\Lambda, G; h)$$

when its complete physical oriented deck has been regenerated; every already exposed proper card and its complete factorization tree have been closed; the complete group remains G; and, on the continuing branch, every surviving nonzero proper card is a G-matching product, while every unresolved lower-arity nonbinary occurrence has been placed earlier in the occurrence induction. An active deck-stable minimum occurrence is automatically relatively deck-stable.

Lemma 6.24 (Relative re-entry)**.** *Let h be a tensor-prime occurrence of even arity in the current normalized state $(\Lambda, \mathcal{M}_m)$, and suppose* $\mathrm{RelStable}(\Lambda, \mathcal{M}_m; h)$*. Then every invocation of deck stability in the preceding active-prime argument remains valid with h in place of the minimum core.*

Proof. The sector-ruling, endpoint, support, six-port, and polarization arguments use minimum arity only to exclude a surviving proper nonmatching card or nonbinary factor. Under the present hypothesis each such object has already produced a resolved leaf or strict stable successor, or has been certified as a current-group matching product. All other inputs—the fixed actual representatives, normalized group, even arity, and tensor primality of h—are unchanged. Thus every local construction and reduction used in the active-prime proof remains licensed. □

Theorem 6.25 (Standard $\mathcal{M}_m$-sector deck rigidity)**.** *Assume*

$$\mathsf{Stable}(\Lambda, \mathcal{M}_m, f), \qquad \mathsf{Even}(\Lambda), \qquad \mathrm{arity}(f) = \nu(\Lambda) = 2k+2, \qquad m \geq 3,$$

where

$$\mathcal{M}_m = \{[\mathsf{R}(t)], [\mathsf{S}(t)] : t \in \mu_m\}.$$

Then $\mathsf{Close}_{\{\mathscr{P}\}}(\Lambda, \mathcal{M}_m, f)$.

Completion of Theorem 6.25. If $\mathrm{arity}(f) = 4$, regenerate and close its complete physical deck. In the standard $\mathcal{M}_m$ case the output of Theorem 5.18 is an invertible dressing of the prime quaternary. It is therefore an actual endpoint-nondegenerate tensor-prime eight-vertex signature. Apply Theorem 1.3 to its retained augmentation. Direct realization and downward inheritance give a hard or $\mathsf{Tract}(K\Lambda)$ leaf for the entire retained set Λ.

For $\mathrm{arity}(f) = 6$, Lemma 6.11 contradicts the tensor primality of the stable core. Now let $n := \mathrm{arity}(f) \geq 8$. At any deleted pair p, two disagreeing nonzero sampled cards would give the six-port output in Lemma 6.6. The first proper closure of the stable core is zero or a matching product, and further monomial contractions preserve that property. This contradicts the two inherited disagreeing nonzero cards. Hence all samples at p share one matching and all factor lines outside one designated edge e. The pairwise argument following Lemma 6.7 gives one common nonzero product P of those fixed factors. In each nonzero sector choose two distinct nonzero phase samples and invert its pencil; an identically zero sector has zero endpoint slices. Thus all four physical slices lie in $\mathbb{C}P \otimes V_e$, proving the exact identity $f = P \otimes q_{p,e}$. Since P contains at least one binary factor, this again contradicts tensor primality.

Consequently no higher-arity stable-core branch remains. The quaternary case already resolves the whole signature set, so no local product-class certificate has to be promoted to a matching-deck certificate for another occurrence. These are precisely the allowed outcomes of $\mathsf{Close}_{\{\mathscr{P}\}}(\Lambda, \mathcal{M}_m, f)$. □

The restriction $m \geq 3$ is essential here. For $m = 2$, two forbidden trace roots can exhaust μ_2, and evaluation on μ_2 does not determine a polynomial of coordinate degree two. That case belongs to the separate normalized V_4 analysis.

6.7 The two nonstandard marked dihedral sectors

For the nonstandard marked groups, the same binary transfers acquire a different edge matrix. We first compute that matrix, then determine which parts of the standard proof survive. The matrices U_τ below change analytic coordinates. The $\mathcal{B}_m$ and $\mathcal{A}_m^0$ proofs additionally give whole-network equivalences; the $\mathcal{A}_m$ proof instead keeps the original physical network and tracks the resulting weights.

Lemma 6.26 (Marked-sector dictionary)**.** *For $\tau = \mathcal{M}, \mathcal{B}$, let $m \geq 3$ and set*

$$U_{\mathcal{M}} = I, \qquad U_{\mathcal{B}} = I - iX.$$

For $\tau = \mathcal{A}$, let $m \geq 4$ be even, fix $\eta \in \mu_{2m} \setminus \mu_m$, choose $s_0 \in \overline{\mathbb{Q}}$ with $s_0^2 = \eta^{-1}$, and set

$$U_{\mathcal{A}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \mathrm{diag}(1, s_0), \quad s_0^2 = \eta^{-1}.$$

Conjugation by U_τ sends $\mathcal{M}_m$ projectively onto the corresponding marked group. Define the transformed edge matrix S_τ by

$$S_\tau = U_\tau^{\mathsf{T}} X U_\tau.$$

Then, projectively,

$$S_{\mathcal{M}} = X, \qquad S_{\mathcal{B}} = I, \qquad S_{\mathcal{A}} = \mathrm{diag}(1, -\eta^{-1}).$$

For $T \in \mathcal{M}_m$, choose a fixed actual marked-sector binary B_T whose sector-coordinate transfer is $[T]$, and set $\widehat{B}_T := U_\tau^{-1} B_T U_\tau^{-\mathsf{T}}$. Then, for $T, U \in \mathcal{M}_m$,

$$\widehat{B}_T \doteq T S_\tau^{-1}, \qquad (T S_\tau^{-1}) S_\tau (U S_\tau^{-1}) = T U S_\tau^{-1}. \tag{50}$$

Thus multiplication and reversal are preserved, and every monomial fusion has the same unique internal lift; consequently such a fusion introduces no cancellation.

Proof.

$$U_{\mathcal{B}} R(t) U_{\mathcal{B}}^{-1} \doteq C(t), \qquad U_{\mathcal{A}} R(t) U_{\mathcal{A}}^{-1} \doteq A(t), \qquad U_{\mathcal{A}} X R(t) U_{\mathcal{A}}^{-1} \doteq Z A(\eta t).$$
$$U_{\mathcal{B}}^{\mathsf{T}} X U_{\mathcal{B}} = -2iI, \qquad U_{\mathcal{A}}^{\mathsf{T}} X U_{\mathcal{A}} = 2\,\mathrm{diag}(1, -\eta^{-1}).$$

Congruence gives $\widehat{B}_T \doteq T S_\tau^{-1}$, and the fusion identity is matrix multiplication; all omitted scalars are fixed nonzero. □

Sector coordinates and available kernels. For an r-ary current-coordinate signature h, write

$$\widehat{h}_\tau = (U_\tau^{-1})^{\otimes r} h$$

for its analytic sector-coordinate table. A *sector-X dressing* is physical port dressing by the fixed actual binary B_X; by Equation (50), its transported transfer fuses as X. Put

$$\beta_{\mathcal{B}} := 1, \qquad \delta := -\eta^{-1}, \quad \beta_{\mathcal{A}} := \delta^{-1}, \qquad \mathscr{R}_\tau := \beta_\tau \mu_m,$$

where the quantities involving η and δ apply only in the $\mathcal{A}$-sector. For a quotient in $\mathscr{R}_\tau$, division by β_τ gives a point of μ_m. This analytic normalization is not a gadget. In one monomial parity plane, a *complementary-axis pencil* has its two coefficient matrices on the two coordinate axes. A sample with exactly one vanishing coordinate is a singular binary.

Lemma 6.27 (Supports of marked quaternaries)**.** *Let $q \neq 0$ be a retained quaternary occurrence in a current $\mathcal{B}_m$-state ($m \geq 3$) or $\mathcal{A}_m$-state (even $m \geq 4$). Suppose its complete physical oriented deck has been regenerated and every member of that deck is zero or a safe current-group binary. Let $\tau = \mathcal{B}$ or $\mathcal{A}$, respectively, and define the sector-coordinate signature*

$$\widehat{q} := (U_\tau^{-1})^{\otimes 4} q.$$

After actual sector-X dressings, for some $K \subseteq \{1,2,3\}$, nonzero a, h, and nonzero b_ℓ, d_ℓ ($\ell \in K$),

$$\widehat{q} = a\delta_{0000} + h\delta_{1111} + \sum_{\ell \in K} \big(b_\ell \delta_{p_\ell} + d_\ell \delta_{\bar{p}_\ell}\big), \tag{51}$$

where p_1, p_2, p_3 represent the three complementary pairs of weight-two words. The assertion includes all six contracted pairs, both orientations, all zero-card patterns, and the fixed representative scalars.

Proof. For an ordered port pair with residual slices (A, B, C, D), the complete card pencils, up to their recorded nonzero representative scalars, are

	$E(t)$	$O(t)$
$\mathcal{B}_m$	$A + tD$	$tB + C$
$\mathcal{A}_m$	$A + \delta t D$	$tB + \delta C, \quad \delta = -\eta^{-1},$

$t \in \mu_m$.

Their effective kernels are $R(t), XR(t)$ for $\mathcal{B}_m$, and

$$SR(t) = \mathrm{diag}(1, \delta t), \qquad SXR(t) = \begin{pmatrix} 0 & t \\ \delta & 0 \end{pmatrix}, \quad S = \mathrm{diag}(1, \delta),$$

for $\mathcal{A}_m$. The diagonal kernels are symmetric. Define the off-diagonal kernel $K^{\mathrm{off}}(t) := \left(\begin{smallmatrix} 0 & t \\ \delta & 0 \end{smallmatrix}\right)$; it satisfies

$$K^{\mathrm{off}}(t)^{\mathsf{T}} = \frac{t}{\delta} K^{\mathrm{off}}(\delta^2/t), \qquad \delta^2/t \in \mu_m; \tag{52}$$

so reversal permutes the labels and restores a known nonzero scalar; the $\mathcal{B}_m$ formula is $\delta = 1$.

With β_τ fixed above, three phases imply that each two-coordinate pencil is zero, lies on a fixed line with ordered ratio in $\mathscr{R}_\tau$, or has its endpoints on the two complementary coordinate axes. Indeed, a zero makes the endpoints proportional. Otherwise the quotient $\phi(t)$ of the two independent affine coordinates is injective on μ_m and takes its values in the fixed coset $\beta_\tau \mu_m$. Hence $\psi(t) = \beta_\tau^{-1}\phi(t)$ permutes μ_m, and Lemma 6.9 gives $\psi(t) = \lambda t$ or λ/t. Cross-multiplication leaves the opposite coordinate axes. Multiplication by β_τ^{-1} is only an analytic scaling of one endpoint coordinate; it is not an available binary or a gadget. Formula Equation (52) covers the opposite orientation with its recorded nonzero scalar.

Apply this three-type pencil classification to both card columns for each of the six deleted pairs. After analytic quotient normalization, the Boolean support constraints are exactly those enumerated in Theorem A.8(ii): every nonempty solution consists of one total parity together with a nonempty subset of its four complementary word pairs. Quotient normalization changes no zero pattern, so the same exact enumeration applies in marked coordinates. Choose a nonzero complementary pair and use actual sector-X dressings, justified by Equation (50), to send it to $0000, 1111$. The other pairs of nonzero entries are then among the three complementary weight-two blocks, giving Equation (51). This includes both orientations and every zero-card pattern; no analytic sector basis is installed as a gadget. □

Lemma 6.28 (Symmetric binary pullback)**.** *Let $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$. If $E, F \in \mathscr{C}$ are nonsingular symmetric binaries, then $EFE \in \mathscr{C}$.*

Proof. Closure under contraction proves the affine case, and the equality/disequality normal forms prove the product case. For the local-affine case, write $E = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$. The diagonal twists in the definition of $\mathscr{L}$ preserve support, which must be affine. Nonsingularity excludes support size one, and affine support excludes size three. Full support is also impossible: the $\alpha = 00$ condition requires $b/a \in \mu_4$, whereas the $\alpha = 01$ twist requires $\zeta_8 b/a \in \mu_4$. On anti-diagonal support symmetry gives $E = bX$, and the same $\alpha = 01$ twist again has a nonzero-entry ratio outside μ_4. Consequently $E = a\,\mathrm{diag}(1, i^k)$, with $a \neq 0$ and $k \in \mathbb{Z}/4\mathbb{Z}$. Conversely these four diagonal forms satisfy both supported twists, $\alpha = 00, 11$, and belong to $\mathscr{L}$. Writing $F = d\,\mathrm{diag}(1, i^\ell)$ in the same way gives $EFE = a^2 d\,\mathrm{diag}(1, i^{2k+\ell}) \in \mathscr{L}$, as required. □

Theorem 6.29 (Signature-set-wide reduction of the $\mathcal{B}_m$ sector)**.** *If*

$$\mathsf{Stable}(\Lambda, \mathcal{B}_m, f), \qquad \mathsf{Even}(\Lambda), \qquad m \geq 3,$$

then $\mathsf{Close}_{\{\mathscr{P}\}}(\Lambda, \mathcal{B}_m, f)$.

The resolved Tract-alternatives here include $\mathsf{Pres}_K(K\Lambda; \mathscr{A})$ and $\mathsf{Pres}_K(K\Lambda; \mathscr{L})$; the product case is the only continuing rigidity leaf.

Proof. Define the transformed signature set $\Gamma := U_{\mathcal{B}}^{-1}\Lambda$ and the transformed active prime

$$\widehat{f} := (U_{\mathcal{B}}^{-1})^{\otimes \operatorname{arity}(f)} f \in \Gamma.$$

This whole-problem change gives

$$U_{\mathcal{B}}^{\mathsf{T}} X U_{\mathcal{B}} = -2iI, \qquad U_{\mathcal{B}}^{-1} I U_{\mathcal{B}}^{-\mathsf{T}} = \frac{i}{2}X, \qquad U_{\mathcal{B}}^{-1} X U_{\mathcal{B}}^{-\mathsf{T}} = \frac{i}{2}I.$$

Thus edge and equality exchange, and Lemma 5.1 gives

$$\text{K-Holant}(\Lambda) \equiv_{\mathrm{T}} \text{Holant}(\Gamma, X) \equiv_{\mathrm{T}} \text{K-Holant}(\Gamma, I).$$

Define the augmented signature set $\Lambda^{\sharp} := \Gamma \cup \{I\}$. Here the *open-network bijection* is the vertexwise change by $U_{\mathcal{B}}^{-1}$, together with the displayed edge/equality exchange, applied also to the ordered open boundary legs; the inverse change restores the original network. It preserves arity, factorization, minimum nonbinary-prime arity, and parity of every prime arity; hence $\mathsf{Even}(\Lambda^{\sharp})$.

For a realizable nonsingular b over $\Lambda^{\sharp}$, its pullback B satisfies

$$B \doteq U_{\mathcal{B}} b U_{\mathcal{B}}^{\mathsf{T}}, \qquad [BX] = [U_{\mathcal{B}} b U_{\mathcal{B}}^{-1}],$$

using $U_{\mathcal{B}}^{\mathsf{T}} X = -2iU_{\mathcal{B}}^{-1}$. Infinite order pulls back to the interpolation terminal of Lemma 3.1; otherwise Lemma 5.3 applies to the pullback transfer $[BX]$. Conjugating by $U_{\mathcal{B}}^{-1}$ first puts $[b]$ in $\mathcal{M}_m$. Since $X \in \mathcal{M}_m$, closure under right multiplication then puts the actual transformed transfer $[bX]$ in $\mathcal{M}_m$.

Conversely, for the chosen actual representative B_T, define its recorded nonzero representative scalar β_T by $B_T X = \beta_T U_{\mathcal{B}} T U_{\mathcal{B}}^{-1}$. Then

$$\widehat{B}_T = U_{\mathcal{B}}^{-1} B_T U_{\mathcal{B}}^{-\mathsf{T}} = \frac{i\beta_T}{2} T.$$

Since $[R(t)X] = [XR(t^{-1})]$ and $[XR(t)X] = [R(t^{-1})]$, every class occurs and

$$G(\Lambda^{\sharp}) = \mathcal{M}_m.$$

The open-network bijection also transports the complete deck: absorbing an equality wire replaces a label by XT, TX, or XTX, which permute $\mathcal{M}_m$. Hence zeros, matching products, factor saturation, and minimum prime arity correspond, so the transformed state is deck-stable. In particular,

$$\mathsf{Stable}(\Lambda^{\sharp}, \mathcal{M}_m, \widehat{f}), \qquad \operatorname{arity}(\widehat{f}) = \nu(\Lambda^{\sharp}) = \operatorname{arity}(f).$$

Apply Theorem 6.25. Complete closure of each transported concrete batch is resolved. On the remaining branch there are $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$ and one $V \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ such that

$$V^{-1}(\Gamma \cup \{I\}) \subseteq \mathscr{C}, \qquad V^{\mathsf{T}} X V \in \mathscr{C}.$$

Define the common witness matrix $W := U_{\mathcal{B}} V$; then $W^{-1}\Lambda \subseteq \mathscr{C}$. With

$$E = V^{\mathsf{T}} X V, \qquad F = V^{-1} I V^{-\mathsf{T}},$$

both E, F are nonsingular symmetric members of $\mathscr{C}$, and Lemma 6.28 gives

$$W^{\mathsf{T}} X W = -2iV^{\mathsf{T}} I V = -2iEFE \in \mathscr{C}.$$

Thus W is common. The $\mathscr{A}, \mathscr{L}$ cases are resolved $\mathsf{Tract}(K\Lambda)$ leaves; $\mathscr{P}$ is the continuing rigidity leaf. The reductions above use $U_{\mathcal{B}}$ for the whole network; $W = U_{\mathcal{B}} V$ records its contribution to the final witness. □

Theorem 6.30 (Literal-equality reduction of the $\mathcal{A}_m^0$ sector)**.** *Assume*

$$\mathsf{Stable}(\Lambda, \mathcal{A}_m^0, f), \qquad \mathsf{Even}(\Lambda), \qquad m \geq 4, \quad 2 \mid m.$$

Then $\mathsf{Close}_{\{\mathscr{P}\}}(\Lambda, \mathcal{A}_m^0, f)$.

Proof. Let

$$U_0 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad U_0^{-1} = \frac{1}{2} U_0.$$

Direct multiplication gives, projectively,

$$U_0 R(t) U_0^{-1} \doteq A(t), \qquad U_0 X R(t) U_0^{-1} \doteq Z A(t), \tag{53}$$

so $U_0^{-1} \mathcal{A}_m^0 U_0 = \mathcal{M}_m$. More importantly,

$$U_0^{\mathsf{T}} X U_0 = 2Z, \qquad U_0^{-1} I U_0^{-\mathsf{T}} = \frac{1}{2} I. \tag{54}$$

Thus this whole-problem change preserves the available literal equality.

Turing-expose the fixed actual representative B_j of the mark $j = [Z]$, retaining its realization scalar and port orientation. Since $[B_j X] = j$, one has $B_j \doteq Y$. Put

$$\Lambda^+ = \Lambda \cup \{B_j\}, \qquad \Gamma = U_0^{-1} \Lambda^+, \qquad \widehat{f} = (U_0^{-1})^{\otimes \operatorname{arity}(f)} f.$$

After recording the two nonzero representative scalars, use literal $I, Y \in \Gamma$; indeed

$$U_0^{-1} I U_0^{-\mathsf{T}} = \frac{1}{2} I, \qquad U_0^{-1} Y U_0^{-\mathsf{T}} = -\frac{1}{2} Y.$$

Write $\mathrm{Holant}^Z(\Gamma)$ for the homogeneous Z-edge half-edge problem. The whole-problem change in Equation (54) gives

$$\text{K-Holant}(\Lambda^+) \equiv_{\mathrm{T}} \mathrm{Holant}^Z(\Gamma),$$

where the factor 2 per edge is known. The exact local identities

$$Z = XIXYX, \qquad X = -ZIZYZ \tag{55}$$

give a signature-preserving open-network equivalence

$$\mathrm{Holant}^Z(\Gamma) \equiv_{\mathrm{T}} \text{K-Holant}(\Gamma). \tag{56}$$

For the forward substitution, replace a Z-edge by three native X-edges with the available ordered vertices I, Y; for the reverse substitution, replace an X-edge by three Z-edges with the same two vertices. Reversing the ordered Y-vertex changes only its recorded nonzero sign. Define

$$\Gamma^\sharp := \Gamma \cup \{Z\}.$$

The first identity in Equation (55) directly realizes Z in the native-X presentation, so adjoining it changes neither the problem nor the open-gadget closure.

The two edge presentations therefore have the same projective set $\mathscr{B}$ of directly realizable nonsingular ordered binary tensors. In the Z-edge presentation, open-network transport and Equation (53) give

$$\{[bZ] : b \in \mathscr{B}\} = U_0^{-1} \mathcal{A}_m^0 U_0 = \mathcal{M}_m.$$

Since m is even, $[Y] = [XR(-1)] \in \mathcal{M}_m$. Hence, in the native-$X$ presentation,

$$G(\Gamma^\sharp) = \{[bX] : b \in \mathscr{B}\} = \{[bZ][ZX] : b \in \mathscr{B}\} = \mathcal{M}_m[Y] = \mathcal{M}_m. \tag{57}$$

The complete deck transports as well. Under U_0^{-1}, the original native-X deck is the corresponding Z-edge deck. Translating any target dressing or closing edge by the second identity in Equation (55), and then absorbing the inserted I, Y binary chain into the adjacent fixed actual representative, merely permutes the $\mathcal{M}_m$ labels. The same argument with the first identity gives the inverse correspondence. Thus zeros, matching products, factorization trees, minimum nonbinary-prime arity, and prime-arity parity correspond. Therefore

$$\mathsf{Stable}(\Gamma^\sharp, \mathcal{M}_m, \widehat{f}), \qquad \operatorname{arity}(\widehat{f}) = \nu(\Gamma^\sharp) = \operatorname{arity}(f), \qquad \mathsf{Even}(\Gamma^\sharp).$$

Apply Theorem 6.25. Every terminal, concrete factor batch, and strict successor transports back through the two open-network equivalences. On a surviving common-presentation leaf, there are $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$ and $V \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ such that

$$V^{-1}\Gamma^\sharp \subseteq \mathscr{C}, \qquad E := V^{\mathsf{T}} X V \in \mathscr{C}.$$

Set $W := U_0 V$. Then $W^{-1}\Lambda \subseteq \mathscr{C}$. Because $Z \in \Gamma^\sharp$, also

$$F := V^{-1} Z V^{-\mathsf{T}} \in \mathscr{C}.$$

The binaries E, F are nonsingular and symmetric, so Lemma 6.28 and $XZX = -Z$ give

$$EFE = V^{\mathsf{T}} XZXV = -V^{\mathsf{T}} Z V, \qquad W^{\mathsf{T}} X W = 2V^{\mathsf{T}} Z V = -2EFE \in \mathscr{C}.$$

Thus W is a common witness for the original problem. The $\mathscr{A}, \mathscr{L}$ cases are resolved $\mathsf{Tract}(K\Lambda)$ leaves and $\mathscr{P}$ is the continuing rigidity leaf. The witness $W = U_0 V$ combines the whole-network equivalence with the transformed common presentation; literal I remains available throughout. □

In Equation (51), write $g_\ell := d_\ell$ for each complementary weight-two block in the sum. Such a block has $b_\ell g_\ell \neq 0$; all other blocks vanish. Ratios below use the displayed order of the two nonzero binary entries.

Lemma 6.31 (The $\mathcal{A}_m$ quaternary coset)**.** *Let $m \geq 4$ be even, let $G(\Lambda) = \mathcal{A}_m$, and let $0 \neq q$ be a retained quaternary occurrence satisfying*

$$\mathrm{RelStable}(\Lambda, \mathcal{A}_m; q).$$

Suppose every member of the complete oriented dressed deck of q is zero or a safe $\mathcal{A}_m$-binary. Then the quaternary analysis yields an established terminal or strict complete-group successor, a genuine factorization of q into two binary signatures whose complete factor batch is retained, or an actual endpoint-nondegenerate eight-vertex signature in the current physical coordinates; in this last alternative the original sector-coordinate signature also satisfies

$$(U_{\mathcal{A}}^{-1})^{\otimes 4} q \in \mathscr{P}.$$

Proof. Apply Lemma 6.27 and fix the resulting actual sector-X dressings. In sector coordinates those dressings are invertible monomial transformations, so they preserve membership in $\mathscr{P}$, decomposability, all flattening ranks, and exact factorization. We may therefore work with Equation (51) and undo the dressings afterward.

Define the phase scalar δ and safe-ratio coset $\mathscr{R}_A$ by

$$\delta = -\eta^{-1}, \qquad \mathscr{R}_A = \delta^{-1}\mu_m = \eta\mu_m.$$

Since $\eta \notin \mu_m$ and m is even,

$$\mathscr{R}_A \cap \mu_m = \varnothing, \qquad \mathscr{R}_A^{-1} = \mathscr{R}_A.$$

By Lemma 6.26, every safe sector binary has ratio in $\mathscr{R}_A$. In Equation (51), denote the endpoint pair by (a, h) and an active central pair by (b_i, g_i).

For one active block, the corresponding diagonal pencil is

$$\operatorname{diag}(a + \delta t g_i,\ b_i + \delta t h), \qquad t \in \mu_m.$$

The root-pencil type classification in Lemma 6.27 excludes a sample with exactly one vanishing coordinate and forces

$$(b_i, g_i) = (a\rho_i, h/\rho_i), \qquad \rho_i \in \mathscr{R}_A, \qquad b_i g_i = ah,$$

whereas an inactive block gives the diagonal card pencil

$$\operatorname{diag}(a, \delta t h), \qquad t \in \mu_m.$$

Its ordered ratio $\delta th/a$ belongs to $\mathscr{R}_A = \delta^{-1}\mu_m$. Since $\delta^2 = \eta^{-2} \in \mu_m$, this is equivalent to

$$h/a \in \delta^{-2}\mu_m = \mu_m.$$

If two blocks j, k are active, the crossing deleted pair and the actual reflections

$$\begin{pmatrix} 0 & t \\ \delta & 0 \end{pmatrix}, \qquad t \in \mu_m$$

give, up to common endpoint factors,

$$ta\rho_j + \delta h/\rho_k, \qquad ta\rho_k + \delta h/\rho_j.$$

Writing

$$L(t) = ta + \frac{\delta h}{\rho_j \rho_k},$$

these entries are $\rho_j L(t)$ and $\rho_k L(t)$. Every nonzero sample therefore has ratio $\rho_j/\rho_k \in \mu_m$, because both ρ_j, ρ_k lie in the same coset $\mathscr{R}_A$. This ratio is outside the safe coset $\mathscr{R}_A$. The nonzero affine function $L(t)$ vanishes for at most one phase, so another actual phase gives a nonsingular outside-group binary. Complete binary closure by Lemma 5.3 gives a terminal or a strict group successor, contradicting a remaining deck-stable branch.

Thus at most one block is active. If exactly one block i is active, then $ah - b_i g_i = 0$, giving a genuine $2 + 2$ factorization exposed by Theorem 2.16 and Lemma 2.20. Its two binaries are then dispatched individually: a new safe class is a strict group successor, and singular, unsafe, or infinite-order factors reach their established terminal.

With no active block, let $\lambda = h/a \in \mu_m$. Before any further rotation, the sector table is already

$$a\delta_{0000} + h\delta_{1111},$$

a generalized equality and hence a member of $\mathscr{P}$. Undoing the fixed sector-X dressings preserves $\mathscr{P}$, proving

$$(U_{\mathcal{A}}^{-1})^{\otimes 4} q \in \mathscr{P}.$$

Moreover $\eta^2/\lambda \in \mu_m$, so it is the product of four available phase parameters. Actual rotations with this product of port phases give the sector tensor

$$a(\delta_{0000} + \eta^2 \delta_{1111}).$$

Since $s_0^4 = \eta^{-2}$,

$$\begin{aligned} U_{\mathcal{A}}^{\otimes 4} a(\delta_{0000} + \eta^2 \delta_{1111}) &= a((1,1)^{\otimes 4} + (1,-1)^{\otimes 4}) \\ &= 2a\, \mathbf{1}[\mathrm{wt}(x) \text{ is even}]. \end{aligned}$$

This is a literal current-coordinate endpoint-nondegenerate eight-vertex signature. The rotations are actual by Equation (50); $U_{\mathcal{A}}$ only verifies the identity. □

An actual quaternary, whether it is the selected occurrence or is produced by a higher-arity marked construction, is not sent directly to Lemma 6.31. First adjoin it with its port order, scalar, and provenance, jointly factor-saturate it, recompute the complete binary group, and regenerate its complete oriented one-copy deck. Test all three physical $2+2$ flattenings. Every rank-one flattening is retained as an exact factor identity. Unary, singular, unsafe, or infinite-order factors reach their established reductions, and a safe binary outside $\mathcal{A}_m$ is a strict group successor. If all exposed binary factors are safe current-group binaries, the child quaternary is a current-group matching product and its originating frozen token is completed; if the quaternary itself is the selected tensor-prime occurrence, such a factorization contradicts its primality.

On the remaining branch the quaternary is tensor-prime, its complete physical deck is zero or safe-group-valued, and the closure just performed gives

$$\mathrm{RelStable}(\Lambda, \mathcal{A}_m; q)$$

in the sense of Definition 6.23. For a later occurrence, closure of its frozen local queue and prior processing of every lower-arity occurrence give this certificate directly. Invoke Lemma 6.31 directly. We do not invoke Lemma 5.5 here: the endpoint support supplied by Lemma 6.27 is an analytic sector-coordinate statement, not a physical endpoint-nondegenerate quaternary.

In the no-active-block branch the physical output is $2a\,\mathbf{1}[\mathrm{wt}(x) \text{ is even}]$, with $a \neq 0$. It is tensor-prime: the weighted generalized equality used to construct it has rank two across every nontrivial partition, and the invertible portwise transformations preserve these ranks. The output is actual, has nonzero endpoints, and is an eight-vertex signature. Apply the already proved Theorem 1.3 to the retained augmentation. This gives a hard or $\mathsf{Tract}(K\Lambda)$ leaf for the entire current signature set, with the reduction supplied by actual realization. This branch does not return merely a local $\mathscr{P}$-certificate for a child. This complete procedure is the *marked-coordinate reduction.*

For the remainder of the $\mathcal{A}_m$-sector, fix

$$\delta := -\eta^{-1}, \qquad D_\delta := \mathrm{diag}(1, \delta),$$

and use the portwise-gauge convention of Section 2.12.2; in particular, $(D_\delta^{(i)} h)(x) = \delta^{x_i} h(x)$.

A finite family of marked-sector cards is a *gauge-coherent comparison batch* if its members use the same ordered contraction multigraph, the same orientations on all contracted edges, and the

same external-port order, and differ only in the chosen phase or product at designated kernels. For a set $\mathscr{S}$ of parent ports write

$$D_\delta^{(\mathscr{S})} := \prod_{i\in\mathscr{S}} D_\delta^{(i)}, \qquad (D_\delta^{(\mathscr{S})}h)(x) = \delta^{\sum_{i\in\mathscr{S}} x_i} h(x).$$

The embedded one-port operators commute. The set $\mathscr{S}$, and hence the gauged parent $D_\delta^{(\mathscr{S})}h$, is fixed throughout one comparison batch. Different contraction skeletons may yield different sets.

Lemma 6.32 (Explicit weighted-gauge transport)**.** *Let $m \geq 4$ be even, $H = \mu_m$, and use the fixed $\delta = -\eta^{-1}$, where $\eta \in \mu_{2m} \setminus \mu_m$, and the diagonal gauge matrix D_δ above. In the analytic $\mathcal{A}_m$-sector the effective diagonal and reflection kernels are*

$$K_t^0 = D_\delta R(t) = \mathrm{diag}(1, \delta t), \qquad K_t^1 = D_\delta X R(t) = \begin{pmatrix} 0 & t \\ \delta & 0 \end{pmatrix}, \qquad t \in H.$$

For every binary kernel K and every two-port slice g on ordered ports (i, j), pointwise in all remaining variables, every occurrence of D_δ can be moved from the kernel to the corresponding endpoint of the parent tensor:

$$\langle D_\delta K, g\rangle_F = \langle K, D_\delta^{(i)} g\rangle_F, \qquad \langle K D_\delta, g\rangle_F = \langle K, D_\delta^{(j)} g\rangle_F, \tag{58}$$

where i and j are the first and second ordered ports and $D_\delta^{(i)}$ means multiplication by δ^{x_i}. Consequently for every gauge-coherent comparison batch there is one fixed set $\mathscr{S}$, independent of the phase and product choices, such that the entire batch is exactly the corresponding standard $\mathcal{M}_m$ batch on the one locally gauged parent $D_\delta^{(\mathscr{S})}h$, followed by invertible row/column gauges and the substitutions

$$\widehat{t} = c\, t^\varepsilon, \qquad c \in \{1, \delta, \delta^{-1}\}, \quad \varepsilon \in \{1, -1\}. \tag{59}$$

In particular:

1. *support, zero or incomplete-support status, matching directions, all flattening ranks, and exact factorization are unchanged by the gauges;*
2. *whenever both product coefficients on a common edge are nonzero and their trace is not identically zero, its trace has the form $\kappa(1 + \theta t^\varepsilon)$ with $\kappa\theta \neq 0$ and $\varepsilon \in \{-1, 1\}$; a one-term trace has no forbidden phase, while an identically-zero product is a zero or incomplete-support dispatch according to the other product. Whenever the corresponding protected minor is defined, its transported value is a nonzero monomial in δ, t times the standard protected determinant;*
3. *reversal sends t to $\delta^2/t \in H$, so both orientations use the same finite phase set, and all safe ordered ratios lie in the single coset $\delta^{-1}H$.*
4. *the four samples $E(t_0), E(t_1), O(t_0), O(t_1)$ at one fixed ordered deleted pair, the two products of one fixed trace-cover skeleton, the phase family producing a quaternary from a six-port tensor, and the complete two-copy polarization grid are gauge-coherent. Hence every joint multi-card comparison in the transported proof is made on one common locally gauged parent.*

Proof. The two displayed kernel identities are direct multiplication. For the first identity in Equation (58), expand the Frobenius pairing: $\sum_{a,b} \delta^a K(a, b) g(a, b)$; the second is identical with the factor δ^b. Transposing or reversing an ordered kernel merely switches i and j and contributes the nonzero scalar in Equation (52). Induction over the contraction edges therefore pushes all D_δ's to vertex ports. For a fixed ordered contraction skeleton, the endpoint at which each D_δ lands is

determined only by the ordered edge, not by its phase or by the choice of diagonal versus reflection product. Hence every member of a gauge-coherent batch has the same parent-port set $\mathscr{S}$. Reversed cards are first put in the common orientation using Equation (52), which only changes the recorded scalar and replaces the phase by δ^2/t. On a matching product the remaining external gauges are row and column gauges of its binary factors, so they preserve support, nonsingularity, ranks, and exact tensor-factor identities. Unrelated contraction skeletons may induce different sets $\mathscr{S}$, but their raw coefficients are never compared: support assertions are gauge-invariant, and coefficient identities pull back with their explicit nonzero product-weight factors.

Pushing the entire gauge to the parent leaves the phase unchanged, so $c = 1$. Alternatively, keeping the raw parent uses $K_t^0 = R(\delta t)$ and $K_t^1 = \delta X R(t/\delta)$, giving $c = \delta$ and $c = \delta^{-1}$, respectively. These are the substitutions in Equation (59); a reversed reflection has δ^2/t, and $\delta^2 = \eta^{-2} \in H$. A trace of two monomial factors is a sum of two nonzero monomials. If the two terms have the same exponent and cancel, that product is identically zero (the full card may consequently be zero or partially supported); otherwise, after extracting one term it is $\kappa(1 + \theta t)$ with $\kappa\theta \neq 0$ (or a one-term trace). A unique-lift fusion multiplies a protected minor by the nonzero entry and orientation scalars; a row/column gauge multiplies a determinant by the product of its diagonal entries. This proves the second item and also shows that a zero or partially supported card is zero or partially supported before and after transport. Finally, $\delta^{-1}H = \eta H$ is disjoint from H and is invariant under inversion because $\eta^{-2} \in H$. This proves the ratio and orientation assertions. □

The six-port argument with weighted kernels. The six-port conclusion of Lemma 6.11 also holds in the marked $\mathcal{A}_m$ sector, for even $m \geq 4$, without a simultaneous normalization of its safe-ratio cosets. Work analytically with $\widehat{h} = (U_{\mathcal{A}}^{-1})^{\otimes 6} h$. The exact sector-ruling argument uses only monomial supports and Segre factorization. In its individual binary pencils, the safe ratios lie in $\delta^{-1}\mu_m$; dividing that one ratio by δ^{-1} gives the same $Z/P/C$ support classification. This is a separate analytic calculation for each pencil, not a common port gauge or an available binary. In the proof above replace the parity kernels I, X by the actual effective kernels

$$D_\delta R(1) = \begin{pmatrix} 1 & 0 \\ 0 & \delta \end{pmatrix}, \qquad D_\delta X R(1) = \begin{pmatrix} 0 & 1 \\ \delta & 0 \end{pmatrix}.$$

They have the same respective supports and nonzero entries. Every cross-pair lift in the P/P and P/C arguments and every injective lift in the support atlas therefore acquires only a nonzero weight; all support counts and projection obstructions are unchanged. A matching coset still has an identically zero opposite-parity sector. The resulting exact factorization of $\widehat{h}$ pulls back by the portwise tensor powers of $U_{\mathcal{A}}$ to an exact factorization of the physical h, with its recorded nonzero scalar. No analytic gauge or factor is used as a gadget.

Lemma 6.33 (Primitive transport in the $\mathcal{A}_m$ sector)**.** *Let*

$$\mathsf{Stable}(\Lambda, \mathcal{A}_m, f_0), \qquad \mathsf{Even}(\Lambda), \qquad m \geq 4 \text{ even},$$

and define the phase group H, phase scalar δ, diagonal gauge matrix D_δ, and safe-ratio coset $\mathscr{R}_A$ by

$$H = \mu_m, \qquad \delta = -\eta^{-1}, \qquad D_\delta = \mathrm{diag}(1, \delta), \qquad \mathscr{R}_A = \delta^{-1}H.$$

Let $h \in \Pi(\Lambda)$ be a selected unresolved nonbinary prime occurrence of even arity at least six. Regenerate and close its complete actual oriented deck over the current retained set. Suppose that this occurrence is relatively deck-stable and has produced neither a resolved complexity leaf nor a

strict stable-tuple successor; in particular, all lower-arity prime occurrences exposed by its proper cards or factors have already been processed, and every surviving binary factor belongs to the current complete group. Then every concrete local comparison batch in the normalized-dihedral proof has a gauge-coherent physical counterpart. Its required invariants are the following:

1. *The actual kernels are monomial, their traces remain affine in $t \in H$, reversal permutes H, and every safe ordered ratio lies in the fixed coset $\mathscr{R}_A$. Analytically dividing that ratio by δ^{-1} gives the pencil normal form. Hence all support, ruling, trace-cover, coefficient-minor, endpoint-type, and bounded-reduction arguments transport with their zero cases; unique-lift monomial fusions remain cancellation-free.*

2. *For a logical coefficient table $c = (c_z)_{z\in\mathbb{F}_2^d}$ on an oriented matching and orientation signs $\epsilon_j \in \{1, -1\}$, the product-weight change $\widetilde{c}_z = \delta^{\sum_j \epsilon_j z_j} c_z$ sends safe logical ratios from $\mathscr{R}_A$ to H while preserving Equation (44) and the cross ratio r. The final physical quaternary is routed through the marked-coordinate reduction.*

3. *Fixed δ-weights do not alter support integration. They multiply the Reed–Muller obstruction by a nonzero scalar. In the polarized determinant, they merely reparameterize the full grid and multiply each tested coefficient by a nonzero product-weight value.*

Consequently every branch either

1. *reaches a resolved complexity leaf or a strict stable-tuple successor;*

2. *returns an exact proper-factor or child-card factor batch to the lower-arity occurrence induction; or*

3. *certifies*
$$(U_{\mathcal{A}}^{-1})^{\otimes \operatorname{arity}(h)} h \in \mathscr{P}.$$

When $h = f_0$, the first two alternatives are resolved in the sense of Definition 5.12. For a later occurrence they are charged to the occurrence-level induction described in Section 6.6; no decrease of the active tuple's Ω is asserted. The bounded outputs are actual quaternary or six-port tensors in the current coordinates. Factor alternatives are exact tensor identities pulled back to those coordinates and exposed through Theorem 2.16 and Lemma 2.20. Their Turing reductions are retained in the worklist. The conclusion concerns the selected occurrence h.

Proof. By Lemma 6.26, an actual sector binary is a nonzero multiple of TD_δ^{-1}, and its effective kernel is $D_\delta T$. The gauge identity Lemma 6.32 moves each kernel weight to a specified parent port. It gives the effective kernels

$$D_\delta R(t) = \operatorname{diag}(1, \delta t), \qquad D_\delta X R(t) = \begin{pmatrix} 0 & t \\ \delta & 0 \end{pmatrix}, \tag{60}$$

so a fiber has pencils $A + \delta t D$ and $tB + \delta C$. For a nonproportional pencil, multiply the safe ratio by δ and use the coset-normalized pencil argument; the resulting ratio is in H, so Lemmas 6.9 and 6.10 applies. This is an analytic rescaling only. Reversal sends t to $\delta^2/t \in H$, and the gauge lemma shows that a common-edge trace has at most one forbidden phase in each product. Hence the two-product trace cover still has an admissible phase because $m \geq 4$, while every zero or partially supported card is preserved exactly by the invertible gauges. Every comparison used here belongs to one of the gauge-coherent batches listed in item 4 of Lemma 6.32. Thus its two samples, two products, or full phase grid are evaluated on one common locally gauged parent. Different batches may carry

different portwise gauges, but their conclusions are either support/rank/factor assertions, which are gauge-invariant, or coefficient identities whose product-weight factors are displayed below. In particular, no equation compares raw coefficients from differently gauged parents. The determinant statement in the gauge lemma gives, with an explicitly nonzero scalar κ,

$$\det_{\mathcal{A}}(t) = \kappa\, \delta^{e} t^{u} \det_{\mathcal{M}}(c\, t^{\varepsilon}), \qquad e, u \in \mathbb{Z}, \quad c \neq 0, \quad \varepsilon \in \{-1, 1\},$$

where $\det_{\mathcal{M}}$ denotes the same protected minor for the locally gauged tensor in the standard sector, so protected tags, the six-to-four Laurent coefficient, and the endpoint minor are nonzero exactly in the corresponding standard calculation. Therefore Lemmas 6.4 to 6.8 and 6.12 to 6.15 transport with the same actual closures. Every bounded quaternary output is routed through the marked-coordinate reduction, and every exact factor identity is pulled back by the corresponding portwise tensor power of $U_{\mathcal{A}}$ and then handled with a recorded Turing reduction. For the active minimum occurrence these transported concrete batches are resolved. For a later occurrence, each batch is completely closed inside that occurrence's frozen local queue and charged to the outer occurrence induction. On the branch surviving the corresponding closure, the standard support, ruling, and coefficient invariants remain available.

Use the product-weight convention of Section 2.12.2; for this logical coefficient cube put

$$\chi(z) := \prod_{j} \gamma_j^{z_j} \qquad (\gamma_j \in \mathbb{C}^{\times}).$$

Multiplication by χ preserves supports and all two-face parallelogram identities. It also preserves every complementary-pair product up to one common nonzero scalar. For oriented matched pairs, a safe logical ratio is $\delta^{-\epsilon_j} h_j$, $h_j \in H$, $\epsilon_j \in \{1, -1\}$. The product-weight change

$$\widetilde{c}_z = \delta^{\sum_j \epsilon_j z_j} c_z$$

puts all ratios in H, multiplies both sides of every relation in Equation (44) equally, and leaves

$$r = \frac{c_{000} c_{110}}{c_{100} c_{010}}$$

unchanged. Here the product-weight change is analytic. Since $\delta^2 \in H$, absorb orientation changes into the phases and write the three initial safe ratios as $\delta^{-1} h_j$, $h_j \in H$. The crossing factor cards used in the standard six-port calculation expose these raw ratios: the extra δt multiplies a column of the card and does not change the ratio on its unclosed matching edge. Actual rotations $R(h_j^{-1})$ therefore leave the physical table

$$c_z = \delta^{-|z|} N_r(z), \qquad N_r(z) = \begin{cases} 1, & |z| \leq 1, \\ r, & |z| \geq 2, \end{cases}$$

after its nonzero realization scalar is recorded. To check $r \in H$ without realizing the analytic weight change, use the standard $X_{12}X_{34}X_{56}$ support and close ports $(1, 3)$ by K_t^1. The resulting physical quaternary, with logical row x_2 and column x_5, has coefficient matrix

$$\begin{pmatrix} 1 & r/\delta \\ t/\delta & tr/\delta^2 \end{pmatrix} = \begin{pmatrix} 1 \\ t/\delta \end{pmatrix} (1, r/\delta).$$

Expose its exact binary factors through the factor theorem. Unless a binary exit occurs, $r/\delta \in \delta^{-1} H$, proving $r \in H$. Closing the third matching pair of the physical table by K_t^1 gives

$$Q_t = \operatorname{diag}(1, \delta^{-1}) \begin{pmatrix} 1+t & r+t \\ r+t & r(1+t) \end{pmatrix} \operatorname{diag}(1, \delta^{-1}), \qquad \det Q_t = \delta^{-2}(r-1)(t^2 - r).$$

If $r = 1$, the displayed six-port table already factors. Otherwise at most two phases satisfy $t^2 = r$; choose another phase in H. The resulting nonzero physical quaternary enters the marked-coordinate reduction with the displayed nonzero minor.

Support integration is Boolean and unchanged. On its sole Reed–Muller residue the raw face, in one orientation, is

$$\begin{pmatrix} A & \delta tB \\ \delta tC & D \end{pmatrix}, \qquad A, B, C, D \neq 0.$$

The actual $t = 1$ matching card gives $AD = \delta^2 BC$, so its determinant is $\delta^2 BC(1 - t^2)$. The complementary exponent pattern gives the same obstruction up to sign and a nonzero scalar. It is nonzero for some $t \in H \setminus \{\pm 1\}$. In polarization, each grid variable is merely reparameterized as $\lambda_j = c_j t_j$, $c_j \in \mathbb{C}^\times$, and the remaining weights form a nonzero product weight χ. The grid is a product of cosets $c_j \mu_m$, so Lemma 6.21 applies without requiring $c_j \in H$. Hence the adjacent coefficient becomes

$$[\lambda^{z+z'}]\Delta_\xi = \chi(z)\chi(z')(u_z v_{z'} - u_{z'} v_z)^2 \qquad \text{for adjacent } z, z'.$$

The cross-slice coefficient is similarly multiplied by $\chi(z)^2$. Thus the full grid and all vanishing tests in Lemmas 6.18, 6.20 and 6.22 remain valid.

The transported support integration therefore gives one global matching support for the selected occurrence, and the transported polarization identities integrate its nonzero coefficients into a product of binary factors on that matching. This is precisely $(U_{\mathcal{A}}^{-1})^{\otimes \operatorname{arity}(h)} h \in \mathscr{P}$. If an intermediate identity factors the selected ancestor h itself, replace its occurrence by the tensor-prime occurrences of its proper factors. If instead only a constructed child card factors, process that child's complete factorization tree inside the selected occurrence's frozen local queue; the token for h remains unresolved until its analysis resumes. Together with the product certificate above, this proves the occurrence-level alternatives. □

Theorem 6.34 ($\mathcal{A}_m$-sector deck rigidity)**.** *Assume*

$$\mathsf{Stable}(\Lambda, \mathcal{A}_m, f), \qquad \mathsf{Even}(\Lambda), \qquad m \geq 4, \quad 2 \mid m.$$

Then $\mathsf{Close}_{\{\mathscr{P}\}}(\Lambda, \mathcal{A}_m, f)$.

Proof. Fix $U_{\mathcal{A}}$ from Lemma 6.26 and let $n = \operatorname{arity}(f)$ be the active minimum arity. If $n = 4$, the marked-coordinate reduction just proved gives a whole-signature-set leaf or an established strict successor. For $n = 6$, the independent weighted six-port support argument above gives a proper factorization, contradicting tensor primality.

Suppose $n \geq 8$, and work analytically with $\widehat{f} = (U_{\mathcal{A}}^{-1})^{\otimes n} f$. The two-card localization proof transports directly to the actual weighted kernels: common-edge traces exclude at most two phases; joining two parity discrepancies gives a fused weight ratio $\kappa t^{\pm 2}$, $\kappa \neq 0$; and cycle shortening or vertex replacement has the same unique-lift matching support. These are the explicit kernel and fusion identities in Lemmas 6.6 and 6.32. As $m \geq 4$, each required actual phase exists. A disagreeing pair of sampled cards would therefore localize to a nonmatching six-port tensor. But the first proper physical closure of the stable core is zero or a product of safe current-group binaries, and subsequent contractions preserve that property by the group fusion law. The inherited nonzero disagreeing cards give a contradiction.

Thus at a fixed deleted pair all nonzero samples have one matching and one common varying edge. The elementary pairwise argument used in the standard completion fixes the other factor lines across both sectors. The weighted pencils $A + \delta tD$ and $tB + \delta C$ are invertible from two distinct

nonzero samples, since $\delta \neq 0$. Their four slices have one common nonzero binary product factor, proving an exact proper factorization of $\widehat{f}$. Pulling it back by the invertible portwise matrices $U_{\mathcal{A}}$ gives a proper factorization of the physical core f, again impossible.

No higher-arity stable-core branch survives, and the quaternary route already resolves the entire retained set. This proves $\mathsf{Close}_{\{\mathscr{P}\}}(\Lambda, \mathcal{A}_m, f)$ without promoting a local product-class certificate to relative matching-deck stability. □

Proof of Theorem 6.1. If $G = \mathcal{M}_m$, then deck stability and $\mathsf{Even}(\Lambda)$ give $\mathrm{arity}(f) = \nu(\Lambda) = 2k + 2$ for some k, so apply Theorem 6.25. If $G = \mathcal{B}_m$, apply Theorem 6.29. If $G = \mathcal{A}_m^0$, apply Theorem 6.30. If $G = \mathcal{A}_m$, apply Theorem 6.34. These are exactly the four cases in the statement. □

7 Proper Cyclic Matching Decks and Reed–Muller Rigidity

Throughout this section, an unadorned 0 or **1** used in a vector context denotes the all-zero or all-one vector in the ambient binary space, whose dimension is clear from context. For $S \subseteq \mathbb{F}_2^r$, write $\mathbf{1}[S]$ for the arity-r indicator signature $x \mapsto \mathbf{1}[x \in S]$.

Here the effective kernels span only $\mathrm{span}\{I, X\}$. In the Walsh coordinates of Equation (11), these kernels impose equality of two bits, with a possible phase on their common value. This restricted access leaves one higher-arity core, RM_8. We first reconstruct that core from its pair contractions, then use networks of it to test every retained signature. The code and self-duality conventions needed for the second step are in Section 2.8.

7.1 Cyclic phase pencils and physical dressings

The cyclic group supplies a finite family of phase contractions. The next three lemmas describe which binary pencils they permit and which changes to a quaternary table their actual port dressings can realize.

Here a tensor or matrix *pencil* is a one-parameter linear family $U + tV$ over the stated field. A *sampled phase pencil* restricts t to a displayed finite root-of-unity group, here usually μ_N, and records zero members as well as the projective classes of nonzero members. Two nonzero members are proportional when one is a nonzero scalar multiple of the other. A monomial matrix has exactly one nonzero entry in each row and column; in dimension two it is diagonal or anti-diagonal.

For each integer $N \geq 3$ and $s \in \mu_N$, define the diagonal phase matrix

$$D_s = \mathrm{diag}(1, s).$$

We use $P = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ for the analytic Walsh change of coordinates. Actual networks continue to use the native X-edge; $P^{\mathsf{T}} X P = 2Z$ gives their edge matrix in the tables below.

For $N \geq 3$ and $a, b, c, d \in \overline{\mathbb{Q}}$, define the relation $\mathcal{R}_N(a, b, c, d)$ by declaring that, for some $\alpha \in \mu_N$, at least one of the following systems holds:

$$c = -\alpha a, \qquad d = -\alpha b, \qquad \text{(proportional)}, \tag{61}$$

$$b = c = 0, \qquad d = \alpha a, \qquad \text{(diagonal monomial)}, \tag{62}$$

$$a = d = 0, \qquad c = \alpha b, \qquad \text{(anti-diagonal monomial)}. \tag{63}$$

The alternatives may overlap and include the all-zero quadruple.

For a quaternary signature q, define its analytic Walsh-transform signature by $\widetilde{q} := (P^{-1})^{\otimes 4} q$. When this analytic tensor is supported on even Hamming weight, define its endpoint and central

Walsh coordinates by

$$p = \widetilde{q}_{0000}, \qquad t = \widetilde{q}_{1111}, \qquad x_{ab} = \widetilde{q}_{\mathbf{1}_{\{a,b\}}}, \qquad 1 \le a < b \le 4. \tag{64}$$

Here $\mathbf{1}_{\{a,b\}} \in \mathbb{F}_2^4$ has ones precisely at ports a, b, so x_{ab} indexes one of the six weight-two words.

Lemma 7.1 (Cyclic phase-pencil rigidity, including zeros)**.** *For every integer $N \ge 3$, the relation $\mathcal{R}_N(a,b,c,d)$ is equivalent to the following condition: for every $s \in \mu_N$,*

$$\operatorname{diag}(a - sb, c - sd)Z \quad \textit{is zero or is projectively } D_\eta \textit{ for some } \eta \in \mu_N.$$

Proof. If $a = b = c = d = 0$, both conditions hold, so assume otherwise. Set

$$u(s) = -c + sd, \qquad v(s) = a - sb.$$

The phase-card condition says that $u(s)$ and $v(s)$ are either both zero or both nonzero, and in the latter case $u(s)/v(s) \in \mu_N$. If the two affine linear forms are dependent, the possibility that exactly one is identically zero is excluded by this condition. Hence $u = \alpha v$ for some $\alpha \in \mu_N$, giving Equation (61). If they are independent, neither form vanishes on μ_N, and $\psi(s) := u(s)/v(s)$ is a nonconstant Möbius transformation. The card condition gives $\psi(\mu_N) \subseteq \mu_N$; injectivity and equality of the finite cardinalities give $\psi(\mu_N) = \mu_N$. Thus Lemma 6.9 applies and gives $\psi(s) = \alpha s$ or α/s, with $\alpha \in \mu_N$. Cross multiplication yields Equations (62) and (63). Direct substitution proves the converse, including the all-zero case. □

For the two physical cyclic lemmas below, fix a finite algebraic retained signature set Λ in K-coordinates, with native edge X, and fix an integer $N \ge 3$. For every $s \in \mu_N$, fix an actual ordered binary gadget C_s over Λ, with a specified orientation, whose analytic Walsh matrix is

$$\widetilde{C}_s = \kappa_s D_s Z, \qquad \kappa_s \in \overline{\mathbb{Q}}^{\times}.$$

The nonzero scalar κ_s is recorded with the gadget. Throughout these lemmas, "actual" and "directly realizable" are relative to this same Λ, and every displayed phase-card matrix is divided by its recorded nonzero scalar. In the proper-cyclic application, analytic Walsh placement puts $G(\Lambda)$ in the standard cyclic form $C_N = \{[D_s] : s \in \mu_N\}$, and the C_s are fixed actual representatives of all its classes.

Lemma 7.2 (Cyclic quaternary phase family)**.** *Let q be an actual quaternary signature over Λ, and suppose $\widetilde{q} = (P^{-1})^{\otimes 4} q$ is supported on the even-weight Walsh sector. Define p, t, x_{ab} by Equation* (64)*. Assume that, for every complementary pair, both orders of its two residual ports, and every $s \in \mu_N$, contraction with the fixed actual representative C_s gives either the zero card or a nonsingular binary card whose Walsh transfer class belongs to $C_N = \{[D_\eta] : \eta \in \mu_N\}$. Equivalently, define*

$$\mathfrak{I} = \{\{1,2\},\{1,3\},\{1,4\}\}, \qquad I^c = [4] \setminus I,$$

and write $x_I := x_{ab}$ for $I = \{a,b\}$. For every $I \in \mathfrak{I}$, $s \in \mu_N$, and $\varepsilon \in \{+,-\}$, the corresponding normalized actual Walsh card has matrix

$$B_{I,+}(s) = \operatorname{diag}(p - s x_I, x_{I^c} - st),$$
$$B_{I,-}(s) = \operatorname{diag}(p - s x_{I^c}, x_I - st).$$

Here + records deletion of I and − deletion of I^c; both orders of the remaining two ports are part of the assumption. Each displayed card is zero or satisfies

$$[B_{I,\varepsilon}(s)Z] \in C_N.$$

The right factor Z is only the Walsh transfer convention and is not an additional gadget. Organize the eight coordinates into the endpoint block $\mathbf{v}_0 = (p, t)$ and the three central complement blocks $\mathbf{v}_I = (x_I, x_{I^c})$, $I \in \mathfrak{I}$. Then exactly one of the following holds:

1. *If $p, t \neq 0$, then $\lambda = t/p \in \mu_N$, and each central block $\mathbf{v}_I$ is zero or*

$$\mathbf{v}_I = (-p\lambda\alpha_I^{-1}, -p\alpha_I), \qquad \alpha_I \in \mu_N. \tag{65}$$

2. *If $p = t = 0$, each central block $\mathbf{v}_I$ is zero or*

$$\mathbf{v}_I = \rho_I(1, \alpha_I), \qquad \rho_I \in \overline{\mathbb{Q}}^{\times}, \quad \alpha_I \in \mu_N.$$

In particular, exactly one of p, t cannot be nonzero. Conversely, every even Walsh table in these alternatives has this zero-or-cyclic property under every phase contraction.

Proof. For a complementary block I, I^c, the phase card is

$$\operatorname{diag}(p - sx_I, x_{I^c} - st), \qquad s \in \mu_N.$$

Apply Lemma 7.1 to $(a, b, c, d) = (p, x_I, x_{I^c}, t)$, and then interchange I, I^c. If $p, t \neq 0$, the anti-diagonal system is impossible. If one of the two complementary-deletion systems is diagonal, then $x_I = x_{I^c} = 0$ and $t/p \in \mu_N$; the other system is then diagonal as well. In the remaining case both systems are proportional. Taking their parameters to be $\alpha, \beta \in \mu_N$ gives

$$x_{I^c} = -\alpha p, \qquad x_I = -\beta p, \qquad t = \alpha\beta p.$$

Thus $t/p = \alpha\beta \in \mu_N$, and choosing $\alpha_I = \alpha$ gives the displayed parametrization. If exactly one endpoint is nonzero, neither monomial system is possible and the two proportional systems force respectively $x_I = 0, x_{I^c} \neq 0$ and $x_{I^c} = 0, x_I \neq 0$, a contradiction. If $p = t = 0$, proportional and diagonal systems give only the zero block; every nonzero block is therefore anti-diagonal, $\mathbf{v}_I = \rho_I(1, \alpha_I)$. Direct substitution checks both complementary deletions and every card arising from a zero complementary block, proving the converse. □

Lemma 7.3 (Actual quaternary cyclic dressing)**.** *Let q be an actual quaternary signature over Λ. For $r_1, r_2, r_3, r_4 \in \mu_N$, attach the fixed representative C_{r_i}, in its specified orientation, to port i of q through one native X-edge, leaving the other port of C_{r_i} exposed. Denote the normalized output signature by h. After division of the direct-gadget boundary tensor by the recorded nonzero scalar $2^4 \prod_i \kappa_{r_i}$, its Walsh coordinates are*

$$\widetilde{h}_y = \widetilde{q}_y \prod_{i=1}^{4} r_i^{y_i}.$$

Define the map $F_N : \mu_N^4 \to \mu_N^4$ by

$$F_N(r_1, r_2, r_3, r_4) = (R, r_{12}, r_{13}, r_{14}), \qquad R = \prod_{i=1}^{4} r_i, \quad r_I = \prod_{i \in I} r_i,$$

and put $(\mu_N)^2 := \{z^2 : z \in \mu_N\}$. A target $(R, u, v, w) \in \mu_N^4$ lies in the image of F_N exactly when

$$\frac{uvw}{R} \in (\mu_N)^2. \tag{66}$$

For odd N, F_N is bijective.

Proof. At port i, the fixed Walsh matrix $\kappa_{r_i} D_{r_i} Z$, followed by the analytically transformed native edge $P^{\mathsf{T}} X P = 2Z$, acts as $2\kappa_{r_i} D_{r_i}$. The four attachments therefore give the stated actual dressing and recorded scalar. Necessity is $uvw/R = r_1^2$; for sufficiency choose

$$r_1^2 = uvw/R, \qquad r_2 = u/r_1, \qquad r_3 = v/r_1, \qquad r_4 = w/r_1.$$

These choices satisfy $F_N(r_1, r_2, r_3, r_4) = (R, u, v, w)$; for odd N, squaring on μ_N is bijective. □

7.2 Quaternary entry and theorem statement

This branch, $\mathscr{K}_G = \mathrm{span}\{I, X\}$, is the marked proper-cyclic row of Xia's taxonomy; see Remark 5.15. Its arity-four boundary is independent.

Lemma 7.4 (Proper-cyclic quaternary boundary)**.** *Let (Λ, G, q) be a deck-stable retained state with the distinguished literal equality I available,* $\mathrm{arity}(q) = \nu(\Lambda) = 4$*, kernel span $\mathscr{K}_G = \mathrm{span}\{I, X\}$, and finite cyclic $G \ni [X]$. Then either an unsafe actual binary is exposed, factor access strictly enlarges the recomputed complete group, or an actual gadget using at most two copies of q realizes a tensor-prime endpoint-nondegenerate eight-vertex signature. The boundary thus returns a resolved leaf, an outer deck successor under Lemma 5.11, or* $\mathsf{Tract}(K\Lambda)$*.*

Proof. Conjugate analytically to Walsh coordinates and continue to write G for the conjugated projective group. The proper algebra is diagonal:

$$G = C_N = \{[D_s] : D_s = \mathrm{diag}(1, s),\ s \in \mu_N\}, \qquad 2 \mid N.$$

Here $[X]$ becomes $[Z] = [D_{-1}]$, and actual representatives are proportional to $D_s Z$. To justify the parity reduction, fix an ordered deleted pair and an odd-weight word y on the two residual ports. Up to a fixed nonzero representative scalar and a possible sign, the corresponding off-diagonal phase-card entry is $u + sv$, where u and v are the contributions of the two Walsh coefficients whose deleted-pair assignments are 00 and 11, respectively, and whose residual word is y. A zero or cyclic card is diagonal, so this entry vanishes. At the two actual phases $s = 1, -1$ we obtain $u + v = u - v = 0$, hence $u = v = 0$. Running through all six deleted pairs and both port orders (equivalently, the complete ordered deck) forces every odd-weight Walsh coefficient to vanish. Thus $\widetilde{q}$ is supported on even Walsh parity, as required by Lemma 7.2. Item 1 of Theorem 5.18 handles $N = 2$; hence assume $N \geq 4$ and use the endpoint and complement-block coordinates of Equation (64), indexed by

$$I_1 = \{1, 2\}, \qquad I_2 = \{1, 3\}, \qquad I_3 = \{1, 4\}.$$

The cyclic quaternary-family result in Lemma 7.2 leaves two endpoint regimes. We treat the zero-endpoint regime first; in the nonzero regime, the dressing-image criterion further separates successful dressing from the residual square obstruction. Define the block phase parameters $\alpha_i := \alpha_{I_i}$ and, for nonzero endpoints, the endpoint ratio $\lambda := t/p$. Call the complementary block indexed by I_i *active* when $(x_{I_i}, x_{I_i^c}) \neq (0, 0)$, and *inactive* otherwise.

For each actual two-copy output below, let Λ^+ be the retained augmentation after preprocessing, complete factor saturation, and group recomputation. Zero, nullary, unary, odd, singular, and infinite-order factors take their established exits. The remaining calculations concern the all-even finite-group branch. Proper factors become available over Λ^+ through the factor theorem, with its Turing reduction and nonzero normalization scalars recorded.

Zero endpoints. Write each complementary block with scale ρ_i as $(x_{I_i}, x_{I_i^c}) = \rho_i(1, \alpha_i)$, taking $\rho_i = 0$ for an inactive block and then choosing any $\alpha_i \in \mu_N$, and define the auxiliary scalar

$X_i := \alpha_i \rho_i^2$. For each i, choose the remaining indices $j < k$, so $\{i, j, k\} = \{1, 2, 3\}$, and order $I_i = (1, i+1)$. Connect the first port of one copy to the first port of the other by the actual representative $D_{\alpha_j \alpha_k} Z$, and connect the second ports by I (with the same orientation on both copies). If Q_i is the $I_i \mid I_i^c$ flattening, let $\widetilde{g}_i$ denote the normalized analytic output

$$\widetilde{g}_i = Q_i^{\mathsf{T}} K_i Q_i, \qquad K_i = (D_{\alpha_j \alpha_k} Z) \otimes I = \operatorname{diag}(1, 1, -\alpha_j \alpha_k, -\alpha_j \alpha_k).$$

Order the residual ports as the two ports of I_i^c in the first copy followed by those in the second copy, each in inherited order. The only potentially nonzero complementary pairs of $\widetilde{g}_i$ are

$$E = (\widetilde{g}_i(0000), \widetilde{g}_i(1111)), \qquad D = (\widetilde{g}_i(0101), \widetilde{g}_i(1010)).$$

Their entries are

$$(E_0, E_1) = (-\alpha_j \alpha_k \rho_i^2,\ \alpha_i^2 \rho_i^2),$$
$$(D_0, D_1) = (\alpha_j (X_j - X_k),\ -\alpha_k (X_j - X_k)),$$

and, whenever both displayed pairs E and D are nonzero,

$$\frac{D_1/D_0}{E_1/E_0} = \left(\frac{\alpha_k}{\alpha_i}\right)^2.$$

Set

$$\Delta_i := (X_j - X_k)^2 - X_i^2.$$

We now verify endpoint nondegeneracy, including the case of an inactive i-block. If $\rho_i \neq 0$, the physical dressing criterion Lemma 7.3 and Equation (66) is applied in the displayed residual-port order. Put $(u, v, w) = (r_{12}, r_{13}, r_{14})$ for these four output ports. The entries at 0101 and 1010 are multiplied by R/v and v, respectively. Take

$$R = (E_1/E_0)^{-1}, \qquad u = 1, \qquad v = -\alpha_j/\alpha_i, \qquad w = R/v.$$

All four parameters lie in μ_N, since N is even, and $uvw/R = 1$. Moreover

$$E_1 R = E_0, \qquad D_0 R/v = D_1 v = \frac{\alpha_j \alpha_k}{\alpha_i}(X_j - X_k).$$

Thus the two entries in each displayed block are equal. Since the Walsh support is even, the two physical endpoints coincide; up to one common nonzero scalar their common value is

$$(X_j - X_k) - X_i.$$

It is nonzero whenever $\Delta_i \neq 0$. If instead $\rho_i = 0$, then $E_0 = E_1 = 0$, while $\Delta_i \neq 0$ implies $D_0 D_1 \neq 0$. Set

$$u = v = 1, \qquad R = D_1/D_0 = -\alpha_k/\alpha_j, \qquad w = R.$$

Here $uvw/R = 1$, so the same criterion applies and makes the two dressed entries of the D-block equal and nonzero. Inverse Walsh transformation again gives two nonzero physical endpoints. Thus every index with $\Delta_i \neq 0$, active or inactive, yields an endpoint-nondegenerate eight-vertex signature. In the displayed four-word matching-code support, the two crossed $2+2$ flattenings have rank at least two; the remaining logical flattening has determinant, up to a nonzero scalar,

$$\alpha_j \alpha_k \Delta_i.$$

If some $\Delta_i \neq 0$, the corresponding actual output has no binary factor. The dressing is a product of nonsingular local maps, so it preserves tensor factorizations. Even parity and its complementary endpoints exclude a unary factor, so the dressed endpoint-nondegenerate eight-vertex signature is tensor-prime.

It remains to exclude simultaneous vanishing of the three minors. If $\Delta_1 = \Delta_2 = \Delta_3 = 0$, choose signs $\epsilon_i \in \{1, -1\}$ such that

$$\begin{aligned} X_2 - X_3 &= \epsilon_1 X_1, \\ X_3 - X_1 &= \epsilon_2 X_2, \\ X_1 - X_2 &= \epsilon_3 X_3. \end{aligned} \tag{67}$$

The coefficient matrix of this system is

$$M_\epsilon = \begin{pmatrix} -\epsilon_1 & 1 & -1 \\ -1 & -\epsilon_2 & 1 \\ 1 & -1 & -\epsilon_3 \end{pmatrix}, \qquad \det M_\epsilon = -(\epsilon_1\epsilon_2\epsilon_3 + \epsilon_1 + \epsilon_2 + \epsilon_3).$$

For the two constant sign triples this determinant is nonzero. For each of the other six, direct substitution gives, up to permutation, precisely the kernel lines $\mathrm{span}_{\mathbb{C}}\{(1,1,0)\}$, $\mathrm{span}_{\mathbb{C}}\{(1,0,1)\}$, and $\mathrm{span}_{\mathbb{C}}\{(0,1,1)\}$. Hence every nonzero solution is a permutation of $(\xi, \xi, 0)$, with $\xi \neq 0$. Since $q \neq 0$, not all three X_i vanish.

Relabel so that $(X_1, X_2, X_3) = (\xi, \xi, 0)$. Then $\rho_1\rho_2 \neq 0$ and $\rho_3 = 0$. In the $I_3 \mid I_3^c$ flattening of $\widetilde{q}$, order the odd assignments on either side as $10, 01$. Its only nonzero block is

$$\begin{pmatrix} \rho_1 & \rho_2 \\ \alpha_2\rho_2 & \alpha_1\rho_1 \end{pmatrix}, \qquad \det = \alpha_1\rho_1^2 - \alpha_2\rho_2^2 = X_1 - X_2 = 0.$$

All four entries are nonzero, so the block has rank one and $\widetilde{q}$ has a genuine binary-by-binary tensor factorization across $I_3 \mid I_3^c$. The Walsh change is a product of invertible local maps, so q has the same genuine factorization. Its factors would be exposed only through factor saturation; under the present deck-stable hypotheses the identity already contradicts that q is the retained tensor-prime core. Therefore some $\Delta_i \neq 0$, as required.

Nonzero endpoints and successful dressing. Define the ratio scalar $R := \lambda^{-1}$ and the dressing scalar $r_I := \epsilon_I \alpha_I \lambda^{-1}$ on each active block, where $\epsilon_I \in \{\pm 1\}$; abbreviate $\epsilon_i := \epsilon_{I_i}$. An inactive block removes the square constraint: its otherwise irrelevant dressing parameter may be chosen so that the criterion's product equation $uvw/R = 1$ holds; with all three active, Lemma 7.3 and Equation (66) applies exactly when $\epsilon_1\epsilon_2\epsilon_3\alpha_1\alpha_2\alpha_3$ is a square. If $N \equiv 2 \pmod 4$, signs remove the obstruction; when $4 \mid N$, this works precisely if $\alpha_1\alpha_2\alpha_3$ is a square. The dressed endpoint block is (p, p); every active complementary block becomes $(-\epsilon_I p, -\epsilon_I p)$, while every inactive block remains $(0, 0)$. Let h be the resulting physical signature. Equality of complementary Walsh blocks makes h even-parity, while its even Walsh support makes it complement-invariant, meaning $h(y) = h(y + \mathbf{1}^4)$ for every $y \in \mathbb{F}_2^4$; it is nonzero because $p \neq 0$. Choose an even support word x with $h(x) \neq 0$, and attach actual X-dressings on exactly the coordinates where x is one. This even translation preserves parity, and the translated signature h' satisfies

$$h'(0000) = h(x) = h(x + 1111) = h'(1111) \neq 0.$$

Thus h' is an endpoint-nondegenerate eight-vertex signature. Every dressing used here is an invertible one-port map, so it also preserves the tensor-primality of q.

Square obstruction. The only remaining case is

$$4 \mid N, \qquad p, t \neq 0, \qquad \text{all three blocks active}, \qquad \alpha_1\alpha_2\alpha_3 \notin (\mu_N)^2.$$

Normalize $p = 1$. For a permutation (i, j, k) of $(1, 2, 3)$, abbreviate

$$(a, b, c) := (\alpha_i, \alpha_j, \alpha_k),$$

and order the $I_i \mid I_i^c$ flattening as $00, 01, 10, 11$:

$$Q_i = \begin{pmatrix} 1 & 0 & 0 & -a \\ 0 & -b & -c & 0 \\ 0 & -\lambda c^{-1} & -\lambda b^{-1} & 0 \\ -\lambda a^{-1} & 0 & 0 & \lambda \end{pmatrix}.$$

Let $D := Z \otimes Z$ be the diagonal four-state matrix, and let S be the permutation matrix that swaps $01, 10$. The actual joins are

$$\widetilde{G}_i^{\parallel} = 4 Q_i D Q_i^{\mathsf{T}},$$
$$\widetilde{G}_i^{\times} = 4 Q_i D S Q_i^{\mathsf{T}}.$$

The crossed one is

$$\frac{1}{4}\widetilde{G}_i^{\times} = \begin{pmatrix} 1 + a^2 & 0 & 0 & -\lambda(a + a^{-1}) \\ 0 & -2bc & -2\lambda & 0 \\ 0 & -2\lambda & -2\lambda^2/(bc) & 0 \\ -\lambda(a + a^{-1}) & 0 & 0 & \lambda^2(1 + a^{-2}) \end{pmatrix}.$$

By Lemma 5.3, an unsafe phase card is already a binary exit. Otherwise the cyclic pencil of the displayed central block forces, when $1 + a^2 \neq 0$,

$$\frac{2\lambda}{1 + a^2} \in \mu_N.$$

Since a is a root of unity, $|1 + a^2| = 2$ gives $a^2 = 1$, whereas $1 + a^2 = 0$ gives $a^2 = -1$. Hence

$$\alpha_i^2 \in \{1, -1\} \qquad (i = 1, 2, 3).$$

If $8 \mid N$, all fourth roots are squares, contradicting the obstruction. Thus $N \equiv 4 \pmod 8$, and an odd number of the three squares equals -1. Choose the permutation (i, j, k) so that $\alpha_i^2 = a^2 = -1$, define the ratio scalar $r := \lambda/(bc)$, and evaluate the straight join. Its sole nonzero block has rank one and factors through the nonsingular anti-diagonal matrix A_r, giving

$$\frac{1}{4}\widetilde{G}_i^{\parallel} = \pm 2 A_r \otimes A_r, \qquad A_r := \begin{pmatrix} 0 & 1 \\ r & 0 \end{pmatrix}, \qquad r \neq 0,$$

on the two residual copies of I_i. Lemma 2.20 exposes the nonsingular anti-diagonal A_r; its transfer obeys

$$(A_r Z)^2 = -rI, \qquad (A_r Z) D_s (A_r Z)^{-1} = s D_{s^{-1}} \quad (s \in \mu_N).$$

Thus $[A_r Z]$ is a safe involution outside C_N and generates its finite dihedral extension. Concretely, close the factor batch first and recompute $G(\Lambda^+)$: nonsingularity follows from $r \neq 0$, the displayed relations prove finite dihedral closure, and the anti-diagonal projective class is not in the diagonal group C_N. Hence this is a strict pulled-back group successor, not an unprocessed fixed-(H, ν) factor. The first two cases pass an actual endpoint-nondegenerate eight-vertex signature tensor to Corollary 4.7, while the third strictly enlarges the complete group. This proves the lemma. □

The preceding lemma supplies the quaternary entry to the branch: arity four is either resolved or passed to a strict outer successor. With that boundary in place, the main statement can isolate the only common-presentation rigidity leaf that remains.

Theorem 7.5 (Reed–Muller deck rigidity)**.** *Assume*

$$\mathsf{Stable}(\mathcal{G}, G, f), \qquad \operatorname{arity}(f) = \nu(\mathcal{G}), \qquad \mathscr{K}_G = \operatorname{span}\{I, X\}.$$

Then $\mathsf{Close}_{\{\mathsf{FLag}\}}(\mathcal{G}, G, f)$*. On its sole rigidity branch,*

$$G = \{[I], [X]\}, \qquad f \sim \mathrm{RM}_8 = \mathbf{1}[RM(1,3)],$$

where $\sim$ *means equality up to a nonzero scalar, a permutation of ports, and actual one-port* X*-dressings.*

7.3 Reconstruction of the minimum Reed–Muller core

Let f be a minimum-arity nonbinary prime of the retained signature set, and let its arity be $n := \operatorname{arity}(f)$. If n is odd, apply Theorem 2.15 to the ordinary-coordinate set $K\mathcal{G}$. Its hardness conclusion pulls back through the holographic K-coordinate equivalence, while its tractable conclusion is exactly $\mathsf{Tract}(K\mathcal{G})$. Thus, on the continuing branch, n is even. Every proper card is zero or a nonsingular G-matching signature; in Walsh coordinates the latter is a weighted-equality product. Define its analytic Walsh coefficients by

$$a_S := \widetilde{f}(\mathbf{1}_S), \qquad S \subseteq [n].$$

For distinct i, j, label and normalize the two actual cards obtained from the I/X-closures as C^+_{ij} and C^-_{ij} so that, for every $T \subseteq [n] \setminus \{i, j\}$,

$$\widetilde{C}^+_{ij}(\mathbf{1}_T) = 2(a_T + a_{Tij}), \qquad \widetilde{C}^-_{ij}(\mathbf{1}_T) = 2(a_T - a_{Tij}), \tag{68}$$

where $Tij = T \cup \{i, j\}$. A nonzero card has the form

$$c \prod_{\{p,q\} \in M} \mathbf{1}[z_p = z_q] r_{pq}^{z_p}, \qquad c \prod_{\{p,q\} \in M} r_{pq} \neq 0, \tag{69}$$

for a perfect matching M; it is nonzero at zero, and its supported weight-two words are exactly the edges of M. Arity four is Lemma 7.4, so assume $n \geq 6$.

Define the zero-card graph Z_f on $[n]$, writing ij for the unordered pair $\{i, j\}$, by

$$ij \in E(Z_f) \quad \Longleftrightarrow \quad C^+_{ij} = C^-_{ij} = 0.$$

By Equation (68),

$$ij \in E(Z_f) \quad \Longleftrightarrow \quad \operatorname{supp} \widetilde{f} \subseteq \{z : z_i + z_j = 1\}.$$

Lemma 7.6 (Elimination of the zero-card graph)**.** *Let* $f \neq 0$ *be the fixed minimum core of arity* n*. Every connected component of* Z_f *is complete bipartite. If* $n > 4$*, then* $E(Z_f) = \varnothing$*.*

Proof. Zero-edge equations alternate along paths, so every nontrivial connected component is complete bipartite. Suppose that $uv \in E(Z_f)$. Since $n > 4$, the set $[n] \setminus \{u, v\}$ has at least three vertices. The graph induced there is a disjoint union of bipartite graphs and therefore is not complete; choose distinct $i, j \notin \{u, v\}$ with $ij \notin E(Z_f)$. At least one of C^+_{ij}, C^-_{ij} is then a nonzero actual card. Because u, v remain external and $\operatorname{supp} \widetilde{f} \subseteq \{z : z_u + z_v = 1\}$, that card vanishes at the all-zero boundary word. This contradicts Equation (69), under which every nonzero card is nonzero at zero. Hence $E(Z_f) = \varnothing$. □

Thus, for the remaining arities, no coordinate pair is invisible to both physical contractions: every pair supports at least one nonzero matching card. We may therefore compare the local card matchings through the common pencil Equation (68). The first global datum extracted from that comparison is the following weight-two graph.

The weight-two graph. Assume henceforth that $n \geq 6$ and $E(Z_f) = \varnothing$. Define the weight-two edge set

$$W_f := \{ij : 1 \leq i < j \leq n,\ a_{ij} \neq 0\}.$$

Let $\mathcal{W}_f := ([n], W_f)$ be the corresponding simple graph. For $S \subseteq [n]$, write

$$W_f[S] := \{pq \in W_f : p, q \in S\}, \qquad W_f[S] - k := W_f[S \setminus \{k\}]$$

for the restricted edge sets.

Lemma 7.7 (Matching consistency)**.** *Let f be the fixed minimum core of arity $n \geq 6$, and assume $E(Z_f) = \varnothing$. Then the following statements hold.*

1. *$a_\varnothing \neq 0$.*
2. *W_f is a matching.*
3. *If both C^+_{ij} and C^-_{ij} are nonzero, their support matchings are equal.*
4. *Every edge $pq \in W_f$ is a global Walsh equality:*

$$\operatorname{supp} \widetilde{f} \subseteq \{z : z_p = z_q\}.$$

Proof. If $a_\varnothing = 0$, then $\widetilde{C}^{\pm}_{ij}(0) = \pm 2a_{ij}$. Since $E(Z_f) = \varnothing$ and matching cards are nonzero at zero, every $a_{ij} \neq 0$. The two ij-cards would then cover every edge on the remaining at least four ports, impossible for two matchings. Hence $a_\varnothing \neq 0$.

For a deleted pair i, j, let P, Q be the support matchings when both cards survive, and write $P \triangle Q$ for their symmetric difference. Their weight-two entries imply

$$W_f[[n] \setminus \{i, j\}] \subseteq P \cup Q, \qquad P \triangle Q \subseteq W_f[[n] \setminus \{i, j\}].$$

Indeed, for $pq \in W_f[[n] \setminus \{i, j\}]$ the two pencil values cannot both vanish; if one whole card vanishes, the other has value $\pm 4a_{pq}$ at pq. Thus, with one nonzero card, $W_f[[n] \setminus \{i, j\}]$ lies in its perfect matching. We use the following elementary graph argument. If a vertex of $\mathcal{W}_f$ had degree at least three, delete two vertices outside it and its three neighbours (possible since $n \geq 6$); the induced graph has degree at least three, whereas the union of two perfect matchings has maximum degree two (and one matching has maximum degree one), a contradiction. Hence every component of $\mathcal{W}_f$ is a path or a cycle.

Suppose a path component has at least two edges. If two vertices lie outside that component, delete them; otherwise (then the component has at least $n - 1 \geq 5$ vertices) delete two consecutive vertices at one end, leaving a path with at least two edges. For this deleted pair, one nonzero card is impossible. If both cards are nonzero, the two edges at the middle of the retained path must lie one in P and one in Q, hence in $P \triangle Q$. Alternating along the path forces an edge of $P \triangle Q$ at an endpoint, but $P \triangle Q$ is a disjoint union of cycles and $P \triangle Q \subseteq W_f[[n] \setminus \{i, j\}]$, whose endpoint has no second edge. This is impossible. A cycle is handled similarly: a triangle is retained by deleting vertices outside it and cannot alternate; for a 4-cycle delete one cycle vertex and one outside vertex, and for a cycle of length at least five delete two adjacent cycle vertices, leaving a path with at least

two edges. Thus every component has at most one edge, so W_f is a matching. Finally, for any pair with both cards nonzero, $P \triangle Q$ is a union of cycles contained in the matching W_f; therefore $P \triangle Q = \varnothing$ and $P = Q$.

For $pq \in W_f$, suppose $z \in \operatorname{supp} \widetilde{f}$ and $z_p \neq z_q$. Choose distinct $i, j \in [n] \setminus \{p, q\}$ with $z_i = z_j$. Both ij-cards vanish on this restriction because every nonzero one contains pq; Equation (68) then kills both lifts, contradicting $z \in \operatorname{supp} \widetilde{f}$. Hence $z_p = z_q$ on the support. □

When $W_f \neq \varnothing$, Lemma 7.7 supplies global equality constraints. The remaining issue is whether the coefficients separate across these pairs. The following lemma recovers that product structure from diagonal restrictions.

Lemma 7.8 (Diagonal-slice integration)**.** *Let $r \geq 3$, and let $h : \mathbb{F}_2^r \to \mathbb{C}^\times$ satisfy $h(0) = 1$. Suppose that, for every $i \neq j$, the restriction $h|_{t_i = t_j}$, viewed as a function of the common bit and the other $r - 2$ bits, is a product of unary functions (equivalently, is a simple tensor). If $r \geq 4$, then*

$$h(t) = \prod_{i=1}^{r} \rho_i^{t_i} \qquad (\rho_i \in \mathbb{C}^\times).$$

If $r = 3$, there are $\rho_1, \rho_2, \rho_3, \kappa \in \mathbb{C}^\times$ such that, for distinct i, j,

$$h(e_i) = \rho_i, \qquad h(e_i + e_j) = \kappa \rho_i \rho_j, \qquad h(111) = \kappa \rho_1 \rho_2 \rho_3. \tag{70}$$

Proof. Use the multiplicative subset expansion

$$h(t) = \prod_{\varnothing \neq S \subseteq [r]} \theta_S^{\prod_{i \in S} t_i}, \qquad \theta_S \in \mathbb{C}^\times. \tag{71}$$

Multiplicative Möbius inversion determines its parameters:

$$\theta_S = \prod_{T \subseteq S} h(\mathbf{1}_T)^{(-1)^{|S|-|T|}}.$$

The parameter θ_S measures the interaction among the coordinates in S; unary factors account exactly for the parameters with $|S| = 1$. Thus the table is simple if and only if $\theta_S = 1$ for every $|S| \geq 2$. On setting $t_i = t_j$, a parameter indexed by a nonempty $U \subseteq [r] \setminus \{i, j\}$ and the common bit is

$$\theta_{U \cup \{i\}} \theta_{U \cup \{j\}} \theta_{U \cup \{i,j\}} = 1. \tag{72}$$

Parameters θ_S with $S \cap \{i, j\} = \varnothing$ and $|S| \geq 2$ must also equal one.

Assume $r \geq 4$. Choosing i, j outside S first gives $\theta_S = 1$ for $2 \leq |S| \leq r - 2$. For $|S| = r - 1$, choose distinct $i, j \in S$ and define the index set $U := S \setminus \{i, j\}$ in Equation (72); the other two factors have index-set cardinality $r - 2$, so $\theta_S = 1$. Taking $U = [r] \setminus \{i, j\}$ then gives $\theta_{[r]} = 1$. Only the unary parameters remain in Equation (71), proving the product formula.

Let $r = 3$. The three instances of Equation (72) imply

$$\theta_{12} = \theta_{13} = \theta_{23} =: \kappa, \qquad \theta_{123} = \kappa^{-2}.$$

Substitution in Equation (71), with $\rho_i = \theta_{\{i\}}$, gives Equation (70). □

Lemma 7.9 (Integration of a nonempty weight-two graph)**.** *Let f be the fixed minimum core of arity $n \geq 6$, with $E(Z_f) = \varnothing$. If $W_f \neq \varnothing$, then either $n = 6$ and a constant-size two-copy gadget realizes the even-parity signature of arity four, or f admits a nontrivial tensor factorization. In particular, a minimum core of arity at least eight has $W_f = \varnothing$.*

Proof. Fix $E_1 = \{u, v\} \in W_f$ and define the residual port set $R := [n] \setminus E_1$. For $k \in R$, let $A_k := C^+_{uk}$. Since $\widetilde{A}_k(0) = 2a_\varnothing \neq 0$, this is a nonzero matching card. The global equality on E_1 gives the unique lift

$$\begin{aligned}
\widetilde{A}_k(z_v, z_{R\setminus\{k\}}) &= 2\widetilde{f}(z_u = z_v, z_k = z_v, z_{R\setminus\{k\}}),\\
\widetilde{A}_k(0) &= 2a_\varnothing \neq 0,\\
\widetilde{A}_k(e_p + e_q) &= 2a_{pq} \quad (p, q \in R \setminus \{k\}),\\
\widetilde{A}_k(e_v + e_q) &= 2a_{\{u,v,k,q\}} \quad (q \in R \setminus \{k\}).
\end{aligned}$$

Its matching therefore consists of $W_f[R] - k$ and at most one edge from v. To justify the last assertion, let U be the set of vertices not covered by W_f; it excludes u, v. For every k, all vertices of $U \setminus \{k\}$ remain on the card boundary and have no available weight-two edge, except that at most one can be paired through v. If $W_f = \{E_1\}$, then $|U| = n - 2 \geq 4$, and the preceding restriction already gives a contradiction. Otherwise a covered vertex of W_f lies in R, so choose $k \in R \setminus U$; hence $|U| \leq 1$. Since $|U| = n - 2|W_f|$ is even, $U = \varnothing$. Thus W_f is perfect; write its edges $E_1, \ldots, E_s$, where $s = n/2$.

By the global equality conclusion of Lemma 7.7, for every word z constant on each E_r, let $t_r(z)$ denote that common edge bit. There is a logical table $h : \mathbb{F}_2^s \to \mathbb{C}$ such that

$$\widetilde{f}(z) = \begin{cases} h(t_1(z), \ldots, t_s(z)), & z|_{E_r} \text{ is constant for every } r,\\ 0, & \text{otherwise.} \end{cases}$$

For two distinct W_f-edges E_i, E_j, close one endpoint of E_i to one endpoint of E_j, forming separately the two cards with the actual $+$ and $-$ kernels. Since cross coefficients between different W_f-edges vanish and $a_\varnothing \neq 0$, both cards are nonzero. Both matching supports impose $t_i = t_j$; the $+/-$ choice changes only the sign character in that common logical bit. Since $s = n/2 \geq 3$, every $t \in \mathbb{F}_2^s$ has a pair $i \neq j$ with $t_i = t_j$; use the plus card for that pair. The boundary word obtained from t by retaining the mates of E_i, E_j and the two ports of every other E_r lies in this card's matching support. The closure equation has a unique compatible lift in the global equality code, namely the word with edge bits t, so the plus-card value is a fixed nonzero scalar times $h(t)$ (there is no second lift with which it could cancel). A nonzero matching product is nonzero at every supported boundary word; hence $h(t) \neq 0$ for every t, and h has full support. Rescale it by $a_\varnothing^{-1}$ (absorbing this nonzero scalar into f), so that $h(0) = 1$. Every cross-pair diagonal slice $t_i = t_j$ is nonzero and rank one. By Lemma 7.8, $s \geq 4$ would make f a product of s nonsingular binaries, which is the factorization alternative of the lemma. Hence, on the remaining branch, $s = 3$. Rename the parameters $\rho_1, \rho_2, \rho_3, \kappa \neq 0$ from Equation (70) as r_1, r_2, r_3, κ. Let $H_i^\pm$ be the 2×2 logical coefficient matrix obtained by closing E_i with the $\pm$ kernel. For $\{i, j, k\} = \{1, 2, 3\}$, one has

$$\det H_i^\pm = r_j r_k (\kappa - 1)(1 - \kappa r_i^2).$$

If $\kappa = 1$, h, and hence f, has the asserted product factorization. On the remaining branch $\kappa \neq 1$, and rank one of the nonzero cards forces

$$\kappa r_1^2 = \kappa r_2^2 = \kappa r_3^2 = 1.$$

In this remaining case, write each matching edge as its two port labels, e.g. $E_i = \{i^0, i^1\}$, in the Walsh-index matching (this notation does not assert that the physical tensor itself is an equality signature). An *actual I/X-link* is the constant-size binary chain allowed by the kernel-span

convention: in Walsh coordinates its kernel is a nonzero scalar times $\epsilon^u \mathbf{1}[u = v]$, with $\epsilon = 1$ for I and $\epsilon = -1$ for X. On a paired join with common bit $u = v$, the two link signs ϵ_0, ϵ_1 therefore contribute $(\epsilon_0\epsilon_1)^u$; choosing their parity realizes either desired character without treating I as a structural edge.

Let E_1', E_2', E_3' denote the corresponding matching edges in the second copy. Use two actual I/X-links for each displayed edge-to-edge join: join the two ports of E_3 in the first copy to the two ports of E_1' in the second copy, and join the two ports of E_2' to those of E_3' inside the second copy; leave both ports of E_1 and E_2 in the first copy external. Evaluating these actual contractions by Equation (68) gives the logical characters below, and every internal matching assignment has a unique lift. Choose the actual I/X-signs

$$\theta = \kappa r_2 r_3, \qquad \eta = -\kappa r_1 r_3,$$

which lie in $\{\pm 1\}$. Writing $x = t_1$ and $y = t_2$ for the two external logical bits, the external table is

$$Q(x, y) = \sum_{a,c \in \mathbb{F}_2} \eta^a \theta^c h(x, y, a) h(a, c, c).$$

Using $\sum_c \theta^c h(a, c, c) = 2r_1^a$, this becomes

$$Q \doteq \frac{\kappa - 1}{\kappa} \begin{pmatrix} 1 & 0 \\ 0 & \kappa r_1 r_2 \end{pmatrix}.$$

The calculation is in the analytic Walsh coordinates of Equation (11); it is not yet the physical boundary tensor. In the logical Walsh variables the quaternary tensor therefore is

$$\widetilde{F} = c(\delta_{0000} + \sigma \delta_{1111}), \qquad c \neq 0, \qquad \sigma = \kappa r_1 r_2 \in \{\pm 1\},$$

where the two nonzero coefficients follow from $\kappa \neq 1$. Applying the inverse analytic change of coordinates, namely $F = P^{\otimes 4} \widetilde{F}$ with $P = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, gives

$$F(x) = c(1 + \sigma(-1)^{x_1 + x_2 + x_3 + x_4}).$$

Define the even-parity signature by

$$\mathrm{Even}_4(x) := \mathbf{1}[x_1 + x_2 + x_3 + x_4 = 0].$$

Thus the actual tensor is a nonzero scalar multiple of Even_4 when $\sigma = 1$, and of its odd-parity coset when $\sigma = -1$; an actual X-dressing on one exposed port toggles the coset. The Walsh matrix is used only for this analytic reconstruction and is not being installed as a physical gadget. For completeness, the equality-anchor witness for Even_4 is explicit: with the diagonal matrix D, Walsh matrix P, and transform matrix U defined by

$$D := \mathrm{diag}(1, i), \qquad P := \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad U := DP^{-1},$$

one has

$$U^{\otimes 4} \mathrm{Even}_4 \doteq \mathrm{Eq}_4, \qquad (U^{-1})^{\mathsf{T}} X U^{-1} = D^{-1} P^{\mathsf{T}} X P D^{-1} = 2I = 2\,\mathrm{Eq}_2.$$

Consequently an actual Even_4 equality anchor is obtained, and Lemma 3.8 applies. □

A nonempty W_f has therefore already produced an equality anchor or a proper factorization. After those alternatives are dispatched, the rigidity branch continues only with

$$W_f = \varnothing.$$

Scale $a_\varnothing = 1$. With all weight-two coefficients absent, the common card matchings determine the entire Walsh support.

Lemma 7.10 (Reed–Muller support reconstruction)**.** *Let f be the fixed minimum core of arity $n \geq 6$, normalized by $a_\varnothing = 1$, and assume $E(Z_f) = \varnothing$ and $W_f = \varnothing$. For $i \neq j$, let M_{ij} be the common support matching of the two ij-cards. Fix an arbitrary ordering of its $n/2-1$ edges and, for $t \in \mathbb{F}_2^{n/2-1}$, write t_e for the coordinate indexed by $e \in M_{ij}$. Let $D := \operatorname{supp} \widetilde{f}$ be the support set. Its equal-ij slice is the $(n/2-1)$-dimensional code*

$$D_{ij} = \left\{ \mathbf{1}_{T(t)} + (\mathrm{wt}(t) \bmod 2)(e_i + e_j) : t \in \mathbb{F}_2^{n/2-1} \right\}, \tag{73}$$

where $T(t) = \bigcup_{e \in M_{ij} : t_e = 1} e$. Every word of D_{ij} has weight divisible by four. Moreover,

$$n = 8, \qquad D = RM(1,3),$$

after a bijective relabelling of the ports by $\mathbb{F}_2^3$.

Proof. By Lemma 7.7, the two ij-cards survive with a common support matching M_{ij}. Normalize them at the zero word $t = 0$ in matching coordinates, and write $r^+_{ij,e}$ and $r^-_{ij,e}$ for the corresponding normalized edge ratios (the $t_e = 1$ factor entry divided by its $t_e = 0$ entry). For each $e \in M_{ij}$, set $\rho_{ij,e} := r^-_{ij,e}/r^+_{ij,e}$. Their quotient on the matching code—the repetition code consisting of assignments constant on every edge of M_{ij}, parametrized by t—is

$$\frac{\widetilde{C}^-_{ij}(t)}{\widetilde{C}^+_{ij}(t)} = \prod_{e \in M_{ij}} \rho_{ij,e}^{t_e}.$$

Each single-edge quotient is -1 by $a_{pq} = 0$ and Equation (68); hence the quotient is $(-1)^{\mathrm{wt}(t)}$. Inverting the pencil selects the $ij = 00$ lift for even $\mathrm{wt}(t)$ and $ij = 11$ for odd $\mathrm{wt}(t)$, proving Equation (73); the selected word has weight $2\,\mathrm{wt}(t)$ in the even case and $2\,\mathrm{wt}(t) + 2$ in the odd case, so its weight is divisible by four in either case.

For $x, y \in D$, some two coordinates have the same pair (x_i, y_i); the linear slice D_{ij} then contains $x, y, x + y$. Thus D is linear. For the linear functional $\ell_{ij} : \mathbb{F}_2^n \to \mathbb{F}_2$ defined by $\ell_{ij}(z) := z_i + z_j$, one has

$$D \cap \ker \ell_{ij} = D_{ij}.$$

This subspace has dimension $n/2 - 1$. If $\dim D = n/2 - 1$, then $D = D_{ij} \subseteq \ker \ell_{ij}$ for every $i \neq j$. Hence every word of D has all coordinates equal, so $\dim D \leq 1$, contradicting $n/2 - 1 \geq 2$. Therefore $\dim D = n/2$. Every word of D lies in one of the displayed slices and hence has weight divisible by four. For $x, y \in D$,

$$0 \equiv \mathrm{wt}(x) + \mathrm{wt}(y) - \mathrm{wt}(x + y) = 2x \cdot y \pmod 4.$$

Thus $D \subseteq D^\perp$, and equal dimensions give

$$D = D^\perp.$$

For distinct i, j, p, the mate q of p in M_{ij} gives the unique weight-four word $\{i, j, p, q\}$. Now Lemma 6.17 yields $n = 8$ and $D = RM(1,3)$. □

Coefficient rigidity and C_2 collapse. The combinatorial reconstruction is complete: the surviving core has arity eight and Reed–Muller support. It remains to control its nonzero coefficients. The next lemma shows that they carry only a linear sign character, so an available X-dressing removes the resulting affine shift.

Lemma 7.11 (Coefficients on $\mathrm{RM}(1,3)$)**.** *Let f be an eight-port signature with $\operatorname{supp}\widetilde{f} = RM(1,3)$ whose proper I/X-cards are zero or have the weighted-equality matching form of Equation* (69)*. There are $c \neq 0$ and $w \in \mathbb{F}_2^8$ such that*

$$\widetilde{f}(z) = c(-1)^{w\cdot z}\mathbf{1}[z \in RM(1,3)]. \tag{74}$$

Consequently, back in the physical K-coordinates,

$$f(x) \doteq \mathbf{1}[x \in w + RM(1,3)]. \tag{75}$$

Proof. Let the support code be $D := RM(1,3)$, and define $b(z) := \widetilde{f}(z)/\widetilde{f}(0)$. For an affine plane $A = a + L \subseteq \mathbb{F}_2^3$, write $\mathrm{inc}(A) \in \mathbb{F}_2^8$ for its incidence vector and identify A with $\mathrm{inc}(A) \in D$.

Fix distinct points $p, q \in \mathbb{F}_2^3$, and let A_1, A_2, A_3 be the three affine planes containing them. Under the identification of the equal-pq plus card with its three logical matching coordinates, the words A_1, A_2, A_3 are the three singleton logical words. Its zero word lifts only to $0 \in D$, because D contains no word of weight two. Normalizing the card at that word, the weighted-product form therefore gives

$$b\left(\sum_{r\in S} A_r\right) = \prod_{r\in S} b(A_r) \qquad (S \subseteq \{1,2,3\}). \tag{76}$$

In particular, $A_1 + A_2 + A_3 = \mathbf{1}^8$. The same argument shows that whenever two affine planes A, B meet in a two-point affine line,

$$b(A + B) = b(A)b(B). \tag{77}$$

For a plane A, write $A^c := \mathbb{F}_2^3 \setminus A$. Now $A_1 + A_2 = A_3^c$ is itself an affine plane. Since $A_2 \cap A_3 = \{p, q\}$, it meets A_2 in the other two points of A_2, hence in a two-point line. Applying Equation (77) first to A_1, A_2 and then to $A_1 + A_2, A_2$ gives

$$b(A_1 + A_2) = b(A_1)b(A_2), \qquad b(A_1) = b(A_1 + A_2)b(A_2).$$

Thus $b(A_2)^2 = 1$. Every affine plane can play the role of A_2, so

$$b(A) \in \{\pm 1\} \qquad \text{for every affine plane } A. \tag{78}$$

For any affine plane A, choose distinct $p, q \in A$ and call the other two planes through them B, C. Then $A^c = B + C$, and Equations (76) and (78) give

$$b(A^c) = b(B)b(C) = b(\mathbf{1}^8)b(A). \tag{79}$$

The code D consists of $0, \mathbf{1}^8$, and the fourteen affine-plane incidence words. We can now check multiplicativity on all of D. The zero case is immediate. Equal planes use Equation (78); distinct noncomplementary planes meet in a two-point line and use Equation (77); complementary planes and sums of $\mathbf{1}^8$ with a plane use Equation (79). Finally, Equations (76) and (78) imply $b(\mathbf{1}^8)^2 = 1$, which handles $\mathbf{1}^8 + \mathbf{1}^8$. Hence

$$b(x + y) = b(x)b(y) \qquad (x, y \in D).$$

Thus b is a sign character of D. Its binary exponent extends to a linear functional on $\mathbb{F}_2^8$, so there is $w \in \mathbb{F}_2^8$ with $b(z) = (-1)^{w\cdot z}$ for $z \in D$. This proves Equation (74). Since $D = D^\perp$, Fourier inversion gives

$$f(x) \doteq \sum_{z\in D} (-1)^{(x+w)\cdot z} = |D|\mathbf{1}[x + w \in D].$$

This is Equation (75). □

Available X-dressings translate the coset, so the standard core RM_8 of Equation (16) is available. In lexicographic order it is the Shao–Cai signature $f_8 = \widehat{f_8}$; see Remark 2.5.

Remark 7.12 (Limited appearance and internal C_2 rigidity). The limited-appearance construction of Shao and Cai [SC20, full version, Section 8.2 and Lemma 8.10] provides a related reduction in their real-valued setting. For the complex-valued signature sets considered here, the present branch is discharged by the independent C_2 rigidity argument below. On the remaining continuation, it first proves that the complete group is exactly $\{[I], [X]\} \cong C_2$, and then determines its internal structure. The exact Reed–Muller network calculus, actual Radon access, localization, affine patching, and recognition prove that the entire retained signature set, together with its edge, has one common flat-Lagrangian presentation.

The final matching-deck synthesis invokes Theorem 7.5 to resolve the proper-C_2 case. The no-inverse result Corollary 7.21 also records why the later H_6-to-RM_8 circuit cannot be reversed using only pure RM_8/Pauli resources. We retain the internal network calculus, Radon access, patching, and recognition arguments below.

Thus the core itself is now fixed, up to the recorded equivalences. The last initial-core issue is the complete transfer group. On the continuing matching-deck branch, the coefficient classification also rules out every safe class beyond $[I]$ and $[X]$.

Lemma 7.13 (Safe-group collapse)**.** *Let $\mathcal{G}$ be a retained normalized signature set, let $G := G(\mathcal{G})$ be its complete transfer group, and assume that its kernel span $\mathscr{K}_G$ is $\mathscr{K}_G = \mathrm{span}\{I, X\}$. Let $f \in \mathcal{G}$ be an irreducible Reed–Muller core such that every safe one-port dressing of f remains in the standing matching-deck branch. Then the complete safe projective transfer group is*

$$\{[I], [X]\} \cong C_2.$$

Proof. Put $D := RM(1, 3)$. By Lemma 7.11, choose $c \neq 0$ and $w \in \mathbb{F}_2^8$ such that $\widetilde{f}(z) = c(-1)^{w\cdot z}$ for $z \in D$, and define $b(z) := \widetilde{f}(z)/c$. Let T be a nonsingular safe physical transfer, and write

$$T^{\mathrm{W}} := P^{-1}TP$$

for its Walsh-coordinate matrix. In the proper algebra, $T^{\mathrm{W}} \doteq \mathrm{diag}(1, r)$ for some $r \neq 0$. Dress one port p by T. This actual constant-size gadget preserves irreducibility, support, and the matching-deck hypotheses; its coefficient table, after division by its value at $z = 0$, is

$$b_T(z) = (-1)^{w\cdot z} r^{z_p} \qquad (z \in D).$$

By Lemma 7.11, b_T, hence also the quotient $r^{z_p} = b_T(z)/b(z)$, is a character of D and takes values ± 1. Evaluating at $\mathbf{1}^8 \in D$ gives $r = \pm 1$, precisely the physical transfer classes $[I]$ and $[X]$. □

7.4 Flat-Lagrangian networks and Radon access

Retain the literal RM_8 to build contexts around an arbitrary signature. A flat-Lagrangian signature is constant on an affine coset $a + L$ with $L = L^{\perp}$; see Sections 2.8.2 and 2.10.1. The next lemma shows both what these networks preserve and what they can construct. Self-duality tracks the boundary after internal variables are summed out; weight-four transvections provide the reverse construction from a basis of L.

Lemma 7.14 (Exact flat-Lagrangian network calculus)**.** *1. Let Ω be a tensor network with ordered boundary B. Suppose that the signature at each tensor vertex u is proportional to $\mathbf{1}[x_u \in a_u + L_u]$, where either $L_u = RM(1,3) \leq \mathbb{F}_2^8$ or $L_u = \mathrm{span}_{\mathbb{F}_2}\{(1,1)\} \leq \mathbb{F}_2^2$, and a_u is a shift in the corresponding ambient space. The binary case includes ordinary I/X-vertices. Suppose every internal connection is labelled by I or X. Then the boundary signature is either zero or*

$$x_B \longmapsto c_B \mathbf{1}[x_B \in a_B + L], \qquad c_B \neq 0, \quad a_B \in \mathbb{F}_2^B, \quad L = L^{\perp} \leq \mathbb{F}_2^B. \tag{80}$$

2. Conversely, for every $a_B \in \mathbb{F}_2^{2k}$ and every $L = L^{\perp} \leq \mathbb{F}_2^{2k}$, a polynomial-size $\mathrm{RM}_8/I/X$-network directly realizes a nonzero scalar multiple of $x_B \longmapsto \mathbf{1}[x_B \in a_B + L]$.
In particular every nonzero boundary produced by a network in item 1 has even arity.

Proof. For item 1, first apply the boundary normalization following Equation (2): remove one-free-endpoint tails and separate isolated I/X-edge boxes. Each isolated box is itself a flat-Lagrangian binary and is restored at the end by a direct sum of support directions. The remaining ordered boundary B therefore consists of unpaired ordinary vertex ports, with no free stub or already paired boundary ports. Let E_{int} be the set of internal half-edge pairs and put

$$H_{\mathrm{int}} := \bigcup E_{\mathrm{int}}, \qquad V_{\mathrm{he}} := \mathbb{F}_2^B \oplus \mathbb{F}_2^{H_{\mathrm{int}}}.$$

For each $e = \{h, h'\} \in E_{\mathrm{int}}$, let $\eta_e = 0$ when e is labelled by I and $\eta_e = 1$ when it is labelled by X. Define

$$E_\eta := \{y \in \mathbb{F}_2^{H_{\mathrm{int}}} : y_h + y_{h'} = \eta_e \text{ for every } e = \{h, h'\} \in E_{\mathrm{int}}\},$$

and let

$$E_{\mathrm{eq}} := \{y \in \mathbb{F}_2^{H_{\mathrm{int}}} : y_h = y_{h'} \text{ for every } \{h, h'\} \in E_{\mathrm{int}}\}, \qquad H_{\mathrm{wire}} := \mathbb{F}_2^B \oplus E_{\mathrm{eq}}.$$

The direction E_{eq} is self-dual in $\mathbb{F}_2^{H_{\mathrm{int}}}$, and hence

$$H_{\mathrm{wire}}^{\perp} = \{\mathbf{0}_B\} \oplus E_{\mathrm{eq}},$$

where, for any finite index set R, $\mathbf{0}_R$ denotes the zero vector in $\mathbb{F}_2^R$.

Let $V(\Omega)$ denote the tensor-vertex set of Ω, and put

$$C_0 := \bigoplus_{u \in V(\Omega)} L_u = C_0^{\perp}, \qquad \mathcal{K} := C_0 \cap H_{\mathrm{wire}}.$$

Let $a_{\mathrm{vert}} := (a_u)_{u \in V(\Omega)}$ be the concatenated vertex shift. The feasible assignments form

$$(a_{\mathrm{vert}} + C_0) \cap (\mathbb{F}_2^B \oplus E_\eta).$$

This intersection is either empty or an affine coset $y_0 + \mathcal{K}$. Define

$$\pi_B : V_{\mathrm{he}} \to \mathbb{F}_2^B, \qquad L := \pi_B(\mathcal{K}), \qquad a_B := \pi_B(y_0).$$

Every nonempty boundary fibre is a coset of $\ker(\pi_B|_{\mathcal{K}})$, so all fibres have size $2^{\dim\ker(\pi_B|_{\mathcal{K}})}$. Multiplying this number by the recorded nonzero vertex scalars gives $c_B \neq 0$ in Equation (80). The support and scalar are therefore determined by affine Gaussian elimination.

To prove $L \subseteq L^\perp$, take $k, k' \in \mathcal{K}$. Since $k, k' \in C_0 = C_0^\perp$, we have $k \cdot k' = 0$. For an internal pair $e = \{h, h'\}$, put

$$t := k_h = k_{h'}, \qquad s := k'_h = k'_{h'}.$$

That pair contributes $ts + ts = 0$ to $k \cdot k'$. Consequently $\pi_B(k) \cdot \pi_B(k') = 0$, proving $L \subseteq L^\perp$.

Moreover,

$$\mathcal{K}^\perp = C_0 + H_{\mathrm{wire}}^\perp.$$

If $b \in L^\perp$, then $(b, \mathbf{0}_{H_{\mathrm{int}}}) \in \mathcal{K}^\perp$, so write

$$(b, \mathbf{0}_{H_{\mathrm{int}}}) = c_0 + d, \qquad c_0 \in C_0, \quad d \in H_{\mathrm{wire}}^\perp.$$

Both $(b, \mathbf{0}_{H_{\mathrm{int}}})$ and d belong to H_{wire}, hence $c_0 \in C_0 \cap H_{\mathrm{wire}} = \mathcal{K}$. Therefore $b = \pi_B(c_0) \in L$, proving $L = L^\perp$.

For the converse, order one RM vertex's input ports as $(000, 100, 010, 001)$ and output ports as $(111, 011, 101, 110)$. Direct evaluation gives

$$y = (I_4 + J_4)x = x + (x_1 + x_2 + x_3 + x_4)\mathbf{1}^4.$$

Here $x, y \in \mathbb{F}_2^4$ are respectively the ordered input and output words, I_4 is the 4×4 identity matrix, and $J_4 := \mathbf{1}^4(\mathbf{1}^4)^{\mathsf{T}}$ is the 4×4 all-ones matrix over $\mathbb{F}_2$. Thus actual RM boxes and port permutations realize every weight-four transvection $\tau_s(x) = x + (s \cdot x)s$; retained I-vertices (or $X - I - X$ chains) compose them. We claim by induction on k that every $L = L^\perp \leq \mathbb{F}_2^{2k}$ is obtained from $L_0 = \mathrm{span}_{\mathbb{F}_2}\{e_1 + e_2, e_3 + e_4, \ldots\}$. The cases $k = 0, 1$ are immediate. For $k \geq 2$, self-orthogonality gives $x \cdot x = 0$ for every $x \in L$, so all words are even and $\mathbf{1} \in L$. Choose $z \in L \setminus \{0, \mathbf{1}\}$; one of $z, z + \mathbf{1}$ has even weight in $[2, k]$. If its weight is larger than two, three support coordinates and one zero coordinate give a weight-four s with $z \cdot s = 1$ and $\mathrm{wt}(z + s) = \mathrm{wt}(z) - 2$. Apply the corresponding transvection to the whole current code and repeat, then permute coordinates, to obtain $p = e_1 + e_2 \in L$. Every codeword has equal first two coordinates. Projection deleting them has kernel $\mathrm{span}_{\mathbb{F}_2}\{p\}$, image dimension $k - 1$, and preserves dot products (the two deleted contributions cancel), so its image is self-dual. Recurse and invert the accumulated actual transvections and permutations. The resulting circuit has size $O(k^2)$. Closing the standard input pairs by I gives one internal lift per boundary word. For each $(a_B)_j = 1$, attach a retained ordinary I-vertex to the corresponding circuit port through one native X-edge and expose the other ordinary port of that vertex. Since $XI = X$, this attachment complements exactly that boundary bit. Leave the other circuit ports unchanged. All exposed ports in this construction are ordinary, so no free-endpoint boundary action cancels the required flip. Applying these dressings independently realizes every affine coset $a_B + L$, rather than only the zero coset. □

An orthogonal map A has a self-dual graph relation $\{(x, Ax) : x \in \mathbb{F}_2^n\}$. The converse of Lemma 7.14 realizes that relation as a network. Attaching a signature to its input and closing one output pair then sums the signature over prescribed two-point orbits. This supplies the following Radon cards for the high-arity argument.

Actual Radon access. We use the symmetric-binary-space terminology and ambient-relative orthogonal notation of Section 2.8.2.

Let g have even arity n. Call $v \in (\mathbf{1}^n)^\perp \setminus \{0, \mathbf{1}^n\}$ *accessible*. Such a v is isotropic. For $\epsilon \in \mathbb{F}_2$, define

$$H_{v,\epsilon} := \{x \in \mathbb{F}_2^n : x \cdot v = \epsilon\}.$$

Translation by v preserves $H_{v,\epsilon}$. For $x \in H_{v,\epsilon}$, write

$$[x]_v := x + \operatorname{span}_{\mathbb{F}_2}\{v\}$$

for its translation orbit, and let $H_{v,\epsilon}/\operatorname{span}_{\mathbb{F}_2}\{v\}$ denote the set of these orbits. Define

$$[u]_v \cdot [w]_v := u \cdot w \qquad (u, w \in v^\perp).$$

This is a well-defined nondegenerate symmetric form on the translation space $v^\perp/\operatorname{span}_{\mathbb{F}_2}\{v\}$; each $H_{v,\epsilon}/\operatorname{span}_{\mathbb{F}_2}\{v\}$ is an affine space over that quotient. Define

$$\mathcal{R}_{v,\epsilon} g([x]_v) = g(x) + g(x+v), \qquad [x]_v \in H_{v,\epsilon}/\operatorname{span}_{\mathbb{F}_2}\{v\}. \tag{81}$$

The right side is invariant under replacing x by $x + v$, so the quotient signature is well defined. The construction below identifies the quotient with $n - 2$ ordinary boundary ports. Thus the quotient sum is an actual arity-reducing operation in the original network model. Write

$$O(n, 2) := \{A \in \mathrm{GL}_n(\mathbb{F}_2) : A^\mathsf{T} A = I_n\}.$$

Lemma 7.15 (Accessible Radon directions)**.** *Let g be a retained signature of even arity $n \geq 4$. For each $v \in (\mathbf{1}^n)^\perp \setminus \{0, \mathbf{1}^n\}$, one can construct and record distinct ports i_v, j_v and an orthogonal matrix $A_v \in O(n, 2)$ such that*

$$A_v^\mathsf{T}(e_{i_v} + e_{j_v}) = v.$$

Define

$$r_v := A_v^\mathsf{T} e_{i_v}, \qquad \Pi_v := \operatorname{span}\{r_v, v\}, \qquad W_v := \Pi_v^\perp,$$

and, for $\epsilon \in \mathbb{F}_2$,

$$\iota_{v,\epsilon} : W_v \longrightarrow H_{v,\epsilon}/\operatorname{span}\{v\}, \qquad w \longmapsto [\epsilon r_v + w]_v.$$

Then $\mathbb{F}_2^n = \Pi_v \perp W_v$, and $\iota_{v,\epsilon}$ is an affine isometry from W_v to $H_{v,\epsilon}/\operatorname{span}\{v\}$. An $O(n^2)$-size network using one copy of g, RM_8, and I/X realizes $\mathcal{R}_{v,\epsilon} g$ in the boundary coordinates supplied by $A_v|_{W_v}$, up to a recorded nonzero scalar. The available directions are exactly the displayed vectors v.

Proof. First take A_v, i_v, j_v as in the statement; their construction is given at the end of the proof. The definitions imply $\Pi_v = A_v^\mathsf{T} \operatorname{span}\{e_{i_v}, e_{j_v}\}$. Then

$$r_v \cdot r_v = 1, \qquad r_v \cdot v = 1, \qquad v \cdot v = 0.$$

Hence Π_v is a nondegenerate two-plane and

$$\mathbb{F}_2^n = \Pi_v \perp W_v, \qquad v^\perp = \operatorname{span}_{\mathbb{F}_2}\{v\} \oplus W_v.$$

The isometry A_v carries Π_v to the physical output-coordinate plane $\operatorname{span}\{e_{i_v}, e_{j_v}\}$ and W_v to its ordinary-coordinate complement. We use A_v to identify these pairs of spaces and call the corresponding output ports the Π_v-ports and the W_v-ports. Consequently $\iota_{v,\epsilon}$ has the asserted domain, range, and isometry property.

For $A \in O(n, 2)$, define

$$\mathrm{Graph}(A) := \{(x, Ax) : x \in \mathbb{F}_2^n\}.$$

Its direction is self-dual in $\mathbb{F}_2^n \oplus \mathbb{F}_2^n$: it is isotropic because $x \cdot y + (Ax) \cdot (Ay) = 0$, and it has half the ambient dimension. Hence Lemma 7.14 realizes a nonzero scalar multiple of

$$(x, y) \longmapsto \mathbf{1}[y = A_v x]$$

by an actual $\mathrm{RM}_8/I/X$-network. Attach one copy of g to the graph input using an effective I-kernel on each input connection: insert one retained ordinary I-vertex between two native X-edges, so that $XIX = I$. The output before closing a pair is therefore $g(A_v^{\mathsf{T}} y)$, with the graph's recorded nonzero scalar. Close the ordered output pair (i_v, j_v) with either actual kernel I or actual kernel X. The two internal lifts differ by v; the choice of I or X fixes $x \cdot v = \epsilon$. Under the isometry $A_v|_{W_v}$, the remaining output coordinates are precisely the W_v-coordinates in the inverse of $\iota_{v,\epsilon}$. Thus the resulting boundary is $\mathcal{R}_{v,\epsilon} g$, up to its recorded nonzero scalar.

The weight-four transvections constructed in Lemma 7.14, together with coordinate permutations, send every nonzero even $v \neq \mathbf{1}^n$ to a weight-two vector. Write the resulting orthogonal map as B, with $Bv = e_{i_v} + e_{j_v}$, and record $A_v = B$. Then $A_v^{\mathsf{T}}(e_{i_v} + e_{j_v}) = B^{-1}Bv = v$, as required. Conversely, orthogonal maps preserve parity and fix $\mathbf{1}^n$, so neither 0 nor $\mathbf{1}^n$ lies in the orbit of a weight-two vector. This proves necessity as well. The transvection construction and graph network use $O(n^2)$ vertices; the associated binary linear systems are solved by Gaussian elimination. □

7.5 Relative flat-Lagrangian closure and completion of Theorem 7.5

We now consider an arbitrary unresolved tensor-prime occurrence g. For arity at most eight, its smaller flat cards are matching products, so the reconstruction above applies. At higher arity, Radon cards compare two quotient layers. A failure to combine them into an affine signature can be retained on at most eight ports; after that obstruction is excluded, affine patching and dimension counts recover a flat-Lagrangian g.

For an n-ary signature h and a shift vector $s \in \mathbb{F}_2^n$, define the translation operator T_s by $(T_s h)(x) := h(x + s)$. Write Id for the identity operator on the current signature space and define the orbit-sum operator

$$\mathsf{P}_s^+ := \mathrm{Id} + T_s.$$

Thus $(\mathsf{P}_s^+ h)(x) = h(x) + h(x + s)$ is the sum over the two-point translation orbit $\{x, x + s\}$. For accessible s, the two quotient signatures $\mathcal{R}_{s,0} h$ and $\mathcal{R}_{s,1} h$ are called the 0- and 1-layer *plus cards* of h in direction s; together they are the two quotient layers of $\mathsf{P}_s^+ h$.

Lemma 7.16 (Low-arity flat signatures are matching products)**.** *Let $L = L^\perp \leq \mathbb{F}_2^{2r}$, where $r \leq 3$. After a port permutation, L is a direct sum of r two-coordinate repetition lines* $\mathrm{span}_{\mathbb{F}_2}\{e_{2j-1} + e_{2j}\}$. *Consequently every flat-Lagrangian signature of arity at most six is a product of projective I/X binaries on a perfect matching.*

Proof. Every vector of L has even weight, so $\mathbf{1} \in L^\perp = L$. If $r > 1$, choose $z \in L \setminus \{0, \mathbf{1}\}$. One of $z, z + \mathbf{1}$ has weight at most r, hence weight two when $2r \leq 6$. Thus $e_i + e_j \in L$ for some pair i, j. Self-duality now gives $x_i = x_j$ for every $x \in L$, so this repetition line splits orthogonally from L. Induction proves the assertion. Translating the affine support changes an equality factor only to its disequality coset. □

Consequently every flat-Lagrangian proper card of an occurrence of arity at most eight is already a matching product. A nonmatching flat-Lagrangian proper card can first occur at arity ten; its appearance is not by itself a protocol exit. This is precisely the remaining case for the Radon localization argument.

Lemma 7.17 (Two flat quotient charts localize nonaffine gluing)**.** *Let g have even arity $n \geq 10$, let v be accessible, and suppose that, for both $\epsilon \in \mathbb{F}_2$, the card $\mathcal{R}_{v,\epsilon}g$ is zero or flat-Lagrangian. Define, only for the analysis, the orbit-sum signature*

$$F_v(x) = g(x) + g(x+v).$$

If $F_v \notin \mathscr{A}$, an $O(n^2)$-size direct $\mathrm{RM}_8/I/X$-context using one copy of g produces a nonzero $H \notin \mathscr{A}$ of arity at most eight.

Proof. Fix the recorded data $(A_v, i_v, j_v, r_v, \Pi_v, W_v)$ from Lemma 7.15, and write $W := W_v$. Before the final output pair is closed, the graph realization gives the orthogonal decomposition

$$\mathbb{F}_2^n = \Pi_v \perp W.$$

By the recorded identification, the two physical output ports i_v, j_v are the Π_v-ports, while the remaining output ports are the W-ports.

Under the isometry $W \simeq v^{\perp}/\operatorname{span}_{\mathbb{F}_2}\{v\}$, the characteristic vector of W corresponds to the class of $\mathbf{1}^n$. This class is nonzero because accessibility excludes $v = \mathbf{1}^n$. Thus W is even-dimensional and nonalternating.

For $\epsilon \in \mathbb{F}_2$, define the analytic quotient chart

$$h_\epsilon(w) := F_v(\epsilon r_v + w), \qquad w \in W.$$

This is well defined because F_v is invariant under translation by v. Each nonzero chart has the form

$$h_\epsilon(w) = c_\epsilon \mathbf{1}[w \in a_\epsilon + L_\epsilon], \qquad c_\epsilon \neq 0, \quad a_\epsilon \in W, \quad L_\epsilon = L_\epsilon^{\perp W} \leq W.$$

If $h_{1-\epsilon} = 0$ and $h_\epsilon \neq 0$, then, in the decomposition $\mathbb{F}_2^n = \Pi_v \perp W$,

$$F_v(x) = c_\epsilon \mathbf{1}[x \in \epsilon r_v + a_\epsilon + (\operatorname{span}_{\mathbb{F}_2}\{v\} \oplus L_\epsilon)].$$

Thus F_v is affine whenever one chart is zero, and the assumed nonaffinity ensures that both displayed charts are nonzero. Here "the charts glue affinely" means that their union, with the layer coordinate restored, is the support coset of one affine signature. This happens exactly when $L_0 = L_1$; once their supports glue, their constant phases glue exactly when $c_1/c_0 \in \mu_4$.

If $L_0 \neq L_1$, take $s \in L_0 \setminus L_1$. Since $s \notin L_1^{\perp W} = L_1$, choose $t \in L_1$ with $s \cdot t = 1$. Elementary completion in a nondegenerate binary symmetric space gives an even-dimensional, nonalternating, nondegenerate subspace

$$U \supseteq \operatorname{span}\{s, t, a_1 - a_0\}, \qquad \dim U \leq 6.$$

To obtain this dimension bound, use the symmetric-space conventions in Section 2.8.2. Put $S := \operatorname{span}\{s, t, a_1 - a_0\}$. Since $s \cdot t = 1$, the plane $\operatorname{span}\{s, t\}$ is nondegenerate, so $\dim \operatorname{rad}(S) \leq 1$. If $\operatorname{rad}(S) = 0$, put $N := S$. Otherwise write $\operatorname{rad}(S) = \operatorname{span}_{\mathbb{F}_2}\{r\}$, choose a nondegenerate complement S_0 of $\operatorname{span}_{\mathbb{F}_2}\{r\}$ in S, and choose $d \in W$ satisfying

$$r \cdot d = 1, \qquad d \in S_0^{\perp}.$$

Put $N := S + \operatorname{span}_{\mathbb{F}_2}\{d\}$. In either case N is nondegenerate and $\dim N \leq 4$. If $\dim N$ is odd, its nondegenerate orthogonal complement has odd dimension and therefore contains an anisotropic vector; adjoining that orthogonal line makes N even-dimensional without exceeding dimension four. If the resulting even-dimensional N is alternating, then $N^{\perp}$ is nonalternating because W is; adjoin

a nonalternating nondegenerate plane from $N^{\perp}$. The resulting U has all the asserted properties and dimension at most six.

If instead $L_0 = L_1$ and $c_1/c_0 \notin \mu_4$, every vector in an even-dimensional nondegenerate nonalternating symmetric binary space lies in a nonalternating nondegenerate plane. For a nonzero isotropic vector z, choose d with $z \cdot d = 1$ and $d \cdot d = 1$; for an anisotropic z, choose d with $z \cdot d = 1$ and $d \cdot d = 0$. For $z = 0$, choose any nonalternating nondegenerate plane. We may therefore choose an even-dimensional, nonalternating, nondegenerate

$$U \supseteq \operatorname{span}\{a_1 - a_0\}, \qquad \dim U \le 2.$$

Put

$$P := U^{\perp_W}, \qquad W = U \perp P,$$

and let π_U, π_P be the corresponding orthogonal projections. The space P is even-dimensional and nondegenerate. By the symmetric-binary Witt decomposition recalled in Section 2.8.2, P contains a half-dimensional totally isotropic subspace J; equivalently,

$$J = J^{\perp_P} \le P.$$

Because $a_1 - a_0 \in U$, the two shifts have the same P-projection; call it p. Choose an isometry $\phi : U \to \mathbb{F}_2^{\dim U}$, and define the affine relation

$$\Gamma_p := \{(u + p + j, \phi(u)) : u \in U,\ j \in J\} \subseteq W \oplus \mathbb{F}_2^{\dim U}.$$

Its direction is self-dual: it has half the ambient dimension, and

$$(u + j) \cdot (u' + j') + \phi(u) \cdot \phi(u') = j \cdot j' = 0$$

for $u, u' \in U$ and $j, j' \in J$. Thus after transporting the W-coordinate through $A_v|_{W_v}$, Lemma 7.14 realizes the corresponding signature

$$(w, z) \longmapsto \mathbf{1}[(w, z) \in \Gamma_p]$$

by an actual $O(n^2)$-size $\mathrm{RM}_8/I/X$-network. Apply this relation to the W-ports and leave the two Π_v-ports external.

In the different-direction case, the boundary points induced by $a_0, a_0 + s, a_1$ survive but the fourth parallelogram point induced by $a_1 + s$ does not. Otherwise there would be $z \in L_1$ with $\pi_U z = s$, and

$$0 = z \cdot t = (\pi_U z) \cdot t = s \cdot t = 1.$$

The output therefore retains a nonaffine parallelogram on at most $\dim U + 2 \le 8$ ports. In the equal-direction case, the two fibers have equal size because their support directions agree. Thus neither cancellation nor projection changes the ratio c_1/c_0 outside μ_4, and the output is again nonaffine.

Let H be the contraction of this actual context with g, and let H^+ be the contraction with $F_v = \mathsf{P}_v^+ g$. Order the boundary with the two retained Π_v-ports first and put

$$w_\partial := (1, 1, \mathbf{0}_{\dim U}).$$

Indeed, translation by v changes the graph output by $e_{i_v} + e_{j_v}$, so it flips exactly the two Π_v-ports and acts trivially on the remaining boundary ports. Consequently, with the same recorded scalar on both terms,

$$H^+ = \mathsf{P}_{w_\partial}^+ H.$$

The missing parallelogram or forbidden layer ratio proves $H^+ \notin \mathscr{A}$. By Lemma 2.7, $H \in \mathscr{A}$ would imply $H^+ = \mathsf{P}^+_{w_\partial} H \in \mathscr{A}$, a contradiction. Therefore $H \neq 0$, $H \notin \mathscr{A}$, and its arity is at most eight. □

In the relative induction, the localized nonaffine signature supplied by the preceding lemma is excluded by the already closed lower-arity cases. Hence every accessible orbit sum F_v is affine. The next lemma patches these hyperplane-wise affine conclusions into a global affine conclusion for g.

Lemma 7.18 (Affine hyperplane patching)**.** *Let $n \geq 10$ be even, and let $g : \mathbb{F}_2^n \to \overline{\mathbb{Q}}$. For every accessible v, define the orbit-sum signature $F_v := \mathsf{P}^+_v g$. If $F_v \in \mathscr{A}$ for every such v, then $g \in \mathscr{A}$.*

Proof. We first record the Walsh-invariance of $\mathscr{A}$ used below. The Walsh kernel $(-1)^{\xi \cdot x}$ has an allowed quadratic modulo-four phase, so its product with an affine signature is affine in the joint variables (ξ, x). Gaussian-eliminate the affine support constraints and sum the remaining internal variables one at a time. If the current phase depends on a free bit y as

$$i^{\alpha y}(-1)^{y\ell}, \qquad \alpha \in \mathbb{Z}_4,$$

where ℓ is an affine linear form in the remaining variables and is independent of y, then summing over y either imposes the affine equation $\ell = \alpha/2 \pmod 2$ when α is even, or contributes a nonzero scalar times a fourth-root phase of the affine linear form ℓ when α is odd. Thus every elimination remains in $\mathscr{A}$, proving that the Walsh transform preserves $\mathscr{A}$. The converse follows from Walsh inversion.

Let $\mathcal{F}_2 g$ denote the (unnormalized) Walsh transform

$$(\mathcal{F}_2 g)(\xi) = \sum_{x \in \mathbb{F}_2^n} (-1)^{\xi \cdot x} g(x),$$

and write $g^{\mathrm{W}} := \mathcal{F}_2 g$. This superscript distinguishes the Walsh transform here from the manuscript's global hat notation for K-coordinates. Since

$$(\mathcal{F}_2 F_v)(\xi) = (1 + (-1)^{\xi \cdot v}) g^{\mathrm{W}}(\xi),$$

each restriction $g^{\mathrm{W}}|_{v^\perp}$ is affine or zero. The claim is immediate when $g^{\mathrm{W}} = 0$, so assume otherwise and choose $\zeta \in \operatorname{supp} g^{\mathrm{W}}$.

Given $x, y \in \operatorname{supp} g^{\mathrm{W}}$, the space

$$E \cap \operatorname{span}\{\zeta, x, y\}^\perp, \qquad E = \mathbf{1}^\perp,$$

has dimension at least $n - 4$, hence contains accessible v. Affinity on $v^\perp$ puts $x + y + \zeta$ in the support, so $\operatorname{supp} g^{\mathrm{W}}$ is a coset; write it as $\zeta + U$.

Choose an isomorphism $\beta : \mathbb{F}_2^{\dim U} \to U$ and define the nowhere-zero coefficient function

$$\psi(u) := g^{\mathrm{W}}(\zeta + \beta u).$$

Fix arbitrary

$$u_0, u_1, u_2, u_3 \in \mathbb{F}_2^{\dim U},$$

where u_0 is the base point and u_1, u_2, u_3 are the three directions of a multiplicative third-difference cube of ψ. The four vectors

$$\zeta + \beta u_0, \quad \beta u_1, \quad \beta u_2, \quad \beta u_3$$

span at most four dimensions. Hence

$$E \cap \operatorname{span}\{\zeta + \beta u_0, \beta u_1, \beta u_2, \beta u_3\}^{\perp}$$

has dimension at least $n - 5 \geq 5$ and contains an accessible direction s. The entire cube lies in $s^{\perp}$, where the nonzero restriction of g^{W} is affine; hence

$$\mathcal{T}_{\psi}(u_0, u_1, u_2, u_3) = 1.$$

By Lemma 2.6, ψ has an allowed quadratic modulo-four phase. Consequently $g^{\mathrm{W}} \in \mathscr{A}$, and the Walsh-invariance proved above gives $g \in \mathscr{A}$. □

Affine patching gives $g \in \mathscr{A}$, but does not yet give the required flat-Lagrangian form. The final recognition lemma uses the dimensions and phases of all actual Radon cards to force a self-dual support direction and constant coefficient on that support.

Lemma 7.19 (Affine-to-flat Radon recognition)**.** *Let $0 \neq h \in \mathscr{A}$ have even arity $2r \geq 8$. Suppose that for every accessible direction v and both $\epsilon \in \mathbb{F}_2$, the actual card $\mathcal{R}_{v,\epsilon}h$ is zero or flat-Lagrangian. Then h is flat-Lagrangian.*

Proof. Write the affine signature h in standard normal form as

$$h(x) = c\mathbf{1}[x \in a + U]i^{Q(x)}, \qquad d := \dim U.$$

Here $c \neq 0$ is the scale, $a + U$ is the affine support, Q is its quadratic phase polynomial, and d is the support dimension. Define the even hyperplane $E := \mathbf{1}^{\perp}$. For $v \notin U$ there is no cancellation: the nonempty card has support dimension d if $v \in U^{\perp}$, and both have dimension $d - 1$ otherwise. A flat card has dimension $r - 1$.

Neither $E \subseteq U$ nor $E \subseteq U^{\perp}$. Suppose first that $E \subseteq U$. Choose $p, q \in E$ so that $p, q, p + q$ are accessible. They exist because $\dim E = 2r - 1 \geq 7$. If

$$\mathsf{P}_p^+ h = \mathsf{P}_q^+ h = 0,$$

then $T_p h = T_q h = -h$, so $T_{p+q} h = h$ and $\mathsf{P}_{p+q}^+ h = 2h \neq 0$. Thus at least one layer plus card arising from the three directions $p, q, p + q$ is nonzero. Let $s \in \{p, q, p + q\}$ be its direction. Passing from h to such a card can lower support dimension by at most one when restricting to its layer hyperplane and by one when quotienting by $\operatorname{span}_{\mathbb{F}_2}\{s\}$. On an overlapping support orbit, factoring out $h(x)$ leaves

$$1 + \zeta(-1)^{\ell(x)}, \qquad \zeta \in \mu_4.$$

Here ℓ is a binary affine-linear form on the current support. For $\zeta \in \{\pm i\}$ this factor is nowhere zero; for $\zeta \in \{\pm 1\}$ its nonzero set is empty, the whole current support, or one affine hyperplane. Since the selected card is nonzero, cancellation therefore lowers support dimension by at most one more. Hence its support dimension is at least

$$d - 3 \geq 2r - 4 > r - 1,$$

contradicting flatness.

If instead $E \subseteq U^{\perp}$, then $d \leq 1$. Choose accessible $v \in E \setminus U$. Here $v \in U^{\perp} \setminus U$, so there is no cancellation and the nonzero card has support dimension d, whereas a flat-Lagrangian card has dimension $r - 1 \geq 3$, again a contradiction.

Thus both $E \cap U$ and $E \cap U^\perp$ are proper subspaces of E. Two proper subspaces cannot cover E; since $\dim E \geq 7$, we may also avoid $0, \mathbf{1}$ and choose accessible $v \notin U \cup U^\perp$. Its two cards have dimension $d-1 = r-1$, so $d = r$.

No accessible vector lies in $U^\perp \setminus U$, since it would give a dimension-r card. Put $A_0 := E \cap U^\perp$. Then

$$A_0 \setminus \{0, \mathbf{1}\} \subseteq U.$$

If $\mathbf{1} \notin A_0$, or if $\mathbf{1} \in A_0 \cap U$, this gives $A_0 \subseteq U$. If instead $\mathbf{1} \in A_0 \setminus U$, choose $z \in A_0 \setminus \{0, \mathbf{1}\}$, which is possible because $\dim A_0 \geq r-1 \geq 3$; then both z and $z + \mathbf{1}$ are accessible and lie in U by the preceding display, forcing $\mathbf{1} \in U$, a contradiction. Hence $A_0 \subseteq U$ in every case. If $U^\perp \subseteq E$, dimensions give $U = U^\perp$. Otherwise $\dim A_0 = r-1$. Since $A_0 \subseteq U$ and $\dim U = r$, choose $u \in U \setminus A_0$. Then $U = A_0 \oplus \operatorname{span}_{\mathbb{F}_2}\{u\}$, and $A_0 \subseteq U^\perp$ makes u orthogonal to A_0. If u were even, then $u \cdot u = 0$, so $u \in U^\perp$ and $U \subseteq U^\perp$; equal dimensions would give $U = U^\perp$, contrary to the present case. Thus u is odd and

$$U = A_0 \oplus \operatorname{span}_{\mathbb{F}_2}\{u\}, \qquad u \cdot u = 1. \tag{82}$$

For accessible $s \in A_0$, put

$$\epsilon_s := a \cdot s.$$

Because $s \in U^\perp$, the support $a + U$ lies wholly in H_{s,ϵ_s}; the other layer plus card is identically zero. The direction of the supported card $\mathcal{R}_{s,\epsilon_s} h$ is $U/\operatorname{span}_{\mathbb{F}_2}\{s\}$, whose orthogonal is $U^\perp/\operatorname{span}_{\mathbb{F}_2}\{s\}$; these differ because $U \neq U^\perp$. Write its phase difference as

$$Q(x+s) - Q(x) = \alpha_s + 2\ell_s(x) \pmod 4, \qquad \alpha_s \in \mathbb{Z}_4,$$

where ℓ_s is affine linear on $a + U$. Thus

$$\frac{h(x+s)}{h(x)} = i^{\alpha_s}(-1)^{\ell_s(x)}.$$

If α_s is odd, the supported plus card is nonzero on the full quotient and has the non-self-dual direction $U/\operatorname{span}_{\mathbb{F}_2}\{s\}$. If α_s is even and ℓ_s is nonconstant, its nonzero support is a codimension-one hyperplane of dimension $r-2$. If ℓ_s is constant, the card is either zero or again has the full non-self-dual direction. The hypothesis therefore forces

$$\mathcal{R}_{s,\epsilon_s} h = 0, \qquad \text{equivalently} \qquad \mathsf{P}_s^+ h = 0.$$

Because $\dim A_0 = r-1 \geq 3$, choose a two-dimensional subspace $B_0 \leq A_0$ with $\mathbf{1} \notin B_0$, and let p, q be a basis of B_0. Then $p, q, p+q$ are all accessible, and $\mathsf{P}_p^+ h = \mathsf{P}_q^+ h = 0$, whereas $T_{p+q} = T_p T_q$ gives $\mathsf{P}_{p+q}^+ h = 2h \neq 0$, a contradiction. Therefore $U = U^\perp$.

For accessible $s \in U = U^\perp$, put $\epsilon_s := a \cdot s$. The support lies wholly in H_{s,ϵ_s}, so $\mathcal{R}_{s,\epsilon_s} h$ is the unique potentially nonzero layer plus card. Use the same phase-difference form as above. If α_s is even and ℓ_s is nonconstant, the nonzero part of the plus card has quotient-support dimension $r-2$, so it cannot be flat-Lagrangian. In every remaining case the plus card is either zero or has the full self-dual quotient support $U/\operatorname{span}_{\mathbb{F}_2}\{s\}$; by the flat-Lagrangian hypothesis it is then constant on that support. Taking $c_s = 0$ in the zero case, we obtain

$$h(x) + h(x+s) = c_s \qquad (x \in a + U).$$

Since $\dim U = r \geq 4$ and $U = U^\perp$, choose a two-dimensional subspace $B_1 \leq U$ with $\mathbf{1} \notin B_1$, and let p, q be a basis. Then $p, q, p+q$ are all accessible. Using $T_{p+q} = T_p T_q$ to compare their three equations gives

$$2h(x) = c_{p+q} - c_q + c_p,$$

so h is constant on $a + U$, hence flat-Lagrangian. □

Completion of Theorem 7.5. We first close the initial minimum core f. If it has odd arity, apply Theorem 2.15 to $K\mathcal{G}$. Its hardness conclusion pulls back through the holographic K-coordinate equivalence, while its tractable conclusion is exactly $\mathsf{Tract}(K\mathcal{G})$. If $\mathrm{arity}(f) = 4$, Lemma 7.4 and the quaternary interface give either a resolved leaf or a strict stable successor, and the complete batch is resolved by Theorem 5.18, Corollary 4.7, Lemmas 5.3 and 5.11, and Definition 5.9. Hence, on the sole remaining initial branch, f has even arity $n \geq 6$, and every proper card is zero or a nonsingular current-group matching signature.

Apply Lemmas 7.6 and 7.7. If $W_f \neq \varnothing$, then Lemma 7.9 gives either the actual Even_4 equality anchor constructed in its proof or a proper tensor factorization. After factor saturation and group recomputation, both alternatives are resolved by Lemmas 2.20, 3.8 and 5.11 and Definition 5.9. Thus the remaining branch has $W_f = \varnothing$. Now Lemmas 7.10 and 7.11 give

$$n = 8, \qquad f(x) = c\mathbf{1}[x \in a + RM(1,3)]$$

after a port permutation, for a recorded $c \neq 0$. Available actual one-port X-dressings remove a, so the retained set contains the literal standard core RM_8. Regenerate the complete deck of every safe one-port dressing of this core. If one leaves the matching branch, its complete batch is resolved by the same case analysis; otherwise Lemma 7.13 gives

$$G = \{[I], [X]\} \cong C_2.$$

This proves the asserted description of the sole initial rigidity branch.

Retain this literal RM_8 and normalized I/X throughout the remaining signature-set-wide closure. At every continuing augmentation, recompute the complete group and process the Reed–Muller safe-dressing deck before re-entering the argument. A decrease of ν or a strict group enlargement is charged to the stable-state measure Ω of Lemma 5.11; on the unchanged remaining branch the preceding safe-dressing argument reapplies and the complete group remains C_2.

We next make the outer induction explicit. Let $\mathcal{W}$ be the finite multiset of unresolved tensor-prime occurrences in the retained factorization trees. When a token u, carrying h_u, first enters the worklist, freeze the following finite Reed–Muller reduction universe $\mathscr{U}_{\mathrm{RM}}(u)$:

1. the complete physical C_2 matching-deck universe of h_u, with every oriented proper contraction and safe I/X-dressing, together with every ordered two-copy I/X-integration network displayed in Lemma 7.9, for all port, orientation, and link-sign choices;

2. for every accessible v and ϵ, the fixed actual Radon network of Lemma 7.15, including its recorded data $(A_v, i_v, j_v, r_v, \Pi_v, W_v)$, and, for every finite parameter tuple (v, U, J, p, ϕ) allowed in Lemma 7.17, its actual localization network applied to h_u using that same recorded Radon data;

3. the complete factorization tree of every output in the preceding two items.

Each network-output key records the ordered physical network, its boundary order, actual scalar, and direct provenance; each attached factor token records its canonical signature and Turing-exposure provenance. The universe is finite because the port set and all of its binary subspaces, vectors, and isometries are finite; every individual listed network has the size bounds proved in the cited lemmas. These choices depend on the fixed retained signature set, not on the input Holant graph. Once selected, each network replaces a vertex by a fixed-size gadget, as required for the stated reductions. Partition it as

$$\mathscr{U}_{\mathrm{RM}}(u) = Q_{u,\mathrm{done}} \sqcup Q_u.$$

For $q \in Q_u$, let $\mathcal{W}_q$ be the multiset of open occurrences in its attached factorization tree, and set

$$\Phi_u := \sum_{q \in Q_u} \Phi(\mathcal{W}_q),$$

using Equation (25). Define

$$\Theta_{\mathrm{RM}}(\mathcal{W}) = \{\{(\mathrm{arity}(h_u), |Q_u|, \Phi_u) : u \in \mathcal{W}\}\},$$

using lexicographic order on each triple and the well-founded multiset extension of that order; explicitly, replacing one triple by any finite multiset of strictly smaller triples is a strict decrease.

Certifying an occurrence removes its triple. If the selected occurrence factors, its triple is replaced by triples of strictly smaller arity. Every batch used below starts from a frozen key: the matching-deck and integration steps use item 1, the Radon steps use the first family in item 2, and the two-chart step uses the second family in item 2. Completing such a key moves it to $Q_{u,\mathrm{done}}$, while an intermediate genuine factor split strictly decreases its summand $\Phi(\mathcal{W}_q)$. Any newly opened child occurrence has smaller arity: proper and Radon cards have arity at most $\mathrm{arity}(h_u) - 2$, localization outputs have arity at most eight when they are invoked from arity at least ten, and every proper factor is smaller than its parent. Thus the old parent triple is replaced by a strictly smaller parent triple together with child triples whose first coordinate is smaller. This strictly decreases Θ_{RM}. No fixed-(H, ν) batch creates a fresh unresolved key, because all of its possible outputs and complete factorization trees were included in $\mathscr{U}_{\mathrm{RM}}(u)$ at the freeze. Consequently this outer induction, rather than a multiset of arities alone, also controls passage to another occurrence of the same arity.

We use the following relative re-entry observation. Suppose a tensor-prime occurrence g has had its complete physical deck regenerated and every proper card and proper factor of smaller arity has already been closed. On the continuing branch, assume every surviving proper card is zero or a current-C_2 matching signature. Then the proofs of Lemmas 7.6, 7.7 and 7.9 to 7.11 apply verbatim with g in place of the initial minimum core. Indeed, minimum arity is used there only to exclude a surviving proper nonmatching card or nonbinary factor; regenerated-deck and prior-closure supply exactly that exclusion. All displayed local networks and coefficient identities are unchanged.

Run induction on Θ_{RM}. Before processing an occurrence, recursively close all its lower-arity proper cards and factors. Unary, odd-arity, and quaternary outputs are dispatched by Theorems 2.15 and 5.18. Hence consider a relatively re-entered nonflat tensor-prime g of even arity at least six.

There are now two possible continuations. First suppose that every surviving proper card of g is zero or a current-C_2 matching signature. The relative re-entry observation and Lemmas 7.6, 7.7 and 7.9 to 7.11 then give a resolved Even_4 equality anchor, a proper factorization, or the flat eight-ary $RM(1,3)$ core. Thus no nonflat occurrence with an all-matching proper deck survives.

Otherwise some nonzero proper card is nonmatching. Factor saturation and prior closure nevertheless make every proper card zero or a tensor product of already certified flat-Lagrangian factors, hence zero or flat-Lagrangian. If $6 \le \mathrm{arity}(g) \le 8$, every such card has arity at most six, and Lemma 7.16 makes it an I/X-matching signature after a port permutation, contradicting the present case. The appearance of a nonmatching flat-Lagrangian card is not itself a protocol exit; it identifies the remaining Radon branch, where $\mathrm{arity}(g) = n \ge 10$.

By Lemma 7.15, every Radon card is actual, is indexed by its frozen item 2 key, and has arity $n-2$. Processing that key includes its complete factor saturation. Every surviving factor has already been certified flat-Lagrangian; tensor closure therefore makes each card zero or flat-Lagrangian. Fix an accessible v. If the analytic orbit sum

$$F_v = \mathsf{P}_v^+ g$$

were nonaffine, Lemma 7.17 would produce an actual nonaffine signature H of arity at most eight. Its parameter tuple and actual context have a frozen item 2 key, whose attached factorization tree is now processed. Factor saturation of H must expose a nonaffine tensor-prime factor, since a tensor product of affine factors is affine. That factor has smaller arity and is processed earlier in the induction, contradicting the remaining branch. Hence $F_v \in \mathscr{A}$ for every accessible v.

It follows from Lemma 7.18 that $g \in \mathscr{A}$. All its actual Radon cards are zero or flat-Lagrangian, so Lemma 7.19 gives $g \in \mathsf{FLag}$, contradicting the choice of a surviving nonflat occurrence. Therefore every retained tensor-prime occurrence is eventually certified flat-Lagrangian.

Every concrete terminal, anchor, factor, or group-change batch above is closed before the next frozen key is selected. A stable-tuple transition strictly decreases Ω; an unchanged-tuple worklist step strictly decreases Θ_{RM}. The two measures therefore have disjoint and well-founded responsibilities, so no re-entry or same-arity cycle is possible.

All certifications were made in the same retained normalized coordinates. The native X-edge is itself flat-Lagrangian, and tensor products of flat-Lagrangian signatures remain flat-Lagrangian by taking direct sums of their self-dual support directions. Retained factor saturation reconstructs every member of the signature set from its certified prime factors. Hence the entire retained set and its edge have one common flat-Lagrangian presentation, which is exactly $\mathsf{Close}_{\{\mathsf{FLag}\}}(\mathcal{G}, G, f)$. □

7.6 Character-flat extensions of the Reed–Muller calculus

Allowing all four Bell binaries adds linear signs to the Reed–Muller network calculus. We describe the resulting class and the corresponding signed Radon cards. These are additional structural results; the standard-V_4 closure in Theorem 8.3 uses the tensor-line argument given there. Orthogonal complements and Witt decomposition retain the conventions of Section 2.8.2.

Recall from Equation (20) that CFlat_{2k} consists of zero together with signatures

$$c(-1)^{\ell(x)}\mathbf{1}[x \in a + L], \qquad c \neq 0, \quad L = L^{\perp}.$$

Thus CFlat differs from FLag by allowing a linear sign character on a self-dual affine support. Here a *pure-Pauli* RM_8*-network* means a tensor network whose nonbinary vertices are retained actual copies of RM_8, whose binary vertices are actual ordered Bell binaries, and whose internal connections are native X-edges. The next lemma gives the exact Pauli network calculus for this class. Its corollary distinguishes the resources H_6 and RM_8: the four-copy circuit in Section 8.3 realizes the latter from the former, while pure Reed–Muller/Pauli networks cannot realize the reverse direction. The remaining lemmas extend Radon localization and recognition from FLag to CFlat.

Lemma 7.20 (Character-flat Pauli network calculus)**.** *1. Every network whose vertices are* RM_8 *and actual ordered Bell binaries* $B_{d,\ell}$*, and whose internal edges are native* X*-edges, has zero boundary or a boundary in* CFlat*.*

2. If RM_8 *and actual nonzero representatives of all four ordered Bell binaries* $B_{d,\ell}$ *are retained, then, for every* $0 \neq f \in \mathsf{CFlat}_{2k}$*, an* $O(k^2)$*-size actual* RM_8*/ordered-Bell network has boundary equal to a nonzero scalar multiple of* f*.*

3. Every nonzero character-flat signature of arity at most six is, after a port permutation, a nonzero scalar multiple of a product of ordered Bell binaries on a perfect matching.

4. CFlat *is closed under nonzero scalars, tensor products, port permutations, one-port ordered-Bell attachments through native* X*-edges, and nonzero tensor factors.*

Proof. Let B be the displayed boundary-port set and let V_{vert} be the tensor-vertex set. As in Lemma 7.14, all displayed boundary ports in this calculus are unpaired ordinary vertex ports; the pairs below are the internal wire pairs. Let H_{int} be the set of internal half-edges. For every vertex let L_v be the linear direction of its affine support. The Reed–Muller direction and the two-coordinate repetition direction of every Bell binary are self-dual. Define the internal equality subspace E_{eq}, the ambient constraint subspace H_{con}, and the vertex-code sum C_0 by

$$\begin{aligned} E_{\mathrm{eq}} &= \{y \in \mathbb{F}_2^{H_{\mathrm{int}}} : y_h = y_{h'} \text{ on every internal pair } \{h, h'\}\}, \\ E_{\mathrm{eq}} &= E_{\mathrm{eq}}^{\perp}, \qquad\qquad H_{\mathrm{con}} = \mathbb{F}_2^{B} \oplus E_{\mathrm{eq}}, \\ C_0 &= \bigoplus_{v \in V_{\mathrm{vert}}} L_v = C_0^{\perp}. \end{aligned}$$

Exactly as in the proof of Lemma 7.14, the feasible boundary direction is $L = \pi_B(C_0 \cap H_{\mathrm{con}}) = L^{\perp}$.

Define the feasible half-edge code $\mathcal{K} := C_0 \cap H_{\mathrm{con}}$ and its internal-fiber subspace

$$J := \mathcal{K} \cap (\{\mathbf{0}_B\} \oplus \mathbb{F}_2^{H_{\mathrm{int}}}),$$

where $\mathbf{0}_B$ is the zero boundary word. The product of the ordered-Bell signs is a linear character $(-1)^{\lambda(z)}$ on the half-edge variables. For a fixed feasible boundary word the internal fiber is $z_0 + J$, and

$$\sum_{j \in J} (-1)^{\lambda(z_0 + j)} = (-1)^{\lambda(z_0)} \sum_{j \in J} (-1)^{\lambda(j)}.$$

If $\lambda|_J \neq 0$, every nonempty boundary fiber cancels; otherwise the same nonzero scalar remains on every fiber and the character descends to the boundary. This proves item 1 and shows that partial cancellation cannot create a non-Lagrangian support.

Conversely, the actual network in Lemma 7.14 realizes the indicator of $a + L$. Attaching a Bell binary through X induces, up to a nonzero sign, one of I, X, Z, Y, so actual ordered-Bell attachments implement the shift and linear character, proving item 2. More explicitly, $B_{d,\ell}X$ is respectively X, Y, I, Z for $(d, \ell) = (0,0), (0,1), (1,0), (1,1)$, up to the fixed orientation signs. Arity zero is the empty matching. At positive arity at most six, Lemma 7.16 splits the self-dual direction into two-coordinate repetition lines. Support shifts and characters turn their factors into ordered Bell signatures, proving item 3.

For the factor claim in item 4, let $f = f_1 \otimes f_2 \neq 0$. The two projections of its affine support are affine; write its Cartesian support as $(a_1 + L_1) \times (a_2 + L_2)$, with $L = L_1 \oplus L_2$. Since $L = L^{\perp}$, each L_j is self-dual in its own coordinate space. Fixing a supported word of the other factor restricts the character of f to a character on $a_j + L_j$, up to a nonzero scalar. Hence f_j is character-flat. The other closure properties follow by the same direct-sum and character identities. □

Corollary 7.21 (No pure-Pauli inverse from RM_8 to H_6)**.** *No network using only* RM_8*, actual ordered Bell binaries, and native* X*-edges has boundary proportional to* H_6*.*

Proof. Every nonzero such boundary lies in CFlat by Lemma 7.20. A character-flat six-port support is a coset of a self-dual subspace of $\mathbb{F}_2^6$, hence has dimension three. By direct expansion of Equation (17), the support of H_6 is the six-bit even-parity space, of dimension five. Thus $\mathrm{H}_6 \notin \mathsf{CFlat}$. Consequently the direct H_6-to-RM_8 circuit constructed below has no inverse using only RM_8, ordered Bell binaries, and native X-edges. □

The support-coset and character data in Equation (20) can be evaluated by affine Gaussian elimination.

For every even n, define the accessible-direction set

$$\mathrm{Acc}_n = (\mathbf{1}^n)^\perp \setminus \{0, \mathbf{1}^n\}.$$

Equivalently, Acc_n consists of the nonzero even-weight vectors other than $\mathbf{1}^n$. In the two smallest even arities,

$$\mathrm{Acc}_2 = \varnothing, \qquad \mathrm{Acc}_4 = \{e_i + e_j : 1 \le i < j \le 4\}.$$

Recall

$$O(n,2) := \{A \in \mathrm{GL}_n(\mathbb{F}_2) : A^\mathsf{T} A = I_n\}.$$

For $A \in O(n,2)$, write

$$\mathrm{Graph}(A) = \{(x, Ax) : x \in \mathbb{F}_2^n\}.$$

As proved in Lemma 7.15, this direction is self-dual; let Γ_A denote a fixed actual $\mathrm{RM}_8/I/X$-network supplied by Lemma 7.14 whose boundary is a nonzero scalar multiple of $\mathbf{1}[\mathrm{Graph}(A)]$.

Lemma 7.22 (Actual signed Radon cards)**.** *Assume* RM_8 *and actual representatives of all four ordered Bell binaries are retained, and let g have even arity n. For each $v \in \mathrm{Acc}_n$, use the recorded data $(A_v, i_v, j_v, r_v, \Pi_v, W_v)$ fixed in Lemma 7.15, and fix the actual graph network Γ_{A_v}. For $d, \ell \in \mathbb{F}_2$, using one copy of g, this graph network, native X-edges, and the actual ordered Bell binary $B_{d,\ell}$, one directly realizes, up to a known nonzero scalar, the signed card*

$$\mathcal{S}_{v,d,\ell} g([x]_v) = (-1)^{\ell (A_v x)_{i_v}} \big(g(x) + (-1)^\ell g(x+v)\big), \qquad [x]_v \in H_{v,d}/\operatorname{span}_{\mathbb{F}_2}\{v\}. \tag{83}$$

The cases $\ell = 0$ and $\ell = 1$ are called respectively the plus *and* minus *signed cards on layer d. The prefactor $(-1)^{\ell (A_v x)_{i_v}}$ is the exterior Bell sign in the chosen quotient coordinates; after stripping this sign and the recorded nonzero scalar, the two orbit numerators are $g(x)+g(x+v)$ and $g(x)-g(x+v)$. The remaining $n-2$ graph-output coordinates give an explicit affine isometry from this quotient to the boundary coordinate space. Replacing (A_v, i_v, j_v) by another such choice changes the displayed analytic card only by a quotient affine isometry, a nonzero scalar, and a linear sign character.*

Proof. Connect g to the input side of Γ_{A_v} through native X-edges, and let x denote the variables at g. The graph input is therefore $x + \mathbf{1}$. Every $A \in O(n,2)$ fixes $\mathbf{1}$: invariance of $z \cdot z = z \cdot \mathbf{1}$ gives $A^\mathsf{T}\mathbf{1} = \mathbf{1}$, and hence $A\mathbf{1} = \mathbf{1}$. Thus the graph output is $z = A_v(x + \mathbf{1}) = A_v x + \mathbf{1}$. Connect output ports i_v, j_v to $B_{d,\ell}$ through their two native X-edges. The ordered Bell arguments are then

$$u = z_{i_v} + 1 = (A_v x)_{i_v}, \qquad w = z_{j_v} + 1 = (A_v x)_{j_v}.$$

Consequently its support equation is

$$u + w = x \cdot A_v^\mathsf{T}(e_{i_v} + e_{j_v}) = x \cdot v = d.$$

For fixed remaining graph-output coordinates there are two lifts, differing by $e_{i_v} + e_{j_v}$ in z and hence by v in x. The Bell sign is $(-1)^{\ell u} = (-1)^{\ell (A_v x)_{i_v}}$ and changes between those lifts by $(-1)^\ell$, giving Equation (83). The graph relation is single-valued before the two-port contraction, so the realization scalar is nonzero.

Each choice identifies the quotient affinely and isometrically with its remaining graph-output coordinates. The transition between two choices is therefore an affine isometry. Their selected-coordinate functionals differ on the quotient by a linear form and a constant, which gives the last assertion. □

Lemma 7.23 (Character-aware two-chart localization)**.** *Assume the resources of Lemma 7.22 are retained. Let g have even arity $n \geq 10$, fix $v \in \mathrm{Acc}_n$, and suppose that the two plus cards $\mathcal{S}_{v,0,0}g$ and $\mathcal{S}_{v,1,0}g$ are zero or character-flat. Define, only for the analysis, the orbit-sum signature*

$$F_v = \mathsf{P}_v^+ g.$$

If $F_v \notin \mathscr{A}$, an actual RM_8/ordered-Bell context using one copy of g realizes a nonzero $H \notin \mathscr{A}$ of arity at most eight.

Proof. Use the recorded data $(A_v, i_v, j_v, r_v, \Pi_v, W_v)$ from Lemma 7.15, and put $W := W_v$. The plane Π_v is nondegenerate, and the map $\iota_{v,\epsilon}$ identifies W isometrically with $H_{v,\epsilon}/\operatorname{span}_{\mathbb{F}_2}\{v\}$. Since F_v is invariant under T_v, its two quotient charts are

$$f_\epsilon(x) := F_v(\epsilon r_v + x), \qquad x \in W.$$

For each nonzero chart choose, without asserting uniqueness, representatives

$$f_\epsilon(x) = c_\epsilon(-1)^{\lambda_\epsilon(x)}\mathbf{1}[x \in a_\epsilon + M_\epsilon], \qquad M_\epsilon = M_\epsilon^{\perp} \leq W. \tag{84}$$

Here a_ϵ is chosen modulo M_ϵ, and λ_ϵ is an ambient extension of the character on the support. Changing either choice changes c_ϵ only by a sign, so membership of c_1/c_0 in μ_4 is independent of these choices.

If one chart is zero, the other gains only one affine layer equation and $F_v \in \mathscr{A}$. Suppose both are nonzero. Their union has affine support exactly when $M_0 = M_1$. When the directions agree, the difference of the two characters is absorbed by an allowed quadratic mixed term with the layer bit, and the layer scales extend to an affine fourth-root phase exactly when $c_1/c_0 \in \mu_4$. Thus $F_v \notin \mathscr{A}$ gives

$$M_0 \neq M_1, \qquad \text{or} \qquad M_0 = M_1 \ \text{ and } \ c_1/c_0 \notin \mu_4. \tag{85}$$

Represent $\lambda_0 + \lambda_1$ by $\rho \in W$, so that $(\lambda_0 + \lambda_1)(x) = \rho \cdot x$. When $M_0 \neq M_1$, choose $s \in M_0 \setminus M_1$ and $t \in M_1$ with $s \cdot t = 1$, and put $\Delta = a_1 - a_0$. The plane $\operatorname{span}_{\mathbb{F}_2}\{s, t\}$ is nondegenerate. Put $S_0 = \operatorname{span}\{s, t, \Delta, \rho\}$. Its radical has dimension at most two; pairing a basis of that radical with dual vectors gives a nondegenerate extension $N \supseteq S_0$ of dimension at most six. If N has odd dimension, then its nondegenerate orthogonal complement is odd-dimensional and contains an anisotropic vector; adjoining that orthogonal line preserves the bound and makes the extension even-dimensional. Denote the result by

$$U \supseteq \operatorname{span}\{s, t, \Delta, \rho\}, \qquad \dim U \leq 6.$$

In the equal-direction case, start from $S_0 = \operatorname{span}\{\Delta, \rho\}$. Pairing its radical needs at most two dual vectors and gives a nondegenerate extension of dimension at most four; if its dimension is odd, adjoin an anisotropic vector from its odd-dimensional nondegenerate complement. This gives an even-dimensional nondegenerate $U \supseteq \operatorname{span}\{\Delta, \rho\}$ with $\dim U \leq 4$. Put

$$P := U^{\perp_W}, \qquad W = U \perp P,$$

and let π_U, π_P be the corresponding orthogonal projections. A Witt decomposition of the even-dimensional nondegenerate space P supplies

$$J \leq P, \qquad J = J^{\perp_P}.$$

Because $\Delta, \rho \in U$, the two shifts have the same P-projection

$$p_0 := \pi_P(a_0) = \pi_P(a_1),$$

and the two characters have the same restriction $\lambda_P := \lambda_0|_P = \lambda_1|_P$.

Let $S := \Pi_v \perp U$, put $m := \dim S$, and choose an isometry $\phi : S \to \mathbb{F}_2^m$. Such an isometry exists because S contains the nonalternating plane Π_v. We have $m \le 8$ in the different-direction case and $m \le 6$ in the equal-direction case. Define

$$R_0 := \{(s_0 + j, \phi(s_0)) : s_0 \in S,\ j \in J\} \le \mathbb{F}_2^n \oplus \mathbb{F}_2^m.$$

For two elements of R_0, their inner product is

$$(s_0 + j) \cdot (s_0' + j') + \phi(s_0) \cdot \phi(s_0') = j \cdot j' = 0.$$

Moreover $\dim R_0 = m + (n-m)/2 = (n+m)/2$, so $R_0 = R_0^\perp$. On R_0, define the linear sign character

$$\chi(s_0 + j, \phi(s_0)) := (-1)^{\lambda_P(j)}.$$

Choose an ambient linear-character extension and denote it again by χ. Hence, for $(x, y) \in \mathbb{F}_2^n \oplus \mathbb{F}_2^m$,

$$C(x, y) := \chi(x - p_0, y)\, \mathbf{1}[(x, y) \in (p_0, 0) + R_0] \tag{86}$$

is character-flat. By Lemma 7.20, choose an actual $O(n^2)$-size RM_8/ordered-Bell realization whose boundary is κC, where the realization scalar $\kappa \in \overline{\mathbb{Q}}^\times$ is recorded. Any fixed coordinate complements contributed by its native X-attachments are absorbed into the displayed affine shift.

Contract this context with F_v. Its boundary is

$$H^+(\phi(s_0)) = \kappa \sum_{j \in J} (-1)^{\lambda_P(j)} F_v(s_0 + p_0 + j), \qquad s_0 \in S. \tag{87}$$

In the different-direction case, write $a_\epsilon = u_\epsilon + p_0$ with $u_\epsilon \in U$. The three boundary points $\phi(u_0)$, $\phi(u_0 + s)$, and $\phi(r_v + u_1)$ give nonzero sums. On each support fiber the context character cancels the common P-restriction of the chart character, so there is no cancellation. The fourth point $\phi(r_v + u_1 + s)$ has value zero: otherwise some $j \in J \subseteq P$ would satisfy $s + j \in M_1$, and then

$$0 = (s + j) \cdot t = s \cdot t = 1.$$

Thus the support of H^+ is not affine. In the equal-direction case, the two corresponding fibers have the same nonzero cardinality $|J \cap M_0|$. Their value ratio differs from c_1/c_0 only by a sign, and therefore remains outside μ_4. In both cases $H^+ \neq 0$ and $H^+ \notin \mathscr{A}$.

Finally, contract the same actual context with g, and call its boundary H. Put $w := \phi(v)$. Translation by v on the input of Equation (86) is translation by w on its boundary, and the context character is trivial on (v, w). Therefore

$$H^+ = \mathsf{P}_w^+ H.$$

If $H \in \mathscr{A}$, then Lemma 2.7 would give $H^+ \in \mathscr{A}$, a contradiction. Thus $H \neq 0$, $H \notin \mathscr{A}$, and its arity m is at most eight. □

Lemma 7.24 (Affine-to-character-flat recognition)**.** *Let $0 \neq g \in \mathscr{A}$ have even arity $2r \ge 10$. If every signed card $\mathcal{S}_{v,d,\ell} g$, for $v \in \mathrm{Acc}_{2r}$ and $d, \ell \in \mathbb{F}_2$, is zero or character-flat, then $g \in \mathsf{CFlat}$.*

Proof. Input translation permutes the two values of d and reindexes the corresponding quotient by an affine isometry. Under this reindexing, it changes a signed card only by a nonzero scalar and a linear sign character. We may therefore translate the affine support to a linear subspace and write

$$g(x) = c\, i^{Q(x)} \mathbf{1}[x \in U], \qquad c \neq 0, \qquad d_U := \dim U. \tag{88}$$

Put $E := \mathbf{1}^\perp$. Every nonzero character-flat card has arity $2r-2$ and hence support dimension $r-1$.

We first prove $U = U^\perp$. If $E \subseteq U$, then $d_U \geq 2r-1$. For an accessible $v \in E$, at least one signed card on a nonempty layer is nonzero. Restriction to the layer and quotienting by $\operatorname{span}_{\mathbb{F}_2}\{v\}$ can each remove at most one dimension. More explicitly, on an overlapping support orbit, factoring out $g(x)$ leaves

$$1 \pm \zeta(-1)^{\theta(x)}, \qquad \zeta \in \mu_4,$$

where θ is a binary linear form. If $\zeta \in \{\pm i\}$, this factor is nowhere zero. If $\zeta \in \{\pm 1\}$, its nonzero set is either empty, the whole current support, or one affine hyperplane. Thus a nonzero signed card loses at most one additional support dimension through cancellation and has support dimension at least

$$d_U - 3 \geq 2r - 4 > r - 1,$$

a contradiction. If $E \subseteq U^\perp$, then $d_U \leq 1$; choosing accessible $v \in E \setminus U$ gives a nonzero card of support dimension d_U, again a contradiction. Hence neither containment holds.

The two proper subspaces $E \cap U$ and $E \cap U^\perp$ do not cover E. Since $\dim E = 2r-1 \geq 9$, choose

$$v \in \mathrm{Acc}_{2r} \setminus (U \cup U^\perp).$$

There is no cancellation because $v \notin U$, and each nonempty card has support dimension $d_U - 1$. Thus $d_U = r$.

No accessible vector can lie in $U^\perp \setminus U$, because such a vector would give a nonzero card of support dimension r. Put $A := E \cap U^\perp$. It follows that

$$A \setminus \{0, \mathbf{1}\} \subseteq U. \tag{89}$$

If $\mathbf{1} \notin A$, or if $\mathbf{1} \in A \cap U$, this gives $A \subseteq U$. If instead $\mathbf{1} \in A \setminus U$, then $\dim A \geq r - 1 \geq 4$; choose $z \in A \setminus \{0, \mathbf{1}\}$. Both z and $z + \mathbf{1}$ are accessible and lie in U by the preceding display, forcing $\mathbf{1} \in U$, a contradiction. Thus $A \subseteq U$ in every case.

If $U^\perp \subseteq E$, equal dimensions give $U = U^\perp$. Otherwise $\dim A = r - 1$. Since $A \subseteq U$ and $\dim U = r$, choose $u \in U \setminus A$, so $U = A \oplus \operatorname{span}_{\mathbb{F}_2}\{u\}$. Because $A \subseteq U^\perp$, the vector u is orthogonal to A. If u were even, then $u \cdot u = 0$, hence $u \in U^\perp$ and $U \subseteq U^\perp$; equal dimensions would again give $U = U^\perp$. Therefore u is odd and

$$U = A \oplus \operatorname{span}_{\mathbb{F}_2}\{u\}, \qquad u \cdot u = 1. \tag{90}$$

Since u is odd, $\mathbf{1} \cdot u = 1$, whereas every vector of A is orthogonal to u; hence $\mathbf{1} \notin A$. Fix $0 \neq s \in A$. Here $\dim A = r - 1 \geq 4$, so s is accessible. For this s, a nonzero signed card has either the full quotient direction $U/\operatorname{span}_{\mathbb{F}_2}\{s\}$, of dimension $r-1$, or a codimension-one subdirection of dimension $r-2$. The first is not self-dual because its orthogonal direction is $U^\perp/\operatorname{span}_{\mathbb{F}_2}\{s\} \neq U/\operatorname{span}_{\mathbb{F}_2}\{s\}$; the second has the wrong dimension. On the supported layer $d = 0$, at least one of the plus and minus cards is nonzero: after stripping their respective exterior Bell signs and recorded nonzero scalars, their orbit brackets are

$$g(x) + g(x+s) \quad \text{and} \quad g(x) - g(x+s),$$

whose sum is $2g(x)$. This contradicts the hypothesis. Therefore

$$U = U^{\perp}.$$

The orthogonal reduction in the proof of Lemma 7.14 sends U to the code

$$\mathsf{Pair}_r := \{x(t) : t \in \mathbb{F}_2^r\}, \qquad x(t) := (t_1, t_1, \ldots, t_r, t_r).$$

Orthogonal maps preserve $\mathbf{1}$ and hence accessible directions. Their induced quotient affine isometries, together with nonzero scalar and linear-character changes, preserve character-flatness, so the signed-card hypothesis is invariant under this coordinate reduction. Put

$$g(x(t)) = c\, h(t),$$

where

$$h(t) = i^{q_0 + \sum_j \alpha_j t_j + 2\sum_{j<k} \beta_{jk} t_j t_k}, \qquad q_0, \alpha_j \in \mathbb{Z}_4, \quad \beta_{jk} \in \mathbb{F}_2. \tag{91}$$

Extend the quadratic coefficients symmetrically by setting $\beta_{jk} := \beta_{kj}$ whenever $k < j$, and put $\beta_{jj} := 0$. Let $\bar{e}_j$ be the j-th basis vector of $\mathbb{F}_2^r$, and define the j-th logical pair and its physical accessible direction by

$$P_j := \{2j-1, 2j\}, \qquad v_j := e_{2j-1} + e_{2j}.$$

Let

$$\mathcal{T}_j := \{t \in \mathbb{F}_2^r : t_j = 0\}.$$

Using the representatives $t \in \mathcal{T}_j$, and stripping the exterior Bell signs and recorded nonzero scalars, the $d = 0$ plus and minus cards in the v_j-direction have coefficient functions

$$A_j(t) = h(t) + h(t + \bar{e}_j), \qquad B_j(t) = h(t) - h(t + \bar{e}_j), \qquad t \in \mathcal{T}_j. \tag{92}$$

Whenever one of these cards has support $\mathcal{T}_j$, character-flatness makes its coefficient function a nonzero scalar times a sign character on $\mathcal{T}_j$.

Define

$$L_j(t) := \sum_{k \neq j} \beta_{jk} t_k \in \mathbb{F}_2, \qquad \frac{h(t + \bar{e}_j)}{h(t)} = i^{\alpha_j} (-1)^{L_j(t)}.$$

Suppose first that some α_j is even. If $L_j \neq 0$, the two functions in Equation (92) are supported on complementary proper hyperplanes, so neither nonzero card is character-flat. Thus $L_j = 0$, which gives $\beta_{jk} = 0$ for every $k \neq j$. One of A_j, B_j is then zero and the other is a nonzero scalar multiple of $h|_{t_j=0}$. Its being a sign character forces

$$\alpha_k \equiv 0 \pmod 2 \quad (k \neq j), \qquad \beta_{km} = 0 \quad (k, m \neq j).$$

Together with the evenness of α_j, this proves that all α_k are even and all β_{km} vanish.

It remains to exclude the alternative in which every α_j is odd. Write $i^{\alpha_j} = \delta_j i$, where $\delta_j \in \{\pm 1\}$. The plus bracket in A_j satisfies

$$1 + \delta_j i (-1)^{L_j} = (1 + \delta_j i)(-\delta_j i)^{L_j}. \tag{93}$$

The multilinear modulo-four expansion of the Boolean parity L_j is

$$L_j(t) \equiv \sum_{k \neq j} \beta_{jk} t_k + 2 \sum_{\substack{k<m \\ k,m \neq j}} \beta_{jk}\beta_{jm} t_k t_m \pmod 4. \tag{94}$$

Since the nonzero full-support card A_j is a sign character, all mixed quadratic coefficients in its phase vanish. Hence, for all distinct j, k, m,

$$\beta_{km} = \beta_{jk}\beta_{jm}. \tag{95}$$

View β as the adjacency matrix of a simple graph. These relations make the graph either empty or complete. Indeed, if $\beta_{ab} = 1$, then for every $c \notin \{a, b\}$,

$$1 = \beta_{ab} = \beta_{ca}\beta_{cb},$$

so c is adjacent to both a and b; applying Equation (95) with $j = a$ joins every remaining pair.

The empty case is impossible: when $L_j = 0$, either nonzero card in Equation (92) is a scalar multiple of $h|_{t_j=0}$, whose ratio in any remaining coordinate k is $i^{\alpha_k} = \pm i$, not the ± 1 ratio of a sign character.

In the complete case, choose distinct logical pairs a, b, c and the physical weight-two direction

$$w := e_{2a-1} + e_{2b-1} \in \mathrm{Acc}_{2r}\,.$$

This vector is not in Pair_r. The support of the plus card $\mathcal{S}_{w,0,0}g$ is the image of $\{t : t_a + t_b = 0\}$, and on the two-point w-translation orbit represented by $x(t)$ its coefficient is $c\,h(t)$, because $x(t)+w \notin U$. Toggling the third logical pair c preserves this support, but its coefficient ratio is

$$\frac{h(t + \bar{e}_c)}{h(t)} = i^{\alpha_c}(-1)^{t_c + \sum_{k\neq c} t_k} \in \{\pm i\}.$$

A character-flat signature has ratio ± 1 along every direction of its support. This contradiction eliminates the complete case.

Thus the all-odd alternative is impossible. The earlier even-α_j case applies, and Equation (91) reduces to a nonzero scalar times a linear sign character on the self-dual support U. This is exactly Equation (20). □

8 Klein-Four Matching Decks

The two marked Klein-four groups require different arguments. For the standard Pauli group, pair contractions detect whether a complex tensor is a scalar multiple of a real tensor; this reduces the remaining case to the real-valued Holant dichotomy. For the exotic group, contractions compare four matrix labels on matching edges and force a proper tensor factor. We use the Bell notation and cores $\mathrm{H}_6, \mathrm{RM}_8$ of Section 2.9. The six-port exotic classification remains the finite certificate in Appendix A.4.

Theorem 8.1 (Klein-four relative rigidity)**.** *Assume*

$$\mathsf{Stable}(\Lambda, G, f), \qquad \mathrm{arity}(f) = \nu(\Lambda), \qquad G \in \{V_4, \mathcal{V}_{\mathrm{ex}}\}.$$

In the standard Pauli case $G = V_4$, assume additionally that $\nu(\Lambda) \geq 6$. Then $\mathsf{Close}_{\{\mathscr{A}\}}(\Lambda, G, f)$.

The two cases are proved in Theorems 8.3 and 8.10. As in the definition of Close, a resolved Tract-leaf completes the branch.

Proof strategy. In the standard Pauli form, pair contractions determine a tensor line above arity four. A retained signature that is not a scalar times a real tensor therefore yields an actual quaternary tensor with the same property. Factor saturation gives a terminal or a strict outer successor. When

every retained signature is a scalar times a real tensor, the real-valued Holant dichotomy closes the entire retained set. The explicit H_6-to-RM_8 circuit is retained for the Platonic analysis.

In the exotic Klein form, the physical exotic calculus and the six-port base first force a common residual ruling, after which fusion constrains the label maps to permutation fibres. Two choices of a dual-cross contraction exclude a nonfactor at arity eight; at arity at least ten, one such contraction suffices. These factor conclusions exclude a higher-arity stable prime, while the quaternary interface gives a resolved leaf or a strict outer successor.

8.1 The standard V_4 form: actual Pauli contractions

Let Λ be retained and factor-saturated, with the distinguished literal equality I and complete group V_4. Let the Bell-label space be $\mathsf{L} := \mathbb{F}_2^2$. For a Bell label $\gamma = (d, \ell) \in \mathsf{L}$, let $B_\gamma = B_{d,\ell}$ be the ordered Bell signature defined in Section 2.9. The chosen actual representative fixed in Section 2.5 is $\beta_\gamma B_\gamma$, where $\beta_\gamma \in \overline{\mathbb{Q}}^\times$ is its known nonzero realization scalar. The formulas below divide actual outputs by these recorded scalars.

Here and below, a *companion* is a retained tensor-prime occurrence other than the distinguished minimum core; its occurrence carries its actual/Turing provenance. For $\alpha = (d, \ell) \in \mathsf{L}$, put

$$P_\alpha := Z^\ell X^d.$$

All contractions below are bilinear, without complex conjugation. For an ordered pair p, define

$$C_p^\gamma(h) := \left\langle B_\gamma^{(p)}, h \right\rangle_p, \qquad \gamma \in \mathsf{L},$$

and omit the argument only when the parent tensor is unambiguous. For a list $\mathbf{p} = (p_1, \dots, p_s)$ of disjoint ordered pairs and $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_s)$, put

$$C_{\mathbf{p}}^{\boldsymbol{\gamma}}(h) := C_{p_s}^{\gamma_s} \cdots C_{p_1}^{\gamma_1}(h).$$

The order is immaterial for disjoint pairs. A *proper Pauli card* means such an actual iterated contraction leaving at least two ports; a *Pauli pair card* has $s = 1$. Dividing the physical output by its recorded nonzero realization scalar gives the displayed canonical card. The phrase *local-Pauli orbit* has the precise projective meaning

$$\mathrm{Orb}_{V_4}(h) := \left\{ c \left(\bigotimes_{j=1}^{\mathrm{arity}(h)} P_{\alpha_j} \right) (\pi h) : c \in \mathbb{C}^\times,\ \alpha_j \in \mathsf{L},\ \pi \in S_{\mathrm{arity}(h)} \right\};$$

port reversal signs and realization scalars are included in the recorded nonzero scalar c.

A *complete Bell block* is an ordered tensor $B_\gamma^{(p)}$ with both ports of p available for attachment. An *I-path* between two equally oriented complete blocks is one copy of the retained equality I joining corresponding endpoints; connecting the blocks uses the two corresponding I-paths, with their recorded realization scalars.

8.2 The Pauli cores

For the fixed ordered pairs $A = (1, 2), B = (3, 4), C = (5, 6)$, and with the order-three Bell-label map Θ defined in Section 2.9, define the six-port Pauli core signature

$$\mathrm{H}_6 = \sum_{\gamma \in \mathsf{L}} B_\gamma^A B_{\Theta\gamma}^B B_{\Theta^2\gamma}^C.$$

Direct expansion gives

$$\operatorname{supp}(\mathrm{H}_6) = \left\{ x \in \mathbb{F}_2^6 : \sum_{j=1}^{6} x_j = 0 \right\},$$
$$\mathrm{H}_6(x) = (-1)^{x_1+x_1x_2+x_1x_6+x_2x_5+x_3x_5+x_3x_6} \quad (x \in \operatorname{supp}(\mathrm{H}_6)).$$

Hence $\mathrm{H}_6 \in \mathscr{A}$, and its even-parity support is tensor-prime. By Remark 2.5, H_6 and RM_8 are the Shao–Cai $\widehat{f_6}$ and $f_8 = \widehat{f_8}$, up to the stated port permutation. Consequently every Pauli pair card of H_6 is a nonzero product of two Pauli binaries, while every Pauli pair card of RM_8 is, up to a nonzero scalar, a product of three copies of the contracted Pauli binary; see [SC20, full version, Definition 6.12 and Lemma 6.13] and [CFS20b, Theorems 13–14].

8.3 The direct H_6-to-RM_8 circuit

In four copies $\mathrm{H}_6^{(v)}$, write the ordered blocks $(\mathsf{A}_v, \mathsf{B}_v, \mathsf{C}_v) = (B_{\gamma_v}, B_{\Theta\gamma_v}, B_{\Theta^2\gamma_v})$ and leave the A_v external. Connect equally oriented complete blocks by

$$\mathsf{B}_1 - \mathsf{B}_2, \qquad \mathsf{C}_2 - \mathsf{C}_3, \qquad \mathsf{B}_3 - \mathsf{B}_4, \qquad \mathsf{C}_4 - \mathsf{C}_1.$$

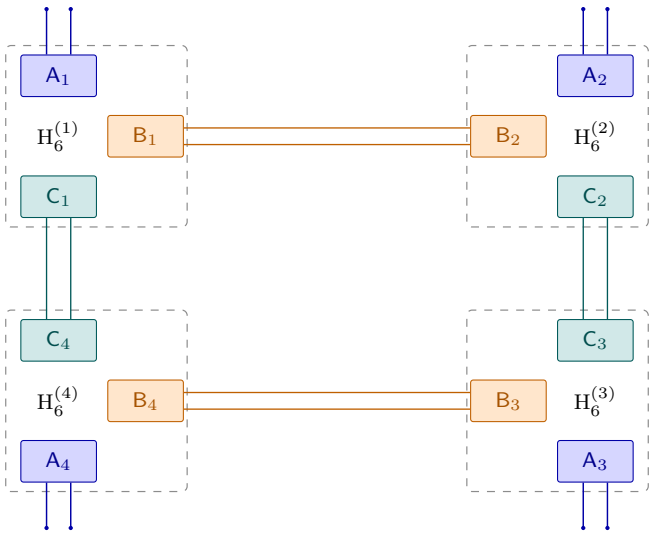


Figure 1: The four-copy H_6-to-RM_8 circuit. Each doubled internal connection represents the two corresponding I-paths between complete blocks. The blue external legs are the eight ports of the resulting RM_8.

Each connection uses two legal I-paths and contributes $2\delta_{\gamma_u,\gamma_v}$, so all labels agree without cancellation:

$$2^4 \sum_{\gamma \in \mathsf{L}} B_\gamma^{\otimes 4} = 32\mathrm{RM}_8. \tag{96}$$

Restoring the known nonzero realization and path scalars gives

$$\text{K-Holant}(\Lambda, \mathrm{RM}_8) \leq_{\mathrm{T}} \text{K-Holant}(\Lambda, \mathrm{H}_6).$$

If H_6 is already actual or Turing-exposed over Λ, then $\text{K-Holant}(\Lambda, \mathrm{H}_6) \leq_{\mathrm{T}} \text{K-Holant}(\Lambda)$ as well. In the actual case, adjoining the circuit output cannot enlarge the complete actual binary group, which remains V_4. The circuit itself only requires H_6 and actual I-paths, so it also applies in an ambient language with a larger complete group. A reverse gadget construction using only RM_8 and pure Pauli resources is excluded by Corollary 7.21.

8.4 The relative implication

The next identity extracts information from all pair contractions without reconstructing a tensor from an enumerated list. It compares two tensors by their antisymmetrized product $u \otimes v - v \otimes u$: this product vanishes exactly when they lie on the same line. Pair contractions will force it to vanish once there are more than four ports.

Lemma 8.2 (Pauli contractions determine a tensor line)**.** *Let $n \geq 5$, and let $u, v \in (\mathbb{C}^2)^{\otimes n}$. If $C^{\gamma}_{rs}(u)$ and $C^{\gamma}_{rs}(v)$ are linearly dependent for every pair $r < s$ and every $\gamma \in \mathsf{L}$, with zero cards allowed, then u, v are linearly dependent. Consequently, if every Pauli pair card of u is zero or a nonzero complex scalar times a real tensor, the same holds for u.*

Proof. Put $W = u \otimes v - v \otimes u$ and group its two copies by physical port, so that $W \in (\mathbb{C}^4)^{\otimes n}$. The hypothesis says

$$(C^{\gamma}_{rs} \otimes C^{\gamma}_{rs})W = 0 \qquad (r < s,\ \gamma \in \mathsf{L}).$$

On the regrouped four-dimensional spaces at r, s, the four contraction matrices are

$$I \otimes I, \quad Z \otimes Z, \quad X \otimes X, \quad (ZX) \otimes (ZX).$$

Use the same orthonormal Bell basis $2^{-1/2}(\operatorname{vec} I, \operatorname{vec} Z, \operatorname{vec} X, \operatorname{vec}(ZX))$ on each regrouped port. These matrices are diagonal in that basis, with eigenvalue rows

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

The rows are independent. Thus the four contraction equations, for any fixed values on the other ports, force each coefficient with equal Bell basis indices at r, s to vanish. Applying this to every pair leaves only words whose n indices are all distinct. There are just four indices, so $n \geq 5$ gives $W = 0$, proving the first assertion.

For the last assertion take $v = \overline{u}$. The Pauli kernels are real, and a vector and its conjugate are dependent precisely when it is zero or a scalar times a real vector: divide by its first nonzero coordinate to see the reverse implication. Conjugation and the Bell basis above are used only to prove this identity; they are not gadgets. □

Theorem 8.3 (Standard Pauli V_4 relative closure)**.** *If*

$$\mathsf{Stable}(\mathcal{G}, V_4, f), \qquad \nu(\mathcal{G}) = \operatorname{arity}(f) \geq 6,$$

then $\mathsf{Close}_{\{\mathscr{A}\}}(\mathcal{G}, V_4, f)$.

Proof. Write $\Lambda = \mathcal{G}$. Dispatch odd, unary, and unsafe binary factors by the established reductions in Theorem 2.15 and Lemmas 2.20, 3.1, 3.4 and 5.3. On the continuing branch every nonbinary prime is even, and all binary prime factors are nonsingular Pauli binaries up to nonzero scalars.

Suppose some retained signature is not a scalar times a real tensor. Factor saturation supplies such a tensor-prime signature g, necessarily of even arity $n \geq \nu(\Lambda) \geq 6$. The contrapositive of Lemma 8.2 selects a Pauli pair card with the same property. Repeat while its arity is at least six, reaching a nonzero quaternary tensor q that is not a scalar times a real tensor. All selected contractions are actual: complete V_4 supplies the four Pauli kernels, including their reversal and

native-edge scalars. Their composition uses one g-vertex and $O(n)$ binary vertices. The intermediate coefficient tables only select this one finite network.

Adjoin q and process its complete factor batch by Definition 5.9. An odd or unsafe factor is resolved; a new finite binary class gives a strict group successor, with the deck regenerated. Otherwise, if all factors were binaries, they would all be Pauli and their product would be a scalar times a real tensor, contrary to the choice of q. Hence an even nonbinary prime remains. It has arity four, so the retained augmentation strictly lowers $\nu(\Lambda) \geq 6$. Retained I excludes the one-sided exception in factor saturation. Thus this entire batch gives an established leaf or a strict outer successor; it opens no new unresolved branch with unchanged group and ν.

It remains to treat the case in which every retained signature is a scalar times a real tensor. Divide each by its first nonzero coefficient and call the resulting real algebraic set Λ_0. These known scalars are accounted for per vertex, and $I \in \Lambda_0$. The equality-wire equivalence Lemma 5.1 gives

$$\text{K-Holant}(\Lambda) \equiv_{\mathrm{T}} \text{K-Holant}(\Lambda_0) \equiv_{\mathrm{T}} \text{Holant}(\mathcal{F}), \qquad \mathcal{F} := \Lambda_0 \cup \{X\}.$$

Apply the real-valued dichotomy of Shao and Cai [SC20, full version, Theorems 1.2 and 2.22] to this whole finite real set. If its condition (T) fails, hardness composes along the displayed reductions. If it holds, the following argument supplies the required $\mathsf{Tract}(K\Lambda)$ leaf in our coordinates.

The alternative $\mathcal{F} \subseteq \langle \mathscr{T} \rangle$ immediately implies $K\Lambda \subseteq \langle \mathscr{T} \rangle$, since invertible one-port maps and nonzero scalars preserve tensor products of unary and binary signatures. In each other alternative there is one algebraic matrix $M \in \mathrm{GL}_2(\overline{\mathbb{Q}})$, by Lemma 2.13, and one $\mathscr{C} \in \{\mathscr{P}, \mathscr{A}, \mathscr{L}\}$ with

$$E := M^{\mathsf{T}} M \in \mathscr{C}, \qquad M^{-1}\mathcal{F} \subseteq \mathscr{C}.$$

In particular $B := M^{-1} X M^{-\mathsf{T}} \in \mathscr{C}$, and

$$M^{\mathsf{T}} X M = EBE.$$

Both E and B are symmetric nonsingular members of $\mathscr{C}$. Therefore Lemma 6.28 gives $EBE \in \mathscr{C}$ for each of the three possible classes. In all cases the same $U = M$ satisfies

$$U^{\mathsf{T}} X U \in \mathscr{C}, \qquad U^{-1}\Lambda \subseteq \mathscr{C}.$$

This is a common native presentation, hence $\mathsf{Tract}(K\Lambda)$, by Definition 2.8.

The real case is therefore a resolved complexity leaf. The other case uses one finite actual batch and strictly decreases the outer parameters unless already resolved. The starting deck is stable; all factors of the new batch are processed before return, and strict successors reapply Lemma 5.8. Thus the queue requirements of Definitions 5.9 and 5.10 hold. There is no remaining common-presentation rigidity leaf to restrict, so the definition of Close gives the stated conclusion. □

8.5 The exotic Klein deck

The exotic Klein form is not $GO(X)$-conjugate to the Pauli form, so we work in its physical basis

$$\begin{aligned} E_0 &= I, & E_1 &= X, \\ E_2 &= \begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}, & E_3 &= \begin{pmatrix} -i & 1 \\ -1 & i \end{pmatrix}. \end{aligned} \tag{97}$$

Identify $0, 1, 2, 3$ with $00, 01, 10, 11 \in V = \mathbb{F}_2^2$, define the exotic projective Klein set $\mathcal{V}_{\mathrm{ex}} := \{[E_a] : a \in V\}$, write addition as $\oplus$, and define the label permutation $\vartheta := (0)(1)(2\ 3)$. A superscript that

is an ordered port pair, as in E_a^{13}, specifies the tensor ports; an upper label, as in E^a, denotes the Frobenius-dual basis element defined below.

For an oriented perfect matching M, an *exotic M-matching product* is a nonzero scalar multiple of

$$\bigotimes_{e\in M} E^e_{\tau_e}, \qquad \tau \in V^M.$$

The products with these representatives form a basis of $(\mathbb{C}^2)^{\otimes 2|M|}$, called the *$M$-basis*; one product spans a *matching line*. Recoupling below means expressing a product on one matching in the basis of another matching. Its coefficient support records how many matching labels must vary.

Lemma 8.4 (Exotic Klein calculus and recoupling)**.** *For the bilinear Frobenius form,*

$$\langle E_a, E_b\rangle_F = d_a \mathbf{1}[b = \vartheta(a)], \qquad (d_0, d_1, d_2, d_3) = (2, 2, 4, 4). \tag{98}$$

Thus define

$$E^a := d_a^{-1} E_{\vartheta(a)}, \qquad \langle E^a, E_b\rangle_F = \mathbf{1}[a = b].$$

This is the Frobenius-dual basis, with an actual contraction up to fixed scalar, and

$$E_a^{\mathsf{T}} \doteq E_{\vartheta(a)}, \qquad E_a E_b \doteq E_{a\oplus b}, \tag{99}$$

with nonzero omitted scalars. Relative to the matching $12|34$*,*

$$\delta^{\times}_{a,b,\gamma} := \vartheta(\gamma) \oplus b \oplus \vartheta(a),$$
$$\delta^{\parallel}_{a,b,\gamma} := \gamma \oplus \vartheta\big(b \oplus \vartheta(a)\big),$$

and define the recoupling coefficients by

$$\rho^{\times}_{a,b}(\gamma) := \left\langle (E^{\gamma})^{12} (E^{\delta^{\times}_{a,b,\gamma}})^{34}, E_a^{13} E_b^{24} \right\rangle_F,$$
$$\rho^{\parallel}_{a,b}(\gamma) := \left\langle (E^{\gamma})^{12} (E^{\delta^{\parallel}_{a,b,\gamma}})^{34}, E_a^{14} E_b^{23} \right\rangle_F.$$

Here the pairing is the bilinear Frobenius pairing on quaternary tensors. The recoupling identities are

$$E_a^{13} E_b^{24} = \sum_{\gamma\in V} \rho^{\times}_{a,b}(\gamma) E_{\gamma}^{12} E^{34}_{\delta^{\times}_{a,b,\gamma}},$$
$$E_a^{14} E_b^{23} = \sum_{\gamma\in V} \rho^{\parallel}_{a,b}(\gamma) E_{\gamma}^{12} E^{34}_{\delta^{\parallel}_{a,b,\gamma}},$$

Every displayed coefficient lies in $\mathbb{Q}(i)^{\times}$. For matchings M, N of $2r$ ports, let $c(M, N)$ count alternating cycles, including common edges. An N-matching product then has exactly

$$4^{r-c(M,N)} \tag{100}$$

nonzero coefficients in the M-basis. Thus support four means one alternating four-cycle and exactly two varying M-edges.

Proof. The displayed matrices satisfy

$$E_1^2 = I, \qquad E_2^2 = E_3^2 = -2I, \qquad E_1E_2 = -E_2E_1 = -iE_3,$$
$$E_0^{\mathsf{T}} = E_0, \qquad E_1^{\mathsf{T}} = E_1, \qquad E_2^{\mathsf{T}} = -E_3, \qquad E_3^{\mathsf{T}} = -E_2,$$
$$\operatorname{tr}(E_a) = 2\mathbf{1}[a = 0].$$

These identities give the fusion and Gram formulas, including all nonzero scalar factors. For example, the coefficient of $E_\gamma^{12}E_\delta^{34}$ in the first recoupling is

$$\operatorname{tr}\Big((E^\delta)^{\mathsf{T}}E_a^{\mathsf{T}}E^\gamma E_b\Big).$$

Fusion makes its matrix product a nonzero multiple of $E_{\delta\oplus\vartheta(a)\oplus\vartheta(\gamma)\oplus b}$. The trace therefore vanishes except at $\delta = \delta^{\times}_{a,b,\gamma}$, and is nonzero there. For the second recoupling the coefficient is

$$\operatorname{tr}\Big((E^\delta)^{\mathsf{T}}E_b^{\mathsf{T}}(E^\gamma)^{\mathsf{T}}E_a\Big),$$

which is nonzero exactly at $\delta = \gamma \oplus \vartheta(b) \oplus a = \delta^{\parallel}_{a,b,\gamma}$. All factors have entries in $\mathbb{Q}(i)$, proving the coefficient assertion without enumerating the labels.

For general M, N, contracting an N-product against a dual M-basis element separates over the alternating cycles of $M \cup N$. On a cycle of length 2ℓ, the coefficient is one trace of a product of the ℓ fixed N-matrices and ℓ dual M-matrices, transposed according to their orientations. Each M-label enters its fused label once, through either the identity or the bijection ϑ. After $\ell - 1$ labels are chosen, the zero-label trace condition determines the last label uniquely. The resulting trace is a nonzero scalar times $\operatorname{tr}(I) = 2$. Thus the cycle contributes exactly $4^{\ell-1}$ nonzero coefficients, including one for a common edge when $\ell = 1$. Multiplying over cycles gives $4^{\sum(\ell-1)} = 4^{r-c(M,N)}$. If this equals four, exactly one cycle has length four and the others are common edges, proving the last assertion. □

Exotic-card convention. For an ordered pair p, define the exotic pair card

$$C_p^a(f) := \partial_p^{E^a} f.$$

For disjoint ordered pairs $p_1, \dots, p_s$, an *exotic card* is the corresponding iterated contraction by fixed actual representatives of the duals $E^{a_1}, \dots, E^{a_s}$; it is *proper* when at least two ports remain. Disjoint contractions commute, and recorded nonzero dual scalars are divided out in the canonical notation. With the fixed representatives, write

$$E_aE_b = \omega(a,b)E_{a\oplus b}, \qquad \omega(a,b) \in \mathbb{Q}(i)^*;$$

the nonzero factors $\omega(a,b)$ and the corresponding transpose factors are recorded. To *dual-cross* two residual matching edges $e = \{u, u'\}$ and $e' = \{v, v'\}$, choose one endpoint u of e, one endpoint v of e', and one fixed $a \in V$; contract the single pair u, v by the fixed actual representative of E^a, and leave u', v' as the endpoints of the fused residual edge. The endpoint orientations are part of the choice, and all ω- and transpose scalars are recorded. A *scale or cancellation locus* is a set of nonzero-card scalar parameters on which a specified card becomes proportional to a matching line or vanishes.

The duals are actual up to the recorded scalars. Tomography

$$f = \sum_{a\in V} E_a^p \otimes C_p^a, \qquad C_p^a = \langle E^a, f\rangle_p, \tag{101}$$

is coordinatewise only; neither its sum nor a matching basis is a gadget.

Lemma 8.5 (The exotic-Klein six-port base)**.** *If every exotic pair card of a nonzero six-port tensor h is zero or a product of two exotic binaries, then h is a product of three exotic binaries. Thus there is no exotic analogue of* H_6*.*

Proof. Normalize a nonzero card to $C_{12}^0 = E_0^{34} E_0^{56}$ by actual dressings and a permutation. The six-port interface Theorem A.13 covers every nonzero/zero, scale, and cancellation locus; all survivors are three-binary products. Undo the normalization. □

Lemma 8.6 (Fixed ruling for exotic cards)**.** *Let $f \neq 0$ have even arity at least eight, and suppose every proper exotic card of f is zero or an exotic matching product. For every deleted pair p, all nonzero cards C_p^a in Equation* (101) *use one residual perfect matching.*

Proof. Fix a deleted pair p, and suppose that two nonzero cards C_p^a and C_p^b use distinct residual matchings $M \neq N$. We construct a sequence of actual exotic contractions that preserves both cards as nonzero matching products and preserves one residual disagreement.

The only local operation needed is the following. If two contracted ports belong to distinct matching edges in a given distinguished card, then, up to the fixed transpose convention, the new edge factor is

$$E_\alpha E_\kappa E_\beta \doteq E_{\alpha \oplus \kappa \oplus \beta}.$$

It is nonzero because every E_γ is nonsingular. Choose two vertices at alternating distance two on an alternating cycle of $M \cup N$. They belong to distinct edges in both distinguished cards, so contracting them shortens that cycle by two vertices while keeping both distinguished cards nonzero. Repeating reduces a marked nontrivial cycle to length four.

Reduce every other nontrivial alternating cycle and splice it into the marked cycle by one cross-contraction; the same three-matrix identity keeps both distinguished cards nonzero and merges the two cycles. A common edge carrying the same label in both cards may be closed with its actual Frobenius dual. If a common edge carries different labels, splice it into the marked cycle instead of closing it, and reduce the enlarged cycle again. After all common edges and all other cycles have been removed, the resulting actual six-port tensor h has two nonzero p-cards with different residual two-edge matchings.

Every exotic pair card of h is zero or a matching product. Indeed, bilinear contractions commute, so any card of h may be reordered so that its final pair contraction is applied first to f. By hypothesis that first card is zero or a matching product, and every subsequent contraction preserves this property by $E_\alpha E_\kappa E_\beta \doteq E_{\alpha \oplus \kappa \oplus \beta}$. Thus Lemma 8.5 applies and makes h a product of three exotic binaries.

For a product of three binaries, all nonzero cards at a fixed deleted pair have one residual matching: if the deleted pair is a factor edge, only its contraction in the dual direction is nonzero; otherwise fusion of the two incident factor edges leaves the same residual matching for every kernel giving a nonzero card. This contradicts the two preserved rulings of h. Hence all nonzero p-cards of f use one residual perfect matching. □

Lemma 8.7 (Fixed-ruling fiber dichotomy)**.** *Let f satisfy Lemma 8.6, fix p and its residual matching M, and suppose f has no nontrivial positive-arity tensor factor. Then all four cards C_p^a are nonzero and have the form*

$$C_p^a = \lambda_a \bigotimes_{e \in M} E^e_{\tau_e(a)},$$

where $\lambda_a \in \mathbb{C}^\times$ are nonzero scalar coefficients and $\tau_e : V \to V$ are label maps; every τ_e is a permutation.

Proof. Let

$$\Gamma = \{a \in V : C_p^a \neq 0\}.$$

Tomography and the fixed ruling give

$$f = \sum_{a\in\Gamma} \lambda_a E_a^p \bigotimes_{e\in M} E^e_{\tau_e(a)}, \qquad \lambda_a \neq 0.$$

For an edge $e \in M$, contraction with the actual Frobenius dual of E_b selects the fibre

$$\Gamma_{e,b} = \{a \in \Gamma : \tau_e(a) = b\}.$$

In the displayed $p \cup (M \setminus \{e\})$-basis, the resulting card has exactly $|\Gamma_{e,b}|$ nonzero coefficients, since distinct a's remain distinct on the p-coordinate. By Equation (100), every nonempty fibre therefore has size one or four.

If $|\Gamma| = 1$, the factor E_a^p is common to the whole tensor, contrary to the no-factor hypothesis. If $|\Gamma| \in \{2, 3\}$, every τ_e is injective on Γ. Choose ports from two distinct edges of M and cross-contract them by any actual exotic kernel. Projective fusion is nonzero in every tomographic summand, and the p-labels remain distinct, so the resulting proper card has exactly $|\Gamma|$ coefficients in the induced matching basis. This is impossible because Equation (100) permits only powers of four. Consequently $\Gamma = V$.

Now every nonempty τ_e-fibre has size one or four. A four-point fibre makes τ_e constant and exposes E_b^e as a genuine edge factor. Since no factor exists, all fibres are singletons. Hence every $\tau_e : V \to V$ is a permutation, proving the stated form. □

Lemma 8.8 (The exotic-Klein eight-port base)**.** *Let $h \neq 0$ have arity eight and suppose every exotic pair card is zero or a product of three exotic binaries. Then h has a genuine tensor factor.*

Proof. Contraction closure under Equation (99) makes every iterated proper exotic card zero or a matching product. For a nonfactor, Lemmas 8.6 and 8.7 gives four cards on one matching. Normalize one actually to $C_{12}^0 = E_0^{34}E_0^{56}E_0^{78}$. Each residual label map is affine because the convention in Section 2.8.1 identifies $\mathrm{AGL}(2, 2)$ with $\mathrm{Sym}(V)$. Actual edge dressings remove translations and absorb fixed nonzero cocycles:

$$h = \sum_{\gamma\in V} \lambda_\gamma E_\gamma^{12} E_{A_1\gamma}^{34} E_{A_2\gamma}^{56} E_{A_3\gamma}^{78}, \qquad A_j \in \mathrm{GL}_2(\mathbb{F}_2), \quad \lambda_\gamma \neq 0. \tag{102}$$

Contract $(3, 5)$ or $(4, 5)$ by $E^0 = I/2$. On the two remaining endpoints of the affected edges, the respective matrices are

$$\tfrac{1}{2} E_{A_1\gamma}^{\mathsf{T}} E_{A_2\gamma}, \qquad \tfrac{1}{2} E_{A_1\gamma} E_{A_2\gamma}.$$

By Equation (99), their labels are

$$D_1\gamma = (\vartheta A_1 + A_2)\gamma, \qquad D_2\gamma = (A_1 + A_2)\gamma,$$

with all sums over $\mathbb{F}_2$ and all omitted scalars nonzero. The two maps cannot both vanish, since that would imply $\vartheta A_1 = A_1$, contrary to the invertibility of A_1 and $\vartheta \neq I$. Choose a nonzero one, denoted by D. In the resulting six-port matching basis the four labels are $(\gamma, D\gamma, A_3\gamma)$. They are distinct because their first coordinates are distinct, and every coefficient is nonzero. All three coordinates vary: the first and third maps are bijective, and $D \neq 0$. This contradicts Equation (100), which allows only two varying edges in a four-term matching product. Thus h has a genuine proper tensor factor. □

Proposition 8.9 (High-arity exotic factor)**.** *Let $f \neq 0$ have arity $2r \geq 10$, and suppose every proper exotic card is zero or an exotic matching product. Then f has a genuine proper tensor factor.*

Proof. Assume no factor. At a fixed pair, Lemmas 8.6 and 8.7 gives four nonzero cards on one matching with permutation label maps. Choose two residual edges and perform one dual-cross contraction as defined above. Every tomographic summand remains nonzero because the three matrices in its endpoint-to-endpoint fusion are nonsingular. The four resulting terms still vary at the fixed pair and at all $r-3$ untouched residual edges, hence in at least $r-2 \geq 3$ coordinates. But Equation (100) allows only two varying coordinates in a four-term matching product. This contradiction proves that a genuine proper factor exists. □

Theorem 8.10 (Exotic-Klein relative rigidity)**.** *Assume*

$$\mathsf{Stable}(\Lambda, \mathcal{V}_{\mathrm{ex}}, f), \qquad \mathrm{arity}(f) = \nu(\Lambda).$$

Then

$$\mathsf{Close}_{\{\mathscr{A}\}}(\Lambda, \mathcal{V}_{\mathrm{ex}}, f).$$

More explicitly, every continuing branch reaches an established resolved leaf or a strict outer deck successor after processing the actual batch from the quaternary interface by Lemma 5.11. No rigidity leaf remains at the current node.

Proof. First dispatch odd-arity factors, including unaries, by Theorem 2.15, and unsafe binaries by Lemmas 3.1 and 5.3. On a continuing branch all retained primes have even arity. Deck stability makes every exotic pair card zero or an exotic matching product. Repeated transpose/fusion using Equation (99) therefore makes every iterated proper exotic card zero or an exotic matching product. At arity four, item 4 of Theorem 5.18 invokes the endpoint-nondegenerate eight-vertex signature interface. By Corollary 4.7, this arity-four branch is a resolved leaf or a strict outer successor, so it contributes no surviving product or local-affine rigidity leaf. At arity six Lemma 8.5 contradicts primality; at arity eight and at least ten, the factor conclusions of Lemma 8.8 and Proposition 8.9 contradict primality as well. These cases exhaust the possible arities of the distinguished nonbinary prime f.

Process any concrete batch from the quaternary interface, including its full factorization, by Definitions 5.9 and 5.12 and Lemma 5.11. The interface supplies a resolved leaf or a strict outer successor, while the higher-arity cases leave no stable tuple. Consequently there is no surviving rigidity leaf at this node. This gives $\mathsf{Close}_{\{\mathscr{A}\}}(\Lambda, \mathcal{V}_{\mathrm{ex}}, f)$. □

Proof of Theorem 8.1. If $G = V_4$, apply Theorem 8.3. If $G = \mathcal{V}_{\mathrm{ex}}$, apply Theorem 8.10. These are exactly the two Klein-four forms in the statement. □

9 Platonic Matching Decks

The finite groups in Theorem 5.14 leave three Platonic cases: tetrahedral, octahedral, and icosahedral. We prove Theorem 9.3 by reducing an arbitrary-arity core to a six-port tensor. A pair contraction is a linear map from the four-dimensional matrix space to residual tensors. Incidence among the group matrices then forces either a common tensor factor or three collinear cards with different matchings. In the latter case four residual ports contain all the variation, and the remaining factors can be closed by actual kernels. The resulting finite classifications are proved in Appendix A.5.

We use the quaternion coordinates of Equation (6), the marked-group classification of Theorem 5.14, and the projective and tensor conventions of Sections 2.1.3 and 2.11.1. Thus $[F]$ is the line spanned by any nonzero matrix or tensor F, including a singular matrix, and $\mathbb{P}(0) = \varnothing$.

The three nonstandard octahedral displays are as follows. For the transposition–transposition form put

$$H_{\mathcal{O}} := \begin{pmatrix} \sqrt{2} & -i\sqrt{2} \\ 1-i & 1+i \end{pmatrix}, \qquad \mathcal{O}' := H_{\mathcal{O}}^{-1}\mathcal{O}H_{\mathcal{O}}. \tag{103}$$

This realizes the marked type $\mathcal{O}'$ of Theorem 5.14 in coefficient coordinates.

For the two mixed fixed-equality forms, put

$$\rho := \sqrt{2}-1, \qquad H_{dt} := Q_0 - \rho Q_1, \qquad H_{td} := Q_0 + \rho Q_3, \tag{104}$$

and, for $\alpha \in \{dt, td\}$, define

$$\theta_\alpha(U) := H_\alpha^{-1} U H_\alpha, \qquad \mathcal{O}_\alpha := \theta_\alpha(\mathcal{O}). \tag{105}$$

Direct quaternion multiplication gives

$$\begin{aligned} &H_{dt}H_{dt}^{\mathsf{T}} = 2\rho(Q_0 - Q_1), && \theta_{dt}(Q_1) = Q_1, && \theta_{dt}(Q_2 - Q_3) = -\sqrt{2}Q_3, \\ &H_{td}H_{td}^{\mathsf{T}} = 2\rho(Q_0 + Q_3), && \theta_{td}(Q_3) = Q_3, && \theta_{td}(Q_1 - Q_2) = \sqrt{2}Q_1. \end{aligned} \tag{106}$$

Thus $\mathcal{O}_{dt}$ has marked class pattern $(dt;t)$, whereas $\mathcal{O}_{td}$ has pattern $(td;t)$. All three matrices $H_{\mathcal{O}}, H_{dt}, H_{td}$ are used as coefficient similarities. Actual contractions and dressings use representatives of the current group; the retained signature set and its literal equality remain in the fixed coordinates.

Throughout this section, for a finite projective group G put

$$G^{\#} := \{[T^{\#}] : [T] \in G\}, \qquad T^{\#} := XT^{\mathsf{T}}X,$$

and call G *reversal-stable* when $G^{\#} = G$. A *marked Platonic state* means a retained, factor-saturated state containing the distinguished literal I, whose recomputed complete group is one of

$$\mathcal{T},\ \mathcal{T}_{\mathrm{ext}},\ \mathcal{O},\ \mathcal{O}',\ \mathcal{O}_{dt},\ \mathcal{O}_{td},\ \mathcal{I},\ \mathcal{I}_{-},$$

with one fixed actual representative and realization record for every projective group class. All statements about group matrices distinguish a matrix representative K from its projective class $[K]$.

Lemma 9.1 (The two marked tetrahedral forms)**.** *Let $G \le \mathrm{PGL}_2(\mathbb{C})$ be a finite reversal-stable complete group in the fixed coordinates containing literal I, and define the physical transfer mark x, the reversal mark j, and the extension group H by*

$$x = [X] \in G, \qquad j = [Z], \qquad H = \langle G, j\rangle.$$

If $G \cong A_4$, then, up to one legal global fixed-I-preserving $GO(X)$-normalization, exactly one of the following holds.

1. *$j \in G$, $H = G$, and $G = \mathcal{T}$, where $\mathcal{T}$ is the standard group in Equation (7).*
2. *$j \notin G$, $H \cong S_4$, and*

$$\begin{aligned} \mathcal{T}_{\mathrm{ext}} = {} & \{[Q_0], [Q_1], [Q_2+Q_3], [Q_2-Q_3]\} \\ & \cup \{[Q_0 + \epsilon Q_1 + \delta\sqrt{2}Q_2], [Q_0 + \epsilon Q_1 + \delta\sqrt{2}Q_3] : \epsilon, \delta \in \{1,-1\}\}. \end{aligned} \tag{107}$$

Here $x = [Q_1] \in \mathcal{T}_{\mathrm{ext}}$, whereas $j = [Q_3] \notin \mathcal{T}_{\mathrm{ext}}$ and $xj = [Q_2] \notin \mathcal{T}_{\mathrm{ext}}$.

Proof. Reversal gives $G \triangleleft H$ and $[H : G] \le 2$. If $j \notin G$, then $H \cong S_4$, with the commuting pair (x, j) marked as a double transposition and a transposition; if $j \in G$, the three involutions form the unique normal Klein four of A_4 by Lemma 2.2. After the corresponding finite-subgroup embedding the remaining ambiguity is

$$C_{\mathrm{PGL}_2(\mathbb{C})}(\langle x, j\rangle) = \langle x, j\rangle,$$

represented inside $GO(X)$. Direct multiplication gives the two stated domains. □

For algebraic comparison only, define the scalar s_{ext}, the matrix R_{ext}, and the coefficient-similarity map θ_{ext} by

$$s_{\mathrm{ext}} := \sqrt{2} - 1, \qquad R_{\mathrm{ext}} := Q_0 + s_{\mathrm{ext}} Q_1, \qquad \theta_{\mathrm{ext}}(U) := R_{\mathrm{ext}}^{-1} U R_{\mathrm{ext}}.$$

On projective sets, apply θ_{ext} representative-wise. Then

$$\theta_{\mathrm{ext}}(\mathcal{T}) = \mathcal{T}_{\mathrm{ext}}, \qquad [R_{\mathrm{ext}}^{\mathsf{T}} X R_{\mathrm{ext}}] = [Q_0 - Q_1] \neq [X].$$

Thus $R_{\mathrm{ext}} \notin GO(X)$: θ_{ext} transports coefficients only; all gadgets and normalizations use actual external representatives.

Write $\mathcal{I}_-$ for the opposite icosahedral coordinate display. The two displays have the same abstract marked classification, but both are retained because their physical representatives are different. Put

$$D_{\mathcal{I}} := Q_0 + Q_3, \qquad D_{\mathcal{I}}^{\mathsf{T}} X D_{\mathcal{I}} = 2X, \qquad \mathcal{I}_- = D_{\mathcal{I}} \mathcal{I} D_{\mathcal{I}}^{-1}. \tag{108}$$

The signed coordinate permutations defining $\mathcal{T}$ also give

$$D_{\mathcal{I}} \mathcal{T} D_{\mathcal{I}}^{-1} = \mathcal{T}, \qquad \mathcal{T} \subseteq \mathcal{I} \cap \mathcal{I}_-.$$

Indeed, conjugation by $D_{\mathcal{I}}$ sends (Q_1, Q_2, Q_3) to $(-Q_2, Q_1, Q_3)$. We use this identity only as the coefficient dictionary below; no retained signature, and in particular no equality signature, is transformed.

Lemma 9.2 (Fixed-equality dictionary for the opposite icosahedral display)**.** *Set $D = D_{\mathcal{I}}$ and*

$$\kappa_D(K) := D^{-\mathsf{T}} K D^{-1}.$$

Let $(\Lambda, \mathcal{I}_-, f)$ be a marked Platonic state in the fixed coordinates containing the supplied literal I, and introduce only for coefficient analysis the tensor $\widehat{f} := D^{-\otimes \mathrm{arity}(f)} f$. For every ordered pair p,

$$\Phi_p^f(\kappa_D(K)) = D^{\otimes(\mathrm{arity}(f)-2)} \Phi_p^{\widehat{f}}(K). \tag{109}$$

The map $[K] \mapsto [\kappa_D(K)]$ bijects the standard $\mathcal{I}$-kernel classes with the actual $\mathcal{I}_-$-kernel classes. Consequently every finite card or network identity proved in the standard display has a direct identity in the opposite display, obtained by replacing each kernel by its κ_D-image and recording the nonzero representative and edge scalars. This dictionary preserves zero cards, tensor factors, all flattening ranks, matching labels, endpoint support, and the property that an output binary transfer lies outside the complete group. It does not change Λ, the active occurrence, or the distinguished literal I.

Proof. Direct contraction gives Equation (109). Moreover $D^{-\mathsf{T}} = \frac{1}{2} X D X$, and hence

$$[\kappa_D(K)] = [X\, D(XK)D^{-1}] \in \mathcal{I}_- \qquad ([K] \in \mathcal{I}).$$

Here $[X], [XK] \in \mathcal{I}$, conjugation sends the latter class into $\mathcal{I}_-$, and $[X] \in \mathcal{T} \subseteq \mathcal{I}_-$. The inverse formula proves bijectivity. Since $D^{\mathsf{T}} X D = 2X$, applying the covariance successively at the vertices of a finite network cancels the internal D, D^{-1} factors across every native edge, leaving only a known nonzero scalar. A residual binary B is carried to DBD^{T}, whose transfer satisfies

$$[DBD^{\mathsf{T}} X] = [D(BX)D^{-1}].$$

Thus nonsingularity and exclusion from the current group are preserved. Finally, D is diagonal and invertible, so it preserves support (hence endpoint nondegeneracy), tensor products, matching labels, and all flattening ranks. All resulting kernels are replaced by the already fixed actual $\mathcal{I}_-$ representatives in their projective classes, with their nonzero realization scalars recorded. At no point is the analytical tensor $\widehat{f}$ adjoined or realized. □

Theorem 9.3 (Platonic deck rigidity)**.** *Assume*

$$\mathsf{Stable}(\Lambda, G, f), \qquad G \in \{\mathcal{T}, \mathcal{T}_{\mathrm{ext}}, \mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}, \mathcal{I}, \mathcal{I}_-\}.$$

Then

$$\begin{cases} \mathsf{Close}_{\{\mathscr{A}\}}(\Lambda, G, f), & G = \mathcal{T}, \\ \mathsf{Close}_{\varnothing}(\Lambda, G, f), & G \in \{\mathcal{T}_{\mathrm{ext}}, \mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}, \mathcal{I}, \mathcal{I}_-\}. \end{cases}$$

9.1 Common card-map interface and quaternary boundary

At a continuing fixed-I deck-stable node, fix for each $g \in G$ a matrix K_g with $[K_g] = g$. Since $[X] \in G$, choose a fixed actual binary representative B_g of transfer class $[XK_g]$. The convention $T_{B_g} = B_g X$ gives $B_g \doteq XK_gX$. Attach its two ordered ports to the selected pair through two native X-edges; the effective bilinear kernel is then $XB_gX \doteq K_g$. Store its nonzero realization scalar and ordered-port record, and define the actual representative set

$$\mathcal{K}_G^{\mathrm{act}} := \{K_g : g \in G\}. \tag{110}$$

Every contraction below uses these fixed representatives. A resulting binary B is tested through its transfer BX. A zero card contributes zero; a nonzero singular B is processed by factor saturation. If B is nonsingular and $[BX] \notin G$, then Lemmas 3.1 and 5.3 gives an infinite-order terminal or a strict enlargement of the complete group. The ordered ports and nonzero realization scalar are retained in each case. These are bounded local operations on the selected card.

For a signature u and an ordered pair $p = (a, b)$ of its ports, define the physical card map

$$\Phi_p^u : \mathrm{Mat}_2(\mathbb{C}) \longrightarrow (\mathbb{C}^2)^{\otimes(\mathrm{arity}(u)-2)}, \qquad \Phi_p^u(K)(y) = \sum_{s,t \in \{0,1\}} u(s,t,y) K(s,t). \tag{111}$$

When the signature is the active f, write $\Phi_p = \Phi_p^f$, and write $\langle K^{(p)}, u \rangle := \Phi_p^u(K)$ for this physical bilinear (not Hermitian) contraction; equivalently $\Phi_p^u(K)(y) = \mathrm{tr}(A_y^{\mathsf{T}} K)$, where $A_y := (u(s,t,y))_{s,t} \in \mathrm{Mat}_2(\mathbb{C})$ is the p-slice matrix. Port permutations supply the other orderings.

For every two-element port set $\{a, b\}$, use the canonical orientation $p = (a, b)$ with $a < b$, and order complementary ports increasingly unless both orientations are explicitly requested. Thus a quaternary has six canonical pairs and a six-port tensor has fifteen; all other orientations are obtained by actual port permutations.

The rank of Φ_p^h is the flattening rank across the selected pair and the remaining ports. Since its domain is $\mathrm{Mat}_2(\mathbb{C})$, it is at most four. We treat ranks below four and rank four separately. All

signatures used as gadgets are directly realized over Λ, with ordered ports, nonzero scalar, and predecessor reduction recorded. The required condition is

$$\Phi_p^h(K_g) = 0 \quad \text{or} \quad \Phi_p^h(K_g) \text{ is } G\text{-matching}, \qquad \text{for every canonical } p \text{ and } K_g \in \mathcal{K}_G^{\mathrm{act}}. \tag{112}$$

Put $\mathcal{D}_G := \{[K] : K \in \mathcal{K}_G^{\mathrm{act}}\}$. A *rich domain line* is a projective line containing at least three points of $\mathcal{D}_G$. Let $\mathscr{G}_G^{\mathrm{dom}}$ be the graph on $\mathcal{D}_G$ joining two points when they lie on a common rich domain line. Such a line is *kernel-avoiding* for a linear map Φ when it is disjoint from $\mathbb{P}(\ker \Phi)$.

For an ancestor of arity at least six, localization retains p and four residual ports. The rank of the ancestor map is used only to choose a kernel-avoiding rich domain line and to bound the resulting six-port rank. After that signature is directly realized, its own card-map rank is recomputed before either the rank-deficient or full-rank finite reduction is invoked.

Quaternion coordinates and quaternary matching products. Recall the quaternion basis from Equation (6). Its Frobenius Gram matrix is

$$\left(\operatorname{tr}(Q_\mu^{\mathsf{T}} Q_\nu)\right)_{\mu,\nu} = 2\,\mathrm{diag}(1,-1,1,-1),$$

so the bilinear dual basis is

$$Q^0 = Q_0/2, \qquad Q^1 = -Q_1/2, \qquad Q^2 = Q_2/2, \qquad Q^3 = -Q_3/2. \tag{113}$$

These dual matrices recover the coefficients of the original tensor: their role is linear reconstruction. For $K \in \mathrm{Mat}_2(\mathbb{C})$, let

$$c_Q(K) = (c_0, c_1, c_2, c_3)^{\mathsf{T}} \quad \Longleftrightarrow \quad K = c_Q(K) \cdot Q := \sum_{\mu=0}^{3} c_\mu Q_\mu.$$

Write U_{ab} for the binary tensor with entries $U(x_a, x_b)$ on the ordered pair (a, b).

Frobenius-dual reconstruction gives, for every signature f and canonical pair p,

$$f = \sum_{\mu=0}^{3} Q_p^\mu \otimes \Phi_p^f(Q_\mu). \tag{114}$$

More generally, for an ordered pair p, a finite-dimensional tensor space W, and a linear map $\Phi : \mathrm{Mat}_2 \to W$, write $f_{\Phi,p} := \sum_{\mu=0}^{3} Q_p^\mu \otimes \Phi(Q_\mu)$. Every assertion below that an abstract card map factors refers to this reconstructed tensor.

On four ordered residual ports define the three matching products

$$\begin{aligned} P_0(U,V) &= U(x_1,x_2)V(x_3,x_4), \\ P_1(U,V) &= U(x_1,x_3)V(x_2,x_4), \\ P_2(U,V) &= U(x_1,x_4)V(x_2,x_3). \end{aligned}$$

For a finite projective transfer group G, define its quaternary projective matching deck by

$$\mathfrak{D}_4(G) = \{[P_r(U,V)] : r \in \{0,1,2\},\ [UX], [VX] \in G\}.$$

Define the nonzero scalar cone $\mathrm{cone}(\mathfrak{D}) := \{\lambda F : \lambda \neq 0, [F] \in \mathfrak{D}\}$.

A *rich deck line* is a projective line containing at least three points of $\mathfrak{D}_4(G)$. It is an s-line when it contains exactly s deck points. Its *matching-label type* is the partition of s formed by the

multiplicities of its residual matching labels. Type (s) is *monochromatic*, type $(1,1,1)$ is a *rainbow three-line*, and every other nonmonochromatic type is *mixed*. A rich deck line is *maximal* under the analogous deck-point containment order. Every deck point has rank triple $(1,4,4)$, up to the fixed order $(12 \mid 34, 13 \mid 24, 14 \mid 23)$. For every matching product F with matching M, and every port subset S,

$$\operatorname{rank} F_{S|\bar{S}} = 2^{c_M(S)}, \tag{115}$$

where $c_M(S)$ counts edges crossing the cut.

For $L \in \mathcal{K}_G^{\mathrm{act}}$, the binary $B_L = XL^{\mathsf{T}}$ has $[B_L X] = [XL^{\mathsf{T}}X] = [L^{\#}] \in G$, and one native edge satisfies $\sum_u X(x,u)B_L(u,y) = L(y,x)$. Hence, for $\mathbf{L} = (L_1, L_2, L_3, L_4) \in (\mathcal{K}_G^{\mathrm{act}})^4$,

$$(\mathscr{D}_{\mathbf{L}} q)(y) = \sum_x q(x) \prod_{j=1}^{4} L_j(y_j, x_j)$$

is a constant-size actual dressing with recorded nonzero scalar.

Lemma 9.4 (Eight-vertex signatures with rank triples $(2,2,2)$ or $(3,3,3)$)**.** *Let Λ be a retained, factor-saturated finite algebraic signature set containing the distinguished literal equality I. If Λ directly realizes an endpoint-nondegenerate eight-vertex signature M of flattening ranks $(2,2,2)$ or $(3,3,3)$, then*

$$\text{K-Holant}(\Lambda) \text{ is } \#\mathsf{P}\text{-hard} \qquad \text{or} \qquad \mathsf{Tract}(K\Lambda) \text{ holds.}$$

Proof. Apply Corollary 4.7 to M. Its first two outcomes are the asserted conclusions. In its sole continuing outcome, M itself has, up to port order and a nonzero scalar, a factorization $E \otimes E'$ into two nonsingular binary signatures. Its matching flattening ranks are therefore $(1,4,4)$, up to permutation, by Equation (115), contradicting either assumed rank triple. The full-even two-copy equality outputs are already resolved inside Corollary 4.6 and are not continuing leaves. Hence only the hard or tractable outcomes remain. □

Lemma 9.5 (The Platonic quaternary boundary)**.** *Let (Λ, G, q) be a deck-stable retained, factor-saturated fixed-I state with nonzero tensor-prime quaternary core q. Let G be the internal A_4 form $\mathcal{T}$, any of the four marked S_4 forms $\mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}$, or a marked A_5 display, and define the transfer-valued pair map*

$$L_p(K) = \Phi_p^q(K)X \in \mathrm{Mat}_2(\mathbb{C}).$$

If every fixed actual representative K_g satisfies

$$L_p(K_g) = 0 \quad \text{or} \quad [L_p(K_g)] \in G \qquad \text{for all six canonical pairs } p,$$

then internal A_4 and either A_5 display have no tensor-prime survivor. In the first S_4 form, there are $\lambda \in \mathbb{C}^{\times}$ and actual matrices U_i with $[U_i] \in \mathcal{O}$ such that

$$q = \lambda\,(U_1 \otimes U_2 \otimes U_3 \otimes U_4)\mathrm{EQ}_4.$$

In the transposition–transposition form, four actual $\mathcal{O}'$-matrices dress q to

$$M_{\mathcal{O}'}(x) = \mathbf{1}[\mathrm{wt}(x) \equiv 0 \pmod 2](-1)^{\sum_{i<j} x_i x_j}, \tag{116}$$

of ranks $(2,2,2)$. In the two mixed forms the normalized rank-two survivor is already an endpoint-nondegenerate eight-vertex signature, with all odd entries zero and, in the even-word order $(0000, 0011, 0101, 0110, 1001, 1010, 1100, 1111)$, coefficient vectors

$$\mathbf{e}(M_{dt}) = (1,1,1,1,1,1,1,1), \qquad \mathbf{e}(M_{td}) = (1,i,i,i,1,1,1,i). \tag{117}$$

Both have ranks $(2,2,2)$. Hence arity four always factors or reaches the unsafe-binary, pure generalized equality, or endpoint-nondegenerate eight-vertex signature terminal and never recurses.

Proof. Let $A_p \in \mathrm{Mat}_4(\mathbb{C})$ represent L_p in the quaternion coordinates above, so that $c_Q(L_p(K)) = A_p c_Q(K)$. Up to scale, the bilinear Frobenius form is $J_{\mathrm{Fr}} = \mathrm{diag}(1,-1,1,-1)$. Write $\bar{p}$ for the complementary pair and define the right-X multiplication map $\mathsf{R}_X : \mathrm{Mat}_2(\mathbb{C}) \to \mathrm{Mat}_2(\mathbb{C})$ and its coordinate matrix $\mathbf{R}_X$ by

$$\mathsf{R}_X(K) := KX, \qquad c_Q(\mathsf{R}_X(K)) = \mathbf{R}_X c_Q(K).$$

Complementary-pair contraction gives

$$L_{\bar{p}} = \mathsf{R}_X \circ L_p^* \circ \mathsf{R}_X, \tag{118}$$

where * is the Frobenius adjoint. Thus

$$A_{\bar{p}} = \mathbf{R}_X A_p^\sharp \mathbf{R}_X, \qquad A_p^\sharp = J_{\mathrm{Fr}} A_p^{\mathsf{T}} J_{\mathrm{Fr}}. \tag{119}$$

Actual right X and reversals turn the six pair conditions into

$$\begin{aligned} &A_p r = 0 \text{ or } [A_p r] \in \mathcal{R}_G, \\ &A_p^\sharp r = 0 \text{ or } [A_p^\sharp r] \in \mathcal{R}_G \end{aligned} \qquad \text{for every } [r] \in \mathcal{R}_G, \qquad \mathcal{R}_G = \{[c_Q(K_g)] : g \in G\}.$$

The exact two-sided classification Theorem A.28(i) has no rank three: rank zero gives $q = 0$, ranks one and four factor, and rank two occurs only for S_4. In the first marked form, actual inverse dressings expose EQ_4, so K-Holant$(\Lambda, \mathrm{EQ}_4) \le_{\mathrm{T}}$ K-Holant(Λ) and Corollary 3.6 and Lemma 3.4 apply. In the second form, its non-$GO(X)$ similarity identifies coefficient tables only; the same proposition checks the dressing to Equation (116), directly in actual $\mathcal{O}'$-representatives, with ranks $(2,2,2)$. For $\mathcal{O}_{dt}, \mathcal{O}_{td}$, the direct reconstructed tensors in Theorem A.28(i) are exactly Equation (117). These three cases invoke Lemma 9.4. Finally, the direct dictionary Lemma 9.2 proves the same no-survivor conclusion for $\mathcal{I}_-$ while leaving the literal equality untouched. □

Lemma 9.6 (Quadratic extension of a fixed matching)**.** *Let Q_0, Q_1, Q_2, Q_3 be the quaternion basis in Equation* (6)*, and let $\mathcal{D} \subset \mathbb{P}(\mathrm{Mat}_2)$ contain the twelve tetrahedral points*

$$[Q_\mu] \ (0 \le \mu \le 3), \qquad [Q_0 + \varepsilon_1 Q_1 + \varepsilon_2 Q_2 + \varepsilon_3 Q_3] \quad (\varepsilon_j \in \{1,-1\}). \tag{120}$$

Let M be a nonempty finite index set, let W_e be a nonzero finite-dimensional complex vector space for every $e \in M$, and let $\Phi : \mathrm{Mat}_2 \to \bigotimes_{e\in M} W_e$ be a nonzero linear map. Define

$$\Sigma_M := \left\{ \left[\bigotimes_{e\in M} w_e \right] : 0 \ne w_e \in W_e \right\} \subseteq \mathbb{P}\left(\bigotimes_{e\in M} W_e \right), \qquad \mathrm{Cone}_0(\Sigma_M) := \{0\} \cup \left\{ \bigotimes_{e\in M} w_e : w_e \in W_e \right\},$$

so that Σ_M and $\mathrm{Cone}_0(\Sigma_M)$ are, respectively, the Segre variety and its affine cone in the terminology of Section 2.11.1. Suppose $\Phi(K) \in \mathrm{Cone}_0(\Sigma_M)$ for every $[K] \in \mathcal{D}$. Then there is one index $e_* \in M$ and fixed nonzero tensors $U_e \in W_e$, $e \ne e_*$, such that

$$\Phi(K) = \left(\bigotimes_{e \ne e_*} U_e \right) \otimes \psi(K) \qquad (K \in \mathrm{Mat}_2) \tag{121}$$

for a linear map $\psi : \mathrm{Mat}_2 \to W_{e_*}$. For the reconstruction conclusion, suppose M partitions the residual physical ports into nonempty blocks, $W_e = (\mathbb{C}^2)^{\otimes e}$, and $\Phi = \Phi_p^f$ under the corresponding port identification. If $|M| \ge 2$, then $f = f_{\Phi,p}$ has a genuine proper tensor factor.

Proof of Lemma 9.6. Every defining Segre minor pulls back to a homogeneous quadratic in the quaternion coordinates. The four axes kill its square coefficients; at the eight sign points the remaining expression is

$$\sum_{0\le\mu<\nu\le3} c_{\mu\nu}\varepsilon_\mu\varepsilon_\nu = 0, \qquad \varepsilon_0 = 1.$$

The six displayed functions are distinct characters of $\{\pm1\}^3$, so every $c_{\mu\nu}$ vanishes. All Segre minors therefore vanish identically, including at zero, and $\operatorname{im}\Phi \subseteq \mathrm{Cone}_0(\Sigma_M)$.

If a linear subspace lies in $\mathrm{Cone}_0(\Sigma_M)$, take two nonzero simple tensors $u = \bigotimes u_e$ and $v = \bigotimes v_e$ in it. If u_i, v_i and u_j, v_j are both independent for $i \neq j$, apply dual functionals on all other factors; a 2×2 flattening minor of $u_i \otimes u_j + t\, v_i \otimes v_j$ is then nonzero, contradicting that $u + tv$ is simple. Thus each pair varies in at most one factor. Fixing one nonzero u, two vectors that vary from u in different factors would violate this pairwise conclusion, so all variation is in one factor. Hence the image lies in one ruling, giving Equation (121). At a test point with $\Phi(K) = 0$, every minor vanishes automatically; the quadratic interpolation never divides by a coordinate. Thus zero values are allowed throughout and no nonzero-test-point assumption was used.

In the card-map applications $\Phi = \Phi_p^f$, Equation (113) is only a coefficient identity, and Frobenius reconstruction gives

$$f = \left(\bigotimes_{e\neq e_*} U_e\right) \otimes \left(\sum_{\mu=0}^{3} Q_p^\mu \otimes \psi(Q_\mu)\right).$$

The fixed factor occupies the nonempty set $\bigcup_{e\neq e_*} e$, while the other factor contains p, so the factorization is proper. Lemma 2.20 exposes the resulting factors under a Turing reduction. □

Lemma 9.7 (Deficient Platonic card maps)**.** *Let (Λ, G, h) be a marked Platonic state, where $G \in \{\mathcal{T}, \mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}, \mathcal{I}, \mathcal{I}_-\}$. Assume $h \neq 0$ is a directly realized six-port signature over Λ, satisfies Equation* (112) *for G, and, for a specified canonical ordered pair p, satisfies* $\operatorname{rank}\Phi_p^h \le 3$. *Then h factors, or an actual binary gives a singular/infinite-order terminal or a strict complete-group successor. No rank-deficient incidence branch remains.*

Proof. For $G = \mathcal{I}_-$, apply the card covariance and direct network dictionary of Lemma 9.2. It identifies the coefficient calculation with the standard $\mathcal{I}$ calculation while replacing every physical kernel by its actual κ_D-image in $\mathcal{I}_-$. Thus its fixed-matching branch uses Lemma 9.6, and its mixed branch uses Theorem A.28(ii)–(iii), directly in the original fixed-I state. Hence assume below that $G \neq \mathcal{I}_-$. Rank zero contradicts $h \neq 0$ by Equation (114), and rank one factors. If all nonzero cards use one matching, apply Lemma 9.6; in any nonstandard S_4 form precompose with its fixed coefficient similarity and pull the varying factor back by the inverse similarity. This analytic pullback, not a gadget, yields $\Phi_p^h(K) = (\bigotimes_{e\neq e_*} U_e) \otimes \psi(K)$; Lemma 2.20 exposes the proper factor. If two nonzero cards use different matchings, the physical interface Theorem A.28(ii) gives an actual rainbow localization and a quaternary whose three flattening ranks are $(3,3,3)$, while the same interface gives its actual outside transfer. The direct-gadget direction Equation (12) and binary dispatch Lemmas 3.1 and 5.3 then apply. □

Lemma 9.8 (Finite Platonic quaternary exits)**.** *Let (Λ, G, f) be a marked Platonic state, and let q be directly realized over Λ, with its canonical port order and nonzero realization scalar recorded. Suppose that q is one of the explicitly enumerated representatives in Theorem A.28*(iii)*, including every normalized output later supplied by a marked rank-four separator. Then* $\text{K-Holant}(\Lambda, q) \le_{\mathrm{T}} \text{K-Holant}(\Lambda)$*, and q directly realizes one of the following: a rank-$(2,2,2)$ endpoint-nondegenerate eight-vertex signature, a quaternary pure generalized equality, or a nonsingular binary B with $[BX] \notin G$. These invoke Lemmas 3.9, 5.3 and 9.4; none recurses.*

Proof. This is exactly the representative-wise three-way direct-realization statement of Theorem A.28(iii). □

Consequently the reductions Lemmas 9.5, 9.7 and 9.8 are bounded local reductions. After their displayed physical output is constructed and its finite factorization tree is processed, every branch is a resolved leaf, a lower-arity factor route, or a strict complete-group successor. An intermediate factor split remains inside the originating card's finite factorization tree and strictly decreases the factor potential of Equation (25). None of these conclusions invokes a queue-termination or restart lemma.

9.2 Rainbow localization and matching-contraction closure

Lemma 9.6 handles cards with one matching. We now show that three collinear cards with different matchings differ on only four ports. Closing their common factors then converts this geometric statement into a six-port gadget.

Lemma 9.9 (Localization of a rainbow matching line)**.** *Let F_0, F_1, F_2 be three distinct nonsingular matching products on one projective line, with pairwise distinct perfect matchings M_0, M_1, M_2. Then there is a unique set S of four ports such that the three matchings restrict on S to the three pairwise distinct perfect matchings and the three tensors have common projective binary factors on S^c. Moreover, this projective line contains exactly three matching-product points. After compatible projective rescaling one may write*

$$F_i = R \otimes E_i \qquad (0 \le i \le 2),$$

where $R \neq 0$ is the fixed matching product on S^c, and the quaternary tensors E_0, E_1, E_2 span a two-dimensional subspace.

In the physical setting, let (Λ, G, f) be a deck-stable marked Platonic state, where f has even arity at least six, and suppose these three tensors occur as the images of one rich domain line under the card map Equation (111). If every common binary factor U satisfies $[UX] \in G$, and the actual kernel set $\mathcal{K}_G^{\mathrm{act}}$ spans $\mathrm{Mat}_2(\mathbb{C})$, those factors can be closed by actual G-kernels without annihilating the three selected cards. This realizes a six-port signature h with one f-vertex and exactly $(\mathrm{arity}(f) - 6)/2$ actual binary kernels, and

$$\text{K-Holant}(\Lambda, h) \le_{\mathrm{T}} \text{K-Holant}(\Lambda). \tag{122}$$

Every other actual G-card of h remains zero or a nonsingular safe quaternary G-matching product.

Proof of Lemma 9.9. For a port set S, let $c_i(S)$ count the edges of M_i crossing $S \mid \bar{S}$. Nonsingularity of every binary factor makes the matching recoverable from flattening ranks and gives

$$\mathrm{rank}(F_i)_{S|\bar{S}} = 2^{c_i(S)}.$$

In particular, proportional nonsingular matching products have the same matching. Collinearity also gives, for every permutation (i, j, k),

$$2^{c_i(S)} \le 2^{c_j(S)} + 2^{c_k(S)}.$$

Decompose $M_0 \cup M_1$ into common edges and alternating cycles. For each component C, both F_0 and F_1 have flattening rank one across $C \mid C^c$, so $\mathrm{rank}_C(F_2) \le 2$. The number $c_2(C)$ is even; hence $2^{c_2(C)} \le 2$ forces $c_2(C) = 0$. Regard the components as tensor factors. A nontrivial linear

combination of two simple tensors is simple only when the two summands differ projectively in at most one factor. Since M_2 is distinct from M_0, M_1, it differs on at least one alternating cycle (a common-edge component has only one internal matching). The simple-sum criterion permits only one such varying component; hence there is exactly one exceptional alternating cycle, and the common-edge factors outside it agree projectively in all three tensors.

The matching M_2 cannot repeat M_0 or M_1 on the exceptional component. Indeed, flatten across the $2 \mid (2\ell - 2)$ cut defined by the two endpoints of an edge of the repeated matching: the two same-matching tensors have rank one, so the third has rank at most two, whereas its different matching has two edges crossing that cut and hence rank four. Thus the three restrictions on the exceptional component are pairwise distinct.

Write the exceptional cycle as 2ℓ cyclic ports. If $\ell \geq 3$, consider its cyclic consecutive triples. Each M_2-edge belongs to at most two of them. If every triple contained an M_2-edge, equality in this incidence count would force every such edge to be a cycle edge and hence $M_2 = M_0$ or M_1. Choose instead a triple T containing no M_2-edge. Then

$$(c_0(T), c_1(T), c_2(T)) = (1, 1, 3),$$

contradicting $8 \leq 2 + 2$. Hence $\ell = 2$, so the exceptional component is the asserted quaternary set and carries the three perfect matchings. It is unique by unique factorization. A fourth matching-product point on the same projective line would repeat one of the three quaternary matchings; applying the preceding repeated-matching rank argument to it and either of the other two points gives a contradiction. Thus the line has exactly three matching-product points, and the displayed decomposition $F_i = R \otimes E_i$ follows.

In the physical card-map setting, for each common factor U, spanning by $\mathcal{K}_G^{\mathrm{act}}$ supplies an actual K with $\mathrm{tr}(U^{\mathsf{T}}K) \neq 0$. Closing the common factors therefore preserves all three selected cards. For any other card, superpose its matching with the closure matching. A closed common edge contributes $\mathrm{tr}(B^{\mathsf{T}}K)$, while a crossed path splices two factors to $B_1^{\mathsf{T}}KB_2$. Put $T_1 := B_1X$, $T_K := KX$, and $T_2 := B_2X$. The latter is nonsingular and, with $T^{\#} = XT^{\mathsf{T}}X$, has transfer

$$[T_{B_1^{\mathsf{T}}KB_2}] = [T_1^{\#}XT_KXT_2] \in G.$$

Internal cycles contribute scalars and may only make the card zero. Iteration over the components proves Equation (112) and Equation (122). □

Lemma 9.10 (Platonic matching-contraction closure)**.** *Let $G \leq \mathrm{PGL}_2$ contain $[X]$ and be reversal-stable. Let F be zero or a product of nonsingular binary factors on a perfect matching, with every factor transfer in G. Contract any disjoint collection of its ports by actual nonsingular G-binaries (including the explicit I- or native-X links). The result is zero or again a product of nonsingular binary factors whose transfers lie in G.*

Proof. Superpose the factor matching with the contraction matching. Components are cycles and paths. A cycle contributes a scalar (possibly zero); a path splices B_1, K, B_2 to $B_1^{\mathsf{T}}KB_2$. Put $T_1 := B_1X$, $T_K := KX$, and $T_2 := B_2X$. Its transfer class is

$$[T_1^{\#}XT_KXT_2], \qquad T^{\#} = XT^{\mathsf{T}}X.$$

Reversal stability, $[X] \in G$, and closure of G keep this class in G; the omitted orientation scalar is nonzero. Iterating over components proves the claim. □

For the second octahedral display, with $H_{\mathcal{O}}$ already fixed in Equation (103), define the coefficient-similarity map

$$\theta_{\mathcal{O}}(U) := H_{\mathcal{O}}^{-1} U H_{\mathcal{O}}, \tag{123}$$

the transformation tuple

$$\mathbf{H}_{\mathcal{O}} := (H_{\mathcal{O}}^{-1}, H_{\mathcal{O}}^{\mathsf{T}}, H_{\mathcal{O}}^{-1}, H_{\mathcal{O}}^{\mathsf{T}}),$$

and its induced quaternary coefficient operator

$$\mathcal{L}_{\mathcal{O}} := \mathscr{D}_{\mathbf{H}_{\mathcal{O}}}, \qquad \mathcal{L}_{\mathcal{O}}(F) = (H_{\mathcal{O}}^{-1} \otimes H_{\mathcal{O}}^{\mathsf{T}} \otimes H_{\mathcal{O}}^{-1} \otimes H_{\mathcal{O}}^{\mathsf{T}}) F.$$

These maps compare coefficient tables under the fixed-coordinate convention at the start of the section.

For $\alpha \in \{dt, td\}$, put

$$S_\alpha := H_\alpha H_\alpha^{\mathsf{T}}, \qquad \mathcal{L}_\alpha := H_\alpha^{-1} \otimes H_\alpha^{\mathsf{T}} \otimes H_\alpha^{-1} \otimes H_\alpha^{\mathsf{T}}.$$

By Equation (106), $[S_\alpha] \in \mathcal{O}$. Ordered contraction gives the three matching formulas

$$\begin{aligned}
\mathcal{L}_\alpha P_0(U,V) &= P_0(\theta_\alpha(U), \theta_\alpha(V)), \\
\mathcal{L}_\alpha P_1(U,V) &= P_1(\theta_\alpha(U S_\alpha^{-1}), \theta_\alpha(S_\alpha V)), \\
\mathcal{L}_\alpha P_2(U,V) &= P_2(\theta_\alpha(U), \theta_\alpha(S_\alpha V S_\alpha^{-1})).
\end{aligned} \tag{124}$$

Thus $\mathcal{L}_\alpha$ bijects the standard and mixed matching decks, preserves their matching labels, and intertwines their actual local group actions after the displayed current-group orientation factors are absorbed. This is a coefficient identity; the physical operations on the right use only actual $\mathcal{O}_\alpha$-representatives.

The incidence counts, actual rainbow-line orbits, and deletion connectivity used below are supplied by Theorem A.20.

Lemma 9.11 (High-arity Platonic rank-deficient localization)**.** *Let (Λ, G, f) be a deck-stable retained, factor-saturated fixed-I state, where $f \neq 0$ is a tensor-prime minimum core of even arity $n \geq 6$ and*

$$G \in \{\mathcal{T}, \mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}, \mathcal{I}, \mathcal{I}_-\}.$$

Fix an ordered pair p, define the ancestor card-map rank $r := \operatorname{rank} \Phi_p^f \in \{2,3,4\}$, and suppose that, for every ordered pair of ports and every actual G-kernel, the corresponding ancestor card is zero or a nonsingular safe G-matching product. Then at least one of the following structural routes applies; when both are available, choose route 1.

1. *All nonzero p-cards use one residual matching, and f has a genuine proper tensor factor.*
2. *One f-vertex and exactly $(n-6)/2$ actual binary kernels directly realize a nonzero six-port signature h, retaining p, such that*

$$2 \leq \operatorname{rank} \Phi_p^h \leq r \leq 4, \tag{125}$$

and every actual G-card of h is zero or a nonsingular safe quaternary G-matching product. In particular, $\text{K-Holant}(\Lambda, h) \leq_{\mathrm{T}} \text{K-Holant}(\Lambda)$.

The second route recomputes the rank of the directly realized six-port card map; the ancestor rank is only the upper bound in Equation (125)*.*

Proof. Use the domain graph $\mathscr{G}_G^{\mathrm{dom}}$ defined above. The kernel-deleted connectivity assertion of Theorem A.20(iii) says that

$$\mathcal{D}_G \setminus \mathbb{P}(\ker \Phi_p^f)$$

is nonempty and connected. If all its nonzero images have one matching, Lemma 9.6 gives a proper factor. For any nonstandard octahedral form, precompose the card map with its fixed invertible coefficient similarity and pull the resulting tensor identity back. For $\mathcal{I}_-$, use Lemma 9.2. These are analytic reconstructions, and factor access is supplied by Lemma 2.20.

Otherwise choose the first matching-changing edge along a path in this deleted graph, and let ℓ be a rich domain line containing that edge. If $r \in \{2, 3\}$ and ℓ met $\mathbb{P}(\ker \Phi_p^f)$, the restriction of Φ_p^f to the two-dimensional vector span of ℓ would have rank at most one. All nonkernel images on ℓ would then be projectively proportional and have the same matching, a contradiction. If $r = 4$, the kernel is zero. Thus ℓ is kernel-avoiding. Let V_ℓ be its two-dimensional vector span. The matching-changing edge gives $\operatorname{rank}(\Phi_p^f|_{V_\ell}) \geq 2$, while kernel-avoidance gives $\operatorname{rank}(\Phi_p^f|_{V_\ell}) \leq 2$; hence the restriction has rank two and induces a projective injection on ℓ. Let F_0, F_1 be the matching-changing images, and choose any third domain point on this rich line, with image F_2. Its matching differs from both of theirs. Indeed, if $M_2 = M_0$, choose an edge of $M_0 \setminus M_1$ and flatten across its two ports. The ranks of F_0, F_2 are one, whereas the rank of F_1 is four, because two nonsingular factors cross this cut. This contradicts collinearity; the case $M_2 = M_1$ is identical. Thus the three images satisfy the general-arity hypotheses of Lemma 9.9. That lemma says their line contains exactly three matching-product points, so injectivity also forces ℓ to have exactly three domain points. The quaternary incidence classification is used only after this localization.

The same lemma supplies a unique quaternary set S and compatible rescalings

$$\Phi_p^f(D_i) = R \otimes E_i, \qquad R = \bigotimes_{j=1}^{(n-6)/2} U_j, \qquad 0 \leq i \leq 2,$$

where the E_i form the residual rainbow deck line and $D_i \in \mathcal{K}_G^{\mathrm{act}}$ are the selected domain representatives. The actual kernels span Mat_2, so for every $U_j \neq 0$ choose an actual K_j with $\operatorname{tr}(U_j^{\mathsf{T}} K_j) \neq 0$. Close the corresponding common-factor ports by these K_j, leaving $p \cup S$. The resulting one-vertex signature h is nonzero because on each selected card the closing scalar is

$$\kappa = \prod_j \operatorname{tr}(U_j^{\mathsf{T}} K_j) \neq 0.$$

Record the retained pair p, selected line ℓ, ordered kernels $(K_j)_j$, output order π, and scalar κ. If C denotes the resulting contraction on the residual ports and P_π the invertible output reindexing induced by the recorded port permutation, then

$$\Phi_p^h = P_\pi \circ C \circ \Phi_p^f.$$

This proves the upper rank bound, while the surviving rainbow deck line proves $\operatorname{rank} \Phi_p^h \geq 2$.

It remains to verify every card, not only the three selected ones. Each is obtained from an ancestor G-matching card by the remaining actual closures, so Lemma 9.10 gives zero or a nonsingular safe G-matching product. The tensors U_j are factors found inside the selected cards; the only gadgets inserted are the actual kernels K_j. □

Whenever localization supplies a rainbow deck line, use actual dressings on the domain pair and on the four residual ports to send the two lines to their specified representatives. The two operations act on disjoint port sets and commute; retain their scalars and port orders. In the standard forms

$G \in \{\mathcal{T}, \mathcal{O}, \mathcal{I}\}$, let Φ be this actual card map after one common nonzero scalar is removed, and put $G_0 = G$. For a nonstandard octahedral form, write $\theta, \mathcal{L}$ for its fixed coefficient maps and Φ_{phys} for the actual normalized map. Its domain and image representatives are the θ- and $\mathcal{L}$-images of the standard ones. Only for coefficient calculations put

$$\Phi = \mathcal{L}^{-1} \circ \Phi_{\text{phys}} \circ \theta, \qquad \Phi_{\text{phys}}(\theta K) = \mathcal{L}\Phi(K), \qquad G_0 = \mathcal{O}. \tag{126}$$

Thus a kernel K in a standard formula means the actual current-form kernel θK, and its physical output is $\mathcal{L}\Phi(K)$. Neither coefficient map is a gadget. The opposite icosahedral display uses its own κ_D-dictionary in Lemma 9.2. In these standard coefficient coordinates define

$$A = P_0(Q_0, Q_0), \qquad B = P_1(Q_0, Q_0), \qquad C = P_2(Q_2, Q_2) = A - B. \tag{127}$$

The two displayed input matrices sum to $2Q_0$. Together with $C = A - B$, this fixes the relative image scalars. Dividing by their one common recorded nonzero scalar, the normalized map satisfies

$$\Phi(Q_0 + Q_1 + Q_2 + Q_3) = A, \qquad \Phi(Q_0 - Q_1 - Q_2 - Q_3) = -B. \tag{128}$$

Thus, defining the image tensors $W_j := \Phi(Q_j)$,

$$\Phi(Q_0) = C/2, \qquad W_1 + W_2 + W_3 = (A + B)/2, \tag{129}$$

and each remaining sign point imposes

$$W_j, \quad A - 2W_j, \quad 2W_j - B \in \{0\} \cup \operatorname{cone}(\mathfrak{D}_4(G_0)). \tag{130}$$

These are coefficient identities of actual cards, read through Equation (126) in the nonstandard forms. Every terminal below is checked on its physical current-form output.

9.3 The four fixed-equality marked S_4 forms

The projective octahedral group has four ordered marked-involution forms in the coordinates containing the supplied literal I. They are the standard $(dd; d)$ form $\mathcal{O}$, the $(tt; d)$ form $\mathcal{O}'$, and the mixed forms $\mathcal{O}_{dt}, \mathcal{O}_{td}$ fixed in Equations (103) and (105). They are exhaustive by Theorem 5.14. The three displayed similarity matrices compare coefficient configurations only; physical operations always use current-form representatives. For a normalized rainbow map, the exact bridge has six rank-deficient solutions, handled on that directly realized six-port map by Lemma 9.7, and the twenty-four orderings of the four frames in Theorem A.27(i).

Retain the even-entry order

$$\mathbf{e}(M) = (M_{0000}, M_{0011}, M_{0101}, M_{0110}, M_{1001}, M_{1010}, M_{1100}, M_{1111}). \tag{131}$$

Lemma 9.12 (Uniform octahedral full-rank exits)**.** *Let Φ be a normalized full-rank map for which $(\Phi(Q_1), \Phi(Q_2), \Phi(Q_3))$ is one of the twenty-four orderings of the four frames in Theorem A.27*(i)*. In a nonstandard form, Φ is the coefficient map associated to the actual Φ_{phys} by Equation* (126)*. Define*

$$K_{\mathcal{O}} := Q_0 + Q_1, \qquad K_{\mathcal{O}'} := \theta_{\mathcal{O}}(K_{\mathcal{O}}), \qquad K_{\mathcal{O}_\alpha} := \theta_\alpha(K_{\mathcal{O}}) \quad (\alpha \in \{dt, td\}).$$

For each marked form, apply its displayed kernel to the actual card map. At most four further actual group binaries, used in the indicated port dressings or pair closure, then directly realize one of: an endpoint-nondegenerate eight-vertex signature of ranks $(2, 2, 2)$, a quaternary pure generalized equality, or a nonsingular binary whose transfer lies outside the current complete group. For $\mathcal{O}_{dt}$, the first-frame entries R_0, R_1, R_3 give the eight-vertex exits and the other six named entries give outside binaries. For $\mathcal{O}_{td}$, R_0, R_3 give the eight-vertex exits, R_1 gives the pure generalized equality, and the other six give outside binaries. Thus every full-rank map is a resolved local exit.

Proof. This is Theorem A.27(i) together with the direct current-coordinate tables in Proposition A.16. Every listed kernel and dressing is an actual representative in its own marked form; the similarities only identify the coefficient tables. Hence Equation (12) gives the stated directions, followed by Lemmas 3.9, 5.3 and 9.4 according to the listed exit. □

For the full-rank six-port statements below, assume

$$\begin{aligned} &\mathsf{Stable}(\Lambda, G, h), \qquad \operatorname{arity}(h) = \nu(\Lambda) = 6, \\ &h \text{ is directly realized over } \Lambda, \qquad \operatorname{rank} \Phi_p^h = 4. \end{aligned} \tag{132}$$

The rank in Equation (132) is always recomputed from the actual coefficients of h, never inherited from an ancestor. Under this condition, put $\Phi := \Phi_p^h$. Varying card matchings give a rainbow deck line by clauses (i)–(iii) of Theorem A.20, together with Equation (115); Lemma 9.9 then supplies the actual normalization. Apply Lemma 9.12 to its normalized coefficient map, using Equation (126) in a nonstandard form, to obtain a resolved eight-vertex, pure-generalized-equality, or outside-binary leaf.

If all nonzero cards use one matching M, define the auxiliary card map $\Psi := \Phi$ in the first form, $\Psi := \Phi \circ \theta_{\mathcal{O}}$ for $\mathcal{O}'$, and $\Psi := \Phi \circ \theta_\alpha$ for $\mathcal{O}_\alpha$, $\alpha \in \{dt, td\}$. The twelve test points map to zero or an M-matching product, so Lemma 9.6 factors Ψ. In a nonstandard form, composing the varying factor with the inverse coefficient similarity pulls this tensor identity back to Φ; it does not use a similarity matrix as a gadget. The reconstructed tensor h has a genuine proper factor.

Theorem 9.13 (Octahedral deck rigidity)**.** *If Equation* (132) *holds and*

$$G \in \{\mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}\},$$

then $\mathsf{Close}_\varnothing(\Lambda, G, h)$.

Proof. If card matchings vary, Lemmas 9.9 and 9.12 gives a resolved local exit. If they are fixed, Lemmas 2.20 and 9.6 exposes a proper factor of the active six-port occurrence, including after the coefficient pullback for any of the three nonstandard forms.

Close that finite factorization tree directly. A nonbinary prime factor has arity below six and therefore lowers ν. A singular or unsafe binary is terminal, and a finite safe binary whose transfer class lies outside G strictly enlarges the recomputed complete group. If every factor were a safe binary whose transfer class already lies in G, the original signature would be a G-matching product, contrary to the retained tensor-prime branch. Hence every branch is a resolved leaf or a strict outer successor, proving $\mathsf{Close}_\varnothing(\Lambda, G, h)$. The rank and factorization used here belong to the realized tensor h. □

9.4 The internal A_4 form

The internal tetrahedral case uses the standard twelve-point model $\mathcal{T}$ in Equation (7). It is closed under multiplication, inverse, and transpose. All normalizations below use actual left and right $\mathcal{T}$-multiplication and port permutations.

Let $\Phi : \mathrm{Mat}_2 \to (\mathbb{C}^2)^{\otimes 4}$ be the full-rank map localized from a tetrahedral rainbow deck line. We use the common normalization Equations (127) to (130) with $G = \mathcal{T}$.

Lemma 9.14 (Tetrahedral bridge)**.** *The simultaneous solution set of Equation* (130) *is*

$$\mathcal{B}_{\mathcal{T}} = \left\{0, \frac{A}{2}, \frac{B}{2}, R_0, R_1, R_3\right\},$$

where

$$R_0 = \frac{1}{2}P_2(Q_0, Q_0), \qquad R_1 = -\frac{1}{2}P_2(Q_1, Q_1), \qquad R_3 = -\frac{1}{2}P_2(Q_3, Q_3).$$

The sum equation leaves exactly twelve ordered maps: the six rank-two permutations of $(0, A/2, B/2)$ *and the six rank-four permutations of* (R_0, R_1, R_3)*.*

Proof. The exact separated-sign and sum calculation, including dependent lines, is Theorem A.27(ii). □

For the rank-four maps write $\sigma = (a, b, c)$ for a permutation of $(0, 1, 3)$. The corresponding six-port ancestor is uniquely reconstructed as

$$f_\sigma = \sum_{\mu=0}^{3} Q_{12}^{\mu} \otimes F_\mu^\sigma, \qquad (F_0^\sigma, F_1^\sigma, F_2^\sigma, F_3^\sigma) = (C/2, R_a, R_b, R_c).$$

Lemma 9.15 (The two permutation classes)**.** *The six rank-four maps split into two sets:*

$$\Sigma_+ = \{(0,1,3), (1,3,0), (3,0,1)\},$$
$$\Sigma_- = \{(0,3,1), (1,0,3), (3,1,0)\}.$$

For $\sigma \in \Sigma_+$*, all fifteen deleted-pair contractions by all twelve actual tetrahedral kernels are nonzero matching products. For* $\sigma \in \Sigma_-$*, contracting ports* $1, 3$ *by the actual kernel* $Q_0 = I$ *gives a directly realizable quaternary* q_σ *whose three flattening ranks are* $(2, 2, 2)$*. For each such* σ*, four actual tetrahedral port dressings transform* q_σ *into an endpoint-nondegenerate eight-vertex signature of the same ranks. The contraction and dressings are direct gadgets from* f_σ*, so Lemma 9.4 applies in the required direction.*

Proof. Theorem A.27(ii) gives the split, common Q_0-card, and actual endpoint-nondegenerate eight-vertex signature dressings. Thus the two direct-gadget reductions point back to f_σ, and Lemma 9.4 applies. □

Lemma 9.16 (The orientation-preserving tensors are affine)**.** *For every* $\sigma \in \Sigma_+$*, the tensor* f_σ *belongs to the affine class* $\mathscr{A}$ *in the same computational basis as all twelve tetrahedral binaries.*

Proof. Theorem A.27(ii) gives affine support and even mixed quadratic coefficients; Equation (7) puts all twelve binaries in that same affine basis. □

To lift this statement to the signature set, define the tetrahedral matrices

$$R_+ = Q_0 + Q_1 + Q_2 + Q_3, \qquad R_- = Q_0 - Q_1 - Q_2 - Q_3.$$

Then

$$R_+R_- = 4I, \qquad R_+Q_1 = -Q_0 + Q_1 - Q_2 + Q_3,$$

so $[R_+], [R_-], [Q_1], [R_+Q_1] \in \mathcal{T}$, and all four matrices have actual inverses up to nonzero scalar. For a matrix M, let $L_j(M)$ denote its physical action $M(y_j, x_j)$ on port j, and let Π_{ab} exchange output ports a, b; products of permutations below act from right to left. Entrywise expansion in the fixed physical port convention gives the identities

$$\begin{aligned}
\Pi_{45}\Pi_{23}\Pi_{34}\big[I \otimes R_+ \otimes I^{\otimes 3} \otimes R_+\big]\mathrm{H}_6 &= -16 f_{(0,1,3)}, \\
\Pi_{34}\big[I \otimes R_- \otimes (R_+Q_1) \otimes R_+ \otimes I^{\otimes 2}\big]\mathrm{H}_6 &= -32 f_{(1,3,0)}, \\
\Pi_{23}\big[I \otimes R_+ \otimes R_+ \otimes I \otimes R_+ \otimes Q_1\big]\mathrm{H}_6 &= 32 f_{(3,0,1)}.
\end{aligned} \tag{133}$$

The brackets are tensor products of the six L_j-actions, not a coefficient-side similarity. Because $R_+^{-1} = R_-/4$ and $Q_1^{-1} = -Q_1$, every local action and permutation is projectively actual-invertible. Thus all three equivalences in Equation (133) are reversible actual identities. Every f_σ, $\sigma \in \Sigma_+$, therefore directly gives literal H_6; four copies then directly give RM_8 by Equation (96). The complete group is A_4, so the signature-set conclusion uses the twelve-card relative theorem Theorem 9.25. For any ambient retained set Λ over which f_σ is directly realizable, both direct-gadget reductions point from K-Holant$(\Lambda, \mathrm{H}_6, \mathrm{RM}_8)$ through K-Holant(Λ, f_σ) to K-Holant(Λ).

After the quaternary boundary, let an ancestor f have even arity at least six. Rank zero is impossible and rank one factors. In ranks two through four, Lemma 9.11 either gives the proper fixed-matching factor or materializes the safe six-port bridge. The bridge rank is then recomputed from its own coefficients before Lemmas 9.14 and 9.15 is invoked; an ancestor rank is never transferred to that reduction.

From one core to the retained signature set. To obtain a common affine basis, every retained occurrence g must be classified with all twelve tetrahedral kernels. A binary factor may have transfer in $\mathcal{T} \setminus V_4$, so the complete family is $\{\Phi_p^g(K) : K \in \mathcal{K}_{\mathcal{T}}^{\mathrm{act}}\}$ for each pair p. The six-port classification supplies the two sets below; an eight-port base and a stabilizer-code shortening argument then handle larger arities.

Define the projective six-port matching set, H-core orbit, and safe union by

$$\mathscr{M}_6 = \left\{ \left[\prod_{\{a,b\} \in M} U_{ab} \right] : M \text{ is a perfect matching of } [6],\ [U_{ab} X] \in \mathcal{T} \right\},$$
$$\mathscr{H}_6 = \mathrm{Orb}_{\mathcal{T}^6 \rtimes S_6}([\mathrm{H}_6]), \qquad \mathscr{S}_6 = \mathscr{M}_6 \,\dot\cup\, \mathscr{H}_6.$$

Define their affine cone by $\mathscr{C}_6 := \{0\} \cup \mathrm{cone}(\mathscr{S}_6)$. Here the local action uses the fixed actual representatives of $\mathcal{T}$, so every orbit equivalence is an actual reversible port dressing. Their disjointness and exact cardinalities are given by Theorem A.23.

Proposition 9.17 (Hereditary tetrahedral eight-port base)**.** *Let $g \neq 0$ be an eight-port signature in the normalized tetrahedral branch whose every physical tetrahedral pair card lies in $\mathscr{C}_6$. Then g factors across a $2|6$ cut or $g \in \mathscr{A}$ in the normalized basis. Thus every tensor-prime eight-port survivor is affine; factors are exposed only through factor saturation.*

Proof. For a pair map $\Phi = \Phi_p^g$, actual dressings normalize a nonzero card to a three-tetrahedral-binary product or H_6. On the four domain lines through that card, opposite endpoints sum to twice it. If both endpoint images are dependent on every line, their four sign differences span the subspace $\mathrm{span}\{Q_1, Q_2, Q_3\}$, so $\mathrm{rank}\,\Phi = 1$; dual-basis reconstruction gives a genuine $2|6$ factor. Otherwise the scale-sensitive exhaustive audit Theorem A.23 says that every nonaffine parent has an actual card outside $\mathscr{C}_6$, contrary to the hypothesis. The remaining parents are affine. Actuality and arbitrary complex scales are part of the cited audit; factors use only Lemma 2.20. □

We use the stabilizer description of affine signatures from Sections 2.13.2 and 2.13.3 to locate the varying part of the twelve cards. The next two lemmas show that this variation occupies at most four logical qubits. The encoding is an analytic description of their coefficient space. The shortening lemma then selects an actual tetrahedral kernel from that description.

Lemma 9.18 (Sharp stabilizer-line bound)**.** *Let u, v be nonproportional nonzero affine signatures of common arity m, and let $c \neq 0$. If $u + cv$ is also nonzero and affine, then, after applying the same analytic Clifford change to both tensors, followed by a common nonzero scalar normalization*

sending u to δ_{0^m}, the support of the transformed v is either a singleton disjoint from zero or a linear subspace of dimension at most two.

Proof. Choose a unitary Clifford C with $Cu = \alpha\delta_{0^m}$, where $\alpha \neq 0$, and apply $\alpha^{-1}C$ to both tensors. The scalar normalization is separate from the Clifford operator. Relabel the resulting second tensor as v. Write v in affine normal form as $v = \lambda\mathbf{1}[a + L]i^Q$, where $\lambda \in \mathbb{C}^\times$ is the scale, $a + L$ is the affine support, Q is the quadratic phase, and $d := \dim L$. If $0 \notin a + L$, the support size $1 + 2^d$ forces $d = 0$. If $0 \in a + L = L$ and the zero coefficient cancels, $2^d - 1$ forces $d = 1$. Otherwise rescale the phase quotient to one off zero. For $d \geq 3$, its multiplicative third difference on a three-cube is one (both phases are quadratic), so its value at zero is also one, contradicting nonproportionality. Hence $d \leq 2$. □

Lemma 9.19 (The tetrahedral card cube has at most four active qubits)**.** *Let $\Phi : \mathrm{Mat}_2(\mathbb{C}) \to (\mathbb{C}^2)^{\otimes m}$ be linear. Suppose that every one of the twelve values $\Phi(K)$, $[K] \in \mathcal{T}$, is zero or affine. Then there are an analytic Clifford operator C on the m output qubits and a subspace $R \leq \mathbb{F}_2^m$, $\dim R \leq 4$, such that*

$$\mathrm{supp}(C\Phi(K)) \subseteq R \qquad (K \in \mathrm{Mat}_2(\mathbb{C})). \tag{134}$$

Equivalently, the image of Φ is contained in one stabilizer code with at most four logical qubits.

Proof. If $\Phi = 0$, take $R = \{0\}$. Otherwise at least one tetrahedral test point has nonzero image because those points span Mat_2. Precompose Φ by an actual tetrahedral multiplication carrying that point to $A := Q_0 - Q_1 - Q_2 - Q_3$, relabel the resulting map as Φ, and hence assume $\Phi(A) \neq 0$. Define

$$B_0 := 2Q_0 - A, \qquad B_j := A + 2Q_j \quad (1 \leq j \leq 3).$$

These are the four opposite-parity tetrahedral sign points paired with A; they form a basis and satisfy

$$A + B_0 = 2Q_0, \qquad B_j - A = 2Q_j \ (1 \leq j \leq 3), \qquad 2A = -B_0 + B_1 + B_2 + B_3.$$

Choose an analytic Clifford C with $C\Phi(A) = \alpha\delta_{0^m}$, where $\alpha \neq 0$. For the following analysis put $\Psi(K) := \alpha^{-1}C\Phi(K)$; this common scalar normalization is separate from the Clifford operator and leaves every support unchanged. By the normalized-delta case in the proof of Lemma 9.18, each nonzero $\Psi(B_j)$ not proportional to δ_{0^m} has, off zero, either one point or the three nonzero points of a two-dimensional $\mathbb{F}_2$-subspace. The last displayed relation makes every point in their union occur at least twice. If its span had dimension at least five, total incidence at most twelve would force sizes $(3, 3, 3, 1)$ or $(3, 3, 3, 3)$. In type $(3, 3, 3, 1)$, double incidence leaves at most five distinct nonzero points, and any three-point support is the nonzero part $\{x, y, x + y\}$ of a two-dimensional $\mathbb{F}_2$-subspace. Its line relation is nontrivial, so those at most five points have span at most four.

In type $(3, 3, 3, 3)$, double incidence leaves at most six points. The only remaining extremal case has six points, each occurring in exactly two of the four supports. Regard a point shared by supports i, j as an edge between vertices i, j. The four supports form a loopless three-regular multigraph on four vertices, and the line relation at each vertex is the sum of its three incident edge labels. The binary vertex–edge incidence matrix has rank at least two: its rank is $4 - c$, where the graph has $c \leq 2$ connected components. Hence the six edge labels satisfy at least two independent relations and again span at most four. Thus their union spans some R with $\dim R \leq 4$. Since the B_j's form a basis and the common nonzero scalar leaves supports unchanged, Equation (134) follows. □

For the physical shortening step, let $\mathcal{O}$ denote the projective one-qubit Clifford group in the quaternion coordinates Equation (6). The normal subgroup $\mathcal{T} \triangleleft \mathcal{O}$ has index two. Each of its two cosets contains the following eight full-support representatives:

$$\begin{aligned}\mathcal{C}_+ &= \{Q_0 + \epsilon_1 Q_1 + \epsilon_2 Q_2 + \epsilon_3 Q_3 : \epsilon_j \in \{\pm 1\}\},\\ \mathcal{C}_- &= \{Q_d + \epsilon Q_o : d \in \{0,3\},\ o \in \{1,2\},\ \epsilon \in \{\pm 1\}\}.\end{aligned} \tag{135}$$

On the open set $K_{01}K_{10} \neq 0$, define the cross-ratio rational function $\chi(K) := K_{00}K_{11}/(K_{01}K_{10})$. Direct use of Equation (6) gives

$$K_{xy} \neq 0, \qquad \chi(K) = -1 \qquad (K \in \mathcal{C}_+ \cup \mathcal{C}_-). \tag{136}$$

Indeed, in $\mathcal{C}_+$ the diagonal and off-diagonal products are 2 and -2, while in $\mathcal{C}_-$ they are 1 and -1. After division by one entry, every such table and its pointwise inverse have entries in μ_4, so both are nowhere-zero affine phase tables.

Lemma 9.20 (Stabilizer erasure and column normal form)**.** *Let $V : (\mathbb{C}^2)^{\otimes k} \to (\mathbb{C}^2)^{\otimes N}$ be a stabilizer isometry and let r be a physical output port. If erasure of r is correctable, meaning $V^\dagger R_r V$ is scalar for every one-qubit operator R, exactly one of the following normal forms applies after a fixed one-qubit Clifford on r:*

1. *there are a one-qubit stabilizer state $|\chi\rangle_r$ and a stabilizer isometry W such that*

$$V|\psi\rangle = |\chi\rangle_r \otimes W|\psi\rangle; \tag{137}$$

2. *there are an auxiliary qubit a and a stabilizer isometry $W : \mathbb{C}^2_a \otimes (\mathbb{C}^2)^{\otimes k} \to (\mathbb{C}^2)^{\otimes (N-1)}$ such that*

$$V|\psi\rangle = 2^{-1/2} \sum_{c \in \mathbb{F}_2} |c\rangle_r W(|c\rangle_a \otimes |\psi\rangle). \tag{138}$$

If no single-site erasure is correctable, then, after postcomposing the outputs by a tensor product of one-qubit Cliffords and precomposing the input by one logical Clifford, there are a vector $c \in \mathbb{F}_2^N$, and a matrix $M = (m_1 \ \cdots \ m_N) \in \mathbb{F}_2^{k \times N}$ such that

$$V|z\rangle = \zeta(z)|c + M^{\mathsf{T}} z\rangle, \qquad \operatorname{rank} M = k, \qquad m_r \neq 0 \quad (1 \le r \le N), \tag{139}$$

where $\zeta(z) = \zeta_0 i^{Q(z)}$ is a nowhere-zero stabilizer quadratic phase in the precise sense of Section 2.13.2.

Proof. By the Hilbert-space and error-correction conventions of Sections 2.13.1 and 2.13.3, correctability of erasure at r is the single-site Knill–Laflamme condition

$$V^\dagger P_r V = \alpha_P I \qquad (P \in \{I, X, -iY, Z\}). \tag{140}$$

These are Hermitian Pauli representatives; recall that $Y = ZX$. Conjugate P_r through a Clifford encoder and then evaluate its fixed ancillas. The three nonidentity expectations in Equation (140) are either all zero or exactly one is ± 1. Thus the input-independent marginal at r is respectively maximally mixed or a pure one-qubit stabilizer state. In the pure case its Schmidt rank is one, which is Equation (137).

In the maximally mixed case a Schmidt decomposition gives

$$V|\psi\rangle = 2^{-1/2}(|0\rangle_r W_0|\psi\rangle + |1\rangle_r W_1|\psi\rangle).$$

Applying Equation (140) to the four matrix units on r gives $W_c^\dagger W_d = \delta_{cd} I$. Hence $W(|c\rangle_a \otimes |\psi\rangle) = W_c|\psi\rangle$ is an isometry. Symplectic Gaussian elimination of the encoder tableau, in the sense fixed in Section 2.13.2, isolates a hyperbolic row pair meeting r; the remaining rows are a stabilizer tableau for W. This proves Equation (138) without inserting the displayed Clifford coordinate changes as gadgets.

Suppose now that no single-site erasure is correctable. For each r, failure of the displayed Knill–Laflamme condition gives a one-site Pauli whose pullback through the stabilizer encoder is nonscalar. By the logical Pauli convention in Section 2.13.2, this is a nontrivial logical Pauli with a physical representative supported on r alone. Representatives on different ports commute, so their logical Paulis commute. Independent physical one-qubit Cliffords send them to Z_r, and one logical Clifford sends the generated commuting logical subgroup to Z-type. Consequently each $V|z\rangle$ is a simultaneous eigenvector of every physical Z_r, and hence is one computational-basis ket times a phase. Its r-th bit is $c_r + m_r \cdot z$; nontriviality of the logical Pauli gives $m_r \neq 0$. If $\operatorname{rank} M < k$, some $0 \neq d \in \ker M^{\mathsf{T}}$ makes $V|z\rangle$ and $V|z+d\rangle$ proportional, contrary to isometry. Thus $\operatorname{rank} M = k$, and the stabilizer amplitude normal form of Section 2.13.2 makes the remaining nowhere-zero phase quadratic modulo four, proving Equation (139). □

Lemma 9.21 (An actual tetrahedral card shortens a nonaffine encoded residue)**.** *Let $t \geq 1$, let $V : (\mathbb{C}^2)^{\otimes k} \to (\mathbb{C}^2)^{\otimes N}$ be a stabilizer isometry with $k \leq 4$ and $N \geq k+2$, and let $\xi \in (\mathbb{C}^2)^{\otimes(t+k)} \setminus \mathscr{A}$. Put $F := (I^{\otimes t} \otimes V)\xi$. Then F factors or one actual tetrahedral contraction of two physical ports is nonzero and nonaffine.*

Proof. Apply Lemma 9.20. If erasure of a code port r is correctable, its pure-marginal case Equation (137) gives a genuine unary factor. In the maximally-mixed case Equation (138) isolates an auxiliary qubit a. Contract r with one of the t untouched ports using the actual kernel Q_0, whose class lies in $\mathcal{T}$. In the analytic local frame of r, this acts on the untouched-port/a coefficient table by an invertible one-qubit Clifford. The residual encoder is a nonzero stabilizer isometry, and a Clifford and its inverse preserve $\mathscr{A}$. Consequently the shortened card is nonaffine.

Assume now that no one-site erasure is correctable. Write the normal form of Equation (139) explicitly as

$$\widetilde{V} = \left(\bigotimes_{j=1}^{N} C_j^{-1}\right) V L, \qquad \widetilde{\xi} = (I^{\otimes t} \otimes L^{-1})\xi,$$

where L is the logical Clifford and the C_j's map normal-form output coordinates back to physical coordinates. Then $\widetilde{\xi} \notin \mathscr{A}$, since Clifford maps and their inverses preserve $\mathscr{A}$, and $F = (I^{\otimes t} \otimes (\bigotimes_j C_j)\widetilde{V})\widetilde{\xi}$. Let c and $M = (m_1 \ \cdots \ m_N)$ be the data for $\widetilde{V}$. Choose a k-column basis of M. Since $N \geq k+2$, choose two positions r, s outside it; deleting their columns leaves rank k. Choose matrix representatives C_r, C_s of the two analytic local Clifford frames used in the normal form, with $[C_r], [C_s] \in \mathcal{O}$. Normality and index two imply that

$$\{[C_r^{\mathsf{T}} K C_s] : [K] \in \mathcal{T}\}$$

is one of the two projective $\mathcal{T}$-cosets in $\mathcal{O}$ (the native X-twist only relabels that coset). By Equations (135) and (136), choose an actual K with $[K] \in \mathcal{T}$ whose effective table $K_{\text{eff}} = C_r^{\mathsf{T}} K C_s$ has full support and cross ratio -1. The analytic frames select this physical K; they are not inserted into the network.

For a logical word z, the two contracted bits are

$$\ell_r(z) = c_r + m_r \cdot z, \qquad \ell_s(z) = c_s + m_s \cdot z.$$

There is a unique physical codeword for each z, so the contraction contains no sum of competing lifts: it multiplies its coefficient by

$$\omega(z) = K_{\mathrm{eff}}(\ell_r(z), \ell_s(z)).$$

The full-support property makes ω nowhere zero. By Equation (136), after multiplication by a common nonzero scalar, $\omega = i^{q(z)}$ for a modulo-four quadratic q, and ω^{-1} has the same property. The undeleted columns retain rank k, so the residual computational-word map is injective. Multiplication by ω, its inverse, and this injective stabilizer encoding preserve affine membership in both directions, as does the normal-form phase $\zeta(z)$. Since $\widetilde{\xi} \notin \mathscr{A}$, the actual two-port K-card is nonzero and nonaffine. □

Proposition 9.22 (Physical tetrahedral localization)**.** *Let g be an even-arity tensor-prime signature in a fixed-I standard marked A_4 state whose recomputed complete group remains $\mathcal{T}$ after factor saturation. Assume the established terminal exits do not apply. If $g \notin \mathscr{A}$ and* $\mathrm{arity}(g) \geq 10$*, some actual tetrahedral pair card of g is nonzero and nonaffine. Iteration therefore reaches a realized nonaffine signature on at most eight ports, unless a factor, terminal, or strict group restart occurs, using one g-vertex and $O(\mathrm{arity}(g))$ actual binaries.*

Proof. If every actual pair card were zero or affine, tomography and Lemma 9.19 would place each card-map image in a stabilizer code with $k \leq 4$ logical qubits. Frobenius reconstruction would then write $g = (I^{\otimes 2} \otimes V)\xi$, with V encoding those k qubits into $N = \mathrm{arity}(g) - 2 \geq 8$ sites and $\xi \notin \mathscr{A}$. Since $N \geq k+2$, Lemma 9.21 gives a factor or a nonaffine actual card, a contradiction. Iterating this two-port descent reaches arity at most eight unless a terminal, factor, or recomputed-group restart occurs. The direct-gadget and factor-saturation directions are Equation (12) and Lemma 2.20. □

Section-local occurrence induction. We close the relative tetrahedral argument by a finite induction on unresolved occurrences and their remaining contractions. On entering the argument, dispatch nonzero odd signatures by Theorem 2.15, unaries by Lemma 3.4, nullaries by preprocessing, and newly exposed binaries by Lemmas 3.1 and 5.3. Thus, on a continuing branch, every unresolved nonbinary tensor-prime occurrence has even arity and every binary transfer belongs to the recomputed group $\mathcal{T}$.

Let $\mathcal{P}$ be the finite multiset of unresolved nonbinary tensor-prime occurrences. When a token u, carrying g_u, is selected, freeze the finite current-coordinate universe consisting of all iterated proper physical tetrahedral pair cards of g_u, all actual dressings and port permutations used by the finite reductions below, and the complete factorization tree of every output. This universe is finite: every proper contraction or factor lowers arity, $\mathcal{T}$ and the reduction list are finite, each arity has finitely many port choices and dressings, and each tensor has a finite prime-factorization tree. Write its remaining keys as Q_u and let $\varphi_u \in \mathbb{N}$ be the sum of their factor potentials from Equation (25). Define

$$\Theta_{\mathrm{Pl}}(\mathcal{P}) = \{\{(\mathrm{arity}(g_u), |Q_u|, \varphi_u) : u \in \mathcal{P}\}\}, \tag{141}$$

using lexicographic order on triples and its well-founded multiset extension.

Before applying the next induction step to u, regenerate its complete oriented current-group deck. Process each proper card and factor, either resolving it or entering it as a lower-arity occurrence with smaller Θ_{Pl}. The queue Q_u then contains only unconsumed finite reduction or certificate keys, not unresolved proper signatures. Before a proper card is returned, consume its unique originating key in Q_u and attach that key, marked completed, to the child's provenance; it is not copied into the child's unfinished key set. If the card factors, its finite factorization tree is processed under that same originating key, and every intermediate split strictly lowers the stored factor potential.

Processing u therefore replaces its triple by a finite multiset of child triples. Every proper child has smaller first coordinate; if a residual copy of u remains, it has the same arity but one fewer frozen card/factor key or smaller factor potential. Thus every inserted triple is strictly smaller than the removed parent triple in lexicographic order. Certification removes the triple altogether. The standard multiset extension therefore makes every same-coordinate local step strictly decrease Θ_{Pl}; a terminal, lower ν, or strict complete-group enlargement is instead an outer exit.

Lemma 9.23 (The quaternary case for Platonic groups)**.** *Let Λ be the current retained factor-saturated fixed-I signature set with recomputed marked group*

$$G \in \{\mathcal{T}, \mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}, \mathcal{I}, \mathcal{I}_-\}.$$

Let $q \in \Lambda$ be a nonzero tensor-prime quaternary occurrence whose complete oriented deck and all proper cards and factors have been processed as above. For each canonical pair p, put $L_p := \mathsf{R}_X \circ \Phi_p^q$. Suppose

$$L_p(K_g) = 0 \quad \text{or} \quad [L_p(K_g)] \in G \qquad (K_g \in \mathcal{K}_G^{\mathrm{act}})$$

for all six canonical pairs. Then q has exactly the factor, generalized equality, endpoint-nondegenerate eight-vertex, or outside-binary outcomes of Lemma 9.5; in particular, no tensor-prime quaternary occurrence continues in the internal tetrahedral or either icosahedral form.

Proof. The proof of Lemma 9.5 uses deck stability only to supply the displayed pair conditions and to exclude an unresolved proper card or factor. Here the pair conditions are assumed explicitly, and the preceding steps have already resolved or entered every such proper object at a smaller value of Θ_{Pl}. The two-sided finite preserver classification and all of its actual terminal gadgets therefore apply verbatim. □

Lemma 9.24 (Stable tetrahedral core augmentation)**.** *Assume Equation* (132) *with $G = \mathcal{T}$, and suppose the full-rank analysis reaches an orientation-preserving tensor f_σ, $\sigma \in \Sigma_+$, by reversible actual tetrahedral dressings and a port permutation. Then nonzero scalars α, β are recorded such that the literal augmentation*

$$\Lambda_{\mathrm{HR}} := \Lambda \cup \{\alpha \mathrm{H}_6, \beta \mathrm{RM}_8\}$$

satisfies K-Holant$(\Lambda_{\mathrm{HR}}) \equiv_{\mathrm{T}}$ K-Holant(Λ)*. Close the finite core batch and recompute its group. On an unchanged $\mathcal{T}$-branch, regenerate the complete physical deck of $\alpha\mathrm{H}_6$, retain and factor-saturate every card and its factorization tree, dispatch all outputs, and recompute the group and deck once more. Every branch is then an outer exit or has a retained residue $\Lambda^\star_{\mathrm{HR}}$ with*

$$\mathsf{Stable}(\Lambda^\star_{\mathrm{HR}}, \mathcal{T}, \alpha\mathrm{H}_6), \qquad \{\alpha\mathrm{H}_6, \beta\mathrm{RM}_8\} \subseteq \Lambda^\star_{\mathrm{HR}}.$$

Only this stable residue is passed to Theorem 9.25.

Proof. The reversible identities Equation (133) produce $\alpha\mathrm{H}_6$, and four copies followed by the forward circuit Equation (96) produce $\beta\mathrm{RM}_8$. Direct realization and literal inclusion give the displayed Turing equivalence. First preprocess and jointly factor-saturate the two cores, dispatch every terminal output, and recompute the complete group. A terminal, lower minimum arity, or strict complete-group enlargement is an outer exit. Otherwise the group remains $\mathcal{T}$ and the minimum nonbinary arity remains six.

Now regenerate the complete oriented physical deck of $\alpha\mathrm{H}_6$. Retain every card with its ordered kernel and scalar provenance, process its entire finite factorization tree, dispatch all resulting binaries and terminals, and recompute the complete group. The core H_6 is tensor-prime by Section 8.2;

its pair cards are matching products because this holds for f_σ by Lemma 9.15 and the displayed equivalence is an actual reversible $\mathcal{T}^6 \rtimes S_6$-dressing. Lemma 9.10 then makes every iterated proper physical card zero or a safe tetrahedral matching product. Thus any new singular or outside binary, lower-arity nonbinary factor, or group change is an outer exit. On the unchanged branch, regenerate the deck once more. It is complete, all of its cards and factors are retained, and every card is zero or a safe $\mathcal{T}$-matching product; hence Definition 5.7 gives the asserted stable tuple. The two cores are certified affine signatures and are excluded from every subsequent unresolved-occurrence set; their augmentation does not create a fresh fixed-(H, ν) token. □

Theorem 9.25 (Tetrahedral relative rigidity)**.** *Assume*

$$\mathsf{Stable}(\Lambda, \mathcal{T}, f), \qquad \{\alpha \mathrm{H}_6, \beta \mathrm{RM}_8\} \subseteq \Lambda, \qquad \alpha\beta \neq 0.$$

Then $\mathsf{Close}_{\{\mathscr{A}\}}(\Lambda, \mathcal{T}, f)$.

Proof. Let $\mathcal{P}$ contain every unresolved nonbinary tensor-prime occurrence of Λ; the two displayed cores are certified affine signatures, not unresolved tokens. Prepare a selected occurrence and induct on $\Theta_{\mathrm{Pl}}(\mathcal{P})$.

Arity four is Lemma 9.23. For the next step, assume the complete oriented deck and all proper cards and factors have been processed as above, and consider a six-port occurrence g. Its quaternary cards have already been closed, so every continuing one is a product of two safe tetrahedral binaries. For a canonical pair p, recompute $\rho = \operatorname{rank} \Phi^g_p$. Rank zero gives $g = 0$, rank one factors, and ranks two or three are closed by Lemma 9.7. At rank four, fixed residual matchings factor by Lemma 9.6; otherwise clauses (i)–(iii) of Theorem A.20, together with Lemmas 9.9, 9.14 and 9.15, give either the resolved endpoint-nondegenerate eight-vertex leaf or an orientation-preserving f_σ. The latter lies in $\mathscr{H}_6$ by Equation (133) and is affine by Lemma 9.16. Because g already has six ports, this rainbow localization closes no common-factor ports: f_σ differs from g only by a nonzero scalar, a port permutation, and reversible actual $\mathcal{T}^6$-dressings. Affine membership and the $\mathscr{H}_6$-orbit conclusion therefore pull back to g. Thus every continuing six-port occurrence lies in $\mathscr{M}_6 \cup \mathscr{H}_6$.

Let g have arity eight. Prepare every physical tetrahedral pair card. For a factored six-port card, process its factorization tree under the originating g-key; a continuing result consists only of safe tetrahedral binaries and hence lies in $\mathscr{M}_6$. Before a tensor-prime six-port card is entered as a child occurrence, consume its originating g-key. Replacing the parent triple by the residual eight-port triple with one fewer key and the six-port child triple is a strict multiset decrease of Θ_{Pl}. The preceding six-port base then places that child in $\mathscr{H}_6$. Therefore all pair cards lie in $\mathscr{C}_6$, and Proposition 9.17 says that g factors or is affine.

Finally let $\operatorname{arity}(g) \geq 10$. If $g \notin \mathscr{A}$, Proposition 9.22 gives a terminal, factor, strict group successor, or a nonzero nonaffine actual card of smaller arity. The first three are outer exits or decrease the current token. In the last case the parent card key is consumed before its factor-saturated child is entered, so Θ_{Pl} strictly decreases. Induction makes every child factor affine; their tensor product would then be affine, contradicting the nonaffine card. Hence g is affine.

After every retained augmentation recompute the complete group. A strict change is an outer exit; otherwise prepare the next occurrence from its regenerated current-group deck. Every continuing step removes or strictly lowers a triple of Θ_{Pl}, so every tensor-prime occurrence is eventually affine. The native edge and all twelve tetrahedral binaries are affine in this basis by Equation (7); H_6 is affine by Section 8.2, and RM_8 is affine because $RM(1, 3)$ in Equation (16) is a linear code. Equivalently their literal relationship is certified by Lemma 9.16 and Equations (96) and (133). Tensor closure, the normalization data, and Definition 2.8 give $\mathsf{Pres}_K(K\Lambda; \mathscr{A})$, proving the claimed relative closure. □

Theorem 9.26 (Tetrahedral deck rigidity)**.** *If Equation* (132) *holds with* $G = \mathcal{T}$*, then* $\mathsf{Close}_{\{\mathscr{A}\}}(\Lambda, \mathcal{T}, h)$.

Proof. Varying card matchings give an actual rainbow localization by Lemma 9.9. The bridge and orientation split then give either the resolved endpoint-nondegenerate eight-vertex leaf or an orientation-preserving tensor f_σ. Fixed matchings expose a proper factor by Lemmas 2.20 and 9.6; closing its finite factorization tree gives a resolved leaf, lower ν, or strict complete-group successor, as in the octahedral proof.

On the orientation-preserving branch, apply Lemma 9.24. Every outer-exit branch is already closed. On its sole stable $\mathcal{T}$-residue, Theorem 9.25 gives the common affine presentation, and the recorded Turing equivalence transports that conclusion back to Λ. The bridge rank throughout is the rank recomputed on the directly realized six-port signature, never an inherited ancestor rank. This proves $\mathsf{Close}_{\{\mathscr{A}\}}(\Lambda, \mathcal{T}, h)$. □

9.5 The external A_4 form

The external form has the same domain incidence as $\mathcal{T}$, but its matching deck has no rainbow line. Its four-port case follows from the two-sided matrix conditions below; at higher arity, connected domain incidence forces all cards to share one matching and hence a proper factor.

Proposition 9.27 (External tetrahedral two-sided preservers)**.** *For a projective matrix set* $\mathcal{S}$*, define its projective coefficient set by*

$$\mathcal{R}_{\mathcal{S}} := \{[c_Q(K)] : [K] \in \mathcal{S}\}.$$

Coordinate matrices act on these projective sets representative-wise. Let $\Theta_{\mathrm{ext}} := c_Q \circ \theta_{\mathrm{ext}} \circ c_Q^{-1}$*, define the Frobenius-form matrix* $J_{\mathrm{Fr}} := \mathrm{diag}(1, -1, 1, -1)$*, and define the associated adjoint map by* $A^\sharp := J_{\mathrm{Fr}} A^{\mathsf{T}} J_{\mathrm{Fr}}$*. A coordinate linear map* A *is a* two-sided preserver *of* $\mathcal{T}_{\mathrm{ext}}$ *if, for every* $[r] \in \mathcal{R}_{\mathcal{T}_{\mathrm{ext}}}$*,*

$$\begin{aligned} Ar = 0 \quad &or \quad [Ar] \in \mathcal{R}_{\mathcal{T}_{\mathrm{ext}}}, \\ A^\sharp r = 0 \quad &or \quad [A^\sharp r] \in \mathcal{R}_{\mathcal{T}_{\mathrm{ext}}}. \end{aligned} \tag{142}$$

This is representative-independent. There is one zero map; up to nonzero scalar, the nonzero maps have rank distribution $(144, 0, 0, 576)$ *in ranks* $(1, 2, 3, 4)$*, respectively. Under the Frobenius-dual reconstruction Equation* (114)*, every rank-one or rank-four map reconstructs a tensor with a binary-pair factorization.*

Proof. The adjoint is taken for the bilinear Frobenius form, without complex conjugation. Since $R_{\mathrm{ext}}^{\mathsf{T}} = R_{\mathrm{ext}}$, conjugation by R_{ext} is self-adjoint, so $\Theta_{\mathrm{ext}}^\sharp = \Theta_{\mathrm{ext}}$. Moreover,

$$[R_{\mathrm{ext}}^2] = [Q_0 + Q_1] \in N_{\mathrm{PGL}_2}(\mathcal{T}), \qquad \Theta_{\mathrm{ext}}^2(\mathcal{R}_{\mathcal{T}}) = \mathcal{R}_{\mathcal{T}}.$$

Put $A' = \Theta_{\mathrm{ext}}^{-1} A \Theta_{\mathrm{ext}}$. If $[r] \in \mathcal{R}_{\mathcal{T}}$, then $[\Theta_{\mathrm{ext}} r] \in \mathcal{R}_{\mathcal{T}_{\mathrm{ext}}}$, so the first condition for A implies the standard first condition for A'. Also

$$(A')^\sharp = \Theta_{\mathrm{ext}} A^\sharp \Theta_{\mathrm{ext}}^{-1}.$$

Because $\Theta_{\mathrm{ext}}^2(\mathcal{R}_{\mathcal{T}}) = \mathcal{R}_{\mathcal{T}}$, one has

$$\Theta_{\mathrm{ext}}^{-1}(\mathcal{R}_{\mathcal{T}}) = \Theta_{\mathrm{ext}}(\mathcal{R}_{\mathcal{T}}) = \mathcal{R}_{\mathcal{T}_{\mathrm{ext}}};$$

the second condition transports in the same way, and the converse follows by reversing the calculation. Thus the external condition is conjugate to the standard tetrahedral two-sided condition, equivalently

the A_4 row of Theorem A.28(i). Rank-one reconstruction gives a binary-pair factorization. In rank four, for some $U, V \in \mathrm{GL}_2(\mathbb{C})$, the standard map is $K \mapsto UKV$ or $K \mapsto UK^{\mathsf{T}}V$. Since $A = \Theta_{\mathrm{ext}} A' \Theta_{\mathrm{ext}}^{-1}$, transport gives, respectively,

$$K \mapsto \theta_{\mathrm{ext}}(U) K \theta_{\mathrm{ext}}(V), \qquad K \mapsto \theta_{\mathrm{ext}}(U) R_{\mathrm{ext}}^{-2} K^{\mathsf{T}} R_{\mathrm{ext}}^{2} \theta_{\mathrm{ext}}(V).$$

Either display reconstructs two crossing nonsingular binaries. These are analytic tensor identities, not actual gadgets. □

Theorem 9.28 (External tetrahedral deck rigidity)**.** *Let $(\Lambda, \mathcal{T}_{\mathrm{ext}}, f)$ be a deck-stable retained, factor-saturated state containing the distinguished literal I and $\mathrm{arity}(f) = \nu(\Lambda)$. Then an established terminal, a group-enlargement restart, or a genuine proper factor of f occurs. Thus this case has no tensor-prime survivor or common-affine leaf.*

Proof. At arity four, use the transfer-valued pair maps $L_p = \mathsf{R}_X \circ \Phi_p^f$. The exact complementary-pair adjoint identity Equations (118) and (119) gives the representative-wise two-sided condition Equation (142). By Proposition 9.27, rank zero makes $f = 0$, ranks one and four factor, and ranks two and three do not occur.

For $\mathrm{arity}(f) \geq 6$, fix a pair and let Φ be its card map. The actual kernel set $\mathcal{K}^{\mathrm{act}}_{\mathcal{T}_{\mathrm{ext}}}$ spans Mat_2, so rank zero is impossible and rank one factors. In ranks two through four, the kernel-deleted rich-line graph is connected by Theorem A.20(iii). A change of matching along an edge would give a kernel-avoiding rich domain line with three distinct images. Their matchings are pairwise distinct: if two repeated a matching, flattening across one of its edges absent from the third would give ranks $1, 1, 4$, contradicting collinearity. Apply Lemma 9.9 before using the quaternary incidence classification. Each common factor has transfer in $\mathcal{T}_{\mathrm{ext}}$, and the actual kernels span Mat_2, so actual closures preserve all three images and produce a quaternary $\mathcal{T}_{\mathrm{ext}}$-rainbow line. This contradicts Theorem A.20(i). Thus all nonzero cards use one matching M.

Now $\Psi = \Phi \circ \theta_{\mathrm{ext}}$ maps the twelve standard test points to zero or an M-matching product. Lemma 9.6 factors Ψ, and composing its varying factor with $\theta_{\mathrm{ext}}^{-1}$ pulls the tensor identity, not a gadget, back to Φ. Since $|M| \geq 2$, dual reconstruction and Lemma 2.20 expose a proper factor; its binaries give the stated terminal or restart. □

9.6 The A_5 form

Recall $\phi = (1 + \sqrt{5})/2$, $\tau = \phi^{-1} = \phi - 1$, and the standard sixty-point model $\mathcal{I}$ from Equation (9). Its axes, tetrahedral sign points, and additional even-permutation points contribute $4, 8, 48$ classes. Theorem A.27(iii) proves directly that these points are closed under multiplication, inverse, and transpose.

For the opposite display, use the fixed-equality card and network dictionary Lemma 9.2. It replaces every standard kernel by its actual κ_D-image while leaving the retained state and the supplied literal I unchanged.

Use the common rainbow normalization and separated-sign system Equations (127), (129) and (130) with $G = \mathcal{I}$.

Lemma 9.29 (Icosahedral bridge and uniform separator)**.** *The separated-sign system for normalized maps $\Phi : \mathrm{Mat}_2(\mathbb{C}) \to (\mathbb{C}^2)^{\otimes 4}$ with $G = \mathcal{I}$ has exactly six rank-deficient solutions and thirty full-rank solutions. For each full-rank solution Φ, the actual kernel*

$$K_{\mathcal{I}} = \phi Q_1 - \tau Q_2 + Q_3, \qquad [K_{\mathcal{I}}] \in \mathcal{I}$$

realizes the quaternary $q_{\mathcal{I}} := \Phi(K_{\mathcal{I}})$*, whose three flattening ranks are* $(3,3,3)$*, and the actual kernel on the canonical pair* $(1,2)$

$$L_{\mathcal{I}} = Q_0 - \tau^2 Q_1 + \tau Q_3, \qquad [L_{\mathcal{I}}] \in \mathcal{I}$$

gives the nonsingular sequential contraction $B_{\mathrm{out}} := \Phi^{q_{\mathcal{I}}}_{(1,2)}(L_{\mathcal{I}})$*, with* $[B_{\mathrm{out}}X] \notin \mathcal{I}$ *in every case. For* $\mathcal{I}_-$*, the corresponding actual kernels are*

$$K_{\mathcal{I}_-} := \kappa_D(K_{\mathcal{I}}), \qquad L_{\mathcal{I}_-} := \kappa_D(L_{\mathcal{I}}).$$

They give the covariant quaternary and a nonsingular binary whose transfer lies outside $\mathcal{I}_-$*, directly in the fixed-I coordinates.*

Proof. The exact line buckets, bridge, five frames, representative-wise thirty-case coverage, and arbitrary scale and cancellation loci are summarized in Theorem A.27(iii). The same interface supplies the uniform actual outsider. Thus Equation (12) gives the required direction; Lemmas 3.1 and 5.3 dispatches infinite order and makes finite order contradict completeness. Lemma 9.2 gives the stated direct physical certificate for the other display. □

For a directly realized six-port localization whose recomputed card-map rank is four, varying matchings and connected incidence supply a mixed line. It cannot be a mixed five-line by Theorem A.20(i), so its rainbow three-line and Lemma 9.29 give the uniform outsider. If instead all nonzero cards use one matching M, the twelve test points map to zero or an M-matching product; since $|M| = 2$, Lemmas 2.20 and 9.6 expose a proper factor.

Theorem 9.30 (Icosahedral deck rigidity)**.** *If Equation* (132) *holds and*

$$G \in \{\mathcal{I}, \mathcal{I}_-\},$$

then $\mathsf{Close}_{\varnothing}(\Lambda, G, h)$*.*

Proof. For $G = \mathcal{I}$, connected incidence and Lemmas 9.9 and 9.29 give the uniform outsider whenever card matchings vary. For $G = \mathcal{I}_-$, the same argument is direct after replacing every kernel and finite network identity by its image under Lemma 9.2. The literal equality remains unchanged.

If matchings are fixed, Lemmas 2.20 and 9.6 exposes a proper factor. Close its finite factorization tree: a nonbinary factor lowers arity, a singular or unsafe binary is terminal, and a finite safe binary whose transfer class lies outside the current group strictly enlarges that group. All-safe current-group binary factors would contradict the tensor-prime branch. Thus every branch is a resolved leaf or strict outer successor, proving the conclusion without a global termination lemma. The separator uses only the rank recomputed on the actual six-port coefficients. □

Proof of Theorem 9.3. Work at the stable tuple in the theorem. First dispatch a nonzero odd signature by Theorem 2.15, a unary by Lemma 3.4, a nullary factor by preprocessing, and every new binary by Lemmas 3.1 and 5.3. Thus a continuing branch is all-even and every binary transfer lies in the recomputed complete group.

If $G = \mathcal{T}_{\mathrm{ext}}$, apply Theorem 9.28. Its factor branch is closed by retained factor saturation. For every other Platonic form, including $\mathcal{I}_-$, use the direct coefficient interfaces already proved above. Hence assume

$$G \in \{\mathcal{T}, \mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}, \mathcal{I}, \mathcal{I}_-\}.$$

At arity four, Lemmas 3.4, 9.4 and 9.5 and Corollary 3.6 and the local-reduction conclusion following Lemma 9.8 close every output. Suppose $n := \mathrm{arity}(f) = \nu(\Lambda) \geq 6$.

Fix a canonical pair p and compute $r = \operatorname{rank} \Phi_p^f$ from the actual coefficients. Rank zero gives $f = 0$ by Equation (114). At rank one there are nonzero tensors U, F_{res} with

$$\Phi_p^f(K) = \operatorname{tr}(U^{\mathsf{T}} K) F_{\mathrm{res}}, \qquad f = U_p \otimes F_{\mathrm{res}},$$

so Lemma 2.20 gives a proper factor route. Closing that finite factorization tree either lowers ν, reaches a binary terminal or strict group successor, or expresses f entirely as safe current-group binaries, contradicting its tensor-prime choice.

Let $r \in \{2, 3, 4\}$. Deck stability gives Equation (112), so Lemma 9.11 applies. The fixed-matching route again gives a proper factor. Close its finite factorization tree by the same alternatives; an all-safe current-group binary product again contradicts tensor primality. The mixed route materializes a nonzero six-port h, using one f-vertex and exactly $(n-6)/2$ actual kernels, with

$$2 \le \operatorname{rank} \Phi_p^h \le r$$

and Equation (112). Form the literal augmentation $\Lambda_h := \Lambda \cup \{h\}$. Direct realization and literal inclusion give

$$\text{K-Holant}(\Lambda_h) \le_{\mathrm{T}} \text{K-Holant}(\Lambda), \qquad \text{K-Holant}(\Lambda) \le_{\mathrm{T}} \text{K-Holant}(\Lambda_h).$$

Preprocess and factor-saturate this entire finite batch, dispatch its newly exposed terminal outputs, and recompute the complete group. If h factors, its selected varying matchings exclude a product entirely of safe current-group binaries, so a lower-arity nonbinary factor or an earlier terminal occurs. If $n > 6$, a retained six-port prime lowers ν; a strict complete-group enlargement is likewise an outer successor. If $n = 6$, the localization has no closing kernels: the contraction C in the proof of Lemma 9.11 is the identity, and h is only the recorded port permutation and nonzero scalar multiple of f. We identify it with the same occurrence and reuse its frozen card/factor key; no fresh fixed-(H, ν) occurrence is inserted.

On the only remaining prime six-port branch with unchanged marked group, regenerate the complete oriented physical deck of h. Retain every card with its ordered kernel, scalar, and predecessor provenance; factor-saturate its complete finite factorization tree; dispatch every resulting binary and terminal; and recompute the complete group. A lower-arity factor or group change is an outer successor. On the unchanged branch regenerate the deck again. Condition Equation (112) and Lemma 9.10 make every card zero or a safe current-group matching product, and all card factors are retained. Consequently Definition 5.7 gives the stable tuple

$$\mathsf{Stable}(\Lambda_h^\star, G, h), \qquad \operatorname{arity}(h) = \nu(\Lambda_h^\star) = 6.$$

Only on this stable residue recompute $\rho_h := \operatorname{rank} \Phi_p^h$. If $\rho_h \le 3$, apply Lemma 9.7. If $\rho_h = 4$, apply, according to the recomputed marked group, Theorems 9.13, 9.26 and 9.30. In the internal tetrahedral orientation-preserving case, Lemma 9.24 first closes the literal $\mathrm{H}_6, \mathrm{RM}_8$ augmentation and re-establishes a stable $\mathcal{T}$-tuple; only then does Theorem 9.25 supply the one signature-set-wide affine presentation. Its common basis is certified by Equations (7), (16), (96) and (133) and Section 8.2.

These reductions exhaust the stable six-port residue. The octahedral and icosahedral forms leave no rigidity survivor, while the internal tetrahedral form leaves only the stated common affine class. No second prime worklist is needed: the internal relative theorem already classifies the retained set, and the other marked forms have no common-presentation leaf. Each construction records its port order and nonzero scalar, and its reduction points to the preceding retained problem. The finite bridges have constant size; localization uses $(n-6)/2$ kernels. These bounds and the local occurrence induction complete the proof. □

10 Synthesis of the Binary-Disequality Dichotomy

We now prove the two dichotomies stated in the introduction. First, Theorem 5.13 combines the four group branches into a terminating classification of an equality-accessible native-X problem. Corollary 10.2 applies that classification to the entire retained set. The final two proofs then remove the factor-saturation hypothesis and use the equality-wire equivalence to recover the ordinary binary-disequality formulation.

Every continuing signature set Λ in this section contains I, so $Z = K^{\otimes 2} I \in K(\Lambda)$. Hence Lemma 2.12 gives

$$\mathsf{Tract}(K(\Lambda)) \quad \Longleftrightarrow \quad \mathsf{Tract}_X(K(\Lambda)).$$

This equivalence identifies the branchwise criterion Tract with the equality-accessible criterion Tract_X.

10.1 Synthesis of the Equality-Accessible Matching-Deck Classification

Lemma 10.1 (Descent in the four group cases)**.** *For every deck-stable tuple (Λ, G, f), work in the fixed coordinate matrix $A = I$, put*

$$H = \overline{G}(I, \Lambda) = G(\Lambda), \qquad \mathscr{U}^{\mathrm{fac}}(I, f, H) = Q_{\mathrm{done}} \sqcup Q,$$

and freeze the displayed partition. Then the unfinished-token batches and the further reductions invoked by the active-core completion proofs in Sections 6 to 9 satisfy Definition 5.10.

Proof. Fix a deck-stable tuple (Λ, G, f) in the fixed-I coordinates and use the displayed frozen partition. By construction, every oriented one-copy proper card of f, with every assignment of frozen actual representatives, has a canonical card token in $\mathscr{U}(I, f, H)$. Factor saturation attaches exactly its finite factorization tree. Consequently, a same-(H, ν) continuation produced by a one-copy card is confined to one token of Q. Closing its stored card and full factorization tree moves that token to Q_{done}; during any intermediate genuine factor split, its stored factor potential strictly decreases.

It remains to check the reductions beyond the frozen one-copy deck. In the standard marked-dihedral branch, the active-core proof of Theorem 6.25 uses an actual tensor-prime eight-vertex output at arity four, closed for the whole retained set by Theorem 1.3. At arity six, Lemma 6.11 excludes the stable prime. At arity at least eight, disagreeing sampled cards would localize by Lemma 6.6 to a nonmatching six-port tensor, although the first proper closure and all later monomial contractions preserve matching products. Thus the samples agree; their common factor lines and the two invertible pencils give a genuine factorization of the active core, again contradicting primality. Thus the dihedral construction closes its selected occurrence by a local factorization or a terminal for the retained set.

For the $\mathcal{B}_m$ form, the local identities

$$U_{\mathcal{B}}^{\mathsf{T}} X U_{\mathcal{B}} = -2iI, \qquad U_{\mathcal{B}}^{-1} I U_{\mathcal{B}}^{-\mathsf{T}} = \frac{i}{2} X, \qquad U_{\mathcal{B}}^{-1} X U_{\mathcal{B}}^{-\mathsf{T}} = \frac{i}{2} I$$

induce an open-network bijection with the $\mathcal{M}_m$ deck. It preserves arity, factorization, ordered actual cards, and their factorization trees; hence it transports the frozen partition bijectively, and the preceding local reduction applies to the transported card and factor tokens. For $\mathcal{A}_m^0$, the literal-I identities Equations (54) and (55) and the deck relabelling in the proof of Theorem 6.30 give the same bijective transport to the $\mathcal{M}_m$ reductions, including their frozen card and factor tokens; the transport is supplied by the displayed local identities. For $\mathcal{A}_m$, the weighted kernel identities and two-card localization in Lemma 6.32 give the same active-core argument, as detailed in the proof

of Theorem 6.34: the six-port prime is excluded, and at higher arity the common factor lines and weighted pencils factor the core. Its arity-four case is the physical whole-set reduction in Lemma 6.31. Neither transport leaves an unclassified fixed-(H, ν) resource.

In the proper-cyclic branch, the C_2 boundary is the common quaternary method Theorem 5.18; its endpoint output is directly realized and sent to Corollary 4.7. For C_N, $N \geq 4$, the independent phase and dressing calculations Lemmas 7.2 and 7.3 and Equation (66) give either an actual tensor-prime endpoint-nondegenerate eight-vertex signature or a nonsingular anti-diagonal binary factor of a direct two-copy output, as in Lemma 7.4. The former is directly realized, factor-saturated, and closed by Theorem 1.3. The latter is Turing-exposed by retained factor saturation; only in that augmented set is it available as a binary signature. After the full unsafe-binary dispatch, its safe transfer class lies outside C_N and strictly enlarges the recomputed complete group, which contains the dihedral subgroup generated by the old cyclic group and this involution. Higher in the Reed–Muller argument, Lemma 7.9 gives either a proper factor or an actual Even_4; the latter is consumed by the explicit equality anchor in Lemma 3.8. Thus every cyclic multi-copy reduction yields a resolved leaf, a lower-arity factor, or a strict complete-group successor.

For the standard Pauli branch, arity four is the common quaternary method followed by Corollary 4.7. At higher arity, Theorem 8.3 either resolves the whole retained set through the real-valued dichotomy or supplies one actual quaternary batch. Its complete factor processing yields a terminal, a strict group enlargement, or a decrease of ν; it has no unresolved same-state return. In the exotic branch, arity four is the common quaternary method followed by Corollary 4.7, while the remaining finite reductions are Lemmas 8.5 and 8.8 and Proposition 8.9. Their proofs use the explicit six- and eight-port classifications and the higher-arity factorization. The tetrahedral base construction is closed by the finite arity induction in Theorem 9.25. Their certified outputs are resolved leaves, proper factors, strict successors, or named cores whose dedicated local theorem closes the invocation; none is appended as a fresh same-state key.

Finally, a Platonic localization is directly realized with its actual kernel tuple, port order, and scalar by Lemma 9.11. If it has arity below the current minimum, it lowers ν. Otherwise its card-map rank is recomputed before re-entry. Rank at most three is closed by Lemma 9.7; rank four is closed, according to the recomputed marked group, by Theorems 9.13, 9.26 and 9.30. Hence no ancestor rank label creates a new same-state resource.

We have verified the reductions using actual contractions, finite arity induction, factorization, and binary dispatch. They therefore supply the hypotheses needed for Lemma 5.11. Every genuinely multi-copy reduction has outcomes 1–3 of Definition 5.10. Any fixed-(H, ν) proper-factor continuation arising from a one-copy reduction remains within its originating unfinished token and satisfies outcome 4: completion decreases $|Q|$, while an intermediate split decreases its stored factor potential. The retained state remains in the same fixed-I coordinates, so no coordinate transport can create or reactivate a token. Thus outcome 4 has the required strict progress, proving the conditions of Definition 5.10. □

Proof of Theorem 5.13. Run the stabilized protocol of Definition 5.9 and Lemma 5.11 in the fixed-I coordinates, with the existing coordinate matrix instantiated as $A = I$. Any normalization used internally by Section 4 is pulled back before a continuing branch returns to the outer protocol. At every continuing state initialize a finite canonical occurrence worklist: one token for each current tensor-prime root, together with the factor-forest and finite card universe attached to that root. A token is removed only after its complete deck queue and factor queue have been exhausted. At each outer successor, dispatch an odd-arity member on the augmented signature set, or choose a minimum-arity nonbinary prime and reapply Lemma 5.8; an interrupted token is copied to the successor rather than silently marked closed. Thus every continuing node consists of a deck-stable

all-even tuple (Λ, H, f), together with its existing queue Q, where $H = \overline{G}(I, \Lambda) = G(\Lambda)$ is finite by Lemma 5.2. Use the previously defined measures $\Xi(I, \Lambda)$ and $\Omega(I, \Lambda, f, Q)$ from Lemma 5.11 and Equation (38). By Lemma 10.1, all four routed batch families satisfy Definition 5.10; hence Lemma 5.11 applies at every restart. Strict group enlargement lowers the first coordinate, a lower minimum arity lowers the second, fixed-(H, ν) closure decreases $|Q|$ in the third, and a genuine factor split lowers Φ_Q in the fourth. A newly exposed safe transfer class outside H is a strict-group successor, whereas a class already in H is handled within the current node and an unsafe binary is a terminal. Reductions compose toward the predecessor, common presentations restrict, and signature-set-wide Tract descends under retained inclusion by Lemma 2.21.

First suppose $\mathrm{arity}(f) = 4$. Apply Theorem 5.14; the following list is exhaustive.

1. If the dressed-kernel algebra is proper and $G = C_2$, use item 1 of Theorem 5.18. If $G = C_N$, with even $N \geq 4$, use Lemma 7.4 and Theorem 1.3.
2. If $G = \mathcal{M}_m$, $m \geq 3$, use item 2 of Theorem 5.18.
3. If $G = \mathcal{B}_m$, with $m \geq 3$, use Theorem 6.29. If $G = \mathcal{A}_m^0$, with even $m \geq 4$, use Theorem 6.30. If $G = \mathcal{A}_m$, with even $m \geq 4$, use Lemma 6.31 and Corollary 4.7.
4. For the standard Pauli or exotic Klein group, use items 3 and 4, respectively, of Theorem 5.18.
5. For $G = \mathcal{T}$, any of $\mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}$, or either of the two $\mathcal{I}$-displays, use Lemma 9.5; for $G = \mathcal{T}_{\mathrm{ext}}$, use Theorem 9.28.

These are physical quaternary interfaces in their stated signature-set-wide problems: $U_{\mathcal{B}}, U_0$, and $U_{\mathcal{A}}$ do not change the continuing outer state, and no analytic sector-coordinate eight-vertex tensor is passed directly to Corollary 4.7. A directly realized endpoint-nondegenerate tensor-prime quaternary is closed by Theorem 1.3; the reduced interface is used only while a genuine factorization or its weighted-equality successor still has to be processed. Once the finite factor/card queue is exhausted, the outcome is a terminal, a successor with smaller Ω, or $\mathsf{Tract}_X(K(\Lambda))$. Indeed, $Z = K^{\otimes 2} I \in K(\Lambda)$, so Lemma 2.12 identifies $\mathsf{Tract}_X(K(\Lambda))$ with $\mathsf{Tract}(K(\Lambda))$. Tensor primeness excludes the arity-two alternative; hence a tractable quaternary leaf has one common $\mathscr{A}$-, $\mathscr{P}$-, or $\mathscr{L}$-presentation. It is retained as a generic $\mathsf{Tract}_X(K(\Lambda))$-terminal, and no local-affine presentation is upgraded to affine.

Assume henceforth that $\mathrm{arity}(f) \geq 6$. Apply Theorem 5.17. If the kernel algebra is $\mathrm{span}\{I, X\}$, then Theorem 7.5, whose arity-four base is Lemma 7.4, gives an established exit or one common flat-Lagrangian K-presentation, which is a structured affine subcase.

It remains to consider the full matrix algebra. Apply Theorem 5.14. Its marked list is exhaustive because

$$G \triangleleft \langle G, [Z] \rangle, \qquad [\langle G, [Z] \rangle : G] \leq 2, \qquad \langle [X], [Z] \rangle \cong V_4.$$

Dispatch the resulting form by

complete marked form	relative theorem
$\mathcal{M}_m, \mathcal{B}_m$ $(m \geq 3)$; $\mathcal{A}_m^0, \mathcal{A}_m$ (even $m \geq 4$)	Theorem 6.1
standard Pauli V_4 and $\mathcal{V}_{\mathrm{ex}}$	Theorem 8.1
all marked Platonic forms	Theorem 9.3

Within the marked-dihedral row, the $\mathcal{B}_m$ branch is a whole-problem equality-wire equivalence. The $\mathcal{A}_m^0$ branch uses the literal-I whole-problem/open-network equivalence Equations (54) and (56); its two edge substitutions use only I, Y, and its final common witness is pulled back inside Theorem 6.30. For $\mathcal{A}_m$, Theorem 6.34 resolves arity four by its physical whole-set reduction and excludes every higher-arity stable prime. The sector matrix $U_{\mathcal{A}}$ expresses the weighted pencils and fusion identities

used in that proof; the realized contractions remain in the current fixed-I coordinates. Common local-affine conclusions in the $\mathcal{B}_m$ and exotic Klein branches are direct generic $\mathsf{Tract}_X(K(\Lambda))$-terminals; they are not promoted to affine presentations. By contrast, the flat-Lagrangian rigidity survivor is already a structured affine subcase.

Each dispatched theorem closes the stabilized protocol. Factor batches create outer successors only by lowering the minimum nonbinary arity or strictly enlarging the recomputed complete group, so Ξ decreases. A safe binary in G is within-node closure; an unsafe binary is terminal; and an actual safe binary outside the asserted complete group forces recomputation by Lemma 5.3. Thus no unresolved branch recurs.

Observability exhausts the two kernel algebras and marked routing exhausts the full-algebra case. Every branch therefore ends in a terminal, a direct $\mathsf{Tract}_X(K(\Lambda))$-leaf, or one common product or affine K-presentation, with the flat-Lagrangian survivor forming a structured affine subcase. The outer synthesis remains in the fixed-I coordinates; any normalization used internally by Section 4 is discharged before a continuing branch re-enters this protocol. Non-$GO(X)$ sector matrices occur only in whole-problem equivalences or as final common-presentation witnesses. By Definition 2.8, such a witness is one basis that simultaneously places the transformed edge and every signature in the retained set in the asserted class. Common product and affine presentations, including the flat-Lagrangian subcase, are clauses of $\mathsf{Tract}_X(K(\Lambda))$; common local-affine conclusions are already direct $\mathsf{Tract}_X(K(\Lambda))$-leaves. Since $Z = K^{\otimes 2}I \in K(\Lambda)$, Lemma 2.12 identifies this criterion with $\mathsf{Tract}(K(\Lambda))$. Hence the two cases in Theorem 5.13 are exhaustive. □

Corollary 10.2 (Equality-accessible signature-set-wide closure)**.** *Let Λ_0 be a retained, factor-saturated finite algebraic signature set containing the distinguished literal equality I. If $\mathsf{Tract}_X(K(\Lambda_0))$ holds, then* K-Holant$(\Lambda_0) \in \mathsf{FP}$*; otherwise* K-Holant$(\Lambda_0)$ *is $\#\mathsf{P}$-hard. Here $\mathsf{Tract}_X(K(\Lambda_0))$ is equivalent to $\mathsf{Tract}(K(\Lambda_0))$, since $Z = K^{\otimes 2}I \in K(\Lambda_0)$. The factor access used in reaching this conclusion is Turing exposure; every deck is regenerated over the current augmented signature set and its recomputed complete group.*

Proof. Because $Z = K^{\otimes 2}I \in K(\Lambda_0)$, Lemma 2.12 identifies $\mathsf{Tract}_X(K(\Lambda_0))$ with $\mathsf{Tract}(K(\Lambda_0))$. If this criterion holds, Theorem 2.19 and the fixed holographic equivalence give K-Holant$(\Lambda_0) \in \mathsf{FP}$.

Suppose the criterion fails. After the odd-arity and unsafe-binary terminals, a state with only unary and binary prime factors would satisfy the arity-two alternative and is therefore impossible on this branch. Otherwise Lemma 5.8 supplies a minimum nonbinary core. Strong induction on its arity handles every lowering successor, while a stable successor is closed by Theorem 5.13; that theorem's Ω-descent also closes all later restarts. Reductions compose toward Λ_0, tractability descends, and common presentations restrict. Thus every nonhard leaf would imply $\mathsf{Tract}(K(\Lambda_0))$, and hence $\mathsf{Tract}_X(K(\Lambda_0))$, contrary to the assumed failure. Every remaining branch is therefore $\#\mathsf{P}$-hard. □

10.2 Proofs of the Two Main Dichotomies

Corollary 10.2 gives polynomial-time evaluation when Tract_X holds and hardness otherwise. Its tractable alternatives are the arity-two tensor closure and the common affine, product, or local-affine classes; the flat-Lagrangian case lies in the affine class. We now match this criterion to the input sets of the two main theorems.

Proof of Theorem 1.2. Preprocess $\mathcal{F}$ according to Section 2.6 and put

$$\Lambda := \mathcal{F}^{\circ}.$$

Because $I \in \mathcal{F}$ is nonzero and has positive arity, $I \in \Lambda$, and the exact preprocessing gives K-Holant($\mathcal{F}$) $\equiv_{\mathrm{T}}$ K-Holant(Λ). The preprocessing convention and Lemma 2.12 give

$$\mathsf{Tract}_X(K(\mathcal{F})) \iff \mathsf{Tract}_X(K(\Lambda)) \iff \mathsf{Tract}(K(\Lambda)),$$

because $Z = K^{\otimes 2} I \in K(\Lambda)$. If this criterion holds, then Theorem 2.19, together with the fixed holographic equivalence, gives K-Holant(Λ) $\in \mathsf{FP}$.

Suppose the criterion fails. The available equality I has support on both charge signs, so Λ is not common one-sided. Apply Lemma 2.20 and put

$$\Lambda^+ = \Lambda \cup \Pi(\Lambda).$$

Then K-Holant(Λ^+) $\equiv_{\mathrm{T}}$ K-Holant(Λ), and Λ^+ is a finite retained factor-saturated set still containing I. If $\mathsf{Tract}_X(K(\Lambda^+))$ held, then Lemma 2.12 and downward inheritance would imply $\mathsf{Tract}(K(\Lambda))$, contrary to the failure above. Hence Corollary 10.2 makes K-Holant(Λ^+), and therefore K-Holant(Λ), #P-hard. Decidability follows from Theorem 2.19. The coordinate convention of the closure theorem is the fixed-I convention used here. □

Proof of Theorem 1.1. Preprocess $\mathcal{F}$ according to Section 2.6 and define

$$\Lambda := \mathcal{F}^\circ \cup \{I\}.$$

Then

$$\mathrm{Holant}(\mathcal{F}, X) \equiv_{\mathrm{T}} \mathrm{Holant}(\mathcal{F}^\circ, X) \equiv_{\mathrm{T}} \text{K-Holant}(\Lambda).$$

The first equivalence performs the exact zero/nullary evaluation, and the second is Lemma 5.1, with the literal equality already included in Λ. The set in the tractability criterion is the fixed holographic image $K(\Lambda)$. Apply Theorem 1.2; the preprocessing and wire substitutions are linear-size, and every known nonzero scalar is restored exactly. Under the preprocessing convention, its tractability condition is precisely

$$\mathsf{Tract}_X(K(\Lambda)) \iff \mathsf{Tract}_X(K(\mathcal{F} \cup \{I\})),$$

which is the criterion stated in Theorem 1.1. □

11 Conclusion

The supplied binary disequality relation yields a decidable FP-versus-#P-hard classification for every fixed finite set of algebraic complex-valued Boolean signatures. Theorems 1.1 and 1.2 express this classification using, respectively, ordinary edges with an X-vertex and native X-edges with an I-vertex. The binary anchor leaves four tractable alternatives: the arity-two tensor closure and common affine, product, or local-affine transformability.

The proof connects contractions of a smallest nonbinary prime to the finite group of available binary transfers. Arity descent, strict group enlargement, and complete factor processing reduce the problem to the four marked group analyses. Theorem 1.3 supplies a signature-set boundary without requiring literal equality, extending the single-signature eight-vertex classification [CF23] in the stated endpoint-nondegenerate, tensor-prime setting. The finite classifications in the appendix support the remaining low-arity cases; their precise scope is stated at each use.

A Finite Classifications and Explicit Contractions

This appendix supplies the finite classifications and explicit contractions used in Sections 4 to 6, 8 and 9. Each subsection specifies the tensors being classified, the equations imposed on them, and the consequence needed in the main proof. Coefficient identities and short finite classifications are given explicitly. Where a completeness claim still relies on exhaustive exact computation, its proof identifies the state space and tests, together with the companion verifier in `certificates/`. The code supplements these stated mathematical arguments and is included with the source.

In a histogram, s^m means m objects of size s. Projective representatives are normalized by their first nonzero coordinate. Rank, span, and equality tests use the exact characteristic-zero field specified in each calculation. We use the projective, tensor, affine, and algebraic conventions of Sections 2.1.3, 2.8.1, 2.8.2, 2.10.4 and 2.11.1.

A.1 Endpoint-Nondegenerate Eight-Vertex Signatures

This subsection gives the two exact finite statements used in Section 4.

A.1.1 The fourth-root odd quaternary residue

Normalize a full-support odd-parity quaternary signature by $a_j = f(e_j)$, $b_j = f(\mathbf{1} - e_j)$, and $a_0 = 1$. Its fourth-root state is

$$a_j = i^{\alpha_j}, \qquad b_j = i^{\beta_j}, \qquad (\alpha_1, \alpha_2, \alpha_3, \beta_0, \dots, \beta_3) \in \mathbb{Z}_4^7. \tag{143}$$

There are $4^7 = 16\,384$ states; write $N(z) = z\bar{z}$ for the Gaussian norm. For $i < j$, with $k < l$ the complementary pair, the six native and six equality-mediated loops are

$$L_{ij}^X = \operatorname{diag}(a_i + a_j, b_i + b_j), \qquad L_{ij}^I = \begin{pmatrix} 0 & a_l + b_k \\ a_k + b_l & 0 \end{pmatrix}. \tag{144}$$

The latter inserts one equality vertex using two native X-wires. A series equality support-flips the former:

$$\operatorname{diag}(P, Q)XI = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix}, \qquad T_{\operatorname{diag}(P,Q)XI} = \operatorname{diag}(P, Q). \tag{145}$$

Thus in either case P/Q is the eigenvalue ratio of an actual transfer.

Proposition A.1 (Deterministic twelve-loop test)**.** *For each state in Equation* (143)*, inspect the twelve ordered pairs in Equation* (144)*:*

(i) *If exactly one coordinate of some pair is zero, declare a rank-one exit.*

(ii) *Otherwise, for every pair with two nonzero coordinates P, Q, compare their Gaussian norms. If $N(P) \neq N(Q)$, declare an infinite-projective-order exit.*

(iii) *Otherwise declare the state remaining, retaining zero–zero pairs.*

The resulting disjoint partition is

$$16\,384 = 15\,792 + 224 + 368, \tag{146}$$

where the summands are respectively rank-one, infinite-order, and remaining states.

Proof. The entries lie in $\mathbb{Z}[i]$. Up to a fourth root of unity, every nonzero sum of two fourth roots is either 2 or $1+i$. Consequently two such sums have the same Gaussian norm precisely when their quotient lies in μ_4. If their norms differ, the quotient has modulus different from one and therefore cannot have finite multiplicative order. Thus, for exactly the nonzero sums occurring here, P/Q is torsion if and only if $N(P) = N(Q)$. Hence Equation (145) makes the three tests exact and disjoint; exhaustive evaluation gives Equation (146). □

Exactly 64 remaining states satisfy

$$b_j = \varepsilon a_j \quad (0 \le j \le 3), \qquad \varepsilon \in \{1, -1\}, \qquad \prod_{j=0}^{3} a_j \in \{i, -i\}. \tag{147}$$

Indeed, after the normalization $a_0 = 1$, exactly half of the 4^3 choices of (a_1, a_2, a_3) have product in $\{i, -i\}$, and ε has two choices; hence the count is $32 \cdot 2 = 64$. Join ports $0, 1$ of two copies A, B by native X-edges. Denote the resulting quaternary signature by Q. In the order (A_2, A_3, B_2, B_3), its coefficients are

x	0000	1111	$0011, 1100$	0101	1010	$0110, 1001$
$Q(x)$	$2a_0a_1$	$2b_0b_1$	$a_0b_0 + a_1b_1$	$2a_3b_2$	$2a_2b_3$	$a_2b_2 + a_3b_3$.

The contraction of two odd-parity signatures along two X-edges has even external parity, and the two endpoint values $2a_0a_1$ and $2b_0b_1$ are nonzero. Under Equation (147), exactly one of $a_0^2 + a_1^2, a_2^2 + a_3^2$ vanishes: indeed, each a_j^2 is a sign and $\prod_j a_j^2 = -1$, so exactly one of the two displayed pairs consists of opposite signs. Hence Q is a support-six eight-vertex signature.

For each other remaining state and $\delta \in \mathbb{Z}_4$, define the phase-shifted signature f_δ by

$$f_\delta(e_j) = i^{\alpha_j}, \qquad f_\delta(\mathbf{1} - e_j) = i^{\beta_j + \delta}$$

and keep $f_\delta(x) = 0$ on every even-weight input. Parameterize the odd coset by

$$x(y) = (y_0, y_1, y_2, 1 \oplus y_0 \oplus y_1 \oplus y_2), \qquad y \in \mathbb{F}_2^3.$$

Define the phase-exponent function $\varphi_\delta : \mathbb{F}_2^3 \to \mathbb{Z}_4$ by $f_\delta(x(y))/f_\delta(x(000)) = i^{\varphi_\delta(y)}$, and define its linear coefficients ℓ_j and mixed-quadratic coefficients q_{jk} by

$$\ell_j = \varphi_\delta(e_j), \qquad q_{jk} = \varphi_\delta(e_j + e_k) - \ell_j - \ell_k.$$

Then f_δ is affine if and only if

$$q_{jk} \in \{0, 2\}\ (j < k), \qquad \varphi_\delta(111) = \ell_0 + \ell_1 + \ell_2 + q_{01} + q_{02} + q_{12} \pmod 4. \tag{148}$$

Indeed, in the Boolean Möbius convention of Section 2.8.1, the expansion of φ_δ over $\mathbb{Z}_4$ has the three q_{jk} as its quadratic coefficients, while the difference between the two sides of the second congruence is its cubic coefficient. A modulo-four phase is affine exactly when the quadratic coefficients are even and that cubic coefficient vanishes. Exactly two δ's pass: $\{0, 2\}$ for 176 states and $\{1, 3\}$ for 128 states.

The complete 512-state affine odd family is

$$(a_0, a_1, a_2, a_3) = (1, A, B, C), \qquad (b_0, b_1, b_2, b_3) = (xyzABC, zBC, yAC, xAB), \tag{149}$$

where $A, B, C \in \mu_4$ and $x, y, z \in \{1, -1\}$; the $4^3 \cdot 2^3$ choices give 512 distinct normalized states. The locus having no rank-one native X-loop is the disjoint union

$$x = y = z \qquad \text{(128 states)}, \tag{150}$$

$$x, y, z \text{ not all equal}, \quad (A^2, B^2, C^2) = (xy, xz, yz) \qquad \text{(48 states)}. \tag{151}$$

The other 336 states have a rank-one loop. For the remaining 176, let G_{0j} join ports $0, j$ of two copies by corresponding native X-edges and retain the other ports in copy order. Call a resulting quaternary *dense product-transformable* when it has the full-even form in Equation (28), namely

$$g(x) = \rho r^{\mathrm{wt}(x)}(-1)^{s\cdot x}\mathbf{1}[\mathrm{wt}(x) \text{ is even}] \quad \text{for some } \rho, r \neq 0,\ s \in \mathbb{F}_2^4.$$

Then

$$\begin{aligned} &G_{01} \text{ is dense product-transformable} \iff x = y \text{ and } B^2 = C^2, \\ &G_{02} \text{ is dense product-transformable} \iff x = z \text{ and } A^2 = C^2, \\ &G_{03} \text{ is dense product-transformable} \iff y = z \text{ and } A^2 = B^2. \end{aligned} \tag{152}$$

In Equation (150) two squares agree; in Equation (151) the unique equal pair of x, y, z has the corresponding equal squares. Hence one test in Equation (152) always succeeds. These formulas are a self-contained mathematical certificate.

A.1.2 Affine endpoint normal forms

The affine classification used in Section 4 reduces to eighteen coefficient transforms and three endpoint-support types. For a nonzero affine signature with support $s + L$, choose the lexicographically first support word as origin and the lexicographically first spanning basis of L as intrinsic coordinates, and divide by the value at that origin. Define the eighteen core matrices

$$\begin{aligned} \mathscr{K}_{\mathrm{aff}} = \{I, X\} &\\ &\cup \left\{ \begin{pmatrix} 1 & r \\ 1 & -r \end{pmatrix} : r \in \mu_8 \right\} \\ &\cup \left\{ \begin{pmatrix} 1 & u(1+\varepsilon\sqrt{2}) \\ 1 & u(1-\varepsilon\sqrt{2}) \end{pmatrix} : u \in \mu_4,\ \varepsilon \in \{1, -1\} \right\}. \end{aligned} \tag{153}$$

Proposition A.2 (Exact affine endpoint-nondegenerate eight-vertex certificate)**.** *For every canonically projectively normalized nonzero arity-four affine signature $f \in \mathscr{A}$ and every $K \in \mathscr{K}_{\mathrm{aff}}$, define the transformed signature q by*

$$q := K^{\otimes 4} f.$$

Retain (K, f) precisely when

$$q(x) = 0 \quad \textit{for odd } \mathrm{wt}(x), \qquad q(\mathbf{0}^4)q(\mathbf{1}^4) \neq 0. \tag{154}$$

Then the following assertions hold.

(i) *Writing each affine support as $s + L$, there are 36 720 normalized projective affine signatures:*

$\dim L$	0	1	2	3	4
# affine supports	16	120	140	30	1
# normalized signatures	16	480	4480	15360	16384.

Among the $36\,720 \cdot 18$ *pairs* (K, f)*, exactly* 4200 *satisfy Equation* (154)*. Six cores retain* 612 *inputs each and the other twelve retain* 44 *inputs each.*

(ii) *Every retained* q *has the form*

$$q = \gamma \, \mathrm{diag}(1, t)^{\otimes 4} h, \qquad \gamma, t \in \overline{\mathbb{Q}}^{\times}, \tag{155}$$

where $h \in \mathscr{A}$*,* $h(\mathbf{0}^4) = 1$*, and*

$$\mathrm{supp}(h) \in \Big\{ \{\mathbf{0}^4, \mathbf{1}^4\},\ \{\mathbf{0}^4, \mathbf{1}^4, \mathbf{1}_I, \mathbf{1}_{I^c}\},\ \{x : \mathrm{wt}(x) \equiv 0 \pmod 2\} \Big\},$$

with $|I| = 2$*. Let* $\mathscr{R}_{\mathrm{aff}}$ *be the finite set of all such normalized affine tensors* h*. Their support-size histogram is*

$$2^4, \qquad 4^{96}, \qquad 8^{512}.$$

The three support cases form, respectively, $1, 24, 256$ *diagonal orbits. Here*

$$g \sim_{\mathrm{diag}} g' \quad \Longleftrightarrow \quad g' = \gamma \, \mathrm{diag}(1, t)^{\otimes 4} g \quad \text{for some } \gamma, t \in \overline{\mathbb{Q}}^{\times}.$$

More explicitly, fix the lexicographic order on the six weight-two words. If $v_1, \dots, v_6$ *are the corresponding central coefficients of an endpoint-nondegenerate eight-vertex signature* g*, define its diagonal-orbit invariant vector by*

$$\mathcal{J}(g) := \left(\frac{v_r v_s}{g(\mathbf{0}^4) g(\mathbf{1}^4)} \right)_{1 \le r \le s \le 6} \tag{156}$$

Then $\mathcal{J}(g)$ *is a complete invariant of these diagonal orbits.*

(iii) *For a support-four reference, let* $I \sqcup I^c = [4]$ *be its two two-port blocks and write its logical table as*

$$F = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad ABCD \neq 0.$$

Here the row bit is the common value on the I*-block, the column bit is the common value on the* I^c*-block, and*

$$A = h(\mathbf{0}^4), \qquad B = h(\mathbf{1}_{I^c}), \qquad C = h(\mathbf{1}_I), \qquad D = h(\mathbf{1}^4).$$

Then $AD/BC \in \{1, -1\}$*. If* $AD = BC$*, the signature is a tensor product of two nonsingular weighted equalities. If* $AD = -BC$*, take two labelled copies, join each of the two physical ports in the* I^c*-block of the first copy to the corresponding port of the second copy by a native* X*-edge, and retain the two* I*-blocks as the four external ports. The resulting logical table is*

$$FXF^{\mathsf{T}} = \begin{pmatrix} 2AB & AD + BC \\ AD + BC & 2CD \end{pmatrix} = \begin{pmatrix} 2AB & 0 \\ 0 & 2CD \end{pmatrix}, \tag{157}$$

and hence the physical output is a nonzero pure generalized equality on its four external ports.

(iv) *The* 512 *full-even references have the disjoint first-witness partition*

$$512 = 16 + 336 + 96 + 48 + 16. \tag{158}$$

The first 16 *have the dense product-transformable form of Equation* (28)*. The next* 336 *have a rank-one native* X*-self-loop. For each remaining state, take two labelled copies and, successively*

for $j = 1, 2, 3$, join ports $0, j$ of the first copy by crossed native X-edges to ports $j, 0$ of the second. Order the external ports as the uncontracted ports of the first copy in increasing order, followed by the corresponding ports of the second copy. The first support-four product occurs for, respectively, $96, 48, 16$ *new states. It has the form*

$$cE_r \otimes E_r, \qquad c \in \mathbb{C}^\times, \qquad E_r := \operatorname{diag}(1, r), \qquad r \in \mu_4,$$

with respect to the induced perfect matching and displayed external-port order. Here c is a nonzero scalar and E_r is the diagonal binary matrix. The histogram in the order $r = 1, -1, i, -i$ is

$$(56, 56, 24, 24).$$

(v) *Every normalized full-support fourth-root odd table p of Equation* (143) *is classified as follows. The twelve actual loops in Equation* (144) *give the partition*

$$16\,384 = 15\,792 + 224 + 368$$

of Proposition A.1. Among the 368 *remaining states, exactly* 64 *satisfy Equation* (147)*; the displayed two-copy gadget there is a support-six endpoint-nondegenerate eight-vertex signature.*

For each of the other 304 *states, let p_δ multiply the four weight-three values of p by i^δ while leaving its weight-one values fixed. Exactly two $\delta \in \mathbb{Z}_4$ make p_δ affine. The valid sets are $\{0, 2\}$ for* 176 *states and $\{1, 3\}$ for* 128 *states. Every such p_δ lies in the* 512*-state family Equation* (149)*. Of that family,* 336 *states have a rank-one native X-loop; every one of the remaining* 176 *states has at least one two-copy output of the dense product-transformable form among the three gadgets in Equation* (152)*.*

Proof. For a support $s + L$, put $d = \dim L$. In the canonical intrinsic coordinates $x = (x_1, \dots, x_d)$, the normalized phases are exactly

$$i^{\sum_{j=1}^{d} \ell_j x_j + 2 \sum_{1 \le j < k \le d} q_{jk} x_j x_k}, \qquad \ell_j \in \mathbb{Z}_4, \qquad q_{jk} \in \mathbb{F}_2.$$

There are

$$2^{4-d} \begin{bmatrix} 4 \\ d \end{bmatrix}_2 4^d 2^{\binom{d}{2}}$$

such signatures of support dimension d: the Gaussian binomial counts the linear spaces, where $\begin{bmatrix} 4 \\ d \end{bmatrix}_2 = \prod_{j=0}^{d-1} (2^4 - 2^j)/(2^d - 2^j)$ and the empty product is one. The factor 2^{4-d} counts their cosets, and the last two factors the linear and mixed phase coefficients. This proves the first table in *(i)*. Applying the eighteen matrices in Equation (153) and testing Equation (154) gives the 4200 retained pairs.

For completeness of the orbit test, normalize the first endpoint to one, write the central layer as v, and write the second endpoint as e. If $v = 0$, two states are diagonally equivalent after choosing $t^4 = e'/e$. Otherwise equality of Equation (156), using one nonzero coordinate of v, gives

$$v' = \lambda v, \qquad e' = \lambda^2 e.$$

Choosing $t^2 = \lambda$ proves Equation (155). The support and orbit histograms then follow from the exhaustive retained list.

On a support-four affine plane, choose the logical coordinates u, v used in the displayed table. After projective normalization its phase is

$$h(u, v) = i^{\ell_1 u + \ell_2 v + 2quv}, \qquad \ell_1, \ell_2 \in \mathbb{Z}_4, \quad q \in \mathbb{F}_2.$$

Therefore $AD/(BC) = (-1)^q \in \{1, -1\}$, proving the asserted cross-ratio; the literal wiring in Equation (157) is then a direct coefficient identity. For full support, the exact replay tests the six literal native self-loops and the three crossed two-copy contractions in the stated first-witness order; this gives Equation (158) and the r-histogram.

Finally it evaluates all 4^7 states of Equation (143), the twelve ordered pairs in Equation (144), the exceptional two-copy output, all four tests in Equation (148), and the three literal gadgets G_{0j}. This gives the last assertion.

The companion exact replay is `certificates/verify_affine_q4.py`. It ranges over every affine support $s + L$, every affine phase in the intrinsic coordinates, all eighteen cores, all 4^7 normalized fourth-root odd states, all twelve loops, all four phase shifts, and all three two-copy contractions listed above. The enumeration regenerates these finite spaces from their definitions and evaluates their identities over $\mathbb{Z}[i, \sqrt{2}]$ and $\mathbb{Q}(i, \sqrt{2})$. □

Remark A.3 (Use in the main proof). Proposition A.2 classifies coefficients for the eighteen displayed cores. The main proof supplies the transformability witness, reduces it to those cores, and implements the required constructions in the retained Holant problem. In particular, the support-four gadget is the literal two-copy wiring in Equation (157). The retained-list and full-support partitions still use exhaustive exact computation in `certificates/verify_affine_q4.py`; the program emits assertions, rather than a separate proof object.

Theorem A.4 (Affine endpoint normal forms)**.** *The core set used by the affine endpoint certificate is exactly the eighteen-element set $\mathscr{K}_{\mathrm{aff}}$ displayed in Equation* (153)*. Let $f \in \mathscr{A}$ be a nonzero canonically projectively normalized arity-four signature, let $K \in \mathscr{K}_{\mathrm{aff}}$, and put $q = K^{\otimes 4} f$. If*

$$q(x) = 0 \quad (\mathrm{wt}(x) \text{ odd}), \qquad q(\mathbf{0}^4) q(\mathbf{1}^4) \neq 0,$$

then there are $\gamma, t \in \overline{\mathbb{Q}}^{\times}$ and a normalized $h \in \mathscr{A}$, with $h(\mathbf{0}^4) = 1$, such that

$$q = \gamma \operatorname{diag}(1, t)^{\otimes 4} h.$$

Moreover, $\mathrm{supp}(h)$ *is exactly one of the following:*

(i) *the complementary endpoint pair $\{\mathbf{0}^4, \mathbf{1}^4\}$;*

(ii) *a four-point plane $\{\mathbf{0}^4, \mathbf{1}^4, \mathbf{1}_I, \mathbf{1}_{I^c}\}$ with $|I| = 2$;*

(iii) *the full even-parity support.*

Proof. This is the direct-use form of Proposition A.2(i)–(ii). The three supports have distinct cardinalities, so the alternatives are disjoint. □

Theorem A.5 (Gadgets from affine reference states)**.** *Let h be supplied by Theorem A.4.*

(i) *If $|\mathrm{supp}(h)| = 2$, then h is a pure generalized equality.*

(ii) *If $|\mathrm{supp}(h)| = 4$, its logical table $F = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ satisfies $AD/BC \in \{1, -1\}$. If $AD = BC$, then h genuinely factors across its two two-port blocks into nonsingular weighted equalities; if $AD = -BC$, the literal two-copy X-edge construction directly realizes a nonzero pure generalized equality.*

(iii) *If h has full even support, then either it has the dense product-transformable form of Equation* (28)*, a native X-self-loop is nonzero of rank one, or one of the three crossed two-copy contractions specified in Proposition A.2*(iv) *directly realizes $cE_r \otimes E_r$, where $c \neq 0$, $E_r = \mathrm{diag}(1, r)$, and $r \in \mu_4$.*

All constructions asserted to be direct use the literal wirings specified in this appendix.

Proof. The support-two assertion is immediate, and the other two assertions are exactly Proposition A.2(iii)–(iv). □

Theorem A.6 (Contractions of fourth-root odd signatures)**.** *Let p be a normalized full-support fourth-root odd table. Its twelve actual loops give exactly one of the following routes: a nonzero rank-one binary, a nonsingular binary of infinite projective transfer order, or a remaining table, with every zero–zero loop retained. In the remaining case, either the exceptional two-copy construction directly realizes a support-six endpoint-nondegenerate eight-vertex signature, or a certified phase shift $p_\delta \in \mathscr{A}$ has a rank-one native loop or a dense product-transformable two-copy output. These alternatives exhaust all fourth-root odd tables and include every scale and cancellation locus.*

Proof. Combine the deterministic test in Proposition A.1 with Proposition A.2(v). □

A.2 Equality-Accessible Matching Decks

This subsection gives the shared finite quaternary certificate statement used in Section 5 and by the later finite-group branches.

Proposition A.7 (Physical equality-accessible quaternary certificate)**.** *Let (Λ, G, q) be a deck-stable retained state, with Λ factor-saturated and containing the distinguished literal equality I, and $q \neq 0$ a quaternary tensor-prime core. Suppose its complete physical deck consists only of zero cards and nonzero scalar multiples of the fixed actual binary representatives of G, and suppose G is $C_2 = \{1, [X]\}$, a monomial dihedral group $\{[\mathrm{diag}(1,t)], [X\,\mathrm{diag}(1,t)] : t \in \mu_m\}$ with $m \geq 3$, the standard Pauli V_4, or the exotic group*

$$\left\{[I], [X], \left[\begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}\right], \left[\begin{pmatrix} -i & 1 \\ -1 & i \end{pmatrix}\right]\right\}.$$

Then an actual gadget using at most two copies of q produces an endpoint-nondegenerate eight-vertex signature. The assertion covers all zero cards, both port orientations, all six unordered contracted pairs, and the fixed nonzero scalars of the actual group representatives. In every case, the endpoint-nondegenerate eight-vertex signature can moreover be chosen not to be a G-matching signature.

Proof. Port dressings and reversal only permute the complete group, multiply cards by invertible factors, and contribute known nonzero scalars. It therefore suffices to check every standard group kernel on the six unordered pairs.

The C_2 case. Use Walsh coordinates only for this calculation, and define the Walsh coefficients

$$a_S = \widetilde{q}(\mathbf{1}_S), \qquad S \subseteq [4].$$

For any Walsh table indexed by subsets of [4], call it *Walsh-complement-symmetric* when $a_{\bar{S}} = a_S$ for every S, where $\bar{S} = [4] \setminus S$. For a deleted pair ij, the two actual I/X-cards satisfy

$$\widetilde{C}^{\pm}_{ij}(\mathbf{1}_T) = 2(a_T \pm a_{Tij}).$$

Here $Tij := T \cup \{i, j\}$, with T a subset of the two residual ports. The safe lines are I, Z. Singleton residual coordinates force $a_S = 0$ for odd $|S|$. For $\{i, j\} \sqcup \{k, l\} = [4]$, define the four Walsh coefficient scalars

$$u = a_\varnothing, \quad v = a_{1234}, \quad x = a_{ij}, \quad y = a_{kl}.$$

Safety for the ij-cards gives

$$(y+v)^2=(u+x)^2, \qquad (y-v)^2=(u-x)^2,$$

and safety for the complementary kl-cards gives

$$(x+v)^2=(u+y)^2, \qquad (x-v)^2=(u-y)^2.$$

Adding and subtracting these four equations gives

$$v^2=u^2, \qquad y^2=x^2, \qquad yv=ux, \qquad xv=uy.$$

If $u\neq 0$, one $\epsilon\in\{1,-1\}$ satisfies

$$v=\epsilon u, \qquad a_{\bar{S}}=\epsilon a_S.$$

When $u=v=0$, call a complementary weight-two block $S,\bar{S}$ *active* if $a_S\neq 0$ (equivalently $a_{\bar{S}}\neq 0$), and call $a_{\bar{S}}/a_S\in\{1,-1\}$ its sign. If all active blocks have one common sign, the same conclusion holds. In both cases Fourier inversion gives one physical parity and complement closure: support only on even-weight Walsh indices gives $q(x)=q(\bar{x})$, while $a_{\bar{S}}=\epsilon a_S$ makes the inverse transform vanish unless $(-1)^{\mathrm{wt}(x)}=\epsilon$. An actual X-dressing then moves a nonzero complementary physical pair to $\mathbf{0}^4,\mathbf{1}^4$. The result is an endpoint-nondegenerate eight-vertex signature and remains tensor-prime, whereas every quaternary C_2-matching signature is a two-binary product.

It remains to treat mixed signs, which necessarily require at least two active complementary blocks. Port permutations permute the three block signs, and an actual X-dressing reverses all of them. We may therefore choose a positive active block and a negative active block, make any third active block positive, and write the Walsh coefficient scalars as x,y,z, with $xz\neq 0$, such that one X-dressing and a permutation give

$$a_{12}=a_{34}=x, \qquad a_{13}=a_{24}=y, \qquad a_{14}=z, \qquad a_{23}=-z, \qquad xz\neq 0,$$

where y may vanish. For $p=12,13,14$, join the p-ports of two copies by the actual path $X-I-X$, and call the output r_p. Its physical kernel is I, its Walsh kernel is $2I$, and direct contraction gives

	common value on $0000,1111$	values on the three complementary weight-two blocks
$\widetilde{r}_{12}/4$	x^2	$(y^2+z^2,0,0)$
$\widetilde{r}_{13}/4$	y^2	$(x^2+z^2,0,0)$
$\widetilde{r}_{14}/4$	z^2	$(x^2+y^2,2xy,0)$.

Every nonzero row is supported only on even-weight Walsh indices and is Walsh-complement-symmetric. The same inverse-transform identities just used therefore make it a complement-symmetric, one-parity physical signature; an actual X-dressing makes its two endpoints nonzero. Hence it becomes an endpoint-nondegenerate eight-vertex signature.

A C_2-matching tensor whose Walsh coefficients are complement-symmetric has Walsh support on the endpoints and exactly one weight-two block, with the two common values differing by a sign. Indeed, in Walsh coordinates each actual C_2 binary factor is projectively I or Z. Their tensor product on a perfect matching has precisely that support, and Walsh-complement symmetry forces the two factor signs to agree.

If $y=0$ and $x^2+z^2\neq 0$, r_{13} has zero Walsh endpoints and is not matching. If $x^2+z^2=0$, the nonzero part of the $13\mid 24$ flattening is

$$\begin{pmatrix} x & -z \\ z & x \end{pmatrix}.$$

All other rows and columns vanish, so this matrix is the complete nonzero block and has rank one, contradicting tensor primeness. The local Walsh transforms are invertible and therefore preserve flattening rank and tensor primeness.

If $y \neq 0$ and all three outputs were matching, the first two rows would give $\sigma, \tau \in \{1, -1\}$ with

$$y^2 + z^2 = \sigma x^2, \qquad x^2 + z^2 = \tau y^2.$$

Since $2xy \neq 0$, the last row forces $x^2 + y^2 = 0$. Then the first equation gives $\sigma = 1, z^2 = 2x^2$, while the second gives $3x^2 = -\tau x^2$, impossible. Thus an actual endpoint-nondegenerate eight-vertex signature output is outside the C_2-matching products.

The monomial dihedral case. For one pair, write its four residual slices as A, B, C, D. Define the even and odd residual-card pencils produced by the standard actual kernels by

$$E(t) := A + tD, \qquad O(t) := C + tB, \qquad t \in \mu_m.$$

The pencil normal form Lemma 6.10 applies verbatim to each of these pencils. It says that the two endpoints are proportional on one monomial line, or, in one fixed parity plane with axes E_0, E_1,

$$U + tV = \lambda(E_0 + \alpha t E_1) \quad \text{or} \quad U + tV = \lambda(tE_0 + \alpha E_1), \qquad \alpha \in \mu_m.$$

Its proof uses the exact Möbius classification Lemma 6.9; in particular, a zero sampled value belongs only to the proportional branch.

Apply this fact to both pencils for all six pairs. The resulting support conditions have exactly the following solutions: choose one total parity and a nonempty subset of its four complementary word pairs. Their size histogram is

$\lvert\operatorname{supp} q\rvert$	2	4	6	8
#	8	12	8	2.

(159)

Hence the support is already one parity and complement-closed. An actual X-dressing makes the same copy an endpoint-nondegenerate eight-vertex signature.

The two Klein cases. In the Pauli Bell basis, one flattening writes

$$q = \sum_{\alpha,\beta \in \mathbb{F}_2^2} t_{\alpha\beta} B_\alpha \otimes B_\beta.$$

Its row and column cards first make $T = (t_{\alpha\beta})$ a partial monomial matrix. Intersecting the four allowed Bell-line preimages for every kernel on the other four pairs leaves exactly the four maximal four-planes

$$V_{\pi,\epsilon} = \{q : q(x) = 0 \text{ if } \operatorname{wt}(x) \not\equiv \pi \pmod 2,\ q(\bar{x}) = \epsilon q(x)\}, \quad \pi \in \mathbb{F}_2, \quad \epsilon \in \{1, -1\}. \tag{160}$$

Thus an actual Pauli X-dressing makes every nonzero member an endpoint-nondegenerate eight-vertex signature.

For $\mathcal{V}_{\text{ex}}$, define the representative matrices

$$E_0 = I, \quad E_1 = X, \quad E_2 = \begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}, \quad E_3 = \begin{pmatrix} -i & 1 \\ -1 & i \end{pmatrix}.$$

Their Frobenius duals are $E_0/2, E_1/2, E_3/4, E_2/4$, so every dual contraction is again an actual group direction. The simultaneous six-pair safe locus is the union of twenty-four maximal two-dimensional

linear subspaces. The support-stratified calculation also produces sixteen one-dimensional solution subspaces, but every one is contained in one of those twenty-four planes and hence contributes no additional component to the safe locus. Actual local exotic dressings and port permutations act transitively on the twenty-four planes. A representative is

$$q_{a,b} = a(E_0 \otimes E_0 + E_1 \otimes E_1) + b(E_2 \otimes E_2 + E_3 \otimes E_3),$$

whose table is

$$q_{a,b}(x) = \begin{cases} a - 2b, & x \in \{0000, 0110, 1001, 1111\}, \\ a + 2b, & x \in \{0011, 0101, 1010, 1100\}, \\ 0, & \mathrm{wt}(x) \text{ odd}. \end{cases} \tag{161}$$

If $a - 2b \neq 0$, this is an endpoint-nondegenerate eight-vertex signature. If $a = 2b \neq 0$, actual X-dressings on the last two ports make both endpoints $4b$. This proves the exotic case.

In each of the latter three cases, the displayed endpoint-nondegenerate eight-vertex signature is obtained from the original q only by invertible group dressings and a port permutation. It is therefore tensor-prime. Since every quaternary $G(\Lambda)$-matching signature is a tensor product of two nonsingular group binaries, these endpoint-nondegenerate eight-vertex signatures are not $G(\Lambda)$-matching. The strengthened conclusion in the C_2 case was proved separately above.

The support and subspace completeness claims in Equations (159) to (161) use the exact enumeration in `certificates/verify_equality_q4.py`. It tests all $2^{16} - 1$ nonempty computational supports in the monomial case and all 209 partial-injection supports in each Bell basis. For the latter, it intersects the allowed preimages for four cross-pairs and four kernels, retaining zero cards and reducing the equations over $\mathbb{Q}$ or $\mathbb{Q}(i)$. The preceding pencil proof supplies the coefficient constraint before this finite support classification. □

Theorem A.8 (Equality-accessible quaternary interface)**.** *Let (Λ, G, q) be a deck-stable retained, factor-saturated state containing the distinguished literal equality I, and let $q \neq 0$ be a tensor-prime quaternary core. Suppose every physical card is zero or a nonzero scalar multiple of one of the fixed actual representatives of G, where G is C_2, a monomial dihedral group of rotation order at least three, the standard Pauli Klein group, or the exotic Klein group. Then an actual gadget using at most two copies of q realizes an endpoint-nondegenerate eight-vertex signature that is not G-matching. More precisely:*

(i) *In the C_2 row, put $a_S := \widetilde{q}(\mathbf{1}_S)$. A complementary weight-two block $S, \bar{S}$ is* active *when $a_S \neq 0$, and its sign is $a_{\bar{S}}/a_S \in \{1, -1\}$. If the two Walsh endpoints $a_\varnothing, a_{[4]}$ are nonzero or all active blocks have one common sign, one dressed copy suffices. Otherwise at least two active blocks have different signs; this is the* mixed-sign *case. For $p \in \{12, 13, 14\}$, let r_p join the p-ports of two copies by the actual path $X - I - X$. One of these three contractions is endpoint-nondegenerate and nonmatching.*

(ii) *In the monomial-dihedral row, the safe support is exactly one total parity together with a nonempty subset of its four complementary word pairs. One actual X-dressing makes the same copy endpoint-nondegenerate.*

(iii) *In either Klein row, actual dressings and port permutations give the asserted tensor-prime endpoint-nondegenerate nonmatching copy.*

All clauses include zero cards, both orientations, all six contracted pairs, and the fixed nonzero scalars of the actual representatives.

Proof. The overall conclusion is Proposition A.7. Its C_2, monomial-dihedral, and two Klein proof cases give clauses (i), (ii), and (iii), respectively. The containment sentence in the exotic-Klein case makes its representative-plane argument exhaustive. □

A.3 Marked Dihedral Matching Decks

The six-port support classification below is used by the standard $\mathcal{M}_m$ branch. The sector identities in Section 6.7 transport it to $\mathcal{B}_m$ and $\mathcal{A}_m$, and Theorem 6.30 transports it to $\mathcal{A}_m^0$. We reduce its 2304 labelled parameter choices to the explicit 48-case table below.

Fix the distinguished pair $p = \{1, 2\}$ and residual ports $R = \{3, 4, 5, 6\}$. For each sector $\epsilon \in \mathbb{F}_2$, choose a perfect matching

$$M_\epsilon = \{e_\epsilon, e'_\epsilon\} \quad \text{of } R,$$

distinguish the fixed edge $e_\epsilon = \{u_\epsilon, v_\epsilon\}$, and order the other edge as $e'_\epsilon = (r_\epsilon, s_\epsilon)$ with $r_\epsilon < s_\epsilon$. This order is canonical and is not an additional choice. Choose $a_\epsilon, b_\epsilon, o_\epsilon \in \mathbb{F}_2$. Define the eight-word support $S \subseteq \mathbb{F}_2^6$ by taking, for every $\epsilon, z \in \mathbb{F}_2$, the two words satisfying

$$\begin{aligned} (x_1, x_2) &= (z, z + \epsilon), \\ x_{u_\epsilon} + x_{v_\epsilon} &= a_\epsilon, \qquad \text{over } \mathbb{F}_2. \\ (x_{r_\epsilon}, x_{s_\epsilon}) &= (o_\epsilon + z, o_\epsilon + z + b_\epsilon), \end{aligned} \tag{162}$$

This is exactly the support form obtained from four complementary flat endpoints: the fixed monomial factor supplies the two words in a slice, and the two endpoints of the variable factor are complementary.

Proposition A.9 (Exact normalized-dihedral six-to-four support lowering)**.** *Among the*

$$3^2 \cdot 2^8 = 2304$$

labelled choices in Equation (162)*, exactly* 96 *give a matching coset: for some perfect matching M and constants $c_e \in \mathbb{F}_2$, the support is*

$$\{x \in \mathbb{F}_2^6 : x_u + x_v = c_{\{u,v\}} \text{ for every } \{u, v\} \in M\}.$$

For every other choice there are a port pair $e = \{e_1, e_2\}$ and $\gamma \in \mathbb{F}_2$ such that projection away from e is injective on

$$S \cap \{x : x_{e_1} + x_{e_2} = \gamma\},$$

and the projected quaternary support has size six or eight. Choosing a largest such projection gives 1152 *support-six cases and* 1056 *support-eight cases.*

Proof. Permute the four residual ports so that the fixed edge in sector zero is 34, and its ordered varying edge is $(5, 6)$. Independently flipping bits on ports $3, 5, 6$ removes a_0, o_0, b_0, respectively: use the flips $a_0, o_0, o_0 + b_0$. Thus sector zero is

$$S_0 = \{(z, z, w, w, z, z) : z, w \in \mathbb{F}_2\}.$$

These permutations and bit flips preserve matching cosets, projection injectivity, and projected support sizes. For each of the $3 \cdot 2 \cdot 2^3 = 48$ choices of sector-zero data, fix one such normalization. It bijects the 48 choices of sector-one data with the following canonical choices: a fixed edge

$e \subseteq \{3,4,5,6\}$, the increasingly ordered complementary edge $e' = (r,s)$, and $(a,b,o) \in \mathbb{F}_2^3$. Their four words satisfy

$$(x_1,x_2) = (z, z+1), \qquad x_u = w, \quad x_v = w+a \quad (e = \{u,v\},\ u<v), \qquad (x_r,x_s) = (o+z, o+z+b).$$

The table gives a projection witness for every canonical choice. An entry ij^s_γ means: retain words with $x_i + x_j = \gamma$, delete ports i,j, and obtain s distinct projected words from exactly s retained words. Thus the projection is injective. M denotes a matching coset.

$e\backslash(a,b,o)$	000	001	010	011	100	101	110	111
34	15^8_0	25^8_0	M	M	15^8_0	25^8_0	15^8_0	16^8_0
35	16^8_0	26^8_0	24^6_1	14^6_1	14^6_0	24^6_0	26^8_0	16^8_0
36	15^8_0	25^8_0	24^6_1	14^6_1	14^6_0	24^6_0	25^8_0	15^8_0
45	16^8_0	26^8_0	23^6_1	13^6_1	13^6_0	23^6_0	26^8_0	16^8_0
46	15^8_0	25^8_0	23^6_1	13^6_1	13^6_0	23^6_0	25^8_0	15^8_0
56	13^6_0	13^6_1	15^6_0	15^6_0	13^6_0	13^6_1	13^6_0	13^6_1

Each entry follows by substituting $(z,w) \in \mathbb{F}_2^2$ in the two displayed sectors. For example, at $e = 35$, $(a,b,o) = 010$, the 24^6_1 projection leaves

$$\{0001, 0100, 0111, 1000, 1011, 1110\}$$

on ports $(1,3,5,6)$, with one preimage per word.

For maximality of the entries marked 6, an eight-word projection would require a pair with constant XOR on both sectors. The pairs with constant XOR on S_0 are $12, 15, 16, 25, 26, 34, 56$. Substitution of the sector-one equations in each 6-entry eliminates these pairs, or leaves a projection that identifies two words. A seven-word parity slice is impossible: on each four-word affine sector its size is $0, 2$, or 4. Thus the displayed six-word images are maximal. At the two M-entries the matching equations are, respectively,

$$x_3 = x_4,\ x_1 = x_5,\ x_2 = x_6; \qquad x_3 = x_4,\ x_1 = x_6,\ x_2 = x_5.$$

A parity contraction of a three-edge matching coset has at most four projected words, so none of the other entries is a matching coset. The canonical table therefore has $2, 24, 22$ cases of the three stated types. Multiplying by the 48 sector-zero choices gives

$$48(2, 24, 22) = (96, 1152, 1056),$$

as required. □

Remark A.10 (From support projection to contraction). The selected parity is implemented by an actual normalized I-kernel for $\gamma = 0$ and an X-kernel for $\gamma = 1$. Injectivity leaves one nonzero summand for each projected word, so the conclusion holds for arbitrary nonzero coefficients on the support. The main proof supplies the sector-ruling hypothesis and exposes any resulting factors by Lemma 2.20.

Theorem A.11 (Six-to-four contractions for normalized dihedral groups)**.** *Let h be a six-port signature whose support is a set S of the form Equation* (162)*, and suppose every coefficient of h on S is nonzero. Then either S is a three-edge matching coset, or there are a port pair $e = \{e_1, e_2\}$ and $\gamma \in \mathbb{F}_2$ for which projection away from e is injective on*

$$S \cap \{x : x_{e_1} + x_{e_2} = \gamma\}$$

and has image of size six or eight. In the latter case, contraction by the actual kernel I for $\gamma = 0$, or X for $\gamma = 1$, directly realizes a cancellation-free quaternary of support size six or eight. That output is not a product of two nonsingular monomial binaries.

Proof. Proposition A.9 gives the exhaustive support alternative and injectivity. The selected parity kernel leaves exactly one nonzero summand above each projected word, proving noncancellation. A product of two nonsingular monomial binaries has support size four. □

A.4 Klein-Four Matching Decks

The following certificate supplies the exotic Klein finite cases used in Section 8.

A.4.1 The exotic Klein form $\mathcal{V}_{\mathrm{ex}}$

Proposition A.12 (Exotic-Klein six- and eight-port certificate)**.** *Use the four exotic binary directions E_a, their Frobenius duals, and the label set $V = \mathbb{F}_2^2$ from Equations* (97) *and* (98)*. The following finite statements hold.*

(i) *Let $h \neq 0$ have arity six, suppose every exotic pair-card is zero or a product of two exotic binaries, and normalize one nonzero card by*

$$C_{12}^{0} = E_0^{34} E_0^{56}.$$

The other three card directions range over the complete normalized state space

$$\big(\{0\} \cup \mathfrak{D}_4(\mathcal{V}_{\mathrm{ex}})\big)^3, \qquad (1+48)^3 = 117{,}649.$$

Over arbitrary algebraic nonzero-card scales, including every cancellation locus in every other card context, exactly seventeen projective scale lines survive. Every one reconstructs a product of three exotic binaries; no tensor-prime arity-six line survives.

(ii) *Let h have arity eight and suppose the analytic argument in the main text has already placed it in the displayed normal form Equation* (102)*, with one matching and all four cards nonzero. If every one of its $28 \cdot 4 = 112$ exotic pair-cards is zero or a product of three exotic binaries, then h has a genuine proper tensor factor. The four specified cards are required to be nonzero.*

Proof. All matching lines and card maps below are generated directly from the four matrices in Equation (97). The calculation first verifies the Gram matrix, the dual basis

$$(E_0/2, E_1/2, E_3/4, E_2/4),$$

and projective fusion closure, so its contractions use the same normalized directions as the main argument.

For part (i), there are $3 \cdot 4^2 = 48$ projective quaternary matching lines. Encode zero by -1, and encode the line with matching $m \in \{0,1,2\}$ and labels $a, b \in V$ by $16m + 4a + b$. Thus a nonnegative code denotes the canonically normalized literal line $P_m(E_a, E_b)$. On each base residual edge 34 and 56, the physical stabilizer contains the paired endpoint dressings with label pairs

$$(1,1), \qquad (2,3), \qquad (3,2),$$

which fix the base E_0-edge projectively. Together with the port permutations

$$(3\,4), \qquad (5\,6), \qquad (3\,5)(4\,6),$$

these generate the physical stabilizer used here. Its induced action on the 48 residual matching lines has order 64. It partitions the full 49^3 card-direction assignments into 4,045 orbits, and the generated orbit-size histogram is

$$1^{27}, \quad 2^{49}, \quad 4^{351}, \quad 8^{521}, \quad 16^{577}, \quad 32^{1830}, \quad 64^{690}.$$

The orbit sizes weighted by these multiplicities sum to 117 649, so the generated orbits cover every state. On one representative of every orbit, exact $\mathbb{Q}(i)$-row reduction computes the union of the preimages of the forty-eight matching lines within each nonbase pair/kernel context, then intersects these unions over all 56 contexts. A zero output lies in every matching-line preimage, and nonzero-card scales remain independent linear variables; in particular, no cancellation locus is divided away.

The only surviving direction triples are

$$(-1,-1,-1), \quad (1,2,3), \quad (1,3,2), \quad (4,8,12), \quad (4,12,8).$$

The first has the single scale line (1). Each of the other four has exactly the four scale lines

$$(1, \epsilon, \delta/2, \epsilon\delta/2), \qquad \epsilon, \delta \in \{1,-1\}.$$

Thus the nonzero-card histogram consists of one line with a single nonzero card and sixteen lines with every card nonzero. An independent literal generation of all $15 \cdot 4^3 = 960$ three-binary product lines reconstructs exactly these seventeen tensors, proving part (i).

For part (ii), write the stipulated normal form as

$$h = \sum_{\gamma \in \mathbb{F}_2^2} \lambda_\gamma E^{12}_\gamma E^{34}_{A_1\gamma} E^{56}_{A_2\gamma} E^{78}_{A_3\gamma}, \qquad A_j \in \mathrm{GL}_2(\mathbb{F}_2), \quad \lambda_\gamma \neq 0.$$

Contracting $(3,5)$ and $(4,5)$ by $E^0 = I/2$ gives, respectively, label maps $D_1 = \vartheta A_1 + A_2$ and $D_2 = A_1 + A_2$, where $\vartheta \neq I$ is the transpose permutation from Equation (99). If both vanished, then $\vartheta A_1 = A_1$, contradicting invertibility of A_1. Choose $D \neq 0$ among them. The resulting six-port card has four nonzero terms with labels

$$(\gamma, D\gamma, A_3\gamma), \qquad \gamma \in \mathbb{F}_2^2.$$

The terms are distinct and all three edge coordinates vary. By Equation (100), a four-term matching product can vary on only two edges. This contradicts the assumed safe-card condition and proves part (ii). This argument applies to arbitrary nonzero λ_γ and uses only two specified actual contractions.

Every scale condition in part (i) is a finite union of homogeneous linear systems over $\mathbb{Q}(i)$. Prescribed liveness is also exact: over an infinite field, a linear space contains a vector avoiding all coordinate hyperplanes exactly when no coordinate vanishes identically on that space. Base change to $\overline{\mathbb{Q}}$, or to $\mathbb{C}$, therefore creates no additional component, and zero target cards are kept as zero-card subspaces. The exact calculation for part (i) regenerates the states, orbits, scale intersections, and factor tensors. The companion exact replay is `certificates/verify_exotic_v4_q6_q8.py`. □

Theorem A.13 (Six-port tensors for the exotic Klein group)**.** *Let $h \neq 0$ have arity six. If every exotic pair-card of h is zero or a product of two exotic binaries, then h is a product of three exotic binaries. Consequently no tensor-prime arity-six survivor exists. This includes arbitrary algebraic scales and every cancellation locus.*

Proof. This is Proposition A.12(i) after undoing the actual normalization of one nonzero card. □

Theorem A.14 (Eight-port tensors for the exotic Klein group)**.** *Let h have arity eight and satisfy the displayed normal form Equation* (102)*, with one matching and all four cards nonzero. If all* 112 *exotic pair-cards of h are zero or products of three exotic binaries, then h has a genuine proper tensor factor.*

Proof. This is Proposition A.12(ii). □

A.5 Platonic Matching Decks

All marked Platonic configurations use the quaternion basis fixed in Equation (6). Whenever a mixed three-point incidence is localized, actual dressings put its rainbow image in the common normalization

$$A = P_0(Q_0, Q_0), \qquad B = P_1(Q_0, Q_0), \qquad C = P_2(Q_2, Q_2) = A - B. \tag{163}$$

For a rich line in a quaternary matching deck, its *matching-label type* is the partition of the line size obtained by counting points on each of the three residual perfect matchings. Thus $(1,1,1)$ is a rainbow three-line, while (3), (4), and (5) are monochromatic lines. We also fix, before their first use,

$$\begin{aligned} R_0 = \tfrac{1}{2}P_2(Q_0, Q_0), \qquad R_1 = -\tfrac{1}{2}P_2(Q_1, Q_1), \qquad R_3 = -\tfrac{1}{2}P_2(Q_3, Q_3), \\ G_+ = Q_0 + Q_1 + Q_2 + Q_3, \qquad G_- = Q_0 - Q_1 - Q_2 - Q_3. \end{aligned} \tag{164}$$

Lemma A.15 (The matching-product bridge)**.** *Let S be the swap matrix, defined by $S(u\otimes v) = v\otimes u$. For $U, V \in \mathrm{GL}_2(\mathbb{C})$ and $\lambda \neq 0$,*

$$\operatorname{rank}\big((U \otimes V^{\mathsf{T}})S - \lambda I_4\big) = 1 \quad \Longleftrightarrow \quad V = \lambda U^{-\mathsf{T}}. \tag{165}$$

For $G \in \{\mathcal{T}, \mathcal{O}, \mathcal{I}\}$, put $\mathscr{C}_G = \{0\} \cup \operatorname{cone}(\mathfrak{D}_4(G))$. Then

$$\{W : W, A-2W, 2W-B \in \mathscr{C}_G\} = \{0, A/2, B/2\} \cup \left\{ \tfrac{1}{2}P_2(J, J^{-\mathsf{T}}) : [J] \in G,\ J^{\mathsf{T}} = J \right\}. \tag{166}$$

Write a symmetric group matrix as $J_u = aQ_0 + bQ_1 + dQ_3$, choosing a nonzero real coefficient vector $u = (a,b,d)$, and put $R_u = \frac{1}{2}P_2(J_u, J_u^{-\mathsf{T}})$. A triple of bridge vectors sums to $(A+B)/2$ precisely when it is a permutation of $(0, A/2, B/2)$, or is (R_u, R_v, R_w) with u, v, w pairwise orthogonal in $\mathbb{R}^3$.

Proof. Conjugation by $U^{-1} \otimes I$ reduces the rank condition to $(I \otimes W^{\mathsf{T}})S - \lambda I_4$, with $W = U^{\mathsf{T}}V = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Its matrix is

$$\begin{pmatrix} a-\lambda & 0 & c & 0 \\ b & -\lambda & d & 0 \\ 0 & a & -\lambda & c \\ 0 & b & 0 & d-\lambda \end{pmatrix}.$$

The second row has nonzero second entry. Rank one forces the first row to vanish, hence $a = \lambda, c = 0$; comparing the second and third rows then gives $b = 0, d = \lambda$. Conversely, $S - I_4$ has rank one. This proves Equation (165).

The three exceptional bridge vectors $0, A/2, B/2$ satisfy the conditions, since $C = A - B = P_2(Y, Y)$ and $[Y] \in G$. For any other solution, all three tested vectors are nonzero. A line with two points of one matching and a point of another would give flattening ranks $1, 1, 4$, which is impossible. If $W = P_0(U, V)$, the condition on $2W - B$ therefore gives matching 2. Its rank-one

flattening and Equation (165) imply $W = \frac{1}{2}P_0(U, U^{-\mathsf{T}})$. The condition on $A - 2W$ has matching 0, so $I \otimes I - U \otimes U^{-\mathsf{T}}$ is simple. The simple-tensor sum criterion forces U to be scalar, giving $W = A/2$, a contradiction. The matching-1 case similarly gives $W = B/2$.

Thus $W = P_2(U, V)$, and $A - 2W$ has matching 1. Its $13 \mid 24$ flattening is $I_4 - 2(U \otimes V^{\mathsf{T}})S$; hence $W = \frac{1}{2}P_2(U, U^{-\mathsf{T}})$. Apply the invertible local matrices $(U^{-1}, I, U^{\mathsf{T}}, I)$, which fix B. They send $2W$ to $P_2(I, U^{-\mathsf{T}}U)$. Since $2W - B$ has matching 0, Equation (165) with $\lambda = 1$ now gives $U^{-\mathsf{T}}U = I$, or $U^{\mathsf{T}} = U$. Conversely, for symmetric U, the two differences are local images of $P_1(Y, Y)$ and $-P_0(Y, Y)$. All leg matrices belong projectively to G, so both differences are in its matching cone. This proves Equation (166), including arbitrary scales.

For the sum equation, polarization of $\frac{1}{2}P_2(J_u, \overline{J}_u)$, where $\overline{J}_u = aQ_0 - bQ_1 - dQ_3$, defines an injective linear map T from the symmetric 3×3 matrices to four-port tensors with

$$T(uu^{\mathsf{T}}) = \tfrac{1}{2}P_2(J_u, \overline{J}_u), \qquad T(I_3) = (A + B)/2, \qquad R_u = T\left(\frac{uu^{\mathsf{T}}}{u^{\mathsf{T}}u}\right).$$

The six coefficient tensors are independent and $C \notin \operatorname{im} T$: in the $14 \mid 23$ binary basis they use only $Q_\mu \otimes Q_\nu$ with $\mu, \nu \in \{0, 1, 3\}$, whereas $C = Q_2 \otimes Q_2$. Also

$$A/2 = \tfrac{1}{2}T(I_3) + C/4, \qquad B/2 = \tfrac{1}{2}T(I_3) - C/4.$$

Comparison of the C-coefficient forces equal numbers of $A/2$ and $B/2$; if they occur, the third summand is zero. Otherwise the trace of $T^{-1}R_u$ is one, so all three summands are nonzero R-vectors. Their sum is $T(I_3)$ exactly when their normalized real coefficient columns form a matrix E with $EE^{\mathsf{T}} = I_3$. This is equivalent to $E^{\mathsf{T}}E = I_3$, proving the orthogonality criterion. The three orthogonal rank-one projectors are independent; together with $C \notin \operatorname{im} T$, they give card-map rank four. The exceptional triple spans A, B and gives rank two. □

A.5.1 The four fixed-equality octahedral forms $\mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}$

For the octahedral certificate, use the standard 24-point set $\mathcal{O}$ in Equations (6) and (8), ordered as the four axes, the eight sign points, and the twelve pair points. It is the quaternion model of S_4 fixed in the preliminaries. The second unmixed distinguished-involution normal form is

$$\mathcal{O}' = H_{\mathcal{O}}^{-1}\mathcal{O}H_{\mathcal{O}}. \tag{167}$$

Use the coefficient-similarity map $\theta_{\mathcal{O}}$, transformation tuple $\mathbf{H}_{\mathcal{O}}$, and induced coefficient operator $\mathcal{L}_{\mathcal{O}}$ fixed in Equations (103) and (123); the similarity induces the corresponding map on projective group classes. Here P'_m denotes the same ordered matching placement as P_m, but in the $\mathcal{O}'$ coefficient coordinates. Writing $X = Q_1$ projectively, direct ordered contraction gives

$$\begin{aligned}
\mathcal{L}_{\mathcal{O}}P_0(U, V) &\doteq P'_0(\theta_{\mathcal{O}}(U), \theta_{\mathcal{O}}(V)),\\
\mathcal{L}_{\mathcal{O}}P_1(U, V) &\doteq P'_1(\theta_{\mathcal{O}}(UX), \theta_{\mathcal{O}}(XV)),\\
\mathcal{L}_{\mathcal{O}}P_2(U, V) &\doteq P'_2(\theta_{\mathcal{O}}(U), \theta_{\mathcal{O}}(XVX)).
\end{aligned} \tag{168}$$

The operator $\mathcal{L}_{\mathcal{O}}$ compares coefficient arrays. Actual dressings use the corresponding $\theta_{\mathcal{O}}(U)$ representatives with their native-wire orientations and nonzero scalars, as in the fixed-coordinate convention of Section 9.

Order the 24 points as the four axes, the eight sign points, and the twelve pair points $Q_\mu \pm Q_\nu$, with pairs lexicographic. The coset classification in Proposition A.19 gives

domain points	three-point lines	four-point lines
24	32	18

Every point lies on seven rich lines. The three-lines and four-lines are single actual left/right orbits, represented by

$$\{[Q_0], [G_+], [G_-]\}$$

and

$$\{[Q_0], [Q_1], [Q_0 + Q_1], [Q_0 - Q_1]\},$$

respectively.

The quaternary deck has $3 \cdot 24^2 = 1728$ projective points. Its complete maximal-rich-line histogram is

line size	matching-label type	number
3	$(1,1,1)$	13 824
3	(3)	4 608
4	(4)	2 592
		21 024

There is no $(2,1)$ line and no mixed four-line. Representatives of the three rows are

$$\{P_0(Q_0,Q_0), P_1(Q_0,Q_0), P_2(Q_2,Q_2)\},$$
$$\{P_0(Q_0,Q_0), P_0(Q_0,G_+), P_0(Q_0,G_-)\},$$
$$\{P_0(Q_0,Q_0), P_0(Q_0,Q_1), P_0(Q_0,Q_0+Q_1), P_0(Q_0,Q_0-Q_1)\},$$

where G_+ and G_- are the two explicitly fixed representatives in Equation (164).

For a mixed three-point incidence, use the normalization (163). For a selected ordered frame, let Φ be the linear card map determined by $\Phi(Q_0) = C/2$ and by $(\Phi(Q_1), \Phi(Q_2), \Phi(Q_3))$ equal to that ordered frame; define the image tensors $W_j := \Phi(Q_j)$, and write $\operatorname{rank}\Phi := \dim \operatorname{im}\Phi$. Before imposing the twelve pair points of (8), the bridge is

$$\{0, A/2, B/2\} \cup \mathcal{R},$$

where the subscripts in $R^{\epsilon\delta}_{\mu\nu}$ specify the two quaternion axes and the superscripts specify the displayed signs:

$$\begin{aligned}\mathcal{R} = \{&R_0, R_1, R_3,\\ &R^{+-}_{01} = \tfrac{1}{4}P_2(Q_0+Q_1, Q_0-Q_1), \quad R^{-+}_{01} = \tfrac{1}{4}P_2(Q_0-Q_1, Q_0+Q_1),\\ &R^{+-}_{03} = \tfrac{1}{4}P_2(Q_0+Q_3, Q_0-Q_3), \quad R^{-+}_{03} = \tfrac{1}{4}P_2(Q_0-Q_3, Q_0+Q_3),\\ &R^{++}_{13} = -\tfrac{1}{4}P_2(Q_1+Q_3, Q_1+Q_3), \quad R^{--}_{13} = -\tfrac{1}{4}P_2(Q_1-Q_3, Q_1-Q_3)\}.\end{aligned}$$

By Lemma A.15, these are all bridge vectors: the symmetric octahedral directions are the three axes and the six $Q_\mu \pm Q_\nu$ directions with $\mu, \nu \in \{0,1,3\}$. An orthogonal triple is either the three axes or a pair of opposite diagonal directions in one coordinate plane together with the remaining axis. There are therefore four unordered frames. The sum equation $W_1 + W_2 + W_3 = (A+B)/2$ leaves 30 ordered maps: the six $\operatorname{rank}\Phi = 2$ maps permute $(0, A/2, B/2)$, and the other twenty-four are the six orderings of each of the four $\operatorname{rank}\Phi = 4$ frames

$$\begin{aligned}&\{R_0, R_1, R_3\}, \qquad \{R_0, R^{++}_{13}, R^{--}_{13}\},\\ &\{R_1, R^{+-}_{03}, R^{-+}_{03}\}, \qquad \{R_3, R^{+-}_{01}, R^{-+}_{01}\}.\end{aligned} \tag{169}$$

The 1,727 deck points other than $[A]$ lie on 1,683 projective lines through $[A]$; the numbers of other deck points on those lines are

$$1^{1645}, \qquad 2^{32}, \qquad 3^{6}.$$

The single actual kernel

$$K_* = Q_0 + Q_1, \qquad [K_*] \in \mathcal{O}$$

has the following property: for every ordered rank-four frame, $q_R = \Phi(K_*) = C/2 + R$ is not a matching-deck point. The 24 supports have sizes

$$2^{4}, \qquad 8^{12}, \qquad 16^{8}.$$

The second normal form is identified analytically by Equation (168) and handled physically by the actual $\mathcal{O}'$-kernels and dressings in Proposition A.16; no naive conjugation of a contraction kernel is used. Invertible local coefficient maps preserve the three ranks.

For the two mixed forms, use the coefficient similarities H_{dt}, H_{td}, groups $\mathcal{O}_{dt}, \mathcal{O}_{td}$, and operators $\mathcal{L}_{dt}, \mathcal{L}_{td}$ fixed in Equations (104), (105) and (124). The three exact transport identities there identify every matching label and absorb their displayed orientation factors into actual matrices of the current mixed group. They are used only to compare coefficient tables; neither similarity is applied to the retained signature set or to the supplied literal I.

Proposition A.16 (Physical octahedral signature-separator certificate)**.** *For every ordering of every frame in Equation* (169)*, use the actual kernel $K_* = Q_0 + Q_1$ in the $\mathcal{O}$ form and the actual kernel $K'_* := \theta_{\mathcal{O}}(K_*)$ in the $\mathcal{O}'$ form. The resulting direct quaternary can be dressed, using only actual local matrices in the current marked octahedral group, to an endpoint-nondegenerate eight-vertex signature of flattening ranks* $(2,2,2)$*. Similarity by $H_{\mathcal{O}}$ is used only to identify coefficient tables; it is never used as a local gadget.*

For $\alpha \in \{dt, td\}$, put $K_{*,\alpha} := \theta_\alpha(K_*)$, so $[K_{*,\alpha}] \in \mathcal{O}_\alpha$, and use the fixed actual kernel in that projective class, with its realization scalar recorded. The resulting direct quaternary has, using only actual current-group contractions and dressings, one of three certified exits: an endpoint-nondegenerate eight-vertex signature of ranks $(2,2,2)$*, a pure generalized equality, or a nonsingular binary B with $[BX] \notin \mathcal{O}_\alpha$. The same three-way conclusion holds for each of the six rank-deficient maps in each mixed form. All representatives, determinants, and outside-transfer directions are given in the proof below.*

Proof. Define the tetrahedral sign matrices by

$$G_{\epsilon_1\epsilon_2\epsilon_3} = Q_0 + \epsilon_1 Q_1 + \epsilon_2 Q_2 + \epsilon_3 Q_3.$$

For the first marked form the separator is $q_R = C/2 + R$, where R is the first member of the ordered frame. In the even-word order Equation (131), exact substitution gives the following actual dressings; every omitted odd-weight coefficient is zero, and the displayed vectors are projective.

R	$\mathbf{L}$	$\mathbf{e}(\mathscr{D}_{\mathbf{L}} q_R)$
R_0	$I^{\otimes 4}$	$(1,1,-1,1,1,-1,1,1)$
R_1	(I, I, Q_1, Q_1)	$(1,0,0,0,0,0,0,1)$
R_3	$I^{\otimes 4}$	$(1,1,-1,-1,-1,-1,1,1)$
R_{01}^{+-}	$G_{+++}^{\otimes 4}$	$(1,-1,1,-i,i,1,-1,1)$
R_{01}^{-+}	$G_{+++}^{\otimes 4}$	$(1,-1,1,i,-i,1,-1,1)$
R_{03}^{+-}	$I^{\otimes 4}$	$(1,1,-1,i,-i,-1,1,1)$
R_{03}^{-+}	$I^{\otimes 4}$	$(1,1,-1,-i,i,-1,1,1)$
R_{13}^{++}	$G_{++-}^{\otimes 4}$	$(1,i,-i,-i,-i,-i,i,-1)$
R_{13}^{--}	$G_{++-}^{\otimes 4}$	$(1,-i,i,i,i,i,-i,-1)$

Both endpoints are nonzero in every row. Since the separator depends only on the first frame member, these nine rows cover all twenty-four ordered maps. Direct row reduction of the three flattenings of every displayed q_R gives $(2,2,2)$.

For the second marked form, use the coefficient-transport operator $\mathcal{L}_{\mathcal{O}}$ above, and define the transported card map Φ' and transported direct card q'_R by

$$\Phi'(\theta_{\mathcal{O}}K) = \mathcal{L}_{\mathcal{O}}\Phi(K), \qquad q'_R := \Phi'(K'_*) = \mathcal{L}_{\mathcal{O}}q_R.$$

Here $[K'_*] \in \mathcal{O}'$, so q'_R is a direct card in the second marked form. With the nonzero comparison scalar $\lambda := 2\sqrt{2}(1-i)$, the identities

$$H_{\mathcal{O}}H_{\mathcal{O}}^{\mathsf{T}} = \lambda X, \qquad H_{\mathcal{O}}^{-\mathsf{T}}H_{\mathcal{O}}^{-1} = \lambda^{-1}X$$

show that $\mathcal{L}_{\mathcal{O}}$ bijects the two coefficient decks. This is only an analytic correspondence. The following port matrices are separately verified to belong projectively to the actual group $\mathcal{O}'$; in the table, $\theta_{\mathcal{O}}$ acts componentwise on every displayed tuple:

R	$\mathbf{L}'$	$\mathbf{e}(\mathscr{D}_{\mathbf{L}'}q'_R)$
R_0	$\theta_{\mathcal{O}}(G_{++-}, G_{++-}, G_{++-}, G_{+-+})$	$(1,-1,-1,-1,-1,-1,-1,1)$
R_1	$\theta_{\mathcal{O}}(I, I, I, I)$	$(1,1,1,-1,-1,1,1,1)$
R_3	$\theta_{\mathcal{O}}(G_{+++})^{\otimes 4}$	$(1,-1,1,1,1,1,-1,1)$
R_{01}^{+-}	$\theta_{\mathcal{O}}(G_{+++}, G_{--+}, G_{--+}, G_{+++})$	$(1,0,0,0,0,0,0,1)$
R_{01}^{-+}	$\theta_{\mathcal{O}}(G_{+++}, G_{+--}, G_{+--}, G_{+++})$	$(1,-1,-1,1,1,-1,-1,1)$
R_{03}^{+-}	$\theta_{\mathcal{O}}(I, I, Q_1, Q_1)$	$(1,1,-1,-1,-1,-1,1,1)$
R_{03}^{-+}	$\theta_{\mathcal{O}}(I, I, Q_2, Q_2)$	$(1,0,0,0,0,0,0,1)$
R_{13}^{++}	$\theta_{\mathcal{O}}(G_{++-}, G_{-++}, G_{++-}, G_{-++})$	$(1,-1,-1,1,1,-1,-1,1)$
R_{13}^{--}	$\theta_{\mathcal{O}}(G_{++-}, G_{---}, G_{+-+}, G_{-++})$	$(1,0,0,0,0,0,0,1)$

Again both endpoints are nonzero. All dressings are invertible, so the three certified ranks remain $(2,2,2)$.

It remains to verify the mixed rows directly in their fixed- I coordinates. For $\alpha \in \{dt, td\}$, absorb the three matching-dependent orientation factors in Equation (124) into the actual current- group representatives. The kernel $K_{*,\alpha}$ first gives the actual card $\mathcal{L}_\alpha q_R$. On ports $2,4$, apply the actual current-group dressing $\theta_\alpha(S_\alpha^{-1})$, where $S_\alpha = H_\alpha H_\alpha^{\mathsf{T}}$. Its product with the H_α^{T} leg of $\mathcal{L}_\alpha$ is H_α^{-1}, so the resulting direct card $q^\alpha_{R,\mathrm{act}}$ has the normalized representative

$$q_R^\alpha := (H_\alpha^{-1})^{\otimes 4}q_R, \qquad q^\alpha_{R,\mathrm{act}} \doteq q_R^\alpha. \tag{170}$$

This equality follows from the displayed actual dressings; it is not a state transformation. When another dressing on ports $2,4$ is listed below, compose the two group elements into one actual representative on that port. The total is at most four dressings, or two dressings and one pair closure for an outside-binary row. All exact determinants below refer to this normalized representative; the corresponding actual output differs by a known nonzero scalar, so only determinant nonvanishing is used. Put

$$\delta_\alpha(B) := [c_Q(H_\alpha(BX)H_\alpha^{-1})].$$

Thus $\delta_\alpha(B) \notin \mathcal{R}_{\mathcal{O}}$ is exactly the assertion $[BX] \notin \mathcal{O}_\alpha$.

For $\mathcal{O}_{dt}$, the three first-frame entries in the following table give endpoint-nondegenerate eight-vertex signatures. All odd coefficients vanish, and the last column is in the even-word order fixed above.

R	actual dressing	$\mathbf{e}(\mathscr{D}q_R^{dt})$
R_0	$\theta_{dt}(G_{--+})^{\otimes 4}$	$(1,-1,1,-1,-1,1,-1,1)$
R_1	$\theta_{dt}(G_{---})^{\otimes 4}$	$(1,1,-1,-1,-1,-1,1,1)$
R_3	identity	$(1,1,-1,-1,-1,-1,1,1)$

The other six possible first entries have the uniform outside-binary exits listed below; the kernel acts on ports $1,2$.

R	actual kernel	$\delta_{dt}(B)$	$\det B$
R_{01}^{+-}	$\theta_{dt}(G_{---})$	$[1,-1,1,0]$	$(51+36\sqrt{2})/64$
R_{01}^{-+}	$\theta_{dt}(G_{---})$	$[1,-1,0,-1]$	$(51+36\sqrt{2})/64$
R_{03}^{+-}	$\theta_{dt}(Q_0)$	$[1,-3,1,-1]$	$(51+36\sqrt{2})/256$
R_{03}^{-+}	$\theta_{dt}(Q_0)$	$[1,-3,-1,1]$	$(51+36\sqrt{2})/256$
R_{13}^{++}	$\theta_{dt}(Q_0)$	$[1,-3,1,1]$	$(51+36\sqrt{2})/256$
R_{13}^{--}	$\theta_{dt}(Q_0)$	$[1,-3,-1,-1]$	$(51+36\sqrt{2})/256$

For $\mathcal{O}_{td}$, no dressing is needed for the two endpoint rows:

$$\mathbf{e}(q_{R_0}^{td}) = (1,i,-i,i,i,-i,i,-1),$$
$$\mathbf{e}(q_{R_3}^{td}) = (1,i,-i,-i,-i,-i,i,-1).$$

Both have zero odd coefficients and ranks $(2,2,2)$. The R_1 row is

$$q_{R_1}^{td} = \frac{3+2\sqrt{2}}{8}(\mathbf{1}[0011] + \mathbf{1}[1100]),$$

so actual X-dressings on ports $3,4$ give $\frac{3+2\sqrt{2}}{8}\mathrm{EQ}_4$, a pure generalized equality. Compose the dressing on port 4 with its preceding correction; together with the corrections on ports $2,4$, this uses only three dressed ports. The remaining rows have the following outside-binary exits on ports $1,2$:

R	actual kernel	$\delta_{td}(B)$	$\det B$
R_{01}^{+-}	$\theta_{td}(Q_0)$	$[0,2,-1,1]$	$(51+36\sqrt{2})/256$
R_{01}^{-+}	$\theta_{td}(Q_0)$	$[0,2,-1,-1]$	$(51+36\sqrt{2})/256$
R_{03}^{+-}	$\theta_{td}(G_{--+})$	$[1,2,0,1]$	$(51+36\sqrt{2})/64$
R_{03}^{-+}	$\theta_{td}(G_{--+})$	$[1,-2,0,-1]$	$(51+36\sqrt{2})/64$
R_{13}^{++}	$\theta_{td}(Q_0)$	$[1,2,-1,0]$	$(51+36\sqrt{2})/256$
R_{13}^{--}	$\theta_{td}(Q_0)$	$[1,-2,1,0]$	$(51+36\sqrt{2})/256$

Finally let q_π^α denote the six normalized mixed reconstructed rank-deficient outputs in the same port convention as Equation (170). Put

$$K_{dt}^{\mathrm{def}} := \theta_{dt}(Q_0 - Q_2), \qquad K_{td}^{\mathrm{def}} := \theta_{td}(Q_2 - Q_3),$$

and let $B_{\pi,\alpha}^{\mathrm{def}} := \partial_{12}^{K_\alpha^{\mathrm{def}}} q_\pi^\alpha$. The fixed actual representative of $[K_\alpha^{\mathrm{def}}]$ on ports $1,2$ works uniformly for all six outputs in each row:

α	K_α^{def}	$\{\delta_\alpha(B_{\pi,\alpha}^{\mathrm{def}}) : \pi \in S_3\}$	$\det B_{\pi,\alpha}^{\mathrm{def}}$
dt	$\theta_{dt}(Q_0 - Q_2)$	$\{[1,-1,0,-1]^2, [1,-3,1,-1]^2, [1,-3,-1,-1]^2\}$	$(51+36\sqrt{2})/256$
td	$\theta_{td}(Q_2 - Q_3)$	$\{[0,2,1,1]^2, [1,2,0,1]^2, [1,2,1,0]^2\}$	$(51+36\sqrt{2})/256.$

Here $[a_0,a_1,a_2,a_3]$ denotes the projective quaternion-coordinate point $[(a_0,a_1,a_2,a_3)^{\mathsf{T}}]$, equivalently the matrix class $[a_0Q_0+a_1Q_1+a_2Q_2+a_3Q_3]$ under c_Q, and an exponent records multiplicity. Every

determinant displayed in the mixed tables is nonzero. Every pulled-back direction has support three, or support four with unequal nonzero magnitudes, whereas the quaternion-coordinate points of $\mathcal{R}_{\mathcal{O}}$ have support one, two, or four with equal nonzero magnitudes. Hence all the listed transfer classes are outside their actual mixed group. Direct row reduction gives rank pattern $(2, 2, 2)$ in every endpoint row. This proves all full-rank and rank-deficient mixed exits using only actual current-form kernels and dressings, while the similarities serve only to write the coefficient arrays. □

For the fixed-matching residue in the standard form, the four axes and eight tetrahedral sign points give the quadratic test set of Lemma 9.6. In any nonstandard form, apply the same test after precomposing Φ with the corresponding one of $\theta_{\mathcal{O}}, \theta_{dt}, \theta_{td}$, and then pull the resulting tensor identity back by its inverse. This is a coefficient pullback only, not a use of any similarity as a gadget. In all four forms, the evaluation matrix of the ten homogeneous quadratic monomials in the four quaternion coordinates has full column rank. Hence every pulled-back Segre minor vanishes identically, giving a genuine common factor rather than a new octahedral survivor.

Proposition A.19 proves the domain and deck incidences by cyclic cosets, tensor factorization, and actual group actions. Lemma A.15 proves the bridge intersections and reduces their sum equations to the displayed orthogonal frames.

A.5.2 The internal tetrahedral permutation classes

For the tetrahedral certificate, order the standard twelve points $\mathcal{T}$ in Equation (7) as the four axes followed by the eight sign points. The coset classification in Proposition A.19 gives 16 three-point lines in one actual left/right A_4-orbit. The quaternary deck has $3 \cdot 12^2 = 432$ distinct projective points. Exactly 1728 of its three-point lines meet all three matching classes, again in one orbit under actual local dressings and port permutations. Thus the common normalization above applies.

Scale the normalized card map $\Phi : \mathrm{Mat}_2(\mathbb{C}) \to (\mathbb{C}^2)^{\otimes 4}$ so that

$$\Phi(G_+) = A, \qquad \Phi(G_-) = -B.$$

Then $\Phi(Q_0) = C/2$; define the image tensors $W_j := \Phi(Q_j)$. Thus $W_1 + W_2 + W_3 = (A + B)/2$. For each $W \in \{W_1, W_2, W_3\}$, require

$$W,\ A - 2W,\ 2W - B \in \{0\} \cup \mathrm{cone}\big(\mathfrak{D}_4(\mathcal{T})\big).$$

The resulting exact bridge is

$$\mathcal{B} = \{0, A/2, B/2, R_0, R_1, R_3\}, \tag{171}$$

where R_0, R_1, R_3 are defined in Equation (164). The remaining bridge vectors obey

W	$A - 2W$	$2W - B$
R_0	$P_1(Q_2, Q_2)$	$-P_0(Q_2, Q_2)$
R_1	$-P_1(Q_3, Q_3)$	$P_0(Q_3, Q_3)$
R_3	$-P_1(Q_1, Q_1)$	$P_0(Q_1, Q_1)$

Lemma A.15 proves completeness of this list. The symmetric tetrahedral directions are exactly Q_0, Q_1, Q_3, so their three coordinate axes form the unique orthogonal frame.

The sum equation on $\mathcal{B}^3$ leaves precisely twelve maps:

$$\begin{aligned} &6 \text{ maps with } \operatorname{rank}\Phi = 2: \quad (W_1, W_2, W_3) \text{ permutes } (0, A/2, B/2), \\ &6 \text{ maps with } \operatorname{rank}\Phi = 4: \quad (W_1, W_2, W_3) \text{ permutes } (R_0, R_1, R_3). \end{aligned} \tag{172}$$

For a permutation $\sigma = (a, b, c)$ of $(0, 1, 3)$, the Frobenius-dual basis is

$$Q^0 = Q_0/2, \quad Q^1 = -Q_1/2, \quad Q^2 = Q_2/2, \quad Q^3 = -Q_3/2,$$

and the uniquely reconstructed six-port ancestor is

$$f_\sigma = \sum_{\mu=0}^{3} Q_{12}^{\mu} \otimes F_\mu^\sigma, \qquad (F_0^\sigma, F_1^\sigma, F_2^\sigma, F_3^\sigma) = (C/2, R_a, R_b, R_c).$$

The duals in this formula are coefficient-reconstruction devices, not available gadgets.

Contract all fifteen port pairs by all twelve actual tetrahedral kernels. The exact 1080-card histogram is

σ	zero or matching-deck cards	nonmatching cards
$(0,1,3), (1,3,0), (3,0,1)$, each	180	0
$(0,3,1), (1,0,3), (3,1,0)$, each	36	144

All 180 cards for each of the three orientations in the first data row are nonzero. For the three rejected orientations, the single actual contraction of ports 1 and 3 by $Q_0 = I$ gives the transparent certificate

σ	support size	three flattening ranks
$(0,3,1)$	8	$(2,2,2)$
$(1,0,3)$	16	$(2,2,2)$
$(3,1,0)$	16	$(2,2,2)$

Normalize each card by its first nonzero entry, as stipulated for the projective enumerations above. In the three σ-rows of the preceding table, respectively, one nonzero 2×2 minor in each of the flattening orders $(12 \mid 34, 13 \mid 24, 14 \mid 23)$ is

$$(-1, -1, -2), \qquad (-2, 2i, 2i), \qquad (-2i, 2, -2i).$$

Thus none of the three outputs is a hidden two-binary matching product.

Proposition A.17 (Tetrahedral permutation classes and eight-vertex signatures)**.** *Let the exceptional orientation set be*

$$\Sigma_- := \{(0,3,1), (1,0,3), (3,1,0)\}.$$

For each $\sigma \in \Sigma_-$, let q_σ be the quaternary obtained by bilinearly contracting ports $1, 3$ of f_σ with the actual kernel $Q_0 = I$. Then q_σ admits actual tetrahedral port dressings to an endpoint-nondegenerate eight-vertex signature. The three flattening ranks remain $(2, 2, 2)$.

Proof. Define the tetrahedral representative matrix $G_{\epsilon_1\epsilon_2\epsilon_3} := Q_0 + \epsilon_1 Q_1 + \epsilon_2 Q_2 + \epsilon_3 Q_3$, and use the even-word order in Equation (131) on the residual ports $(2, 4, 5, 6)$. Direct expansion gives

σ	$\mathbf{L}$	$\mathbf{e}(\mathscr{D}_{\mathbf{L}} q_\sigma)$
$(0,3,1)$	$(Q_0, G_{++-}, G_{+++}, Q_1)$	$(1,0,0,0,0,0,0,-i)$
$(1,0,3)$	(Q_0, Q_0, G_{+++}, Q_0)	$(1,1,-i,i,1,-1,i,i)$
$(3,1,0)$	$(Q_0, G_{+++}, Q_0, G_{++-})$	$(1,-1,1,1,-i,-i,i,-i)$

Every omitted odd entry is zero and both endpoints are nonzero. Each displayed dressing is actual and invertible, so it preserves the certified flattening ranks. □

For the three surviving orientations, the following coefficient formulas prove membership in the same affine class. Let $f_{013}, f_{130}, f_{301}$ be the three surviving six-port signatures and $q_{013}, q_{130}, q_{301}$ their quadratic phase polynomials, and put

$$x = (x_1, \dots, x_6) \in \mathbb{F}_2^6.$$

In the display below, the indicator constraints and their additions are over $\mathbb{F}_2$, the exponents q_σ are read modulo four, and the displayed prefactors are complex scalars:

$$\begin{aligned}
f_{013}(x) &= -\frac{i}{4}\,\mathbf{1}[x_3 + x_4 + x_5 + x_6 = 0]\, i^{q_{013}(x)},\\
q_{013} &= 2x_3 + 3x_4 + x_5 + 2(x_1x_2 + x_1x_3 + x_1x_5 + x_2x_3 + x_2x_4 + x_4x_5),\\
f_{130}(x) &= -\frac{i}{4}\,\mathbf{1}[x_3 + x_4 + x_5 + x_6 = 0]\, i^{q_{130}(x)},\\
q_{130} &= 3x_1 + x_2 + 3x_4 + x_5 + 2(x_1x_2 + x_1x_3 + x_1x_4 + x_2x_3 + x_2x_5 + x_3x_4 + x_3x_5),\\
f_{301}(x) &= \frac{1-i}{4}\,\mathbf{1}[\substack{x_3+x_4+x_5+x_6=0,\\ x_1+x_2+x_4+x_5=1}]\, i^{q_{301}(x)},\\
q_{301} &= 2 + x_1 + 2x_2 + x_5 + x_6 + 2(x_1x_2 + x_1x_3 + x_1x_4 + x_3x_4).
\end{aligned}$$

The displayed supports are affine and every mixed phase coefficient is even, so the three ancestors are affine. Separately, direct expansion of the twelve actual binary representatives KX, $[K] \in \mathcal{T}$, verifies that they are affine in this same normalized basis.

A.5.3 The icosahedral forms $\mathcal{I}$ and $\mathcal{I}_-$

For the icosahedral certificate, use the standard set $\mathcal{I}$ in Equations (6) and (9), ordered as the four axes, the eight tetrahedral sign points, and the 48 distinct even-permutation points. Exact arithmetic over $\mathbb{Q}(\phi, i)$ gives

$$|\mathcal{I}| = 60, \qquad \#\{g : \mathrm{ord}(g) = 1, 2, 3, 5\} = (1, 15, 20, 24). \tag{173}$$

The set is closed under multiplication, inverse, and transpose, and hence is the projectivization of the binary icosahedral group, hence the order-60 projective group A_5. The apparent opposite chirality gives no additional distinguished-involution case. If $\mathcal{I}_-$ is obtained by using odd coordinate permutations in the additional part of (9), then, for $D = Q_0 + Q_3$,

$$D^{\mathsf{T}} X D = 2X, \qquad \mathcal{I}_- = D\mathcal{I}D^{-1}, \qquad (DQ_1D^{-1}, DQ_2D^{-1}, DQ_3D^{-1}) = (-Q_2, Q_1, Q_3). \tag{174}$$

Lemma 9.2 uses this identity to translate kernel and card formulas while preserving the retained fixed-I problem.

The maximal-cyclic-coset classification in Proposition A.19 gives 200 three-point C_3-coset lines and 72 five-point C_5-coset lines. Every point is on ten three-lines and six five-lines. The quaternary matching-product deck has $3 \cdot 60^2 = 10\,800$ distinct projective points, and its complete maximal-rich-line histogram is

line size	matching-label type	number
3	$(1, 1, 1)$	216 000
3	(3)	72 000
5	(5)	25 920

There is no $(2,1)$ line and no mixed five-line.

Normalize a mixed three-point incidence by (163), and scale its card map so that $\Phi(G_+) = A$ and $\Phi(G_-) = -B$. Define the image tensors $W_j := \Phi(Q_j)$. Then

$$\Phi(Q_0) = C/2, \qquad W_1 + W_2 + W_3 = (A+B)/2. \tag{175}$$

The other six tetrahedral sign points require each of $W_j, A-2W_j, 2W_j-B$ to lie in the zero/matching-product cone

$$\mathscr{C}_{\mathcal{I}} := \{0\} \cup \{\lambda P_m(U,V) : \lambda \in \mathbb{C}^\times,\ m \in \{0,1,2\},\ [U],[V] \in \mathcal{I}\}.$$

Define the following 15-point slice in coefficient projective space:

$$\begin{aligned}
&\mathcal{H} = \{[a:b:d] \in \mathbb{P}^2 : [aQ_0 + bQ_1 + dQ_3] \in \mathcal{I}\}, && |\mathcal{H}| = 15,\\
&J_u = aQ_0 + bQ_1 + dQ_3, && \bar{J}_u = aQ_0 - bQ_1 - dQ_3,\\
&R_u = \frac{P_2(J_u, \bar{J}_u)}{2(a^2+b^2+d^2)}.
\end{aligned}$$

Here $u = (a,b,d)$ is any nonzero representative of $[a:b:d] \in \mathcal{H}$. The displayed quotient is unchanged when u is rescaled, so R_u is well defined by the projective coefficient point. Moreover, $a^2+b^2+d^2 = \det J_u \neq 0$: the equality follows directly from the quaternion matrices in Equation (6), and nonvanishing follows from $[J_u] \in \mathcal{I} \subset \mathrm{PGL}_2(\mathbb{C})$. Thus the displayed denominator never vanishes. The exact bridge intersection is the eighteen-point set

$$\mathcal{B} := \{W : W, A-2W, 2W-B \in \mathscr{C}_{\mathcal{I}}\} = \{0, A/2, B/2\} \cup \{R_u : [u] \in \mathcal{H}\}. \tag{176}$$

Lemma A.15 proves this equality. The symmetric icosahedral directions are precisely the three axes with no Q_2 component and the twelve additional points in that coordinate plane. For such a direction, $J_u^{-1} = \overline{J}_u/(a^2+b^2+d^2)$, which gives the displayed formula for R_u. Thus the list includes every complex scale and every cancellation case. As a finite completeness check, the lines through A determined by the other 10,799 deck points have intersection-size buckets

$$1^{10\,591}, \qquad 2^{80}, \qquad 4^{12}. \tag{177}$$

The sum equation on $\mathcal{B}^3$ has exactly 36 ordered solutions. Six permute $(0, A/2, B/2)$; the other thirty are the six orderings of the following five unordered frames, where a vector u represents the tensor R_u:

$$\begin{aligned}
\mathcal{F}_0 &= \{(0,0,1),(0,1,0),(1,0,0)\},\\
\mathcal{F}_1 &= \{(1,-\tau^2,\tau),(1,-\phi,-\phi^2),(1,\phi,-\tau)\},\\
\mathcal{F}_2 &= \{(1,-\tau^2,-\tau),(1,-\phi,\phi^2),(1,\phi,\tau)\},\\
\mathcal{F}_3 &= \{(1,-\phi,\tau),(1,\phi,\phi^2),(1,\tau^2,-\tau)\},\\
\mathcal{F}_4 &= \{(1,-\phi,-\tau),(1,\phi,-\phi^2),(1,\tau^2,\tau)\}.
\end{aligned} \tag{178}$$

Lemma A.15 reduces the nondegenerate sum solutions to orthogonal triples in $\mathcal{H}$. The five displayed frames contain the fifteen points of $\mathcal{H}$ once each. Using $\phi^2 = \phi + 1$, their dot products show that each point is orthogonal to precisely the other two points in its displayed frame. These are therefore all orthogonal triples. This proves the $6 + 5 \cdot 6 = 36$ solutions without a search through $\mathcal{B}^3$; the eighteen-point bridge is proved by the same lemma.

One legal group element suffices for all thirty orderings:

$$K_* = \phi Q_1 - \tau Q_2 + Q_3, \qquad [K_*] \in \mathcal{I}.$$

For an ordering (u_1, u_2, u_3) of any frame, its direct card is

$$q_{u_1,u_2,u_3} = \phi R_{u_1} - \tau R_{u_2} + R_{u_3}. \tag{179}$$

Exact row reduction of the thirty explicit tensors gives, uniformly,

$$(\mathrm{rank}_{12|34}, \mathrm{rank}_{13|24}, \mathrm{rank}_{14|23}) = (3, 3, 3), \tag{180}$$

with support-size histogram $8^6, 12^4, 16^{20}$. Each output is therefore a direct nonmatching quaternary card.

For the no-mixed-line branch, first discard rank at most one, which is already rank lowering. On the remaining branch $\dim \ker \Phi \le 2$. Along every rich line avoiding $\mathbb{P}(\ker \Phi)$, the absence of a mixed line forces all images to use one matching. The kernel-deleted connectedness assertion Equation (183) propagates this matching to all nonzero card images among the sixty domain points, placing them in the fixed-matching Segre variety: the projective simple-tensor locus Σ_M of Lemma 9.6, with the two matching edges as its factors. Because the icosahedral domain contains the test set in Equation (120), Lemma 9.6 then gives, after a port permutation,

$$\Phi(K) = U_* \otimes \psi(K) \quad \text{for all } K, \qquad \text{or} \qquad \Phi(K) = \psi(K) \otimes U_*. \tag{181}$$

Here U_* is a fixed nonzero binary factor and $\psi : \mathrm{Mat}_2(\mathbb{C}) \to \mathrm{Mat}_2(\mathbb{C})$ is a binary-valued linear map. Combining the rank-two solutions, (180), and (181) gives the bounded-interface trichotomy: rank lowering, an actual nonmatching quaternary, or a genuine common factor.

The group parametrization and Lemma A.15 prove completeness of the five frames. The rejecting kernel and rank assertions are evaluated on the thirty displayed outputs over $\mathbb{Q}(\phi, i)$. The common-factor alternative follows from Lemma 9.6.

Lemma A.18 (Actual rainbow-line orbit certificate)**.** *For a marked physical group G, let Γ_G be the finite permutation group of the projective deck induced by the actual local operations*

$$\mathscr{L}_{j,g} : [F] \longmapsto [(I^{\otimes(j-1)} \otimes K_g \otimes I^{\otimes(4-j)})F] \quad (g \in G,\ 1 \le j \le 4), \qquad (12), (23), (34),$$

where the first operations mean the physical attachment of an actual binary $B_g \doteq X K_g^{\mathsf{T}}$ in the orientation with effective leg map $B_g^{\mathsf{T}} X \doteq K_g$, which induces the displayed projective action on that leg and records its nonzero representative scalar; the last three are port permutations. Let the standard representative rainbow line be

$$\ell_0^{\mathrm{std}} = \{[P_0(Q_0, Q_0)], [P_1(Q_0, Q_0)], [P_2(Q_2, Q_2)]\}.$$

Write $\mathscr{S}_G$ for the displayed finite generating set (all $\mathscr{L}_{j,g}$ together with the three adjacent swaps), so that $\Gamma_G = \langle \mathscr{S}_G \rangle$. Here K_g is the fixed actual representative of g, not a coefficient-space point. For the second octahedral form, write $\theta := \theta_{\mathcal{O}}$ and use $\mathcal{L}_{\mathcal{O}}$ only to name the initial projective line

$$\ell_0^{\mathcal{O}'} = \{[P_0'(\theta(Q_0), \theta(Q_0))], [P_1'(\theta(Q_1), \theta(Q_1))], [P_2'(\theta(Q_2), \theta(Q_2))]\},$$

which is $\mathcal{L}_{\mathcal{O}}(\ell_0^{\mathrm{std}})$ projectively by the explicit matching formulas below. The operator $\mathcal{L}_{\mathcal{O}}$ is not a generator. Define $\ell_0^G := \ell_0^{\mathrm{std}}$ for $G \in \{\mathcal{T}, \mathcal{O}, \mathcal{I}\}$. The orbit of ℓ_0^G under these actual operations has the following size:

marked form	*deck points*	*all rainbow three-lines*	*orbit size*
$\mathcal{T}$	*432*	*1 728*	*1 728*
$\mathcal{O}$	*1 728*	*13 824*	*13 824*
$\mathcal{O}'$	*1 728*	*13 824*	*13 824*
$\mathcal{I}$	*10 800*	*216 000*	*216 000*
$\mathcal{T}_{\mathrm{ext}}$	*432*	*0*	*0*

For $\mathcal{O}'$, the representative in the orbit column is $\ell_0^{\mathcal{O}'}$; for the other rows with a nonzero rainbow count it is ℓ_0^{std}. Consequently every rainbow line in the first four rows is sent to that row's representative by a sequence of actual local dressings and port permutations. The corresponding scalar and ordered port permutation are part of the finite witness; no coefficient similarity is used in this assertion. For $\alpha \in \{dt, td\}$, define the initial mixed line as the $\mathcal{L}_\alpha$-image of ℓ_0^{std}. The three identities in Equation (124) *intertwine every standard orbit generator with the corresponding actual $\mathcal{O}_\alpha$-representative and preserve its nonzero scalar and port order. Hence each mixed form has the same* 1 728 *deck points and* 13 824 *rainbow three-lines, in one actual local-dressing orbit. These two rows are omitted from the table only because their three numerical entries coincide with the two displayed octahedral rows.*

Proof. First let G contain $[X]$ and be reversal-stable, so it is closed under inverse transpose. Every binary factor of a G-matching tensor has its matrix class in G, because its transfer does and $[X] \in G$. Consider any rainbow line. Actual local dressings send its matching-0 point to $A = P_0(I, I)$; write its matching-1 point as $P_1(U, V)$. The further dressing $(U^{-1}, U^{\mathsf{T}}, I, I)$ fixes A and sends that point to $P_1(I, W)$, where $W = U^{\mathsf{T}}V$.

The third point has rank one across $14 \mid 23$. Since the three points are distinct, both coefficients of its linear combination of A and $P_1(I, W)$ are nonzero. Equation (165) therefore forces W to be scalar, and the line is $\{[A], [B], [C]\}$, with $C = A - B = P_2(Y, Y)$. Thus every rainbow line has an actual local normalization to this line. It exists exactly when $[YX] = [Z] \in G$. In particular, the external form has none by Lemma 9.1.

For $G \in \{\mathcal{T}, \mathcal{O}, \mathcal{I}\}$, the local action of G^4 preserves the three matching labels. A stabilizer of the normalized line therefore fixes its three points, and the first two give

$$T_2 = T_1^{-\mathsf{T}}, \qquad T_3 = T_1^{-\mathsf{T}}, \qquad T_4 = T_1$$

projectively. Conversely, $(T, T^{-\mathsf{T}}, T^{-\mathsf{T}}, T)$ fixes all three, using $TYT^{\mathsf{T}} = \det(T)Y$. The stabilizer has $|G|$ elements, so the complete rainbow orbit has $|G|^3 = 1728, 13824, 216000$ elements, respectively. This establishes both transitivity and the counts directly. For $\mathcal{O}'$, recall that $\mathbf{H}_{\mathcal{O}}$ is the transformation tuple and $\mathcal{L}_{\mathcal{O}} = \mathscr{D}_{\mathbf{H}_{\mathcal{O}}}$ is its induced coefficient operator. The identity $H_{\mathcal{O}} H_{\mathcal{O}}^{\mathsf{T}} \doteq X$ gives a matching-dependent but explicit bijection

$$\mathcal{L}_{\mathcal{O}} P_M(U, V) \doteq P'_M(U'_M, V'_M),$$

where P'_M denotes the same ordered matching placement in the $\mathcal{O}'$ coordinates. More explicitly, writing $\theta = \theta_{\mathcal{O}}$ and $X = Q_1$ projectively, the three cases are

$$\begin{aligned}
\mathcal{L}_{\mathcal{O}} P_0(U, V) &\doteq P'_0(\theta(U), \theta(V)),\\
\mathcal{L}_{\mathcal{O}} P_1(U, V) &\doteq P'_1(\theta(UX), \theta(XV)),\\
\mathcal{L}_{\mathcal{O}} P_2(U, V) &\doteq P'_2(\theta(U), \theta(XVX)).
\end{aligned}$$

These formulas are the ordered $H_{\mathcal{O}}^{-1}/H_{\mathcal{O}}^{\mathsf{T}}$ pair contractions; the displayed X's record the fixed orientation factors. Thus this analytic map preserves matching labels and rich-line incidence. It

also conjugates the local group $\mathcal{O}^4$ to $(\mathcal{O}')^4$: on an H^{-1} leg this is $U \mapsto H^{-1}UH$, while on an H^{T} leg it is

$$H^{\mathsf{T}}UH^{-\mathsf{T}} = H^{-1}(SUS^{-1})H, \qquad S = HH^{\mathsf{T}}, \quad [S] \in \mathcal{O}.$$

The same argument applies to H_{dt}, H_{td}, with their already verified S-matrices. Hence all three nonstandard octahedral forms have the same transitive local orbit, starting at their displayed $\mathcal{L}$-image of the standard line. The operator $\mathcal{L}$ only proves this correspondence; each conjugated leg matrix belongs to the actual current group. For $\mathcal{T}_{\text{ext}}$, the preceding rank-one argument and $[Z] \notin \mathcal{T}_{\text{ext}}$ exclude every rainbow line.

To obtain an actual normalization witness, choose coset representatives as follows. In the standard forms, right multiplication by a stabilizing tuple gives each local coset a unique representative (I, T_2, T_3, T_4). Enumerate these $|G|^3$ tuples and their images; each rainbow line occurs once. The nonstandard correspondence gives the same enumeration in its actual current-group representatives. Applying the inverse recorded tuple sends any selected line to the prescribed one. The port order and all nonzero representative scalars are obtained by these exact matrix products. □

A.5.4 The complete Platonic incidence atlas

Proposition A.19 (Exact Platonic incidence atlas)**.** *For the standard tetrahedral, first octahedral, external tetrahedral, and icosahedral marked configurations, the domain and quaternary matching decks have the following complete maximal-rich-line data.*

configuration	*domain points and lines*	*deck points*	*deck lines by matching-label type*
$\mathcal{T}$	$12;\ 16_{[3]}$	432	$1728(1,1,1),\ 1152(3)$
$\mathcal{O}$	$24;\ 32_{[3]}, 18_{[4]}$	1728	$13824(1,1,1),\ 4608(3),\ 2592(4)$
$\mathcal{T}_{\text{ext}}$	$12;\ 16_{[3]}$	432	$1152(3)$
$\mathcal{I}$	$60;\ 200_{[3]}, 72_{[5]}$	10800	$216000(1,1,1),\ 72000(3),\ 25920(5)$

In the second column the entry before the semicolon is the number of domain points, and $n_{[s]}$ denotes n rich lines of size s. A parenthesized partition in the last column is the matching-label type. Thus the standard tetrahedral deck has only rainbow or monochromatic three-lines; the external tetrahedral deck has only monochromatic three-lines; and the octahedral and icosahedral decks have no type $(2,1)$ line and no mixed four- or five-line, respectively. The octahedral row also gives the abstract incidence of $\mathcal{O}' = H_{\mathcal{O}}^{-1}\mathcal{O}H_{\mathcal{O}}$ under the coefficient transport in Equation (167). Since $H_{\mathcal{O}} \notin GO(X)$, this transport is not a gadget assertion; every bridge kernel and port dressing used in the second marked form is checked separately in actual $\mathcal{O}'$-representatives. Explicitly, the coefficient operator $\mathcal{L}_{\mathcal{O}} = \mathscr{D}_{\mathbf{H}_{\mathcal{O}}}$ satisfies $\mathcal{L}_{\mathcal{O}}P_M(U,V) \doteq P'_M(U'_M, V'_M)$. Writing $\theta = \theta_{\mathcal{O}}$ and $X = Q_1$ projectively, the three explicit instances are

$$\mathcal{L}_{\mathcal{O}}P_0(U,V) \doteq P'_0(\theta(U), \theta(V)),$$
$$\mathcal{L}_{\mathcal{O}}P_1(U,V) \doteq P'_1(\theta(UX), \theta(XV)),$$
$$\mathcal{L}_{\mathcal{O}}P_2(U,V) \doteq P'_2(\theta(U), \theta(XVX)).$$

The X's are the fixed orientation factors from the ordered $H_{\mathcal{O}}^{-1}/H_{\mathcal{O}}^{\mathsf{T}}$ pair contractions, and each displayed matrix lies in the current $\mathcal{O}'$-representatives. Thus $\mathcal{L}_{\mathcal{O}}$ preserves matching labels and rich-line incidence as an analytic bijection only. The actual $\mathcal{O}'$ port witnesses are supplied by Lemma A.18 and the separate bridge tables.

For $\mathcal{O}_{dt}$ and $\mathcal{O}_{td}$, the direct transport Equation (124) likewise preserves every matching label, rich-line incidence, and kernel-deleted adjacency, while intertwining the generators with actual representatives of the current mixed form. Thus both mixed rows have the octahedral counts and

connectivity in the table. As in the preceding paragraph, the coefficient operator is only the proof dictionary; the physical witnesses are the transported actual kernels and dressings.

For $\mathcal{T}$, each domain point lies on four three-lines. For $\mathcal{O}$, each lies on seven rich lines. For $\mathcal{I}$, each lies on ten three-lines and six five-lines. In each standard marked form, every displayed domain-line family is an actual left/right group orbit. The intersection graphs of rich domain lines—vertices are rich lines and adjacency means nonempty intersection—are connected; their edge counts for $\mathcal{T}, \mathcal{O}, \mathcal{I}$ are respectively

$$72, \qquad 504, \qquad 7200. \tag{182}$$

For the localization interface, define the actual projective domain set

$$\mathcal{D}_G = \{[K_g] : g \in G\}$$

and join two distinct domain points when they lie on a common rich line. For every complex subspace $W \leq \mathrm{Mat}_2$ with $\dim W \leq 2$, the induced graph on

$$\mathcal{D}_G \setminus \mathbb{P}(W) \tag{183}$$

is nonempty and connected. The distinct deletion sets tested for $\mathcal{T}, \mathcal{O}, \mathcal{I}$ number, respectively,

$$\begin{aligned} 47 &= 1 + 12 + (18 + 16), \\ 147 &= 1 + 24 + (72 + 32 + 18), \\ 783 &= 1 + 60 + (450 + 200 + 72). \end{aligned} \tag{184}$$

Here the parenthesized summands count the distinct two-point spans by their numbers of domain points: 18 of size two and 16 of size three for $\mathcal{T}$; $72, 32, 18$ of sizes two, three, four for $\mathcal{O}$; and $450, 200, 72$ of sizes two, three, five for $\mathcal{I}$. The same assertion for $\mathcal{O}'$ and $\mathcal{T}_{\mathrm{ext}}$ is transported by their respective invertible coefficient similarities. Each $\mathcal{O}'$ rainbow line used below nevertheless has a separately checked normalization witness from actual $(\mathcal{O}')^4 \rtimes S_4$ port operations; the similarity is never installed as a gadget.

Finally, the actual kernel representatives span Mat_2. One may take

$$\begin{aligned} \mathcal{T}, \mathcal{O}, \mathcal{I} &: Q_0, Q_1, Q_2, Q_3, \\ \mathcal{O}' &: \theta_{\mathcal{O}}(Q_0), \theta_{\mathcal{O}}(Q_1), \theta_{\mathcal{O}}(Q_2), \theta_{\mathcal{O}}(Q_3), \\ \mathcal{T}_{\mathrm{ext}} &: Q_0, Q_1, Q_2 + Q_3, Q_2 - Q_3. \end{aligned} \tag{185}$$

The tetrahedral incidence counts, and the rainbow deck-line normalizations used in the octahedral and icosahedral bridges, are realized by actual independent port dressings and port permutations. For any finite configuration in $\mathbb{P}(\mathscr{V})$, and distinct points $[F_0], [F]$, define the quotient direction of $[F]$ at $[F_0]$ *to be*

$$[F + \mathrm{span}_{\mathbb{C}}\{F_0\}] \in \mathbb{P}(\mathscr{V} / \mathrm{span}_{\mathbb{C}}\{F_0\}),$$

using arbitrary nonzero representatives. This class is independent of those representatives, and two points have the same quotient direction exactly when they lie with $[F_0]$ on one projective line. For $\mathcal{T}_{\mathrm{ext}}$, actual external-group port actions are transitive on the 432 deck points and permute the three matching labels; at $P_0(Q_0, Q_0)$ the quotient-direction histogram is

$$1^{415}, \qquad 2^8, \tag{186}$$

and all eight rich lines there are monochromatic. No use of the analytic twist R is made in this deck transitivity assertion.

For completeness, at the normalized octahedral and icosahedral deck points the exact quotient-direction histograms are, respectively,

$$1^{1645},\ 2^{32},\ 3^{6}, \qquad 1^{10591},\ 2^{80},\ 4^{12}. \tag{187}$$

The first gives all octahedral rich-line types through the base point; the second gives all icosahedral rich-line types through it. Hence the global counts in the table follow by actual transitivity and incidence double counting, not by assuming an abstract group action.

Proof. We first classify every domain secant. Let ℓ contain distinct $[g], [h] \in G$. Left multiplication by g^{-1} sends it to $\mathbb{P}(\mathrm{span}\{I, T\})$, where $[T] = [g^{-1}h] \neq 1$. A nonidentity finite-order projectivity has two distinct eigenlines, so T is diagonal in their basis. Every matrix in this two-dimensional span is diagonal there. The nonsingular group points on the line form $C = G \cap \mathbb{T}$, where $\mathbb{T}$ is that diagonal torus. A finite subgroup of $\mathbb{C}^{\times}$ is cyclic, so C is cyclic. It is maximal cyclic in G: any cyclic extension containing its nonidentity element $[T]$ has the same eigenlines. Conversely, every maximal cyclic subgroup lies on its eigenline torus's projective line. Thus the domain secants are exactly the left cosets of maximal cyclic subgroups, with line size equal to the subgroup order.

The permutation cycle types of A_4, S_4, A_5 give the following numbers of maximal cyclic subgroups:

	$\lvert C\rvert = 2$	$\lvert C\rvert = 3$	$\lvert C\rvert = 4$	$\lvert C\rvert = 5$
A_4	3	4	0	0
S_4	6	4	3	0
A_5	15	10	0	6.

For example, the three double transpositions of S_4 are squares of its three cyclic order-four subgroups; only the six transpositions contribute maximal order-two subgroups. The order-three and order-five counts divide their numbers of generators by two and four, respectively. A subgroup of order d has $|G|/d$ left cosets, and distinct maximal cyclic subgroups give distinct coset sets. Hence the complete secant counts are

$$A_4 : (18, 16), \qquad S_4 : (72, 32, 18), \qquad A_5 : (450, 200, 72),$$

ordered by the nonzero subgroup-order columns above. Each point belongs to one coset of each subgroup. This also proves every stated number of rich lines through a point. Conjugation in G is transitive on the maximal cyclic subgroups of each displayed order, and left multiplication moves between their cosets, proving the asserted actual left/right line orbits.

The deck-line classification then follows structurally. A line containing three matching products either has one matching or has all three: repetition of one matching together with another would give flattening ranks $1, 1, 4$, contradicting collinearity. In the latter case, Lemmas 9.9 and A.18 gives exactly three points and the stated $|G|^3$ count in each standard form. For a line with one matching, the simple-tensor sum criterion in Lemma 9.6 implies that exactly one binary factor varies. For each domain line there are three matchings, two choices of varying factor, and $|G|$ fixed factors. Thus each monochromatic line count is $6|G|$ times its domain-line count. This proves the complete deck table from the domain table.

We next prove kernel-deleted connectivity. The domain-point graph has

$$(n, \delta, s) = (12, 8, 3),\ (24, 17, 4),\ (60, 44, 5),$$

where n is its number of vertices, δ its degree, and s the maximum line size. The degree is the sum of $t - 1$ over the rich lines of size t through a point: respectively $4 \cdot 2$, $4 \cdot 2 + 3 \cdot 3$, and $10 \cdot 2 + 6 \cdot 4$.

A subspace of dimension at most two removes $k \le s$ points. The remaining graph has $n-k>0$ vertices and minimum degree at least $\delta-k$. If it were disconnected, two components would each have at least $\delta-k+1$ vertices. But every displayed triple satisfies

$$2(\delta-k+1) > n-k \qquad (0 \le k \le s),$$

a contradiction. This proves Equation (183) without graph traversal. The deletion counts follow from the domain secant counts by adding the empty and singleton deletions, as displayed in Equation (184). Every complex subspace is covered: an intersection with at least two domain points spans its two-dimensional vector space; smaller intersections are empty or singletons. The rich-line intersection graphs are also connected: a path in the connected domain-point graph gives a sequence of rich lines with consecutive nonempty intersections. Their edge counts $72, 504, 7200$ follow by summing $\binom{d_v}{2}$ over domain points, where $d_v = 4, 7, 16$, respectively. Distinct projective lines meet in at most one point, so this counts every edge once.

The coefficient transports preserve these domain and deck incidences. Lemma A.18 supplies actual local normalization matrices. The spanning assertions in Equation (185) follow from the displayed four independent matrices in each row.

For the external form, the coefficient similarity θ_{ext} transports the twelve domain points and their sixteen three-lines. Its deck has no rainbow line by Lemma A.18. The monochromatic count is therefore $6 \cdot 12 \cdot 16 = 1152$. Actual independent port actions are transitive on each matching class, and port permutations exchange the three classes, so the full action is transitive on the 432 deck points. At each point, the two factor choices and four domain lines through each factor give eight rich three-lines. These account for 16 of the other 431 points, proving the quotient-direction histogram $1^{415}, 2^8$.

The octahedral and icosahedral quotient-direction histograms follow in the same way. Rainbow transitivity gives $|G|$ rainbow lines through each deck point, while domain incidences give the monochromatic lines. The remaining points each determine a separate two-point line. The programs `certificates/verify_a4_s4_certificates.py`, `certificates/verify_a4_external_form.py`, `certificates/verify_a5_certificate.py`, and `certificates/verify_platonic_extended.py` also reproduce the complete finite data. The coset, rank-one, and degree arguments above prove this proposition independently of those computations. □

Theorem A.20 (Incidence of Platonic kernel lines)**.** *For the marked Platonic domain and quaternary matching decks, the following statements hold.*

(i) *The four entries in each row below count, in order, rainbow three-lines and monochromatic lines of sizes three, four, and five:*

$\mathcal{T}$	$(1728, 1152, 0, 0)$
$\mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}$	$(13824, 4608, 2592, 0)$
$\mathcal{T}_{\mathrm{ext}}$	$(0, 1152, 0, 0)$
$\mathcal{I}, \mathcal{I}_-$	$(216000, 72000, 0, 25920)$.

In particular, no deck has a line of type $(2,1)$*, no octahedral deck has a mixed four-line, neither icosahedral deck has a mixed five-line, and the external tetrahedral deck has no rainbow line.*

(ii) *Every rainbow line for* $G \in \{\mathcal{T}, \mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}, \mathcal{I}, \mathcal{I}_-\}$ *lies in one orbit under actual local* G*-dressings and port permutations. A witness includes the ordered port permutation and its nonzero realization scalar; no coefficient similarity is used as a gadget.*

(iii) *Let $\mathcal{D}_G$ be the actual projective domain set for any of the marked rows above. For every complex subspace $W \le \mathrm{Mat}_2(\mathbb{C})$ with $\dim W \le 2$, the rich-line domain graph induced on $\mathcal{D}_G \setminus \mathbb{P}(W)$ is nonempty and connected.*

Proof. The complete line counts and kernel-deleted connectivity are Proposition A.19; actual rainbow transitivity, including the separate $\mathcal{O}'$ witnesses, is Lemma A.18. For the mixed rows, Equation (124) gives the same counts and intertwines the standard orbit generator-by-generator with actual current-form matrices. The fixed-equality dictionary Lemma 9.2 gives both icosahedral statements in $\mathcal{I}_-$, with actual κ_D-kernels and nonzero scalars, without changing the retained literal I. □

Remark A.21 (Computational completeness and physical interpretation). The finite preserver and hereditary eight-port completeness claims retain an exact-computation component over the stated number fields. The bridge classifications have the structural proof in Lemma A.15. The incidence classification and deletion connectivity have the structural proof in Proposition A.19. The companion programs regenerate the configurations and test the stated predicates; they emit assertions rather than independent proof objects. The main lemmas supply the physical interpretation: actual kernels and dressings, factor exposure, and reductions of the retained Holant problem. The mixed-octahedral exits are given by the explicit coefficient and contraction tables in Proposition A.16.

A.5.5 Tetrahedral eight-port tensors with safe pair cards

Proposition A.22 (Tetrahedral eight-port classification)**.** *Let $\mathscr{M}_6, \mathscr{H}_6, \mathscr{S}_6$, and $\mathscr{C}_6$ be the six-port sets defined before Proposition 9.17; let the base matching-product signature be M_0 and the base H-core signature be H_0:*

$$M_0 = (Q_0)_{12}(Q_0)_{34}(Q_0)_{56}, \qquad H_0 = \mathrm{H}_6.$$

Then

$$|\mathscr{M}_6| = 25920, \qquad |\mathscr{H}_6| = 31104, \qquad |\mathscr{S}_6| = 57024, \tag{188}$$

and the union defining $\mathscr{S}_6$ is disjoint. Through either $[M_0]$ or $[H_0]$, the quotient-direction histogram of the other safe points is

$$1^{56855}, \qquad 2^{84}. \tag{189}$$

Here two safe points $F, F' \neq F_0$ have the same quotient direction at $[F_0]$ when $[F + \mathrm{span}_{\mathbb{C}}\{F_0\}] = [F' + \mathrm{span}_{\mathbb{C}}\{F_0\}]$ in the projectivized quotient. Thus the histogram means that 56,855 directions contain one other safe point and 84 directions contain two. An $\mathscr{X}\mathscr{Y}$-endpoint line means that its two nonbase safe points lie in $\mathscr{X}$ and $\mathscr{Y}$, with repetition allowed. The 84 rich lines through $[M_0]$ split into 48 $\mathscr{M}_6\mathscr{M}_6$-endpoint lines and 36 $\mathscr{H}_6\mathscr{H}_6$-endpoint lines, with orbit sizes $36, 36, 12$ under the actual stabilizer subgroup specified below. Those through $[H_0]$ split into 60 $\mathscr{H}_6\mathscr{M}_6$-endpoint lines and 24 $\mathscr{H}_6\mathscr{H}_6$-endpoint lines, with orbit sizes $24, 12, 8, 12, 12, 4, 12$ under its specified actual stabilizer subgroup.

For each of the ten resulting line-orbit representatives and either endpoint orientation, denote the oriented endpoint tensors by A, D and the base safe tensor by F_0, scaled so that $A + D = 2F_0$; define the bridge set

$$\mathcal{B}(A, D) = \{W : W,\ A - 2W,\ D + 2W \in \mathscr{C}_6\}. \tag{190}$$

Then $|\mathcal{B}(A, D)| = 6$, and exactly twelve ordered triples in $\mathcal{B}(A, D)^3$ sum to $(A - D)/2$. The resulting $10 \cdot 2 \cdot 12 = 240$ reconstructed eight-port parents are projectively distinct. For a reconstructed parent g, let

$$\Phi_g : \mathrm{Mat}_2(\mathbb{C}) \longrightarrow (\mathbb{C}^2)^{\otimes 6}, \qquad \Phi_g(K) := \Phi_{12}^g(K),$$

be its distinguished-pair card map. Testing the $\binom{8}{2} \cdot 12 = 336$ *actual tetrahedral cards in a fixed order, stopping only after an outside witness and exhausting all cards for every parent without one, gives*

$\operatorname{rank} \Phi_g$	*outcome*	*number*
2	*an actual card outside* $\mathscr{C}_6$	120
3	*no parent*	0
4	*an actual card outside* $\mathscr{C}_6$	60
4	*all* 336 *cards in* $\mathscr{C}_6$	60.

Every one of the final sixty parents is affine in the normalized computational basis; twenty-four also have an exact nontrivial rank-one flattening. For each of the other 180 *parents there is a specified physical pair and actual tetrahedral kernel producing a card outside* $\mathscr{C}_6$*. The count* 180 *is a count of parents with an outside witness.*

Proof. Both safe sets are generated over $\mathbb{Q}(i)$: matching products are canonicalized directly, while the H_6-orbit is generated to closure under actual local tetrahedral matrices and port permutations. Exact projective comparison gives Equations (188) and (189). The subgroups used for this audit are generated by the following actual operations. At M_0, use the pair stabilizers $(U, U^{-\mathsf{T}})$, $U \in \mathcal{T}$, and the permutations preserving its three-pair matching. At H_0, first enumerate the local two-port tetrahedral actions that permute the four projective Bell signatures. Retain the triples of such pair-block actions that fix $[H_0]$, and add all residual-port permutations that directly fix $[H_0]$. Every selected generator fixes its base tensor projectively. Close their induced permutations on the complete set of 84 rich lines. These explicitly generated stabilizer subgroups give the three plus seven orbits and sizes stated above. In a fixed lexicographic ordering of normalized coefficient vectors, the first line of each orbit is the deterministic line-orbit representative used below; no unlisted generator is assumed. Applying the inverse stabilizer word transports every bridge solution, reconstructed parent, and recorded outside-card witness from that representative to every other line in its orbit, with the physical pair, kernel, port order, and nonzero scalar transported along the same actual operation. Thus the ten representative calculations cover every rich line through either base point, not merely one line from each abstract incidence class.

Fix representatives over $\mathbb{Q}(i)$ on one such line. Since its endpoints are independent, $A + D = 2F_0$ uniquely fixes both endpoint scales. To solve Equation (190), the case $W = 0$ is tested separately. If W and $A - 2W$ are nonzero and nonproportional, their directions lie in an exhaustively generated rich-line bucket and the equation fixes both scales. In the remaining nonzero case write $W = tA$ for a nonzero scalar t; then $D + 2tA \neq 0$, and the third safe direction on $\langle A, D \rangle$ fixes t. Thus the six bridge points and twelve sum triples are a complete symbolic solve over $\mathbb{Q}(i)$, including all zero and complex-scale cases.

For every ordered bridge triple (W_1, W_2, W_3), put $F_j = W_j$ for $1 \leq j \leq 3$, retain the displayed base tensor as F_0, and use the Frobenius-dual quaternion basis to reconstruct

$$g = \sum_{\mu=0}^{3} Q_{12}^{\mu} \otimes F_{\mu}.$$

Exact projective comparison gives 240 distinct parents. The companion replay `certificates/verify_a4_b8.py`, invoked with `-fresh` by the suite runner, then scans physical pairs in lexicographic order and the twelve actual kernels in the point order of Equation (7). For each card it tests zero first and otherwise tests projective membership in the definition-level regenerated set $\mathscr{S}_6$. For an outside parent it records and rechecks the first witness; for a safe parent it necessarily

checks all 336 cards. Affineness of each survivor is tested from the definition: affine-coset support, fourth-root ratios, linear and even mixed quadratic phase coefficients, and the recovered phase on every support point. Factor cuts are tested by exact flattening minors. This exhaustive calculation regenerates the two orbits, their line buckets, bridges, parents, and outside witnesses over $\mathbb{Q}(i)$. The finite classification of the safe parents remains computational. □

Theorem A.23 (Tetrahedral eight-port contractions)**.** *Let $\mathscr{M}_6$ and $\mathscr{H}_6$ be the actual tetrahedral orbits of the base matching product and* H_6*, and put $\mathscr{S}_6 = \mathscr{M}_6 \sqcup \mathscr{H}_6$ and $\mathscr{C}_6 = \{0\} \cup \mathrm{cone}(\mathscr{S}_6)$. Then*

$$|\mathscr{M}_6| = 25920, \qquad |\mathscr{H}_6| = 31104, \qquad |\mathscr{S}_6| = 57024.$$

Every normalized eight-port parent reconstructed from a rich line through either safe base point is actual-equivalent to one of the 240 *audited parents. No audited parent has distinguished card-map rank three;* 180 *have a recorded actual tetrahedral card outside $\mathscr{C}_6$, and the remaining* 60 *have all* 336 *actual cards safe and belong to the affine class $\mathscr{A}$. These assertions include arbitrary nonzero complex scales and both endpoint orientations.*

Proof. This is the finite-level conclusion of Proposition A.22. Its actual-stabilizer covariance transports the ten representative line calculations, their reconstructed parents, and their recorded card witnesses to every rich line through either base point. The actual tetrahedral port dressings and port permutations preserve $\mathscr{A}$, because the corresponding actual tetrahedral binaries are affine, so the affine classification transports as well. □

A.5.6 Deficient mixed-card localization

Proposition A.24 (Platonic contractions of rank less than four)**.** *Let Λ be a signature set, let G be the standard marked tetrahedral form, either unmixed octahedral form, or the marked icosahedral form, and let h be a localized six-port signature directly realizable from Λ, whose card map Φ has rank two or three. Suppose every actual G-card is zero or a nonsingular G-matching product and that two nonzero cards use different matchings. Then the nonkernel domain points contain a rich line disjoint from the projectivized kernel of Φ whose three images are nonzero and use the three distinct residual matchings. For the standard tetrahedral, first octahedral, and icosahedral forms, actual contractions produce one of six support-ten quaternaries q_π, indexed by $\pi \in S_3$, each satisfying*

$$(\mathrm{rank}_{12|34}\, q_\pi, \mathrm{rank}_{13|24}\, q_\pi, \mathrm{rank}_{14|23}\, q_\pi) = (3,3,3). \tag{191}$$

For the second octahedral form they instead produce the corresponding transported quaternaries q'_π. These need not have support size ten, but invertible local transport preserves the same three ranks. Writing $q^\star_\pi = q_\pi$ in the standard forms and $q^\star_\pi = q'_\pi$ in the second octahedral form, one has

$$\text{K-Holant}(\Lambda, q^\star_\pi) \le_{\mathrm{T}} \text{K-Holant}(\Lambda, h) \le_{\mathrm{T}} \text{K-Holant}(\Lambda).$$

The statement includes the second octahedral form in its own actual representatives; similarity by $H_{\mathcal{O}}$ is not used as a gadget.

Proof. Apply the kernel-deleted connectivity assertion Equation (183) to $W = \ker \Phi$, whose dimension is two or one. Along a path between two nonzero cards with different matchings, choose the first matching-changing edge and a rich line containing it. That line cannot meet $\mathbb{P}(\ker \Phi)$: its two-dimensional vector span would otherwise map with rank at most one, making all nonkernel images projectively proportional. The exact deck-line types in Proposition A.19 exclude type $(2,1)$ and mixed four- or five-lines. The selected line is therefore a kernel-avoiding rainbow three-line.

Apply Lemma 9.9. It gives an actual six-port map Ψ with the normalized rainbow line and $\operatorname{rank}\Psi \le 3$. The three exact bridge classifications have only six rank-two maps and their listed rank-four frames, so Ψ has rank two. In the common coefficient normalization,

$$(F_0^\pi, F_1^\pi, F_2^\pi, F_3^\pi) = (C/2, \pi(0, A/2, B/2)), \qquad \pi \in S_3.$$

Using the Frobenius-dual quaternion basis, reconstruct the six-port signature

$$h_\pi := \sum_{\mu=0}^{3} Q_{12}^\mu \otimes F_\mu^\pi.$$

Contract ports $1, 3$ by the actual kernel $Q_0 = I$, and order the surviving ports $(2, 4, 5, 6)$ as $(1, 2, 3, 4)$. Direct expansion gives support size ten for every q_π, and exact row reduction gives Equation (191). All preceding normalizations and the final contraction are actual; undoing them gives the displayed reduction in the three standard forms. In the second octahedral form the coefficient transport identifies $q'_\pi = (H_{\mathcal{O}}^{-1})^{\otimes 4} q_\pi$. The corresponding localizations and contraction kernels are checked separately in actual $\mathcal{O}'$-representatives. Local invertibility preserves the three flattening ranks, but no support-size claim is transported. □

A.5.7 Two-sided root geometry at the minimum quaternary core

For a four-port tensor, contracting either of two complementary pairs gives adjoint linear maps on the matrix space, as in Equation (118). Thus the group-card condition becomes a two-sided preservation condition on a finite set of quaternion-coordinate lines. We classify these maps by rank and reconstruct the original tensor from each case. This is the finite geometric input to Lemma 9.5; the root-system terminology is defined in Section 2.11.3. The fixed actual representatives from Equation (110), written in quaternion coordinates, give the following concrete root-point configurations:

$$\begin{aligned}
\mathcal{R}_{A_4} &:= \{[c_Q(K)] : [K] \in \mathcal{T}\} \cong \mathcal{R}(D_4),\\
\mathcal{R}_{S_4} &:= \{[c_Q(K)] : [K] \in \mathcal{O}\} \cong \mathcal{R}(F_4),\\
\mathcal{R}_{A_5} &:= \{[c_Q(K)] : [K] \in \mathcal{I}\} \cong \mathcal{R}(H_4).
\end{aligned}$$

where the displayed coordinate models are those of Equations (7) to (9) and the displayed isomorphisms mean projective identification in the sense of Sections 2.1.3 and 2.11.3. Their sizes are $12, 24, 60$. Here D_4 is the root system, not the order-four dihedral group $D_4 \cong V_4$ used earlier. Recall the Frobenius-form matrix and define the associated adjoint map by

$$J_{\mathrm{Fr}} := \operatorname{diag}(1, -1, 1, -1), \qquad A^\sharp := J_{\mathrm{Fr}} A^{\mathsf{T}} J_{\mathrm{Fr}}.$$

All calculations are exact over $\mathbb{Q}(i)$ for D_4, F_4 and over $\mathbb{Q}(\phi, i)$ for H_4.

Proposition A.25 (Exact Platonic two-sided preserver classification)**.** *For $G \in \{A_4, S_4, A_5\}$, let $\mathcal{R}_G$ denote the corresponding root-line configuration and define*

$$\mathscr{E}_G = \{A \in \operatorname{End}(\mathbb{C}^4) :\ \forall [r] \in \mathcal{R}_G,\ (Ar = 0 \text{ or } [Ar] \in \mathcal{R}_G),\ (A^\sharp r = 0 \text{ or } [A^\sharp r] \in \mathcal{R}_G)\}. \tag{192}$$

Here r denotes any nonzero representative of $[r]$; the two conditions are independent of that choice. Up to nonzero scalar, the nonzero members of $\mathscr{E}_G$, together with the single zero map recorded in the

rank-zero column, have the exact rank distribution

G	rank 0	rank 1	rank 2	rank 3	rank 4
A_4	1	144	0	0	576
S_4	1	576	2592	0	1152
A_5	1	3600	0	0	7200.

(193)

Rank one reconstructs a binary-pair tensor factor. Rank two occurs only for S_4; all 2592 maps form one orbit under actual local octahedral actions in each current marked form. In the first normal form the reconstructed tensors form the actual local-$\mathcal{O}$ orbit of EQ_4; in the second normal form they admit actual local-$\mathcal{O}'$ dressings to the endpoint-nondegenerate eight-vertex signature

$$M_{\mathcal{O}'}(x) = \mathbf{1}[\mathrm{wt}(x) \equiv 0 \pmod 2](-1)^{\sum_{i<j} x_i x_j}.$$

In the mixed forms the normalized directly reconstructed survivors have the even-entry vectors in Equation (117), all odd entries zero, and ranks $(2,2,2)$. Every rank-four map has one of the forms

$$K \longmapsto UKV \qquad or \qquad K \longmapsto UK^{\mathsf{T}}V, \tag{194}$$

and reconstructs a product of two nonsingular binaries on a crossing matching. The rank counts hold for all four fixed-equality octahedral forms $\mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}$, while their actual rank-two terminals are the ones just stated.

Proof. We first exclude rank three without enumerating arbitrary linear maps. In the following table, the secants are the distinct projective lines spanned by pairs of points of $\mathcal{R}_G$, while the plane sections are the distinct intersections with planes spanned by noncollinear triples of those points; exponents record the number of distinct lines or sections of each size. Exact row reduction gives the complete line and plane-section data

G	secant-line sizes	plane sections
A_4	$2^{18}, 3^{16}$	$3^{12}, 6^{12}$
S_4	$2^{72}, 3^{32}, 4^{18}$	$4^{96}, 9^{24}$
A_5	$2^{450}, 3^{200}, 5^{72}$	$4^{600}, 6^{660}, 15^{60}$.

(195)

For a projective point $k = [\mathbf{k}]$, let $d(k)$ be the number of distinct nonzero projective classes $[\mathbf{r} + \mathrm{span}_{\mathbb{C}}\{\mathbf{k}\}]$ in $\mathbb{P}(\mathbb{C}^4/\mathrm{span}_{\mathbb{C}}\{\mathbf{k}\})$, as $r = [\mathbf{r}]$ ranges over the root points other than k. If k lies on at most one secant containing two root points, the maximum line sizes $3, 4, 5$ give

$$d(k) \geq 10, \qquad d(k) \geq 21, \qquad d(k) \geq 56,$$

respectively. The centers lying on at least two secants are the finite set of pairwise secant intersections. Exact quotient row reduction gives

A_4	$7^{12}, 9^{12}$
S_4	$13^{24}, 19^{96}$
A_5	$31^{60}, 49^{300}, 51^{360}, 55^{600}$.

(196)

A rank-three map is projection from its kernel point to its image plane, and the induced map from the quotient is injective. Hence its quotient directions must fit into a plane section of the target

configuration. The bounds and every count in Equation (196) exceed the plane maxima $6, 9, 15$ in Equation (195). This proves the rank-three column of Equation (193).

We next classify rank two. For $A \in \mathscr{E}_G$, define the projective image line $W := \mathbb{P}(\operatorname{im} A^\sharp)$. Because the root points span $\mathbb{C}^4$, W is a secant line. Moreover $\ker A = (\operatorname{im} A^\sharp)^{\perp_{J_{\mathrm{Fr}}}}$. Projecting the root points modulo this kernel gives the following complete table:

G	$(\lvert W \cap \mathcal{R}_G\rvert, \#$ quotient directions, $\#$ killed points$)$
A_4	$(2,4,2)^{18},\ (3,6,0)^{16}$
S_4	$(2,6,2)^{72},\ (3,6,3)^{32},\ (4,4,4)^{18}$
A_5	$(2,12,2)^{450},\ (3,12,3)^{200},\ (5,10,5)^{72}.$

The image of A is also a projective line, containing at most $3, 4, 5$ root points. The middle entry in every row is larger than that bound except for $(4,4,4)^{18}$ in the octahedral row. The eighteen surviving four-lines form one orbit under actual left and right $\mathcal{O}$-actions. There are eighteen choices for $\mathbb{P}(\operatorname{im} A^\sharp)$, eighteen for $\mathbb{P}(\operatorname{im} A)$. Each surviving four-point set is harmonic in the cross-ratio convention of Section 2.11.3, and exactly eight projective isomorphisms between any two such lines preserve its harmonic pairings. Thus there are exactly eight harmonic projectivities, and consequently

$$18 \cdot 18 \cdot 8 = 2592$$

projective rank-two maps survive. They form one actual local-octahedral orbit of the rank-two endomorphism

$$A_0 := \operatorname{diag}(1,1,0,0).$$

Bilinear-dual reconstruction, including the fixed output X in the transfer convention $L_p(K) = \Phi_p(K)X$, gives an actual local-octahedral equivalent of the quaternary signature

$$q_0(x) := \frac{1}{2}\mathbf{1}[x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod 2].$$

The actual octahedral matrix $W_0 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, with $[W_0] = [Q_1 + Q_3] \in \mathcal{O}$, satisfies $W_0^{\otimes 4}\mathrm{EQ}_4 = 2\mathbf{1}[x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod 2]$ and $W_0^{\otimes 4} q_0 = 4\mathrm{EQ}_4$. It follows that every rank-two survivor has, for some actual matrices U_j with $[U_j] \in \mathcal{O}$,

$$q = \lambda(U_1 \otimes U_2 \otimes U_3 \otimes U_4)\mathrm{EQ}_4, \qquad \lambda \neq 0.$$

Conversely, direct substitution shows that this orbit satisfies the two-sided condition. The complete four-line orbit above leaves no additional projective rank-two degeneration; every surviving tensor has flattening ranks $(2,2,2)$.

For the second octahedral form, analytic conjugation by $H_{\mathcal{O}}$ transports the two-sided linear equations and therefore preserves the count 2592 and the rank pattern, but it does not transport actual local gadgets. Indeed, the actual $(\mathcal{O}')^{\otimes 4}$-orbit of the second-form survivor has 2592 projective tensors with support histogram

$$8^{16}, \qquad 10^{128}, \qquad 12^{384}, \qquad 16^{2064},$$

and hence does not meet the actual $(\mathcal{O}')^{\otimes 4}$-orbit of the support-two signature EQ_4. For the normalized second-form rank-two survivor, use the four actual matrices

$$\begin{aligned} L_1 &= \theta_{\mathcal{O}}(Q_0 + Q_2), \\ L_2 = L_3 &= \theta_{\mathcal{O}}(Q_0 - Q_1 + Q_2 - Q_3), \\ L_4 &= X\theta_{\mathcal{O}}(Q_2 - Q_3). \end{aligned} \tag{197}$$

Exact substitution gives the endpoint-nondegenerate eight-vertex signature even-entry vector

$$(1,-1,-1,-1,-1,-1,-1,1), \tag{198}$$

with every odd entry zero and ranks $(2,2,2)$. Acting before and after this certificate by the actual group proves the stated terminal for the entire second-form orbit. No use of $H_{\mathcal{O}}$ as a gadget is made.

For $\alpha \in \{dt, td\}$, complementary-pair reconstruction of the same normalized coefficient map uses the invertible coefficient operator

$$\mathscr{T}_\alpha = H_\alpha^{\mathsf{T}} \otimes H_\alpha^{-1} \otimes H_\alpha^{-1} \otimes X H_\alpha^{\mathsf{T}} X.$$

Direct exact substitution in q_0 gives, projectively,

$$\mathbf{e}(\mathscr{T}_{dt} q_0) = (1,1,1,1,1,1,1,1), \qquad \mathbf{e}(\mathscr{T}_{td} q_0) = (1,i,i,i,1,1,1,i),$$

with all odd entries zero. These are the direct current-coordinate survivors, not tensors obtained by installing H_α as a gadget. Exact row reduction gives ranks $(2,2,2)$; acting by actual $\mathcal{O}_\alpha$-matrices covers the entire rank-two orbit.

The rank-zero member is the zero map. Because the quaternion basis spans $\mathrm{Mat}_2(\mathbb{C})$, its reconstructed quaternary is zero. For rank one, choose nonzero column vectors $u, v \in \mathbb{C}^4$ and write $A = uv^{\mathsf{T}}$. The two-sided condition forces $[u], [J_{\mathrm{Fr}} v] \in \mathcal{R}_G$; conversely every such pair works. This gives $|G|^2 = 144,576,3600$ projective maps and a rank-one $p \mid \bar{p}$ flattening of the reconstructed tensor.

Finally, a rank-four member of $\mathscr{E}_G$ permutes $\mathcal{R}_G$. Exact root-incidence automorphism classification gives precisely Equation (194), with

$$[V] \in N_{\mathrm{PGL}_2}(G), \qquad [UV] \in G.$$

The projective counts of the two displayed families are

$$2|G|\,|N_{\mathrm{PGL}_2}(G)| = 2 \cdot 12 \cdot 24, \quad 2 \cdot 24^2, \quad 2 \cdot 60^2,$$

namely $576, 1152, 7200$. If the physical card map is $K \mapsto UKV$, undoing the output X gives $q_{uvyz} = U_{yu}(VX)_{vz}$; the transpose case gives $q_{uvyz} = U_{yv}(VX)_{uz}$. In either case the quaternary is a product of two nonsingular binaries on a crossing matching.

The use of $A^\sharp$ in Equation (192) is indispensable. The one-sided A_4 preserver equations have 1536 projective rank-two solutions, but every one fails the complementary $A^\sharp$ equations. Thus the zero in the A_4 rank-two column is a two-sided conclusion, not a one-sided assertion.

The coefficient similarities identify the four rank tables. The formulas Equations (197) and (198) and the two direct mixed reconstructions supply the required physical terminals in the respective current forms.

For all three rows, exact enumeration generates the root configurations and then recovers every secant, plane section, multiple-secant center, and quotient direction by characteristic-zero row reduction: over $\mathbb{Q}(i)$ for D_4, F_4, and over $\mathbb{Q}(\phi, i)$ for H_4. It enumerates all projectivities between the surviving octahedral four-lines, giving $18^2 \cdot 8 = 2592$. For the full-rank case it constructs the complete finite incidence hypergraph of root points, rich lines, and plane sections, enumerates its automorphisms, solves for the inducing projective linear map, and retains the map only after checking both A and $A^\sharp$ on every root point. The resulting automorphism counts equal the cardinalities of the explicit left–right/transpose families for D_4, F_4, H_4, proving completeness of all three full-rank lists rather than merely verifying their displayed members. □

A.5.8 Terminal certificates for the Platonic quaternaries

This subsection records the finite terminal calculation used in Lemma 9.8. Ports of every quaternary are numbered $1, 2, 3, 4$ in their displayed order. If ports r, s are contracted by an actual kernel K, the surviving ordered binary is denoted $B = \partial_{rs}^{K} q$. The pairing is bilinear, with no conjugation. Every group-membership test below is made on the transfer class

$$T_B = BX,$$

and never on $[B]$ alone.

Proposition A.26 (Platonic quaternary exit certificate)**.** *For each quaternary in the following list, let G denote the marked Platonic group from whose row the quaternary is obtained. Every one of the six rank-deficient quaternaries q_π constructed in Proposition A.24 and satisfying Equation* (191)*, one of the three orientation-reversing tetrahedral outputs q_σ specified in Proposition A.17, one of the twenty-four octahedral separator outputs (in any of the four marked forms) specified in Proposition A.16, or one of the thirty icosahedral frame outputs q_{u_1,u_2,u_3} in Equations* (178) *and* (179) *has one of the following directly realized signatures:*

1. *an endpoint-nondegenerate eight-vertex signature with flattening ranks $(2, 2, 2)$;*
2. *a pure generalized equality of arity four;*
3. *a nonsingular ordered binary B whose transfer class $[BX]$ is outside G.*

For either mixed octahedral form the list also includes the six direct current-coordinate counterparts of the rank-deficient outputs. This holds for all four octahedral distinguished-involution normal forms and for both icosahedral coordinate displays. Every such signature uses one quaternary vertex and at most four actual group binaries.

Proof. First consider the six support-ten rank-deficient outputs in Proposition A.24. For each fixed output, exact enumeration of all six port pairs and all actual kernels gives the following table. "Inside" means $[BX]$ is in the displayed group, and "singular" counts nonzero singular cards; zero cards are kept separate.

group	cards per q	zero	inside	outside	singular	one uniform outsider
A_4	72	3	9	60	0	$\partial_{12}^{Q_0} q$
S_4	144	6	108	30	0	$\partial_{24}^{Q_0} q$
A_5	360	3	36	321	0	$\partial_{12}^{Q_0} q$

Thus the same displayed pair and kernel work for all six outputs in each row. The first two rows are computed over $\mathbb{Q}(i)$, and the A_5 row is computed over $\mathbb{Q}(\phi, i)$.

For the second octahedral form, define $\sigma_{\mathcal{O}} := \sqrt{2}$ and the transport matrix $S_{\mathcal{O}} := H_{\mathcal{O}}^{-1}$. The transported rank-deficient outputs are $q'_\pi = S_{\mathcal{O}}^{\otimes 4} q_\pi$. The single actual kernel

$$K'_{\mathrm{def}} = H_{\mathcal{O}}^{-1}(Q_0 - Q_1 - Q_2 - Q_3)H_{\mathcal{O}}, \qquad [K'_{\mathrm{def}}] \in \mathcal{O}' \tag{199}$$

on ports $1, 2$ works for all six. Direct expansion gives $\det(\partial_{12}^{K'_{\mathrm{def}}} q'_\pi) = -3/512$. As π ranges over S_3, the pulled-back transfer directions $H_{\mathcal{O}}(B'X)H_{\mathcal{O}}^{-1}$, in quaternion coordinates, form the multiset

$$[3, -1, 1, 1],\ [1, -1, 0, 1],\ [3, -1, 1, 1],\ [1, -1, 0, 1],\ [3, -1, -1, 1],\ [3, -1, -1, 1].$$

None is in $\mathcal{O}$: its support has size three, or size four with unequal nonzero magnitudes. Hence every corresponding $[B'X]$ is outside $\mathcal{O}' = H_{\mathcal{O}}^{-1}\mathcal{O}H_{\mathcal{O}}$.

For $\mathcal{O}_{dt}$ and $\mathcal{O}_{td}$, the final exact table in Proposition A.16 gives one actual current-group kernel on ports $1,2$, a common nonzero determinant, and all six pulled-back outside-transfer directions. This proves the rank-deficient rows directly in the retained fixed-I coordinates.

The notation here is compatible with the earlier separator transport $q'_R = \mathcal{L}_{\mathcal{O}} q_R$, but the two formulas use different displayed port-orientation conventions. The matching-dependent X-factors in Equations (168) and (200) convert the alternating $H_{\mathcal{O}}^{-1}/H_{\mathcal{O}}^{\mathsf{T}}$ convention of $\mathcal{L}_{\mathcal{O}}$ into the uniform $S_{\mathcal{O}}^{\otimes 4}$ convention after those fixed orientation factors are absorbed into the current $\mathcal{O}'$-representatives. Thus both notations parametrize the same marked second-form coefficient families in their stated port orders; neither equality is being used as a physical gadget.

For the three tetrahedral outputs q_σ specified in Proposition A.17, the one actual kernel

$$K_{\mathcal{T}}^{\mathrm{out}} = Q_0 + Q_1 - Q_2 + Q_3, \qquad [K_{\mathcal{T}}^{\mathrm{out}}] \in \mathcal{T}$$

on ports $1,2$ gives, projectively,

σ	$B = \partial_{12}^{K_{\mathcal{T}}^{\mathrm{out}}} q_\sigma$	BX
$(0,3,1)$	$\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$
$(1,0,3)$	$\left(\begin{smallmatrix} 1 & -i \\ -i & 1 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} -i & 1 \\ 1 & -i \end{smallmatrix}\right)$
$(3,1,0)$	$\mathrm{diag}(1,i)$	$\left(\begin{smallmatrix} 0 & 1 \\ i & 0 \end{smallmatrix}\right)$

All three determinants are nonzero, and exact projective comparison with the twelve points in Equation (7) excludes all three transfer classes.

We next treat the first octahedral form. Define the octahedral sign matrices by

$$S_{\epsilon_1\epsilon_2\epsilon_3} = Q_0 + \epsilon_1 Q_1 + \epsilon_2 Q_2 + \epsilon_3 Q_3.$$

For an ordered frame in Equation (169), its separator is $q = C/2 + W_1$, where W_1 is the first entry. The twelve ordered outputs with

$$W_1 \in \{R_0, R_3, R_{03}^{+-}, R_{03}^{-+}\}$$

are endpoint-nondegenerate eight-vertex signatures on all eight even-parity words and have ranks $(2,2,2)$. The four with $W_1 = R_1$ have support $\{0011, 1100\}$; actual X-dressings on the last two ports turn them into pure generalized equalities. The remaining eight outputs occur twice for each row of the following table:

W_1	kernel on 12	B	$[BX]$
R_{13}^{++}	S_{++-}	$\left(\begin{smallmatrix} 1+i & -i \\ -i & 1-i \end{smallmatrix}\right)$	$[Q_0 + Q_1 - Q_2]$
R_{13}^{--}	S_{+++}	$\left(\begin{smallmatrix} 1-i & -i \\ -i & 1+i \end{smallmatrix}\right)$	$[Q_0 + Q_1 + Q_2]$
R_{01}^{+-}	S_{+++}	$\left(\begin{smallmatrix} 1-i & -1 \\ 1 & 1+i \end{smallmatrix}\right)$	$[Q_1 + Q_2 - Q_3]$
R_{01}^{-+}	S_{++-}	$\left(\begin{smallmatrix} 1+i & -1 \\ 1 & 1-i \end{smallmatrix}\right)$	$[Q_1 - Q_2 - Q_3]$

Every determinant equals 3. The points of $\mathcal{O}$ in Equation (8) have quaternion-coordinate support size one, two, or four, never three, so all four displayed transfer directions are outside $\mathcal{O}$.

For the second octahedral form, we compute the actual kernels in its fixed coordinates. Define the nonzero scalar $c_{\mathcal{O}} := 2\sigma_{\mathcal{O}}(1-i)$. Direct multiplication gives

$$H_{\mathcal{O}}^{\mathsf{T}} X H_{\mathcal{O}} = c_{\mathcal{O}} I, \qquad S_{\mathcal{O}} U S_{\mathcal{O}}^{\mathsf{T}} = c_{\mathcal{O}}^{-1} H_{\mathcal{O}}^{-1} (UX) H_{\mathcal{O}}. \tag{200}$$

Thus $S_{\mathcal{O}}^{\otimes 4}$ transports every $\mathcal{O}$-matching factor to an $\mathcal{O}'$-matching factor. More precisely, if h is the standard-form six-port ancestor with card map Φ, define, in its displayed port order,

$$h' := (H_{\mathcal{O}}^{\mathsf{T}} \otimes H_{\mathcal{O}}^{-1} \otimes S_{\mathcal{O}}^{\otimes 4}) h.$$

The card map Φ' of h' is parametrized by

$$\Phi'(H_{\mathcal{O}}^{-1}KH_{\mathcal{O}}) = S_{\mathcal{O}}^{\otimes 4}\Phi(K). \tag{201}$$

This invertible coefficient correspondence exhausts the second-form tensors. The contractions below use the listed actual current-group kernels, with their ordered ports and nonzero realization scalars.

Define the conjugated octahedral sign matrices by $S'_{\epsilon_1\epsilon_2\epsilon_3} := H_{\mathcal{O}}^{-1}S_{\epsilon_1\epsilon_2\epsilon_3}H_{\mathcal{O}}$. For the twelve ordered outputs in the next table, direct contraction of $q' = S_{\mathcal{O}}^{\otimes 4}q$ on ports $1, 2$ gives $\det B' = -3/256$:

W_1	actual kernel	$H_{\mathcal{O}}(B'X)H_{\mathcal{O}}^{-1}$
R_{13}^{++}	S'_{+++}	$[2Q_0 + Q_1 + Q_2]$
R_{13}^{--}	S'_{+-+}	$[Q_1 + Q_2 + 2Q_3]$
R_{03}^{+-}	S'_{++-}	$[Q_0 + 2Q_1 + Q_3]$
R_{03}^{-+}	S'_{+++}	$[Q_0 + 2Q_2 + Q_3]$
R_{01}^{+-}	S'_{++-}	$[2Q_0 + Q_1 - Q_2]$
R_{01}^{-+}	S'_{+++}	$[2Q_0 - Q_1 + Q_2]$

Each row occurs for two orderings. Every pulled-back direction has three nonzero quaternion coordinates of magnitudes $2, 1, 1$, so it is outside $\mathcal{O}$, proving $[B'X] \notin \mathcal{O}'$.

The remaining twelve ordered $\mathcal{O}'$-outputs have $W_1 = R_0, R_1, R_3$, four of each. Actual local dressings turn them into endpoint-nondegenerate eight-vertex signatures. In the even-word order

$$(0000, 0011, 0101, 0110, 1001, 1010, 1100, 1111),$$

the exact outputs are

W_1	dressing	even coefficients, up to scale
R_0	$(S'_{++-})^{\otimes 4}$	$(1, 1, -1, 1, 1, -1, 1, 1)$
R_1	identity	$(1, -1, 1, 1, 1, 1, -1, 1)$
R_3	$(S'_{+++})^{\otimes 4}$	$(1, -1, 1, 1, 1, 1, -1, 1)$

All odd-weight coefficients vanish, both endpoints are nonzero, and local invertible dressings preserve ranks $(2, 2, 2)$.

The two mixed full-rank rows are the direct tables in Proposition A.16. For $\mathcal{O}_{dt}$ they partition into endpoint and outside-binary exits; for $\mathcal{O}_{td}$ they partition into endpoint, pure-generalized-equality, and outside-binary exits. The tables cover every possible first frame entry, hence all twenty-four orderings, and use only actual matrices of the current mixed group.

Finally, for each of the thirty icosahedral outputs in Equation (179), use the one actual representative

$$L_{\mathcal{I}} = Q_0 - \tau^2 Q_1 + \tau Q_3 \qquad \text{with} \qquad [L_{\mathcal{I}}] \in \mathcal{I}$$

on ports $1, 2$. Exact arithmetic over $\mathbb{Q}(\phi, i)$ proves that all thirty resulting binaries are nonsingular, their thirty projective transfers $[BX]$ are distinct, and none belongs to the complete sixty-point set $\mathcal{I}$. Enumerating all $30 \cdot 6 \cdot 60 = 10800$ actual cards gives

$$\text{zero } 180, \qquad \text{inside } 1260, \qquad \text{outside } 9360, \qquad \text{nonzero singular } 0.$$

For the opposite display, use the fixed-equality dictionary Lemma 9.2: replace the two standard kernels by their actual κ_D-images. Card covariance conjugates the complete transfer group and preserves nonsingularity and group exclusion, while the retained literal I is untouched. This exhausts every family in the proposition.

The displayed kernels and dressing tables prove the listed bounded exits by exact matrix multiplication. The complete terminal histograms and the completeness claims used earlier are also reproduced by `certificates/verify_a4_s4_certificates.py`, `certificates/verify_s4_second_form.py`, `certificates/verify_a5_certificate.py`, and `certificates/verify_platonic_extended.py` over the number fields specified above. In particular, the full preserver and hereditary eight-port classifications retain the exhaustive computations described in their own proofs. □

Theorem A.27 (Platonic bridge interface)**.** *Under the normalized rainbow and separated-sign hypotheses of Equations* (127)*,* (129) *and* (130)*, the following complete bridge statements hold.*

(i) *For each marked octahedral form, there are exactly six rank-deficient maps and* 24 *full-rank maps, the latter being the six orderings of each of*

$$\{R_0, R_1, R_3\}, \quad \{R_0, R_{13}^{++}, R_{13}^{--}\}, \quad \{R_1, R_{03}^{+-}, R_{03}^{-+}\}, \quad \{R_3, R_{01}^{+-}, R_{01}^{-+}\}.$$

The fixed actual kernel is $K_{\mathcal{O}} = Q_0 + Q_1$ in the first form and $K_{\mathcal{O}'} = \theta_{\mathcal{O}}(K_{\mathcal{O}})$ in the second, while $K_{\mathcal{O}_{dt}} = \theta_{dt}(K_{\mathcal{O}})$ and $K_{\mathcal{O}_{td}} = \theta_{td}(K_{\mathcal{O}})$ are the two mixed-form kernels. In the two unmixed forms, actual current-group dressings send every full-rank map to an endpoint-nondegenerate eight-vertex signature of ranks $(2, 2, 2)$*. In either mixed form, the fixed kernel and the direct current-coordinate tables give one of the three exits: such an eight-vertex signature, a pure generalized equality, or a nonsingular outside-group binary.*

(ii) *For the internal tetrahedral form, the bridge set is $\{0, A/2, B/2, R_0, R_1, R_3\}$, and the sum equation has exactly six rank-deficient and six full-rank solutions. Write*

$$\Sigma_+ = \{(0,1,3), (1,3,0), (3,0,1)\}, \qquad \Sigma_- = \{(0,3,1), (1,0,3), (3,1,0)\}.$$

For every $\sigma \in \Sigma_+$, all 180 *actual pair cards of the reconstructed parent are safe and the parent lies in $\mathscr{A}$. For every $\sigma \in \Sigma_-$, contraction of ports* $1, 3$ *by the fixed actual kernel $Q_0 = I$, followed by four actual dressings, gives an endpoint-nondegenerate eight-vertex signature of ranks* $(2, 2, 2)$*.*

(iii) *The sixty displayed projective classes form a projective group isomorphic to A_5 and are closed under induced projective multiplication, inverse, and transpose. Its bridge has exactly six rank-deficient and* 30 *full-rank solutions, the latter being the six orderings of the five displayed frames $\mathcal{F}_0, \ldots, \mathcal{F}_4$. The fixed actual kernel $K_{\mathcal{I}} := \phi Q_1 - \tau Q_2 + Q_3$ gives a quaternary of ranks* $(3, 3, 3)$*, and the fixed subsequent actual kernel $L_{\mathcal{I}} := Q_0 - \tau^2 Q_1 + \tau Q_3$ gives a nonsingular binary whose transfer lies outside $\mathcal{I}$, for all* 30 *solutions. The statement includes arbitrary complex scales and cancellations. For $\mathcal{I}_-$, the actual kernels $K_{\mathcal{I}_-} = \kappa_D(K_{\mathcal{I}})$ and $L_{\mathcal{I}_-} = \kappa_D(L_{\mathcal{I}})$ give the corresponding direct fixed-I identities.*

Proof. Clause (i) is the exact octahedral bridge Equation (169) together with the separate physical certificate Proposition A.16. Clause (ii) is Equations (171) and (172) and the all-card, affine-phase, and actual-orientation checks in Appendix A.5.2, including Proposition A.17. Clause (iii) is the exact group, bridge, frame, and rank calculation in Appendix A.5.3, followed by the uniform physical exit in Proposition A.26; Lemma 9.2 gives the opposite display with actual κ_D-kernels and leaves the supplied equality unchanged. □

Theorem A.28 (Classification of Platonic quaternaries)**.** *Let $G \in \{\mathcal{T}, \mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}, \mathcal{I}, \mathcal{I}_-\}$. The finite quaternary certificates have the following directly usable form.*

(i) *Let q be quaternary and define $L_p(K) = \Phi_p^q(K)X$. If, for every one of the six pairs p and every fixed actual representative K_g,*

$$L_p(K_g) = 0 \qquad or \qquad [L_p(K_g)] \in G,$$

then, for each pair p, put $r_p := \operatorname{rank} L_p$. One has $r_p \neq 3$. If $r_p = 0$, then $q = 0$; if $r_p \in \{1, 4\}$, then q has a binary-pair factorization; and $r_p = 2$ can occur only for the octahedral forms: it is actually dressable to EQ_4 for $\mathcal{O}$, and to an endpoint-nondegenerate eight-vertex signature of ranks $(2, 2, 2)$ for $\mathcal{O}'$, $\mathcal{O}_{dt}$, and $\mathcal{O}_{td}$, with the mixed vectors in Equation (117).

(ii) *Let $h \neq 0$ have arity six and let Φ_p^h have rank two or three. Suppose, for every $g \in G$, the card $\Phi_p^h(K_g)$ formed with its fixed actual representative is zero or a nonsingular G-matching product, with two nonzero cards in this p-family using different residual matchings. Actual localization produces a quaternary of ranks $(3, 3, 3)$, and an actual contraction of that quaternary gives a nonsingular binary B with $[BX] \notin G$. This includes the separately verified $\mathcal{O}'$ and mixed representatives and both icosahedral displays.*

(iii) *For each fixed G, every corresponding enumerated rank-deficient quaternary has the following conclusion. The same conclusion holds for the tetrahedral-orientation outputs when $G = \mathcal{T}$, the octahedral-frame outputs when $G \in \{\mathcal{O}, \mathcal{O}', \mathcal{O}_{dt}, \mathcal{O}_{td}\}$, and the icosahedral-frame outputs when $G \in \{\mathcal{I}, \mathcal{I}_-\}$. Each such quaternary directly realizes one of: an endpoint-nondegenerate eight-vertex signature of ranks $(2, 2, 2)$, a pure generalized equality of arity four, or a nonsingular binary B with $[BX] \notin G$. Each realization uses the quaternary and at most four actual G-binaries.*

Proof. For clause (i), use Lemma 9.2 for the $\mathcal{I}_-$ coefficient calculation, replacing every physical kernel by its actual κ_D-image and leaving the literal I fixed. Fix a pair p, and let A_p be the matrix of L_p in the corresponding quaternion root coordinates. The card conditions at p give the first condition in Equation (192). For the complementary pair $\bar{p}$, let $\mathbf{R}_X$ be the coordinate matrix of right multiplication $K \mapsto KX$. Direct complementary-pair reshaping gives $A_{\bar{p}} = \mathbf{R}_X A_p^{\sharp} \mathbf{R}_X$; the fixed X-reversals permute the root configuration, so the card conditions at $\bar{p}$ give the second condition. Hence A_p belongs to the corresponding one of $\mathscr{E}_{A_4}, \mathscr{E}_{S_4}, \mathscr{E}_{A_5}$, and Proposition A.25 gives all stated rank and reconstruction alternatives, including the separately verified $\mathcal{O}'$ terminal and the direct mixed reconstructions above.

Clause (ii) combines the actual localization and rank calculation in Proposition A.24 with the corresponding uniform outside-binary row of Proposition A.26. Clause (iii) is the representative-wise terminal statement of the latter proposition. The card and network covariance in Lemma 9.2 supplies all three clauses for $\mathcal{I}_-$ directly with actual κ_D-kernels. □

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